



OptiX Metro 6100 WDM Multi-Service Transmission System
V100R008

Product Description

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About This Document

Related Versions

The following table lists the product versions related to this document.

Product Name	Version
OptiX Metro 6100	V100R008
OptiX iManager T2000	V200R007C03

Intended Audience

The intended audiences of this document are:

- Network Planning Engineer
- Data Configuration Engineer
- System Maintenance Engineer

Symbol Conventions

The following symbols may be found in this document. They are defined as follows

Symbol	Description
 DANGER	Indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.
 WARNING	Indicates a hazard with a medium or low level of risk which, if not avoided, could result in minor or moderate injury.

Symbol	Description
 CAUTION	Indicates a potentially hazardous situation that, if not avoided, could cause equipment damage, data loss, and performance degradation, or unexpected results.
 TIP	Indicates a tip that may help you solve a problem or save you time.
 NOTE	Provides additional information to emphasize or supplement important points of the main text.

GUI Conventions

Convention	Description
Boldface	Buttons, menus, parameters, tabs, window, and dialog titles are in boldface . For example, click OK .
>	Multi-level menus are in boldface and separated by the ">" signs. For example, choose File > Create > Folder .

Update History

Updates between document versions are cumulative. Therefore, the latest document version contains all updates made to previous versions.

Updates in Issue 09 (2012-06-30) Based on Product Version OptiX Metro 6100 V100R008

Some bugs in the manual of the previous version are fixed.

The description of "Automatic Laser Shutdown" has been updated.

Updates in Issue 08 (2010-08-15) Based on Product Version OptiX Metro 6100 V100R008

The update of contents is described as follows.

Some bugs in the manual of the previous version are fixed.

Add "Impact of PMD" in "Networking and Design Considerations".

Add the information about C9LWC1, CBETMXS, CBTMRS, CBELOGS, CBLBFS, C9TMX, C9TMXS, C9ELOG, C9ELOGS, C9LOG, C9LOGS, CAETMX, CAETMXS, C9SC1, C9SC2, L2TC1 and L2TC2.

Chapter	Update Description
Chapter 13 Technical Specifications	Updates the board and system parameters.

Updates in Issue 07 (2009-10-30) Based on Product Version OptiX Metro 6100 V100R008

The update of contents is described as follows.

Some bugs in the manual of the previous version are fixed.

Chapter	Update Description
Chapter 13 Technical Specifications	Updates the board and system parameters.

Updates in Issue 06 (2009-06-30) Based on Product Version OptiX Metro 6100 V100R008

The update of contents is described as follows.

Some bugs in the manual of the previous version are fixed.

Chapter	Update Description
Chapter 13 Technical Specifications	Updates the board and system parameters.

Updates in Issue 05 (2009-04-15) Based on Product Version OptiX Metro 6100 V100R008

The update of contents is described as follows.

Some bugs in the manual of the previous version are fixed.

Chapter	Update Description
Chapter 3 Product Features	● Upgrades the description in 40G transmission system. ● Adds the description about DQPSK technology.
Chapter 4 Hardware Architecture	● Adds the descriptions of the C9LU40S, C9LUR40S, C9TMX40S, C9CP40, C9M40, C9D40, C9V40 board. ● Adds the descriptions of the C9TMX, C9TMXS, CAETMX, CAETMXS, C9FIU, C9OAU, C9OPU, C9OBU board.
Chapter 4 Hardware Architecture	Upgrades the description of the board in "Function Boards".
Chapter 9 Protection	Upgrades the description of extended intra-board wavelength protection.

Updates in Issue 04 (2008-11-15) Based on Product Version OptiX Metro 6100 V100R008

The update of contents is described as follows.

Some bugs in the manual of the previous version are fixed.

Chapter	Update Description
Chapter 13 Technical Specifications	Updates the board and system parameters.

Updates in Issue 03 (2008-08-07) Based on Product Version OptiX Metro 6100 V100R008

The update of contents is described as follows.

Chapter	Update Description
Chapter 3 Product Features	Adds the descriptions of the LQM2 board.
Chapter 4 Hardware Architecture	Adds the descriptions of the LQM2 board. Upgrade the version map of the board.
Chapter 13 Technical Specifications	Updates the board and system parameters.

Updates in Issue 02 (2008-03-14) Based on Product Version OptiX Metro 6100 V100R008

The update of contents is described as follows.

Chapter	Update Description
Chapter 2 System Functions	Updates the service type that the system can access.
Chapter 3 Product Features	Adds the descriptions of optical layer supervisory, ntp technology, package loading, simulated package loading.
Chapter 4 Hardware Architecture	Adds the descriptions of the LW40, LR40, WSMD4 and ITL boards.
Chapter 11 Operation, Administration and Maintenance	Adds the descriptions of NE security management features.
Chapter 13 Technical Specifications	Updates the board and system parameters.

Updates in Issue 01 (2007-09-22) Based on Product Version OptiX Metro 6100 V100R008

This is the first commercial release of the OptiX Metro 6100 V100R008.

One new chapter is added: Chapter 12 "Factors in Networking Design."

Chapter 4 "System Architecture" of the last version is divided into Chapter 4 "Hardware Architecture" and Chapter 5 "Software Architecture."

Chapter 5 "DWDM System Configuration" and Chapter 6 "CWDM System Configuration" are renamed as Chapter 6 "NE Types and Signal Flow of a DWDM System" and Chapter 7 NE Types and Signal Flow of a CWDM System."

The update of contents is described as follows.

Chapter 8 "APE, ALC and IPA Application" is renamed as Chapter 10 "Optical Power Management."

Chapter 9 "Application and Networking of the ROADM and GE ADM Features" is renamed as Chapter 8 "Grooming of Wavelengths and Services."

The update of contents is described as follows.

Chapter	Update Description
Chapter 6 NE Types and Signal Flow of a DWDM System	Adds the configuration rules..

Chapter	Update Description
Chapter 7 NE Types and Signal Flow of a CWDM System	Adds the configuration rules..
Chapter 4 Hardware Architecture	Adds the descriptions of the ST1, ST2, ELOGS and LOMS boards.
Chapter 9 Protection	Adds the descriptions of OWSP (DCP). Adds the description of VLAN SNCP protection. Adds the description of the application scheme about the LAM.
Chapter 13 Technical Specifications	Updates the board and system parameters.

Updates in Issue 06 (2008-07-15) Based on Product Version OptiX Metro 6100 V100R007 and OptiX Metro 6040 V300R002

Board and system parameters have been updated

Some bugs in the manual of the previous version are fixed.

Updates in Issue 05 (2007-12-03) Based on Product Version OptiX Metro 6100 V100R007 and OptiX Metro 6040 V300R002

Some bugs in the manual of the previous version are fixed.

Updates in Issue 04 (2007-09-30) Based on Product Version OptiX Metro 6100 V100R007 and OptiX Metro 6040 V300R002

Some bugs in the manual of the previous version are fixed.

Updates in Issue 03 (2007-06-20) Based on Product Version OptiX Metro 6100 V100R007 and OptiX Metro 6040 V300R002

One new chapter is added: Chapter 9 "Application and Networking of the ROADM and GE ADM Features."

The update of contents is described as follows.

Chapter	Update Description
Chapter 2 System Functions	Adds the descriptions of LPT and PRBS. Adds three communication modes of the system. Updates the transmission distance.
Chapter 3 Product Features	Adds more details of the SAN feature. Adds the descriptions of ODB technology. Updates the descriptions of Ethernet management capability.
Chapter 4 System Structure	Adds the descriptions of the ETMX, ETMXS, ELOG, LQM, LBF, TBE, WSM9, WSD9, RPC and DCP.
Chapter 5 DWDM System Configuration	Adds the system configuration of ROADM with the WSD9 and RMU9 boards.
Chapter 7 Protection	Adds more details of the application for each protection, such as trigger conditions, dependent alarms, and configuration rules.
Chapter 8 APE, ALC and IPA Application	Updates the application of the IPA.
Chapter 10 Technical Specifications	Updates the board and system parameters.

Updates in Issue 02 (2007-04-23) Based on Product Version OptiX Metro 6100 V100R007 and OptiX Metro 6040 V300R002

Some bugs in the manual of the previous version are fixed.

Updates in Issue 01 (2007-02-14) Based on Product Version OptiX Metro 6100 V100R007 and OptiX Metro 6040 V300R002

This is the first commercial release of the OptiX Metro 6100 V100R007 and OptiX Metro 6040 V300R002.

The technical details of the OptiX Metro 6040 are added.

The system configuration is divided into two chapters according to the applications of the DWDM system and the CWDM system.

Three new chapters are added: Chapter 8 "APE, ALC and IPA Application", Appendix D "Generic Technical Principles" and Appendix F "Uninterruptible Power Modules."

The update of contents is described as follows.

Chapter	Update Description
Chapter 3 Product Features	Adds more details of the 40G transmission system. Adds the descriptions of IP over DCC. Adds the descriptions of the Raman amplification technology. Updates the descriptions of the automatic laser shutdown.
Chapter 4 System Structure	Adds the descriptions of the ETMX, ETMXS, ELOG, LQM, LBF, TBE, WSM9, WSD9, RPC and DCP. Adds the 40G OTU. Deletes the descriptions of the AP4 and EC8.
Chapter 5 DWDM System Configuration	Adds the system configuration of ROADM with the WSD9 and WSM9 boards.
Chapter 7 Protection	Adds the details of the application for each protection.
Chapter 10 Technical Specifications	Updates the board and system parameters.

Updates in Issue 03 (2007-07-02) Based on Product Version OptiX Metro 6100 V100R006

Some bugs in the manual of the previous version are fixed.

Updates in Issue 02 (2006-10-10) Based on Product Version OptiX Metro 6100 V100R006

Some bugs in the manual of the previous version are fixed.

The board and system parameters are updated.

Updates in Issue 01 (2006-06-18) Based on Product Version OptiX Metro 6100 V100R006

This is the first commercial release of the OptiX Metro 6100 V100R006.

Two new chapters are added: Chapter 7 "Operation, Administration and Maintenance" and Appendix C "Compliant Standards."

The update of contents is described as follows.

Chapter	Update Description
Chapter 1 Network Application	Adds more details of the technology that is used to achieve the flexible networking.
Chapter 2 System Functions	Adds the descriptions of SNMP and NTP. Adds the service convergence capability of the FDG, L4G and ESG.
Chapter 3 Product Features	Adds more details of the GE ADM and ROADM. Adds the descriptions of the Ethernet management capability. Adds the descriptions of the OTN processing capability. Adds more details of the IPA, ALC, and APE.
Chapter 4 System Structure	Adds the descriptions of the FDG, L4G and EGS. Deletes the descriptions of the TC1 and TC2.
Chapter 5 System Configuration	Adds the signal flow and board information of the OADM station with the DWC.
Chapter 6 Protection	Deletes the principle of the outdate optical channel protection.
Chapter 8 Technical Specifications	Updates the board and system parameters.

Updates in Issue 06 (2006-12-10) Based on Product Version OptiX Metro 6100 V100R005

Several bugs in the manual of the previous version are fixed.

The specifications of the boards are updated.

A new chapter is added: Chapter 07 "Operation, Administration and Maintenance."

Updates in Issue 05 (2006-04-30) Based on Product Version OptiX Metro 6100 V100R005

The former manual version is T2-040206-20060430-C-5.05.

Several bugs in the manual of the previous version are fixed.

The description of the DWDM OADM node with OADM boards (or M40/V40 and D40 boards) and DWC boards is added.

The description of the ALS function is added.

Updates in Issue 04 (2006-01-10) Based on Product Version OptiX Metro 6100 V100R005

The former manual version is T2-040206-20060110-C-5.04.

Several bugs in the manual of the previous version are fixed.

Updates in Issue 03 (2005-11-10) Based on Product Version OptiX Metro 6100 V100R005

The former manual version is T2-040206-20051110-C-5.03 The description of ROADM is added.

The description of optical wavelength shared protection is added.

The description of inter-subrack 1+1 optical channel protection is added.

Several bugs in the manual of the previous version are fixed.

Updates in Issue 02 (2005-09-05) Based on Product Version OptiX Metro 6100 V100R005

The former manual version is T2-040206-20050905-C-5.02.

Several bugs in the manual of the previous version are fixed.

Updates in Issue 01 (2005-02-25) Based on Product Version OptiX Metro 6100 V100R005

The former manual version is T2-040206-20050225-C-5.01.

This is the first commercial release of the OptiX Metro 6100 V100R005.

1 Network Application

About This Chapter

This chapter describes the position and basic topologies of the product.

[1.1 Position in Networks](#)

The product applies to metropolitan area backbone network, local network, broadband data network and storage area network.

[1.2 Classification of System Types](#)

To meet the requirements of different areas, users and investing environments, the system is available in two types.

[1.3 Classification of Bands](#)

The system splits wavelength into different bands according to a certain principle. Systems of different types use different bands.

[1.4 Networking and Applications](#)

The product can apply to the following networking topologies: point-to-point network, chain network, ring network. The product can also work with other equipments.

1.1 Position in Networks

The product applies to metropolitan area backbone network, local network, broadband data network and storage area network.

The OptiX Metro 6100 WDM Multi-Service Transmission System (the OptiX Metro 6100 for short) uses dense wavelength division multiplexing (DWDM) or coarse wavelength division multiplexing (CWDM) to achieve transparent transmission with a wide bandwidth and a large capacity.

The OptiX Metro 6100 supports a variety of networking modes, including:

- Point-to-point network
- Chain network
- Ring network

It may also work with the OptiX Metro 6040 to offer a complete Metro WDM solution.

With each of its nodes capable of wavelength grooming, the OptiX Metro 6100 has the following features:

- Easy capacity expansion
- Flexible service access
- High bandwidth utility
- High reliability

Currently, the OptiX Metro 6100 can multiplex up to 40 service channels in a fiber. It can transmit up to 40 carrier signals of different wavelengths.

The normal rate of each signal is 10 Gbit/s or lower and the maximum rate of each signal can be 40 Gbit/s. The OptiX Metro 6100 achieves 1600 Gbit/s transmission in two directions with two fibers.

The OptiX Metro 6100 transmits the unidirectional services over one fiber. A bidirectional transmission is achieved by two fibers. One of them is used to transmit and the other is used to receive.

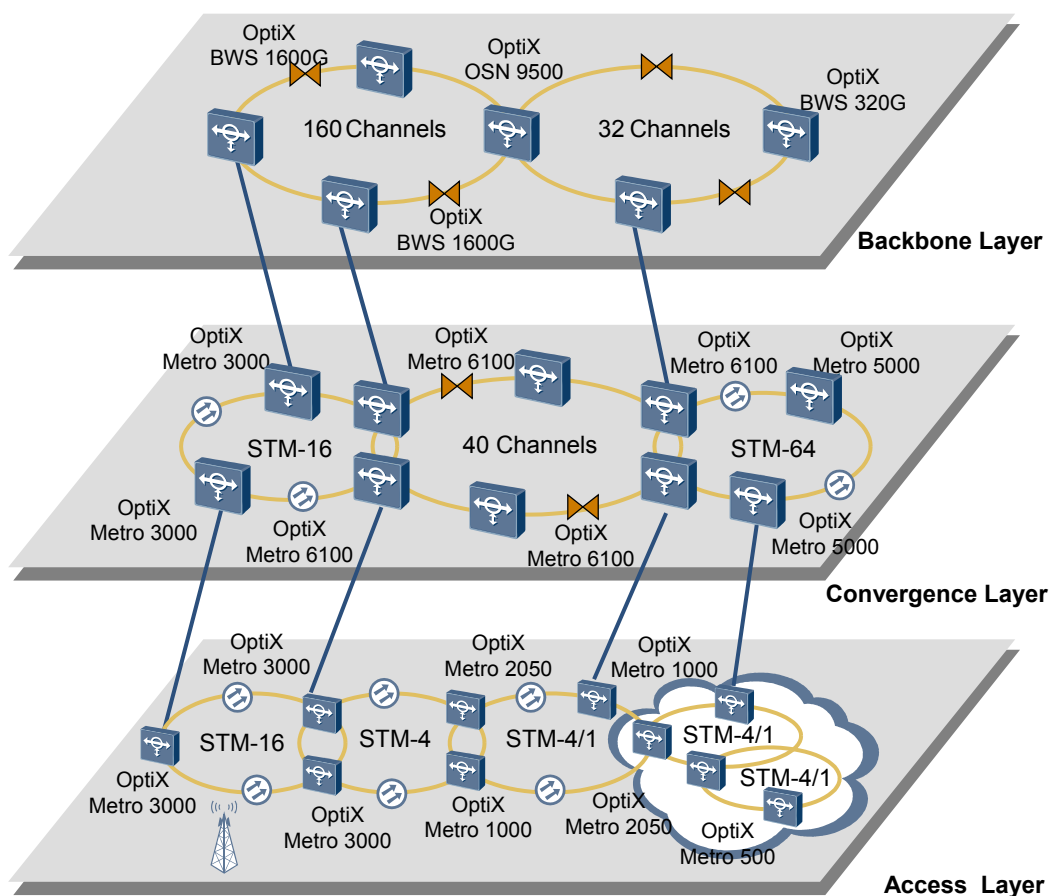
The OptiX Metro 6100 is highly reliable.

It also supports topologies such as chain and ring. The flexible networking is achieved by using:

- Reliable multiplexer/demultiplexer
- Erbium-doped optical fiber amplifier
- Channel equalization technology
- SuperWDM technology
- Dispersion compensation technology
- Universal and centralized network management system

Figure 1-1 shows the position of the OptiX Metro 6100 in the overall hierarchy of a network.

Figure 1-1 Position of the OptiX Metro 6100 system in the network hierarchy



1.2 Classification of System Types

To meet the requirements of different areas, users and investing environments, the system is available in two types.

To meet requirements from different users, regions and investment environment, the OptiX Metro 6100 is classified into the following two systems:

- Dense wavelength division multiplexing (DWDM) system
- Coarse wavelength division multiplexing (CWDM) system

1.3 Classification of Bands

The system splits wavelength into different bands according to a certain principle. Systems of different types use different bands.

The OptiX Metro 6100 DWDM system adopts the C band in the fiber communication window. The minimum channel spacing in C band is 100 GHz. The number of available wavelengths is 40. The wavelength range is from 192.10 THz to 196.00 THz (1529.55 nm to 1560.61 nm).

The OptiX Metro 6100 CWDM system adopts the channel spacing of 20 nm. The operating wavelength range for 2.5 Gbit/s CWDM system is from 1311 nm to 1611 nm.

1.4 Networking and Applications

The product can apply to the following networking topologies: point-to-point network, chain network, ring network. The product can also work with other equipments.

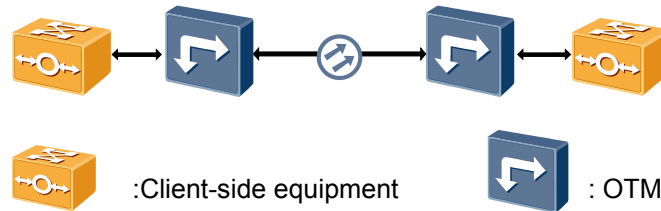
1.4.1 Point-to-Point Network

Point-to-point networking is the simplest networking mode for the product.

It is used for end-to-end transmission of services. The point-to-point networking mode is the basic mode. Other network modes are generated based on this mode.

Figure 1-2 shows a point-to-point network that is composed of the OptiX Metro 6100.

Figure 1-2 Point-to-point network



NOTE

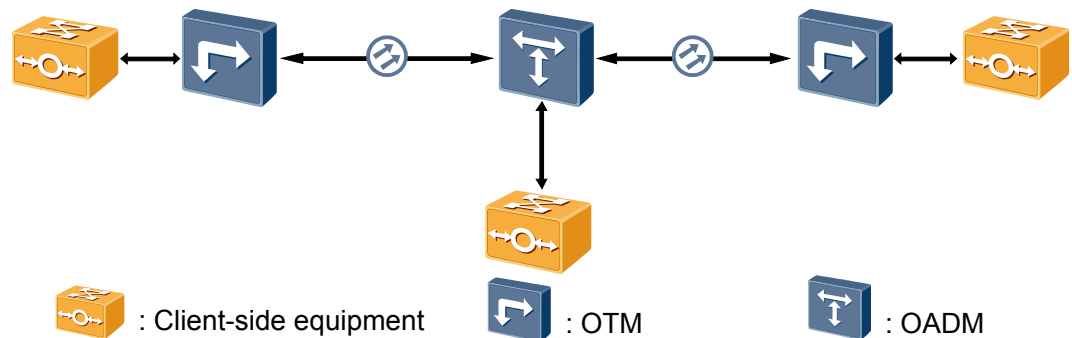
The OLA is omitted in **Figure 1-2**.

1.4.2 Chain Network

A chain network with OADM(s) is suitable when it is required to add or drop some wavelengths while passing others on.

Figure 1-3 shows a chain network that is composed of the OptiX Metro 6100.

Figure 1-3 Chain network





NOTE

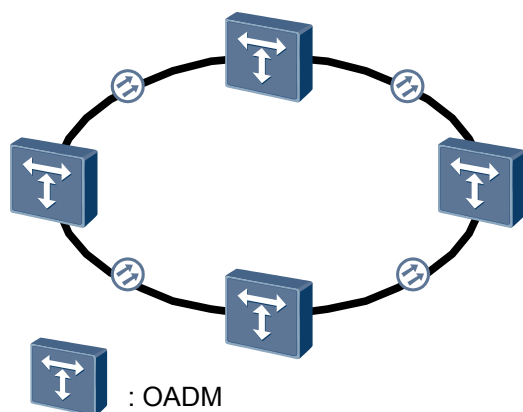
The OLA is omitted in [Figure 1-3](#).

1.4.3 Ring Network

Network security and reliability are two key factors that indicate the quality of services. Because of its excellent survivability, the ring network is a dominant networking mode in metropolitan area network (MAN) DWDM network planning. Many complicated networks can be built based on the ring network, for example, tangent rings, intersecting rings, and ring with chain.

[Figure 1-4](#) shows a ring network that is composed of the OptiX Metro 6100.

Figure 1-4 Ring network

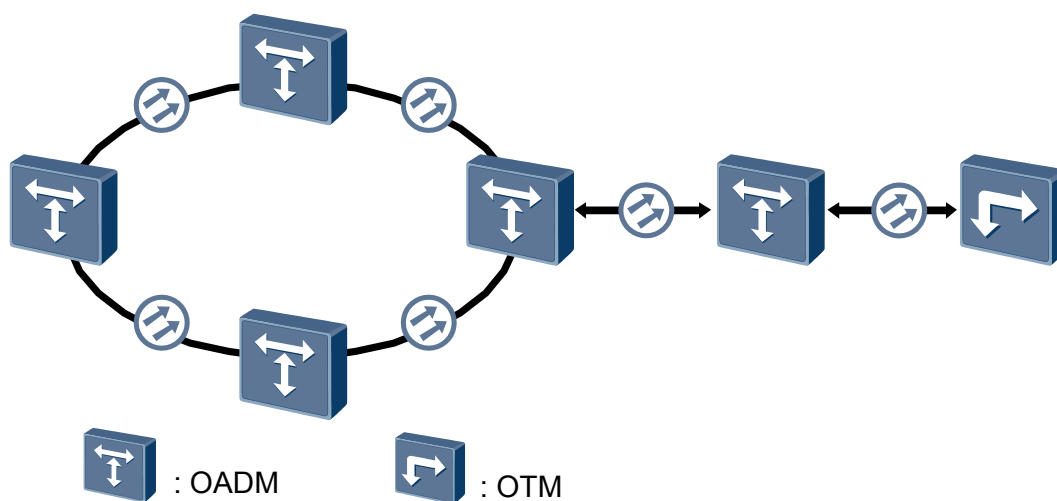


NOTE

The OLA is omitted in [Figure 1-4](#).

[Figure 1-5](#) shows the ring-with-chain network that is composed of the OptiX Metro 6100.

Figure 1-5 Ring with chain

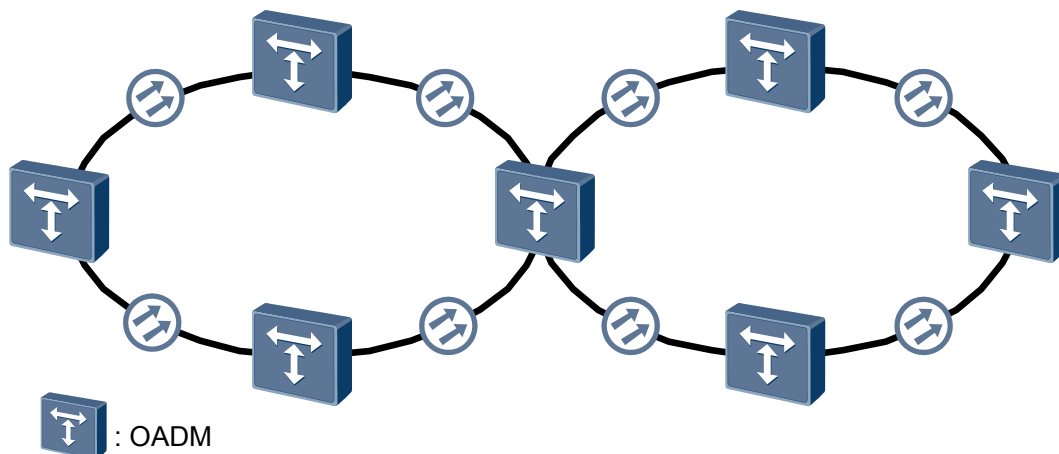


 NOTE

- The OptiX Metro 6100 serves as the central node in a ring with chain network.
- The OLA is omitted in [Figure 1-5](#).

[Figure 1-6](#) shows the tangent-ring network that is composed of the OptiX Metro 6100.

Figure 1-6 Tangent rings

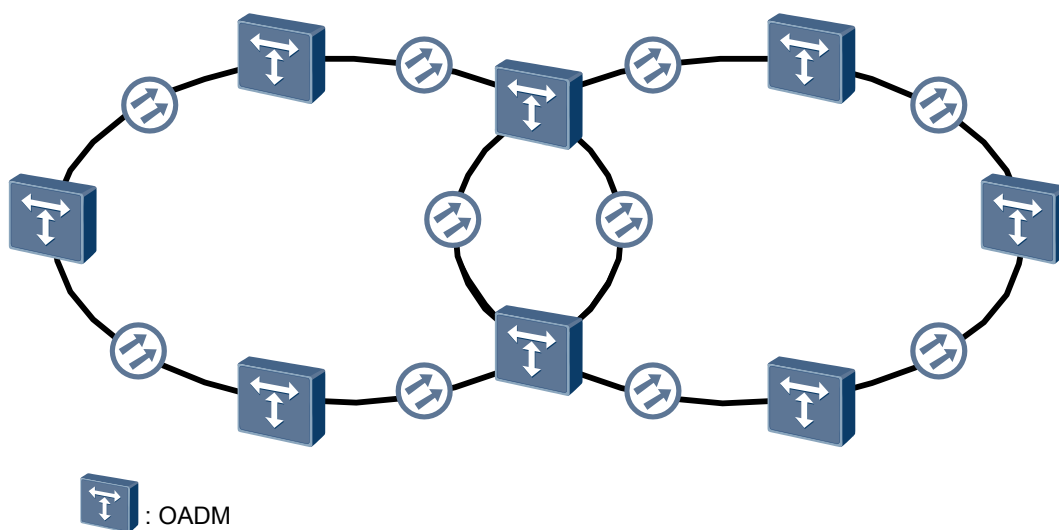


 NOTE

- The OptiX Metro 6100 serves as the central node in a tangent-ring network.
- The OLA is omitted in [Figure 1-6](#).

[Figure 1-7](#) shows the intersecting-ring network composed of the OptiX Metro 6100 system.

Figure 1-7 Intersecting rings



 NOTE

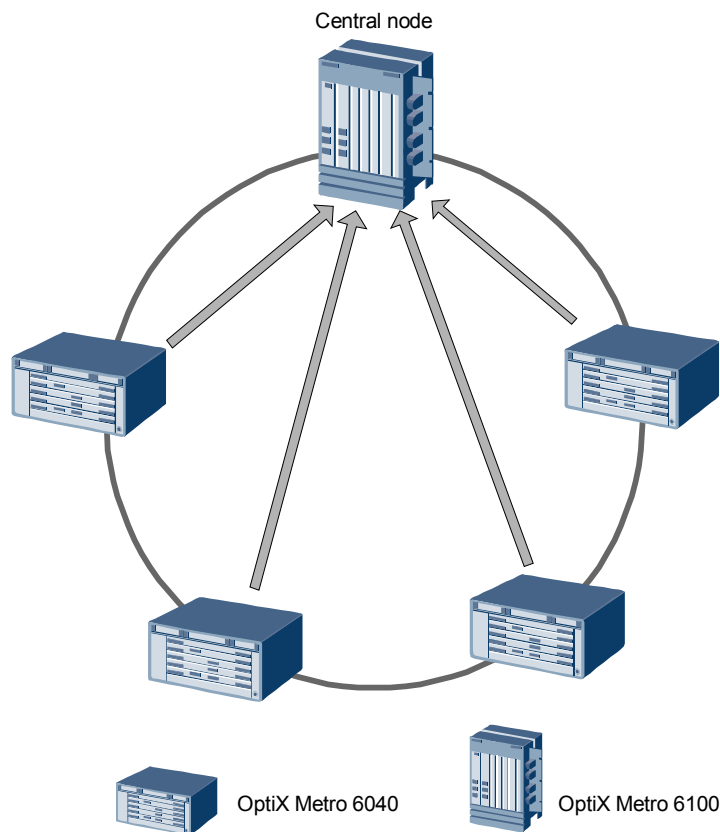
- The OptiX Metro 6100 serves as the central node in an intersecting-ring network.
- The OLA is omitted in [Figure 1-7](#).

1.4.4 Hybrid Networking Between the OptiX Metro 6100 and the OptiX Metro 6040

The OptiX Metro 6100 and the OptiX Metro 6040 can coexist in one network. The common networking mode is hub. Also, the OptiX Metro 6040 can serve as the extension of an OptiX Metro 6100 network.

In a star network, as shown in [Figure 1-8](#), the traffic of all the non-central nodes is aggregated to the central node. The star network is often used in Ethernet networks due to the large traffic volume at the central node.

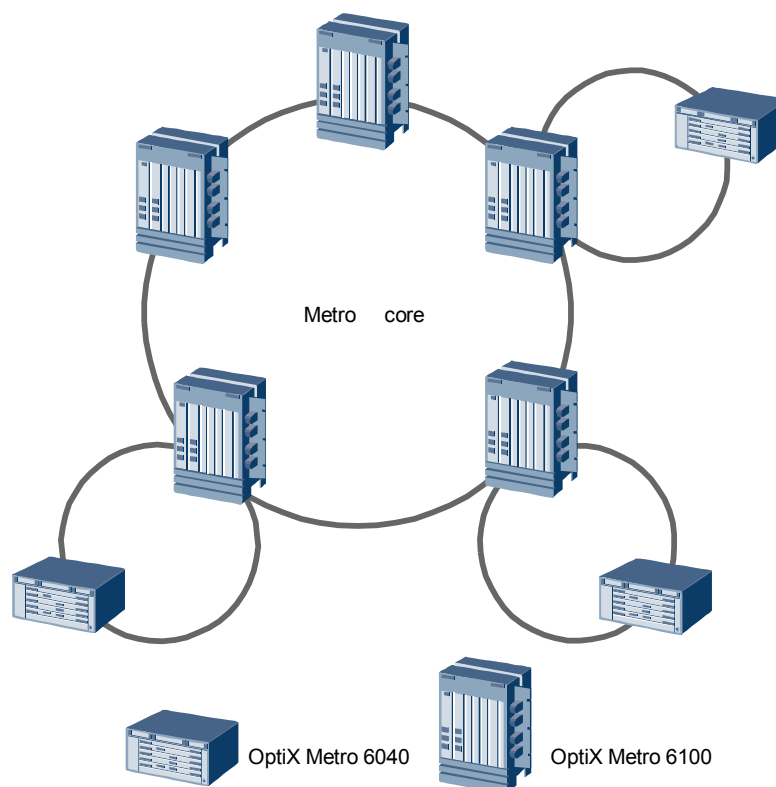
Figure 1-8 Star network composed of the OptiX Metro 6100 system and the OptiX Metro 6040 system



The other common networking mode in which the OptiX Metro 6100 and the OptiX Metro 6040 coexist is described as follows.

The OptiX Metro 6040 system can also form the extension of Metro core in the hybrid networking with the OptiX Metro 6100 system, as shown in [Figure 1-9](#).

Figure 1-9 Metro core and its extensions



1.4.5 Hybrid Networking Between the OptiX Metro 6100 and Other Equipment

The OptiX OSN 900A Compact Container WDM System (Type A) (the OptiX OSN 900A for short) is used to access and transmit low-end services in MANs and in local networks. It can interwork with equipment in which the ITU-T G.694.2 CWDM and ITU-T G.692 DWDM optical interfaces are used.

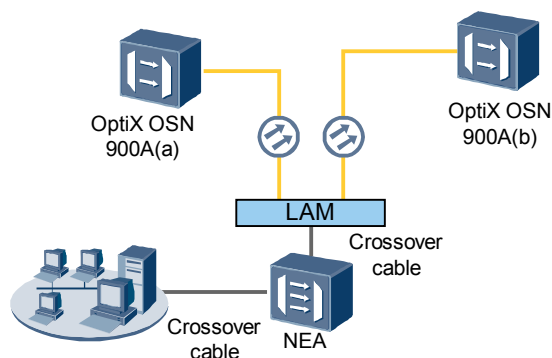
The OptiX OSN 900A can be used in local, point-to-point and ring networking comprising the OptiX Metro 6100.

In a hybrid network comprising the OptiX OSN 900A and the OptiX Metro 6100, the LAM board supports the interworking with a maximum of eight optical supervisory channels of the OptiX OSN 900A. The LAM board provides the function to realize the conversion between electrical supervisory channels and optical supervisory channels. The converted electrical supervisory channels can be accessed into the DCN network of customers or the Ethernet interfaces (ETHERNET1 and ETHERNET2) on the interface area of the OptiX Metro 6100 subrack for transmission.

Local Networking

In the network, the OptiX OSN 900A connects directly to the OptiX Metro 6100 that is connected to the T2000 server. One OptiX Metro 6100 system is used to access the optical supervisory channels of the OptiX OSN 900A in multiple directions. **Figure 1-10** shows the local networking.

Figure 1-10 Local networking



NOTE

The version of the SCC board in the OptiX Metro 6100 must be C8 or higher. The SCC board with version C6 does not support the local networking mode.

Point-to-Point Networking

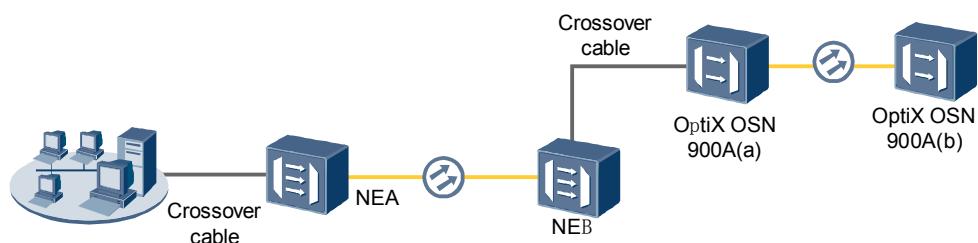
In a network, the OptiX Metro 6100 systems are used to build a point-to-point network.

Through the OSC/ESC communication mode, the IP over DCC mode is used for the communication between the OptiX Metro 6100 system. The OptiX OSN 900A connects to one of the OptiX Metro 6100 systems.

A network cable can be used to connect the OptiX OSN 900A to a non-gateway NE (GNE). This is applicable when one OptiX OSN 900A system is used. In addition, the LAM board can be used to connect the OptiX OSN 900A to a GNE through fibers. This is applicable when several OptiX OSN 900A systems are used or where the OptiX OSN 900A system is accessed remotely.

When one OptiX OSN 900A system is accessed, the OptiX OSN 900A NE is on the same station as the OptiX Metro 6100 NE to which the OptiX OSN 900A is directly connected. **Figure 1-11** shows the point-to-point networking mode.

Figure 1-11 Point-to-point networking (1)

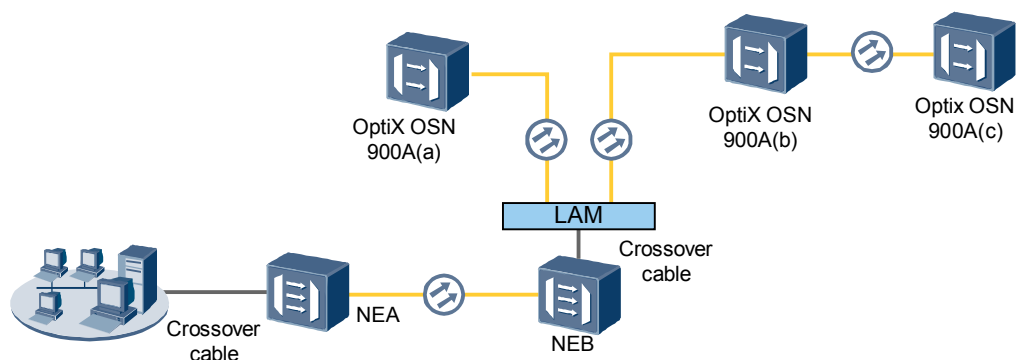


NOTE

In **Figure 1-11**, the OptiX Metro 6100 at NE B and the OptiX OSN 900A (a) are in the same station

Figure 1-12 shows the point-to-point networking mode when several OptiX OSN 900A systems are accessed or where the OptiX OSN 900A is accessed remotely.

Figure 1-12 Point-to-point networking (2)



In the network, if the several OptiX OSN 900A systems need to be configured in a cascading manner, the OSC communication mode is applicable when one OptiX OSN 900A is used, and the extended transfer mode through network cables is applicable when one OptiX OSN 900A system is to be cascaded. In [Figure 1-11](#) and [Figure 1-12](#), the cascaded OptiX OSN 900A systems adopt the OSC communication mode.

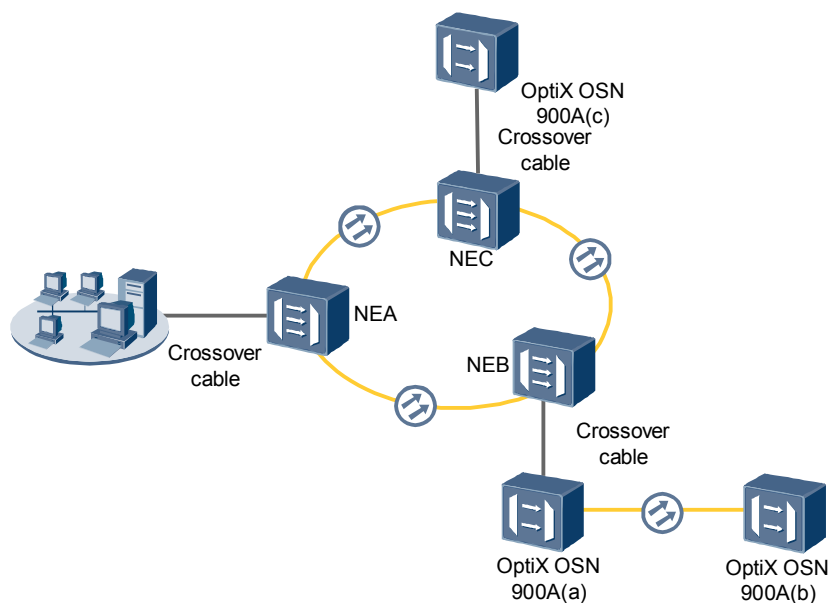
Ring Networking

In a network, the OptiX Metro 6100 is used to build a ring network.

The requirement of ring networking is the same as that of point-to-point networking.

When one OptiX OSN 900A is accessed, the OptiX OSN 900A NE is on the same station as the OptiX Metro 6100 NE to which the OptiX OSN 900A is directly connected. [Figure 1-13](#) shows the ring network.

Figure 1-13 Ring networking (1)

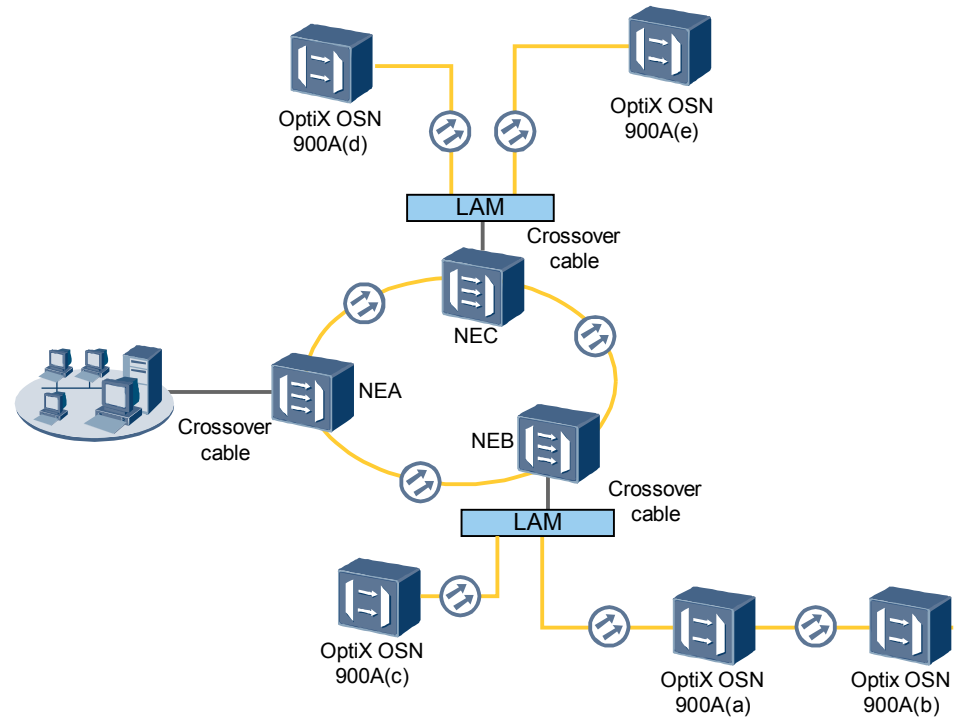


 NOTE

In **Figure 1-13**, the OptiX Metro 6100 at NE B and the OptiX OSN 900A (a) are in the same station. The OptiX Metro 6100 at NE C and the OptiX OSN 900A (c) are in the same station.

Figure 1-14 shows the ring network when several OptiX OSN 900A systems are accessed or where the OptiX OSN 900A is accessed remotely.

Figure 1-14 Ring networking (2)



In **Figure 1-13** and **Figure 1-14**, the cascaded OptiX OSN 900A systems adopt the OSC communication mode.

2 Product Functions

About This Chapter

This chapter describes the functions of the product.

[2.1 Basic Specification](#)

The product supports various optical layer technologies.

[2.2 Service Access](#)

The product can access services of different types by using different optical transponder boards (OTUs).

[2.3 Management and Auxiliary Interfaces](#)

The product provides management and auxiliary interfaces for the maintenance.

[2.4 Guaranteed Reliability](#)

The product provides equipment level protection and network level protection.

[2.5 Network Management Tools and Protocols](#)

By using the Qx and CORBA interfaces, the NM system manages alarm, performance, configuration, communication, security, and topology of the entire optical transmission system.

[2.6 Transmission of Network Management Information](#)

There are three communication modes: HWECC, IP over DCC and OSI over DCC.

2.1 Basic Specification

The product supports various optical layer technologies.

2.1.1 Grooming Ability

The product provides the following wavelength resource allocation solutions: optical signal grooming capability and electrical signal grooming capability.

Optical Signal Grooming Capability

These two solutions are applicable to the OptiX Metro 6100 DWDM systems (DWDM system for short). Only FOADM is applicable to the OptiX Metro 6100 CWDM systems (CWDM system for short).

In a DWDM system, the optical signal grooming that is realized by the ROADM technology is classified into two types:

- Two-dimensional grooming (the wavelength grooming in two directions)
- Multi-dimensional grooming (the wavelength grooming in multiple directions)

With the ROADM technology and the full-band tunable optical module on the OTU board, the DWDM system realizes the dynamic networking of optical signals, and supports many types of complex networks, such as tangent rings, chain, and intersecting rings. In this way, the system ensures flexible grooming of optical signals in or between networks.

For details of optical signal grooming, refer to [8 Grooming of Wavelengths and Services](#).

Electrical Signal Grooming Capability

The OptiX Metro 6100 provides two layers of electrical signal processing capabilities.

On layer 1, the GE ADM technology ensures the electrical signal grooming of GE services between the OTU boards at 2.5, 5, or 10 Gbit/s. Hence, the following features are achieved:

- Better wavelength utilization
- Longer transmission distance without dispersion compensation
- Arbitrary cross-connection capability

On layer 2, the L2 switching technology enables the convergence of a maximum of 32 GE services into one wavelength. Hence, the local convergence of GE services in a piece of equipment is achieved.

For details of electrical signal grooming, refer to [8 Grooming of Wavelengths and Services](#).

2.1.2 Technical Specifications

The product supports DWDM and CWDM technical specifications.

[Table 2-1](#) lists the transmission capacity and the upgrade and expansion ability for DWDM and CWDM technical specifications.

Table 2-1 Transmission capacity and upgrade and expansion ability

Item	DWDM System	CWDM System
Maximum capacity (Gbit/s)	1600 ^a	40
Used band	C-EVEN: 1529.55 nm-1560.61 nm	1311nm-1611 nm
Channel spacing (nm)	0.8	20
Frequency spacing (GHz)	100	-
Maximum channel count	40	16
Maximum accessed rate for each channel (Gbit/s)	40 ^a	2.5
Maximum add/drop traffic ^b (wavelength count)	40	16
Whether support optical amplifier	Yes	No
Applied fiber	G.652/G.653	G.652
Upgrade and expansion	Supporting upgrade of each wavelength of C band at 2.5 Gbit/s, 5 Gbit/s, 10 Gbit/s and 40 Gbit/s rates.	Supporting upgrade of each wavelength at 2.5 Gbit/s rate.
a: This value is based on the scenario where the DWDM system uses only 40 Gbit/s wavelengths to access signals. b: The maximum add/drop traffic is the add/drop traffic in the OADM station that is formed by back-to-back OTMs. The DWDM system can block and pass through any channel through the DWC board or the WSD9, WSM9, RMU9 and WSMD4 boards. The DWC is the dynamic wavelength control board. The WSD9 is the nine-port wavelength-selective switching demultiplexer board. The WSM9 is the nine-port wavelength-selective switching multiplexer board. The RMU9 is the nine-port ROADM multiplexing board. The WSMD4 is the 4-port wavelength selective switching demultiplexing/multiplexing board. In this way, the system can dynamically add/drop wavelengths to realize the ROADM.		

2.1.3 Transmission Capacity

The transmission capacity varies with the system type.

- A DWDM system can access a maximum of 40 wavelengths. The rate of each can be 2.5 Gbit/s, 5 Gbit/s, 10 Gbit/s, or 40 Gbit/s.
- A CWDM system can access a maximum of 16 wavelengths. The rate of each can be 2.5 Gbit/s.

2.1.4 Transmission Distance

The transmission distance varies with the system type.

Transmission Distance Specification in DWDM Systems

According to the type of fibers, the DWDM system uses the forward error correction (FEC), SuperWDM, dispersion compensation and optical equilibrium technologies by option and can support the long haul transmission without an REG.

The transmission distance specifications on the G.652 fiber of the DWDM system are as follows:

- When the FEC and SuperWDM (DRZ coding) technologies are used, the systems allow an attenuation of 20 x 22 dB for transmission without an REG.
- When 40-wavelength access is used, the DWDM system allows an attenuation of 46 dB for long-haul single-span transmission.

Transmission Distance Specification in CWDM Systems

On the G.652 fiber, the CWDM system allows the 80 km single-span transmission.

2.1.5 Networking Capability

The product provides a variety of networking mode.

The OptiX Metro 6100 supports point-to-point, chain and ring networking modes.

The OptiX Metro 6100 can also be used as equipment for the convergence layer and work with the OptiX Metro 6040 to offer a complete Metro WDM solution.

2.1.6 Integrated System and Open System Compatibility

There are two types of DWDM systems: integrated DWDM system and open DWDM system.

The open DWDM system is configured with the OTU boards to convert a wavelength into a G.694.1-compliant wavelength.

The integrated DWDM system does not need the OTU boards when its client-side equipment (for example, the SDH equipment) has optical transmitter interfaces that comply with G.694.1.

2.2 Service Access

The product can access services of different types by using different optical transponder boards (OTUs).

2.2.1 Types of Service Access

The product can access almost all types of services that are rated from 16 Mbit/s to 40 Gbit/s.

Table 2-2 lists the types of the services that can be accessed by the OptiX Metro 6100.

Table 2-2 Types of Service Access

Classification	Service Types
Standard SDH/POS/ ATM service	STM-256 standard or concatenation service STM-64 standard or concatenation service STM-16 standard or concatenation service STM-4 standard or concatenation service STM-1 standard service
Standard SONET service	OC-768 standard or concatenation service OC-192 standard or concatenation service OC-48 standard or concatenation service OC-12 standard or concatenation service OC-3 standard service
POS service	Packet Over SDH/SONET service
Ethernet service	10 GE LAN service 10 GE WAN service Fast Ethernet (FE) service Gigabit Ethernet (GE) service
SAN service	Enterprise Systems Connection (ESCON) service FC 100 service FC 200 service FC 400 service Fiber Connection (FICON) service FICON Express service
Any 16 Mbit/s–2.5 Gbit/s service	Fiber Distributed Data Interface (FDDI) service PDH (34 Mbit/s, 45 Mbit/s, 140 Mbit/s) service Digital video broadcasting-asynchronous serial interface (DVB- ASI) service HDTV service
OTN service	OTU1 service OTU2 service OTU3 service

2.2.2 Ability of Service Access

The product provides multi-types of the service access capability.

Table 2-3 lists the service access capabilities.

Table 2-3 Service access capability

Service Type	Max. Number of Services Accessed on a Board	Max. Number of Services Accessed in a Subrack
GE	8	96
10GE	1	12
STM-256/OC-768	1	4
STM-64/OC-192	4	24
STM-16/OC-48	4	48
STM-4/STM-1/OC-12/OC-3	8	96
Services at any rate (16 Mbit/s to 2.5 Gbit/s)	8	96
OTU1	4	48
OTU2	4	24
OTU3	1	4

2.3 Management and Auxiliary Interfaces

The product provides management and auxiliary interfaces for the maintenance.

Table 2-4 lists the management and auxiliary interfaces in the OptiX Metro 6100.

Table 2-4 Management and auxiliary interfaces.

Interface Type	Description
Management interface	Operation administration and maintenance (OAM) interfaces (SERIAL): located in the subrack interface area. Commissioning interface (COM): located in the subrack interface area NM cascading interfaces (ETHERNET1/ETHERNET2) ^a : located in the subrack interface area NM interfaces (ETHERNET1/ETHERNET2) ^a : located in the subrack interface area
Alarm interface	Alarm input/output interface (ALM): located in the subrack interface area Subrack alarm output and concatenation interfaces (LAMP1/LAMP2): located on the PMU board
a: For the C6SCC board, ETHERNET1 is the full duplex Ethernet electrical interface at 100 Mbit/s and ETHERNET2 is the half duplex Ethernet electrical interface at 10 Mbit/s. For the C8SCC board, ETHERNET1 and ETHERNET2 are the full duplex Ethernet electrical interfaces at 100 Mbit/s.	

2.4 Guaranteed Reliability

The product provides equipment level protection and network level protection.

2.4.1 Equipment Level Protection

The product provides various equipment level protection schemes to enhance the system reliability.

Each board in the OptiX Metro 6100 accesses two external –48 V DC working power supplies. The two power supplies serve as mutual backup. When any of the two fails, the board remains in the normal working state.

2.4.2 Network Level Protection

The product provides various network level protection schemes to enhance the system reliability.

The OptiX Metro 6100 provides the following network protection schemes:

- Optical channel protection
- Optical line protection
- Wavelength cross-connection protection (WXCP)
- Tribute protection switching (TPS) and double path protection switching (DPPS)
- VLAN SNCP protection
- Optical wavelength shared protection (OWSP)
- Optical wavelength shared protection (DCP)

There are five schemes for optical channel protection:

- Inter-board wavelength protection
- Intra-board wavelength protection
- Extended intra-board wavelength protection
- Inter-subrack 1+1 optical channel protection
- 1+1 wavelength protection at client

2.4.3 Configuration Data Backup

The SCC board provides an external storage device to back up network element (NE) configuration data.

It serves to restore the data after the SCC board is replaced.

2.4.4 Performance Monitory of Access Services

The product monitors various performances of the services. This helps in the routine maintenance and troubleshooting.

The OptiX Metro 6100 provides multiple monitoring functions, including:

- B1 bit error monitoring

- Ethernet performance monitoring
- Optical power monitoring

The monitoring covers services on the WDM side and on the client side as well as the converged sub-rate services.

Table 2-5 shows the performance monitoring based on the service types.

Table 2-5 Performance monitoring of the accessed services

Service Sort	Monitorable Performance Item	Service Types
Data	Ethernet performance RMON statistic	GE/10GE/FE
SDH/SONET	B1 bit error	STM-1/STM-4/STM-16/ STM-64/STM-256 OC-3/OC-12/OC-48/ OC-192/OC-768
OTN	SM-BIP8 bit error PM-BIP8 bit error	OTU1/OTU2/OTU3

2.4.5 In-Service Optical Performance Monitoring

The system provides the performance monitoring based on the network.

The optical monitoring interfaces are provided on the multiplexer, demultiplexer, and the optical amplifier of the DWDM system. The optical spectrum analyzer, multi-wavelength meter or MCA board can be directly connected to these interfaces to measure performance parameters at reference points without service loss.

2.5 Network Management Tools and Protocols

By using the Qx and CORBA interfaces, the NM system manages alarm, performance, configuration, communication, security, and topology of the entire optical transmission system.

The NM system also provides end-to-end management according to the requirements of the user. The NM system improves the network quality, lowers the maintenance cost, and ensures reasonable utilization of the network resource.

The NM system provides user friendly interfaces and comprehensive functions. Its software system adopts component technology and object-oriented technology so that the application sub-systems can be tailored according to the requirements of the user. This facilitates system expansion.

2.5.1 T2000

OptiX iManager T2000 (T2000 for short) is a subnet management system (SNMS). Subnet management system of the new generation can manage and control NEs and the area network. In the telecommunication management network (TMN) architecture, SNMS is located between

the NE level and network level. Therefore, the T2000 supports all functions of NE-level and part of the network-level management functions.

The T2000 provides the users with single-layer management network solutions for small and medium-sized transmission networks. The T2000 can assist network layer management system and service layer management system in managing large-scale transmission networks together with the upper-level network management system (through the standard external interfaces).

2.5.2 Simple Network Management Protocol

The system provides the simple network management protocol (SNMP).

The SNMP is a standard protocol based on user datagram protocol (UDP). With an SNMP compatible management interface, any NM system can query the alarms and performances of the equipment.

2.6 Transmission of Network Management Information

There are three communication modes: HWECC, IP over DCC and OSI over DCC.

- HWECC: The management information is transmitted by using the HWECC protocol sharing method.
- IP over DCC: The management information is transferred by using the IP protocol sharing method.
- OSI over DCC: The management transmitted is transferred by using the OSI protocol sharing method.

3 Product Features

About This Chapter

This chapter describes the technical features and the features of upgrade and maintenance of the system.

[3.1 Service Processing and Grooming](#)

The product adopts the OTN technology and provides the dynamic optical signal grooming and electrical signal cross-connection.

[3.2 Features of WDM Transmission Technology](#)

The product provides WDM transmission features, such as FEC, tunable wavelength, automatic laser shutdown and optical power management.

[3.3 Features of Ethernet](#)

The product supports Ethernet service switching based on GE ADM and L2 switching function.

[3.4 Features of Upgrade and Maintenance](#)

The product has the following upgrade and maintenance features: software package loading, PRBS function, and pluggable optical modules.

3.1 Service Processing and Grooming

The product adopts the OTN technology and provides the dynamic optical signal grooming and electrical signal cross-connection.

3.1.1 Optical Transport Network (OTN) Signal Processing

OTN is the new generation optical transmission system specified by the International Telecommunication Union (ITU). The OTN integrates the capacity advantages of the WDM network, with the flexibility of SDH/SONET network and the convenience of Ethernet. It supports all existing services. Through OTN, various networks and services can be integrated into a universal future-oriented architecture.

Technical Background

The optical transport network (OTN) is a brand-new optical transmission technology mechanism that is defined by ITU-T G.872, G.798, and G.709. It comprises the optical layer and electrical layer and provides corresponding mechanisms to manage and monitor the networks at different layers and to ensure the network survival. The idea of OTN derives from the SDH/SONET mechanisms (such as mapping, multiplexing, cross-connection, embedded overhead, protection, and FEC). The OTN applies the operable and manageable abilities of the SDH/SONET to WDM systems, possessing the reliability and flexibility of the SDH/SONET and large capacity of the WDM.

Technical Advantages

With OTN-related technologies, the OptiX Metro 6100 has technical advantages in the following aspects:

- The OTN supports interfaces with the rates of OTUk.
- The OTN transmits various service data signals transparently, such as SDH/SONET, Ethernet, ATM, IP, MPLS, and OTN (ODUk).
- The OTN provides the standard FEC function. This function decreases the optical signal-to-noise ratio (OSNR) tolerance of optical channels, extends the electrical regeneration distance, reduces the number of system stations, and lowers the total cost for networking.
- The OTN provides various maintenance signals to isolate faults and to suppress alarms. This helps to analyze and locate faults in a network, and relieves the burden of system maintenance.
- Support end-to-end service performance monitoring.
- The end-to-end management of the wavelengths at the optical layer is realized through the supervisory overheads at the OTSn, OMSn, and OCh layers.

Implementation Scheme

The OptiX Metro 6100 supports the OTN technology and the implementation scheme consists of the following:

- OTN WDM line interfaces that realize the operation, administration and maintenance (OAM) of transmission lines

The OptiX Metro 6100 provides the OTU1, OTU2 and OTU3 WDM line interfaces for some OTU boards to support some management overhead bytes that are defined in ITU-T G.709.

The boards owning the OTN processing capability on the WDM side are the LWC1, LWC1D, FDG, ELOG, ELOGS, ETMX, ETMXS, LBE, LBES, LBF, LBFS, LOG, LOGS, LQM, LQM2, LWF, LWFS, TMX, TMXS, TMR, TMRS, TRC1, TRC2, LOM, LOMS, LU40S, LUR40S and TMX40S boards.

The TRC2 board can provide bidirectional electrical regeneration to OTU1 services. The system uses the general communication channel (GCC) byte to realize ESC management. For some boards, you can select the GCC0 byte to serve as the transmission channel of the management information. The system supports performance monitoring and reporting of section monitoring (SM) and path monitoring (PM), FEC and correction result reporting, and querying of the optical wavelength information.

- Direct access of client services

At the client-side optical interface of some OTU boards of the OptiX Metro 6100, OTN services, whose levels contain OTU1 and OTU2 signals, can be directly accessed. This function can realize transparent transmission or convergence of the OTN client-side services.

The boards owning the OTN accessing capability on the client side are the LWC1, LWX, LBF, LBFS, ETMX, ETMXS, LU40S and TMX40S.

- End-to-end management of wavelengths at the optical layer is realized.

The system defines the supervisory overheads at the OTSn, OMSn and OCh layers according to the OTN standard. The working status of the network can be monitored by processing of these supervisory overheads. In this manner, the supervisory function at the OTN optical layer is achieved.

The supervisory function at the optical layer mainly involves the following aspects:

- Fiber connection management
- Supervisory of the continuity at the optical layer
- Supervisory of the maintenance signals at the optical layer

3.1.2 ROADM Technology

The ROADM is reconfigurable optical add/drop multiplexer. When the ROADM works with tunable lasers, flexible grooming of wavelengths can be achieved. In this way, capacity expansion without interrupting services can be realized. The ROADM can be realized by configuring the wavelength blocker (WB) boards or the wavelength selective switch (WSS) boards.

Technical Background

Most of the optical add/drop multiplexers (OADMs) laid in DWDM networks are static. Once they are laid, the configuration of channels is fixed and cannot be altered.

To avoid service interruption during expansion, it is needed to plan wavelengths and adopt wavelength reservation. However, the increase in actual services is hard to estimate. The wavelength capacity of certain nodes may require adjustment. The manual adjustment or adding of equipment leads to the interruption of services, affecting the existing services.

The expansion at a single OADM station can cause change of insertion loss. It is needed to make the engineering budget and perform the manual adjustment again, which slows down the service

providing, increases the costs of operation, and leads to bit errors as well as periodical interruption of services.

Advantages in Application

The OptiX Metro 6100 DWDM system adopts the reconfigurable optical add and drop multiplexer (ROADM) technology and realizes dynamic wavelength grooming inside a ring network or between ring networks.

The ROADM technology has the following advantages:

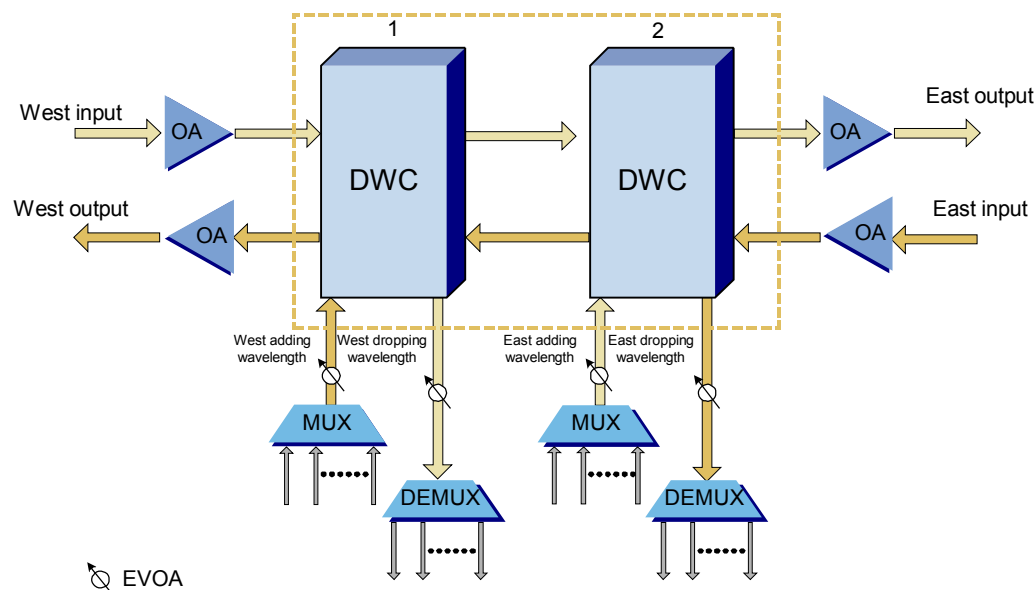
- Expansion does not affect the existing services.
A wavelength at each ROADM station can be selected to pass through or be terminated. With the ROADM technology, any wavelengths can be added and dropped at any station without affecting the existing services. The wavelength plan is not required. Nor are the services be interrupted during expansion. The quality of service is well guaranteed.
- Channels are changed rapidly and effectively.
When bandwidth allocation is to be adjusted, such as to add or delete wavelengths or change the position where wavelengths are added and dropped, the ROADM technology can be adopted to change the channels by using a remote network management system and to rapidly adjust the adding and dropping of wavelengths. It is not needed to make the network engineering budget for the second time. The manual maintenance is averted, which saves the costs of operation.
- Power equilibrium function is embedded.
The ROADM technology supports the power equilibrium function. It supports the delicate channel-level power equilibrium and the wavelength-level power equilibrium and control, flattening the spectrum waveform of the signals within the working bandwidth.

Scheme for Realization

The DWDM system can provide the ROADM function in the following ways.

- ROADM function being realized through DWC boards
The ROADM function can be realized through the configuration of DWC boards. The configuration of a typical ROADM station is displayed in [Figure 3-1](#).

Figure 3-1 Configuration diagram of an ROADM station realized through the DWC boards



The concatenation of two DWC boards realizes adding and dropping of any wavelength in the east and west directions.

The DWC can block arbitrary channels. Thus, any wavelengths can be terminated at any node on a ring or chain network. Since the pass-through channels and add/drop channels are isolated from each other, the adjustment of add/drop channels does not affect the services carried by the main path. Besides, the power budget for the line is not needed to be made once again after any change of wavelength allocation.

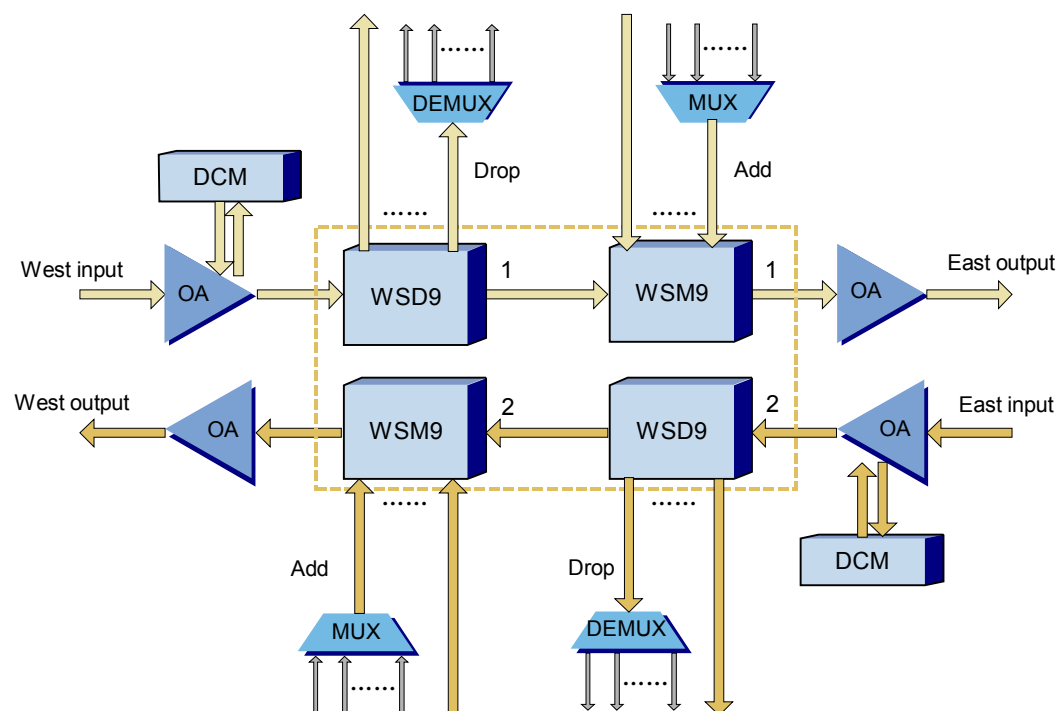
For the application of the ROADM node constituted by the DWC and the networking signal flow direction, refer to [8.1.1 Intra-Ring Wavelength Grooming by DWC Boards](#).

- ROADM function being realized through the WSM9 and WSD9 boards

The ROADM function can be realized also through the configuration of the WSD9 and the WSM9 boards.

Combining the WSD9 and WSM9 boards to achieve the intra-ring ROADM function is considered an example. One ROADM node is constituted by two WSD9 and two WSM9 boards, as shown in [Figure 3-2](#).

Figure 3-2 Configuration diagram of an ROADM station realized by the WSD9 and WSM9 boards



The ROADM node that consists of the WSD9 and WSM9 boards realizes separation of east and west services, and provides dynamic wavelength grooming inside a ring network or between ring networks. If multiple wavelengths are output from an add/drop port, you can configure on the T2000 to connect the WSS board of other service flows. In this way, you can smoothly extend the wavelength dimension, realize mesh extension, and achieve dynamic wavelength grooming between fiber line rings.

If one ROADM node consists of more than four WSM9s and the same number of WSD9s, this ROADM node can multi-dimensionally groom signals.

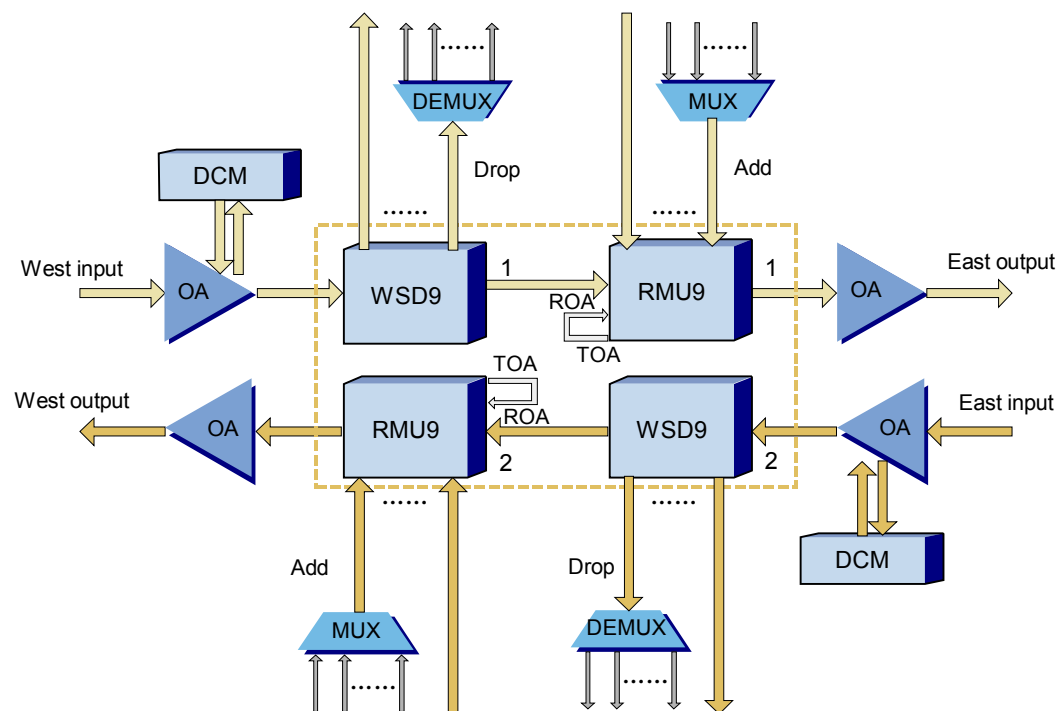
For the application of the ROADM node constituted by the WSD9 and WSM9 boards, and the networking signal flow direction, refer to [8.1.3 Intra-Ring Wavelength Grooming by WSD9 Boards and WSM9 Boards](#) and [8.1.6 Inter-Ring Grooming by WSD9 Boards and WSM9 Boards](#).

- ROADM function being realized through the RMU9 and WSD9 boards

The ROADM function can be realized also through the configuration of the WSD9 and the RMU9 boards. The ROADM can dynamically add/drop the wavelengths within the ring or groom the wavelength among the rings. It also supports the inter-ring wavelength extended grooming.

Combining the WSD9 and RMU9 boards to achieve the intra-ring ROADM function is considered an example. One ROADM node is constituted by two WSD9 and two RMU9 boards, as shown in [Figure 3-3](#).

Figure 3-3 Configuration diagram of an ROADM station realized by the RMU9 and WSD9 boards



The RMU9 board is mainly used to add wavelengths. The port for adding wavelength can be interworked with the tunable OTU to realize the fully dynamical input of eight wavelengths. The port for adding wavelength can also be connected to the multiplexer. The client-side signals are multiplexed in the multiplexer and then input from the port for adding wavelength of the RMU9.

The WSD9 is mainly used to configure any wavelengths to any interfaces. A node on the ring or chain network can transmit any wavelength combination to any interface so as to achieve the dynamic allocation of wavelengths.

The ROADM node constituted by the RMU9 and WSD9 boards can serve as the central node or edge node. The ROADM station supports the flexible and convenient expansion without affecting any services and has a low operation cost. The adding, dropping or pass-through state of wavelengths can be directly adjusted through the network management software so as to realize the remote dynamic adjustment.

If one ROADM node consists of more than four RMU9s and the same number of WSD9s, this ROADM node can multi-dimensionally groom signals.

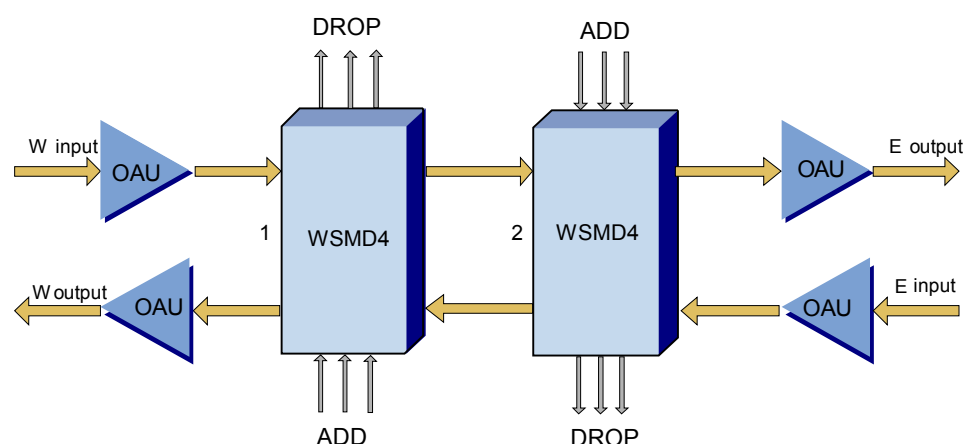
For the application of the ROADM node constituted by the RMU9 and WSD9 boards and the networking signal flow direction, refer to [8.1.2 Intra-Ring Wavelength Grooming by WSD9 Boards and RMU9 Boards](#) and [8.1.5 Inter-Ring Grooming by WSD9 Boards and RMU9 Boards](#).

- ROADM function being realized through the WSMD4 boards

Multiple WSMD4 boards can also form the ROADM node that provides the dynamic wavelength grooming function.

The intra-ring ROADM function in which one ROADM node consists of two WSMD4s is considered an example, as shown in [Figure 3-4](#).

Figure 3-4 Configuration of the ROADM node formed by two WSMD4s



Each WSMD4 adds and drops wavelengths. The combination of two WSMD4s can add or drop a maximum of 40 wavelengths. The ROADM node formed by two WSMD4s provides two-dimensional inter-ring dynamic wavelength grooming. The state of wavelength add/drop and passthrough can also be remotely and dynamically adjusted on the T2000.

NOTE

- The drop and pass-through interfaces of the WSMD4 board output four equal multiplexed optical signals. In the drop channel, even when there is only one wavelength signal, the WSMD4 need to be connected to a demultiplexing board before it is connected to the OTU.
- The AMx and DMx optical interfaces can be connected in tandem to cascade multiple WSMD4s. The AMx and DMx can also add or drop service signals to or from the WSMD4.

If one ROADM node consists of more than four WSMD4s, this ROADM node can multi-dimensionally groom signals.

For the application of the ROADM node that consists of multiple WSMD4s and the networking signal flow, refer to [8.1.4 Intra-Ring Wavelength Grooming by WSMD4 Boards](#) and [8.1.7 Inter-Ring Grooming by WSMD4 Boards](#).

3.1.3 Electrical Signal Cross-Connection Grooming

The product provides grooming of GE service based electrical signals at Layer 1 (L1) and Layer 2 (L2). It is capable of processing electrical signal services of two layers.

Grooming at L1 dispatches and orients the GE service flows between different stations.
Grooming at L2 converges the local GE service flows.

For the technology that realizes electrical signal cross-connection grooming, refer to [3.3 Features of Ethernet](#).

For the networking and application of electrical signal cross-connection grooming, refer to [8.2 Application and Networking of the GE ADM Feature](#).

3.1.4 SAN Service Feature

The product can access multi-types of SAN service to create direct connection between the storage unit and the server or client for the purpose of storage.

Technical Background

Storage area network (SAN) is a new storage connection topology structure and is called the second network besides local area network (LAN). SAN provides powerful and exclusive functions including data centralization, backup, restoration, protection, mirror, and load equalization.

Advantages in Application

In a SAN network, each node should be separated from each other in geography, for the consideration of data centralization, remote backup and remote disaster recovery. Hence, a complete solution of storage and connection is required.

The OptiX Metro 6100 provides the SAN service interface to interconnect two or more remote SAN networks. The SAN service interface ensures network performance and provides complete SAN network connection mode.

To ensure that the service bandwidth does not decrease during distance extension of FC services, the OptiX Metro 6100 uses special flow control mechanism. In this mechanism, false response signal is provided for the service transmit end. Hence, the service transmit end regards that the data is received and replied by the receive end.

Implementation Scheme

The OptiX Metro 6100 accesses and converges the SAN services, including FC, FICON, and ESCON. It can interconnect two or more remote SAN networks and ensures network performance.

Table 3-1 lists the OTU boards that are used to access the SAN services.

Table 3-1 OTU boards that can access the SAN services

Service Type	Corresponding Board
ESCON	AP8, LQM, LQM2, LWX
FICON, FICON Express	LOM, LOMS, LQM, LQM2
FC100, FC200	AP8, LQM, LQM2, LWX, FCE, LOG, LOGS, ELOG, ELOGS, LOM, LOMS
FC400	LOM, LOMS

NOTE

When the AP8, LQM, LQM2, LWX, FCE, LOM, LOMS, LOG, LOGS, ELOG, and ELOGS boards are used to access the FC services, the FC switch is recommended to cooperate with the boards so as to ensure the high-reliability transmission of the memory services. In the process of networking, confirm the transmission distance supported by the port protocol layer of the FC switch. Ensure that the actual transmission distance of the services is less than the maximum transmission distance allowed by the port of the FC switch.

The FCE, LOM and LOMS boards apply a flow control mechanism between FC service client-side equipment and between two FCE/LOM/LOMS boards to provide the distance extension function of FC services. This function ensures that the signal bandwidth does not decrease during long haul transmission of FC services.

- For FC100 services, the extension transmission distance on the WDM side of the FCE, LOM and LOMS board is 3000 km ideally.
- For FC200 services, the extension transmission distance on the WDM side of the FCE, LOM and LOMS board is 1500 km ideally.
- For FC400 services, the extension transmission distance on the WDM side of the LOM and LOMS boards is 750 km ideally.

3.1.5 LAN Protocol Processing Feature

The product supports the LAN service accessing to carry Ethernet services of different types of granularity and interconnect between high-speed network equipment.

Technical Background

One development trend of the broadband network is that more and more users and carriers use the Ethernet interface to realize interconnection of broadband data services. The Ethernet technology has become the main networking technology in the metropolitan area network (MAN). The common goal of various carriers is to create economical and effective MAN that supports multiple types of services. How to solve the problem of transmitting Ethernet services of large granularity becomes the key consideration of building the MAN.

The LAN technology is used to access users in the MAN, to converge services at the central node of the MAN, and to build the broadband data network.

Advantages in Application

The OptiX Metro 6100 provides boards that can access different LAN services. It realizes transparent transmission of the L2 and L3 protocol services that conform to the industry standard. Some boards support Ethernet performance monitoring of services.

Implementation Scheme

The OptiX Metro 6100 provides boards that support line transparent transmission of LAN signals of the L2 and L3 protocols, which conform to the universe standard. The boards are the LBE, LBES, LOG, LOGS, ELOG, ELOGS, LOM, LOMS, LBF, LBFS, LQG, FDG, LDG, AP8, LQM, LQM2, LWX, L4G, EGS8, TBE and LAM.

Table 3-2 lists the LAN protocol that is supported by the OptiX Metro 6100 and the corresponding boards.

Table 3-2 LAN protocol types supported by the OptiX Metro 6100 system and the corresponding boards

Protocol	LBE/ LBES	ELOG/ ELOGS/ LOG/ LOGS/ LOM/ LOMS	LBE/ LBFS	LQG	FDG/ LDG	AP8/ LQM/ LQM2/ LWX	L4G/ EGS8	TBE	LAM
IEEE 802.3z (L1 protocol)	C	C	C	C	C	C	C	C	C
IEEE 802.3ae (L1 protocol)	NA	C	NA	C	C	C	C	C	-
IEEE 802.1D	C	NA	C	NA	NA	NA	NA	NA	NA
IEEE 802.1P	T	T	T	T	T	T	T	T	T
IEEE 802.1Q	T	T	T	T	T	T	C	T	NA
RIPv1&v2	T	T	T	T	T	T	T	T	NA
OSPF	T	T	T	T	T	T	T	T	-
IGMP	T	T	T	T	T	T	T	T	-
IGRP	T	T	T	T	T	T	T	T	T
STP	T	T	T	T	T	T	T	T	-
C= Compliant (performance monitoring) T= Transparent transport(non performance monitoring) NA=Not applicable									

3.2 Features of WDM Transmission Technology

The product provides WDM transmission features, such as FEC, tunable wavelength, automatic laser shutdown and optical power management.

3.2.1 40G Transmission System

The system provides the 40 x 40G and 40G inverse multiplex transmission solution, meeting operators' requirements for large capacity and high performance of the network.

Basic Concept

The 40G transmission system directly accesses signals from the 40G OTU board based on the existing system platform, which meets operators' requirements for expanding transmission capacity and configuring high performance of the optical network. This system realizes seamless expansion of the transmission capacity on a 40G basis without affecting the existing low rate services. Thus, the capacity expansion cost is reduced by utilizing the existing investment. The 40G transmission system is mainly characterized by the following features:

- Supports the hybrid transmission of C-band 40 channels x 40G/10G.
- Accesses 1 x 40G (STM-256/OC768/OTU3) service on the client side of the 40G OTU.
- Accesses 4 x 10G (STM-64/OC192/OTU2/10GE) service on the client side of the 40G OTU.
- Supports eDQPSK modulation on the WDM side of the 40G OTU.
- Provides an advanced modulation format to ensure that the optical spectrum can pass the existing filter (10G MUX/DMUX maintained).
- In the case of eDQPSK modulation, the system supports a maximum of 1500 km transmission without regeneration.
- Supports smooth upgrade from the 10G DWDM system to the 40G DWDM system through the plug-and-play mode without interrupting existing services.

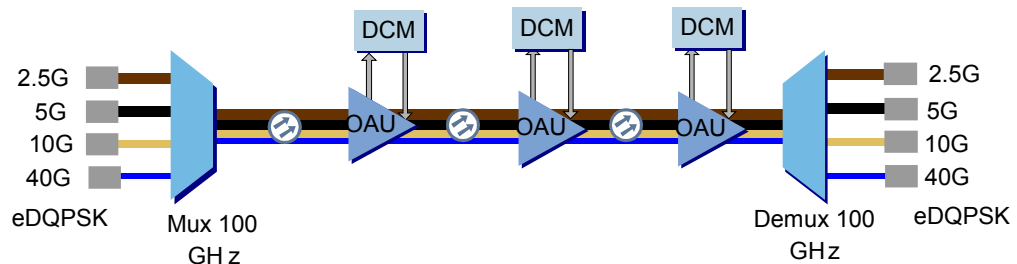
Advantages in Application

The OptiX Metro 6100 DWDM system provides 40 channels x 40G transmission solutions. The 40G optical transponder units (OTUs) can be directly used in the OptiX Metro 6100 DWDM system. This meets the operators' demands to expand capacity in ordinary channels and to configure optical networks of high performance. It allows the operators to perform seamless expansion without affecting existing low-rate services. The existing investment is safe and the expansion costs less.

Scheme for Realization

Figure 3-5 shows the typical application of the 40G system of the OptiX Metro 6100 DWDM system.

Figure 3-5 Structure of the 40G eDQPSK system



The 40G transmission system of the OptiX Metro 6100 DWDM system provides flexible hardware/software interfaces of WDM solution. The 40 Gbit/s OTU and the 10 Gbit/s OTU adopt the same EDFA, dispersion compensation module and multiplex/demultiplex unit. The

40 Gbit/s services can be added/dropped at OTMs and OADMs/ROADMs and are managed by the same NM system as the 2.5 Gbit/s, 5 Gbit/s and 10 Gbit/s services.

3.2.2 Expansion

The product adds/drops service through optical terminal multiplexer nodes and optical add/drop multiplexer nodes. The capacity expansion is flexible and convenient.

The OptiX Metro 6100 system adds and drops services through OTM and OADM nodes. The capacity expansion is flexible and convenient.

- If the OptiX Metro 6100 system uses OADM boards, the initial investment is small and the capacity expansion can be done by adding hardware later. At most 40 add/drop channels are supported.
- If the OptiX Metro 6100 system uses M40/D40 boards, capacity expansion does not interrupt the existing services. At most 40 add/drop channels are supported.

3.2.3 Supervisory Channel

NEs in the product exchanges supervisory signals by supervisory channels.

Two kinds of supervisory channel are available in the OptiX Metro 6100 system:

- Optical supervisory channel (OSC)
- Electric supervisory channel (ESC)

Principle of the OSC

The OSC mainly carries orderwire and network management information. The OptiX Metro 6100 transmits supervisory signals at 1510 nm, with the rate of 2.048 Mbit/s.

The SC1/SC2 board can provide the supervisory channel rated at 2 Mbit/s. The TC1/TC2/ST1/ST2 board can provide not only the supervisory channel rated at 2 Mbit/s but also the clock transmission channel rated at 2048kbit/s or 2MHz and the transmission channel for FE signals.

Principle of the ESC

In this mode, the OTU multiplexes the supervisory information into the service channel for transmission. The 2.5 Gbit/s OTU realizes ESC transmission by DCC bytes. The 10 Gbit/s OTU realizes ESC transmission by associated GCC bytes that are compliant to ITU-T G. 709.

The ESC reduces the investment of the OSC. It also deletes the insertion loss of the FIU. This lowers the cost and the power budget of optical channels.

NOTE

- If an OTU board is configured with the ESC, set the **Laser State** of the optical interface on the WDM side of the board to **Enabled** and the **Automatic Laser Shutdown** to **Disabled** on the T2000.
- The OTU board using the ESC can be configured with the inter-board wavelength protection. However, the interface attributes on the working and protection OTU must be the same.
- The networking with the LAM board that requires the ESC to be accessed can be achieved by configuring **Board Mode** of the LAM board on the T2000.

3.2.4 Single-Fiber Bi-Directional Transmission

The CWDM system supports the single-fiber bidirectional transmission mode.

In the single-fiber bidirectional transmission mode, the optical signals at different wavelengths in the receive and transmit directions travel over the same fiber. As a result, any wavelength(s) among 16 wavelengths in the CWDM band (1311 nm–1611 nm) can be added or dropped.

 NOTE

When the system operates in the CWDM band, it is recommended to configure the latter eight wavelengths (1431–1611 nm) of the band for the system.

The maximum dispersion limit of the 8-channel CWDM system is 40 km; the maximum dispersion limit of the 16-channel CWDM system is 20 km.

In the single-fiber bidirectional transmission mode, the SBM1 or SBM2 board is required.

3.2.5 FEC Function

The product provides the OTU that has the FEC or advanced FEC (AFEC) function.

The forward error correction (FEC) technology adopts Reed-Solomon coding. FEC can correct a maximum of eight byte errors in any location per 255 bytes. Hence, FEC is highly capable of error correction. The OTU can process the overheads that comply with ITU-T G.709.

FEC coding can correct the bit errors that occur during signal transmission. Hence, the FEC coding elevates the optical signal-to-noise ratio (OSNR) tolerance at the receive end and extends the distance of the repeater section.

Compared with the FEC coding, the AFEC coding adopts a two-level coding mode for higher coding gain and stronger error correction capability.

3.2.6 SuperWDM Technology

The product provides the SuperWDM.

The Differential Return to Zero (DRZ) coding and special phase modulating used in the OptiX Metro 6100 DWDM system is adopted to decrease the non-linear effect in transmission of a long distance. This improves the tolerance of the system against optical noises.

The boards support the SuperWDM technology are the LWFS, LBES, LBFS, ETMXS, LRFS, TMXS, TMRS, LOGS, ELOGS and LOMS.

3.2.7 ODB Technology

The product provides the ODB technology.

The OptiX Metro 6100 DWDM system adopts optical duobinary (ODB) code to realize SuperWDM technology. The ODB codes the input NRZ on the WDM side of a board and outputs a code sequence to drive the modulator for signal modulation. In this way, the ODB realizes the modulation of optical signals. The optical signals received on the WDM side of the board, however, are not specifically processed. It is only required that the normal light density detector perform O-E conversion for the signals.

If the ODB and NRZ have same dispersion and root mean square (RMS) spectrum, the power spectrum of the modulated signals of the ODB code is more centralized than that of the NRZ code. Hence, the system can reduce the output signal bandwidth to realize transmission over longer distance. The ODB 3 dB cut-off frequency is only a quarter of the NRZ. Therefore, the ODB can realize 4800 ps dispersion tolerance and 200 km compensation-free transmission.

3.2.8 DQPSK Technology

The 40 Gbit/s OTU supports the enhanced differential quadrature phase shift keying (eDQPSK) modulation technology.

DQPSK is a new modulation format. On the transmit side, the input electrical signals are differentially encoded. Then, the modulator performs quadrature phase shift keying modulation and outputs the optical signals in four phases: 0, $\pi/2$, π , and $3\pi/2$. On the receive side, the demodulator differentially decodes the optical signals; the signals are received in an equalized manner.

DQPSK is a multilevel modulation format in which the bit rate is two times of the baud rate. Hence, DQPSK is highly suitable for the 40G high-speed transmission system. In the DQPSK modulation format, the spectrum width is narrow and the output spectrum is smooth. As a result, the DQPSK modulation format can effectively suppress various nonlinear effects of the fiber. The phase shift helps reduce the phase-related nonlinear effects (such as SPM, XPM, and FWM) and enhance the tolerance to chromatic dispersion and polarization mode dispersion. Hence, DQPSK is a critical modulation format used in the long-haul, high-speed, and large-capacity optical transmission.

Huawei enhanced DQPSK (eDQPSK) technology is a coding technology that adopts the chirped RZ modulation technique on the basis of the DQPSK technology. eDQPSK further elevates the nonlinearity tolerance of the system and thus is currently the best coding technology for 40G ultra long haul transmission.

3.2.9 Tunable Wavelengths

The product supports tunable wavelengths.

The OptiX Metro 6100 system adopts 40 Gbit/s, 10 Gbit/s, 5 Gbit/s and 2.5 Gbit/s OTUs that support tunable wavelengths. The boards providing the tunable wavelengths are listed in [Table 3-3](#).

The 40 Gbit/s, 10 Gbit/s, 5 Gbit/s and 2.5 Gbit/s OTUs support tunable wavelengths in up to 40 channels with 100 GHz spacing.

Besides, the tunable wavelength OTUs can also act as spare parts to substitute OTUs of different wavelengths. This reduces the amount of OTUs and lowers the cost.

Table 3-3 List of boards with tunable wavelengths

Signal Rate on the WDM side	Board
40 Gbit/s	LU40S, LUR40S, TMX40S
10 Gbit/s	LWF, LWFS, LRF, LRFS, LBE, LBES, TMR, TMRS, TMX, TMXS, LOG, LOGS, ELOG, ELOGS, LBF, LBFS, ETMX, ETMXS, LOM, LOMS
5 Gbit/s	L4G, LQG
2.5 Gbit/s	LWX, LWM, LWXR, LWMR, LWC1, TRC1, TRC2, FDG, LQS, AS8, AP8, FCE, LDG, LQM, LQM2

3.2.10 EDFA Technology

The product uses mature erbium-doped fibre amplifier (EDFA) technology for the amplification of C-band signals, and the implement of long haul transmission.

The OptiX Metro 6100 DWDM system uses mature Erbium-doped fiber amplifier (EDFA) technology for long haul transmission with no regenerator.

EDFA adopts gain locking technology and transient control technology to make the gain of each channel independent of the number of channels.

Adding or dropping channels does not bring in burst bit error in the existing channels.

3.2.11 Raman Amplification

The product adopts both the Raman amplifier and the erbium-doped fiber amplifier (EDFA) to realize wideband flat gain.

Besides the EDFA amplification, the OptiX Metro 6100 DWDM system also supports the Raman fiber amplification. The hybrid application of the Raman and EDFA achieves broad gain bandwidth and low system noise, and reduces the interference of non-linearity on the system, which thus greatly stretches the transmission distance.

3.2.12 Jitter Suppression Function

The product provides the excellent jitter suppression function.

With a jitter suppression unit between the optical receive module and the optical transmit module, the OptiX Metro 6100 system has an excellent jitter suppression function.

3.2.13 Unidirectional Electrical Regeneration

The DWDM system can loopback the overhead bytes of the bidirectional supervisory channel at an REG station.

Some electrical regenerating boards used in the DWDM system are able to transmit signals only in one direction. But no matter the optical supervisory channel in the system uses the OSC or ESC, it needs to process signals in two directions: the transmit direction and the receive direction. Thus, the overhead bytes used by the NM to manage NEs cannot be looped back.

In the OptiX Metro 6100 DWDM system, two of the same electrical regenerating boards (for example, the LRF, LRFS, TMR, TMRS and TRC1) can be installed in two paired slots in a subrack at an REG station so that the overhead bytes of the bidirectional supervisory channel can be looped back. In this way, the NM is able to manage the NEs on the link.

The slots of unidirectional electrical regeneration in the OptiX Metro 6100 subrack is: IU1 and IU8, IU2 and IU9, IU3 and IU10, IU4 and IU11, IU5 and IU12, and IU6 and IU13.

NOTE

Unidirectional electrical regeneration can be used between the electrical regenerating boards with SuperWDM and the electrical regenerating boards without SuperWDM. For example, an LRF and an LRFS can be configured in one unidirectional electrical regeneration unit.

3.2.14 Automatic Laser Shutdown

The OTU board in the system provides the automatic laser shutdown (ALS) function.

Functionality Implementation

NOTE

The ALS function provided by the OptiX WDM products has no any relationship with the ALS mentioned in ITU-T G.664. The repetition in terms of name and acronym is just a coincidence.

-
- When the system adopts the ESC, the ALS function for the WDM side of each OTU board must be set to **Disabled**, because the supervisory signals have already been multiplexed into the transmission channel for service signals by the OTU boards.
- As for the OTU board that accesses the OTN services, the ALS function for the WDM side is **Disabled** by default.

With the ALS function, the OTU board can automatically shut down or turn on the laser based on the condition of the input optical signals.

The ALS function applies only to the output optical interfaces on the WDM side and the client side of the OTU board. This function can be **Enabled** or **Disabled** through the network management system, as shown in [Figure 3-6](#) and [Figure 3-7](#). The ALS function is implemented by using the following methods:

- When a receive optical interface on the client side of OTU A board receives no optical signal, OTU B board automatically shuts down the laser on the corresponding transmit optical interface on the client side. See [Figure 3-6](#) (a) and [Figure 3-7](#) (a).
- When a receive optical interface on the WDM side of OTU B board receives no optical signal, OTU B board automatically shuts down the lasers on the transmit optical interfaces on the client side if ALS of the optical interfaces are set to **Enabled**. See [Figure 3-6](#) (b) and [Figure 3-7](#) (b).

Figure 3-6 ALS function diagram (OTU board without service non-convergence function)

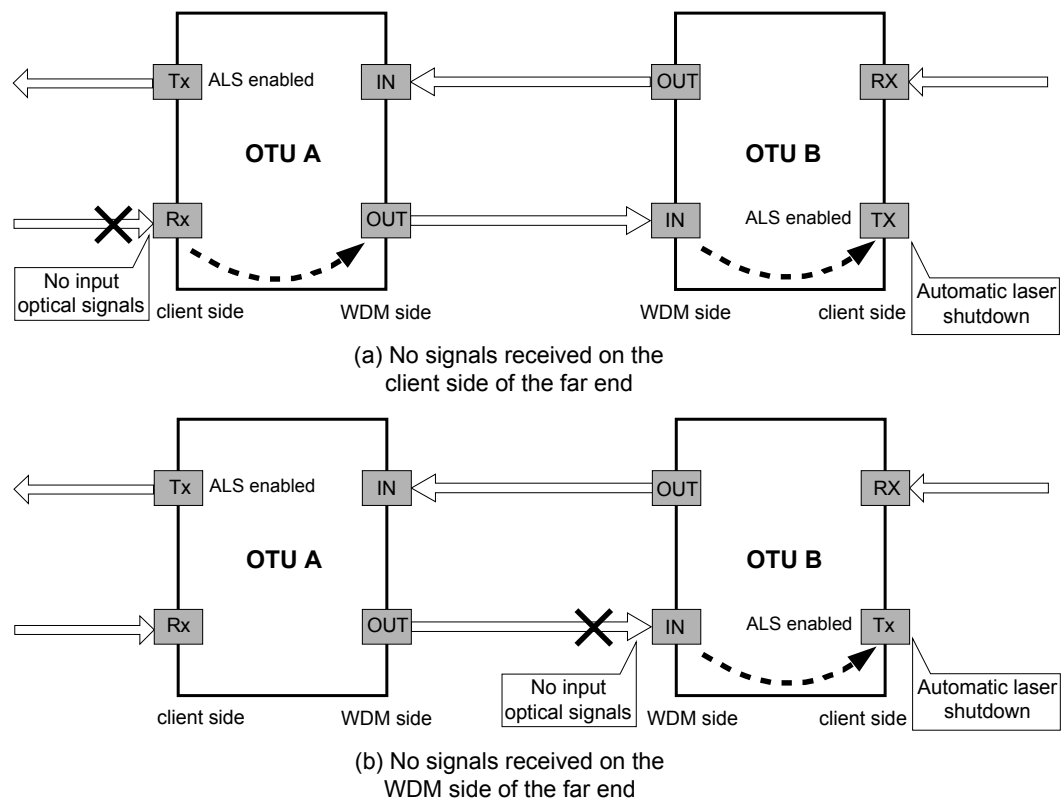
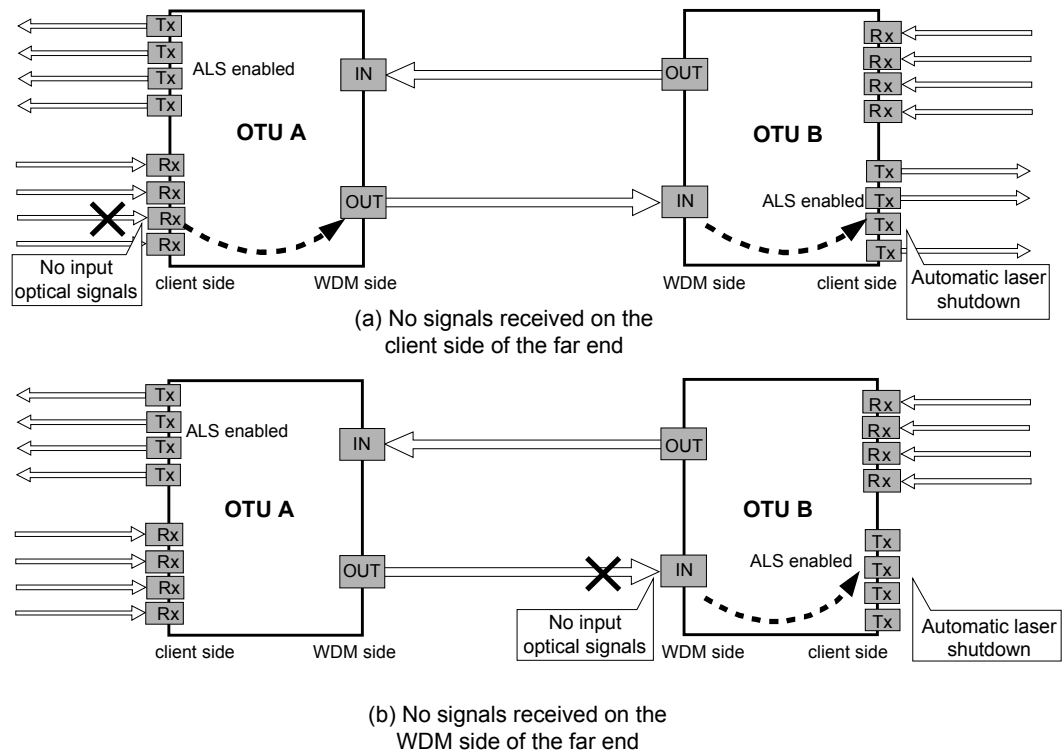


Figure 3-7 ALS function diagram (OTU board with service convergence function)



Default ALS State and Laser State

Table 3-4 describes the default ALS state and the default laser state on the client and WDM sides of the OTU board.

Table 3-4 Default ALS state and laser state of the OTU

Board Name	Default ALS State		Default Laser State	
	Client Side	WDM Side	Client Side	WDM Side
Convergence OTU board	Enabled	/	Close	Open
Non-convergence OTU board	Enabled	/	Close	Open
Regeneration board	/	/	/	Open

Matching Relation Between the ALS Status and the Status of Optical Interfaces

The state of an optical interface on an OTU board varies depending on the type of the board, the scenario where the board is used, and the ALS state of the optical interface. This section describes the matching relation between the ALS status and the status of an optical interface in terms of the OTU type and the scenario where the board is used.

The matching relation between the ALS status and the status of an optical interface is described according to the following three scenarios.

Application Scenario 1

A receive optical interface on the client side of OTU A, which is a convergence board, has no input optical power.

Figure 3-8 Schematic diagram of the ALS function (convergence OTUs)

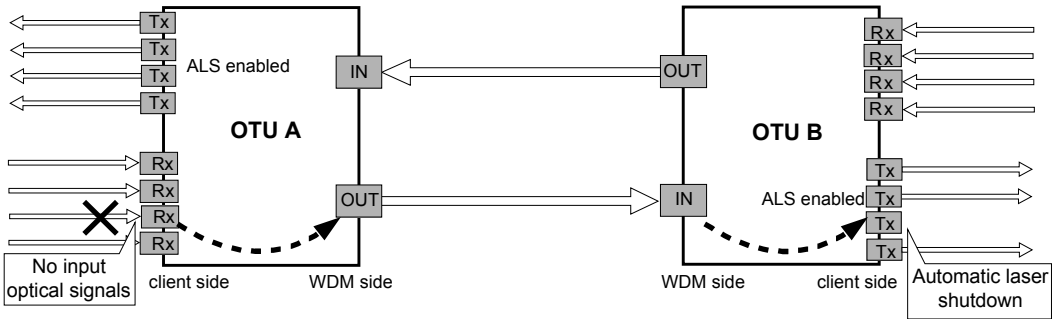


Table 3-5 Status of the optical interfaces on the OTUs (scenario 1)

Status of the WDM-Side Optical Interface on OTU A	Status of the WDM-Side Optical Interface on OTU B	Status of the Client-Side Optical Interface on OTU B (ALS Disabled)	Status of the Client-Side Optical Interface on OTU B (ALS Enabled)
The laser is open.	No alarm is generated.	The laser is open and an REM_SF alarm is reported.	The laser is closed.

Application Scenario 2

All receive optical interfaces on the client side of OTU A, which is a convergence board, have no input optical power.

Figure 3-9 Schematic diagram of the ALS function (convergence OTUs)

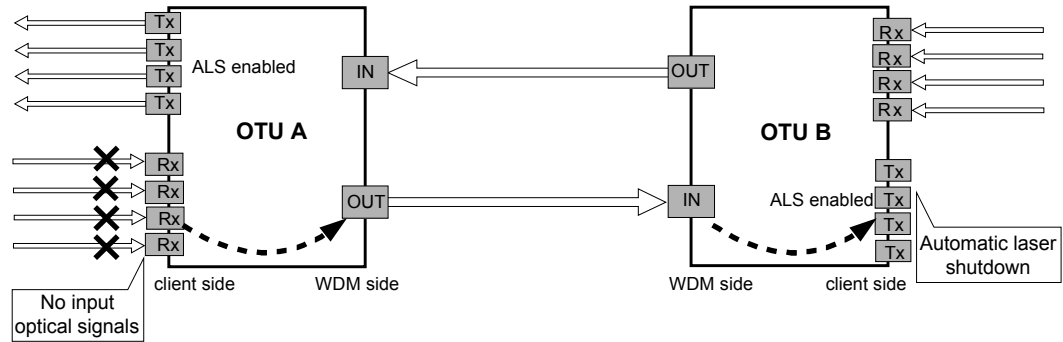


Table 3-6 Status of the optical interfaces on the OTUs (scenario 2)

Status of the WDM-Side Laser on OTU B When WDM-Side Laser on OTU A Transmits Light Forcibly	Status of the WDM-Side Laser on OTU B When Client-Side Laser on OTU A Transmits Light Forcibly ^a
The laser is open.	The laser is closed and an REM_SF alarm is generated.
a: The client-side ALS on OTU B is Disabled , which is the default state.	

Application Scenario 3

A receive optical interface on the client side of OTU A, which is a non-convergence board, has no input optical power.

Figure 3-10 Schematic diagram of the ALS function (non-convergence OTUs)

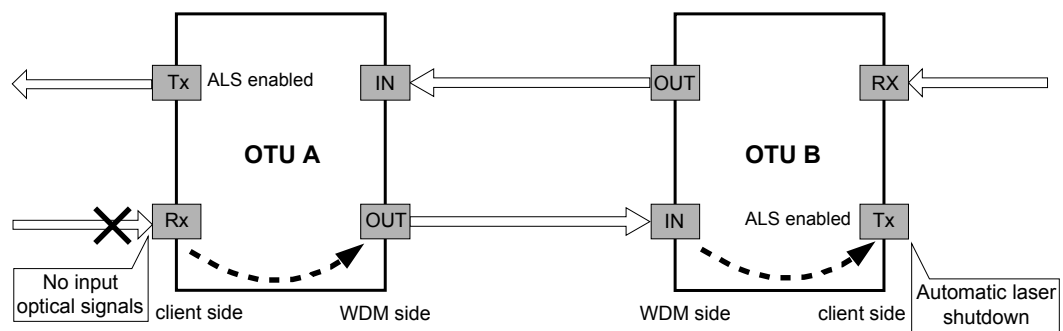


Table 3-7 Status of the optical interfaces on the OTUs (scenario 3)

Status of the WDM-Side Laser on OTU B When WDM-Side Laser on OTU A Transmits Light Forcibly	Status of the WDM-Side Laser on OTU B When Client-Side Laser on OTU A Transmits Light Forcibly ^a
The laser is open.	The laser is closed and an REM_SF alarm is generated.
a: The client-side ALS on OTU B is Dnabled , which is the default state.	

3.2.15 LPT Protocol Check

When the overhead byte supporting the link state pass through (LPT) protocol is added to the frame format of the WDM side signals, the system can monitor the running status of the network access point or the service network.

When the overhead byte supporting the link state pass through (LPT) protocol is added to the frame format of the WDM side signals, the OptiX Metro 6100 can monitor the running status of the network access point or the service network.

Normally, the OTU board at the upstream station transmits the LPT protocol information that indicates normal WDM side transmission line to the OTU board at the downstream station. When the status of the upstream WDM side transmission line changes, for example, a fault occurs or a fault is removed, the OTU board at the upstream station transmits the LPT packet that indicates network status change to the OTU board at the downstream station. When the downstream station knows that the status of the transmission line changes, it enables or disables the standby transmission line to ensure that services on the transmission line are available.

The OTU boards with multi-port provide the LPT base on the port and the status of the ports are individual.

The boards that support LPT contain the LOG, LOGS, LQG, LDG, FDG, LQM, LQM2, LBF, LBFS, LOM, LOMS, ELOG and ELOGS.

3.2.16 Optical Power Management

The system provides APE, ALC and IPA functions, realizing the management of optical power.

Automatic Power Equilibrium

When the transmission distance reaches a certain length, the powers of channels in the DWDM differ sharply, which causes optical signal-to-noise ratio (OSNR) degrade of signal at the receive end and affects the receiving performance of the system. See [Figure 3-11](#).

The OptiX Metro 6100 DWDM provides the automatic power equilibrium (APE) function. The system automatically adjusts the optical power on the transmission end of each channel and ensures that the flatness of the optical power on the receive end is close to the value obtained during the deployment commissioning, which then ensures the OSNR. See [Figure 3-12](#).

Figure 3-11 Flatness of optical power at the receive end when the APE is not enabled

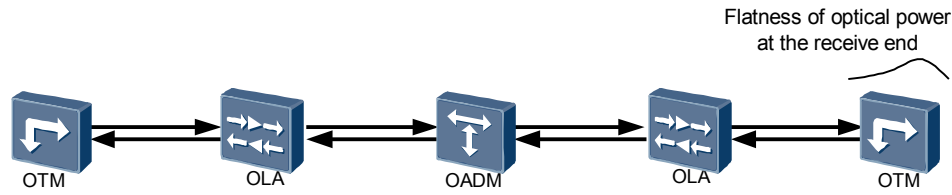
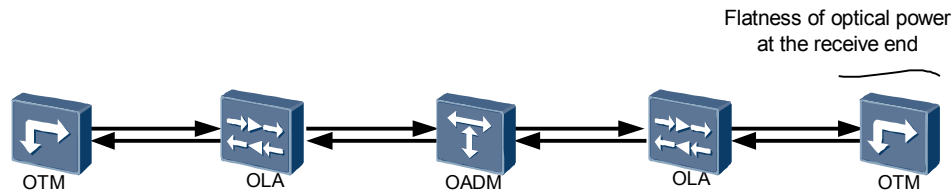


Figure 3-12 Figure 3-18 Flatness of optical power at the receive end when the APE is enabled



The APE function is mainly used in the situations where there are many spans.

The APE function facilitates the deployment commissioning of a WDM system and the network maintenances afterward. You can set the APE mode to APE with operator or network maintainer involved. In this way, the operator or network maintainer can decide whether to enable APE according to the specific network conditions.

For details about the APE function and application, refer to [10.4 Automatic Power Equilibrium](#).

Automatic Level Control

In DWDM system application, aging of optical fibers and optical connectors and human factors may result in abnormal line attenuation. For the system in which the optical amplifier (OA) is only in the gain control mode, when the attenuation of a span increases, the input and output powers of all downstream OAs decrease, the OSNR of the system degrades, and the received power by the receiver also decreases, thus greatly affecting the receiving performance. The closer the attenuated line is to the transmit end, the more impact on OSNR there is.

The OptiX Metro 6100 DWDM system provides the automatic level control (ALC) function. For the system that adopts the ALC mode, when the attenuation of a span increases, only the input power of the OA in this span decreases. Its output power, the input power and output power of other downstream OAs do not change. Hence, the influence on the OSNR is much less, and the receive optical power of the receiver does not change. As a result, the quality of the transmission signal and maintenance of the equipment are improved. [Figure 3-13](#) and [Figure 3-14](#) show respectively the power changes at the optical line amplifier (OLA) relay stations of the system in the gain control mode and the power control mode in the case of abnormal line attenuation.

Figure 3-13 Power change of the system in the gain control mode in the case of abnormal line attenuation

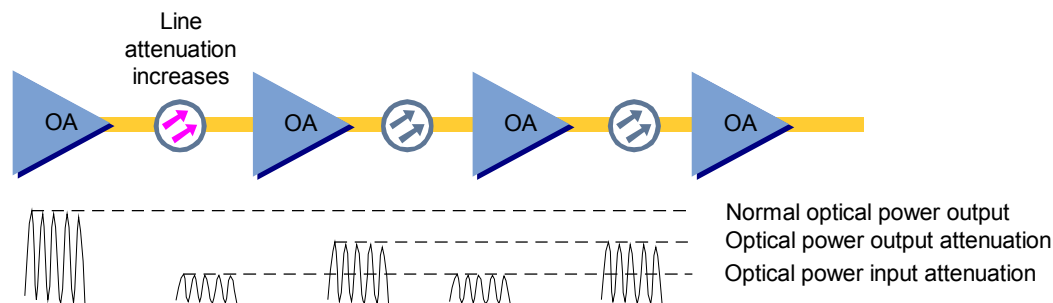
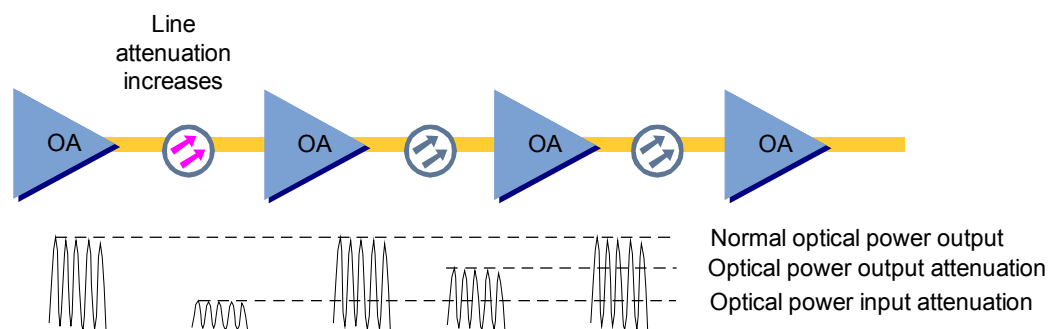


Figure 3-14 Power change of the system in the power control mode in the case of abnormal line attenuation



During normal operation, there are two factors that would change the input power of the OA.

- Increase or decrease of the channel number (possibly many channels are added/dropped at the same time)
- Abnormal line attenuation

The ALC adjustment usually takes several minutes. The abnormal line attenuation also takes a long time and the system redundancy design allows the occurrence of abnormal line attenuation. If the attenuation is within the system design range, the minute-level power adjustment period is able to ensure the normal operation of the system.

The ALC function has two modes of realization: wave number detection and power reference.

For details about the ALC function and application, refer to [10.3 Automatic Level Control](#).

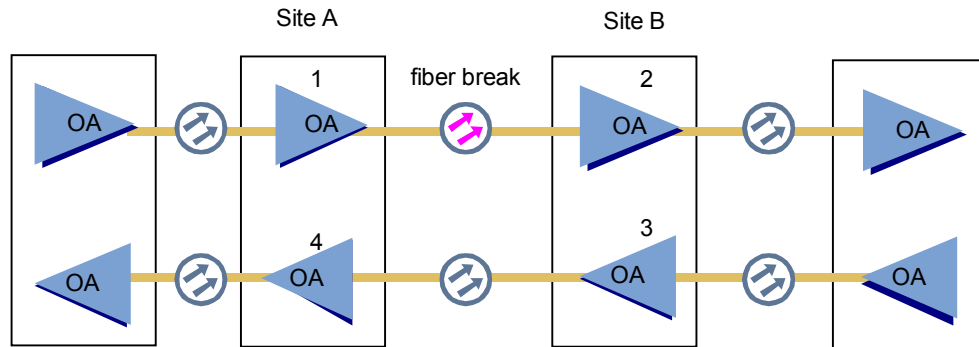
Intelligent Power Adjustment

As required in ITU-T G.664, the OptiX Metro 6100 DWDM system provides the intelligent power adjustment (IPA) function. When the optical signals are lost in one or multiple optical relay spans on the working path because of fiber cut, equipment downgrade, or poor connection of the connector, the system IPA check unit detects the loss of optical signals on the line, that is, the system detects alarms such as loss of signal (LOS), loss of frame (LOF), and LOC. Then, the system promptly decreases the output optical power of one OA in the upstream under the

safety level to prevent harm to maintainers. When the optical signals are restored, the system restores the OA.

Figure 3-15 shows the IPA function of the OptiX Metro 6100 DWDM system. When fiber cut occurs on the line, and the OA 2 on Station B detects an alarm of LOS, LOF, or LOC, station B shuts down OA 3. In the same way, OA 4 on station A detects an alarm of LOS, LOF, or LOC, and station A shuts down OA 1. When the optical signal restores, OA 3 and OA 1 are restarted and they continue to run.

Figure 3-15 IPA function



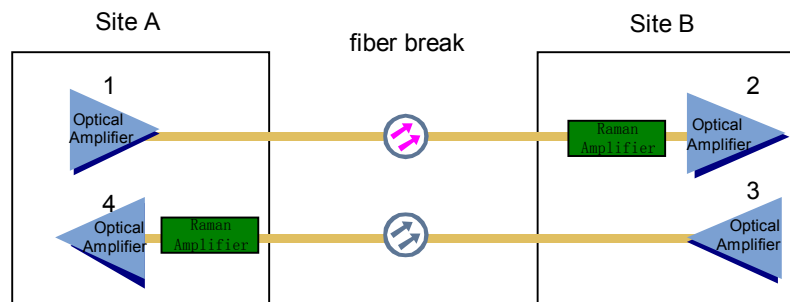
For details about the IPA function and application, refer to [10.1 Intelligent Power Adjustment](#).

Intelligent Power Adjustment in a Raman System

As required in ITU-T G.664, the OptiX Metro 6100 DWDM system provides the IPA function. When the system is configured with the Raman amplifier, the optical power that overflows from the fiber cut end face is too large, normal IPA cannot properly ensure the safety of maintainers during operations on this system. Hence, the Raman system IPA function is derived from the normal IPA.

When the OptiX Metro 6100 DWDM system is configured with the Raman amplifier, the pump optical power that is reversely output is very high. Hence, when a fiber cut is detected, the Raman amplifier should also be shut down to ensure that the line optical power is safe. See [Figure 3-16](#). When the optical signal restores, OA 3 and OA 1 are restarted and they continue to run.

Figure 3-16 Raman System IPA



For details about the Raman system IPA function and application, refer to [10.2 Intelligent Power Adjustment of Raman System](#).

3.2.17 NTP Technology

The system supports the Network Time Protocol (NTP). The NTP is used to synchronize the distributed time server and the client.

Basic Concept

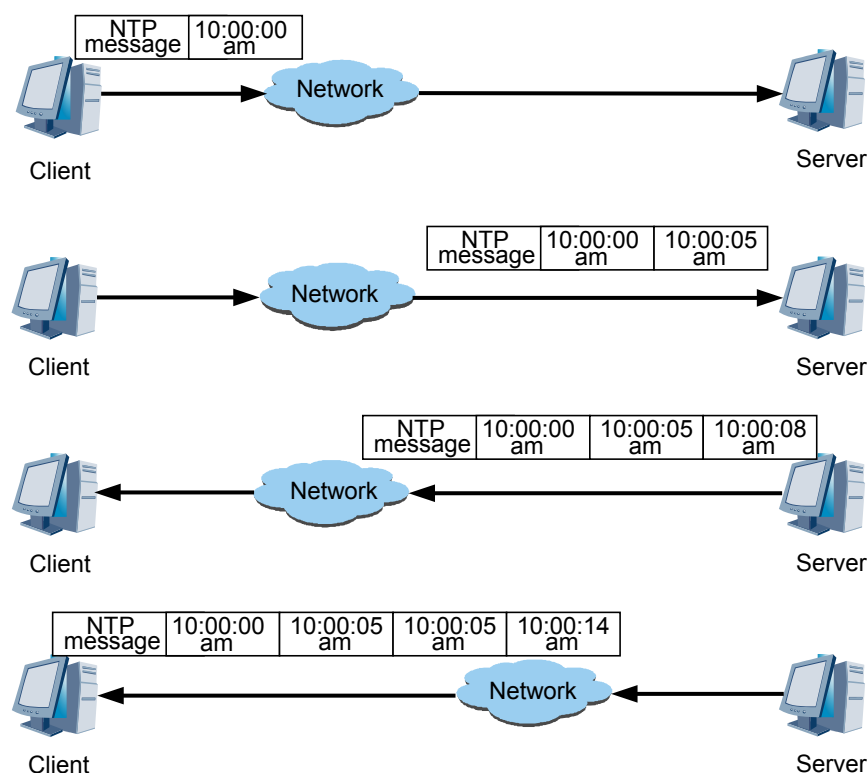
The NTP defines the data formats, algorithms, entities and protocols used during the realization of the protocol:

- The NTP is based on Internet Protocol (IP) and User Datagram Protocol (UDP). It can also be used by other protocols.
- The NTP develops from time protocol and Internet Control Message Protocol (ICMP) timestamp message. It has special design for correctness and robustness.
- The NTP defines the mechanism of time synchronization. In theory, the accuracy can reach billionth second.
- The NTP specifies the features of the local clock, time server, and the method used to estimate the time difference between the local clock and time server.
- The NTP describes the clock-filter algorithm and clock select algorithm during the realization of the protocol. When there are multiple time servers in the network, the system selects the algorithm to calculate the time offset of each time server to improve the accuracy of the local clock.

Function Implementation

For the working principle of the NTP, see [Figure 3-17](#).

Figure 3-17 Principle of the NTP



The process of the system synchronization is as follows:

- The client sends an NTP message to the server. The message capsule has a timestamp that records the time when the capsule leaves the client. The timestamp is 10:00:00 am.
- When the NTP message capsule reaches the server, the server adds its own timestamp to the message capsule. This timestamp is 10:00:05 am.
- When the NTP message capsule leaves the server, the server adds its own timestamp to the message capsule once again. This timestamp is 10:00:08 am.
- Receiving the returned message capsule, the client adds a new timestamp. This timestamp is 10:00:14 am.

After this process, the client has enough information to calculate two important parameters:

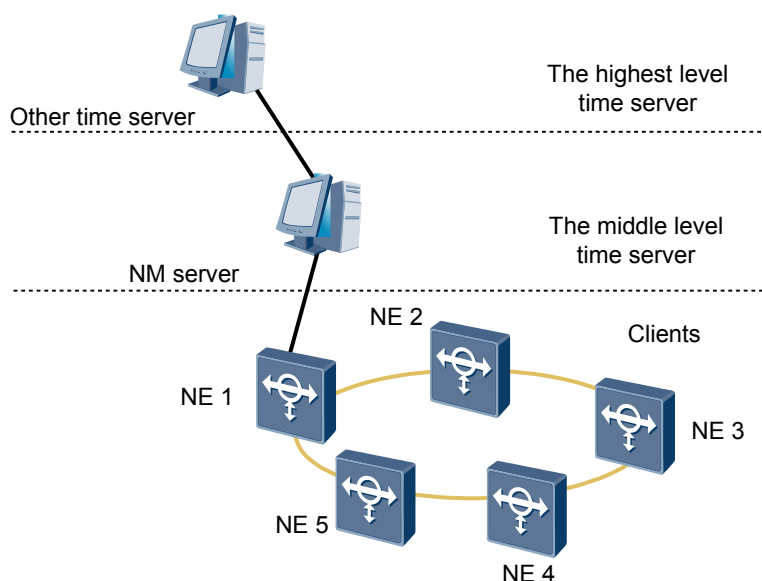
- Round trip delays of an NTP message
- Clock offsets between the client and the server

Thus, the client can set its own clock and keep synchronization with the server based on the information.

Application

For the synchronization of the network, see [Figure 3-18](#).

Figure 3-18 Synchronization of the network



As shown in [Figure 3-18](#), the equipment in the synchronized network can be classified into three categories:

- The highest level time server, referring to the 0-level time server.
- The middle level time server, referring to the 1- or 2-level time server that obtain time from the higher-level time server and provide time services for the lower-level time server.
- Clients, obtaining time without providing time services.

In application, choose the server and the client in the following way:

- Choose the network management server as the time server for the NE equipment. The server can be Windows 2000 Server or Solaris 10. The network management server can be set as the highest-level time server or set to obtain time from other time servers.
- The OptiX Metro 6100 NE can be only the client, obtaining time from the specified time server.

3.2.18 DCN Management

The system supports HWECC, IP over DCC, or OSI over DCC to realize the DCN management.

HWECC

The HWECC protocol is used to transfer the management information between Huawei optical equipment. When Huawei equipment interworks with third-party equipment, the HWECC protocol is unable to read the management information of the third-part equipment. However, the HWECC protocol can transparently transmit the management information. Users have centralized management to equipment with the available DCC resources.

The HWECC protocol has the following features:

- Supports flexible networking.

- Provides HWECC communication after the optical interfaces or network ports between NEs are connected.
- Transmits the management information of third-party equipment transparently.

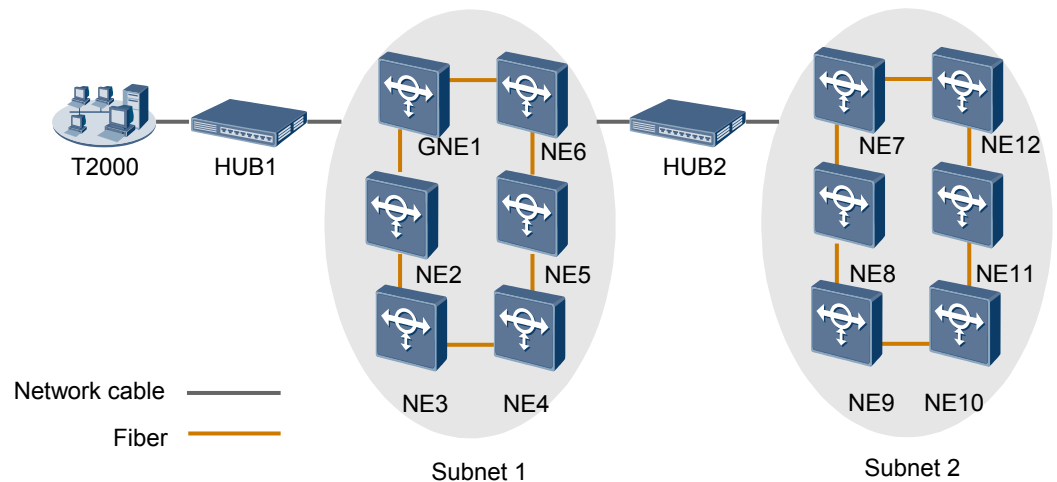
The HWECC protocol has three typical applications depending on the networking.

- Network management information is transmitted between OptiX WDM equipment only
When management information is transmitted between OptiX WDM equipment only, a gateway NE (GNE) is required to communicate with the NM. The NM connects to the GNE by using a Qx interface. In this way, the NM tests, manages and maintains the entire network.

With the NM, the network service quality is improved, the maintenance cost is decreased, and the proper use of network resources is guaranteed. Non-gateway NEs connect to the GNE by using ECC channels, which transmits management information.

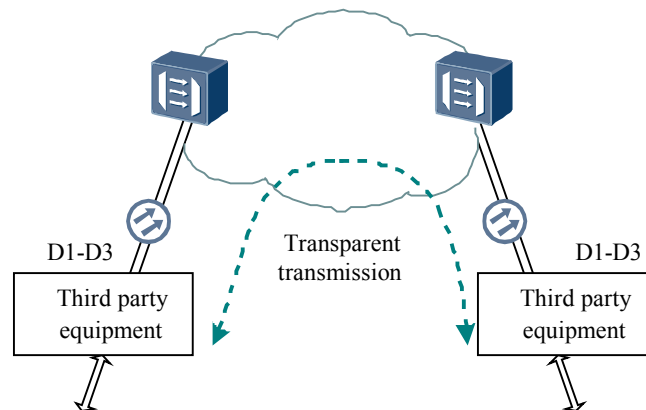
See [Figure 3-19](#) In certain applications, the Ethernet port can be used to achieve extended ECC communication between NEs.

Figure 3-19 Network with extended ECC



- Network management information of the third-party equipment is transmitted transparently
When there is OptiX WDM equipment between third-party equipment, the D4–D12 bytes in OptiX WDM equipment can be used to transmit the management information of third-party equipment. See [Figure 3-20](#).

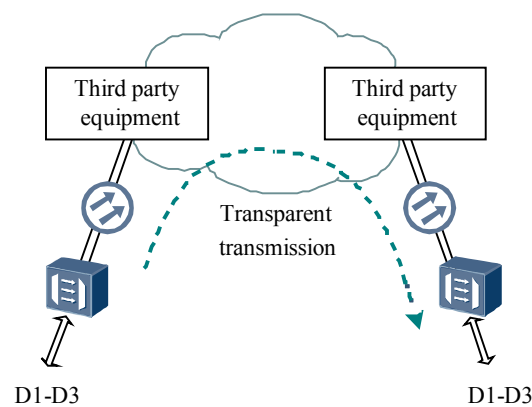
Figure 3-20 Transparent transmission of management information of third-party equipment (ECC)



- Network management information is transmitted transparently by the third-party equipment

When there is third-party equipment between OptiX WDM equipment, the D4–D12 bytes in third-party equipment can be used to transmit the management information of OptiX WDM equipment. See [Figure 3-21](#).

Figure 3-21 Management information transmitted transparently by third-party equipment (ECC)



For the networking planning principles of the HWECC protocol, refer to [12.11.4 HWECC Planning Rules](#).

IP over DCC

The OptiX Metro 6100 supports remote operation and maintenance through the IP over DCC.

The IP over DCC follows TCP/IP standards and controls remote NEs through the Internet. It uses the D-byte in overheads for communication. The default D-byte is D1–D3. At present, the OptiX Metro 6100 supports dynamic and static routing functions.

The scheme of IP over DCC uses the network layer protocol for NM information transmission. It is required that the GNE, external DCN and element management system (EMS) all support

IP. Thus, the network composed of the third party equipment and that composed of Huawei equipment (such as the OptiX Metro 6100) can form a DCN based on IP protocol.

The following are features of IP over DCC:

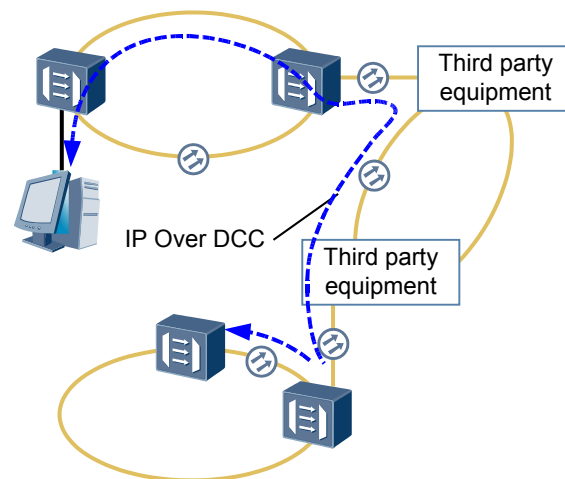
- Adopts the standard TCP/IP protocol, which makes Huawei equipment compatible with third-party equipment and simplifies the network management.
- Adopts the third layer function in the protocol stack, which makes the extra overheads or service paths unnecessary.
- Brings flexible networking models.
- Supports multiple application layer protocols.

Two networking models are available for IP over DCC.

- Network management information is transmitted transparently by the third-party equipment

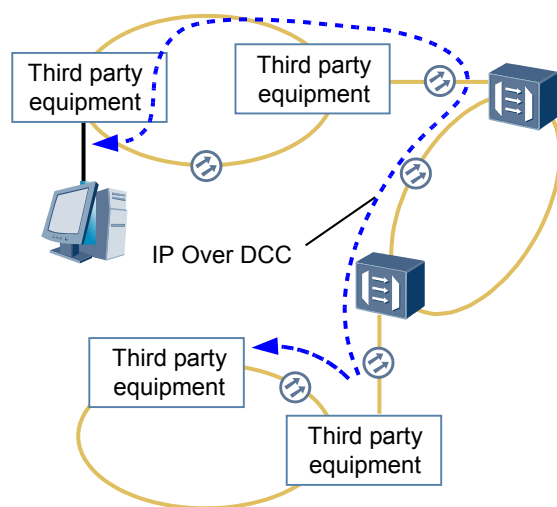
When there is third-party equipment between OptiX WDM equipment, management information can be transparently transmitted by the third-party equipment by using IP over DCC. See [Figure 3-22](#).

Figure 3-22 Management information transmitted transparently by third-party equipment (IP)



- Network management information of the third-party equipment is transmitted transparently
- When there is OptiX WDM equipment between third-party equipment, management information can be transparently transmitted by the OptiX WDM equipment by using IP over DCC. See [Figure 3-23](#).

Figure 3-23 Transparent transmission of management information of third-party equipment (IP)



For the networking planning principles of the IP over DCC protocol, refer to [12.11.5 IP over DCC Planning Rules](#).

OSI over DCC

OSI (open systems interconnection) over DCC is used when the OptiX WDM equipment interworks with other optical network equipment that supports OSI over DCC feature.

OSI over DCC adopts standard OSI protocols (also called TP4) to transmit NM information at network layer.

The following are features of OSI over DCC:

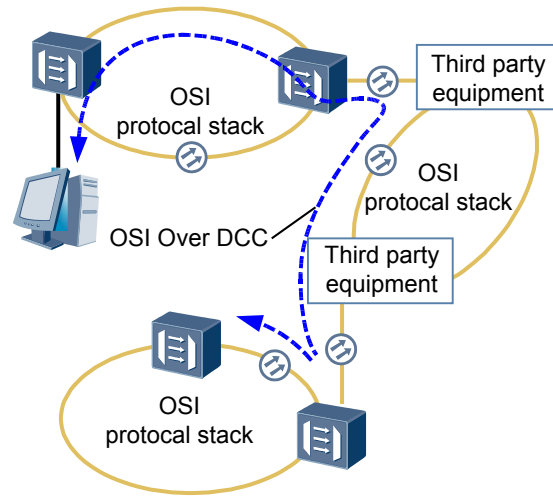
- When third-party equipment exists in the network, the OSI over DCC feature erases the limits on the networking of multi-vendor equipment. OSI over DCC enables more flexible networking by transmitting the management information transparently at the network layer.
- Users do not need to create extra DCN channels. The available DCC resource is enough to realize the centralized management to the equipment from different vendors.

The OSI over DCC has two typical applications depending on the networking.

- Network management information of OptiX WDM equipment is forwarded by the third-party equipment

When there is third-party equipment between OptiX WDM equipment, network management information can be transparently transmitted by the third-party equipment by using OSI over DCC. See [Figure 3-24](#). Huawei equipment is located on the edge of the network. Third-party equipment is located in the core network. The management information between the T2000 and equipment need be forwarded by the third-party equipment. In this case, at least one GNE is required in each subnet comprising Huawei equipment.

Figure 3-24 Management information transmitted transparently by third-party equipment (OSI)



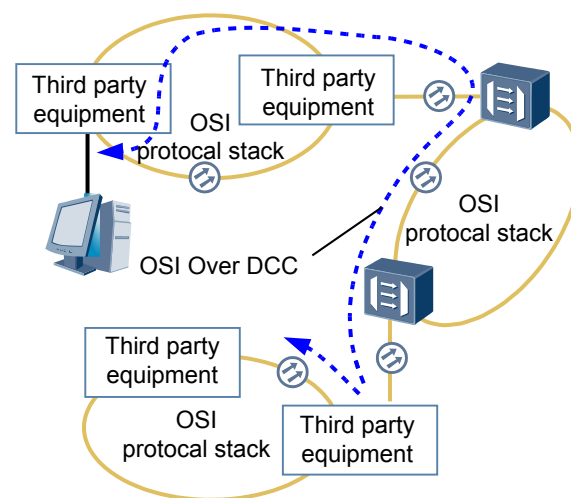
- Network management information of third-party equipment is forwarded by OptiX WDM equipment

When there is OptiX WDM equipment between third-party equipment, management information can be transparently transmitted by the OptiX WDM equipment by using OSI over DCC. See [Figure 3-25](#). Huawei equipment is located in the core network. Third-party equipment is located on the edge of the network. The management information between the NM and equipment of other vendors need be forwarded by Huawei equipment.

NOTE

In actual applications, the network is not classified in such a detail. The hybrid network where equipment from different vendors are scattered from the core to the edge of the network is most common.

Figure 3-25 Transparent transmission of management information of third-party equipment (OSI)



For the networking planning principles of the OSI over DCC protocol, refer to [12.11.6 OSI over DCC Planning Rules](#).

3.3 Features of Ethernet

The product supports Ethernet service switching based on GE ADM and L2 switching function.

3.3.1 GE ADM

The product provides the add/drop multiplexing (ADM) function for GE service and supports cross grooming for GE service granules.

Technical Background

At the convergence layer of the MAN, the transmission and protection of large gigabit Ethernet (GE) services need be considered. If the Metro WDM transmission equipment realizes service grooming at the sub-wavelength level, the WDM network can be developed from a static network to a network that can be configured dynamically. In this situation, pass-through, adding/dropping, and loopback of each GE service can be performed independently at any station and these operations do not affect services in other channels. Automatic GE service configuration can be realized by using remote management.

Advantages in Application

The OptiX Metro 6100 DWDM provides the add/drop multiplexing (ADM) function for GE services. It has the capability of cross-connection grooming for GE service granularity. It realizes electrical signal based service convergence and grooming at L1, and provides flexible and reliable networking configuration solution of data service application in MAN.

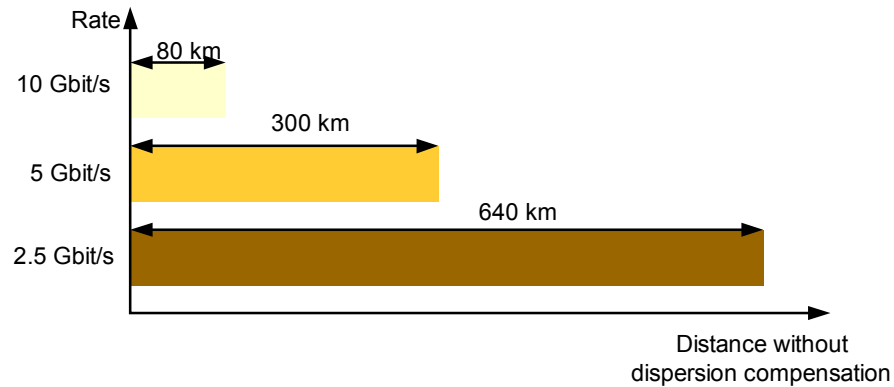
The OptiX Metro 6100 DWDM system uses the LOM, LOMS, LQG, ELOG, ELOGS, LOG, LOGS, L4G, TBE and EGS8 boards to realize GE ADM.

The GE ADM technology has the following features and advantages:

- 5 Gbit/s line rate

The line rate of the LQG and L4G boards are 5 Gbit/s. These boards support transmission of 300 km without dispersion compensation. Expensive EDC or any other line coding solution is not required. Compared with the traditional line rates of 2.5 Gbit/s and 10 Gbit/s, the 5 Gbit/s line rate realizes the best transport cost in each bit unit distance. See [Figure 3-26](#).

Figure 3-26 Comparison of distance without dispersion compensation between the 2.5 Gbit/s, 5 Gbit/s, and 10 Gbit/s line rates



- **WXCP function**
The GE ADM technology provides the function of wavelength cross-connection protection (WXCP) and realizes switching between the active and standby services by using the cross-connection grooming. The WXCP function has the advantages of high wavelength utilization, flexible configuration, fast switching, and high stability and reliability.
- **Dynamic network**
The GE ADM technology realizes grooming at the sub-wavelength level. It dynamically configures the network structure and transport routes, optimizes the configuration according to the network resources, and develops the WDM network from static to dynamic. If a network contains preserved bandwidth resources, you only need to specify the source and sink ports on the T2000. The system automatically creates the best route path and provides services fast.
- **Electrical regeneration**
The GE ADM technology realizes pass-through at the electrical layer of services at the sub-wavelength level. It also realizes the 3 R functions of the electrical regenerator. Hence, special electrical regenerator board is not required and the initial investment is decreased.
- **Low expansion cost**
During data network expansion, the cross-connection grooming of the GE ADM technology ensures smooth service upgrade and lowers the expansion cost.
- **High wavelength utilization**
The GE ADM technology shares the bandwidth of the same wavelength between different nodes and increases the bandwidth utilization of each wavelength.
- **End-to-end configuration and management**
The GE ADM technology allows remote end-to-end configuration, management, and monitoring of GE services on the T2000. Hence, the maintenance cost decreases.
- **Reliable QoS**
The GE ADM provides the monitoring of performance and bit errors on the WDM side and the client side. The system can monitor the status and quality of service transmission in real time.

Implementation Scheme

The backplanes of the OptiX Metro 6100 use the high-speed data bus. With the large-capacity space division cross-connection technology and powerful processing capacity of the Ethernet

Layer 2, the OptiX Metro 6100 DWDM system can independently distribute, converge, and groom the GE services at the wavelength or sub-wavelength level of single equipment. Hence, the cross-connection of each wavelength and the end-to-end management of the services at the sub-wavelength level can be realized inside single equipment.

With the remote configuration and management of the T2000, the GE services accessed into the OptiX Metro 6100 DWDM system can be groomed, multiplexed, protected, looped back at each node without affecting services in other channels.

The backplane bus with the GE ADM feature provided by the OptiX Metro 6100 DWDM system covers the slots of the subrack. For the more details about distribution of the cross-connection slots, refer to [8.2 Application and Networking of the GE ADM Feature](#).

3.3.2 L2 Switching Capability

The product supports Ethernet service switching based on L2 switching. It also realizes basic Ethernet management, including: the Ethernet private line (EPL) services, the Ethernet virtual private line (EVPL) services and the quality of service (QoS).

Technical Background

In the traditional transmission system, many wavelengths that carry GE services in the network are not saturated. The actual traffic bandwidth may be only 100 Mbit/s to 200 Mbit/s. In this type of system, normally, multiple links are converged to one MAN convergence node, multiple convergence nodes are further converged to a central node, and the central node transmits services to the router of the transmission equipment at the backbone layer to process. This processing method simply converges the services level by level, and does not require complicated data processing capability.

The system works together with the LAN switch equipment, which provides the L2 switching feature. Hence, the system converges VLAN traffics and improves the performance of service grooming.

Advantages in Application

The OptiX Metro 6100 DWDM configures the OTU board of L2 switching and enables built-in VLAN traffic convergence capability. Hence, operations of wavelength integration, sub-wavelength multiplexing, grooming, protection, and data traffic convergence are realized at a single station on single equipment, without the LAN switch.

The OptiX Metro 6100 DWDM system provides the following L2 service processing capabilities:

- VLAN traffic convergence

The OptiX Metro 6100 DWDM system provides the L4G board of the Ethernet L2 switching feature.

The board processes the VLAN labels of services. It accesses multiple GE services whose bandwidth is not full from the client side optical interface, attaches VLAN labels to the services, and converges services into one or more GE services. The converged GE services are then multiplexed into the standard wavelength by using the GE ADM function for line transmission.

At the opposite station, the process is reverse. The standard wavelength is obtained from the optical transmission line, and then GE services are demultiplexed from the standard wavelength by using the GE ADM function. The system obtains different VLAN labels,

distinguishes multiple GE services whose bandwidth is not full according to the VLAN labels, and then outputs the GE services at the client side optical interface.

The L4G board has the L2 processing capability and converges/deconverges multiple GE services into/from one GE service. The L4G board also has the function of GE ADM, GE service mapping, framing, and conversion between optical and electrical signals. The line rate on the WDM side is 5 Gbit/s.

As the tributary board of the L4G, the EGS8 provides functions of VLAN convergence, bandwidth convergence, and GE ADM. The EGS8 provides even more powerful GE service convergence and realizes flexible bandwidth utilization. It is mainly used to extend the number of client side optical interfaces on the L4G board. As a result, the OptiX Metro 6100 DWDM system can process a maximum of 16-channel L2 GE services.

- VLAN broadcast

The L4G and EGS8 boards support IPTV application.

In an IPTV network, the traffic of the uplink service is small and each node only processes the IGMP packets, and transports the IP packets that contain relevant authentication information to the broadband remote access server (BRAS) by using the networkwide route. For large service traffics of signals like television programs, the downlink service copies two service flows of the same VLAN ID, and uses the specific VLAN or GE channels to transport the service streams in the two directions of the ring. Dropping and pass-through of downlink service stream are realized at each station. Then, the station transmits the service stream to the downstream station. As a result, the broadcasting or multicasting of signals like television program is realized.

The IPTV services that are transmitted in two directions of the ring serve as the protection service of each other and are selectively received at each node.

- VLAN SNCP protection

The OptiX Metro 6100 DWDM system realizes protection based on a specific VLAN flow.

The upstream node that is configured with the L4G and EGS8 can use the broadcasting mode of dual-fed services with different VLAN IDs to transmit OAM frames periodically at the out port of the broadcasting. The protection port of the downstream node periodically checks the OAM frame information. The L4G and EGS8 trigger VLAN protection switching according to the status of the OAM frame. If the frame information is normal, the L4G and EGS8 configure the service on the active channel. If the frame information is abnormal, the L4G and EGS8 delete the service on the active channel and configure the service on the standby channel.

- Ethernet Management

The L4G and EGS8 boards have the basic Ethernet management capability. The boards provide functions of Ethernet service mounting management, QoS management, basic port attribute configuration, port aggregation management, test frame management, and multi-station wavelength sharing.

Implementation Scheme

Based on the following schemes, the OptiX Metro 6100 DWDM system supports two typical data applications, which are Ethernet private line (EPL) and Ethernet virtual private line (EVPL). The system also provides the following two Ethernet networking modes:

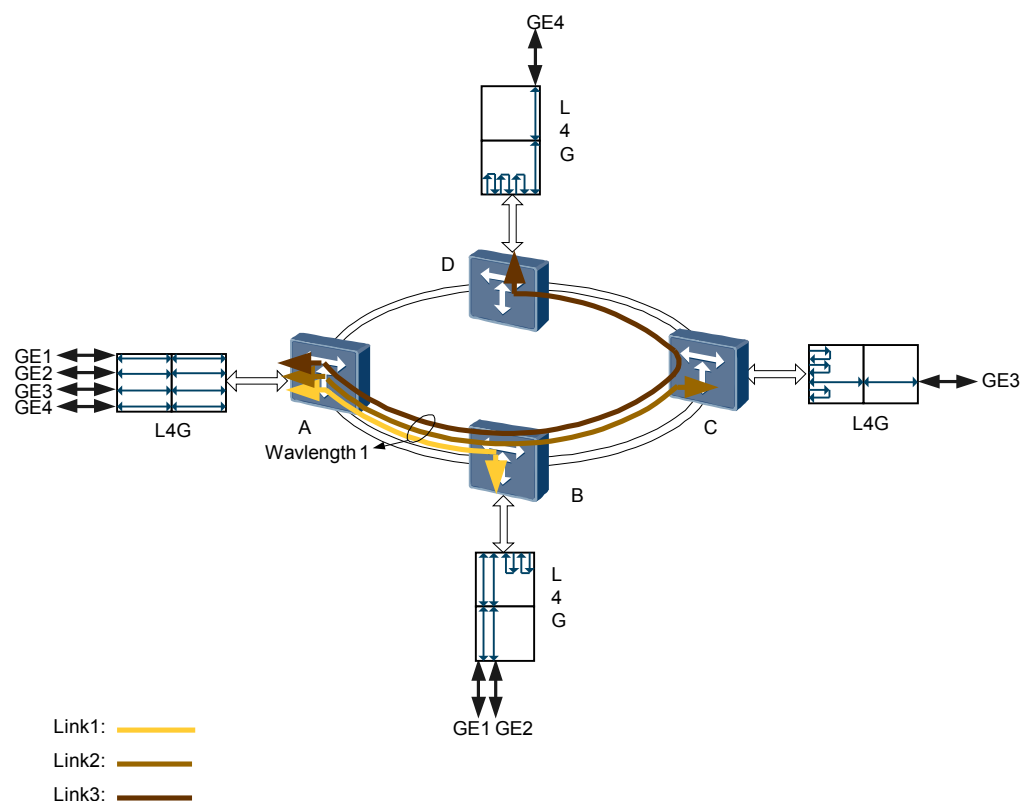
- Convergence from multiple GE paths to the central node

Project A is a ring network formed by stations A, B, C, and D. GE1, GE2, GE3 and GE4 services are transmitted by the L4G to nodes B, C, and D after passing through node A.

GE1, GE2, GE3 and GE4 services are transmitted to nodes B, C and D by sharing wavelength 1. The L4G of node A configures GE1 and GE2 services as Link1, GE3 service as Link2, and GE4 service as Link3. The L4Gs on nodes B, C and D respectively identify the labels of Link1, Link2 and Link3, and complete the add/drop of GE1, GE2, GE3 and GE4 services on each node. See [Figure 3-27](#).

The project uses the EPL data network. The L4G configures the client side port of the EPL link with the transparent transmission mode. It also multiplexes/demultiplexes GE services to/from line wavelength by using the cross-connect structure. The L4G has no restriction on service type or VLAN label. It supports point-to-point, ring and chain networks.

Figure 3-27 Convergence from multiple GE paths to the central node

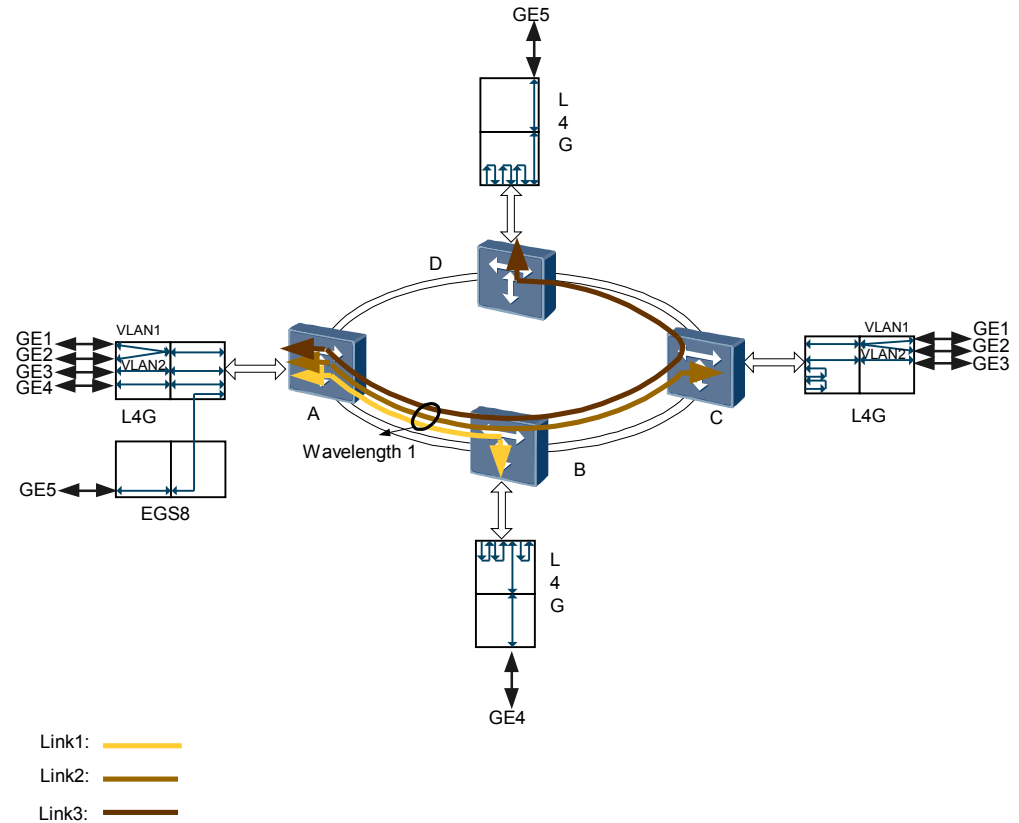


- Point-to-point private line of the VLAN or GE path

Project B is a ring network formed by stations A, B, C and D. GE1, GE2 and GE3 services are transmitted by the L4G to the node C after passing through node A. GE1 and GE2 services are converged into a full-bandwidth GE services. GE4 is transmitted by the L4G to node B after passing through node B. GE5 is accessed from the EGS8 of node A, connected to the L4G through GE ADM cross-connection, and then transmitted to node D.

GE1, GE2, GE3, GE4 and GE5 services are transmitted to nodes B, C and D by sharing wavelength 1. The L4G of node A configures the GE4 service as Link1, GE1, GE2 and GE3 services as Link2 after adding the VLAN1 and VLAN2 labels to GE1 and GE2 respectively, GE5 service as Link3. The L4Gs in nodes B, C and D respectively identify the labels of Link1, Link2 and Link3, and complete the add/drop of GE1, GE2, GE3, GE4 and GE5 services on each node. See [Figure 3-28](#).

Figure 3-28 Point-to-point private line of the VLAN or GE path



Project B uses the EPL and EVPL data networks. In this networking, if the accessed GE services are with VLAN labels, the L4G allows the use of the original label. If the accessed GE services are without VLAN labels, the L4G adds VLAN labels corresponding to the input and output ports to the services. The VLAN labels of all accessed services must be unique in the entire network according to the networking requirements.

NOTE

To clearly indicate the networking of service flows, [Figure 3-27](#) and [Figure 3-28](#) show the L4G boards with only one service flow direction of one wavelength. However, in the actual networking of the GE ADM, each wavelength with the GE ADM feature must be configured with one east OTU and one west OTU.

3.4 Features of Upgrade and Maintenance

The product has the following upgrade and maintenance features: software package loading, PRBS function, and pluggable optical modules.

3.4.1 Software Package Loading

Software upgrade by package loading refers to a process in which all NE software and board software of an NE are loaded at a time to replace the original software. This loading mode avoids the repetitive loading actions for the boards one by one and thus improves the upgrade efficiently.

Software package loading includes two modes: Non-diffusion mode and diffusion mode

- When you adopts non-diffusion mode, you can load all NE software and board software on the NE at the same time, so that you need not repeat the operation of loading software for one board at a time.
- When you adopts diffusion mode, you can load the software package to the first node NE only. The first node NE automatically diffuses the software package to downstream nodes according to the diffusion relation. This greatly improves the loading efficiency.

To ensure that the upgrade is successful, perform a physical check on the OptiX Metro 6100 before the upgrade, such as checking NE alarms and NE software.

Software package loading has the following features:

- The loading process is based on only the desired NE and is performed in one graphic user interface (GUI).
- The NE can be automatically managed. The software of the newly seated board is automatically updated when it does not match the software of the NE. So the efficiency of the software upgrade is improved.
- Software package loading is an incremental loading process in which only the files that need be updated are loaded.
- Software package loading supports the rollback function. When the software or hardware of the system is faulty, the loading fails, and the NE software is restored to the status before loading.

Software package loading applies to the following scenarios:

- Upgrade of NE software
- Replacement of equipment software version

3.4.2 PRBS Error Detection Function

Some OTU boards of the product supports pseudo random bit sequence (PRBS) check. You can enable or disable the PRBS test at the client side port of the OTU board on the T2000, to perform PRBS test during deployment without mounting extra instrument on the equipment.

Basic Concept

By starting or stopping on the T2000 a PRBS bit error test at the client-side interface of the OTU, the bit error test of the transmission link can be performed without attaching an extra meter to the equipment during equipment deployment.

Function Implementation

This function can be realized by using the combination of the PRBS signal generator and PRBS signal monitor. The PRBS signal generator of the OTU that supports PRBS bit error detection generates and transmits PRBS signals. The PRBS signal monitor monitors the PRBS codes transmitted from the PRBS signal generator and the PRBS codes looped back from the opposite station. In other words, the PRBS signal monitor compares the transmitted signals with the looped-back signals and determines whether the equipment or transmission line is normal.

Application

The OptiX Metro 6100 provides two kinds of the PRBS application.

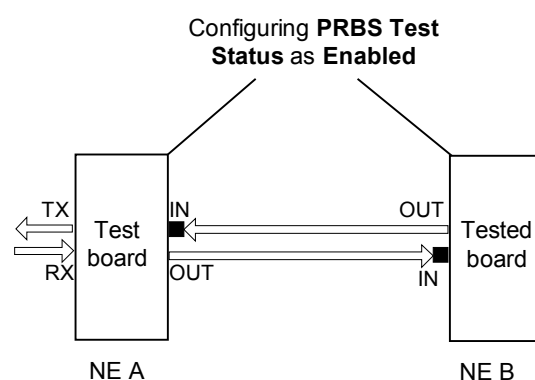
- PRBS detection on the WDM side

The WDM side of the following boards supports PRBS detection: LQM, LOM, ELOG, LOG, LWX and LBE.

Figure 3-29 shows the PRBS detection functional diagram on the WDM side. The PRBS realization process is as follows:

- An idle OTU on NE A, which supports PRBS detection on the WDM side, serves as the test board. That is, the PRBS signal generator and the PRBS signal monitor.
- An idle OTU on NE B, which supports PRBS detection on the WDM side, serves as the tested board. That is, the PRBS signal monitor.
- Use a fiber to connect the WDM side interfaces of the test board in NE A and the tested board in NE B.
- On the T2000, configure the WDM side interface of the tested board in NE B as **Outloop**.
- On the T2000, set the parameters of the test board and those of the tested board. Start the test.
- On the T2000, obtain in real time the PRBS test data of the channel. Evaluate the quality of the line according to the bit error count reported.

Figure 3-29 PRBS detection functional diagram on the WDM side



■ : Fixed optical attenuator

- PRBS detection on the WDM side and client side

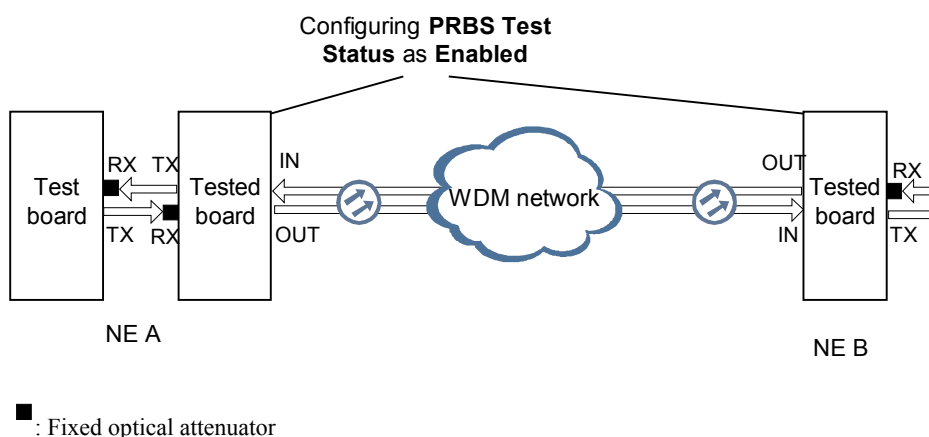
The WDM side and client side of the following boards support PRBS detection: TMX, LWF, ETMX, LBF, LWC1 and LWC1D.

Figure 3-30 shows the PRBS detection functional diagram on the WDM side and client side. The PRBS realization process is as follows:

- An idle OTU on NE A, which supports PRBS detection on the WDM side and client side, serves as the test board. That is, the PRBS signal generator and the PRBS signal monitor.
- Another OTU between NEs A and B, one of whose wavelength channels supports PRBS detection, serves as the tested board. The tested boards are connected to each other on the WDM side.

- On NE A, use a fiber to connect the client-side interface on the test board to that on the tested board.
- On NE B, use a fiber jumper to loop back the transmit and receive ends of the client-side interface of the tested board.
- On the T2000, set the parameters of the test board and those of the tested board. Start the test.
- On the T2000, obtain in real time the PRBS test data of the channel. Evaluate the quality of the line according to the bit error count reported.

Figure 3-30 PRBS detection functional diagram on the WDM side and client side



For the method to configure the PRBS detection function on the T2000, refer to the *Configuration Guide*.

3.4.3 Small Form-Factor Pluggable Module

There are two types of pluggable optical modules: the small form-factor pluggable (SFP) and the 10 Gbit/s small form-factor pluggable (XFP).

The OptiX Metro 6100 supports the following three pluggable optical modules:

- Small Form-Factor Pluggable (SFP)
- 10 Gbit/S Small Form-Factor Pluggable (XFP)

When it is required to adjust the type of accessed services or replace a faulty optical module, the optical module can be directly replaced without replacing its dominant board.

4 Hardware Architecture

About This Chapter

The hardware of the system includes the cabinet, subracks and functional boards.

[4.1 Cabinet](#)

The product consists of the cabinet, the subrack, the OADM frame, the DCM frame, the fiber-spooling frame, and the boards.

[4.2 Subrack](#)

The subrack is the basic working unit of the product. One subrack accesses two external power supplies for the subrack.

[4.3 OADM frame](#)

The OptiX Metro 6100 provides the OADM frame for the optical add and drop boards. The CTL board is the control board of the OADM frame.

[4.4 Function Boards](#)

The system provides different types of functional units.

4.1 Cabinet

The product consists of the cabinet, the subrack, the OADM frame, the DCM frame, the fiber-spooling frame, and the boards.

4.1.1 Structure

This section describes the appearance of the cabinet and system parameters.

As the basic working unit of the OptiX Metro 6100, the subrack can access two external power inputs for itself.

The subrack can be installed in a 300 mm ETSI cabinet. It can also be installed in a 600 mm ETSI cabinet, a 19-inch or 23-inch cabinet, or a 19-inch open rack as an integrated subrack.

In typical configuration, the OptiX Metro 6100 is installed in a 300 mm ETSI cabinet, as shown in [Figure 4-1](#).

The cabinet has a front door, a rear panel that is fixed with screws, and side panels at both sides.

A power box is installed on the top of the cabinet. The external $-48\text{V}/-60\text{V}$ DC power supply provides power for the equipment through the power box. The power box supports $-48\text{V}/-60\text{V}$ DC power dual-backup mode. It also provides interfaces for 16 channels of external alarm inputs and four channels of cabinet alarm outputs to facilitate the equipment management.

The cabinet has the following features in terms of design:

- The cabinet reserves sufficient space for fiber and cable routing to facilitate cabling and daily maintenance.
- The cabinet has slide rails on the top and at the bottom of the side door to support sliding installation.
- The cabinet has ventilation holes at the rear panel and at the top for equipment cooling.

Figure 4-1 Appearance of an 300 mm ETSI cabinet

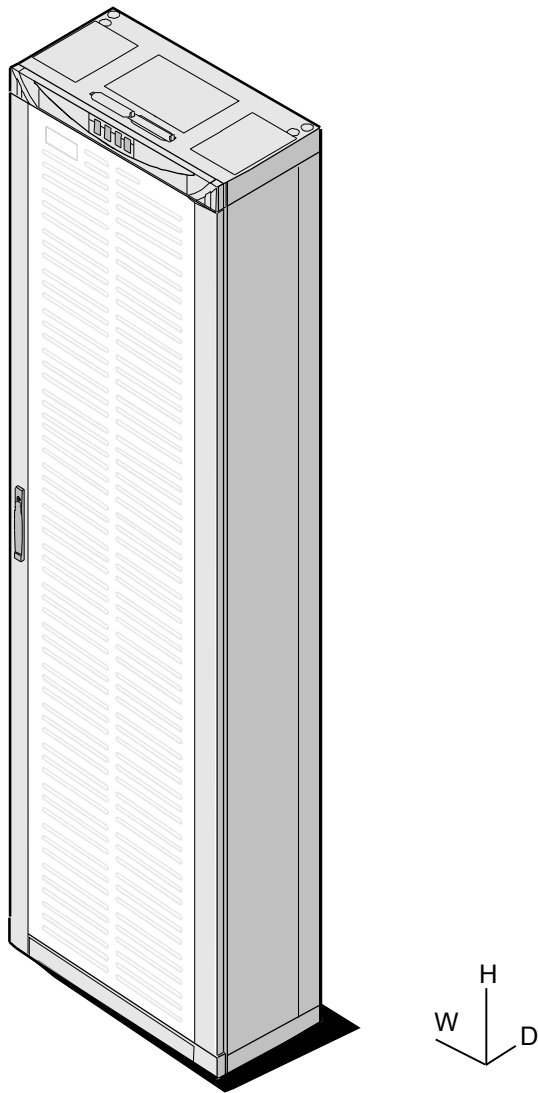


Table 4-1 lists the parameters of the ETSI 300-mm cabinet.

Table 4-1 Parameters of the ETSI 300-mm cabinet

ItemsItem	Parameters of 2.2 m-High Cabinet	Parameters of 2.6 m-High Cabinet
Dimensions	2200 mm (H) x 600 mm (W) x 300 mm (D)	2600 mm (H) x 600 mm (W) x 300 mm (D)
Weight	69 kg	80 kg
Maximum power consumption	2000 W	2000 W
Standard working voltage	-48 V/-60 V DC	-48V /-60 V DC

ItemsItem	Parameters of 2.2 m-High Cabinet	Parameters of 2.6 m-High Cabinet
Range of working voltage	−38.4 V to −72 V DC	−38.4 V to −72 V DC

4.1.2 Configuration of the Integrated Cabinet

This section describes the configuration principle of the integrated cabinet.

The system mechanical structure design is reflected in high integration. [Table 4-2](#) lists the full configuration of ETSI 300 mm cabinets of different heights. When the cabinet is not fully configured, configure the work subracks from bottom to top.

Table 4-2 Full configuration of 300 mm deep cabinets of different heights

Cabinet Height	Power Box Count	Subrack Count	DCM Frame Count	HUB Frame Count
2.2 m	1	3 (or two subracks and two OADM frames)	1 ^a (Or 4 in special cases)	1 ^a
2.6 m	1	3 (or two subracks and two OADM frames)	1 (Or 4 in special cases)	1
a: The 2.2 m cabinet can only contain either one DCM frame or one HUB frame.				

4.2 Subrack

The subrack is the basic working unit of the product. One subrack accesses two external power supplies for the subrack.

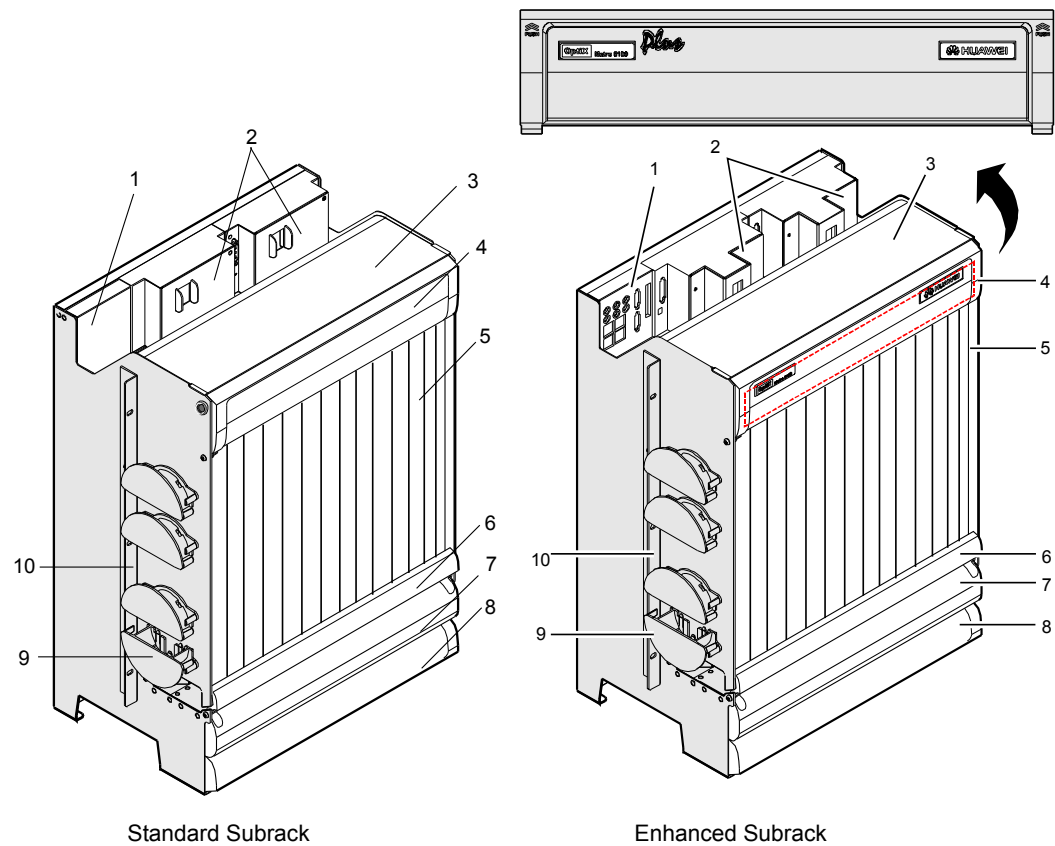
4.2.1 Structure

The subrack is divided into four areas from top to down: interface area in the upper part where electrical interfaces are accessed, board area in the middle, fiber routing area and fan area in the lower part.

Structure

[Figure 4-2](#) shows the structure of an OptiX Metro 6100 subrack.

Figure 4-2 Subrack structure



NOTE

The hardware platform of the OptiX Metro 6100 V100R005 uses the standard subrack.

"Plus" printed on the cover of the air exhaust vent indicates that this subrack is just for the enhanced subrack.

The hardware platform of the OptiX Metro 6100 V100R006 or above uses the enhanced subrack. Identify the subrack type during the operation and maintenance. There are some differences on DPFU and cross-connection capacity between the enhanced subrack and the standard subrack.

For details on the subrack, refer to the *Hardware Description*.

Technical Parameter

Table 4-3 shows the technical parameters of the OptiX Metro 6100 standard subrack.

Table 4-3 Technical parameters of the standard subrack

Item	Parameter
Dimensions	625.0 mm (H) x 440.0 mm (W) x 290.0 mm (D) ^a
Weight (empty subrack)	18.0 kg
Maximum power consumption (full configuration)	650.0 W
Minimum power consumption (only configuring with the SCC, the PMU and the fan tray assembly)	65.5 W
Power consumption of a fan tray assembly	43.0 W
Rated working current	16 A
Nominal working voltage	–48 V DC or –60 V DC
Working voltage range	–38.4 V to –72 V DC
Fuse capacity	20 A
a: H = Height, W = Width, D = Depth	

Table 4-4 shows the technical parameters of the OptiX Metro 6100 enhanced subrack.

Table 4-4 Technical parameters of the enhanced subrack

Item	Parameter
Dimensions	625.0 mm (H) x 440.0 mm (W) x 290.0 mm (D) ^a
Weight (empty subrack)	18.0 kg
Maximum power consumption (full configuration)	800.0 W
Minimum power consumption (only configuring with the SCC, the PMU and the fan tray assembly)	65.5 W
Power consumption of a fan tray assembly	43.0 W
Rated working current	16 A
Nominal working voltage	–48 V DC or –60 V DC
Working voltage range	–38.4 V to –72 V DC
Fuse capacity	30 A
a: H = Height, W = Width, D = Depth	

Table 4-5 shows the technical parameters of the common units.

Table 4-5 Power consumption of the common units

Unit Name	Maximum Power Consumption	Remarks
OTU subrack	449.5W	It is the power consumption when the subrack is installed with twelve LWMs (single-fed board), one SCC, one PMU, and one fan tray assembly.
OTM subrack	361.5W	It is the power consumption when the subrack is installed with eight LWMs (single-fed board), one M40, one D40, one SCC, one PMU, and one fan tray assembly.
OLA subrack	179.5W	It is the power consumption when the subrack is installed with two OAUs, two OBUs, two FIUs, one SC2, one SCC, one PMU, and one fan tray assembly.

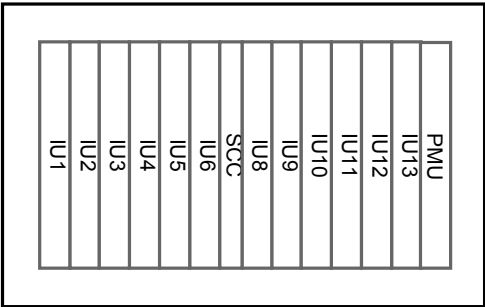
4.2.2 Slot Distribution

There are 14 slots in the board area of the subrack (from left to right when viewed from the front of the subrack). IU7 is for the SCC board. IU14 is for the PMU.

Figure 4-3 shows the slot distribution in the board area of the subrack.

The slot backplane of the OptiX Metro 6100 subrack adopts the distributed cross-connection. For details, refer to 8.2.1 Description.

Figure 4-3 Slot distribution of the subrack of the OptiX Metro 6100



4.2.3 Integrated Subrack

To install the OptiX Metro 6100 subrack into a 600 mm ETSI cabinet, a 19-inch/23-inch cabinet or a 19-inch open rack, you need to use the integrated subrack.

An integrated subrack is an assembly of the subrack, OADM frame and fiber-spooling frame, as shown in Figure 4-4.

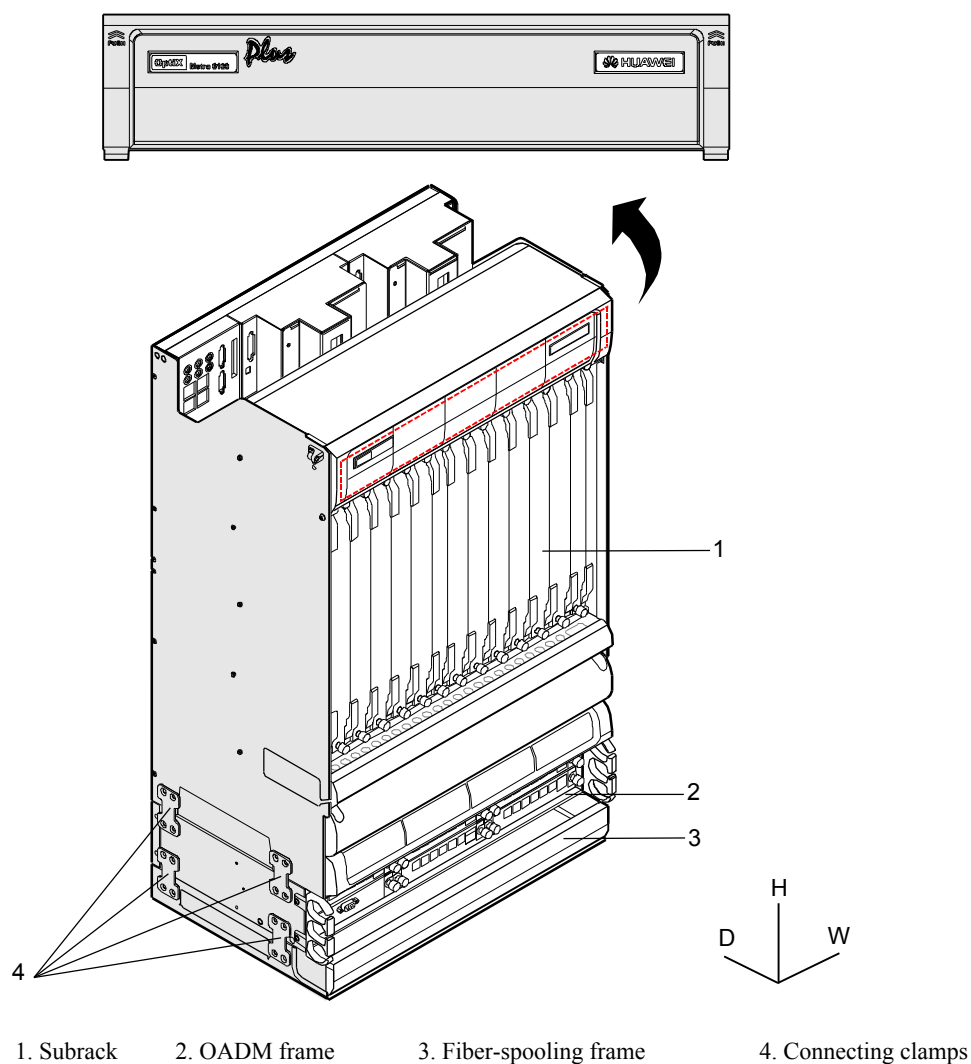
Figure 4-4 Integrated subrack**NOTE**

Figure 4-4 takes the enhanced subrack used in the hardware platform of the OptiX Metro 6100 V100R006 or above as example.

Table 4-6 shows the maximum of configurable integrated subracks in cabinets of different types.

Table 4-6 Maximum of configurable integrated subracks in different types of OptiX Metro 6100 cabinets

Cabinet Type	1600 mm (H) x 600 mm (W) x 300 mm (D)	2000 mm (H) x 600 mm (W) x 300mm (D)	2200 mm (H) x 600 mm (W) x 300 mm (D)	2600 mm (H) x 600 mm (W) x 300mm (D)
Open subrack	2	2	2	-
19-inch standard cabinet	2	2	2	2

Cabinet Type	1600 mm (H) x 600 mm (W) x 300 mm (D)	2000 mm (H) x 600 mm (W) x 300mm (D)	2200 mm (H) x 600 mm (W) x 300 mm (D)	2600 mm (H) x 600 mm (W) x 30 0mm (D)
23-inch standard cabinet	2	2	2	2
ETSI 600-mm cabinet	-	-	2	2

Dimensions of an integrated subrack of the OptiX Metro 6100 is 757.4 mm (H) × 440 mm (W) × 290 mm (D).

For working parameters, refer to [Table 4-3](#).

4.2.4 Installation Mode

The subrack of the OptiX Metro 6100 can be fed with power supplies independently. It can be installed in an ETSI 300-mm/600-mm cabinet, a 19-inch/23-inch cabinet, or a 19-inch open rack. The OptiX Metro 6100 can also adopt the integrated subrack mode and be installed in an ETSI 600-mm cabinet, a 19-inch/23-inch cabinet, or a 19-inch open rack.

For details on how to install the OptiX Metro 6100, refer to the *Installation Guide*.

4.3 OADM frame

The OptiX Metro 6100 provides the OADM frame for the optical add and drop boards. The CTL board is the control board of the OADM frame.

4.3.1 Structure

The use of the OADM frame saves the slots in the OptiX Metro 6100 subrack, and increases the system integrity. When used with an OADM frame, one subrack can access a maximum of 16 channels.

[Figure 4-5](#) shows the appearance of an OADM frame.

Figure 4-5 OADM frame

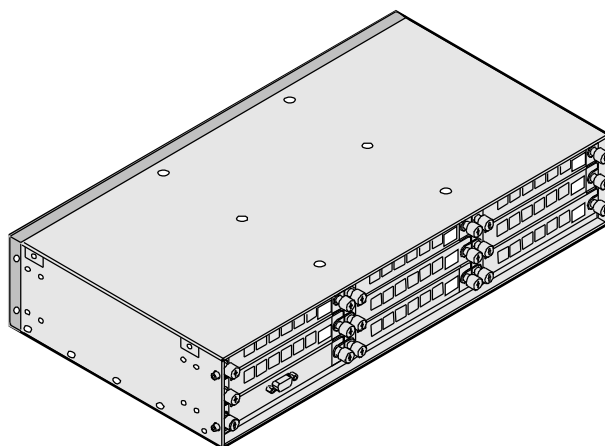


Table 4-7 shows the mechanical specifications of an OADM frame.

Table 4-7 Mechanical specifications of an OADM frame

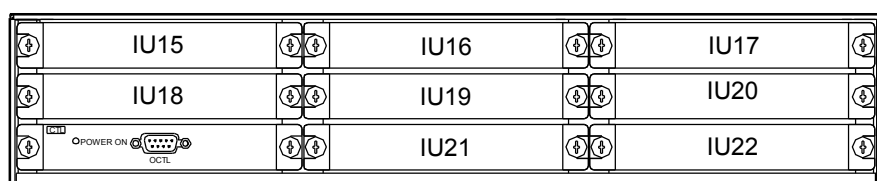
Item	Specification
Dimensions	86.0 mm (H) x 440.0 mm (W) x 290.0 mm (D) ^a
Weight	4.8 kg
a: H = Height, W = Width, D = Depth	

4.3.2 Slots in OADM Frame

The OADM frame provides nine slots.

The OADM frame is shown in **Figure 4-6**. The slot on lower left is used to house the CTL board. The other slots are defined as IU15 – IU22, to house the OADM boards such as the MR2, SBM2, SBM1 and EFIU boards.

Figure 4-6 Slots in the OADM frame



4.4 Function Boards

The system provides different types of functional units.

The boards can be divided into nine functional units, as shown in **Table 4-8**.

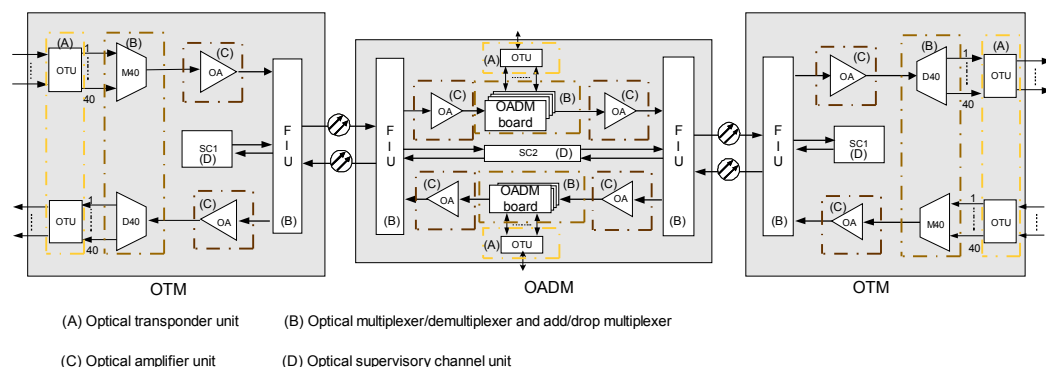
Table 4-8 Functional units of the OptiX Metro 6100 system

Functional Units	Boards
Optical transponder board	LU40S, LUR40S, TMX40S, LWF, LWFS, LRF, LRFS, LBE, LBES, ETMX, ETMXS, TMX, TMXS, TMR, TMRS, ELOG, ELOGS, LOG, LOGS, LWC1, TRC1, LWM, LWMR, LWX, LWXR, AS8, LQS, LQG, LDG, FDG, AP8, LQM, LQM2, FCE, LBF, LBFS, LOM, LOMS, L4G, EGS8, TBE, TRC2, LAM

Functional Units	Boards
Optical multiplexer and demultiplexer board	M40, V40, D40, FIU, EFIU
Optical add/drop multiplexer board	MR4, MR2, SBM1, SBM2, WSD9, WSM9, RMU9, WSMD4, DWC
Optical amplifier board	OAU, OBU, OPU, RPC
Optical supervisory channel and timing transmission board	SC1, SC2, TC1, TC2, ST1, ST2
System control, supervision and communication board	SCC, PMU
Optical protection board	OLP, SCS, OWSP, DCP, CP40
Spectrum analyzer board	MCA
Variable optical attenuator board	VOA, VA2, VA4

Figure 4-7 shows the position of each functional unit.

Figure 4-7 Position of the functional units in the OptiX Metro 6100



4.4.1 Optical Transponder Board

The optical transponder board (OTU) converges or converts the signals to output a standard DWDM wavelength compliant with G.694.1 or a standard CWDM wavelength compliant with G.694.2. It accesses one or multiple channels.

In this way, it helps the multiplexer board to perform wavelength division multiplexing on signals of different wavelengths.

All OTU boards of the OptiX Metro 6100 are transceivers and can perform the preceding process as well as its reverse process at the same time.

Table 4-9 Board name and category of OTUs

Service Type	Board	Board Name
40 Gbit/s OTU without service convergence	C9LU40S	40Gbit/s wavelength conversion board with AFEC function (Super WDM)
	C9LUR40S	40G transmit-receive line regenerating wavelength conversion unit with AFEC function (SuperWDM)
40 Gbit/s OTU with service convergence	C9TMX40S	40G tributary multiplexing/demultiplexing wavelength conversion board with AFEC function (Super WDM)
10 Gbit/s OTU without service convergence	L2LWF C9LWF	STM-64 transmit-receive line wavelength conversion board (FEC)
	C7LWF C8LWF CALWF	STM-64 transmit-receive line wavelength conversion board (AFEC)
	C6LWFS C9LWFS	STM-64 transmit-receive line wavelength conversion board (FEC, Super WDM)
	C7LWFS C8LWFS CALWFS	STM-64 transmit-receive line wavelength conversion board (AFEC, Super WDM)
	L2LRF	STM-64 regenerating wavelength conversion board with FEC function
	C6LRFS	STM-64 regenerating wavelength conversion board with FEC Function, Super WDM
	C6LBE C8LBE CALBE	10GE LAN transmit-receive line wavelength conversion board
	C6LBES C8LBES CALBES	10GE LAN transmit-receive line wavelength conversion board (Super WDM)
	C6TMR C8TMR C9TMR	Line regenerating wavelength conversion board for 10G (AFEC)
	C6TMRS C8TMRS C9TMRS	Line regenerating wavelength conversion board for 10G (AFEC, Super WDM)

Service Type	Board	Board Name
	CBTMRS	Line regenerating wavelength conversion board for 10G (AFEC-2, Super WDM)
	C8LBF C9LBF	10G universal transmit-receive line wavelength conversion board
	C8LBFS C9LBFS	10G universal transmit-receive line wavelength conversion board (AFEC, Super WDM)
	CBLBFS	10G universal transmit-receive line wavelength conversion board (AFEC-2, Super WDM)
10 Gbit/s OTU with service convergence	C8ETMX C9ETMX CAETMX	Enhanced 4 channels STM-16/OTU1 asynchronous MUX OTU-2 wavelength conversion board (AFEC)
	C8ETMXS C9ETMXS CAETMXS	Enhanced 4 channels STM-16/OTU1 asynchronous MUX OTU-2 wavelength conversion board (AFEC, Super WDM)
	CBETMXS	Enhanced 4 channels STM-16/OTU1 asynchronous MUX OTU-2 wavelength conversion board (AFEC-2, Super WDM)
	C6TMX C7TMX C9TMX	4 channels STM-16 asynchronous MUX OTU-2 wavelength conversion board (AFEC)
	C6TMXS C7TMXS C9TMXS	4 channels STM-16 asynchronous MUX OTU-2 wavelength conversion board (AFEC, Super WDM)
	C8ELOG C9ELOG	Enhanced 8-port Gigabit Ethernet multiplexing & wavelength conversion board (AFEC)
	C8ELOGS C9ELOGS	Enhanced 8-port Gigabit Ethernet multiplexing & wavelength conversion board (AFEC, Super WDM)
	CBELOGS	Enhanced 8-port Gigabit Ethernet multiplexing & wavelength conversion board (AFEC-2, Super WDM)
	C6LOG C9LOG	8-port Gigabit Ethernet multiplex and optical wavelength conversion board (AFEC)

Service Type	Board	Board Name
	C6LOGS C9LOGS	8-port Gigabit Ethernet multiplex and optical wavelength conversion board (AFEC, Super WDM)
	C8LOM	8-port multi-service multiplexing & optical wavelength conversion board (AFEC)
	C8LOMS	8-port multi-service multiplexing & optical wavelength conversion board (AFEC, Super WDM)
5 Gbit/s OTU with service convergence	C6LQG C9LQG	4-port Gigabit Ethernet service convergence board (FEC)
	C7L4G	Line wavelength conversion board with 4 x Gigabit Ethernet line capacity
OTU without service convergence at the rate of 2.5 Gbit/s or low	C6LWC1	STM-16 transmit-receive line wavelength conversion board
	C8LWC1 C8LWC1D C9LWC1	STM-16/OTU1 transmit-receive line wavelength conversion board
	C6TRC1 C8TRC1	OTU1 line regenerating wavelength conversion board
	C8TRC2	OTU1 bidirectional line regenerating wavelength conversion board
	C6LWM C8LWM	Multirate (STM16/4/1) wavelength conversion board
	C6LWMR C8LWMR	Multirate (STM16/4/1) line regenerating wavelength conversion board
	C6LWX C8LWX	Arbitrary bit rate (16Mbit/s~2.5Gbit/s) wavelength conversion board
	C6LWXR C8LWXR	Arbitrary bit rate line regenerating wavelength conversion board
2.5 Gbit/s OTU with service convergence	C6LQS C7LQS	4 x STM-1/4 multiplex wavelength conversion board
	C6LDG C8LDG	2-port Gigabit Ethernet wavelength conversion board
	C6FDG C8FDG	2-port Gigabit Ethernet wavelength conversion board with FEC
	C6FCE	Fiber channel distance extension board

Service Type	Board	Board Name
	C8LQM C9LQM	4-channel protocol-independent service convergence board
	C9LQM2	Double 4-channel protocol-independent service convergence board or 8-channel protocol-independent service convergence board
	L2AS8 C7AS8	8-channel any SDH convergence board
	C6AP8	8-channel protocol-independent service convergence board
Other OTU	C7EGS8	8 x Gigabit Ethernet switching processing board
	C8TBE	10 Gigabit Ethernet tributary board
	C7LAM	Any protocol & multi-channel line wavelength conversion board

Figure 4-7 shows the position of the OTU boards in the system.

Functions of OTUs boards or units are listed in [Table 4-10](#), [Table 4-11](#), [Table 4-12](#), [Table 4-13](#), [Table 4-14](#), [Table 4-15](#), [Table 4-16](#) and [Table 4-17](#).

Table 4-10 Major functions of 40 Gbit/s OTUs without service convergence

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C9LU40S	- Wavelength conversion - Line code: eDQPSK	- Supports the tunable wavelengths - Supports FEC/AFEC^a - Supports SuperWDM	- Client-side optical signals: 1xSTM-256/OC-768/40G POS/40G WAN/OTU3 - WDM-side optical signals: OTU3	C9LUR40S	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C9LUR40S	3R (reshaping, retiming and regeneration) - Line code: eDQPSK	- Supports the tunable wavelengths - Supports FEC/AFEC^a - Supports SuperWDM	- Client-side optical signals: NA - WDM-side optical signals: OTU3/OTU3e	-	DWDM
a: The default working mode of the board is AFEC, which can be set or modified on the T2000. b: The decoding and encoding of the signals comply with ITU-T G.975.1. c: The overhead processing of the signals comply with ITU-T G.709.					

Table 4-11 Major functions of 40 Gbit/s OTUs with service convergence

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C9TMX40S	- Wavelength conversion^{b c} - Line code: eDQPSK	- Supports the tunable wavelengths - Supports FEC/AFEC^a - Supports SuperWDM	- Client-side optical signals: 4xSTM-64 / OC-192/10GE WAN/10GE LAN/OTU2/OTU2e - WDM-side optical signals: OTU3/OTU3e	C9LUR40S	DWDM
a: The default working mode of the board is AFEC, which can be set or modified on the T2000. b: The decoding and encoding of the signals comply with ITU-T G.975.1. c: The overhead processing of the signals comply with ITU-T G.709.					

Table 4-12 Major functions of 10 Gbit/s OTUs without service convergence

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
L2LWF C9LWF	- Wavelength conversion - Line code: NRZ	- Supports the tunable wavelengths - Supports FEC	- Client-side optical signals: 1xSTM-64/OC-192/10G-WAN - WDM-side optical signals: OTU2	L2LRF/ C9TMR	DWDM
C7LWF C8LWF	- Wavelength conversion - Line code: NRZ/ODB	- Supports the tunable wavelengths - Supports AFEC	- Client-side optical signals: 1xSTM-64/OC-192/10G-WAN - WDM-side optical signals: OTU2	C6TMR/ C8TMR/ C9TMR	DWDM
CALWF	- Wavelength conversion - Line code: NRZ	- Supports the tunable wavelengths - Supports AFEC - Supports XFP	- Client-side optical signals: 1xSTM-64/OC-192/10G-WAN - WDM-side optical signals: OTU2	C6TMR/ C8TMR/ C9TMR	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C6LWFS	● Wavelength conversion ● Line code: DRZ	● Supports the tunable wavelengths ● Supports FEC ● Supports SuperWDM	● Client-side optical signals: 1xSTM-64/OC-192/10G-WAN ● WDM-side optical signals: OTU2	C6LRFS/C9TMRS	DWDM
C9LWFS	● Wavelength conversion ● Line code: RZ, DRZ	● Supports the tunable wavelengths ● Supports FEC ● Supports SuperWDM	● Client-side optical signals: 1xSTM-64/OC-192/10G-WAN ● WDM-side optical signals: OTU2	C9TMRS	DWDM
C7LWFS C8LWFS	● Wavelength conversion ● Line code: DRZ	● Supports the tunable wavelengths ● Supports AFEC ● Supports SuperWDM	● Client-side optical signals: 1xSTM-64/OC-192/10G-WAN ● WDM-side optical signals: OTU2	C6TMRS/C8TMRS	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
CALWFS	- Wavelength conversion - Line code: RZ, DRZ	- Supports the tunable wavelengths - Supports AFEC - Supports SuperWDM - Supports XFP	- Client-side optical signals: 1xSTM-64/OC-192/10G-WAN - WDM-side optical signals: OTU2	C9TMRS	DWDM
L2LRF	- Unidirectional 3R (reshaping, retiming and regeneration) - Line code: NRZ	- Supports the tunable wavelengths - Supports FEC - Regenerating rate: 10.71Gbit/s	- Client-side optical signals: NA - WDM-side optical signals: OTU2	–	DWDM
C6LRFS	- Unidirectional 3R (reshaping, retiming and regeneration) - Line code: DRZ	- Supports the tunable wavelengths - Supports FEC - Supports SuperWDM - Regenerating rate: 10.71Gbit/s	- Client-side optical signals: NA - WDM-side optical signals: OTU2	–	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C6LBE C8LBE	- Wavelength conversion - Line code: NRZ/ODB	- Supports the tunable wavelengths - Supports AFEC^f	- Client-side optical signals: 1x10GE LAN - WDM-side optical signals: OTU2	C6TMR/ C8TMR/ C9TMR	DWDM
CALBE	- Wavelength conversion - Line code: NRZ	- Supports the tunable wavelengths - Supports AFEC^f - Supports XFP	- Client-side optical signals: 1x10GE LAN - WDM-side optical signals: OTU2	C6TMR/ C8TMR/ C9TMR	DWDM
C6LBE S C8LBE S	- Wavelength conversion - Line code: DRZ	- Supports the tunable wavelengths - Supports AFEC^f - Supports SuperWDM	- Client-side optical signals: 1x10GE LAN - WDM-side optical signals: OTU2	C6TMRS/ C8TMRS	DWDM
CALBE S	- Wavelength conversion - Line code: RZ, DRZ	- Supports the tunable wavelengths - Supports AFEC^f - Supports SuperWDM - Supports XFP	- Client-side optical signals: 1x10GE LAN - WDM-side optical signals: OTU2	C9TMRS	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C6TMR	● 3R (reshaping, retiming and regeneration) ● Line code: NRZ	● Supports the tunable wavelengths ● Supports FEC/AFEC^b ● Regenerating rate: 10.71Gbit/s	● Client-side optical signals: NA ● WDM-side optical signals: OTU2	–	DWDM
C8TMR	● Bidirectional 3R (reshaping, retiming and regeneration) ● Line code: NRZ/ODB	● Supports the tunable wavelengths ● Supports FEC/AFEC^b ● Regenerating rate: 10.71Gbit/s or 11.1Gbit/s	● Client-side optical signals: NA ● WDM-side optical signals: OTU2	–	DWDM
C9TMR	● Bidirectional 3R (reshaping, retiming and regeneration) ● Line code: NRZ	● Supports the tunable wavelengths ● Supports FEC/AFEC^b ● Regenerating rate: 10.71Gbit/s or 11.1Gbit/s	● Client-side optical signals: NA ● WDM-side optical signals: – OTU2 – OTU2v	–	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C6TMR S C8TMR S	● Bidirectional 3R (reshaping, retiming and regeneration) ● Line code: DRZ	● Supports the tunable wavelengths ● Supports AFEC^b ● Supports SuperWDM	● Client-side optical signals: NA ● WDM-side optical signals: OTU2	–	DWDM
C9TMR S CBTMR S	● Bidirectional 3R (reshaping, retiming and regeneration) ● Line code: RZ, DRZ	● Supports the tunable wavelengths ● Supports AFEC^g ● Supports SuperWDM	● Client-side optical signals: NA ● WDM-side optical signals: – OTU2 – OTU2v	–	DWDM
C8LBF	● Wavelength conversion ● Line code: NRZ/ODB	● Supports the tunable wavelengths ● Supports FEC/AFEC^a	● Client-side optical signals: 1x10GE LAN/10GE WAN/STM-64/OC-192/OTU2 ● WDM-side optical signals: OTU2	C8TMR	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C9LBF	- Wavelength conversion - Line code: NRZ	- Supports the tunable wavelengths - Supports FEC/AFEC^a	- Client-side optical signals: 1x10GE LAN/10GE WAN/STM-64/OC-192/OTU2 - WDM-side optical signals: - OTU2 - OTU2v	C9TMR	DWDM
C8LBFS	- Wavelength conversion - Line code: DRZ	- Supports the tunable wavelengths - Supports AFEC^a - Supports SuperWDM	- Client-side optical signals: 1x10GE LAN/10GE WAN/STM-64/OC-192/OTU2 - WDM-side optical signals: OTU2	C8TMRS	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C9LBFS	- Wavelength conversion^{c d} - Line code: RZ, DRZ	- Supports the tunable wavelengths - Supports AFEC^a - Supports SuperWDM	- Client-side optical signals: 1x10GE LAN/10GE WAN/STM-64/OC-192/OTU2/FC10G - WDM-side optical signals: - OTU2 - OTU2v	C9TMRS	DWDM
CBLBFS	- Wavelength conversion^{c d} - Line code: RZ, DRZ	- Supports the tunable wavelengths - Supports AFEC^h - Supports SuperWDM	- Client-side optical signals: 1x10GE LAN/10GE WAN/STM-64/OTU2/FC10G - WDM-side optical signals: - OTU2 - OTU2v	CBTMRS	DWDM
a: The default working mode of the board is AFEC, which can be set or modified on the NM. b: The working mode of the board is adaptive to the FEC mode of the accessed signals. c: The overhead processing of the signals comply with ITU-T G.709. d: The decoding and encoding of the signals comply with ITU-T G.975.1. e: The decoding and encoding of the signals comply with ITU-T G.975. f: G.975-based Huawei AFEC codes are adopted. g: CBTMRS supports the AFEC-2 encoding, C6TMRS, C8TMRS and C9TMRS support the AFEC encoding. The AFEC encoding and AFEC-2 encoding cannot be interconnected. h: CBLBFS supports the AFEC-2 encoding, C8LBFS and C9LBFS support the AFEC encoding. The AFEC encoding and AFEC-2 encoding cannot be interconnected.					

Table 4-13 Major functions of 10 Gbit/s OTUs with service convergence

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C8ETMX C9ETMX CAETMX	- Service convergence - Wavelength conversion - Line code: NRZ	- Supports the tunable wavelengths - Supports AFEC	- Client-side optical signals: 4xSTM-16/OC-48/OTU1 - WDM-side optical signals: OTU2	C6TMR/ C8TMR/ C9TMR	DWDM
C8ETMXS C9ETMXS CAETMXS	- Service convergence - Wavelength conversion - Line code: DRZ	- Supports the tunable wavelengths - Supports AFEC - Supports SuperWDM	- Client-side optical signals: 4xSTM-16/OC-48/OTU1 - WDM-side optical signals: OTU2	C6TMRS/ C8TMRS/ C9TMRS	DWDM
CBETMXS	- Service convergence - Wavelength conversion - Line code: DRZ	- Supports the tunable wavelengths - Supports AFEC^c - Supports SuperWDM	- Client-side optical signals: 4xSTM-16/OTU1 - WDM-side optical signals: OTU2	CBTMRS	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C6TMX C7TMX C8TMX C9TMX	- Service convergence - Wavelength conversion a b - Line code: NRZ/ODB	- Supports the tunable wavelengths - Supports AFEC	- Client-side optical signals: 4xSTM-16/OC-48 - WDM-side optical signals: OTU2	C6TMR/ C8TMR/ C9TMR	DWDM
C6TMXS C7TMXS C8TMXS C9TMXS	- Service convergence - Wavelength conversion a b - Line code: DRZ	- Supports the tunable wavelengths - Supports AFEC - Supports SuperWDM	- Client-side optical signals: 4xSTM-16/OC-48 - WDM-side optical signals: OTU2	C6TMRS/ C8TMRS/ C9TMRS	DWDM
C8ELOG C9ELOG	- Service convergence - Wavelength conversion a b - Line code: NRZ/ODB	- Supports the tunable wavelengths - Supports AFEC - Supports GE service cross-connection	- Client-side optical signals: 8xGE/FC100/FC200 - WDM-side optical signals: OTU2	C6TMR/ C8TMR/ C9TMR	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C8ELOG S C9ELOG S	- Service convergence - Wavelength conversion^{a b} - Line code: DRZ	- Supports the tunable wavelengths - Supports AFEC - Supports SuperWDM - Supports GE service cross-connection	- Client-side optical signals: 8xGE/FC100/FC200 - WDM-side optical signals: OTU2	C6TMRS/ C8TMRS/ C9TMRS	DWDM
CBELOG S	- Service convergence - Wavelength conversion^{a b} - Line code: DRZ	- Supports the tunable wavelengths - Supports AFEC^d - Supports SuperWDM	- Client-side optical signals: 8xGE/FC100/FC200 - WDM-side optical signals: OTU2	CBTMRS	DWDM
C6LOG C9LOG	- Service convergence - Wavelength conversion^{a b} - Line code: NRZ	- Supports the tunable wavelengths - Supports AFEC - Supports GE service cross-connection	- Client-side optical signals: 8xGE/FC100/FC200 - WDM-side optical signals: OTU2	C6TMR/ C8TMR/ C9TMR	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C6LOGS C9LOGS	- Service convergence - Wavelength conversion - Line code: DRZ	- Supports the tunable wavelengths - Supports AFEC - Supports SuperWDM - Supports GE service cross-connection	- Client-side optical signals: 8xGE/FC100/FC200 - WDM-side optical signals: OTU2	C6TMRS/ C8TMRS/ C9TMRS	DWDM
C8LOM	- Service convergence - Wavelength conversion - Line code: NRZ/ODB	- Supports the tunable wavelengths - Supports AFEC - Supports FC service distance extension - Supports GE service cross-connection	- Client-side optical signals: 8xGE/FC100/FC200/FC400/FICON/FICON Express - WDM-side optical signals: OTU2	C6TMR/ C8TMR/ C9TMR	DWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C8LOMS	- Service convergence - Wavelength conversion^{a b} - Line code: DRZ	- Supports the tunable wavelengths - Supports AFEC - Supports SuperWDM - Supports FC service distance extension - Supports GE service cross-connection	- Client-side optical signals: 8xGE/FC100/FC200/FC400/FICON/FICON Express - WDM-side optical signals: OTU2	C6TMRS/C8TMRS/C9TMRS	DWDM
a: The overhead processing of the signals comply with ITU-T G.709. b: The decoding and encoding of the signals comply with ITU-T G.975.1. c: CBETMXS supports the AFEC-2 encoding, C8ETMXS, C9ETMXS and CAETMXS support the AFEC encoding. The AFEC encoding and AFEC-2 encoding cannot be interconnected. d: CBELOGS supports the AFEC-2 encoding, C8ELOGS and C9ELOGS support the AFEC encoding. The AFEC encoding and AFEC-2 encoding cannot be interconnected.					

Table 4-14 Major functions of 5 Gbit/s OTUs with service convergence

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C6LQG C9LQG	● Service convergence ● Wavelength conversion ● Line code: NRZ	● Supports the tunable wavelengths ● Supports FEC ● Supports GE service cross-connection	● Client-side optical signals: 4xGE ● WDM-side optical signals: 5Gbit/s line signal	–	DWDM
C7L4G	● Service convergence ● Wavelength conversion ● Line code: NRZ	● Supports the tunable wavelengths ● Supports FEC ● Supports GE service conversion ● Supports GE service cross-connection ● Supports Ethernet L2 switching	● Client-side optical signals: up to 8xGE ● WDM-side optical signals: 5Gbit/s line signal	–	DWDM

Table 4-15 Major functions of OTUs without service convergence at the rate of 2.5 Gbit/s or low

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C6LWC1	- Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards - Wavelength conversion - Line code: NRZ	- Supports the tunable wavelengths - Supports FEC	- Client-side optical signals: 1xSTM-16/OC-48 - WDM-side optical signals: OTU1	C6TRC1/ C8TRC1/ C8TRC2/ C9LQM/ C9LQM2	DWDM CWDM
C8LWC1 C8LWC1D C9LWC1	- C8LWC1 is the single fed and single receiving board - C8LWC1D is dual fed and selective receiving board - Wavelength conversion - Line code: NRZ	- Supports the tunable wavelengths - Supports FEC	- Client-side optical signals: 1xSTM-16/OC-48/OTU1 - WDM-side optical signals: OTU1	C6TRC1/ C8TRC1/ C8TRC2/ C9LQM/ C9LQM2	DWDM CWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C6TRC1 C8TRC1	● Unidirectional 3R (reshaping, retiming and regeneration) ● Line code: NRZ	● Supports the tunable wavelengths ● Supports FEC	● Client-side optical signals: NA ● WDM-side optical signals: OTU1	–	DWDM CWDM
C8TRC2	● Bidirectional 3R (reshaping, retiming and regeneration) ● Line code: NRZ	● Supports the tunable wavelengths ● Supports FEC	● Client-side optical signals: NA ● WDM-side optical signals: OTU1	–	DWDM CWDM
C6LWM C8LWM	● Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards ● Wavelength conversion ● Line code: NRZ	● Supports the tunable wavelengths	● Client-side optical signals: 1xSTM–1/OC–3, STM–4/OC–12, STM–16/OC–48 ● WDM-side optical signals: the same as the accessed client-side optical signals	C6LWMR/ C8LWMR	DWDM CWDM

Board Name	Function	Feature	Client/ WDM-Side Optical Signal	Regenerati ng Board	WDM Technical Specificati on
C6LWM R C8LWM R	● Bidirection al 3R (reshaping, retiming and regeneratio n) ● Line code: NRZ	● Supports the tunable wavelengt hs	● Client- side optical signals: NA ● WDM- side optical signals: 155.5Mbi t/s, 622.2Mbi t/s, 2.5Gbit/s	–	DWDM CWDM
C6LWX	● Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards ● Wavelengt h conversion ● Line code: NRZ	● Supports the tunable wavelengt hs	● Client- side optical signals: 1x34Mbit /s to 2.7Gbit/s ● WDM- side optical signals: the same as the accessed client- side optical signals	C6LWXR	DWDM CWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C8LWX	- Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards - Wavelength conversion - Line code: NRZ	Supports the tunable wavelengths	- Client-side optical signals: 1x16Mbit/s to 2.5Gbit/s - WDM-side optical signals: the same as the accessed client-side optical signals	C8LWXR	DWDM CWDM
C6LWXR	- Bidirectional 3R (reshaping, retiming and regeneration) - Line code: NRZ	Supports the tunable wavelengths	- Client-side optical signals: NA - WDM-side optical signals: 34Mbit/s to 2.7Gbit/s	–	DWDM CWDM
C8LWXR	- Bidirectional 3R (reshaping, retiming and regeneration) - Line code: NRZ	Supports the tunable wavelengths	- Client-side optical signals: NA - WDM-side optical signals: 34Mbit/s to 2.7Gbit/s	–	DWDM CWDM
a: The overhead processing of the signals comply with ITU-T G.709.					

Table 4-16 Major functions of 2.5 Gbit/s OTUs with service convergence

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C6LQS C7LQS	- Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards - Service convergence - Wavelength conversion - Line code: NRZ	Supports the tunable wavelengths	- Client-side optical signals: 4xSTM-1/STM-4 - WDM-side optical signals: STM-16	C6LWMR/ C8LWMR	DWDM CWDM
C6LDG C8LDG	- Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards - Service convergence - Wavelength conversion - Line code: NRZ	Supports the tunable wavelengths	- Client-side optical signals: 2xGE - WDM-side optical signals: STM-16	C6LWMR/ C8LWMR	DWDM CWDM

Board Name	Function	Feature	Client/ WDM-Side Optical Signal	Regenerati ng Board	WDM Technical Specificati on
C6FDG C8FDG	- Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards - Service convergence - Wavelength conversion a - Line code: NRZ	- Supports the tunable wavelengths - Supports FEC	- Client-side optical signals: 2xGE - WDM-side optical signals: OTU1	C6TRC1/ C8TRC1/ C8TRC2/ C9LQM/ C9LQM2	DWDM CWDM
C6FCE	- Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards - Service convergence - Wavelength conversion b - Line code: NRZ	- Supports the tunable wavelengths - Supports FC service distance extension	- Client-side optical signals: 2xFC100 or 1xFC200 - WDM-side optical signals: STM-16	C6LWMR/ C8LWMR	DWDM CWDM

Board Name	Function	Feature	Client/ WDM-Side Optical Signal	Regenerati ng Board	WDM Technical Specificati on
C8LQM C9LQM	- Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards - Service convergence - Wavelength conversion - Line code: NRZ	- Supports the tunable wavelengths - Supports FEC - C9LQM provides the regeneration function	C8LQM - Client-side optical signals: up to 4x125Mb it/s to 2.50Gbit/s (the maximum bandwidth is less than 2.5 Gbit/s) - WDM-side optical signals: OTU1 C9LQM - Client-side optical signals: up to 4x125Mb it/s to 2.67Gbit/s (the maximum bandwidth is less than 2.67 Gbit/s) - WDM-side optical signals: OTU1	C6TRC1/ C8TRC1/ C8TRC2/ C9LQM/ C9LQM2	DWDM CWDM

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C9LQM2	2LQM mode - Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards - Service convergence - Wavelength conversion - Line code: NRZ	- Supports the tunable wavelengths - Supports FEC - Provides the regeneration function	- Client-side optical signals: up to 4x125Mbit/s to 2.67Gbit/s (the maximum bandwidth is less than 2.67 Gbit/s) - WDM-side optical signals: 2xOTU1	C6TRC1/ C8TRC1/ C8TRC2/ C9LQM/ C9LQM2	DWDM CWDM
	AP8 mode - Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards - Service convergence - Wavelength conversion - Line code: NRZ	- Supports the tunable wavelengths - Supports FEC - Provides the regeneration function	- Client-side optical signals: up to 8x100Mbit/s to 2.5Gbit/s (the maximum bandwidth is less than 2.5 Gbit/s) - WDM-side optical signals: 1xOTU1	C6TRC1/ C8TRC1/ C8TRC2/ C9LQM/ C9LQM2	DWDM CWDM

Board Name	Function	Feature	Client/ WDM-Side Optical Signal	Regenerati ng Board	WDM Technical Specificati on
L2AS8 C7AS8	● Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards ● Service convergence ● Wavelength conversion^b ● Line code: NRZ	● Supports the tunable wavelengths	● Client-side optical signals: 2xSTM-1/OC-3 or 4xSTM-4/OC-12 ● WDM-side optical signals: STM-16	C6LWMR/ C8LWMR	DWDM CWDM
C6AP8	● Provides the single fed and single receiving boards as well as the dual fed and selective receiving boards ● Service convergence ● Wavelength conversion^b ● Line code: NRZ	● Supports the tunable wavelengths	● Client-side optical signals: 8x200Mbit/s to 2.12Gbit/s (the maximum bandwidth is less than 2.5 Gbit/s) ● WDM-side optical signals: STM-16	C6LWMR/ C8LWMR	DWDM CWDM
a: The overhead processing of the signals comply with ITU-T G.709. b: The overhead processing of the signals comply with ITU-T G.783.					

Table 4-17 Major functions of other OTUs

Board Name	Function	Feature	Client-/WDM-Side Optical Signal	Regenerating Board	WDM Technical Specification
C7EGS8	Provides extended access interfaces for eight GE service	- Supports GE service conversion - Supports GE service cross-connection - Supports Ethernet L2 switching	- Client-side optical signals: up to 8xGE - WDM-side optical signals: NA	-	-
C8TBE	provides extended access interfaces for four GE service	- Chooses ten channels of signals from the 48xGE services at most that are cross-connected to it and converges the chosen signals to 1x10GE signals - Supports GE service conversion - Supports GE service cross-connection	- Client-side optical signals: up to 4xGE - WDM-side optical signals: NA	-	-

Board Name	Function	Feature	Client/ WDM-Side Optical Signal	Regenerati ng Board	WDM Technical Specificati on
C7LAM	● Service convergence ● Wavelength conversion ● Line code: NRZ	● Converges eight FE services into one FE service and supports the communication between the OptiX OSN 900A and the OptiX Metro 6100 . ● Converges eight FE services into one GE service. ● Converges seven FE services into one GE service and provides the dual-fed and selective receiving protection at GE ports. ● Accesses a maximum of four services at any rate from 16 Mbit/s to 2.5 Gbit/s and provides	● Client-side optical signals: up to 8x16 Mbit/s to 2.5 Gbit/s (the maximum bandwidth is less than 2.5 Gbit/s) ● WDM-side optical signals: 2.5Gbit/s line signal	-	DWDM CWDM

Board Name	Function	Feature	Client/ WDM-Side Optical Signal	Regenerati ng Board	WDM Technical Specificati on
		one OSC channel. ● Accesses a maximum of three services at any rate from 16 Mbit/s to 2.5 Gbit/s and provides three dedicated OSC channels. ● Accesses a maximum of two services at any rate from 16 Mbit/s to 2.5 Gbit/s, realizes the dual-fed and selective receiving protection of two WDM-side services and provides two dedicated OSC channels. ● Accesses a maximum of four GE services and provides one OSC or			

Board Name	Function	Feature	Client/ WDM-Side Optical Signal	Regenerati ng Board	WDM Technical Specificati on
		ESC channel. ● Accesses a maximum of three GE services and provides three dedicated OSC or ESC channels. ● accesses a maximum of two GE services, realize the dual-fed and selective receiving protection of two WDM-side services and two dedicated OSC or ESC channels. ● Regenerate s two bidirection al services at any rate from 16 Mbit/s to 2.5 Gbit/s and provides two OSC channels.			

4.4.2 Optical Multiplexer and Demultiplexer Board

The optical multiplexer and demultiplexer board can multiplex or demultiplex optical signals of different wavelengths.

Figure 4-7 shows the position of the two units in the system.

The optical multiplexer and demultiplexer boards are included in **Table 4-18**.

Table 4-18 Board name and category of the multiplexer and demultiplexer board

Board	Board Name
C6M40 C9M40	40-channel multiplexing board
C6V40 C9V40	40-channel multiplexing board with VOA
C6D40 C9D40	40-channel demultiplexing board
C6FIU C7FIU C9FIU	Fiber interface unit
C6EFIU	External fiber interface unit

Table 4-19 briefs the application and functions of the above boards. For more details, refer to *Hardware Description*.

Table 4-19 Application and description of the multiplexer and demultiplexer board

Board	Application	Function
C6M40 C9M40	● Applies in the C_EVEN band in DWDM system with 100G channel space.	● Multiplexes 40-channel optical signals coming from the OTU, or accesses equipment into the main path. ● Provides an in-service monitoring port "MON", so that the optical performance of optical signals can be checked in service through the MCA board or an optical spectrum analyzer.

Board	Application	Function
C6V40 C9V40	● Applies in the C_EVEN band in DWDM system with 100G channel space.	● Multiplexes 40-channel optical signals coming from the OTU, or accesses equipment into the main path. ● Adjusts the optical power of each channel for power pre-equilibrium. ● Provides an in-service monitoring port "MON", so that the optical performance of optical signals can be checked in service through the MCA board or an optical spectrum analyzer.
C6D40 C9D40	● Applies in the C_EVEN band in DWDM system with 100G channel space.	● Demultiplexes the main path optical signal into 40-channel optical signals. ● Provides an in-service monitoring port "MON", so that the optical performance of optical signals can be checked in service through the MCA board or an optical spectrum analyzer.
C6FIU C7FIU C9FIU	● Applies in the C_EVEN band in DWDM system with 100G channel space. ● Only used in the subrack.	● Performs the multiplexing and demultiplexing of main path signals and supervisory channel signals. In the transmit direction, the FIU accesses optical supervisory signals. In the receive direction, the FIU extracts optical supervisory signals. ● Provides in-service monitoring of optical interfaces, monitoring the spectrum of the main path without bringing service interruption. ● C7FIU provides an in-service monitoring port "MON", so that the optical performance of optical signals can be checked in service through the MCA board or an optical spectrum analyzer.
C6EFI U	● Applies in the C_EVEN band in DWDM system with 100G channel space. ● Only used in the OADM frame.	● Performs the multiplexing and demultiplexing of main path signals and supervisory channel signals. In the transmit direction, the EFIU accesses optical supervisory signals. In the receive direction, the EFIU extracts optical supervisory signals. ● Provides in-service monitoring of optical interfaces, monitoring the spectrum of the main path without bringing service interruption. ● Provides an in-service monitoring port "MON", so that the optical performance of optical signals can be checked in service through the MCA board or an optical spectrum analyzer.

4.4.3 Optical Add/Drop Multiplexer Board

In the optical add/drop multiplexer board, single-wavelength optical signals are added to and dropped from the multiplexed signals and sent to an OTU.

Figure 4-7 shows the position of the OADM board in the system.

The optical add/drop multiplexer boards are included in **Table 4-20**.

Table 4-20 Board name and category of the optical add/drop multiplexer board

Board	Board Name
C6DWC	Dynamic wavelength add/drop control board
C8WSD9 C9WSD9	9-port wavelength selective switching demultiplexing board
C8WSM9 C9WSM9	9-port wavelength selective switching multiplexing board
C8RMU9	9-port ROADM multiplexing board
C9WSMD4	4-port wavelength selective switching demultiplexing/ multiplexing board
CM6MR4	4-channel optical add/drop multiplexing board
C6MR2	2-channel optical add/drop multiplexing board
L2SBM2	Single fiber bidirectional 2-channel optical add/drop multiplexing configuration board
L2SBM1	Single fiber bidirectional 1-channel optical add/drop multiplexing configuration board

Table 4-21 briefs the application and functions of the above boards. For more details, refer to *Hardware Description*.

Table 4-21 Application and description of the optical add/drop multiplexer board

Board	Application	Functions
C6DWC	● The board mainly applies to ROADM. It provides a function of dynamic configurable. ● Only used in DWDM system. ● Only used in two-fiber bidirectional system.	● It is to realize the ROADM function. ● Blocks the dropped service in the pass-through direction. ● Multiplexes the added service with the pass-through services by a multiplexer. Then sends them to the WDM line for transmission.

Board	Application	Functions
C8WSD9 C9WSD9	● The board mainly applies to ROADM. It provides a function of dynamic configurable. ● Only used in DWDM system. ● Only used in two-fiber bidirectional system.	● Achieves the dynamically configurable multiplexing of any wavelength from any port. ● Cooperates with the WSM9, optical multiplexer and demultiplexer board or optical add/drop multiplexer board, a node on the ring or chain network can receive any wavelengths at the local station through any interfaces to achieve the dynamic allocation of wavelengths.
C8WSM9 C9WSM9	● The board mainly applies to ROADM. It provides a function of dynamic configurable. ● Only used in DWDM system. ● Only used in two-fiber bidirectional system.	● Achieves the dynamically configurable multiplexing of any wavelength from any port. ● Cooperates with the WSD9, optical multiplexer and demultiplexer board or optical add/drop multiplexer board, a node on the ring or chain network can receive any wavelengths at the local station through any interfaces to achieve the dynamic allocation of wavelengths.
C8RMU9	● The board mainly applies to ROADM. It provides a function of dynamic configurable. ● Only used in DWDM system. ● Only used in two-fiber bidirectional system.	● Adds eight channels of signals. ● Its adding interfaces can cooperate with tunable OTU boards to realize full dynamic input of eight channels of signals.
C9WSM D4	● The board mainly applies to ROADM. It provides a function of dynamic configurable. ● Only used in DWDM system. ● Only used in two-fiber bidirectional system.	● Achieves services broadcasting function, and the dynamic and configurable multiplexing and demultiplexing of any wavelengths to any ports. ● A node on the ring or chain network can receive any wavelengths at the local station through any ports. It also can transmit any wavelength combination to any interface, so as to achieve the dynamic allocation of wavelengths.

Board	Application	Functions
CM6MR4	● The board mainly applies to OADM or OTM. ● Only used in DWDM and CWDM system. ● Only used in two-fiber bidirectional system.	● Adds/drops and multiplexes four serial wavelengths to/from the multiplexed signals.
C6MR2	● The board mainly applies to OADM or OTM. ● Only used in DWDM and CWDM system. ● Only used in two-fiber bidirectional system.	● Adds/drops and multiplexes any two wavelengths to/from the multiplexed signals.
L2SBM2	● The board mainly applies to OADM or OTM. ● Only used in CWDM system. ● Only used in single-fiber bidirectional.	● Adds/drops two wavelengths to/from the multiplexed signals and multiplexes the other two wavelengths into the multiplexed signals. ● The added or dropped signals must be in the different wavelengths.
L2SBM1	● The board mainly applies to OADM or OTM. ● Only used in CWDM system. ● Only used in single-fiber bidirectional.	● Adds/drops one wavelength to/from the multiplexed signals and multiplexes another wavelength into the multiplexed signals. ● The added or dropped signals must be in the different wavelengths.

4.4.4 Optical Amplifier Board

The optical amplifier board amplifies the power of the multiplexed signals to extend the transmission distance.

Figure 4-7 shows the position of the optical amplifier board in the system.

The optical amplifier boards are included in **Table 4-22**.

Table 4-22 Board name and category of the optical amplifier board

Board	Board Name
C6OAU C9OAU	Dual stage WDM optical booster amplifier board

Board	Board Name
C6OBU C8OBU C9OBU	WDM optical booster amplifier board
C6OPU C8OPU C9OPU	WDM pre-amplifier board
C8RPC	Raman pump amplifier board for C-band

Table 4-23 briefs the application and functions of the above boards. For more details, refer to *Hardware Description*.

Table 4-23 Application and description of the optical amplifier board

Board	Application	Functions
C6OAU C9OAU	- Amplifies at most 40 channels (the channel spacing being 100 GHz) at the same time - The C6OAU includes C6OAU01A, C6OAU01B, C6OAU02A, C6OAU02B, C6OAU03A, C6OAU03B, C6OAU05A. - The C9OAU includes C9OAU01, C9OAU02, C9OAU03, C9OAU05A.	- The OAU board can amplify the input optical signal, compensate for the fiber loss, and increase the receive-end sensitivity budget. - C6OAU01A, C6OAU02A, C6OAU03A, C6OAU05A, C9OAU01, C9OAU02, C9OAU03, C9OAU05A support EVOA. - C6OAU01B, C6OAU02B, C6OAU03B do not support EVOA. - The OAU board uses the automatic gain control technique to realize the gain locking function.
C6OBU C8OBU C9OBU	- Amplifies at most 40 channels (the channel spacing being 100 GHz) at the same time. - The C6OBU includes C6OBU01, C6OBU03, C6OBU05. - The C8OBU includes C8OBU03. - The C9OBU includes C9OBU03 and C9OBU05.	- The OBU board can amplify the optical signal power. - The OBU board uses the automatic gain control technique to realize the gain locking function. - The OBU dose not support EVOA.

Board	Application	Functions
C6OPU C8OPU C9OPU	● Used alone or together with the OBU and applied to the C-band. ● Amplifies at most 40 channels (the channel spacing being 100 GHz) at the same time and features small noise figure. ● The C6OPU includes C6OPU01, C6OPU02, C6OPU03. ● The C8OPU includes C8OPU02, C8OPU04. ● The C9OPU includes C9OPU03.	● Features small noise figure, used to improve the receiver sensitivity budget. ● Uses the automatic gain control technique for gain locking.

RPC is provided only by the DWDM system.

Table 4-24 lists the functions of the Raman pump amplifier board.

Table 4-24 Functions of the Raman pump amplifier board

Board	Application	Functions
C8RPC	● Amplifies the signal in C-band. ● Always used together with the EDFA. ● Used at the receive end of the DWDM system, it amplifies signals during transmission by sending high-power pump light to the transmission fiber.	● Raman pump amplifier boards realize long-haul, broad bandwidth, low noise, and distributed online optical signal amplification. ● These units can automatically lock the pump power, receive the SCC command to switch on/off the pump source, separate the signal light, report performances and alarms, and protect the pump laser.

4.4.5 System Control, Supervision and Communication Board

The system control, supervision and communication board is the control center for the equipment. It helps the NM system to manage the boards of the equipment and enables the equipment to communicate with each other.

The system control, supervision and communication boards are included in **Table 4-25**.

Table 4-25 Board name and category of the system control and communication board

Board	Board Name
C6SCC C8SCC	System control & communication unit and EOW unit
C6PMU C8PMU	Power and environment monitoring unit

Table 4-26 briefs the application and functions of the above boards. For more details, refer to *Hardware Description*.

Table 4-26 Application and description of the system control and communication board

Board	Application	Functions
C6SCC C8SCC	Applicable for every subrack.	● Manages and supports the equipment to communicate with each other. ● Provides an interface for the equipment and the NM system. ● Processes the related overheads.
C6PMU C8PMU	Applicable for IU14 in every subrack.	● Monitors ambient conditions such as voltage and temperature, input/output alarm values. ● Provides audible alarms.

 NOTE

One OptiX Metro 6100 NE node may include multiple OptiX Metro 6100 subracks. An SCC board should be inserted in slot IU7 for each subrack to manage the subrack and the communication between the subrack and other subracks. For more details, refer to **11.2.1 Supervision and Administration Module**.

4.4.6 Optical Supervisory Channel and Timing Transmission Board

The main function of the optical supervisory channel and timing transmission board is to transmit and process the system overhead information.

Figure 4-7 shows the position of the optical supervisory channel and timing transmission board in the system.

The optical supervisory channel and timing transmission boards are included in **Table 4-27**.

Table 4-27 Board name and category of the optical supervisory channel and timing transmission board

Board	Board Name
L2SC1 C9SC1	Unidirectional optical supervisory channel board
L2SC2 C9SC2	Bidirectional optical supervisory channel board
L2TC1 ^a	Unidirectional optical supervisory channel and timing transmission unit
L2TC2 ^a	Bidirectional optical supervisory channel and timing transmission unit
C9ST1 ^b	Unidirectional optical supervisory channel and clock transmitting board
C9ST2 ^b	Bidirectional optical supervisory channel and clock transmitting board
a: The TC1 and TC2 are only applicable to the subrack of the OptiX Metro 6100 with versions of V100R005. b: The ST1 and ST2 are applicable to the subracks of the OptiX Metro 6100 with versions of V100R006 or above.	

Table 4-28 briefs the application and functions of the above boards. For more details, refer to *Hardware Description*.

Table 4-28 Application and description of the optical supervisory channel and timing transmission board

Board	Application	Function
L2SC1 C9SC1	Applies to OTM	- Transmits and receives the optical supervisory signal in one transmission direction and processes the overheads. - The carrier wavelength of the optical supervisory channel is 1510 nm.
L2SC2 C9SC2	Applies to OADM, OLA, REG	- Transmits and receives the optical supervisory signal in two transmission directions and processes the overheads. - The carrier wavelength of the optical supervisory channel is 1510 nm.

Board	Application	Function
L2TC1	Applies to OTM	● Accomplish the processing and regeneration of the supervisory channel as the SC1. Besides, the TC1 also provide the clock transmission function. ● Add or drop 3-channel E1 clock service, and provide the electrical interface for the external synchronous signal and synchronous equipment timing source. The clock interface has the 2.048 Mbit/s or 2048 kHz interface physical characteristics defined by the ITU-I G.703 Recommendation. ● Support three west clock inputs/outputs, three clock external inputs/outputs, and also support external clock input, clock output and unidirectional clock transmission. ● Process the synchronous information status byte: judge the synchronous timing quality level according to S1 byte content, and report synchronous status information. If the upper stream clock signal is missing, add "clock invalid" information to notify clock receiving equipment downstream. ● Supervisory information and clock signals are transmitted in both 1510 nm.
L2TC2	Applies to OADM, OLA, REG	● Accomplish the processing and regeneration of the supervisory channel as the SC2. Besides, the TC2 also provide the clock transmission function. ● Add or drop 3-channel E1 clock service, and provide the electrical interface for the external synchronous signal and synchronous equipment timing source. The clock interface has the 2.048 Mbit/s or 2048 kHz interface physical characteristics defined by the ITU-I G.703 Recommendation. ● Support three west clock inputs/outputs, three east clock inputs/outputs, three clock external inputs/outputs, and also support external clock input, clock output, bi-directional clock transmission and clock pass-through. ● Process the synchronous information status byte: judge the synchronous timing quality level according to S1 byte content, and report synchronous status information. If the upper stream clock signal is missing, add "clock invalid" information to notify clock receiving equipment downstream. ● Supervisory information and clock signals are transmitted in 1510 nm.

Board	Application	Function
C9ST1	Applies to OTM	● Transmits and receives a channel of optical supervisory signals, two channels of 2048 kbit/s or 2 MHz clock signals, and an FE service. The 10M/100M FE service is encapsulated into an E1 service for transmission over an E1 path. ● Provides signals of normal power and of large power. The ST1 of large power is used in a network where line attenuation is large. ● Supervisory information and clock signals are transmitted in 1510 nm.
C9ST2	Applies to OADM, OLA, REG	● Transmits and receives a channel of optical supervisory signals in each of the two directions, two channels of 2048 kbit/s or 2 MHz clock signals, and an FE service. The 10M/100M FE service is encapsulated into an E1 service for transmission over an E1 path. ● Provides signals of normal power and of large power. The ST2 of large power is used in a network where line attenuation is large. ● Supervisory information and clock signals are transmitted in 1510 nm.

4.4.7 Optical Protection Board

The optical protection board protects the networks with an system in self-healing mode.

The optical protection boards are included in [Table 4-29](#).

Table 4-29 Board name and category of the optical protection board

Board	Board Name
C8DCP	Double channel protection board
C6OLP C8OLP L2OLP	Optical line protection board
L2SCS	Sync optical channel separator board
L2OWSP	Optical wavelength shared protection board
C9CP40	40G optical channel protect board

[Table 4-30](#) briefs the application and functions of the above boards. For more details, refer to *Hardware Description*.

Table 4-30 Application and description of the optical protection board

Board	Application	Functions
C8DCP	● Applies to inter-subrack 1+1 optical channel protection, located between client equipment and the optical transponder board. ● Applies to extended intra-board wavelength protection, located between the optical multiplexer and demultiplexer board (or optical add/drop multiplexer board) and the optical transponder board. ● Applies to optical wavelength shared protection (DCP)^a, located between the optical multiplexer and demultiplexer board (or optical add/drop multiplexer board) and the optical transponder board. ● C8DCP01 supports single-mode optical interface at 1310 nm and 1550 nm and provides the extended intra-board wavelength protection. ● C8DCP02 supports multi-mode optical interface at 850 nm and does not provide the extended intra-board wavelength protection.	● Performs dual-fed selective receiving of two channels of optical signals. ● Realizes extended intra-board wavelength protection, which adopts on OTU board and on DCP board that supports dual-fed selective receiving of optical signals. ● Realizes inter-subrack 1+1 optical channel protection and optical wavelength shared protection (DCP). The signals automatically switch to protection channel when the working channel degrades.
C6OLP C8OLP L2OLP	● Applies to optical line protection, located between the FIU board and the WDM side. ● Applies to inter-subrack 1+1 optical channel protection, located between the client equipment and the optical transponder board. ● Applies to extended intra-board wavelength protection, located between the optical multiplexer and demultiplexer board (or optical add/drop multiplexer board) and the optical transponder board.	● Performs dual-fed selective receiving of one channel of optical signals. Uses the OLP board for optical line protection. Able to automatically switch the traffic to the standby fiber when the performance of the active fiber degrades. ● Realizes optical line protection and inter-subrack 1+1 optical channel protection. The signals automatically switch to protection channel when the working channel degrades. ● Realizes extended intra-board wavelength protection, which adopts on OTU board and on OLP board that supports dual-fed selective receiving of optical signals.

Board	Application	Functions
L2SCS	● Applies to inter-board wavelength protection, located between the client equipment and the optical transponder board. ● Applies to client-side 1+1 wavelength protection, located between the client equipment and the optical transponder board. ● Applies to protection between tributary boards, located between the client equipment and the optical transponder board.	● Achieves dual-fed for optical signals. ● Helps to realize the inter-board 1+1 channel protection, the client-side 1+1 optical channel protection and protection between tributary boards. Is able to automatically switch the traffic to the standby fiber when the signal quality in the active fiber degrades.
L2OWSP	● Applies to optical wavelength shared protection (OWSP) ● Located between the optical multiplexer and demultiplexer board and the optical transponder board.	● Protects the service with wavelength shared in ring network being configured with distributed service.
C9CP40	● Applies to extended intra-board wavelength protection for 40G OTU board. ● Located between the optical multiplexer and demultiplexer board and the 40G OTU board.	● Realizes dual fed and selective receiving of one channel of optical signals. ● Realizes 1+1 40G wavelength protection, in which the signals can be automatically switched to the protection channel when the performance of the working channel degrades.
a: The optical wavelength shared protection (DCP) is applicable to subracks of the OptiX Metro 6100 with versions of V100R006 or above.		

4.4.8 Spectrum Analyzer Board

The spectrum analyzer board is used to monitor the optical spectrum characteristics and optical power.

The spectrum analyzer boards are included in [Table 4-31](#).

Table 4-31 Board name and category of the spectrum analyzer board

Board	Board Name
L2MCA C7MCA	Multi-channel spectrum analyzer board

Table 4-32 briefs the application and functions of the above boards. For more details, refer to *Hardware Description*.

Table 4-32 Application and description of spectrum analyzer board

Board	Application	Function
L2MCA C7MCA	L2MCA: in-service monitoring of eight optical channels. C7MCA: in-service monitoring of four optical channels.	- Provides built-in in-service optical spectrum analyzing. - Monitoring function, with which the central wavelength, optical power and optical signal-to-noise ratio of the optical signals on eight or four different points in the system can be monitored in-service.

4.4.9 Variable Optical Attenuator Board

The variable optical attenuator board is used to adjust the optical power.

The variable optical attenuator board are included in **Table 4-33**.

Table 4-33 Board name and category of the variable optical attenuator board

Board	Board Name
L2VOA C9VOA	Variable optical attenuator board
C9VA2	2-channel variable optical attenuator board
C6VA4 C9VA4	4-channel variable optical attenuator board

Table 4-34 briefs the application and functions of the above boards. For more details, refer to *Hardware Description*.

Table 4-34 Application and description of variable optical attenuator board

Board	Application	Function
L2VOA C9VOA	- Adjusts the optical power of the one channel of the optical signal. - The L2VOA supports input optical power reporting, but the C9VOA does not.	Adjusts the optical power of one optical channel according to the control command sent by the SCC.

Board	Application	Function
C9VA2	● Adjusts the optical power of the two channels of the optical signal. ● Always applies to the OADM and adjust the power of the add/drop channel optical signal, ensuring power equalization for the main path signal.	Adjusts the optical power of two optical channels according to the control command sent by the SCC.
C6VA4 C9VA4	● Adjusts the optical power of the four channels of the optical signal. ● Always applies to the OADM and adjust the power of the add/drop channel optical signal, ensuring power equalization for the main path signal. ● The C6VA4 supports input optical power reporting, but the C9VA4 does not.	Adjusts the optical power of four optical channels according to the control command sent by the SCC.

5 Software Architecture

About This Chapter

The software of the system includes the board software, NE software and network management system.

[5.1 Overview](#)

The software system is of a modular design. Each module provides specific functions and works with other modules.

[5.2 Communication Protocols and Interfaces](#)

The Qx interface is used for communication. Complete protocol stack and messages of the Qx interface are described in ITU-T G.773, Q.811 and Q.812.

[5.3 Board Software](#)

The board software runs on each board and it manages, monitors and controls the operation of the board.

[5.4 NE Software](#)

The NE software manages, monitors and controls the board operations in the NE. In addition, the NE software functions as a communication service unit between the T2000 and the boards, so that the T2000 can control and manage the NE.

[5.5 Network Management System](#)

The NM system implements a unified management over the optical transmission network, and maintains all OSN, SDH, Metro, SLM, DWDM NE equipment in the network.

5.1 Overview

The software system is of a modular design. Each module provides specific functions and works with other modules.

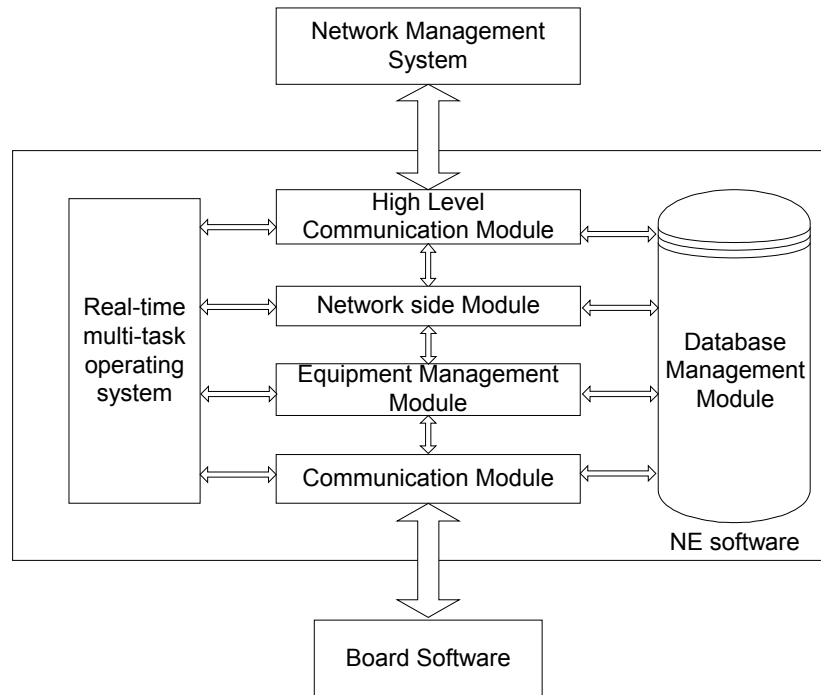
The entire software is distributed in three modules including board software, NE software and NM system.

The software system is designed with hierarchical structure. Each layer performs specific functions and provides service for the upper layer.

The system software architecture is shown in **Figure 5-1**.

In the diagram, all modules are NE software except "Network Management System" and "Board Software".

Figure 5-1 Software architecture



5.2 Communication Protocols and Interfaces

The Qx interface is used for communication. Complete protocol stack and messages of the Qx interface are described in ITU-T G.773, Q.811 and Q.812.

The Qx interface is mainly used to connect the mediation device (MD), Q adaptation (QA) and NE (NE) equipment with the operating system (OS) through local communication network (LCN).

At present, QA is provided by the NE management layer. MD and OS are provided by the NM layer. They are connected to each other through the Qx interface.

According to the Recommendations, the Qx interface provided by the system is developed on the basis of TCP/IP connectionless network layer service (CLNS1) protocol stack.

In addition, to support remote access of the NM through Modem, the IP layer uses serial line internet protocol (SLIP).

5.3 Board Software

The board software runs on each board and it manages, monitors and controls the operation of the board.

It receives the command issued from the NE software and reports the board status to the NE software through performance events and alarm.

The specific functions include:

- Alarm management
- Performance management
- Configuration management
- Communication management

The drive control over functional circuits is board software-specific. The board software realizes under the control of the NE software the ITU-T compliant functions.

5.4 NE Software

The NE software manages, monitors and controls the board operations in the NE. In addition, the NE software functions as a communication service unit between the T2000 and the boards, so that the T2000 can control and manage the NE.

According to ITU-T M.3010, NE software is at unit management layer in telecom management network, performing NE function (NEF), partial mediation function (MF) and OS function at network unit layer.

Data communication function (DCF) provides communication channel between NE and other equipment (including NM and other NEs).

- Real-time multi-task operating system

The NE software offers real-time multi-task operating system to manage public resources and support application programs.

It isolates the application programs from the processor and provides an application program execution environment, which is independent of the processor hardware.

- Communication module

The communication module is the interface module between NE software and board software.

According to related protocol, communication function between the NE software and the board software is for information exchange and maintenance of the equipment.

Through the communication, board maintenance and operation commands from the NE software are sent to the boards. On the other hand, the state, alarm and performance events of the board are reported to the NE software.

- Network Side (NS) Module

The NS module is between the communication module and the equipment management module. It converts the data format between the user operation side (at the application layer) and the NE equipment management layer, and provides security control for the NE layer.

Functionally, the NS module is divided into the following three submodules:

- Qx interface module
- Command line interface module
- Security management module
- Equipment management module

The equipment management module is the core of the NE software for the NE management. It includes administrator and agent.

Administrator can send NM operation commands and receive events.

Agent can respond to the NM operation commands sent by the administrator, implement the operations of the managed object, and send up events according to the change of status of the managed object.

- High-level communication module

The high-level communication module exchanges management information among NEs and between the NM system and the NE.

It consists of network communication module, serial communication module and ECC communication module.

- Database management module

The database management module is a part of the NE software.

It includes two independent parts: data and program.

The data are organized in the form of database, including network database, alarm database, performance database and equipment database.

The program manages and accesses the data in the database.

5.5 Network Management System

The NM system implements a unified management over the optical transmission network, and maintains all OSN, SDH, Metro, SLM, DWDM NE equipment in the network.

In compliance with ITU-T Recommendations, it is an NM system that integrates standard management information model as well as object-oriented management technology.

It exchanges information with the NE software through the communication module to monitor and manage the network equipment.

The NM software runs on a workstation or PC, managing the equipment and the transmission network to help to operate, maintain and manage the transmission equipment.

The management functions of the NM software include:

- Alarm management: collects, prompts, filters, browses, acknowledges, checks, clears, and counts in real time; fulfills alarm insertion, alarm correlation analysis and fault diagnosis.
- Performance management: sets performance monitoring; browses, analyzes and prints performance data; forecasts medium-term and long-term performance; and resets performance register.

- Configuration management: configures and manages interfaces, clocks, services, trails, subnets and time.
- Security management: provides NM user management, NE user management, NE login management, NE login lockout, NE setting lockout and local craft terminal (LCT) access control of the equipment.
- Maintenance management: provides loopback, board resetting, automatic laser shutdown (ALS) and optical fiber power detection, and collects equipment data to help the maintenance personnel in troubleshooting.

6 DWDM System Configuration

About This Chapter

The product can be configured as any of the following four equipment types: optical terminal multiplexer (OTM), fixed optical add/drop multiplexer (FOADM), reconfigurable optical add/drop multiplexer (ROADM) and optical line amplifier (OLA).

6.1 OTM

The OTM is a terminal station of a network. An OTM is divided into the transmit end and the receive end.

6.2 OLA

The DWDM system can be configured as the OLA comprising optical amplifier boards.

6.3 FOADM

The FOADM equipment adds/drops optical wavelength signals at the intermediate node.

6.4 ROADM

The ROADM equipment dynamically adds/drops and cross-connects optical wavelength signals at the intermediate node.

6.5 REG

The REG equipment is an electrical regenerator and is used to further extend the optical transmission distance.

6.1 OTM

The OTM is a terminal station of a network. An OTM is divided into the transmit end and the receive end.

The OptiX Metro 6100 DWDM OTM node can be formed in the following two modes:

- The OTM comprising the optical multiplexing board (M40/V40) and optical demultiplexing board (D40)
- The OTM comprising optical add/drop multiplexing boards

Select the OTM type based on the initial service distribution, cost and future expansion requirements.

6.1.1 OTM Node with the M40/V40 and D40 Boards

This section describes the signal flow, formation method, typical configurations, and configuration principles of the OTM node with the M40/V40 and D40 boards.

Functions

The OptiX Metro 6100 DWDM OTM node is used at the terminal station, and is logically divided into the following two directions:

- Transmit direction
- Receive direction

In the transmit direction, the OTM node converges or transforms and then amplifies client-side signals. Then, the signals are multiplexed with the optical supervisory signals. At last, the multiplexed signals are sent to the line for transmission. In the receive direction, the OTM node performs the converse process.

If more than 16 wavelengths are accessed in the early phase, the OTM comprising optical multiplexing units and optical demultiplexing units is normally adopted. This type of OTM equipment supports the expansion to a maximum of 40 wavelengths without interrupting services.

Functional Units

An OptiX Metro 6100 DWDM OTM node consists of the following functional units:

- Optical transponder board (OTU)
- Optical amplifier board (OA)
- Optical multiplexing board (M40/V40)
- Optical demultiplexing board (D40)
- Unidirectional OSC board (SC1/TC1/ST1)
- Fiber interface unit (FIU)

For the boards used in each unit, refer to [4.4 Function Boards](#).

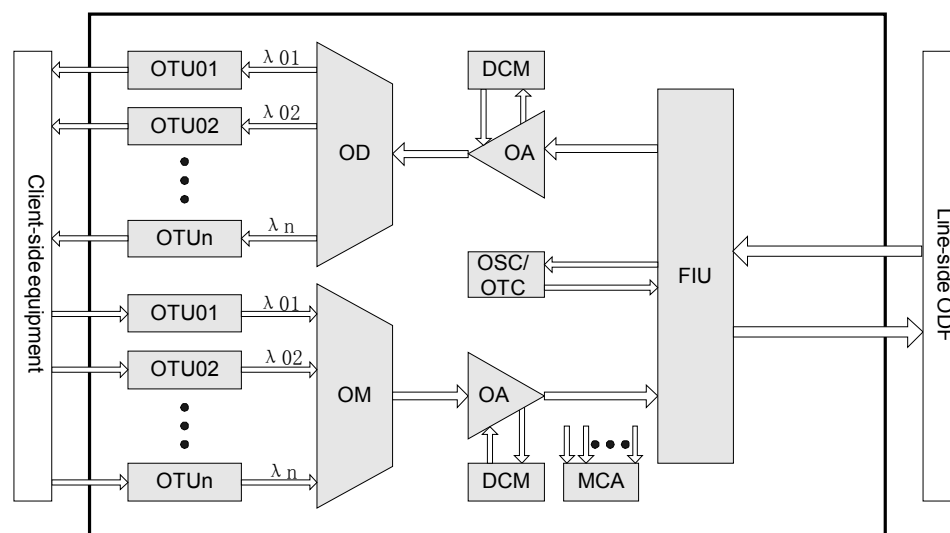
Signal Flow

In the transmit direction, through the OTU, the OTM node converges or transforms the accessed signals into signals at ITU-T G.694.1-compliant DWDM wavelengths. After that, the signals are multiplexed by the optical multiplexing board into the main optical path. Then, the main path signals are amplified and then are multiplexed with the optical supervisory signals. At last, the multiplexed signals are sent to the line for transmission.

In the receive direction, the optical supervisory signals and the main path signals are separated. The optical supervisory signals are sent to the optical supervisory channel (OSC) board for processing. After being amplified, the main path signals are demultiplexed by the optical demultiplexing board into signals at different wavelengths. Then, the signals are sent to the corresponding client-side equipment after being transformed or divided by the OTUs.

The diagram of this OTM node type is shown in **Figure 6-1**.

Figure 6-1 Schematic diagram of the OptiX Metro 6100 DWDM OTM node with M40 and D40 boards



OTU: optical transponder board

OD: optical demultiplexing board

FIU: fiber interface unit

MCA: multi-channel spectrum analyzer unit

OM: optical multiplexing board

OSC/OTC: optical supervisory channel board

OA: optical amplifier board

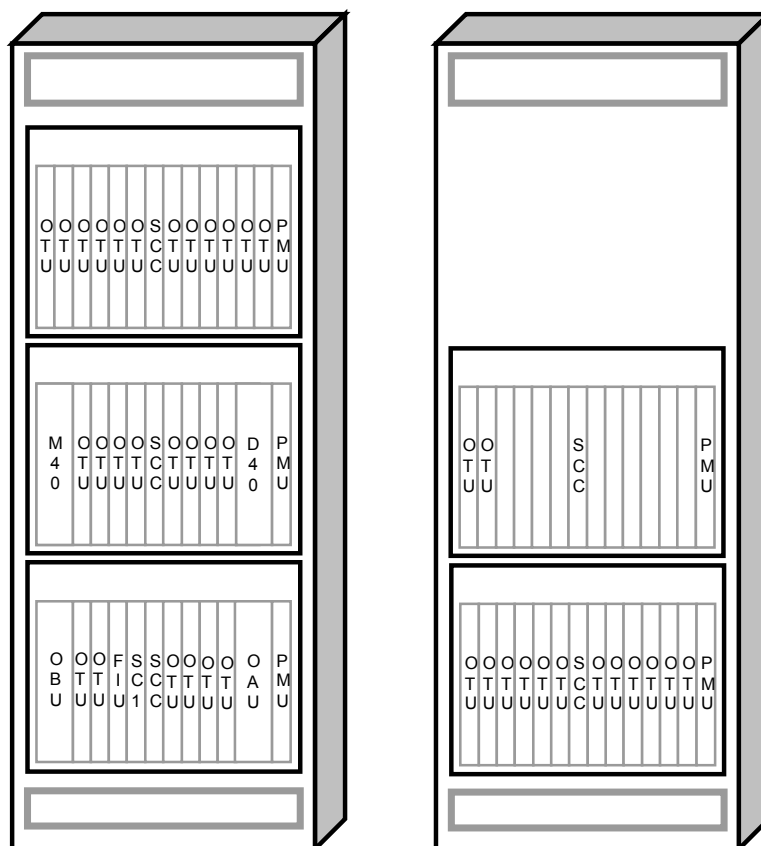
DCM: dispersion compensation module

Typical Configuration

The 40-wavelength OptiX Metro 6100 DWDM OTM node with M40 and D40 boards is taken as an example, as shown in **Figure 6-2**.

Five subracks and two cabinets are used.

Figure 6-2 Configuration diagram of the 40-wavelength OptiX Metro 6100 DWDM OTM node with M40 and D40 boards



Configuration Rules

The following are the rules for configuring the M40/V40 and D40 boards:

- In an open system or a mixed system, if more than 16 wavelengths are used to carry accessed services, use the M40/V40 board.
- The M01–M40 optical interfaces on the M40/V40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1 THz to 196.0 THz.
- The D01–D40 optical interfaces on the D40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1 THz to 196.0 THz.
- If the power pre-equilibrium function is required, use the V40 together with the MCA.
- If the power pre-equilibrium function is not required, use only the M40.
- If the east and west cabinets are separated in a station, install the west M40 (or V40) and D40 into IU1 and IU12 in the subrack in west cabinet, and install the east M40 (or V40) and D40 into IU1 and IU12 in the subrack in east cabinet.

The following are the rules for configuring the OTU boards:

- The OTU board with a smaller frequency is configured first. When there are multiple subracks, the OTU board is configured in a lower subrack first. The OTU board is configured in the left slot first.

- The 40G OTU must be inserted into the middle subrack and lower subrack.
- In a station where GE ADM is configured, the east-west separation principle is adopted. The OTUs in different directions and with mutual cross-connections must be installed in paired slots or on the same cross-connect plane in the same subrack.
- In the client-side 1+1 protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUS from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the inter-board wavelength protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUS from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the extended intra-board wavelength protection mode, the OTU and OLP boards must be configured in the same subrack and the OLP must be just next to the OTU.
- In the WXCP protection mode, the working OTU and the protection OTU must be installed on the same cross-connect plane or in paired cross-connect slots. On the same cross-connect plane, install the OTUS from left to right following the sequence of west OTU1, east OTU1, west OTU2, and east OTU2. When the OTUs are located in paired cross-connect slots, west OTU1 and west OTU2, and east OTU1 and east OTU2 in paired cross-connect slots are installed from left to right.
- When the DPPS and TPS are configured, the working and protection OTUs in DPPS protection and the active and standby TBEs in TPS protection must be located on the same cross-connect plane. They must be installed from left to right in a sequence that is the same as the OTUs in WXCP protection.

The following is the rule for configuring the SCC boards:

- The SCC board is installed in IU7 in each subrack.

The following is the rule for configuring the PMU boards:

- The PMU board is installed in IU14 in each subrack.

The following is the rule for configuring the supervisory channel boards:

- IU6 is preferred to house the supervisory channel board. If IU6 houses another board, install the supervisory channel board into IU8.

The following are the rules for configuring the amplifier boards:

- Install the FIU board into IU5 first, and then install optical amplifier boards one by one. If IU5 houses another board, install the FIU board into IU9.
- The west and east optical amplifier boards (the OAU, OBU and OPU) are installed on the left and right sides of the subrack respectively.
- When there are multiple optical amplifier boards in a service flow, install west optical amplifier boards on the left side of the subrack and install them from right to left along the service flow. Install east optical amplifier boards on the right side of the subrack and install them from left to right along the service flow.
- When a Raman amplifier is used, the Raman amplifier must be installed in the subrack where the optical amplifier board in the same direction is located.

The following are the rules for configuring the protection boards:

- When configuring the optical line protection, make the OLP board located close to the FIU board.
- When configuring the intra-board wavelength protection, configure the OTU board with dual-fed and selective receiving function for 2.5 Gbit/s services. In the case of 5 Gbit/s and 10 Gbit/s services, adopt the extended intra-board wavelength protection and configure the OLP board. In the case of 40 Gbit/s services, adopt the extended intra-board wavelength protection and configure the CP40 board.
- When configuring the client-side 1+1 wavelength protection, use the SCS board when the working and protection OTUs are in the same subrack. When the client-side 1+1 wavelength protection is configured in the system, the working and protection channels take different directions and are configured into a ring network.
- When configuring the inter-subrack wavelength protection, configure the DCP or OLP board as the dual-fed and selective receiving unit.

6.1.2 OTM Node with the OADM Boards

This section describes the signal flow, formation method, typical configurations, and configuration principles of the OTM node with the OADM boards.

Functions

The OTM node is used at the terminal station, and is logically divided into the following two directions:

- Transmit direction
- Receive direction

In the transmit direction, the OTM node converges or transforms and then amplifies client-side signals. Then, the signals are multiplexed with the optical supervisory signals. At last, the multiplexed signals are sent to the line for transmission. In the receive direction, the OTM node performs the converse process.

If less than 16 wavelengths are accessed in the early phase, the OTM comprising optical add/drop multiplexing boards is normally adopted, to lower the cost.

Functional Units

An OptiX Metro 6100 DWDM OTM node consists of the following functional units:

- Optical transponder board (OTU)
- Optical amplifier board (OA)
- Optical add/drop multiplexing board (OADM board)
- Unidirectional OSC board (SC1/TC1/ST1)
- Fiber interface unit (FIU)

For the boards used in each unit, refer to [4.4 Function Boards](#).

Signal Flow

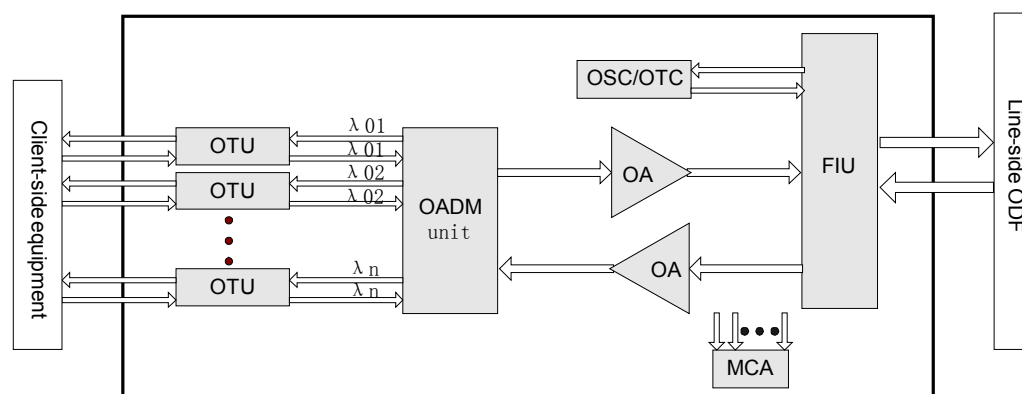
In the transmit direction, through the OTU, the OTM node converges or transforms the accessed signals into signals at ITU-T G.694.1-compliant DWDM wavelengths. After that, the signals are multiplexed by the optical add/drop multiplexing board into the main optical path. Then, the

main path signals are amplified and then are multiplexed with the optical supervisory signals. At last, the multiplexed signals are sent to the line for transmission.

In the receive direction, the optical supervisory signals and the main path signals are separated. The optical supervisory signals are sent to the OSC board for processing. After being amplified, the main path signals are demultiplexed by the optical add/drop multiplexing board into signals at different wavelengths. Then, the signals are sent to the corresponding client-side equipment after being transformed or divided by the OTUs.

The diagram of this OTM node type is shown in **Figure 6-3**.

Figure 6-3 Schematic diagram of the OptiX Metro 6100 DWDM OTM node with OADM boards



OTU: optical transponder board

OA: optical amplifier board

OSC/OTC: optical supervisory channel board

OADM board: optical add/drop multiplexing board

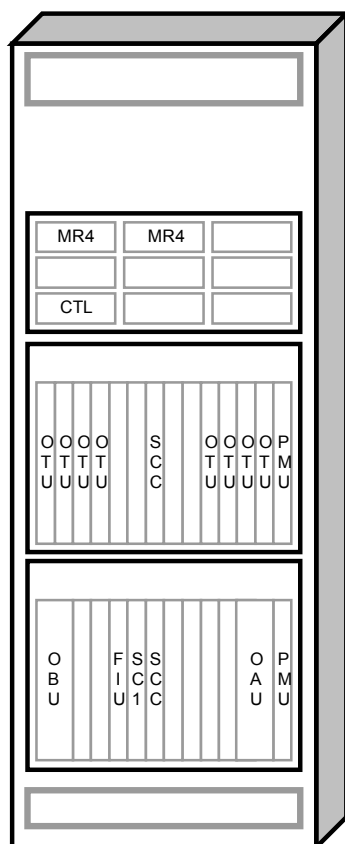
FIU: fiber interface unit

Typical Configuration

The eight-wavelength OptiX Metro 6100 DWDM OTM node with OADM boards is taken as an example, as shown in **Figure 6-4**.

Two subracks, one OADM frame and one cabinet are used.

Figure 6-4 Configuration diagram of the eight-wavelength OptiX Metro 6100 DWDM OTM node with OADM boards



Configuration Rules

The following are the rules for configuring the MR4 and MR2 boards:

- When the number of added/dropped wavelengths is smaller than 16, use the OADM board.
- Install the OADM boards in the order of wavelength and install them from left to right and from top to down.
- The MR2 and MR4 boards can be configured as an OTM.
- Install the MR4 or MR2 board into IU15–IU22 in the OADM frame in an ascending order of frequency. The optical interfaces of the boards are arranged in an ascending order of frequency.

The following are the rules for configuring the OTU boards:

- The OTU board with a smaller frequency is configured first. When there are multiple subracks, the OTU board is configured in a lower subrack first. The OTU board is configured in the left slot first.
- The 40G OTU must be inserted into the middle subrack and lower subrack.
- In a station where GE ADM is configured, the east-west separation principle is adopted. The OTUs in different directions and with mutual cross-connections must be installed in paired slots or on the same cross-connect plane in the same subrack.

- In the client-side 1+1 protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the inter-board wavelength protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the extended intra-board wavelength protection mode, the OTU and OLP boards must be configured in the same subrack and the OLP must be just next to the OTU.
- In the WXCP protection mode, the working OTU and the protection OTU must be installed on the same cross-connect plane or in paired cross-connect slots. On the same cross-connect plane, install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, and east OTU2. When the OTUs are located in paired cross-connect slots, west OTU1 and west OTU2, and east OTU1 and east OTU2 in paired cross-connect slots are installed from left to right.
- When the DPPS and TPS are configured, the working and protection OTUs in DPPS protection and the active and standby TBEs in TPS protection must be located on the same cross-connect plane. They must be installed from left to right in a sequence that is the same as the OTUs in WXCP protection.

The following is the rule for configuring the SCC boards:

- The SCC board is installed in IU7 in each subrack.

The following is the rule for configuring the PMU boards:

- The PMU board is installed in IU14 in each subrack.

The following is the rule for configuring the supervisory channel boards:

- IU6 is preferred to house the board. If IU6 houses another board, install the supervisory channel board into IU8.

The following are the rules for configuring the amplifier boards:

- Install the FIU board into IU5 first, and then install optical amplifier boards one by one. If IU5 houses another board, install the FIU board into IU9.
- The west and east optical amplifier boards (the OAU, OBU and OPU) are installed on the left and right sides of the subrack respectively.
- When there are multiple optical amplifier boards in a service flow, install west optical amplifier boards on the left side of the subrack and install them from right to left along the service flow. Install east optical amplifier boards on the right side of the subrack and install them from left to right along the service flow.
- When a Raman amplifier is used, the Raman amplifier must be installed in the subrack where the optical amplifier board in the same direction is located.

The following are the rules for configuring the protection boards:

- When configuring the optical line protection, make the OLP board located close to the FIU board.
- When configuring the intra-board wavelength protection, configure the OTU board with dual-fed and selective receiving function for 2.5 Gbit/s services. In the case of 5 Gbit/s and

10 Gbit/s services, adopt the extended intra-board wavelength protection and configure the OLP board. In the case of 40 Gbit/s services, adopt the extended intra-board wavelength protection and configure the CP40 board.

- When configuring the client-side 1+1 wavelength protection, use the SCS board when the working and protection OTUs are in the same subrack. When the client-side 1+1 wavelength protection is configured in the system, the working and protection channels take different directions and are configured into a ring network.
- When configuring the inter-subrack wavelength protection, configure the DCP or OLP board as the dual-fed and selective receiving unit.

6.2 OLA

The DWDM system can be configured as the OLA comprising optical amplifier boards.

Functions

The OLA is used at the optical amplifier station to amplify the optical signals in two directions.

Functional Units

An OptiX Metro 6100 DWDM OLA node consists of the following functional units:

- Optical amplifier board (OA)
- Bidirectional OSC board (SC2/TC2/ST2)
- Fiber interface unit (FIU)

For the boards used in each unit, refer to [4.4 Function Boards](#).

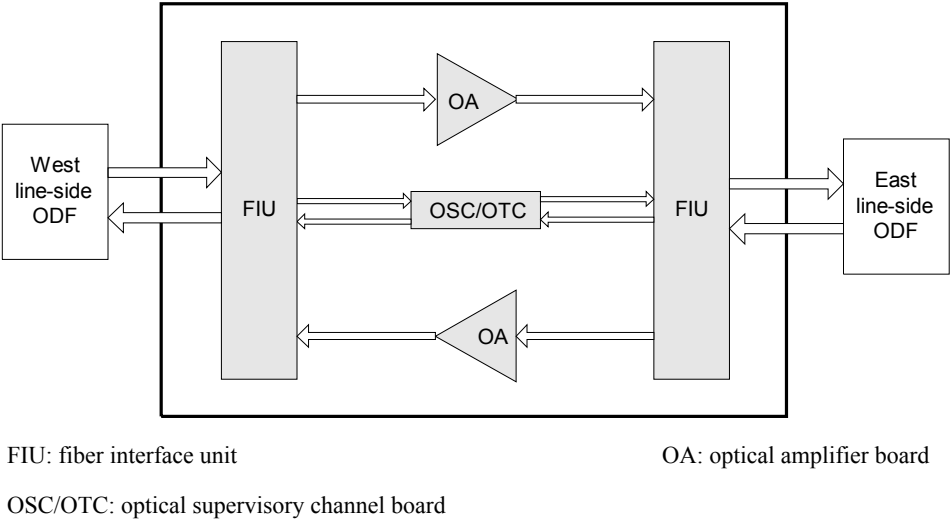
Signal Flow

It separates the optical supervisory signal from the signals in the main path and sends the former to the OSC unit for processing.

The signals in the main path are amplified by the amplifier board and multiplexed with the OSC signals that has already been processed, and then sent to the line fiber for transmission.

The diagram of OptiX Metro 6100 DWDM OLA node is illustrated in [Figure 6-5](#).

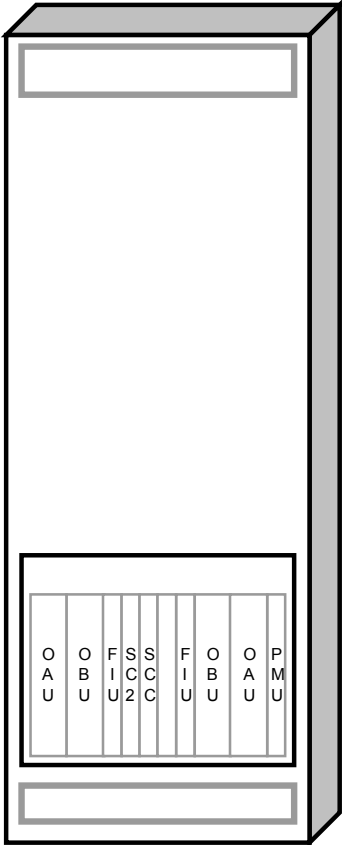
Figure 6-5 Schematic diagram of the OptiX Metro 6100 DWDM OLA node



Typical Configuration

The typical configuration of OptiX Metro 6100 DWDM OLA node is shown in Figure 6-6. One subrack and one cabinet are used.

Figure 6-6 Configuration diagram of the OptiX Metro 6100 DWDM OLA node



Configuration Rules

The following are the rules for configuring the amplifier boards:

- Install the FIU boards into IU5 and IU9, and then install optical amplifier boards one by one.
- The west and east optical amplifier boards (the OAU, OBU and OPU) are installed on the left and right sides of the subrack respectively.
- When there are multiple optical amplifier boards in a service flow, install west optical amplifier boards on the left side of the subrack and install them from right to left along the service flow. Install east optical amplifier boards on the right side of the subrack and install them from left to right along the service flow.
- The optical amplifiers from west to east can be inserted into IU1–IU4, and the IU1 is for the OAU. The optical amplifiers from east to west can be inserted into IU8–IU12, and the IU12 is for the OAU.
- When a Raman amplifier is used, the Raman amplifier must be installed in the subrack where the optical amplifier board in the same direction is located.

The following is the rule for configuring the SCC boards:

- The SCC board is installed in IU7 in each subrack.

The following is the rule for configuring the PMU boards:

- The PMU board is installed in IU14 in each subrack.

The following is the rule for configuring the supervisory channel boards:

- IU6 is preferred to house the board. If IU6 houses another board, install the supervisory channel board into IU8.

6.3 FOADM

The FOADM equipment adds/drops optical wavelength signals at the intermediate node.

The OptiX Metro 6100 DWDM FOADM node can be formed in the following two modes:

- The FOADM comprising the optical multiplexing board (M40/V40) and optical demultiplexing board (D40)
- The FOADM comprising optical add/drop multiplexing boards

6.3.1 FOADM Node with Optical Multiplexer Board and Optical Demultiplexer Board

This section describes the signal flow, construction method, typical configurations, and configuration principles of the FOADM equipment with optical multiplexer board and optical demultiplexer board.

Functions

The FOADM node performs the add/drop multiplexing of fixed wavelengths from the multiplexed signals. The FOADM node comprising optical multiplexing and optical

demultiplexing boards is normally used at a central station. It is formed by two back-to-back OTMs. The advantage is that services are not interrupted in expansion.

Functional Units

An FOADM node formed by the M40/V40 and D40 consists of the following functional units:

- Optical transponder board (OTU)
- Optical amplifier board (OA)
- Optical multiplexing board (M40/V40)
- Optical demultiplexing board (D40)
- Bidirectional OSC board (SC2/TC2/ST2)
- Fiber interface unit (FIU)

For the boards used in each unit, refer to [4.4 Function Boards](#).

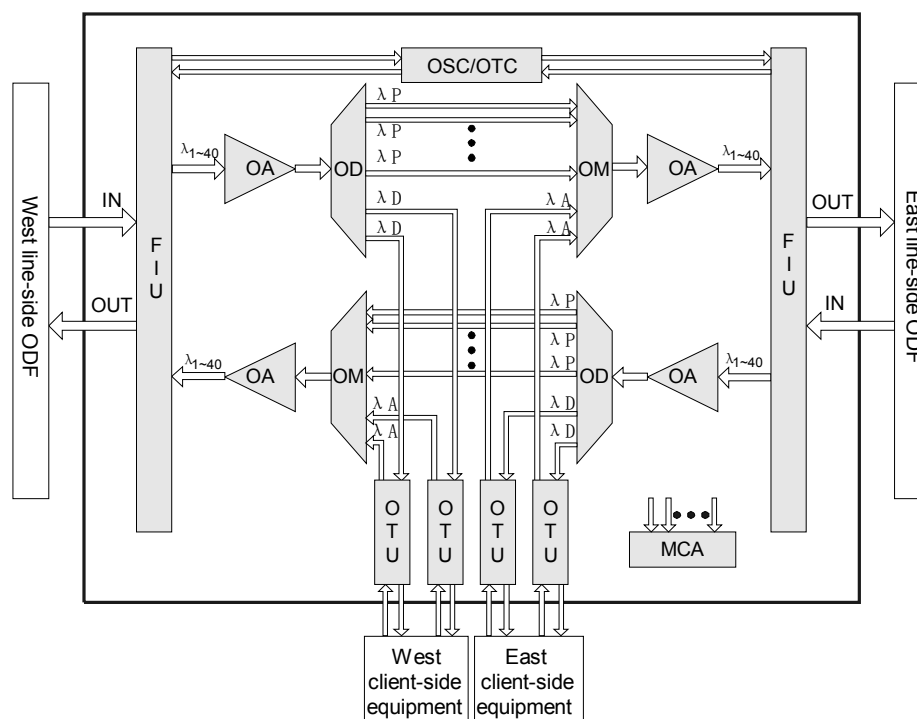
Signal Flow

The FOADM node processes the optical signals in two directions.

The optical supervisory signals and the main path signals are separated from the signals from the line. The optical supervisory signals are sent to the OSC board for processing. After being amplified, the main path signals are sent to the optical demultiplexing board. Some wavelengths separated from the main path signals are sent to the OTU and then to the local client-side equipment. Other wavelengths are not dropped on the node. Pass-through wavelengths and the added wavelengths are multiplexed by the optical multiplexing board and then are amplified. Then, the signals are multiplexed with the processed optical supervisory signals. At last, the multiplexed signals are sent to the line for transmission.

The diagram of this FOADM node type is shown in [Figure 6-7](#).

Figure 6-7 Schematic diagram of the OptiX Metro 6100 DWDM FOADM node with M40 and D40 boards



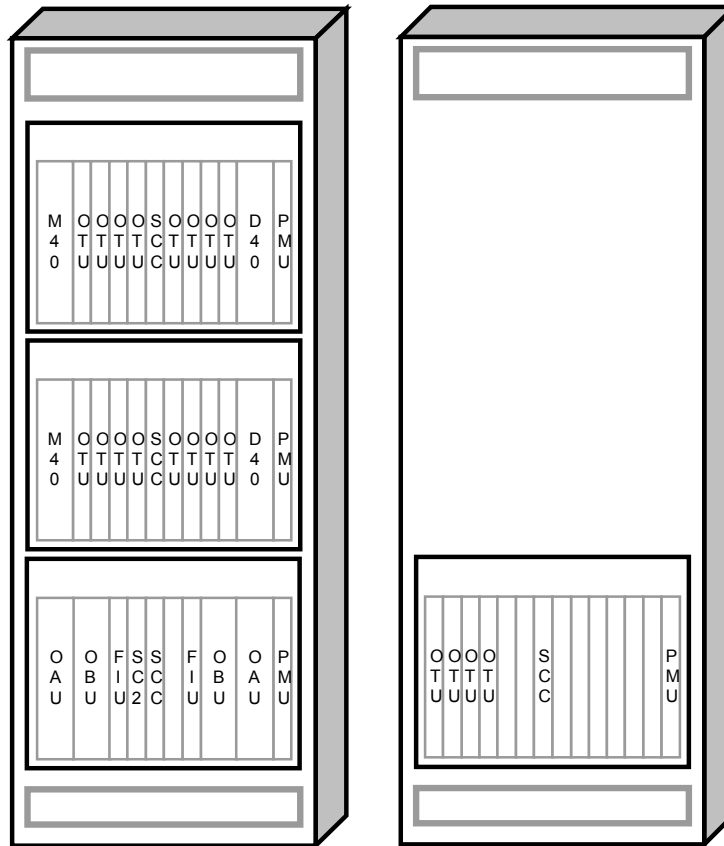
OTU: optical transponder	OA: optical amplifier board	OM: optical multiplexing board
OD: optical demultiplexing board	OSC/OTC: optical supervisory channel board	MCA: multi-channel spectrum analyzer board
λP: Pass-through service	λA: Added service	λD: Dropped service

Typical Configuration

The 10-wavelength FOADM node with the M40 and D40 boards is taken as an example, as shown in [Figure 6-8](#).

This FOADM node can add and drop 10 channels respectively in two transmission directions. Four subracks and two cabinets are used.

Figure 6-8 Configuration diagram of the 10-wavelength OptiX Metro 6100 DWDM FOADM node with the M40 and D40 boards



Configuration Rules

The following are the rules for configuring the M40 and D40 boards:

- When the number of wavelengths for the accessed services is larger than 16, use the M40/V40 and D40 boards.
- The M01–M40 optical interfaces on the M40/V40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1THz to 196.0THz.
- The D01–D40 optical interfaces on the D40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1 THz to 196.0 THz.
- If the east and west cabinets are separated in a station, install the west M40/V40 and D40 into IU1 and IU12 in the subrack in west cabinet, and install the east M40/V40 and D40 into IU1 and IU12 in the subrack in east cabinet.

The following are the rules for configuring the OTU boards:

- The OTU board with a smaller frequency is configured first. When there are multiple subracks, the OTU board is configured in a lower subrack first. The OTU board is configured in the left slot first.
- The 40G OTU must be inserted into the middle subrack and lower subrack.

- In a station where GE ADM is configured, the east-west separation principle is adopted. The OTUs in different directions and with mutual cross-connections must be installed in paired slots or on the same cross-connect plane in the same subrack.
- In the client-side 1+1 protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the inter-board wavelength protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the extended intra-board wavelength protection mode, the OTU and OLP boards must be configured in the same subrack and the OLP must be just next to the OTU.
- In the WXCP protection mode, the working OTU and the protection OTU must be installed on the same cross-connect plane or in paired cross-connect slots. On the same cross-connect plane, install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, and east OTU2. When the OTUs are located in paired cross-connect slots, west OTU1 and west OTU2, and east OTU1 and east OTU2 in paired cross-connect slots are installed from left to right.
- When the DPPS and TPS are configured, the working and protection OTUs in DPPS protection and the active and standby TBEs in TPS protection must be located on the same cross-connect plane. They must be installed from left to right in a sequence that is the same as the OTUs in WXCP protection.
- When unidirectional electrical regeneration is configured, the unidirectional electrical regeneration boards (the TRC1, LRF, LRFS, TMR or TMRS) must be of the same type and be installed in paired slots (IU1 and IU8, IU2 and IU9, IU3 and IU10, IU4 and IU11, IU5 and IU12, and IU6 and IU13). On the IU1–IU6 plane, the receive optical interface of the unidirectional electrical regeneration board should be defined as west, and the transmit optical interface as east. On the IU8–IU13 plane, the receive optical interface of the unidirectional electrical regeneration board should be defined as east, and the transmit optical interface as west.
- If there are more than two unidirectional electrical regeneration boards in a subrack, install the board at smaller wavelength first and insert them into slots from left to right.

The following is the rule for configuring the SCC boards:

- The SCC board is installed in IU7 in each subrack.

The following is the rule for configuring the PMU boards:

- The PMU board is installed in IU14 in each subrack.

The following is the rule for configuring the supervisory channel boards:

- IU6 is preferred to house the board. If IU6 houses another board, install the supervisory channel board into IU8.

The following are the rules for configuring the amplifier boards:

- Install the FIU boards into IU5 and IU9, and then install optical amplifier boards one by one.

- The west and east optical amplifier boards (the OAU, OBU and OPU) are installed on the left and right sides of the subrack respectively.
- When there are multiple optical amplifier boards in a service flow, install west optical amplifier boards on the left side of the subrack and install them from right to left along the service flow. Install east optical amplifier boards on the right side of the subrack and install them from left to right along the service flow.
- When a Raman amplifier is used, the Raman amplifier must be installed in the lower subrack where the optical amplifier board in the same direction is located.

The following are the rules for configuring the protection boards:

- When configuring the optical line protection, make the OLP board located close to the FIU board.
- When configuring the intra-board wavelength protection, configure the OTU board with dual-fed and selective receiving function for 2.5 Gbit/s services. In the case of 5 Gbit/s and 10 Gbit/s services, adopt the extended intra-board wavelength protection and configure the OLP board. In the case of 40 Gbit/s services, adopt the extended intra-board wavelength protection and configure the CP40 board.
- When configuring the client-side 1+1 wavelength protection, use the SCS board when the working and protection OTUs are in the same subrack. When the client-side 1+1 wavelength protection is configured in the system, the working and protection channels take different directions and are configured into a ring network.
- When configuring the inter-subrack wavelength protection, configure the DCP or OLP board as the dual-fed and selective receiving unit.

6.3.2 FOADM Node with OADM Boards

This section describes the signal flow, construction method, typical configurations, and configuration principles of the FOADM equipment with OADM boards.

Functions

The FOADM node performs the add/drop multiplexing of fixed wavelengths from the multiplexed signals. The FOADM node comprising optical add/drop multiplexing boards is normally used at an edge station. It features low insertion loss, flexible expansion and low initial cost.

Functional Units

The FOADM node formed by optical add/drop multiplexing boards consists of the following functional units:

- Optical transponder board (OTU)
- Optical amplifier board (OA)
- Optical add/drop multiplexing board (OADM board)
- Bidirectional OSC board (SC2/TC2/ST2)
- Fiber interface unit (FIU)

For the boards used in each unit, refer to [4.4 Function Boards](#).

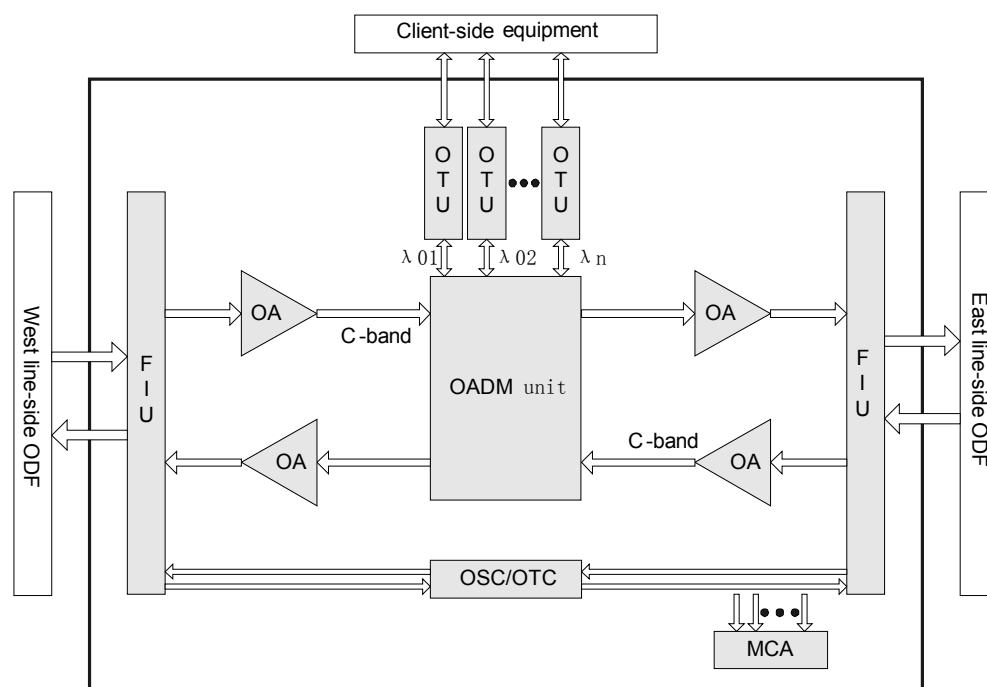
Signal Flow

The FOADM node processes the optical signals in two directions.

The optical supervisory signals and the main path signals are separated from the signals from the line. The optical supervisory signals are sent to the OSC board for processing. After being amplified, the main path signals are sent to the optical add/drop multiplexing board. Some wavelengths separated from the main path signals are sent to the OTU and then to the local client-side equipment. No add/drop multiplexing is performed in other wavelengths on the node. Pass-through wavelengths and the added wavelengths are multiplexed and then are amplified. Then, the signals are multiplexed with the processed optical supervisory signals. At last, the multiplexed signals are sent to the line for transmission.

The functional modules of the FOADM node are shown in **Figure 6-9**.

Figure 6-9 Schematic diagram of the FOADM node comprising optical add/drop multiplexing boards



OTU: optical transponder board

OA: optical amplifier board

OSC/OTC: optical supervisory channel board

OADM board: optical add/drop multiplexing board

MCA: multi-channel spectrum analyzer board

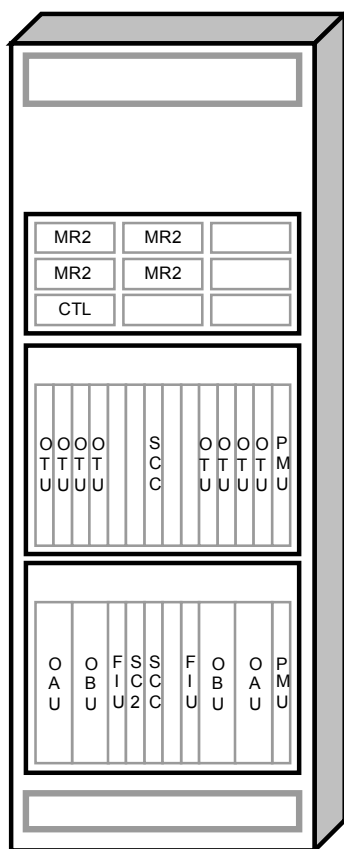
Typical Configuration

The four-wavelength FOADM node with OADM boards is taken as an example, as shown in **Figure 6-10**.

This FOADM node can add and drop four channels respectively in two transmission directions.

Two subracks, one OADM frame and one cabinet are used.

Figure 6-10 Configuration diagram of the four-wavelength OptiX Metro 6100 DWDM FOADM node with OADM boards



Configuration Rules

The following are the rules for configuring the MR4 and MR2 boards:

- When the number of added/dropped wavelengths is smaller than 16, use the OADM board.
- Normally, the OADM frame is located in the middle or upper subrack position in a cabinet. In addition, it is suggested to make the OADM frame located in the same cabinet as the subrack housing the corresponding OTU board.
- When one OADM frame is to be configured and the number of OADM boards is smaller than 6, IU15–IU17 and IU21 are defined as west, and IU18–IU20 and IU22 are defined as east. IU21 is for the west EFIU. IU22 is for the east EFIU. IU15–IU17 are for the west OADM boards. IU18–IU20 are for the east OADM boards.
- It is suggested to make the OADM frame located in the same cabinet as the subrack housing the corresponding OTU board.
- To decrease the insertion loss, drop 10 Gbit/s services first and then 2.5 Gbit/s services.
- The MR2 or MR4 board can be configured as an OTM.
- Install the MR4 or MR2 board into IU15–IU22 in the OADM frame in an ascending order of frequency. The optical interfaces of the boards are arranged in an ascending order of frequency.

- When the MR2 boards are configured as an OTM, set the maximum number of wavelengths in one OADM frame to 12. If more wavelengths are to be added/dropped, add another OADM frame.
- If one OADM subrack corresponds to only one direction, IU21 is preferred to house the EFIU.

The following are the rules for configuring the OTU boards:

- The OTU board with a smaller frequency is configured first. When there are multiple subracks, the OTU board is configured in a lower subrack first. The OTU board is configured in the left slot first.
- The 40G OTU must be inserted into the middle subrack and lower subrack.
- When the MR2 board is used (in serial or in parallel manner), try to make the OTU boards in the same band located in the same subrack.
- In a station where GE ADM is configured, the east-west separation principle is adopted. The OTUs in different directions and with mutual cross-connections must be installed in paired slots or on the same cross-connect plane in the same subrack.
- In the client-side 1+1 protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the inter-board wavelength protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the extended intra-board wavelength protection mode, the OTU and OLP boards must be configured in the same subrack and the OLP must be just next to the OTU.
- In the WXCP protection mode, the working OTU and the protection OTU must be installed on the same cross-connect plane or in paired cross-connect slots. On the same cross-connect plane, install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, and east OTU2. When the OTUs are located in paired cross-connect slots, west OTU1 and west OTU2, and east OTU1 and east OTU2 in paired cross-connect slots are installed from left to right.
- When the DPPS and TPS are configured, the working and protection OTUs in DPPS protection and the active and standby TBEs in TPS protection must be located on the same cross-connect plane. They must be installed from left to right in a sequence that is the same as the OTUs in WXCP protection.
- When unidirectional electrical regeneration is configured, the unidirectional electrical regeneration boards (the TRC1, LRF, LRFS, TMR or TMRS) must be of the same type and be installed in paired slots (IU1 and IU8, IU2 and IU9, IU3 and IU10, IU4 and IU11, IU5 and IU12, and IU6 and IU13). On the IU1–IU6 plane, the receive optical interface of the unidirectional electrical regeneration board should be defined as west, and the transmit optical interface as east. On the IU8–IU13 plane, the receive optical interface of the unidirectional electrical regeneration board should be defined as east, and the transmit optical interface as west.
- If there are more than two unidirectional electrical regeneration boards in a subrack, install the board at smaller wavelength first and insert them into slots from left to right.

The following is the rule for configuring the SCC boards:

- The SCC board is installed in IU7 in each subrack.

The following is the rule for configuring the PMU boards:

- The PMU board is installed in IU14 in each subrack.

The following is the rule for configuring the supervisory channel boards:

- IU6 is preferred to house the board. If IU6 houses another board, install the supervisory channel board into IU8.

The following are the rules for configuring the amplifier boards:

- Install the FIU boards into IU5 and IU9, and then install optical amplifier boards one by one.
- The west and east optical amplifier boards (the OAU, OBU and OPU) are installed on the left and right sides of the subrack respectively.
- When there are multiple optical amplifier boards in a service flow, install west optical amplifier boards on the left side of the subrack and install them from right to left along the service flow. Install east optical amplifier boards on the right side of the subrack and install them from left to right along the service flow.
- When a Raman amplifier is used, the Raman amplifier must be installed in the lower subrack where the optical amplifier board in the same direction is located.

The following are the rules for configuring the protection boards:

- When configuring the optical line protection, make the OLP board located close to the FIU board.
- When configuring the intra-board wavelength protection, configure the OTU board with dual-fed and selective receiving function for 2.5 Gbit/s services. In the case of 5 Gbit/s and 10 Gbit/s services, adopt the extended intra-board wavelength protection and configure the OLP board. In the case of 40 Gbit/s services, adopt the extended intra-board wavelength protection and configure the CP40 board.
- When configuring the client-side 1+1 wavelength protection, use the SCS board when the working and protection OTUs are in the same subrack. When the client-side 1+1 wavelength protection is configured in the system, the working and protection channels take different directions and are configured into a ring network.
- When configuring the inter-subrack wavelength protection, configure the DCP or OLP board as the dual-fed and selective receiving unit.

6.4 ROADM

The ROADM equipment dynamically adds/drops and cross-connects optical wavelength signals at the intermediate node.

The OptiX Metro 6100 DWDM ROADM node can be formed in the following four modes:

- The ROADM comprising the DWC boards
- The ROADM comprising the WSD9 and WSM9 boards
- The ROADM comprising the RMU9 and WSD9 boards
- The ROADM comprising the WSMD4 boards

6.4.1 ROADM Node with DWC Boards

The ROADM node comprising the DWC boards can be used to add/drop a maximum of 40 wavelengths through the specified optical interfaces.

Functions

The DWC board provides functions such as optical power detection, channel optical power equilibrium, and the dynamic add/drop, pass-through and blocking of service wavelengths. It dynamically grooms service wavelengths in the ring network.

The ROADM node comprising the DWC boards can be used at a central station, or an edge station. It features flexible expansion, no service interruption in expansion, and low operation cost. On the ROADM node, the network management software is used to allocate the add/drop and pass-through states of wavelengths and to adjust the states of wavelengths dynamically and remotely.

Functional Units

An ROADM node comprising the optical add/drop multiplexing boards or optical multiplexing and demultiplexing boards and the DWC boards consists of the following functional units:

- Optical transponder board (OTU)
- Optical amplifier board (OA)
- Bidirectional OSC board (SC2/TC2/ST2)
- Fiber interface unit (FIU)
- Optical add/drop multiplexing board or optical multiplexing and demultiplexing board (OM/OD)
- Dynamic wavelength control board (DWC)
- Electrical variable optical attenuator (VOA)

For the boards used in each unit, refer to [4.4 Function Boards](#).

Signal Flow

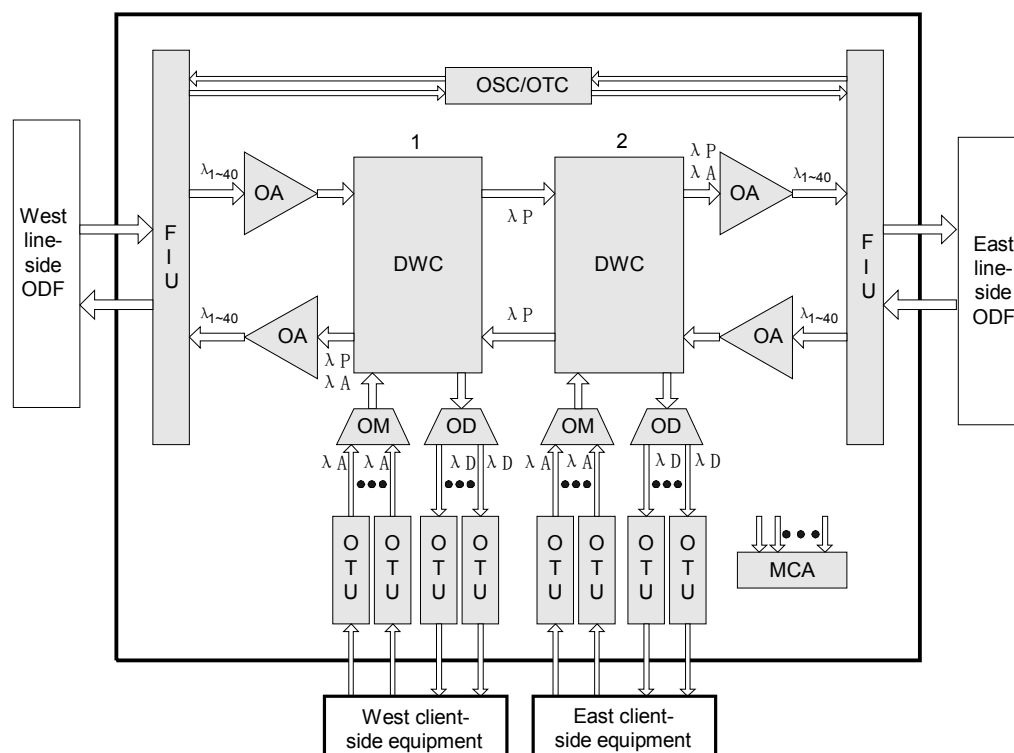
The ROADM node processes the optical signals in two directions.

The optical supervisory signals and the main path signals are separated from the signals from the line. The optical supervisory signals are sent to the OSC board for processing. After being amplified, the main path signals are sent to the DWC board. The wavelengths to be dropped are demultiplexed by the demultiplexing board or the optical add/drop multiplexing board. Then, the wavelengths are sent to the OTU and then to the local client-side equipment. The wavelengths where no add/drop multiplexing is required are pass-through wavelengths. They and the added wavelengths are multiplexed and then are amplified. Then, the signals are multiplexed with the processed optical supervisory signals. At last, the multiplexed signals are sent to the line for transmission.

For the application of the ROADM node comprising the DWC boards, and the networking signal flow, refer to [8.1.1 Intra-Ring Wavelength Grooming by DWC Boards](#).

The functional modules of this ROADM node are shown in [Figure 6-11](#).

Figure 6-11 Schematic diagram of the ROADM node comprising the DWC boards



OA: optical amplifier board

OSC/OTC: optical supervisory channel board

OM/OD: optical add/drop multiplexing board or optical multiplexing and demultiplexing board

OTU: optical transponder board

DWC: dynamic wavelength control board

MCA: multi-channel spectrum analyzer board

λP : Pass-through service

λA : Added service

λD : Dropped service

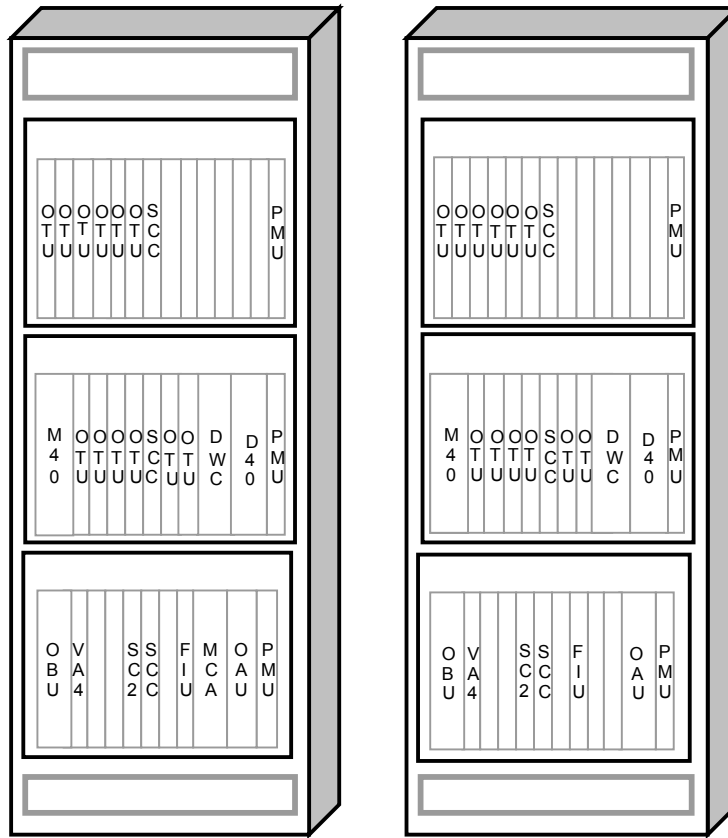
Typical Configuration

The 12-wavelength ROADM node with M40, D40 boards and DWC boards is taken as an example, as shown in [Figure 6-12](#).

This ROADM node can add and drop 12 channels respectively in two transmission directions.

Six subracks and two cabinets are used.

Figure 6-12 Configuration diagram of the 12-wavelength OptiX Metro 6100 DWDM ROADM node with M40, D40 boards and DWC boards



NOTE

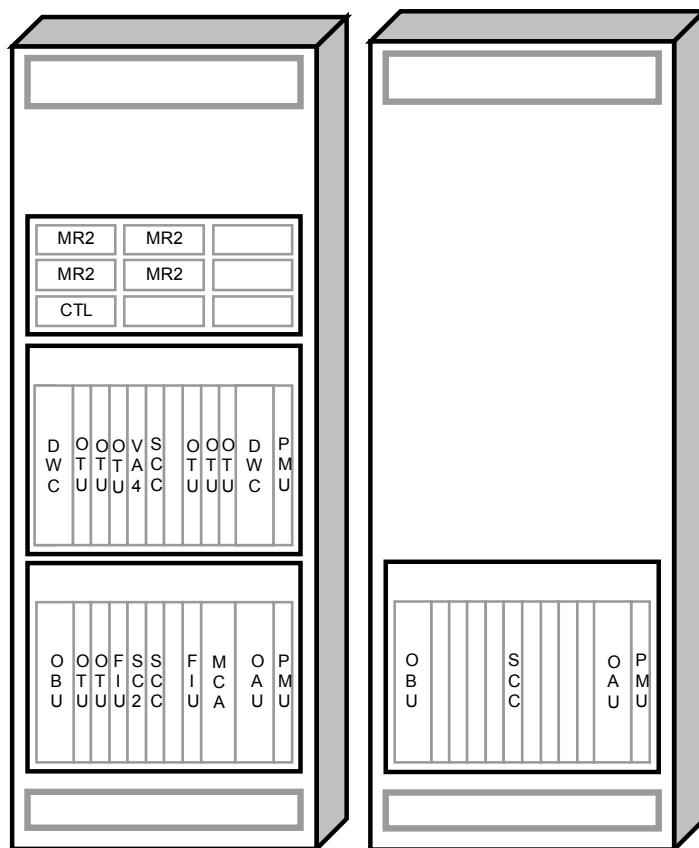
In the above typical configuration, two SC2 boards used in one node are for enhancing the processing capability of the optical supervisory channel.

The four-wavelength ROADM node with OADM boards and DWC boards is taken as an example, as shown in [Figure 6-13](#).

This ROADM node can add and drop four channels respectively in two transmission directions.

Three subracks, one OADM frame and two cabinets are used.

Figure 6-13 Configuration diagram of the four-wavelength OptiX Metro 6100 DWDM ROADM node with OADM boards and DWC boards



Configuration Rules

The following are the rules for configuring the DWC boards:

- In the case of a node where a large number of wavelengths are added/dropped and where two-dimensional grooming is performed, the DWC board is recommended to realize the intra-ring wavelength grooming.
- If the OADM boards are used in the station and the east and west subracks are in the same cabinet, install the DWC boards into IU1 and IU12 of the middle subrack.
- If the M40/V40 and D40 boards are used and the east and west cabinets are separated in the station, install the west DWC board into IU1 in the middle subrack in west cabinet, and install the east DWC board into IU12 in the middle subrack in east cabinet.

The following are the rules for configuring the M40/V40 and D40 boards:

- When the number of wavelengths for the accessed services is larger than 16, use the M40/V40 and D40 boards.
- The M01–M40 optical interfaces on the M40/V40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1THz to 196.0THz.
- The D01–D40 optical interfaces on the D40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1 THz to 196.0 THz.

The following are the rules for configuring the OADM frame:

- When the number of added/dropped wavelengths is smaller than 16, use the OADM board.
- Normally, the OADM frame is located in the middle or upper subrack position in a cabinet. In addition, it is suggested to make the OADM frame located in the same cabinet as the subrack housing the corresponding OTU board.
- When one OADM frame is to be configured and the number of OADM boards is smaller than 6, IU15–IU17 and IU21 are defined as west, and IU18–IU20 and IU22 are defined as east. IU21 is for the west EFIU. IU22 is for the east EFIU. IU15–IU17 are for the west OADM boards. IU18–IU20 are for the east OADM boards.
- It is suggested to make the OADM frame located in the same cabinet as the subrack housing the corresponding OTU board.
- To decrease the insertion loss, drop 10 Gbit/s services first and then 2.5 Gbit/s services.
- The MR2 or MR4 board can be configured as an OTM.
- Install the MR4 or MR2 board into IU15–IU22 in the OADM frame in an ascending order of frequency. The optical interfaces of the boards are arranged in an ascending order of frequency.
- When the MR2 boards are configured as an OTM, set the maximum number of wavelengths in one OADM frame to 12. If more wavelengths are to be added/dropped, add another OADM frame. Normally, the ACS is inserted into IU22.
- If one OADM subrack corresponds to only one direction, IU21 is preferred to house the EFIU.

The following are the rules for configuring the OTU boards:

- The OTU board with a smaller frequency is configured first. When there are multiple subracks, the OTU board is configured in a lower subrack first. When there is only one subrack, the OTU board is configured in the left slot first.
- The 40G OTU must be inserted into the middle subrack and lower subrack.
- In a station where GE ADM is configured, the east-west separation principle is adopted. The OTUs in different directions and with mutual cross-connections must be installed in paired slots or on the same cross-connect plane in the same subrack.
- In the client-side 1+1 protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the inter-board wavelength protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the extended intra-board wavelength protection mode, the OTU and OLP boards must be configured in the same subrack and the OLP must be just next to the OTU.
- In the WXCP protection mode, the working OTU and the protection OTU must be installed on the same cross-connect plane or in paired cross-connect slots. On the same cross-connect plane, install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, and east OTU2. When the OTUs are located in paired cross-connect slots, west OTU1 and west OTU2, and east OTU1 and east OTU2 in paired cross-connect slots are installed from left to right.
- When the DPPS and TPS are configured, the working and protection OTUs in DPPS protection and the active and standby TBEs in TPS protection must be located on the same

cross-connect plane. They must be installed from left to right in a sequence that is the same as the OTUs in WXCP protection.

The following is the rule for configuring the SCC boards:

- The SCC board is installed in IU7 in each subrack.

The following is the rule for configuring the PMU boards:

- The PMU board is installed in IU14 in each subrack.

The following is the rule for configuring the supervisory channel boards:

- IU6 is preferred to house the board. If IU6 houses another board, install the supervisory channel board into IU8.

The following are the rules for configuring the amplifier boards:

- Install the FIU boards into IU5 and IU9, and then install optical amplifier boards one by one.
- The west and east optical amplifier boards (the OAU, OBU and OPU) are installed on the left and right sides of the subrack respectively.
- When there are multiple optical amplifier boards in a service flow, install west optical amplifier boards on the left side of the subrack and install them from right to left along the service flow. Install east optical amplifier boards on the right side of the subrack and install them from left to right along the service flow.
- When a Raman amplifier is used, the Raman amplifier must be installed in the lower subrack where the optical amplifier board in the same direction is located.

The following are the rules for configuring the protection boards:

- When configuring the optical line protection, make the OLP board located close to the FIU board.
- When configuring the intra-board wavelength protection, configure the OTU board with dual-fed and selective receiving function for 2.5 Gbit/s services. In the case of 5 Gbit/s and 10 Gbit/s services, adopt the extended intra-board wavelength protection and configure the OLP board. In the case of 40 Gbit/s services, adopt the extended intra-board wavelength protection and configure the CP40 board.
- When configuring the client-side 1+1 wavelength protection, use the SCS board when the working and protection OTUs are in the same subrack. When the client-side 1+1 wavelength protection is configured in the system, the working and protection channels take different directions and are configured into a ring network.
- When configuring the inter-subrack wavelength protection, configure the DCP or OLP board as the dual-fed and selective receiving unit.

6.4.2 ROADM Node with WSD9 and WSM9 Board

The multi-dimensional WSS-based ROADM function realizes the full dynamic adding/dropping of wavelengths in a ring. The inter-ring extension is supported. A maximum of eight dimensions of wavelength grooming is supported.

Functions

The WSD9 board is used to demultiplex any wavelength and allocate it to any port. The demultiplexing is dynamic and configurable. On a node in a ring or chain, the board is able to

combine any wavelengths in the line and output the multiplexed wavelength. In addition, the board is able to allocate the output wavelength to any port. In this manner, the board realizes full dynamic allocation of wavelengths.

The WSM9 board is used to multiplex any wavelength and allocate it to any port. The demultiplexing is dynamic and configurable. On a node in a ring or chain, the board is able to combine any wavelengths in the line and input the multiplexed wavelength. In addition, the board is able to allocate the input wavelength to any port. In this manner, the board realizes full dynamic allocation of wavelengths.

The ROADM node comprising the WSD9 and WSM9 boards can be used at a central station, or an edge station. It features flexible expansion, no service interruption in expansion, and low operation cost. On the ROADM node, the network management software is used to allocate the add/drop and pass-through states of wavelengths and to adjust the states of wavelengths dynamically and remotely.

Functional Units

An ROADM node comprising the optical add/drop multiplexing boards or optical multiplexing and demultiplexing boards and the WSD9 and WSM9 boards consists of the following functional units:

- Optical transponder board (OTU)
- Optical amplifier board (OA)
- Bidirectional optical supervisory channel board (SC2/TC2/ST2)
- Fiber interface unit (FIU)
- Optical add/drop multiplexing board or optical multiplexing and demultiplexing board (OM/OD)
- Wavelength selective switching multiplexing board (WSM9)
- Wavelength selective switching demultiplexing board (WSD9)

For the boards used in each unit, refer to [4.4 Function Boards](#).

Signal Flow

- Two-dimensional grooming

The ROADM node processes the optical signals in two directions.

The optical supervisory signals and the main path signals are separated from the signals from the line. The optical supervisory signals are sent to the OSC board for processing. After being amplified, the main path signals are sent to the WSD9 board.

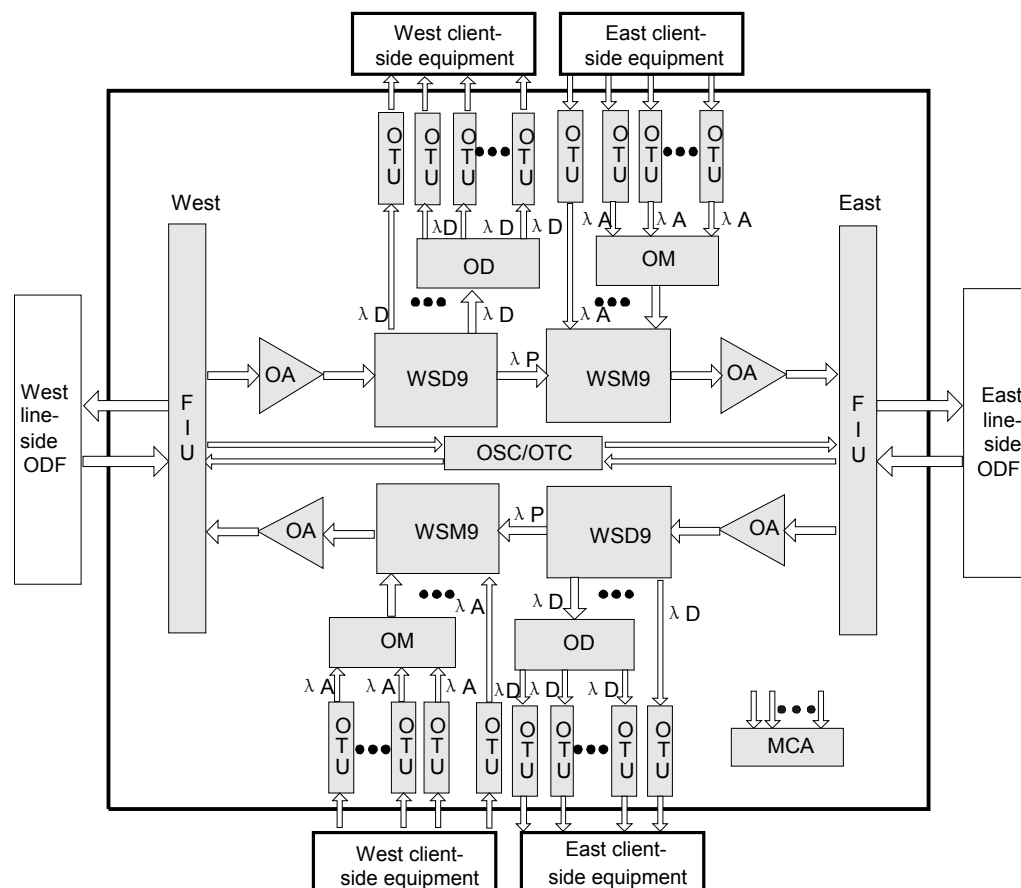
The wavelengths to be dropped are output from the specified port according to the configuration. In the case of the multiplexed signals, the signals are demultiplexed into individual wavelengths by the demultiplexing board or the optical add/drop multiplexing board. Then, the wavelengths are sent to the OTU and then to the local client-side equipment. In the case of a single wavelength, it can be directly sent to the OTU and then to the local client-side equipment.

The wavelengths where no add/drop multiplexing is required are pass-through wavelengths. They and the added wavelengths in the WSM9 board are multiplexed and then are amplified. Then, the signals are multiplexed with the processed optical supervisory signals. At last, the multiplexed signals are sent to the line for transmission.

For the application of the two-dimensional grooming ROADM node comprising the WSD9 and WSM9 boards and the networking signal flow, refer to [8.1.3 Intra-Ring Wavelength Grooming by WSD9 Boards and WSM9 Boards](#).

The functional modules of this ROADM node are shown in [Figure 6-14](#).

Figure 6-14 Schematic diagram of the Two-dimensional grooming ROADM node comprising the WSD9 and WSM9 boards



FIU: fiber interface unit

OA: optical amplifier board

OSC/OTC: optical supervisory channel board

OD: OADM or optical demultiplexer

OM: OADM or optical multiplexer

OTU: optical transponder board

WSD9: wavelength selective switching demultiplexing board

WSM9: wavelength selective switching multiplexing board

λP: Pass-through service

λA: Added service

λD: Dropped service

- Multi-dimensional grooming

One ROADM node that consists of more than two WSM9s and the same number of WSD9s provides multi-dimensional grooming. The signal grooming from west to east, south and north is considered an example. The signal grooming from east, south and north to the other three directions is the same as that from west to east, south and north.

[Figure 6-15](#) shows the functional block of the ROADM node.

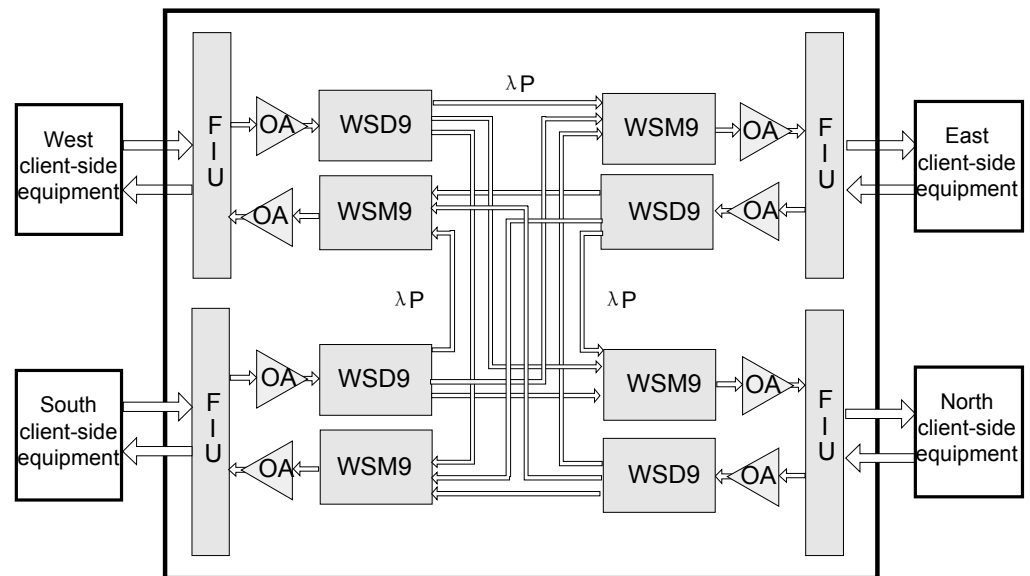
The ROADM node shown in **Figure 6-15** transmits the monitoring byte in ESC mode. Initially, the optical amplifier amplifies the line signals received from the west line side and sends them to the west WSD9.

The WSD9 splits the main path signals into up to eight equal optical signals. If the service signals need be output eastward, the signals from west is input through the east WSM9. This signals are multiplexed with the wavelengths groomed from other directions. The multiplexed wavelengths are amplified and are further sent to the line for transmission.

The signal flow of the service signals output southward or northward is the same as that eastward.

For the application of the multi-dimensional grooming ROADM node comprising the WSD9 and WSM9 boards and the networking signal flow, refer to **8.1.6 Inter-Ring Grooming by WSD9 Boards and WSM9 Boards**.

Figure 6-15 Schematic diagram of the Multi-dimensional grooming ROADM node comprising the WSD9 and WSM9 boards



FIU: fiber interface unit

OA: optical amplifier board

WSD9: wavelength selective switching demultiplexing board

WSM9: wavelength selective switching multiplexing board

λP: Pass-through service

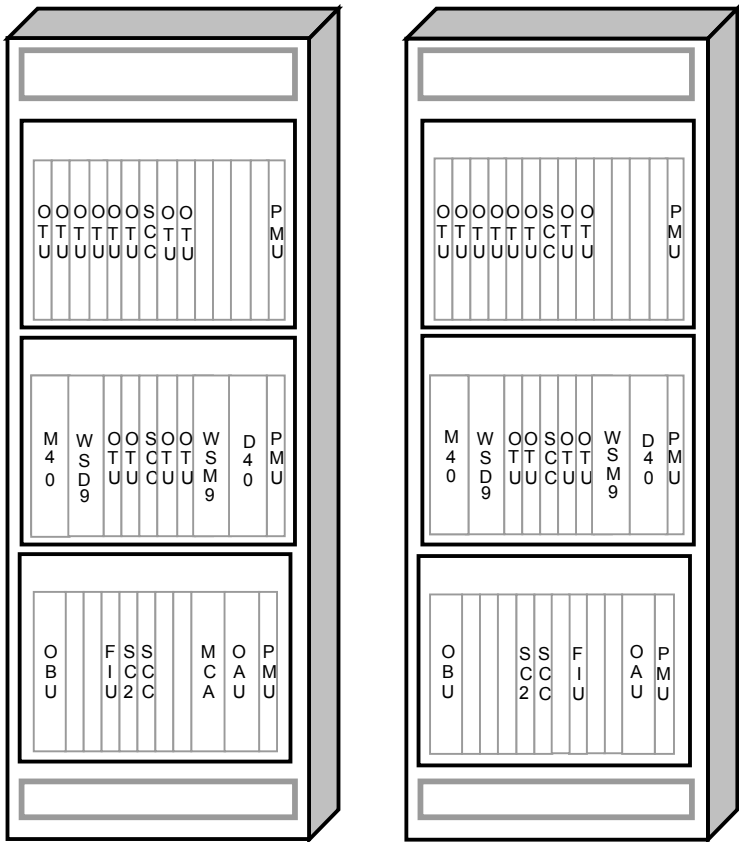
Typical Configuration

The 12-wavelength ROADM node with M40, D40 boards, WSD9 and WSM9 boards is taken as an example, as shown in **Figure 6-16**.

This ROADM node can add and drop 12 channels respectively in two transmission directions.

Six subracks and two cabinets are used.

Figure 6-16 Configuration diagram of the 12-wavelength OptiX Metro 6100 DWDM ROADM node with M40, D40 boards, WSD9 and WSM9 boards

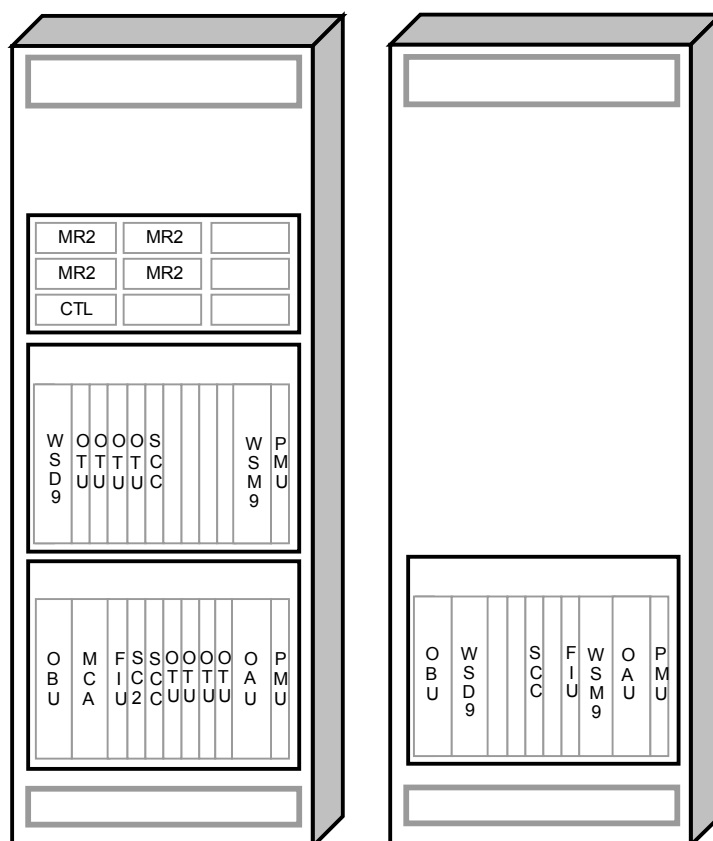


The four-wavelength ROADM node with OADM boards, WSD9 and WSM9 boards is taken as an example, as shown in [Figure 6-17](#).

This ROADM node can add and drop four channels respectively in two transmission directions.

Three subracks, one OADM frame and two cabinets are used.

Figure 6-17 Configuration diagram of the four-wavelength OptiX Metro 6100 DWDM ROADM node with OADM boards, WSD9 and WSM9 boards



Configuration Rules

The following are the rules for configuring the WSD9 and WSM9 boards:

- In the case of the node that requires multi-dimensional grooming in the future, the wavelength grooming scheme realized by the WSM9 and WSD9 boards is recommended.
- In the case of the node that requires the grooming in more than four dimensions between rings, the wavelength grooming scheme realized by the WSD9 and WSM9 boards is recommended. The WSM9 board can provide the wavelength selection function.
- IU2 and IU13 in the lower subrack are preferred to house the WSD9 and WSM9. If there are no enough idle slots, install them into IU2 and IU13 in the middle subrack.

The following are the rules for configuring the M40/V40 and D40 boards:

- When the number of wavelengths for the accessed services is larger than 16, use the M40/V40 and D40 boards.
- The M01–M40 optical interfaces on the M40/V40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1THz to 196.0THz.
- The D01–D40 optical interfaces on the D40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1 THz to 196.0 THz.

The following are the rules for configuring the OADM frame:

- When the number of added/dropped wavelengths is smaller than 16, use the OADM board.
- Normally, the OADM frame is located in the middle or upper subrack position in a cabinet. In addition, it is suggested to make the OADM frame located in the same cabinet as the subrack housing the corresponding OTU board.
- When one OADM frame is to be configured and the number of OADM boards is smaller than 6, IU15–IU17 and IU21 are defined as west, and IU18–IU20 and IU22 are defined as east. IU21 is for the west EFIU. IU22 is for the east EFIU. IU15–IU17 are for the west OADM boards. IU18–IU20 are for the east OADM boards.
- It is suggested to make the OADM frame located in the same cabinet as the subrack housing the corresponding OTU board.
- To decrease the insertion loss, drop 10 Gbit/s services first and then 2.5 Gbit/s services.
- The MR2 or MR4 board can be configured as an OTM.
- Install the MR4 or MR2 board into IU15–IU22 in the OADM frame in an ascending order of frequency. The optical interfaces of the boards are arranged in an ascending order of frequency.
- When the MR2 boards are configured as an OTM, set the maximum number of wavelengths in one OADM frame to 12. If more wavelengths are to be added/dropped, add another OADM frame.
- If one OADM subrack corresponds to only one direction, IU21 is preferred to house the EFIU.

The following are the rules for configuring the OTU boards:

- The OTU board with a smaller frequency is configured first. When there are multiple subracks, the OTU board is configured in a lower subrack first. When there is only one subrack, the OTU board is configured in the left slot first.
- The 40G OTU must be inserted into the middle subrack and lower subrack.
- At ROADM station comprising the WSD9 and WSM9 boards, the east-west separation principle is not adopted. That is, the east and west are in the same subrack. On the west, install the boards into IU1–IU6 from left to right. On the east, install the boards into IU8–IU13 from left to right.
- In a station where GE ADM is configured, the east-west separation principle is adopted. The OTUs in different directions and with mutual cross-connections must be installed in paired slots or on the same cross-connect plane in the same subrack.
- In the client-side 1+1 protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUS from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the inter-board wavelength protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUS from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the extended intra-board wavelength protection mode, the OTU and OLP boards must be configured in the same subrack and the OLP must be just next to the OTU.
- In the WXCP protection mode, the working OTU and the protection OTU must be installed on the same cross-connect plane or in paired cross-connect slots. On the same cross-connect plane, install the OTUS from left to right following the sequence of west OTU1, east OTU1, west OTU2, and east OTU2. When the OTUs are located in paired cross-connect slots, west

OTU1 and west OTU2, and east OTU1 and east OTU2 in paired cross-connect slots are installed from left to right.

- When the DPPS and TPS are configured, the working and protection OTUs in DPPS protection and the active and standby TBEs in TPS protection must be located on the same cross-connect plane. They must be installed from left to right in a sequence that is the same as the OTUs in WXCP protection.

The following is the rule for configuring the SCC boards:

- The SCC board is installed in IU7 in each subrack.

The following is the rule for configuring the PMU boards:

- The PMU board is installed in IU14 in each subrack.

The following is the rule for configuring the supervisory channel boards:

- IU6 is preferred to house the board. If IU6 houses another board, install the supervisory channel board into IU8.

The following are the rules for configuring the amplifier boards:

- Install the FIU boards into IU5 and IU9, and then install optical amplifier boards one by one.
- The west and east optical amplifier boards (the OAU, OBU and OPU) are installed on the left and right sides of the subrack respectively.
- When there are multiple optical amplifier boards in a service flow, install west optical amplifier boards on the left side of the subrack and install them from right to left along the service flow. Install east optical amplifier boards on the right side of the subrack and install them from left to right along the service flow.
- When a Raman amplifier is used, the Raman amplifier must be installed in the lower subrack where the optical amplifier board in the same direction is located.

The following are the rules for configuring the protection boards:

- When configuring the optical line protection, make the OLP board located close to the FIU board.
- When configuring the intra-board wavelength protection, configure the OTU board with dual-fed and selective receiving function for 2.5 Gbit/s services. In the case of 5 Gbit/s and 10 Gbit/s services, adopt the extended intra-board wavelength protection and configure the OLP board. In the case of 40 Gbit/s services, adopt the extended intra-board wavelength protection and configure the CP40 board.
- When configuring the client-side 1+1 wavelength protection, use the SCS board when the working and protection OTUs are in the same subrack. When the client-side 1+1 wavelength protection is configured in the system, the working and protection channels take different directions and are configured into a ring network.
- When configuring the inter-subrack wavelength protection, configure the DCP or OLP board as the dual-fed and selective receiving unit.

6.4.3 ROADM Node with WSD9 Board and RMU9 Board

The ROADM node comprising the WSD9 and RMU9 boards realizes the full dynamic adding/dropping of wavelengths in a ring. The inter-ring extension is supported. The extension of a maximum of eight dimensions is supported.

Functions

The WSD9 board is used to demultiplex any wavelength and allocate it to any port. The demultiplexing is dynamic and configurable. On a node in a ring or chain, the board is able to combine any wavelengths in the line and output the multiplexed wavelength. In addition, the board is able to allocate the output wavelength to any port. In this manner, the board realizes full dynamic allocation of wavelengths.

The RMU9 board is used to add wavelengths. The wavelength adding port can work with the wavelength tunable OTU board, to realize the full dynamic input of eight wavelengths. Each of the wavelength adding ports on the RMU9 board can be connected to a multiplexing board. In this manner, after being multiplexed by the multiplexing board, client-side signals are input through the wavelength adding port on the RMU9 board.

The ROADM node comprising the WSD9 and RMU9 boards can be used at a central station, or an edge station. It features flexible expansion, no service interruption in expansion, and low operation cost. On the ROADM node, the network management software is used to allocate the add/drop and pass-through states of wavelengths and to adjust the states of wavelengths dynamically and remotely.

Functional Units

An ROADM node comprising the optical add/drop multiplexing boards or optical multiplexing and demultiplexing boards and the WSD9 and RMU9 boards consists of the following functional units:

- Optical transponder board (OTU)
- Optical amplifier board (OA)
- Bidirectional optical supervisory channel board (SC2/TC2/ST2)
- Fiber interface unit (FIU)
- Optical add/drop multiplexing board or optical multiplexing and demultiplexing board (OM/OD)
- ROADM multiplexing board (RMU9)
- Wavelength selective switching demultiplexing board (WSD9)

For the boards used in each unit, refer to [4.4 Function Boards](#).

Signal Flow

- Two-dimensional grooming

The ROADM node processes the optical signals in two directions.

The optical supervisory signals and the main path signals are separated from the signals from the line. The optical supervisory signals are sent to the OSC board for processing. After being amplified, the main path signals are sent to the WSD9 board.

The wavelengths to be dropped are output from the specified port according to the configuration. In the case of the multiplexed signals, the signals are demultiplexed into individual wavelengths by the demultiplexing board or the optical add/drop multiplexing board. Then, the wavelengths are sent to the OTU and then to the local client-side equipment. In the case of a single wavelength, it can be directly sent to the local client-side equipment.

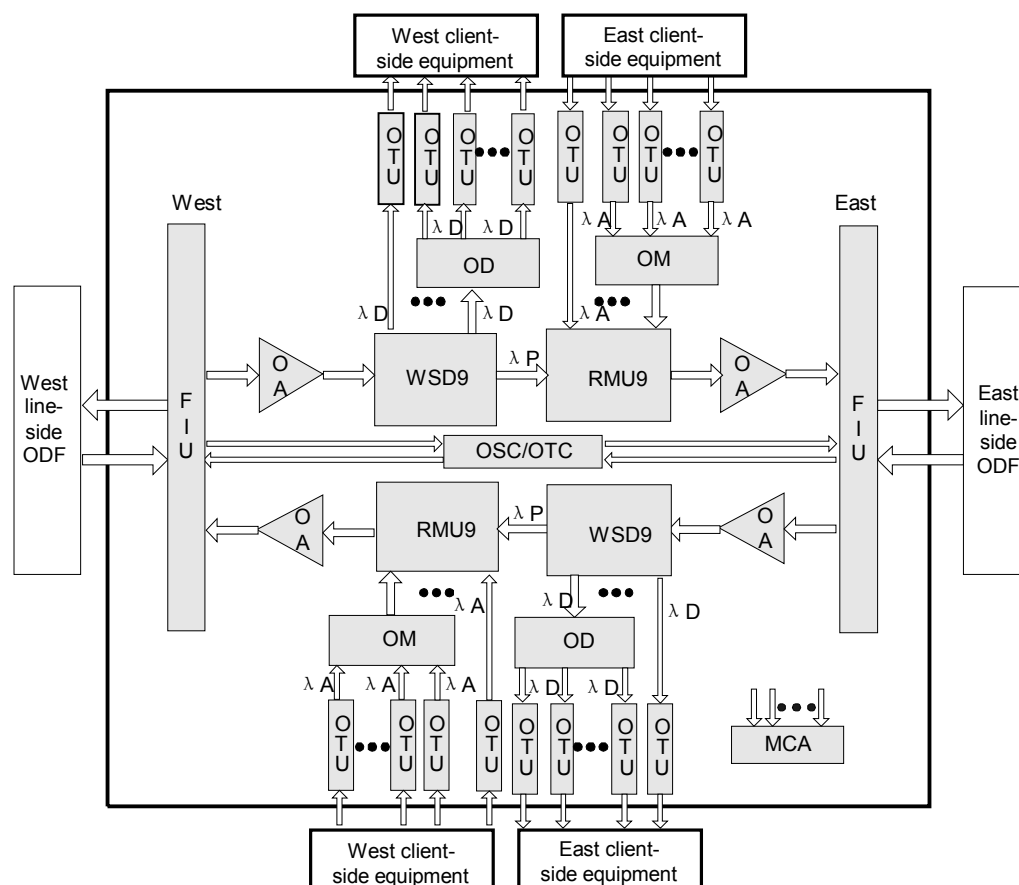
The wavelengths where no add/drop multiplexing is required are pass-through wavelengths. They and the added wavelengths in the RMU9 board are multiplexed and

then are amplified. Then, the signals are multiplexed with the processed optical supervisory signals. At last, the multiplexed signals are sent to the line for transmission.

For the application of the two-dimensional grooming ROADM node comprising the RMU9 and WSD9 boards and the networking signal flow, refer to [8.1.2 Intra-Ring Wavelength Grooming by WSD9 Boards and RMU9 Boards](#).

The functional modules of this ROADM node are shown in [Figure 6-18](#).

Figure 6-18 Schematic diagram of the two-dimensional grooming ROADM node constituted by the RMU9 and WSD9 boards



FIU: fiber interface unit

OA: optical amplifier board

OSC/OTC: optical supervisory channel board

OD: optical demultiplexer

OTU: optical transponder board WSD9: wavelength selective switching demultiplexing board

RMU9: ROADM multiplexing board OM: optical multiplexing board

λp : pass-through wavelength

λA : added wavelength

λD : dropped wavelength

- Multi-dimensional grooming

One ROADM node that consists of more than two RMU9s and the same number of WSD9s provides multi-dimensional grooming. This kind of ROADM node always owns a large number of the signals required to be groomed. But its way of signal grooming is the same

with the multi-dimensional grooming ROADM contributed by the WSM9 and WSD9 boards.

The combination of the WSD9 and WSM9 boards or the combination of the WSD9 and RMU9 boards can be chosen to form the ROADM station. The structure in two modes is the same.

For the application of the multi-dimensional grooming ROADM node comprising the RMU9 and WSD9 boards and the networking signal flow, refer to [8.1.5 Inter-Ring Grooming by WSD9 Boards and RMU9 Boards](#).

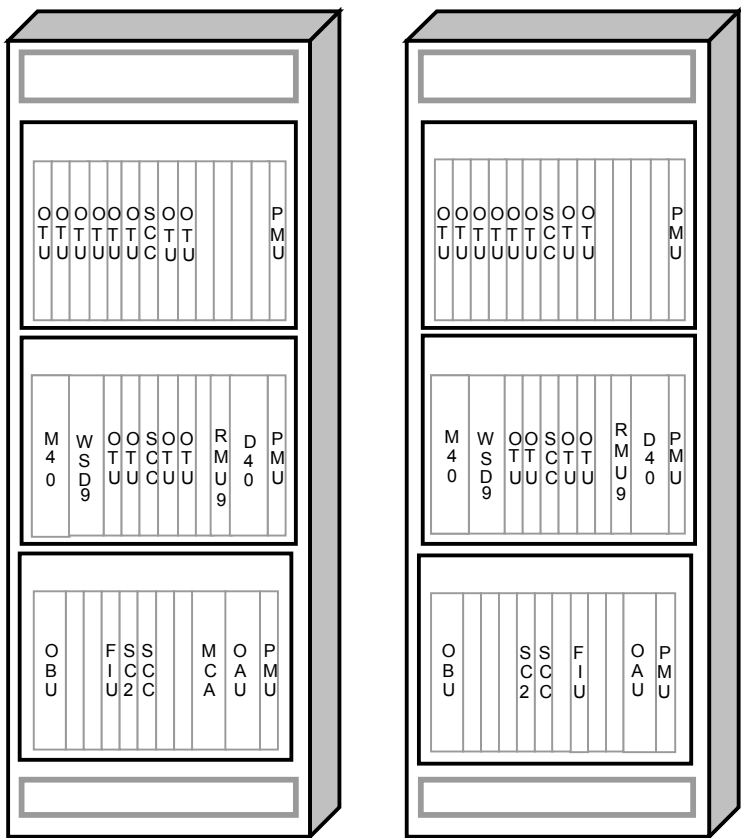
Typical Configuration

The 12-wavelength ROADM node with M40, D40 boards, RMU9 and WSD9 boards is taken as an example. This ROADM node can add and drop 12 channels respectively in two transmission directions.

The typical configuration of the OptiX Metro 6100 DWDM system is shown in [Figure 6-19](#).

Two cabinets and six subracks are used.

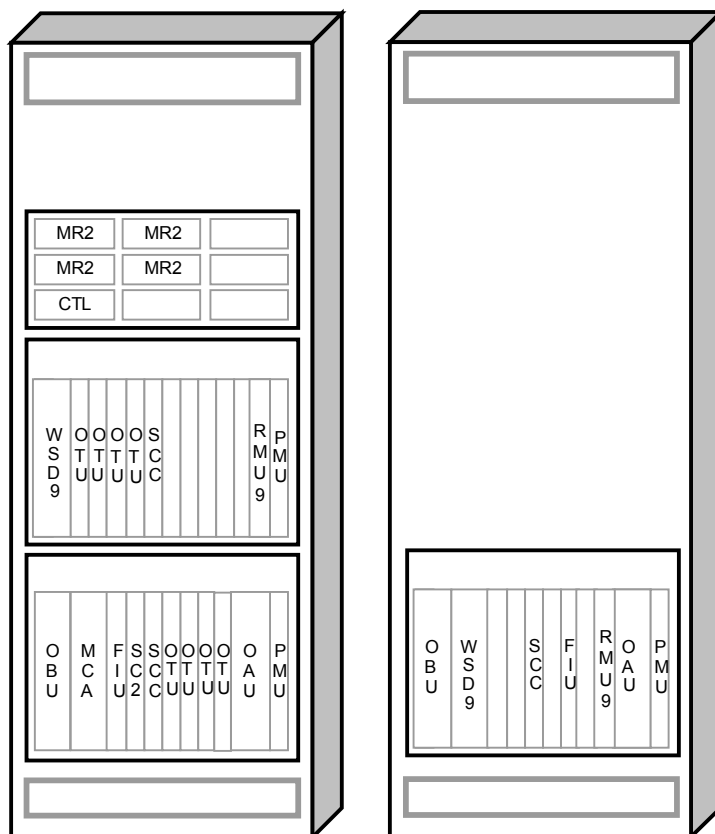
Figure 6-19 Configuration diagram of the 12-wavelength OptiX Metro 6100 DWDM ROADM node with M40, D40 boards, RMU9 and WSD9 boards



The four-wavelength ROADM node with OADM boards, RMU9 and WSD9 boards is taken as an example. This ROADM node can add and drop four channels respectively in two transmission directions.

Two cabinets, three subracks and one OADM frame are used.

Figure 6-20 Configuration diagram of the four-wavelength OptiX Metro 6100 DWDM ROADM node with OADM boards, RMU9 and WSD9 boards



Configuration Rules

The following are the rules for configuring the WSD9 and RMU9 boards:

- In the case of the node that requires no relation between add/drop wavelengths and optical interfaces in a ring, the wavelength grooming scheme realized by the WSD9 and RMU9 boards is recommended.
- In the case of the node that requires the grooming in less than four dimensions between rings, the wavelength grooming scheme realized by the WSD9 and RMU9 boards is recommended. The inter-ring ROADM node comprising the WSD9 and RMU9 boards decrease the networking cost. Compared with the WSM9 board, however, the RMU9 board does not provide the wavelength selection function.
- The lower subrack is preferred to house the RMU9 and WSD9 boards. If there are no enough idle slots, install them into the middle subrack
- Install the RMU9 board into IU1 or IU13, and the WSD9 board into IU13 or IU2.

The following are the rules for configuring the M40/V40 and D40 boards:

- When the number of wavelengths for the accessed services is larger than 16, use the M40/V40 and D40 boards.
- The M01–M40 optical interfaces on the M40/V40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1THz to 196.0THz.
- The D01–D40 optical interfaces on the D40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1 THz to 196.0 THz.

The following are the rules for configuring the OADM frame:

- When the number of added/dropped wavelengths is smaller than 16, use the OADM board.
- Normally, the OADM frame is located in the middle or upper subrack position in a cabinet. In addition, it is suggested to make the OADM frame located in the same cabinet as the subrack housing the corresponding OTU board.
- When one OADM frame is to be configured and the number of OADM boards is smaller than 6, IU15–IU17 and IU21 are defined as west, and IU18–IU20 and IU22 are defined as east. IU21 is for the west EFIU. IU22 is for the east EFIU. IU15–IU17 are for the west OADM boards. IU18–IU20 are for the east OADM boards.
- It is suggested to make the OADM frame located in the same cabinet as the subrack housing the corresponding OTU board.
- To decrease the insertion loss, drop 10 Gbit/s services first and then 2.5 Gbit/s services.
- The MR2 or MR4 board can be configured as an OTM.
- Install the MR4 or MR2 boards into IU15–IU22 in the OADM frame in an ascending order of frequency. The optical interfaces of the boards are arranged in an ascending order of frequency.
- When the MR2 boards are configured as an OTM, set the maximum number of wavelengths in one OADM frame to 12. If more wavelengths are to be added/dropped, add another OADM frame.
- If one OADM subrack corresponds to only one direction, IU21 is preferred to house the EFIU.

The following are the rules for configuring the OTU boards:

- The OTU board with a smaller frequency is configured first. When there are multiple subracks, the OTU board is configured in a lower subrack first. When there is only one subrack, the OTU board is configured in the left slot first.
- The 40G OTU must be inserted into the middle subrack and lower subrack.
- At ROADM station comprising the RMU9 and WSD9 boards, the east-west separation is not adopted. That is, the east and west are in the same subrack. On the west, install the boards into IU1–IU6 from left to right. On the east, install the boards into IU8–IU13 from left to right.
- In a station where GE ADM is configured, the east-west separation principle is adopted. The OTUs in different directions and with mutual cross-connections must be installed in paired slots or on the same cross-connect plane in the same subrack.
- In the client-side 1+1 protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUS from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the inter-board wavelength protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUS from

left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.

- In the extended intra-board wavelength protection mode, the OTU and OLP boards must be configured in the same subrack and the OLP must be just next to the OTU.
- In the WXCP protection mode, the working OTU and the protection OTU must be installed on the same cross-connect plane or in paired cross-connect slots. On the same cross-connect plane, install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, and east OTU2. When the OTUs are located in paired cross-connect slots, west OTU1 and west OTU2, and east OTU1 and east OTU2 in paired cross-connect slots are installed from left to right.
- When the DPPS and TPS are configured, the working and protection OTUs in DPPS protection and the active and standby TBEs in TPS protection must be located on the same cross-connect plane. They must be installed from left to right in a sequence that is the same as the OTUs in WXCP protection.

The following is the rule for configuring the SCC boards:

- The SCC board is installed in IU7 in each subrack.

The following is the rule for configuring the PMU boards:

- The PMU board is installed in IU14 in each subrack.

The following is the rule for configuring the supervisory channel boards:

- IU6 is preferred to house the board. If IU6 houses another board, install the supervisory channel board into IU8.

The following are the rules for configuring the amplifier boards:

- Install the FIU boards into IU5 and IU9, and then install optical amplifier boards one by one.
- The west and east optical amplifier boards (the OAU, OBU and OPU) are installed on the left and right sides of the subrack respectively.
- When there are multiple optical amplifier boards in a service flow, install west optical amplifier boards on the left side of the subrack and install them from right to left along the service flow. Install east optical amplifier boards on the right side of the subrack and install them from left to right along the service flow.
- When a Raman amplifier is used, the Raman amplifier must be installed in the lower subrack where the optical amplifier board in the same direction is located.

The following are the rules for configuring the protection boards:

- When configuring the optical line protection, make the OLP board located close to the FIU board.
- When configuring the intra-board wavelength protection, configure the OTU board with dual-fed and selective receiving function for 2.5 Gbit/s services. In the case of 5 Gbit/s and 10 Gbit/s services, adopt the extended intra-board wavelength protection and configure the OLP board. In the case of 40 Gbit/s services, adopt the extended intra-board wavelength protection and configure the CP40 board.
- When configuring the client-side 1+1 wavelength protection, use the SCS board when the working and protection OTUs are in the same subrack. When the client-side 1+1 wavelength protection is configured in the system, the working and protection channels take different directions and are configured into a ring network.

- When configuring the inter-subrack wavelength protection, configure the DCP or OLP board as the dual-fed and selective receiving unit.

6.4.4 ROADM Node with WSMD4 Boards

The product can dynamically add and drop wavelengths in a ring network by using the multi-dimensional ROADM function that multiple WSMD4s provide. The system also supports inter-ring expansion to provide the wavelength grooming of a maximum of four dimensions.

Functions

The WSMD4 is used to demultiplex any dynamically configurable wavelengths to any ports. In a ring or chain network, any wavelength locally added from any node can be input through any port, and any wavelength can be dropped through any port. In this way, a completely dynamic allocation of wavelengths is achieved.

The ROADM node formed by WSMD4s can serve as a central node or an edge node. The ROADM station supports the flexible and easy expansion without interrupting services and has a low cost for operation and maintenance. The add, drop or pass-through state of wavelengths can be remotely and dynamically adjusted on the T2000.

Functional Units

An ROADM node comprising the optical add/drop multiplexing boards or optical multiplexing and demultiplexing boards and the WSMD4 boards consists of the following functional units:

- Optical transponder board (OTU)
- Optical amplifier board (OA)
- Bidirectional optical supervisory channel board (SC2/TC2/ST2)
- Fiber interface unit (FIU)
- Optical add/drop multiplexing board or optical multiplexing and demultiplexing board (OM/OD)
- 4-port wavelength selective switching demultiplexing/multiplexing board (WSMD4)

For the boards used in each unit, refer to [4.4 Function Boards](#).

Signal Flow

- Two-dimensional grooming

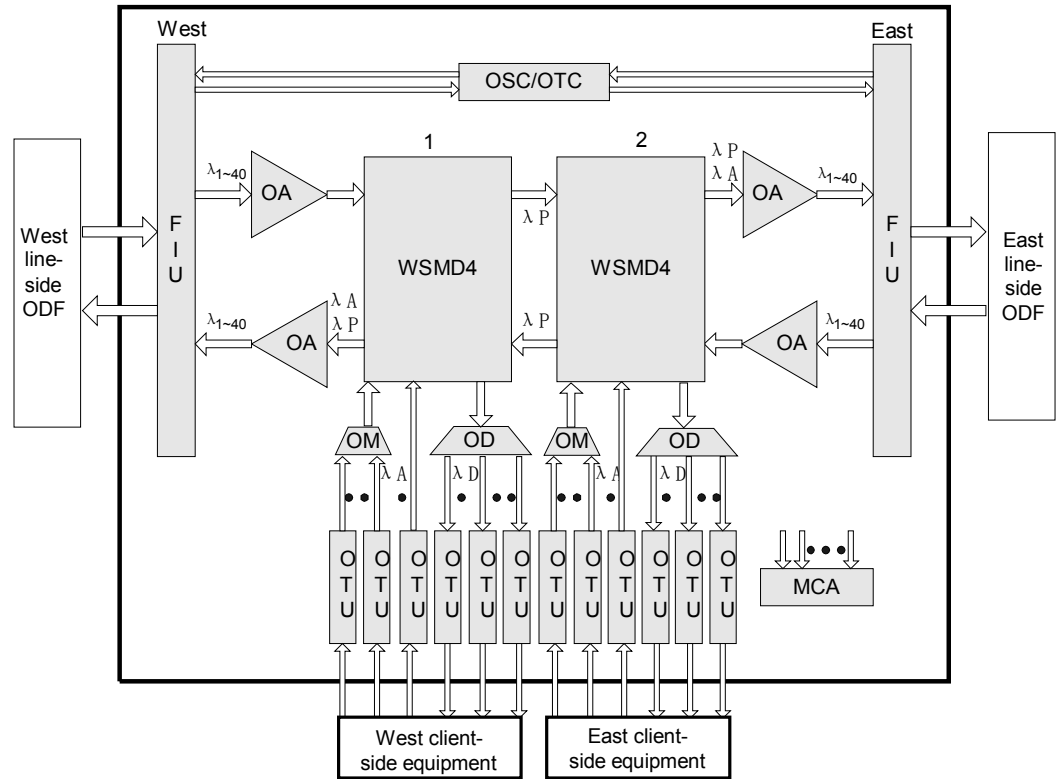
One ROADM node consists of two WSMD4s.

Figure 6-21 shows the functional block of the ROADM node.

The ROADM node processes optical signals in two transmission directions. Initially, optical supervisory signal and optical signal in the main channel are separated from the line signal received. The optical supervisory signal is sent to the optical supervisory unit for processing, and the optical signal in the main channel is sent to the WSMD4 after being amplified. Some wavelengths are demultiplexed by the demultiplexing board or OADM board, sent to the OTU, and then sent to the local client equipment. Other wavelengths pass through without being locally multiplexed or demultiplexed. They are multiplexed with the wavelengths that are locally added, and the multiplexed wavelength is amplified. Finally, those wavelengths are multiplexed with the optical supervisory signal that is processed and sent to the line for transmission.

For the application of the two-dimensional grooming ROADM node comprising the WSMD4 boards and the networking signal flow, refer to [8.1.4 Intra-Ring Wavelength Grooming by WSMD4 Boards](#).

Figure 6-21 Schematic diagram of the two-dimensional grooming ROADM node constituted by two WSMD4s boards



FIU: fiber interface unit

OA: optical amplifier board

OSC/OTC: optical supervisory channel board

OD: optical demultiplexer

OTU: optical transponder board OM: optical multiplexing board

WSMD4: 4-port wavelength selective switching demultiplexing/multiplexing board

λ_P : pass-through wavelength

λ_A : added wavelength

λ_D : dropped wavelength

- Multi-dimensional grooming

One ROADM node that consists of more than two WSMD4s provides multi-dimensional grooming. The signal grooming from west to east, south and north is considered an example. The signal grooming from east, south and north to the other three directions is the same as that from west to east, south and north.

Figure 6-22 shows the functional block of the ROADM node.

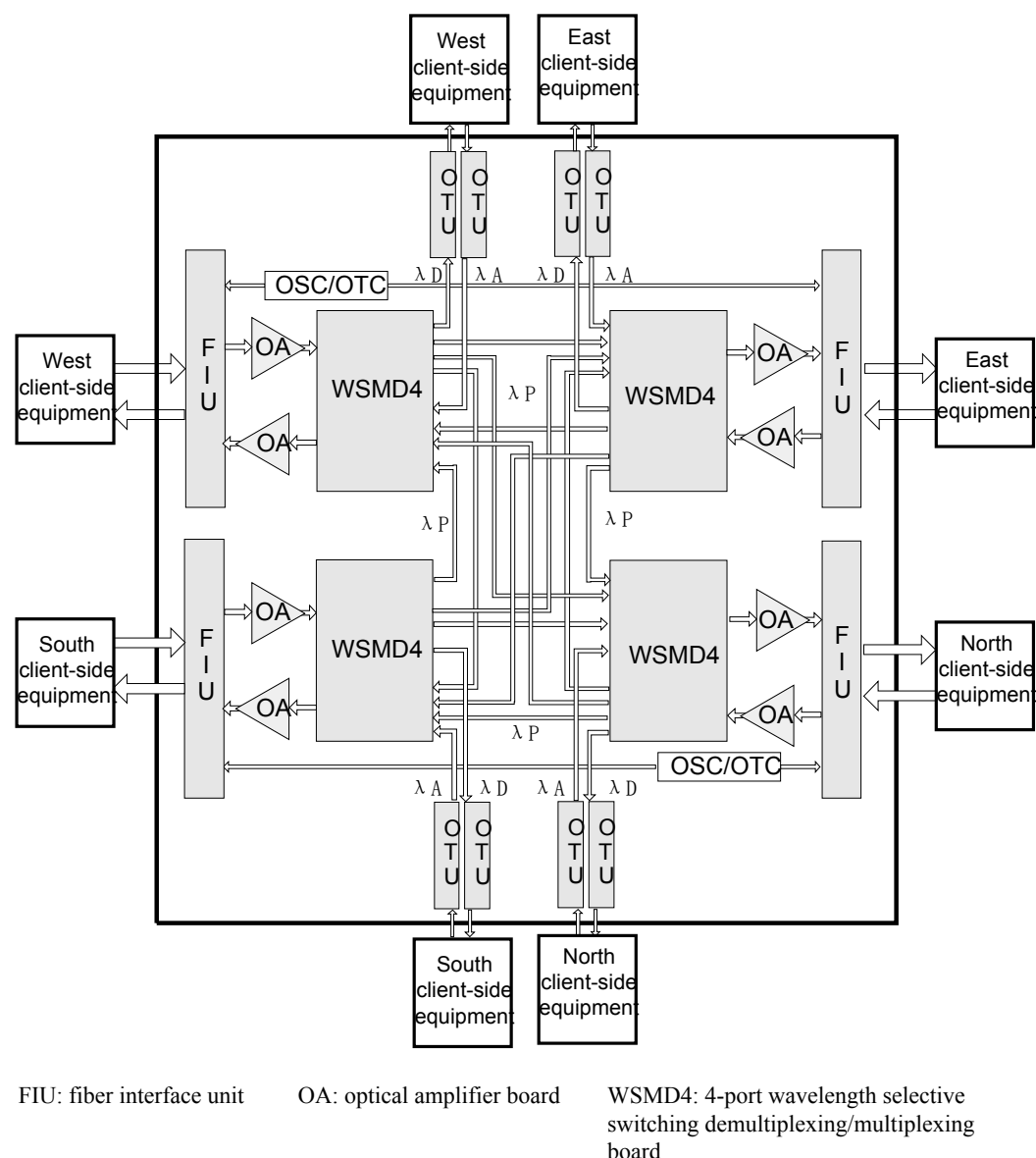
Initially, the optical supervisory signal and optical signal in the main channel are separated from the line signal received. The optical supervisory signal is sent to the optical supervisory unit for processing, and the optical signal in the main channel is sent to the WSMD4 after being amplified.

The WSMD4 splits the main path signals into four equal optical signals. The optical demultiplexing board demultiplexes the single wavelengths that need be output from the local station. If the service signals need be output eastward, the signals from west is input through the east WSMD4. The added wavelengths that need be output eastward are added through an input port selected by the optical switch on the east WSMD4. The added wavelengths are multiplexed with the wavelengths groomed from west. Finally, those wavelengths are multiplexed with the optical supervisory signal that is processed and sent to the line for transmission.

The signal flow of the service signals output southward or northward is the same as that eastward.

For the application of the multi-dimensional grooming ROADM node comprising the WSMD4 boards and the networking signal flow, refer to [8.1.7 Inter-Ring Grooming by WSMD4 Boards](#).

Figure 6-22 Schematic diagram of the multi-dimensional grooming ROADM node constituted by WSMD4s boards



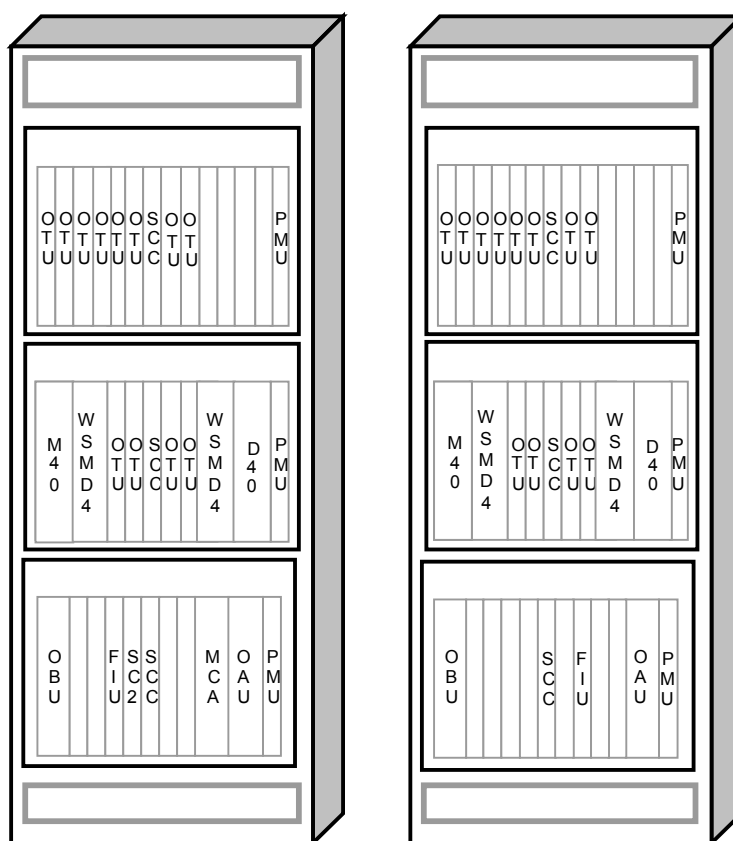
OD: optical demultiplexer OTU: optical transponder board OM: optical multiplexing board

λ_p : pass-through wavelength λ_A : added wavelength λ_D : dropped wavelength

Typical Configuration

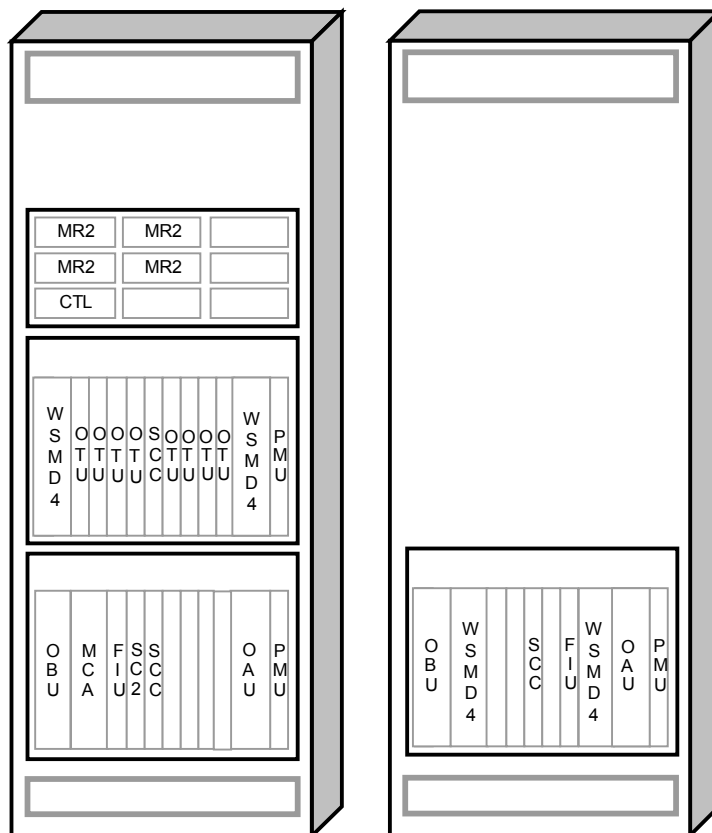
The 12-channel ROADM node formed by the M40 (or V40), D40 and WSMD4 boards is taken as an example. This ROADM node can add and drop 12 channels respectively in two transmission directions. **Figure 6-23** shows the typical configuration of the OptiX Metro 6100 DWDM system with the 12-channel ROADM node. Two cabinets and six subracks are used in the system.

Figure 6-23 Configuration diagram of the four-wavelength OptiX Metro 6100 DWDM ROADM node with the M40 (or V40), D40 and WSMD4 boards



The four-channel ROADM node formed by the OADM and WSMD4 boards is considered as an example. This ROADM node can add and drop four channels respectively in two transmission directions. **Figure 6-24** shows the typical configuration of the OptiX Metro 6100 DWDM system with the four-channel ROADM node. Two cabinets, three subracks and one OADM frame are used in the system.

Figure 6-24 Configuration diagram of the four-wavelength OptiX Metro 6100 DWDM ROADM node with the OADM and WSMD4 boards



Configuration Rules

The following rules for configuring the WSMD4 are as follows:

- At a node, in a ring network that has less than four grooming dimensions, where wavelengths need be dynamically added or dropped, it is recommended to use the wavelength grooming scheme realized by using the WSMD4.
- The lower subrack is preferred for holding the WSMD4. If the lower subrack has no enough slots, the WSMD4 can be configured in the middle subrack.
- It is recommended to configure a pair of WSMD4s in slots IU1 and IU12 of the lower subrack. If the lower subrack has no enough slots, the WSMD4 pair can be configured in slots IU1 and IU12 of the middle subrack.

The following are the rules for configuring the M40/V40 and D40 boards:

- When the number of wavelengths for the accessed services is larger than 16, use the M40/V40 and D40 boards.
- The M01–M40 optical interfaces on the M40/V40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1THz to 196.0THz.
- The D01–D40 optical interfaces on the D40 are arranged in an ascending order of frequency. The frequency of the interfaces is increased from 192.1 THz to 196.0 THz.

The following are the rules for configuring the OADM frame:

- When the number of added/dropped wavelengths is smaller than 16, use the OADM board.
- Normally, the OADM frame is located in the middle or upper subrack position in a cabinet. In addition, it is suggested to make the OADM frame located in the same cabinet as the subrack housing the corresponding OTU board.
- When one OADM frame is to be configured and the number of OADM boards is smaller than 6, IU15–IU17 and IU21 are defined as west, and IU18–IU20 and IU22 are defined as east. IU21 is for the west EFIU. IU22 is for the east EFIU. IU15–IU17 are for the west OADM boards. IU18–IU20 are for the east OADM boards.
- It is suggested to make the OADM frame located in the same cabinet as the subrack housing the corresponding OTU board.
- To decrease the insertion loss, drop 10 Gbit/s services first and then 2.5 Gbit/s services.
- The MR2 or MR4 board can be configured as an OTM.
- Install the MR4 or MR2 boards into IU15–IU22 in the OADM frame in an ascending order of frequency. The optical interfaces of the boards are arranged in an ascending order of frequency.
- When the MR2 boards are configured as an OTM, set the maximum number of wavelengths in one OADM frame to 12. If more wavelengths are to be added/dropped, add another OADM frame.
- If one OADM subrack corresponds to only one direction, IU21 is preferred to house the EFIU.

The following are the rules for configuring the OTU boards:

- The OTU board with a lower frequency is configured first. When there are multiple subracks, the OTU board is configured in a lower subrack first. When there is only one subrack, the OTU board is configured in the left slot first.
- The 40G OTU must be inserted into the middle subrack and lower subrack.
- At the ROADM station comprising the WSMD4 boards, the east-west separation is not adopted. That is, the east and west are in the same subrack. On the west, install the boards into IU1–IU6 from left to right. On the east, install the boards into IU8–IU13 from left to right.
- In a station where GE ADM is configured, the east-west separation principle is adopted. The OTUs in different directions and with mutual cross-connections must be installed in paired slots or on the same cross-connect plane in the same subrack.
- In the client-side 1+1 protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the inter-board wavelength protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the extended intra-board wavelength protection mode, the OTU and OLP boards must be configured in the same subrack and the OLP must be just next to the OTU.
- In the WXCP protection mode, the working OTU and the protection OTU must be installed on the same cross-connect plane or in paired cross-connect slots. On the same cross-connect plane, install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, and east OTU2. When the OTUs are located in paired cross-connect slots, west

OTU1 and west OTU2, and east OTU1 and east OTU2 in paired cross-connect slots are installed from left to right.

- When the DPPS and TPS are configured, the working and protection OTUs in DPPS protection and the active and standby TBEs in TPS protection must be located on the same cross-connect plane. They must be installed from left to right in a sequence that is the same as the OTUs in WXCP protection.

The following is the rule for configuring the SCC boards:

- The SCC board is installed in IU7 in each subrack.

The following is the rule for configuring the PMU boards:

- The PMU board is installed in IU14 in each subrack.

The following is the rule for configuring the supervisory channel boards:

- IU6 is preferred to house the board. If IU6 houses another board, install the supervisory channel board into IU8.

The following are the rules for configuring the amplifier boards:

- Install the FIU boards into IU5 and IU9, and then install optical amplifier boards one by one.
- The west and east optical amplifier boards (the OAU, OBU and OPU) are installed on the left and right sides of the subrack respectively.
- When there are multiple optical amplifier boards in a service flow, install west optical amplifier boards on the left side of the subrack and install them from right to left along the service flow. Install east optical amplifier boards on the right side of the subrack and install them from left to right along the service flow.
- When a Raman amplifier is used, the Raman amplifier must be installed in the lower subrack where the optical amplifier board in the same direction is located.

The following are the rules for configuring the protection boards:

- When configuring the optical line protection, make the OLP board located close to the FIU board.
- When configuring the intra-board wavelength protection, configure the OTU board with dual-fed and selective receiving function for 2.5 Gbit/s services. In the case of 5 Gbit/s and 10 Gbit/s services, adopt the extended intra-board wavelength protection and configure the OLP board. In the case of 40 Gbit/s services, adopt the extended intra-board wavelength protection and configure the CP40 board.
- When configuring the client-side 1+1 wavelength protection, use the SCS board when the working and protection OTUs are in the same subrack. When the client-side 1+1 wavelength protection is configured in the system, the working and protection channels take different directions and are configured into a ring network.
- When configuring the inter-subrack wavelength protection, configure the DCP or OLP board as the dual-fed and selective receiving unit.

6.5 REG

The REG equipment is an electrical regenerator and is used to further extend the optical transmission distance.

Functions

We have already discussed that the OLA can extend the optical transmission distance without regeneration. However, when the distance is longer, such factors as dispersion, optical noise, non-linear effect, or PMD will affect the transmission performance. In this case, we need to regenerate the original signals. An REG implements the 3R function: reshaping, re-timing and regenerating. This is to improve the signal quality and to extend the transmission distance.

Functional Units

An OptiX Metro 6100 REG node consists of the following functional units:

- Optical transponder board (OTU)
- Optical amplifier board (OA)
- Optical multiplexing board (M40)
- Optical demultiplexing board (D40)
- Bidirectional OSC board (SC2/TC2/ST2)
- Fiber interface unit (FIU)

For the system where the electrical regeneration is needed, each functional unit of the systems consists of boards of different types or a combination of the boards of the same type.

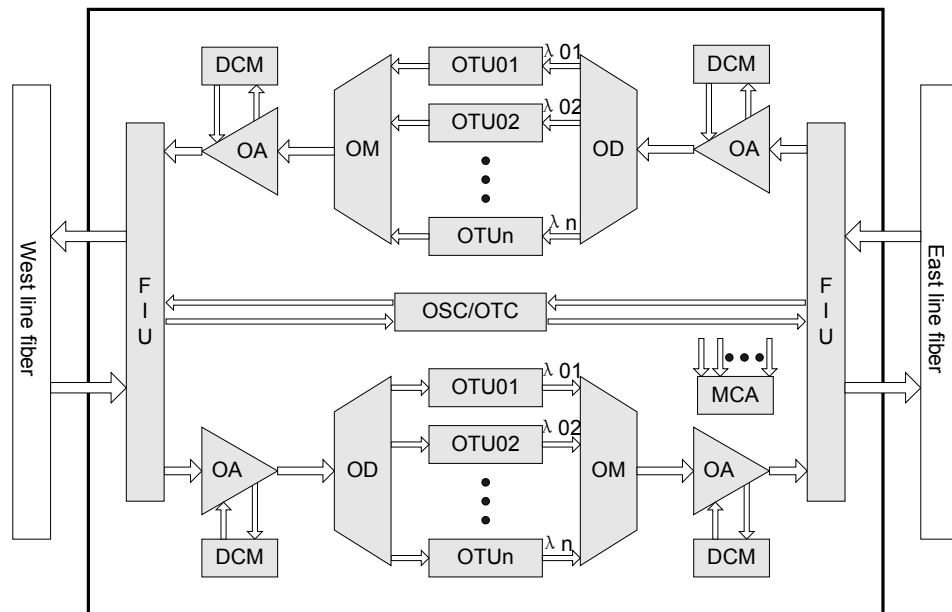
The structure of the OM, OD, and OA of the system is the same as that of the OTM equipment.

For the boards used in each unit, refer to [4.4 Function Boards](#).

Signal Flow

[Figure 6-25](#) shows the block diagram of the REG signal flow.

Figure 6-25 Schematic diagram of the OptiX Metro 6100 REG node



OTU: optical transponder board

OM: optical multiplexing board

OD: optical demultiplexing board	OSC/OTC: optical supervisory channel board
FIU: fiber interface unit	OA: optical amplifier board
MCA: multi-channel spectrum analyzer board	DCM: dispersion compensation module

The signal flow of the REG is similar to that of back-to-back OTMs, except that no signal is added/dropped. Signals are regenerated through the regenerating OTU.

Typical Configuration

The configuration of the REG is basically equivalent to that of two back-to-back OTMs, following the same configuration rule.

Difference:

- The REG needs to be configured with a bidirectional OSC/OTC.
- The REG needs to be configured with two FIU boards.
- The REG needs the regenerating OTU.

The configuration of the REG application is the same as that shown in [Figure 6-2](#).

Configuration Rules

The configuration principle of the REG is the same as that of the OTM.

7 CWDM System Configuration

About This Chapter

The CWDM system can be configured as any of the following two equipment types: optical terminal multiplexer (OTM) and fixed optical add/drop multiplexer (FOADM).

The CWDM technology does not support optical amplifiers. Normally, CWDM systems adopt ESC channels to lower down the cost. Hence, the OptiX Metro 6100 CWDM systems can be configured as simpler equipment types. These types of equipment are consisted of OTU boards and optical add/drop multiplexing boards supporting CWDM specifications.

The OptiX Metro 6100 CWDM system supports single-fiber bidirectional transmission mode. That is, on a transmission line, the optical signals in the transmit and receive directions are transmitted over the same fiber. The principle and functional units of the single-fiber bidirectional CWDM system are similar to those of the two-fiber bidirectional CWDM system. This chapter describes only the two-fiber bidirectional system.

7.1 OTM

The CWDM system can be configured as the OTM comprising optical add/drop multiplexing boards.

7.2 FOADM

The CWDM system can be configured as the FOADM comprising optical add/drop multiplexing boards.

7.1 OTM

The CWDM system can be configured as the OTM comprising optical add/drop multiplexing boards.

Functions

The OTM node is used at the terminal station, and has the following two logical directions:

- Transmit direction
- Receive direction

In the transmit direction, the OTM node converges or transforms the client-side signals. Then, the signals are multiplexed with the optical supervisory signals by an optical add/drop multiplexing board. Finally, the multiplexed signals are sent to the line for transmission. In the receive direction, the OTM node performs the converse process.

Functional Units

An OptiX Metro 6100 CWDM OTM node has the following functional units:

- Optical transponder board (OTU)
- Optical add/drop multiplexing board (OADM board)

For the boards used in each unit, refer to [4.4 Function Boards](#).

Signal Flow

The OptiX Metro 6100 CWDM OTM node is used at the terminal station, and has the following two logical directions:

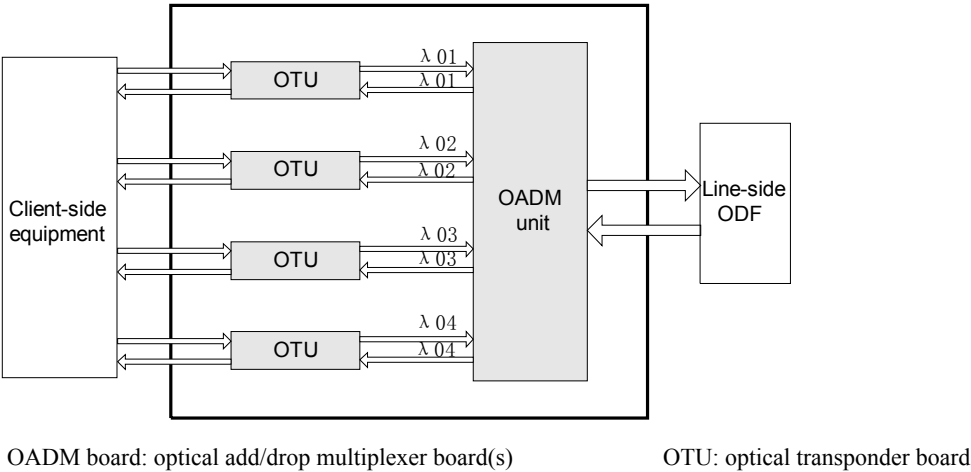
- Transmit direction
- Receive direction

In the transmit direction, through the OTU, the OTM node converges/transforms the accessed signals into signals with ITU-T G.694.2-compliant CWDM wavelengths. After that, the signals are multiplexed by the OADM board into the main optical path and sent to the line.

In the receive direction, the line signal is demultiplexed by the OADM board into signals of different wavelengths, and then sent to the corresponding client-side equipment after being transformed and divided by the OTUs.

The diagram of CWDM OTM node is shown in [Figure 7-1](#).

Figure 7-1 Schematic diagram of the OptiX Metro 6100 CWDM OTM node

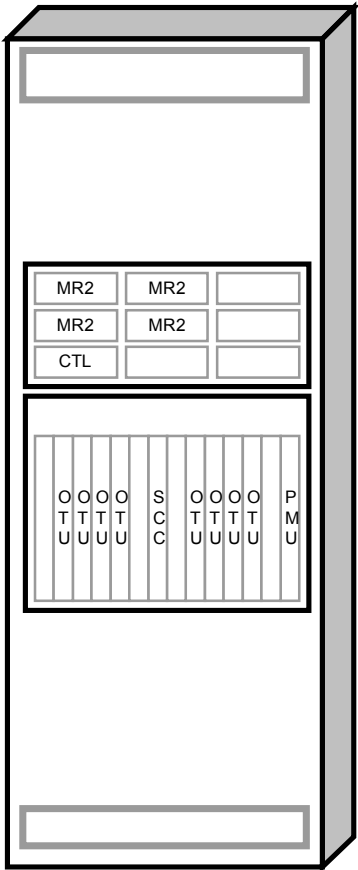


Typical Configuration

Take the 8-wavelength OptiX Metro 6100 CWDM OTM node as an example, as shown in Figure 7-2.

One subrack, one OADM frame and one cabinet are used.

Figure 7-2 Configuration diagram of the 8-wavelength OptiX Metro 6100 CWDM OTM node



Configuration Rules

The following are the principles of configuring the OADM boards:

- When a subrack and an OADM frame are configured, install the subrack in the middle of the cabinet. The optical interfaces on the OADM board are arranged in an ascending order of frequency. For example, if the wavelength that can be dropped from the MR2 board is 1511/1531, the A01/D01 on the board corresponds to 1511 nm and the A02/D02 to 1531 nm.
- Install OADM boards into slots from left to right in an ascending order of frequency. On the west, slots are IU15–IU17 and IU21. On the east, slots are IU18–IU20 and IU22. For example, install the boards at 1511/1531 nm into IU15 and IU18 first. Then, install the boards at 1551/1571 nm into IU16 and IU19.

The following are the principles of configuring the OTU boards:

- The OTU board with a smaller frequency is configured first. When there are multiple subracks, the OTU board is configured in a lower subrack first. When there is only one subrack, the OTU board is configured in the left slot first. On the west, install boards into slots started with IU1 from left to right. On the east, install boards into slots started with IU8 from left to right.
- In the client-side 1+1 protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the inter-board wavelength protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUs from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the intra-board wavelength protection mode, the left side of the subrack is preferred to house the dual-fed and selective receiving OTU board.

The following are the principles of configuring the SCC boards:

- The SCC board is installed in IU7 in each subrack.

The following are the principles of configuring the PMU boards:

- The PMU board is installed in IU14 in each subrack.

7.2 FOADM

The CWDM system can be configured as the FOADM comprising optical add/drop multiplexing boards.

Functions

The FOADM node performs the add/drop multiplexing of fixed wavelengths from the multiplexed signals.

Functional Units

An OptiX Metro 6100 CWDM FOADM node has the following functional units:

- Optical transponder board (OTU)
- Optical add/drop multiplexing board (OADM board)

For the boards used in each unit, refer to [4.4 Function Boards](#).

Signal Flow

The OptiX Metro 6100 CWDM FOADM node processes the optical signals from two directions.

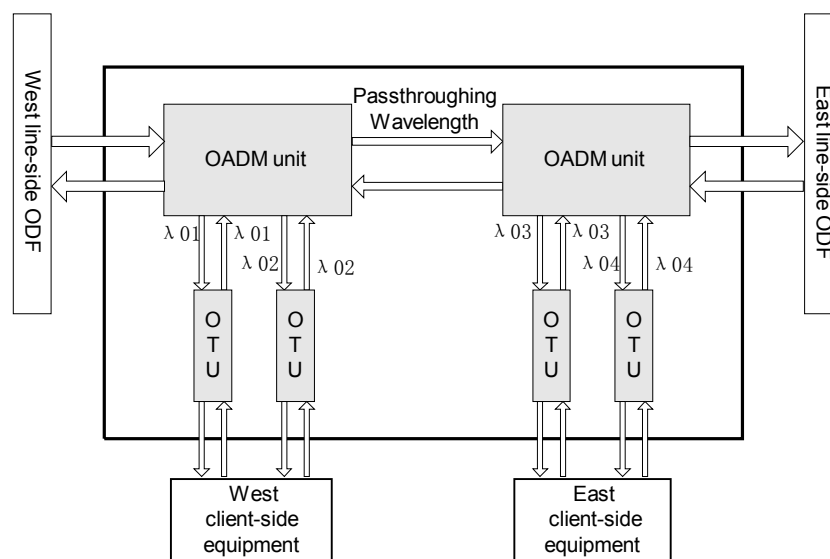
It receives and sends line signals to the OADM board, where some wavelengths are dropped to the OTUs and then to the client side.

The other wavelengths just pass through the OADM and are multiplexed with the wavelengths added locally.

Then the multiplexed wavelengths are sent into the line for transmission.

The diagram of OptiX Metro 6100 CWDM FOADM node is shown in [Figure 7-3](#).

Figure 7-3 Schematic diagram of the OptiX Metro 6100 CWDM FOADM node



OADM board: optical add/drop multiplexer board(s)

OTU: optical transponder board

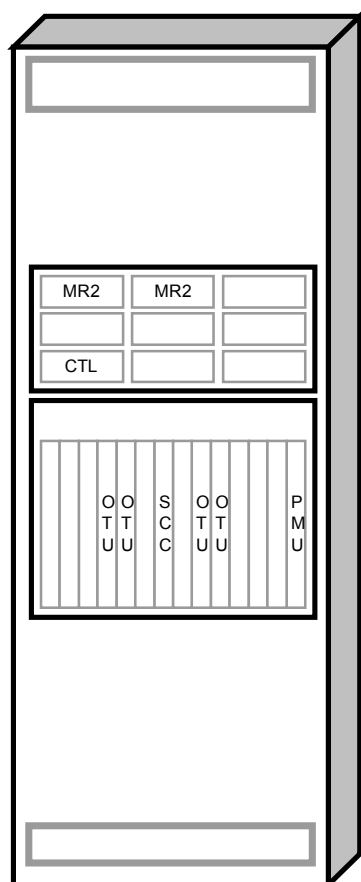
Typical Configuration

Take the two-wavelength OptiX Metro 6100 CWDM FOADM node as an example, as shown in [Figure 7-4](#).

This OADM can add and drop two channels respectively in two transmission directions.

One subrack, one OADM frame and one cabinet are used.

Figure 7-4 Configuration diagram of the 2-wavelength OptiX Metro 6100 CWDM FOADM node



Configuration Rules

The following are the principles of configuring the OADM boards:

- When a subrack and an OADM frame are configured, install the subrack in the middle of the cabinet.
- The optical interfaces on the OADM board are arranged in an ascending order of frequency. For example, if the wavelength that can be dropped from the MR2 board is 1511/1531, the A01/D01 on the board corresponds to 1511 nm and the A02/D02 to 1531 nm.
- Install OADM boards into slots from left to right in an ascending order of frequency. On the west, slots are IU15–IU17 and IU21. On the east, slots are IU18–IU20 and IU22. For example, install the boards at 1511/1531 nm into IU15 and IU18 first. Then, install the boards at 1551/1571 nm into IU16 and IU19.

The following are the principles of configuring the OTU boards:

- The OTU board with a smaller frequency is configured first. When there are multiple subracks, the OTU board is configured in a lower subrack first. When there is only one subrack, the OTU board is configured in the left slot first. On the west, install boards into slots started with IU1 from left to right. On the east, install boards into slots started with IU8 from left to right.

- In the client-side 1+1 protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUS from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the inter-board wavelength protection mode, the east and west OTUs must be configured in the same subrack and the left side of the subrack is preferred to house the OTUs. Two OTUs in mutual backup mode are installed in an adjacent manner. Install the OTUS from left to right following the sequence of west OTU1, east OTU1, west OTU2, east OTU2 and SCS.
- In the intra-board wavelength protection mode, the left side of the subrack is preferred to house the dual-fed and selective receiving OTU board.

The following are the principles of configuring the SCC boards:

- The SCC board is installed in IU7 in each subrack.

The following are the principles of configuring the PMU boards:

- The PMU board is installed in IU14 in each subrack.

8 Grooming of Wavelengths and Services

About This Chapter

This chapter describes the networking application of optical signal grooming features, the signal flow and electrical signal grooming features of the product.

[8.1 Dynamic Optical Layer Grooming](#)

The ROADM supported by the DWDM system is mainly realized by the DWC, WSD9, WSM9, RMU9 and WSMD4 boards.

[8.2 Application and Networking of the GE ADM Feature](#)

The GE ADM service grooming is used in the product.

8.1 Dynamic Optical Layer Grooming

The ROADM supported by the DWDM system is mainly realized by the DWC, WSD9, WSM9, RMU9 and WSMD4 boards.

For the details of the related board functions, refer to the *Hardware Description*.

8.1.1 Intra-Ring Wavelength Grooming by DWC Boards

The product provides the intra-ring wavelength grooming by DWC boards.

Function

The ROADM node by two DWC boards can block any wavelength path, terminal any wavelength at any node in the chain or ring network, and then add the fixed filter corresponding to the drop wavelength in the drop wavelength path. The pass-through path and add/drop wavelength path are completely separated. Hence, adjusting the fixed filter for adding/dropping wavelength has no impact on the services of the main optical path.

Application

For the two-dimensional grooming node that adds/drops a lot of wavelengths, the intra-ring wavelength grooming realized through DWC boards is recommended.

Node Structure

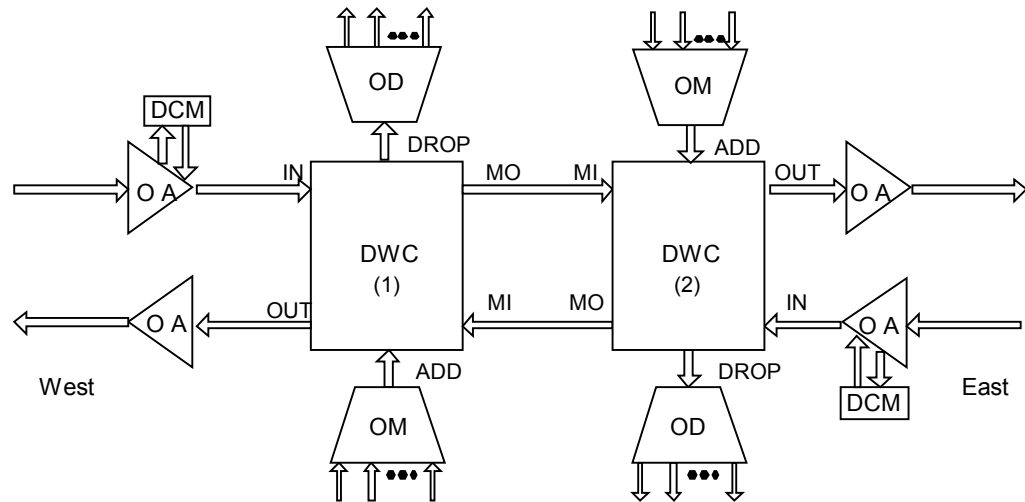
An ROADM node is constituted by two DWC boards, as shown in [Figure 8-1](#).

The main path signals from west are amplified and input through the IN optical interface of the DWC (1) board. Then the main optical signals are divided into the same two channels of optical signals. The optical signals that need to be dropped locally are output from the DROP optical interface to the optical demultiplexer (OD), and then they are dropped from the optical interfaces 01–40 of the OD to the local station. At the same time, the optical signals dropped from the DWC (1) are blocked in the DWC (1). The pass-through optical signals are output from the MO optical interface.

The pass-through optical signals of the DWC (1) are input from the IN optical interface of the DWC (2). The optical signals added locally are accessed from the optical interfaces 01–40 of the optical multiplexer (OM), and then input from the ADD optical interface of the DWC (2) to the DWC (2) for dynamic add/drop multiplexing. The pass-through optical signals are multiplexed with those optical signals added locally and are together output from the OUT optical interface. Then the main optical signals are amplified and output eastward.

The grooming of wavelength signals transmitted from east to west is in the same way.

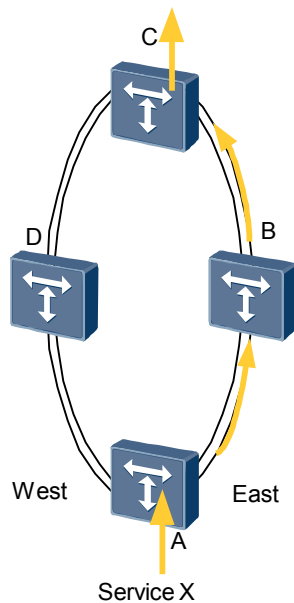
Figure 8-1 ROADM node by DWC boards



Signal Flow

Figure 8-2 shows engineering project T which is a ring network by station A, B, C, and D. All of the four stations are ROADM nodes by DWC boards.

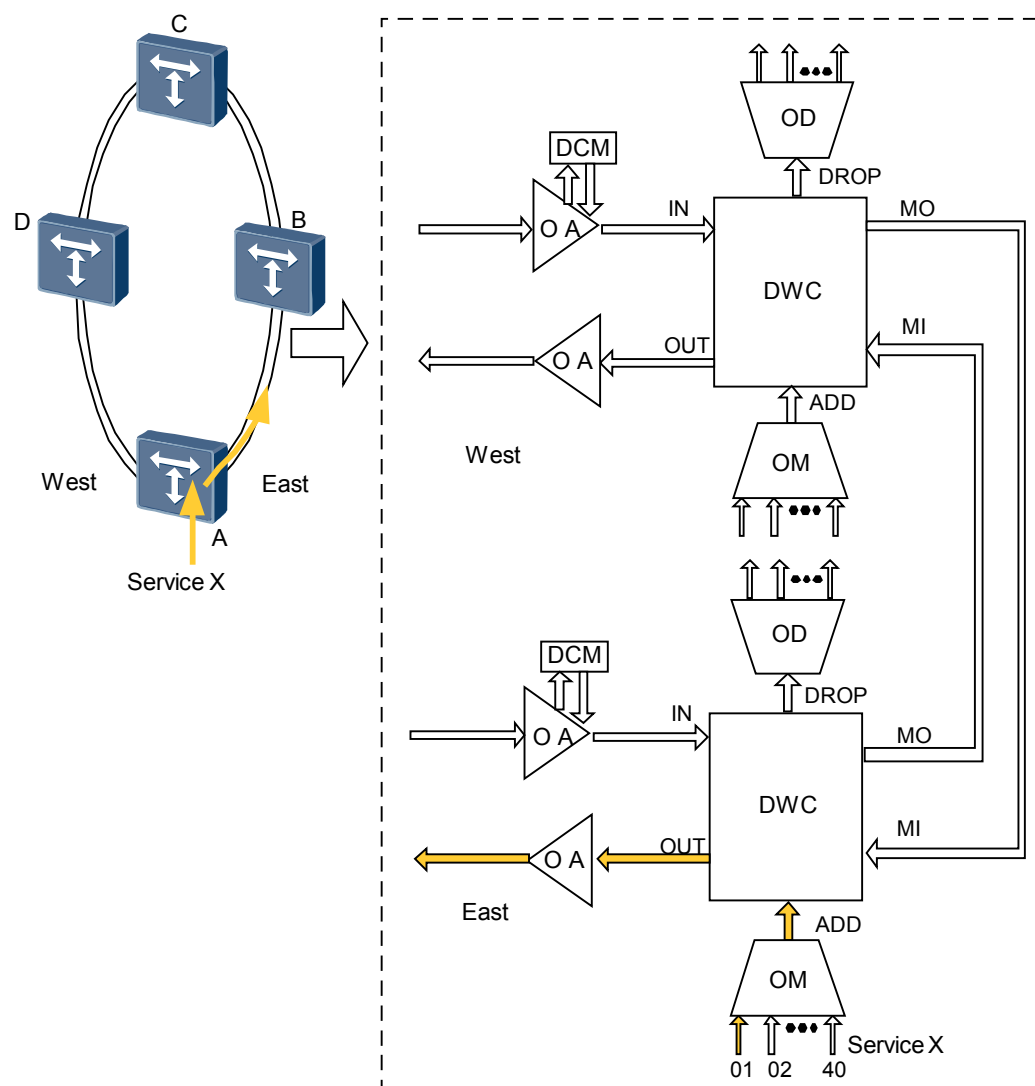
Figure 8-2 Intra-ring grooming diagram (DWC+DWC)



Suppose that service X is input at node A, passes through node B, and is to be output at node C. The grooming process is as follows.

- Service X is input at node A and output eastward.
Service X is input at node A. It uses the wavelength 1 in the engineering project T.

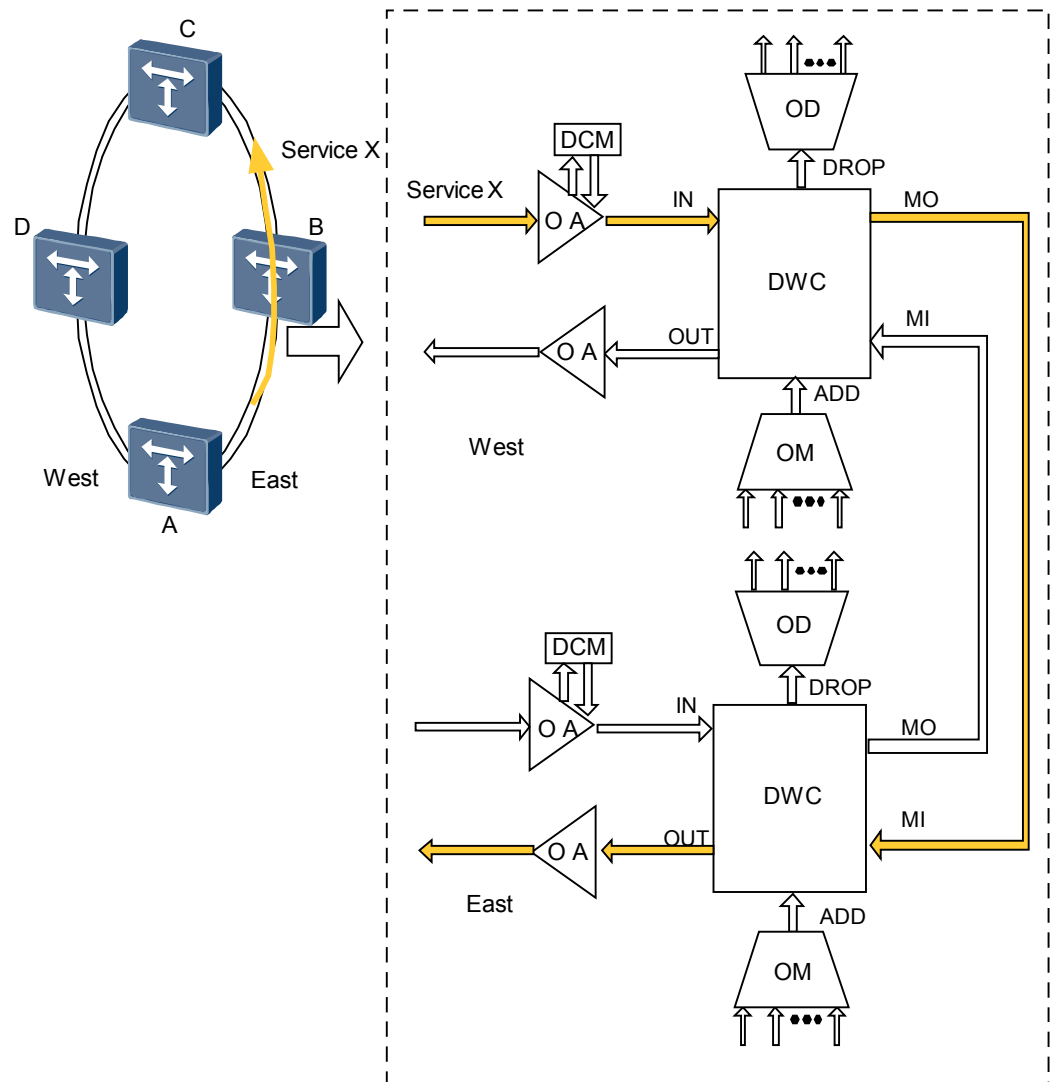
Figure 8-3 Service grooming at node A



The pass through service X is input from the MI optical interface of the east DWC board and is output from the OUT optical interface. The main optical signals containing Service X are amplified and output at node B eastward.

Figure 8-4 shows the signal flow of node B.

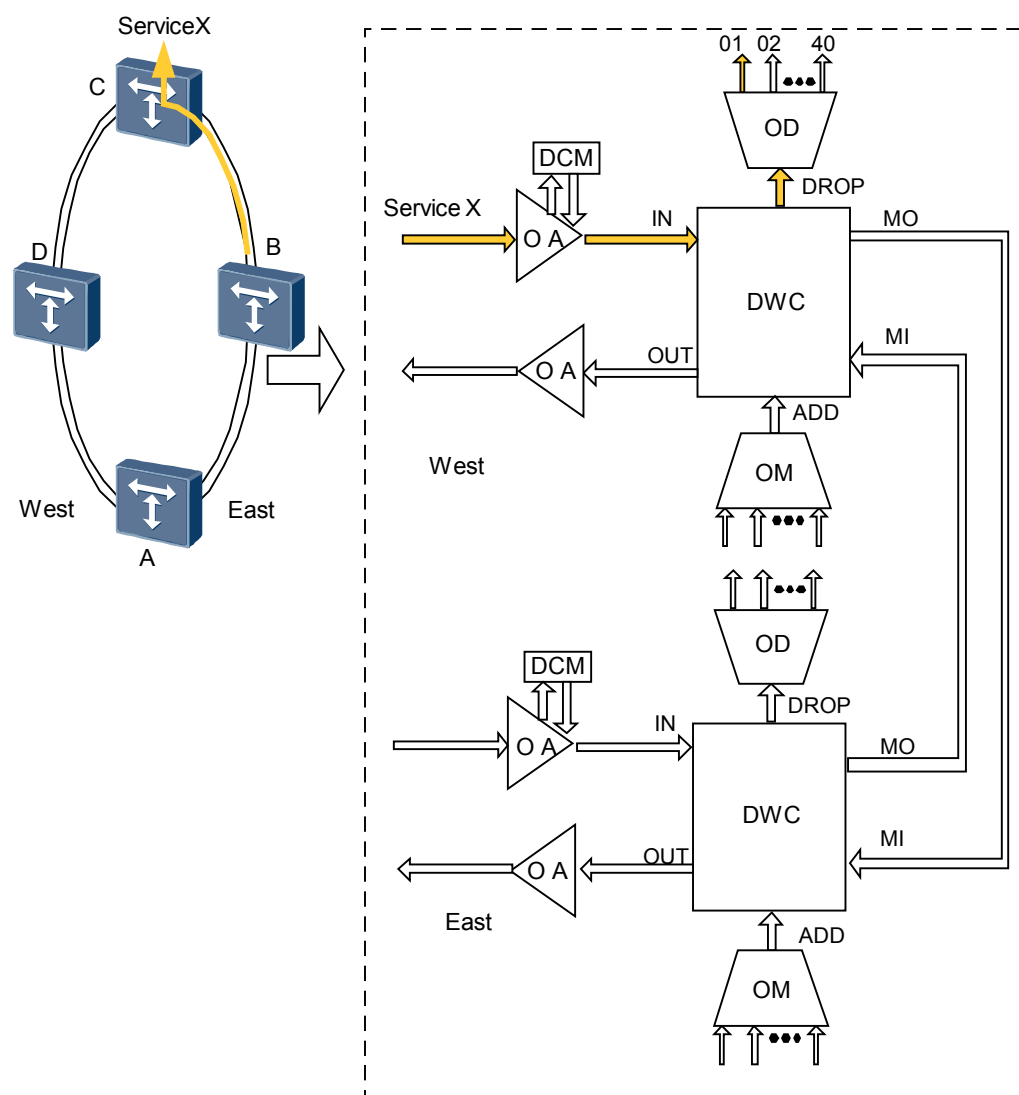
Figure 8-4 Service grooming at node B



- The service is input at node C from west and is output.

The main optical signals containing Service X are input at node C from west. Then they are amplified and input from the IN optical interface of the DWC board. Service X to be dropped is output through the DROP optical interface. And the wavelength 1 carrying the Service X is received locally in the D40 board.

Figure 8-5 shows the signal flow at node C.

Figure 8-5 Service grooming at node C

Dynamic Grooming

Suppose that the requirement is changed. The service X needs to be input at node A, pass through node B and node C, and to be output at node D. You only need to change the corresponding configurations by using the T2000. Ensure that the Service X passes through node B and node C, and is output at node D to local station.

8.1.2 Intra-Ring Wavelength Grooming by WSD9 Boards and RMU9 Boards

The product provides the intra-ring wavelength grooming by WSD9 boards and RMU9 boards.

Function

ROADM nodes by two WSD9 boards and the same number of RMU9 boards can realize full dynamic wavelength grooming in a ring network. They can also remotely and dynamically adjust the status of wavelength adding/dropping and passing through using NE software.

Application

ROADM nodes by WSD9 boards and RMU9 boards are recommended for nodes whose adding/dropping wavelengths need to be irrelevant with the optical port.

Node Structure

One ROADM node consists of two WSD9 boards and two RMU9 boards as shown in [Figure 8-6](#). Main path signals from west are amplified and input through the IN port of the WSD9 board.

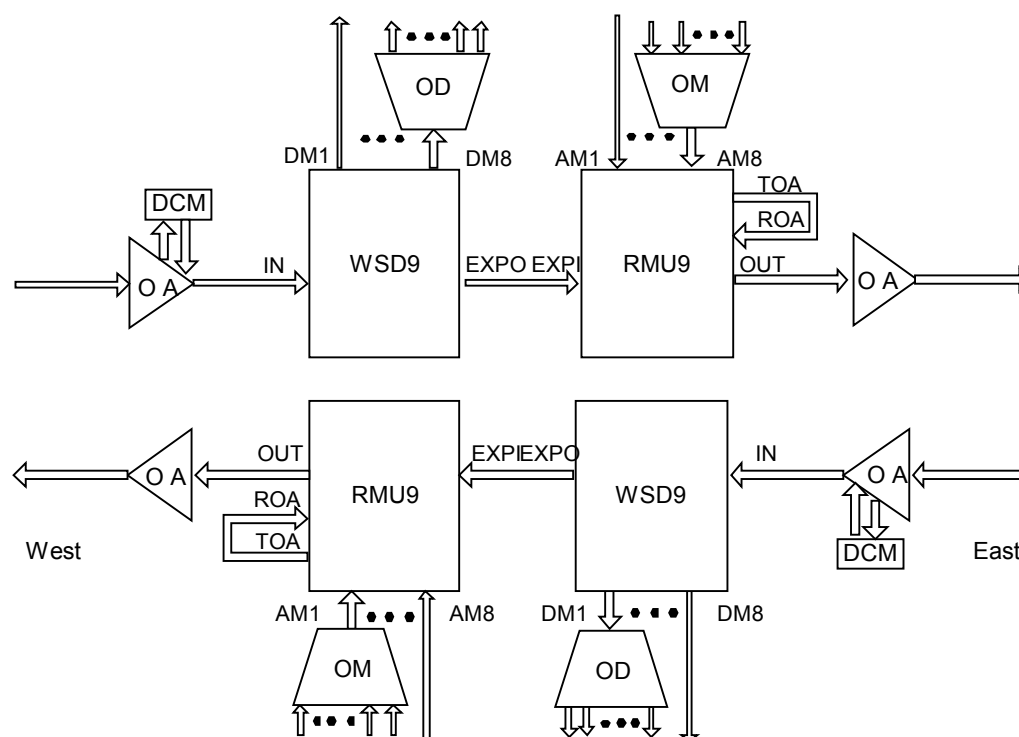
The wavelengths that need to be output locally are output from the ports given in the configuration. Output multiplexed signals are demultiplexed to single wavelengths by the demultiplexer before sending to the local client device by OTU. Output single wavelengths can be directly sent to the local client device by OTU.

Optical wavelengths not to be added/dropped locally are output through the EXPO port of the WSD9.

The locally input optical signals are multiplexed by the RMU9 board and output through the TOA port.

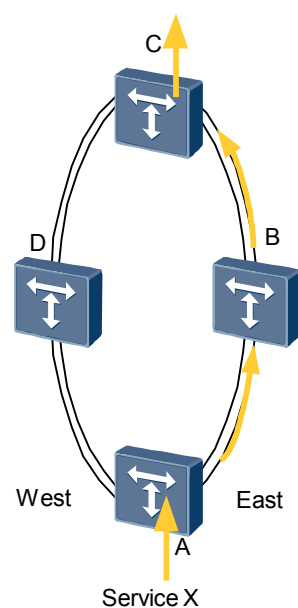
The passing through main optical signals are input through the EXPI port of the RMU9 board. Short the TOA port and ROA port. The wavelengths input through ROA port are multiplexed with those input through EXPI port and are together output through OUT port. Then the main path optical signals are amplified and sent to the line for transmission.

The grooming of wavelength signals transmitted from east to west is in the same way.

Figure 8-6 ROADM node by WSD9 boards and RMU9 boards

Signal Flow

Figure 8-7 shows engineering project T which is a ring network by station A, B, C, and D. All of the four stations are ROADM nodes by WSD9 boards and RMU9 boards.

Figure 8-7 Intra-ring grooming diagram (WSD9+RMU9)

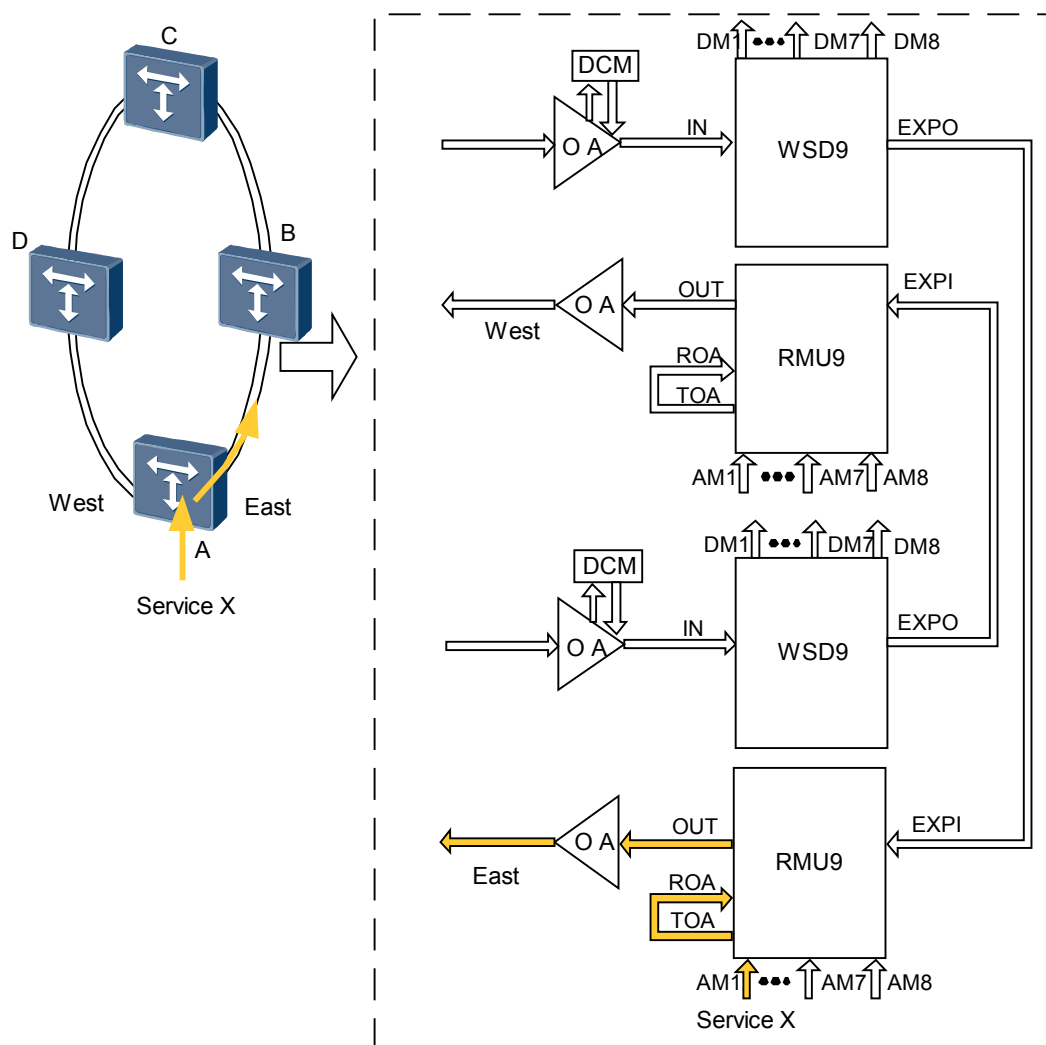
Suppose that service X is input at node A, passes through node B, and is to be output at node C. The grooming process is as follows.

- Service X is input at node A and output eastward.

Service X is input at node A. It is input through the adding port of the RMU9 board and multiplexed before it is output through TOA port. Short the TOA port and the ROA port, then the service is output through OUT port. The main optical signals containing service X are amplified and output at node A eastward.

Figure 8-8 shows the service grooming of node A.

Figure 8-8 Service grooming at node A



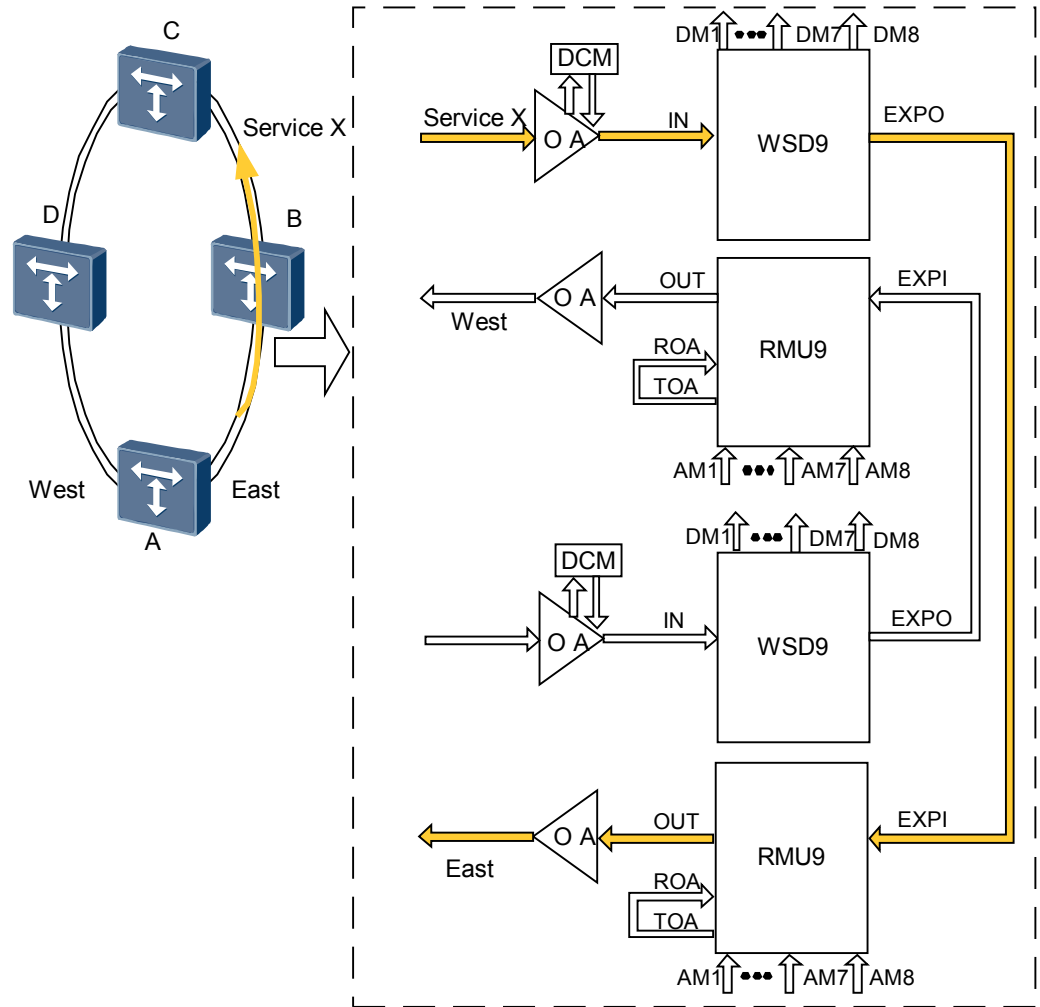
- Service X is input at node B and output eastward.

Service X is input at B from the west and amplified before entering the IN port of the WSD9 board. Then it passes through and is output through the EXPO port.

The passing through service X is input through the EXPI port of the east RMU9 board and is output through the OUT port. The main optical signals containing service X are amplified and output at node B eastward.

Figure 8-9 shows the service grooming of node B.

Figure 8-9 Service grooming at node B

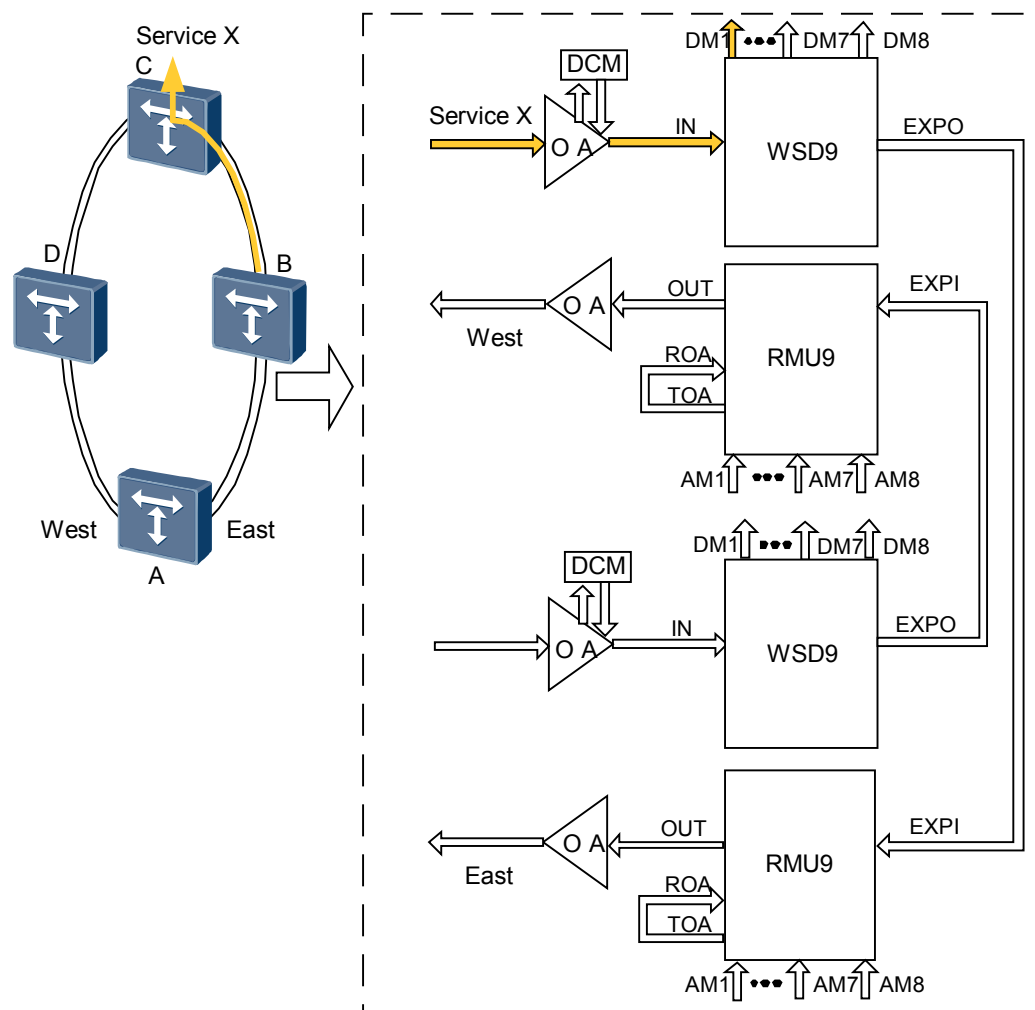


- The service is input at node C from west and is output.

The optical signals containing service X are input at node C from west. The main optical signals are amplified and input through the IN port of the WSD9 board. Service X to be dropped is output through the dropping port and is terminated in the WSD9.

Figure 8-10 shows the service grooming at node C.

Figure 8-10 Service grooming at node C



Dynamic Grooming

Suppose that the requirement is changed. The service X needs to be input at node A, pass through node B and node C, and to be output at node D. You only need to change the corresponding configurations using the T2000.

8.1.3 Intra-Ring Wavelength Grooming by WSD9 Boards and WSM9 Boards

The product provides the intra-ring wavelength grooming by WSD9 boards and WSM9 boards.

Function

ROADM nodes by two WSD9 boards and the same number of WSM9 boards can realize full dynamic wavelength grooming in a ring network. They can also remotely and dynamically adjust the status of wavelength adding/dropping and passing through using NE software.

Application

ROADM nodes by WSD9 boards and WSM9 boards are recommended for nodes with possible requirement for multi-dimensional maintenance.

Node Structure

One ROADM node consists of two WSD9 boards and two WSM9 boards as shown in [Figure 8-11](#).

Main path signals from west are amplified and input through the IN port of the WSD9 board.

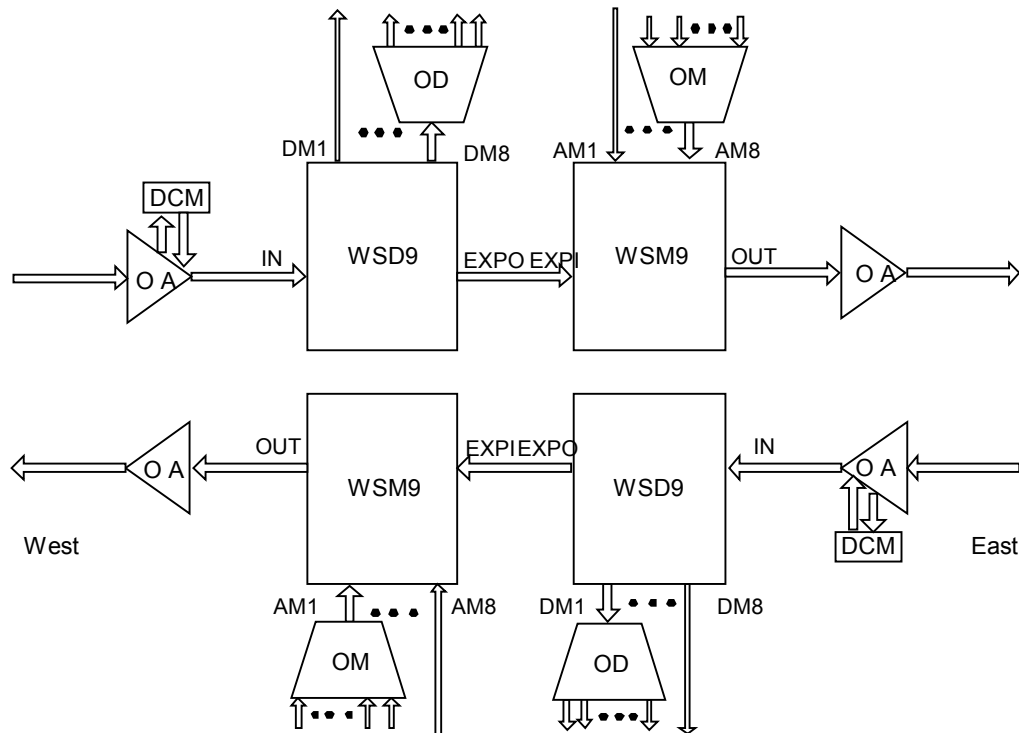
The wavelengths that need to be output locally are output from port DM1 to DM8 according to the configuration. Output multiplexed signals are demultiplexed by the demultiplexer into separate wavelengths. Then they go through the OTU and are sent to the local client-side equipment. Output single wavelengths can be directly sent to the local client-side equipment.

Optical wavelengths not to be added/dropped locally are output through the EXPO port of the WSD9.

The optical signals input locally are input through port AM1 to AM8 of the WSM9 board. The passing through main optical signals are input through the EXPI port of the WSM9 board. The wavelength input through port AM1 to AM8 are multiplexed with that input through EXPI port and output through OUT port. Then the main path optical signals are amplified and sent to the line for transmission.

The grooming of wavelength signals transmitted from east to west is in the same way.

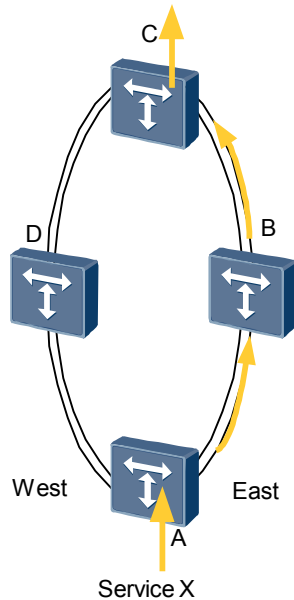
Figure 8-11 ROADM node by WSD9 boards and WSM9 boards



Signal Flow

Figure 8-12 shows engineering project T which is a ring network by station A, B, C, and D. All of the four stations are ROAMD nodes by WSD9 boards and WSM9 boards.

Figure 8-12 Intra-ring grooming diagram (WSD9+WSM9)



Suppose that service X is input at node A, passes through node B, and is to be output at node C. The grooming process is as follows.

- Service X is input at node A and output eastward.

Service X is input at node A. It is input through the adding port of the WSM9 board and multiplexed with other services before it is output through OUT port. The main optical signals containing service X are amplified and output at node A eastward.

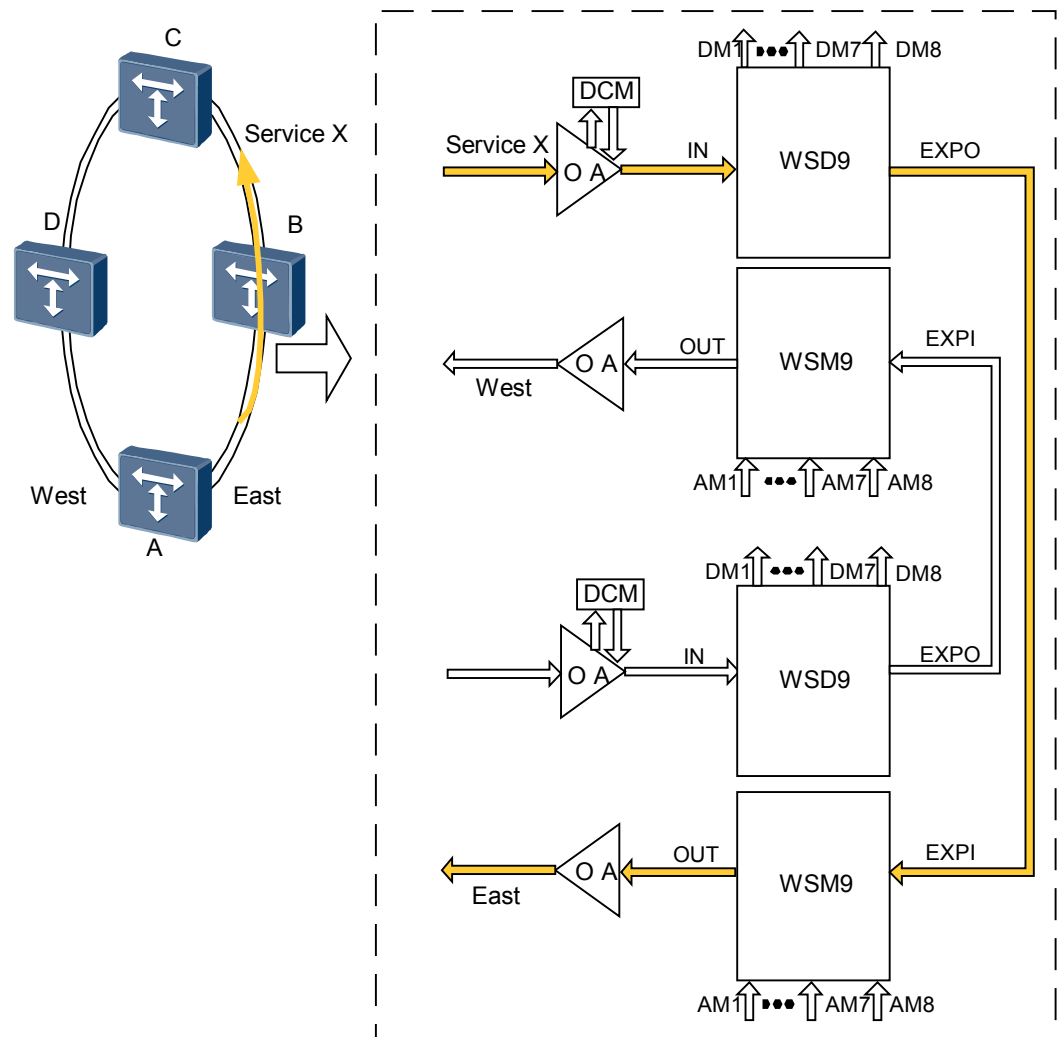
Figure 8-13 shows the service grooming of node A.

- Service X is input at B from the west and amplified before entering the IN port of the WSD9 board. Then it passes through and is output through the EXPO port.

The passing through service X is input through the EXPI port of the east WSM9 board and is output through the OUT port. The main optical signals containing service X are amplified and output at node B eastward.

Figure 8-14 shows the service grooming of node B.

Figure 8-14 Service grooming at node B

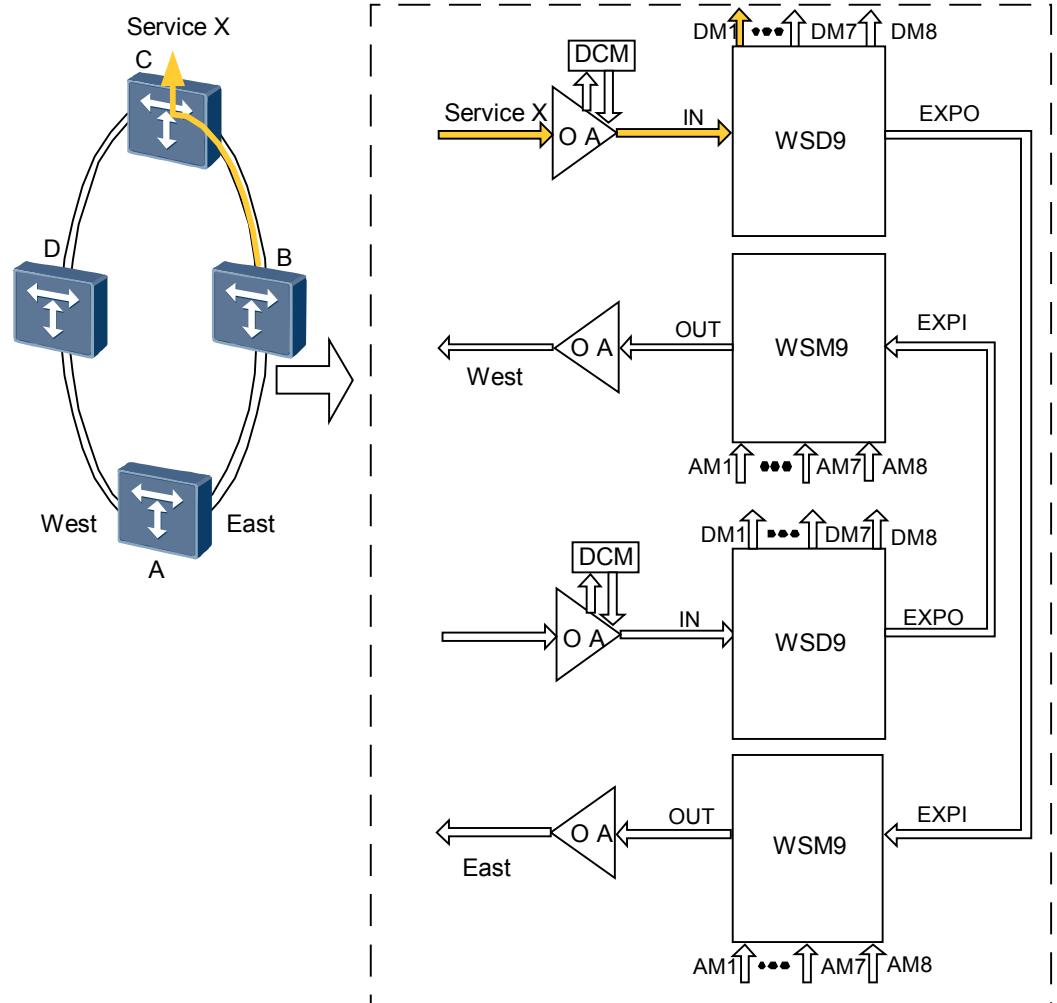


- The service is input at node C from west and is output.

The optical signal containing service X is input at node C from west. The main optical signals are amplified and input through the IN port of the WSD9 board. Service X to be dropped is output through the dropping port.

Figure 8-15 shows the service grooming of node C.

Figure 8-15 Service grooming at node C



Dynamic Grooming

Suppose that the requirement is changed. The service X needs to be input at node A, pass through node B and node C, and to be output at node D. You only need to change the corresponding configurations using the T2000.

8.1.4 Intra-Ring Wavelength Grooming by WSMD4 Boards

The product provides the intra-ring wavelength grooming by WSMD4 boards.

Function

Two WSMD4s can form an intra-ring grooming node to realize the dynamic wavelength grooming in the ring network. Each WSMD4 adds and drops wavelengths. The WSMD4 + WSMD4 combination supports the adding or dropping of 40 wavelengths with 100 GHz spacing, and provides any adding wavelength combination according to the requirement.

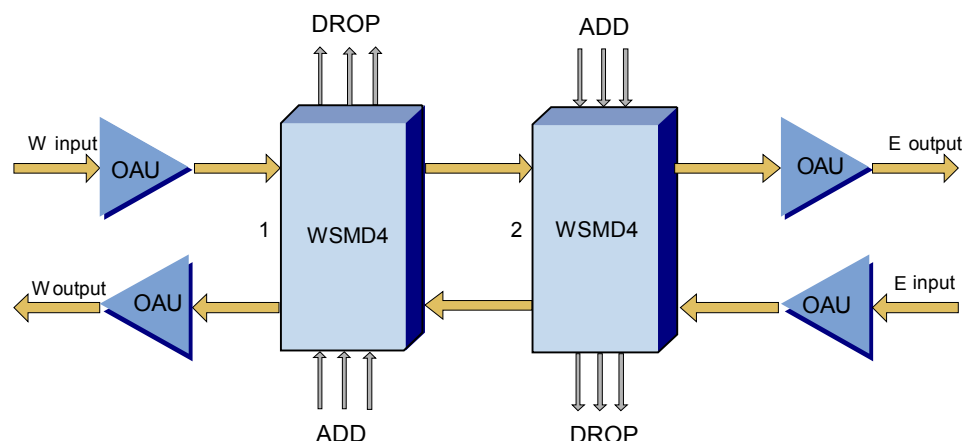
Application

In the case of the two-dimensional grooming node that requires dynamic wavelength grooming, it is recommended to use the intra-ring wavelength grooming scheme realized by using two WSMD4s

Node Structure

Figure 8-16 shows the structure of the intra-ring grooming node formed by two WSMD4s.

Figure 8-16 Intra-ring grooming ROADM node formed by two WSMD4s



The signal flow from west to east is considered as an example.

The input signals from west travel over the optical splitter of WSMD4 board 1 and then are output through the pass-through and drop interfaces of this WSMD4. The dropped wavelengths can be output through one output interface of the demultiplexing board that is properly configured.

The pass-through signals from WSMD4 board 1 pass WSMD4 board 2 directly. Each added wavelength to be output to east is added through an input interface selected by an optical switch on WSMD4 board 2. Such added wavelengths are multiplexed with the pass-through multi-wavelength signals into one multiplexed signal. This signal is input to an optical amplifier; the amplified signal is output to the line side.

The realization of the ROADM function in the signal flow from east to west is the same as that from west to east. The two WSMD4s are combined to add/drop any 40 wavelengths in the east and west directions.

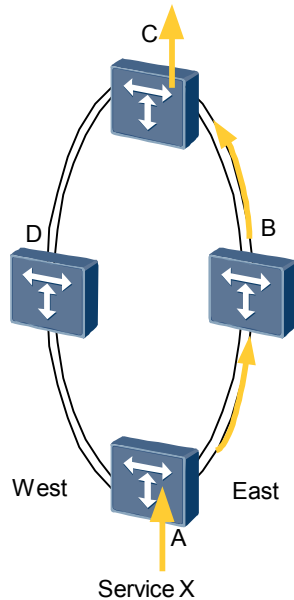
NOTE

- The drop and pass-through interfaces of the WSMD4 board output four equal multiplexed optical signals. In the drop channel, even when there is only one wavelength signal, the WSMD4 need be connected to a demultiplexing board and then to the OTU.
- Connecting the AMx and DMx optical interfaces in tandem can realize the cascading of multiple WSMD4s and adding/dropping of signals to/from the WSMD4.

Signal Flow

Figure 8-17 shows project T which is a ring network formed by nodes A, B, C and D. Each of them is an ROADM node formed by two WSMD4s.

Figure 8-17 Intra-ring grooming diagram (WSMD4+WSMD4)

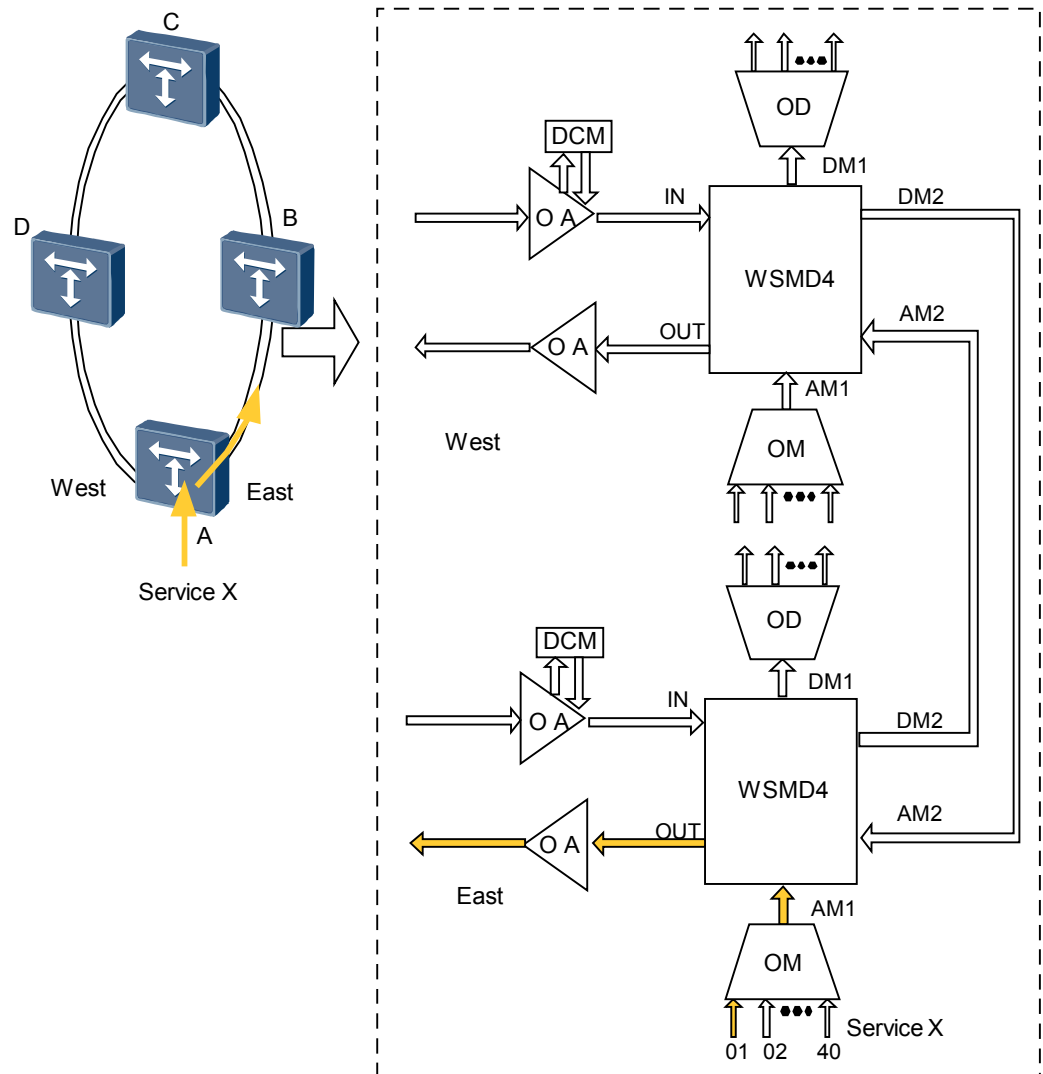


Suppose that service X is input at node A, passes through node B, and is to be output at node C. The grooming process is as follows.

- Service X is input at node A and output eastward.
Service X is input at node A and uses wavelength 1 in project T. Service X is input from the optical interface 01 of the M40 and then is accessed through the AM1 optical interfaces of the east WSMD4 before it is output from the OUT optical interface. The main optical signals containing service X are amplified and output at node A eastward.

Figure 8-18 shows the signal flow of node A.

Figure 8-18 Service grooming at node A



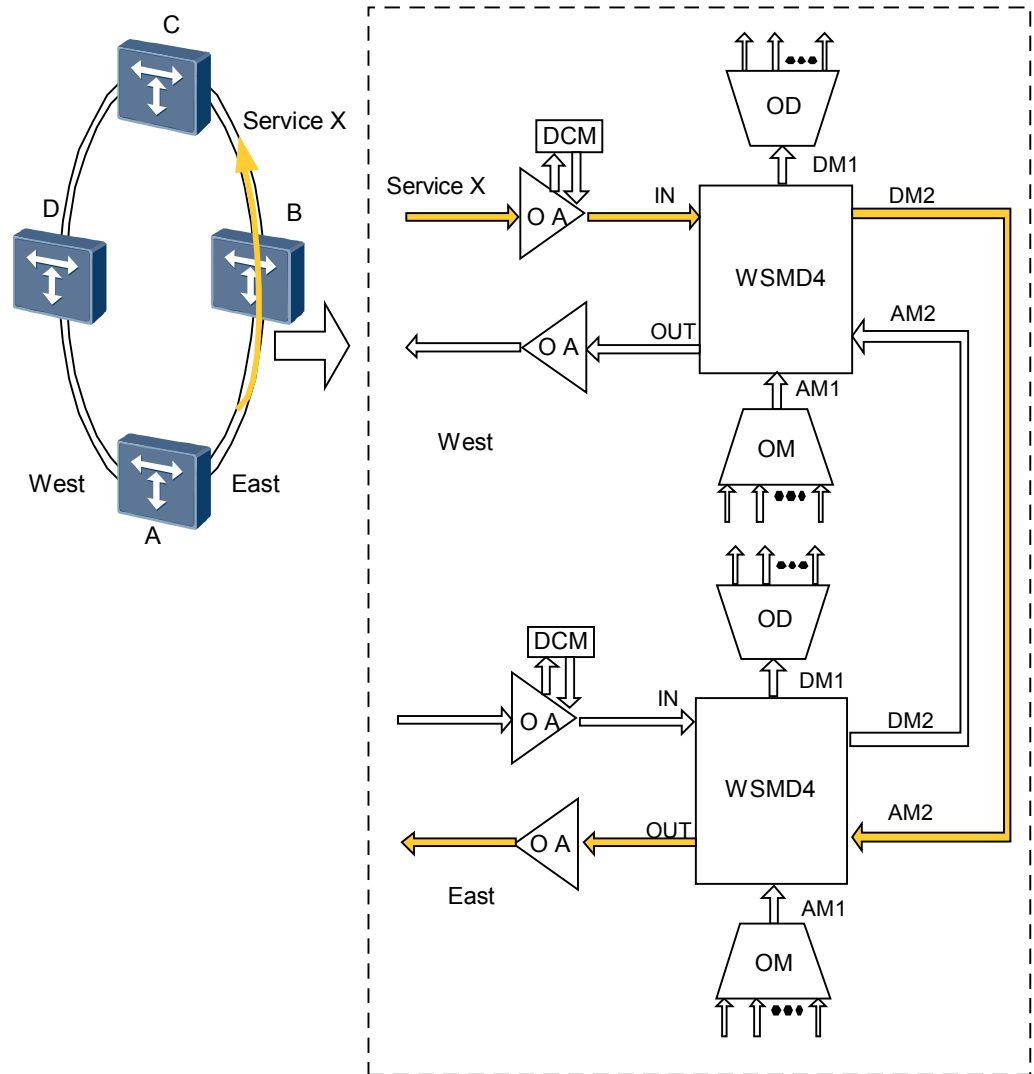
- Service X is input from west at node B and output eastward.

The main optical signals containing service X are input from west at node B and amplified before entering the WSMD4 through the IN optical interface. Service X passes through the WSMD4 and is output through any of the DM1–DM4 optical interfaces.

The pass-through service X is input through any of the AM1–AM4 optical interfaces of the east WSMD4 and then is output from the OUT optical interface. The main optical signals containing service X are amplified and output at node B eastward.

Figure 8-19 shows the signal flow of node B.

Figure 8-19 Service grooming at node B



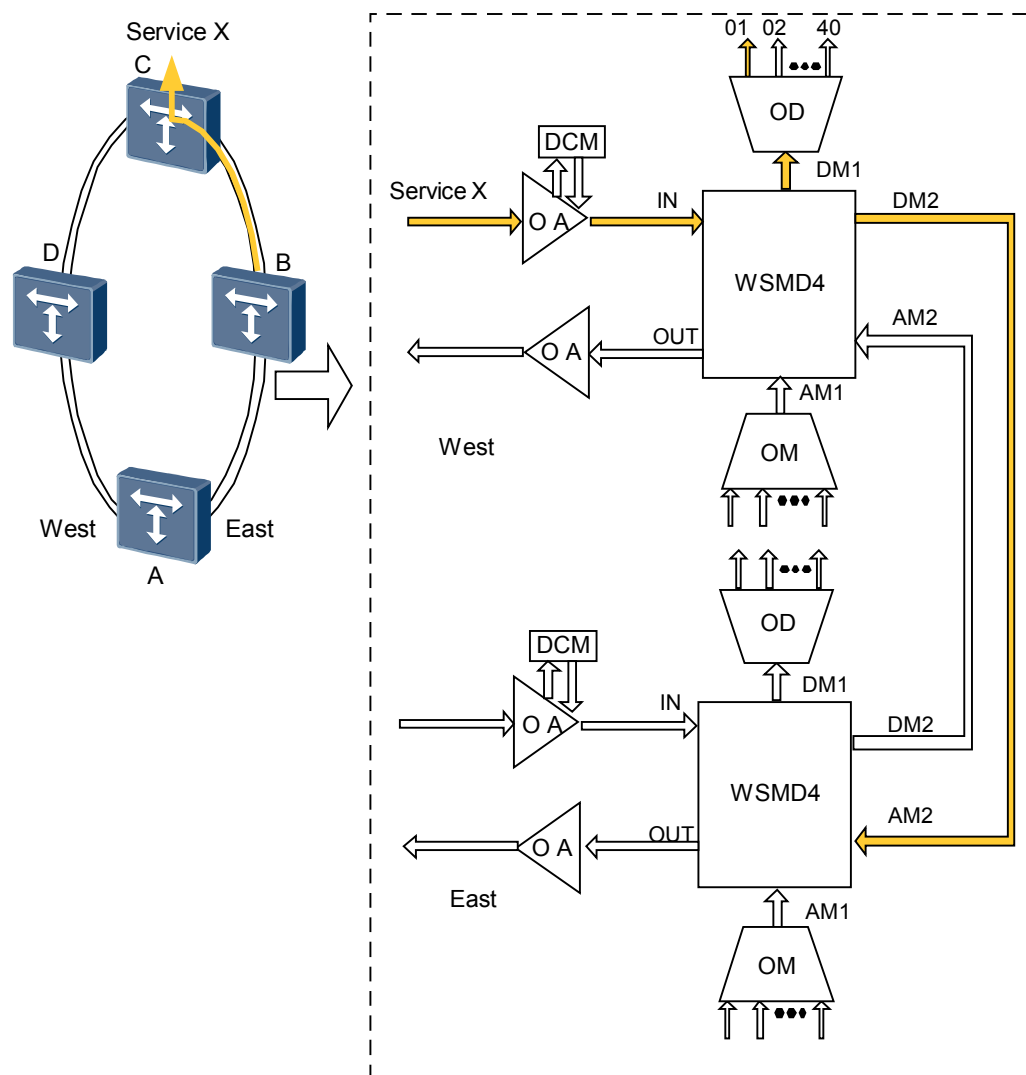
- The service is input at node C from west and is output.

The main optical signals containing service X are input at node C from west. The signals are then amplified and input through the IN optical interface of the west WSM4. Service X to be locally dropped is output through any of the DM1–DM4 optical interfaces.

The pass-through service X is input through any of the AM1–AM4 optical interfaces of the east WSM4. The east WSM4 locally terminates wavelength 1 that bears service X.

Figure 8-20 shows the signal flow of node C.

Figure 8-20 Service grooming at node C



Dynamic Grooming

Suppose that the requirement is changed. The service X needs to be input at node A, pass through node B and node C, and to be output at node D. You only need to change the corresponding configurations by using the T2000. Ensure that service X passes through node B and node C, and is output at node D to the local station.

8.1.5 Inter-Ring Grooming by WSD9 Boards and RMU9 Boards

The product provides the inter-ring wavelength grooming by WSD9 boards and RMU9 boards.

Function

ROADM nodes by four WSD9 boards and the same number of RMU9 boards can realize four-dimensional inter-ring full-dynamic wavelength grooming. They can also remotely and dynamically adjust the status of wavelength adding/dropping and passing through using NE software.

Application

For nodes where the optical power and the OSNR of the signals meet the requirements, wavelength grooming by the WSD9 and the RMU9 is recommended.

ROADM nodes by WSD9 boards and RMU9 boards can decrease the networking cost. But the RMU9 board does not provide the function of wavelength selection compared with the WSM9 board.

Node Structure

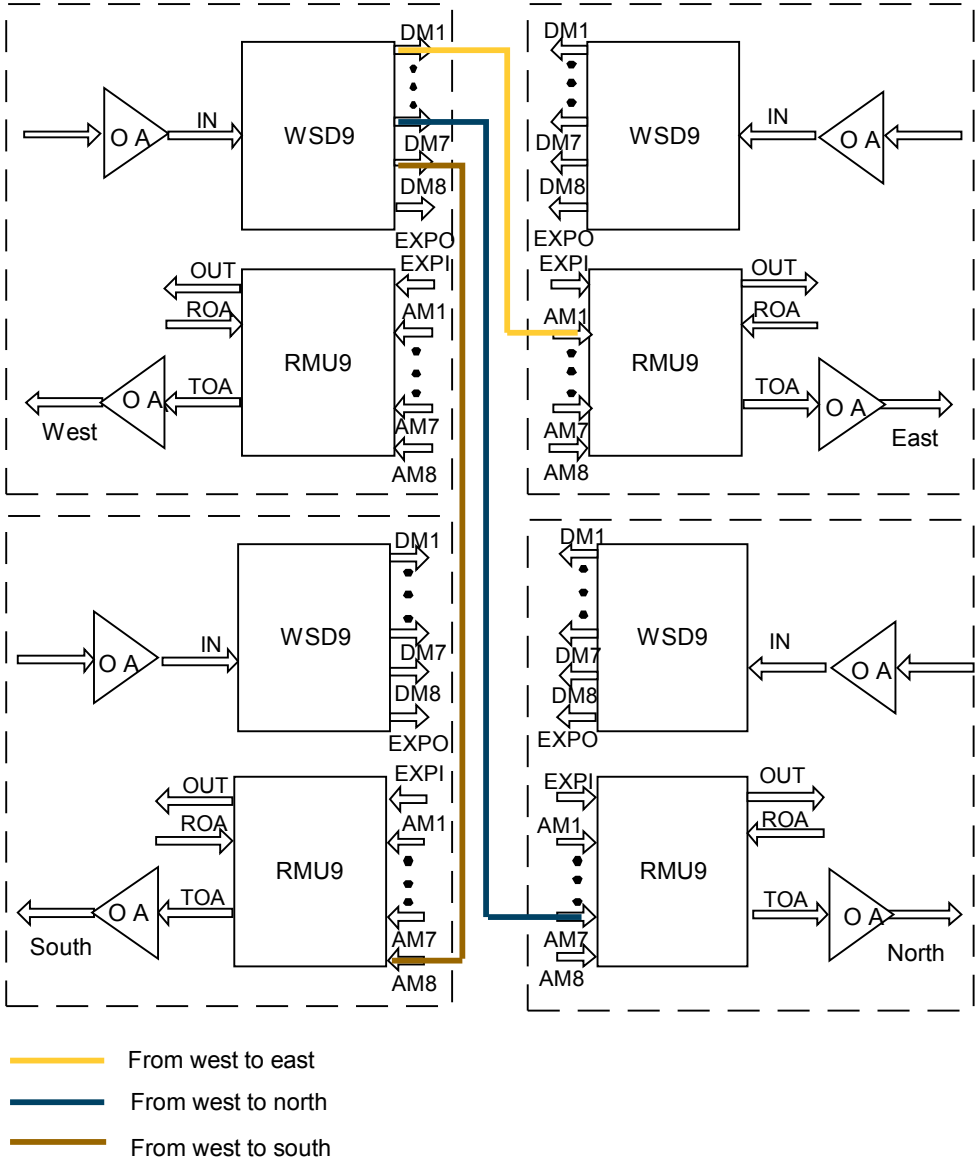
One ROADM node consists of four WSD9 boards and four RMU9 boards as shown in [Figure 8-21](#). The signal grooming from west to east, south, and north is taken for example. Grooming of the signals from the east, south, and north are the same.

Main path signals from west are amplified and input through the IN port of the WSD9 board. Wavelengths to be groomed are output through the EXPO port or any of the DM1 to DM8 of the WSD9 board.

If the service signals need to be output eastward, the signals from west are input through any of the AM1 to AM8 of the east RMU9 board. The input optical signals are multiplexed by the RMU9 board and output through the TOA port. The signals are amplified and output eastward.

If the service signals need to be output southward or northward, the signal flow is the same with those that are output eastward.

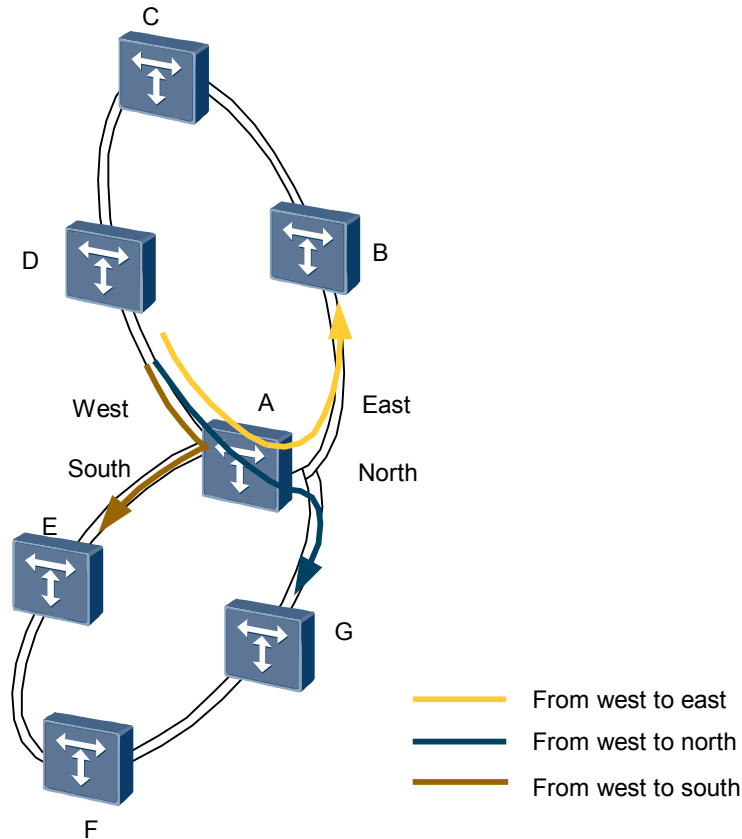
Figure 8-21 ROADM node with inter-ring grooming (WSD9+RMU9)



Signal Flow

Figure 8-22 shows engineering project T which is a tangent ring network by station A, B, C, D, E, F and G. Station A is ROAMD node by WSD9 boards and RMU9 boards.

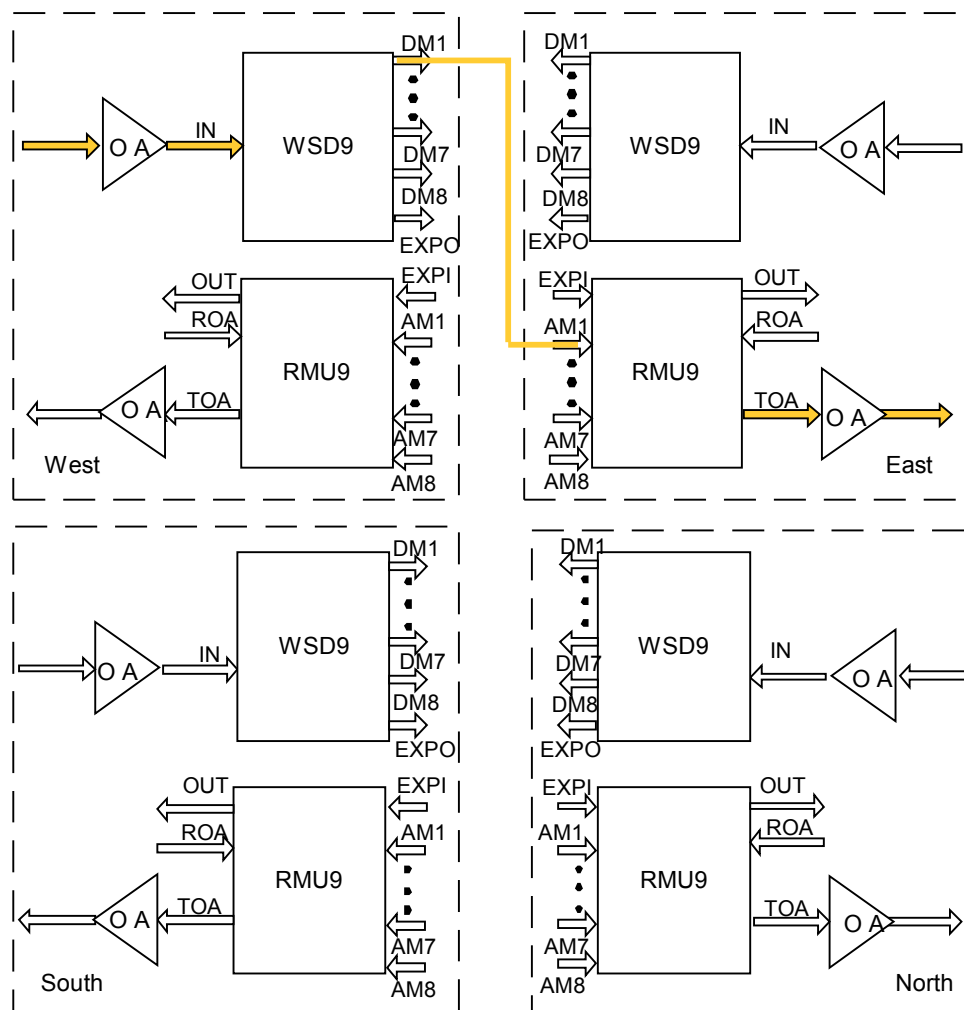
Figure 8-22 Networking with inter-ring grooming (WSD9 + RMU9)



- Suppose that service X is input at Node A from west and needs to be output eastward.
The main path signals containing service X are input at Node A westward. The main path signals are amplified and input through port IN of the WSD9 board. Wavelengths to be groomed are output through the EXPO port or any of the DM1 to DM8 of the WSD9 board.
If the service signals need to be output eastward, the signals from west are input through any of the AM1 to AM8 of the east RMU9 board. The input optical signals are multiplexed by the RMU9 board and output through the TOA port. The signals are amplified and output eastward.

Figure 8-23 shows the signal flow from west to east.

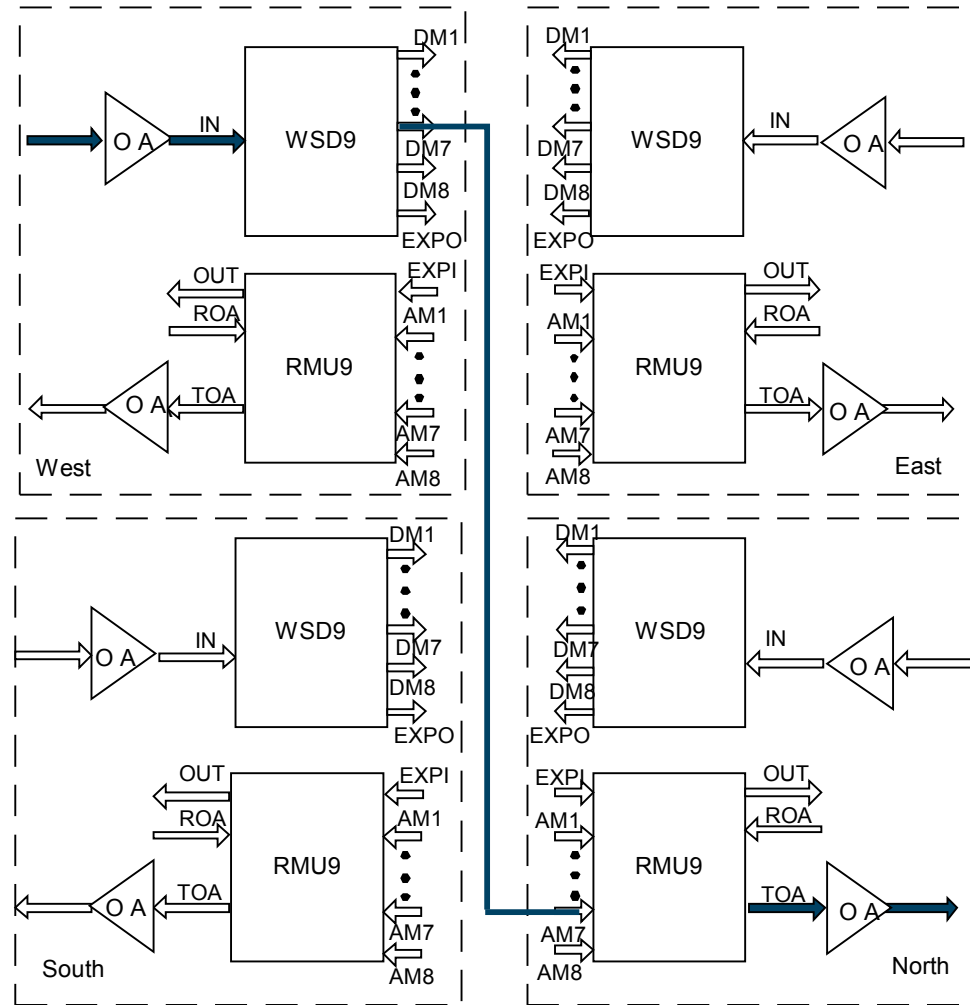
Figure 8-23 Grooming from west to east



- Suppose that service X is input at Node A from west and needs to be output northward.
The main path signals containing service X are input at Node A westward. The main path signals are amplified and input through port IN of the WSD9 board. Wavelengths to be groomed are output through the EXPO port or any of the DM1 to DM8 of the WSD9 board.
If the service signals need to be output northward, the signals from west are input through any of the AM1 to AM8 of the north RMU9 board. The input optical signals are multiplexed by the RMU9 board and output through the TOA port. The signals are amplified and output northward.

Figure 8-24 shows the service grooming from west to north.

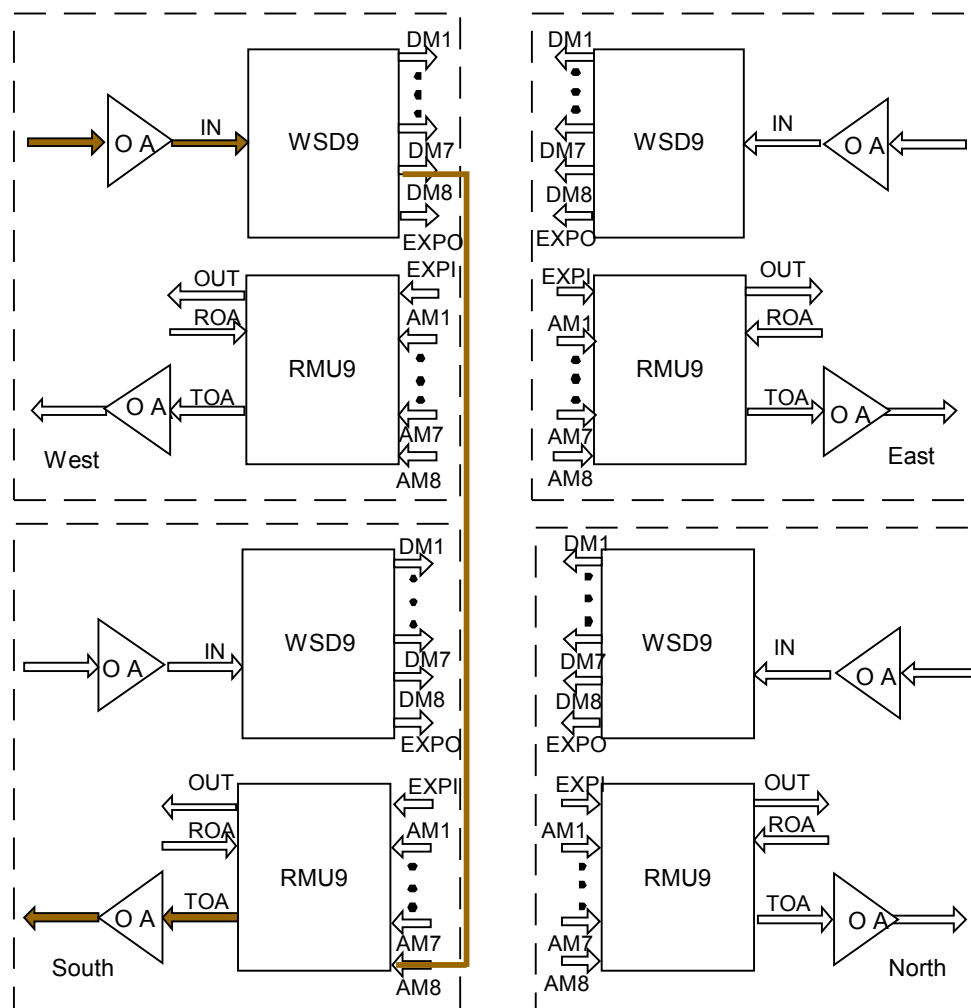
Figure 8-24 Grooming from west to north



- Suppose that service X is input at Node A from west and needs to be output southward. The main path signals containing service X are input at Node A southward. The main path signals are amplified and input through port IN of the WSD9 board. Wavelengths to be groomed are output through the EXPO port or any of the DM1 to DM8 of the WSD9 board. If the service signals need to be output southward, the signals from west are input through any of the AM1 to AM8 of the east RMU9 board. The input optical signals are multiplexed by the RMU9 board and output through the TOA port. The signals are amplified and output southward.

Figure 8-25 shows the service grooming from west to south.

Figure 8-25 Grooming from west to south



Dynamic Grooming

Suppose that the requirement is changed. The service X needs to be input from east and output northward. You only need to change the corresponding configurations using the T2000.

8.1.6 Inter-Ring Grooming by WSD9 Boards and WSM9 Boards

The product provides the inter-ring wavelength grooming by WSD9 boards and WSM9 boards.

Function

ROADM nodes by four WSD9 boards and the samenumber of WSM9 boards can realize a maximum of eight-dimensional inter-ring full dynamic wavelength grooming. They can also remotely and dynamically adjust the status of wavelength adding/dropping and passing through by using T2000.

The grooming process is the same as that of the ROADM nodes by WSD9 boards and RMU9 boards.

Application

For nodes where the optical power and the OSNR of the signals is insufficient, wavelength grooming by the WSD9 and the WSM9 is recommended.

The WSM9 board can provide the function of wavelength selection.

Node Structure

One ROADM node consists of more than four WSM9 boards and the same number of WSD9 boards. The signal grooming increases accompanying with the amount of single flows. The grooming mode is the same as the inter-ring grooming of the ROADM nodes by WSD9 boards and RMU9 boards.

Signal Flow

The inter-ring grooming of the ROADM nodes by WSD9 boards and WSM9 boards can groom more services than the ROADM nodes by WSD9 boards and RMU9 boards. The signal flow direction of the two kinds of ROADM nodes are the same.

Dynamic Grooming

Changing the corresponding configuration by using the T2000 according to the requirement change can realize the dynamic grooming.

8.1.7 Inter-Ring Grooming by WSMD4 Boards

The product provides the inter-ring wavelength grooming by WSMD4 boards.

Function

The ROADM node formed by four WSMD4s realizes the four-dimensional inter-ring dynamic wavelength grooming function. You can also directly use the T2000 to remotely and dynamically adjust the status of wavelength adding/dropping and passing through.

Application

At a node with a maximum of four grooming dimensions, it is recommended to use the WSMD4 to realize wavelength grooming.

Node Structure

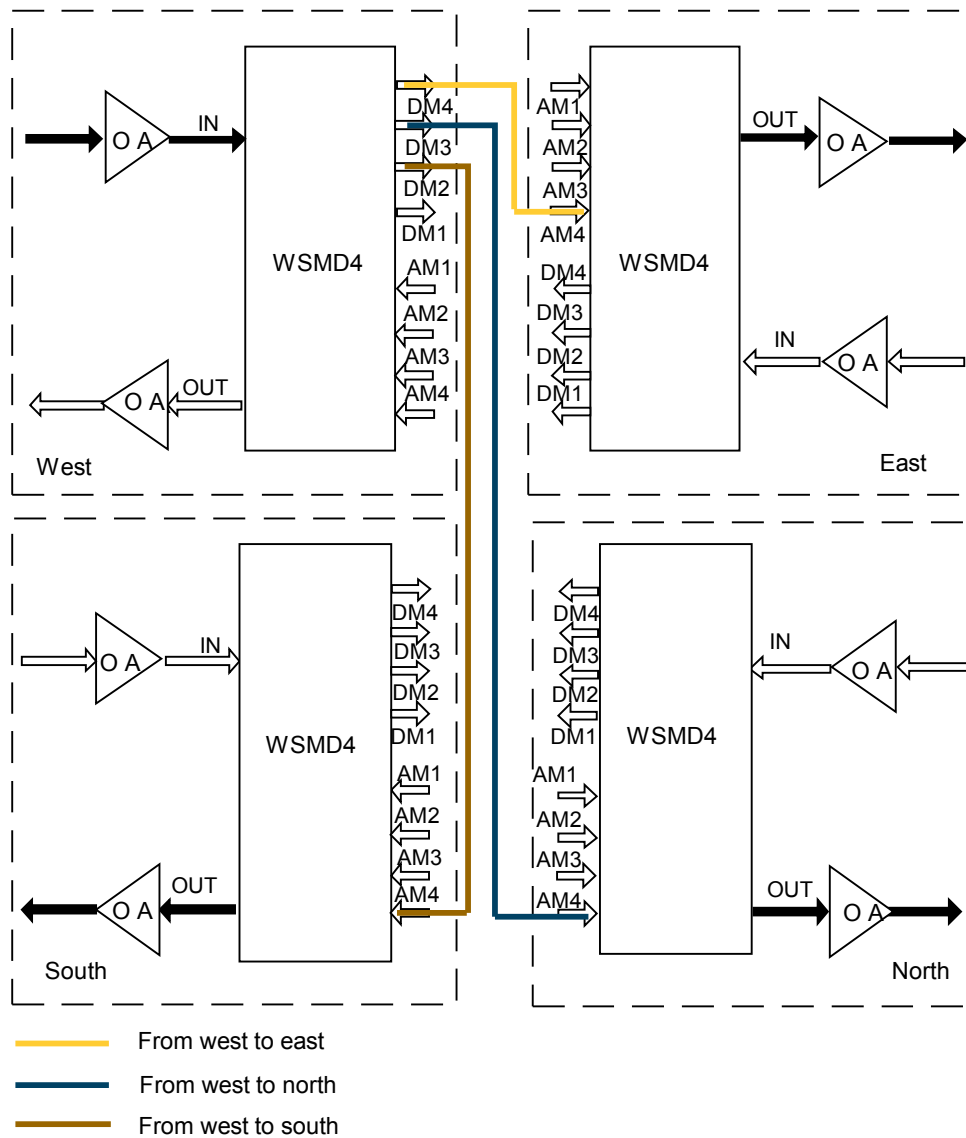
One ROADM node consists of more than four WSMD4s, See [Figure 8-26](#). The signal grooming from west to east, south, and north is considered as an example. The signal grooming from east, south and north to the other three directions is the same as that from west to east, south and north.

The OA amplifies the main path signals input from west and feeds them to the WSMD4 through the IN optical interface. The optical signals to be groomed are output through any of the DM1–DM4 optical interfaces of the WSMD4.

If the service signals need be output eastward, the signals from west should be input to the east WSMD4 through any of the AM1–AM4 optical interfaces. The WSMD4 multiplexes the input signals and output them through the OUT optical interface. The signals are amplified and output eastward.

The signal flow of the service signals output southward or northward is the same as that eastward.

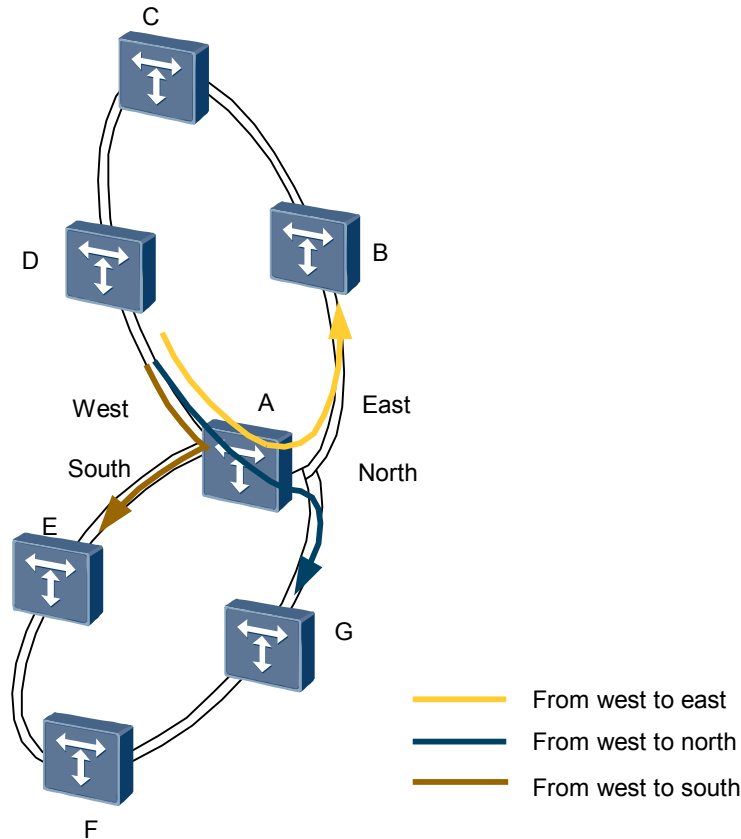
Figure 8-26 ROADM node with inter-ring grooming (WSMD4+WSMD4)



Signal Flow

Figure 8-27 shows project T which is a tangent ring network formed by nodes A, B, C, D, E, F and G. Node A is an ROADM node that uses the WSMD4.

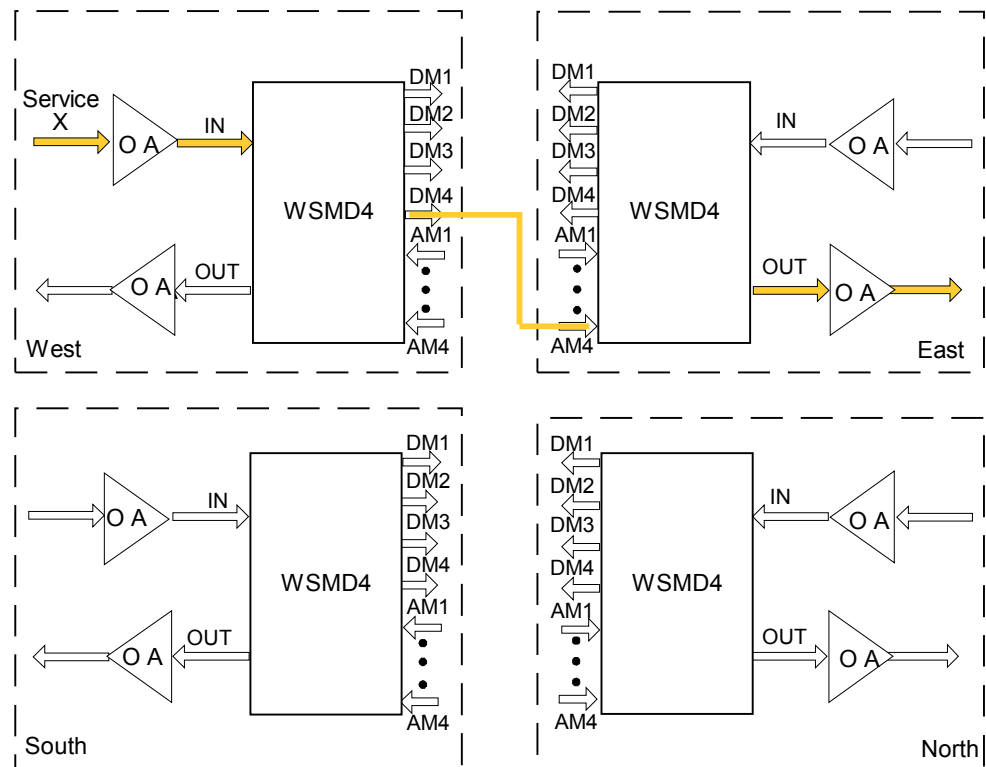
Figure 8-27 Networking with inter-ring grooming (WSMD4)



- Suppose that service X is input at Node A from west and needs to be output eastward.
The main path signals containing service X is input at Node A from west. The OA amplifies the main path signals and feeds them to the WSMD4 through the IN optical interface. The optical splitter equally splits the signals into four channels of optical signals. The optical signals to be dropped from the local node are output through any of the DM1–DM4 optical interfaces of the WSMD4. The optical signals to be groomed are output through any of the DM1–DM4 optical interfaces.

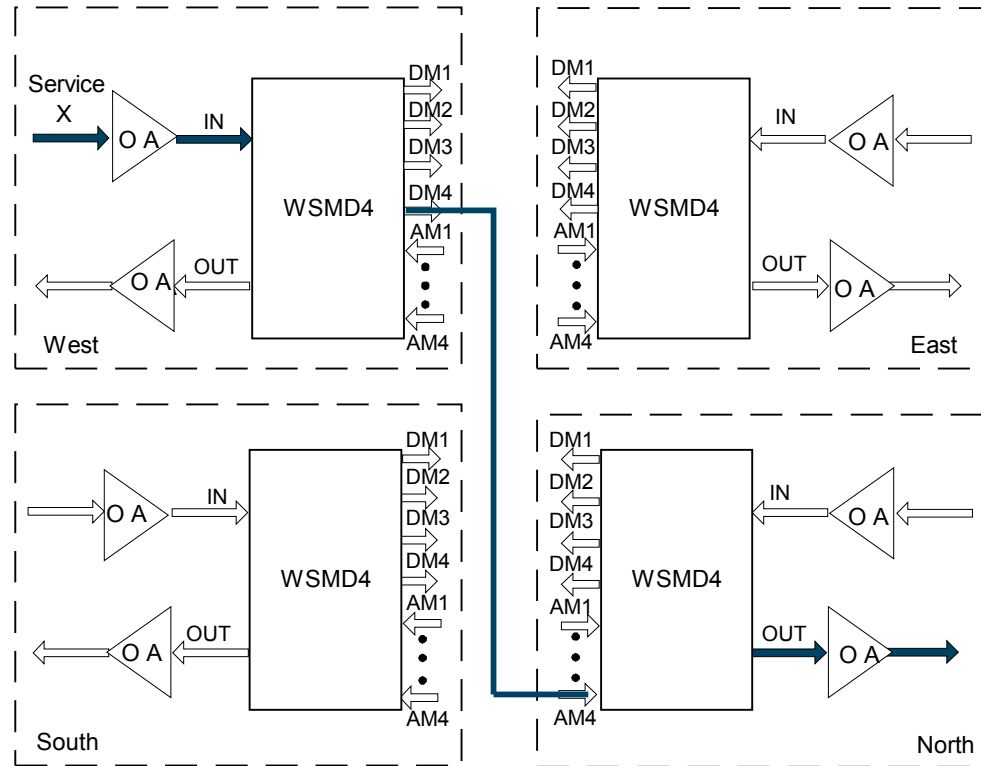
If the service signals need be output eastward, the signals from west should be input to the east WSMD4 through any of the AM1–AM4 optical interfaces. The optical signals to be added to the local node are input through any of the AM1–AM4 optical interfaces. The WSMD4 multiplexes the added optical signals with the optical signals groomed from west and outputs the multiplexed signals through the OUT optical interface. The signal is amplified and output eastward. **Figure 8-28** shows the signal flow from west to east.

Figure 8-28 Grooming from west to east



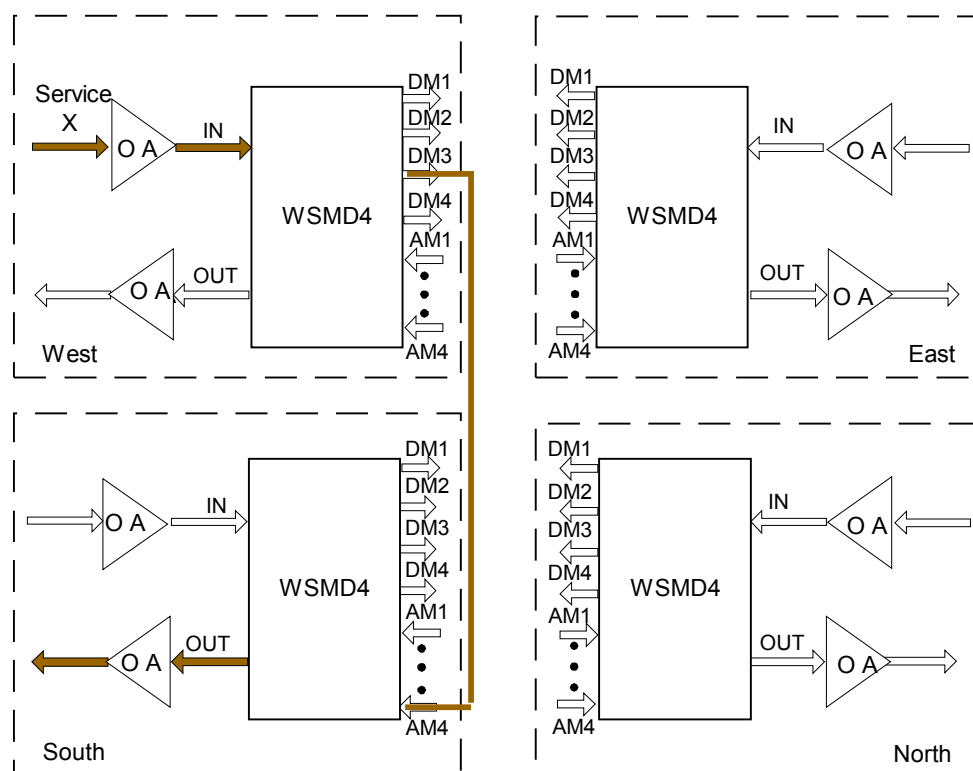
- Suppose that service X is input at Node A from west and needs to be output northward.
The main path signals containing service X is input at Node A from west. The OA amplifies the main path signals and feeds them to the WSMD4 through the IN optical interface. The optical splitter splits the signals into four channels of equal optical signals. The optical signals to be dropped from the local node are output through any of the DM1–DM4 optical interfaces of the WSMD4. The optical signals to be groomed are output through any of the DM1–DM4 optical interfaces.
If the service signals need be output northward, the signals from west should be input to the east WSMD4 through any of the AM1–AM4 optical interfaces. The optical signals to be added to the local node are input through any of the AM1–AM4 optical interfaces. The WSMD4 multiplexes the added optical signals with the optical signals groomed from west and outputs the multiplexed signals through the OUT optical interface. The signal is amplified and output northward. **Figure 8-29** shows the signal flow from west to north.

Figure 8-29 Grooming from west to north



- Suppose that service X is input at Node A from west and needs to be output southward. The main path signals containing service X is input at Node A from west. The OA amplifies the main path signals and feeds them to the WSMD4 through the IN optical interface. The optical splitter splits the signals into four channels of equal optical signals. The optical signals to be dropped from the local node are output through any of the DM1–DM4 optical interfaces of the WSMD4. The optical signals to be groomed are output through any of the DM1–DM4 optical interfaces. If the service signals need be output southward, the signals from west should be input to the east WSMD4 through any of the AM1–AM4 optical interfaces. The optical signals to be added to the local node are input through any of the AM1–AM4 optical interfaces. The WSMD4 multiplexes the added optical signals with the optical signals groomed from west and outputs the multiplexed signals through the OUT optical interface. The signal is amplified and output southward. [Figure 8-30](#) shows the signal flow from west to south.

Figure 8-30 Grooming from west to south



Dynamic Grooming

Suppose that the requirement is changed to such a case that the service X needs to be input from east and output northward. To meet the changed requirement, you just need to change the corresponding settings on the T2000.

8.2 Application and Networking of the GE ADM Feature

The GE ADM service grooming is used in the product.

8.2.1 Description

Besides GE service convergence, GE ADM services are added with GE service cross-connection and end-to-end management abilities. Hence, the cross-connection between wavelengths is realized. The L2 switching ability enables the convergence and grooming of services at the sub-wavelength level.

The OptiX Metro 6100 subrack employs the distributed cross-connection. According to the types of the subrack, the distributions of the cross-connection slots are different.

Standard Subrack

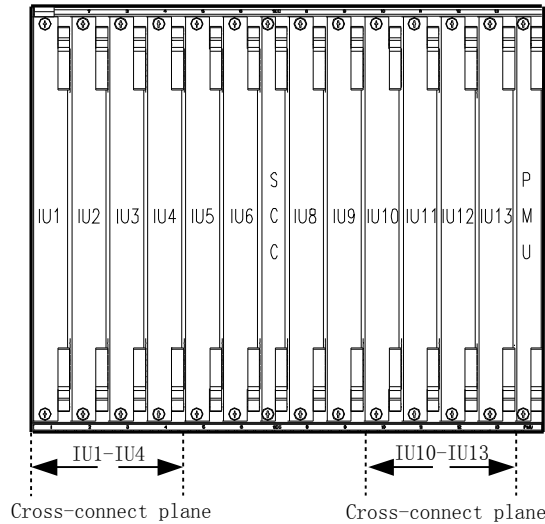
For the standard subrack is separated into two cross-connect planes by the SCC, IU1–IU4 and IU10–IU13 cross-connect planes, as shown in [Figure 8-31](#).

NOTE

The hardware platform of the OptiX Metro 6100 V100R005 uses the standard subrack.

The maximum connection bandwidth between any two boards in one plane of the OptiX Metro 6100 subrack is 8 x GE. The maximum connection bandwidth of the paired slots between the two planes is also 8 x GE.

Figure 8-31 Slots in the cross-connect planes of the standard subrack



Enhanced Subrack

For the enhanced subrack is separated into two cross-connect planes by the SCC, IU1–IU6 and IU8–IU13 cross-connect planes, as shown in [Figure 8-32](#).

NOTE

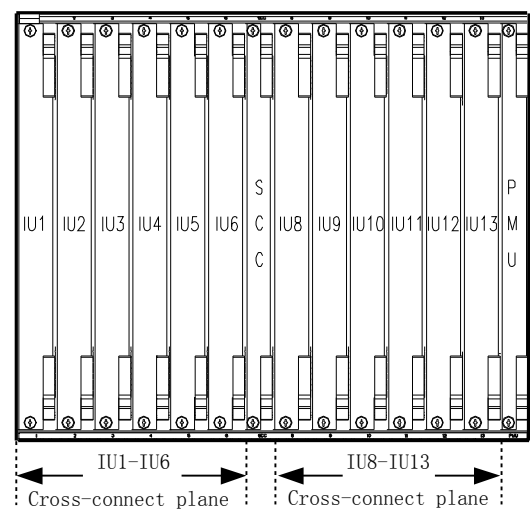
The hardware platform of the OptiX Metro 6100 V100R006 or above uses the enhanced subrack.

In the enhanced subrack, between any two service boards in the same plane, the 8 x GE full cross-connection is employed. The two planes employ the same cross-connection design. The service connection of different planes can be realized by the 8 x GE connection of the paired slots between the planes. The paired slots between the planes are described as follows:

- IU1 and IU8
- IU2 and IU9
- IU3 and IU10
- IU4 and IU11
- IU5 and IU12
- IU6 and IU13

The maximum connection bandwidth between any two boards in one plane of the OptiX Metro 6100 subrack is 8 x GE. The maximum connection bandwidth of the paired slots between the two planes is also 8 x GE.

Figure 8-32 Slots in the cross-connect planes of the enhanced subrack



Distribution of the Cross-Connection Slots

For the OptiX Metro 6100 system, the LQG, LOG, LOGS, L4G, EGS8, TBE, ELOG, ELOGS, LOM, LOMS can realize the cross-connection function. But according to the types of the subrack and the boards, the distribution of the cross-connection slots are different.

The distribution of the cross-connection slots is given in Table 8-1.

Table 8-1 Distribution of the cross-connection slots

Board	Slots for intra-board cross-connection		Slots for inter-board cross-connection	
	Standard subrack	Enhanced subrack	Standard subrack	Enhanced subrack
C6LQG	IU1–IU5, IU9–IU13	IU1–IU6, IU8–IU13	IU1–IU4, IU10–IU13	IU1–IU4, IU8–IU11
C9LQG	IU1–IU5, IU9–IU13	IU1–IU6, IU8–IU13	IU1–IU4, IU10–IU13	IU1–IU6, IU8–IU13
LOG/LOGS	IU1–IU5, IU9–IU12	IU1–IU5, IU8–IU12	IU1 and IU3, IU10 and IU12	IU1 and IU3, IU8 and IU10
L4G/EGS8/TBE/ELOG/ELOGS	IU1–IU5, IU9–IU13	IU1–IU6, IU8–IU13	IU1–IU4, IU10–IU13	IU1–IU6, IU8–IU13 or IU1 and IU8, IU2 and IU9, IU3 and IU10, IU4 and IU11, IU5 and IU12, IU6 and IU13

Board	Slots for intra-board cross-connection		Slots for inter-board cross-connection	
	Standard subrack	Enhanced subrack	Standard subrack	Enhanced subrack
LOM/LOMS	IU2–IU5, IU9–IU13	IU2–IU6, IU9–IU13	IU2 and IU4, IU11 and IU13	IU2–IU6, IU9–IU13 or IU2 and IU9, IU3 and IU10, IU4 and IU11, IU5 and IU12, IU6 and IU13

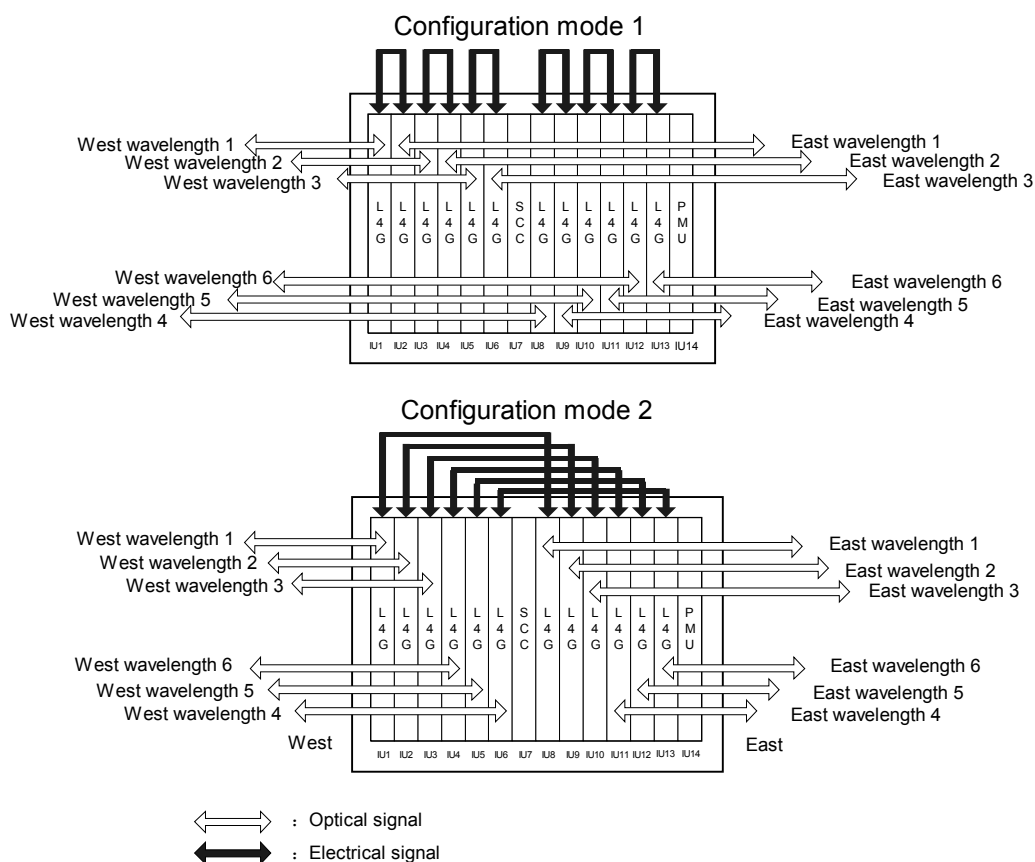
8.2.2 Networking Configuration

In the OptiX Metro 6100 DWDM systems configured with the GE ADM grooming, each wavelength must be configured with one east OTU and one west OTU to realize the optical channel protection of sub-wavelength level.

Take the L4G board and the enhanced subrack in the OptiX Metro 6100 DWDM system as an example. Two configuration modes are recommended, as shown in [Figure 8-33](#).

- In configuration mode 1, the left plane and right plane are respectively configured. Each plane supports the GE service cross-connection among three wavelengths.
- In configuration mode 2, a single cross-connect plane employs the full cross-connection of six wavelengths. At that time, the two boards, one east and one west, of the same wavelength need to be respectively configured to the corresponding paired slots that are IU1–IU8, IU2–IU9, IU3–IU10, IU4–IU11, IU5–IU12 and IU6–IU13.

Figure 8-33 GE ADM networking configuration



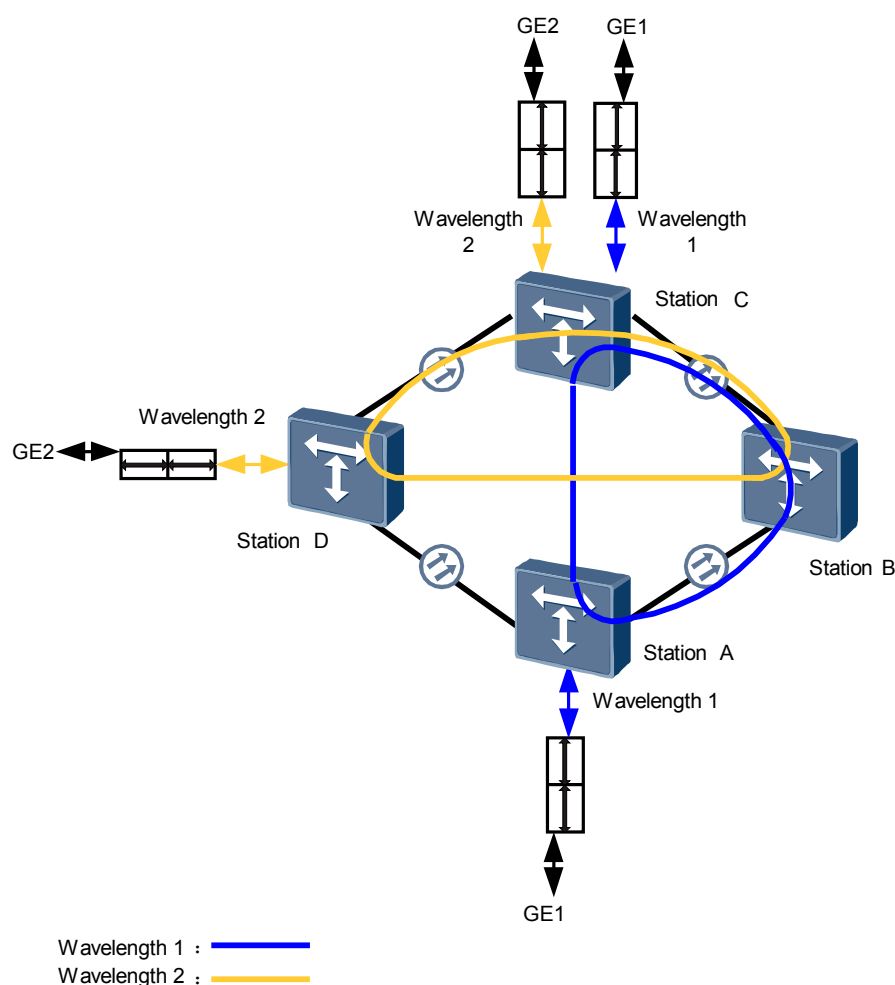
8.2.3 Application

With the cross-connection feature of the GE ADM service grooming, the product allows using multiple wavelength end to end in the network. In addition, the wavelength transformation and service grooming functions are realized. If the idle channel resource exists in the transmission line, the GE ADM service grooming can use the idle channel of multiple wavelengths among different nodes to fast create the end-to-end GE services.

Networking Example 1

The service grooming among different wavelengths of different OTUs in the same equipment can be realized through the GE ADM function. As a result, the wavelength resource can be maximally used.

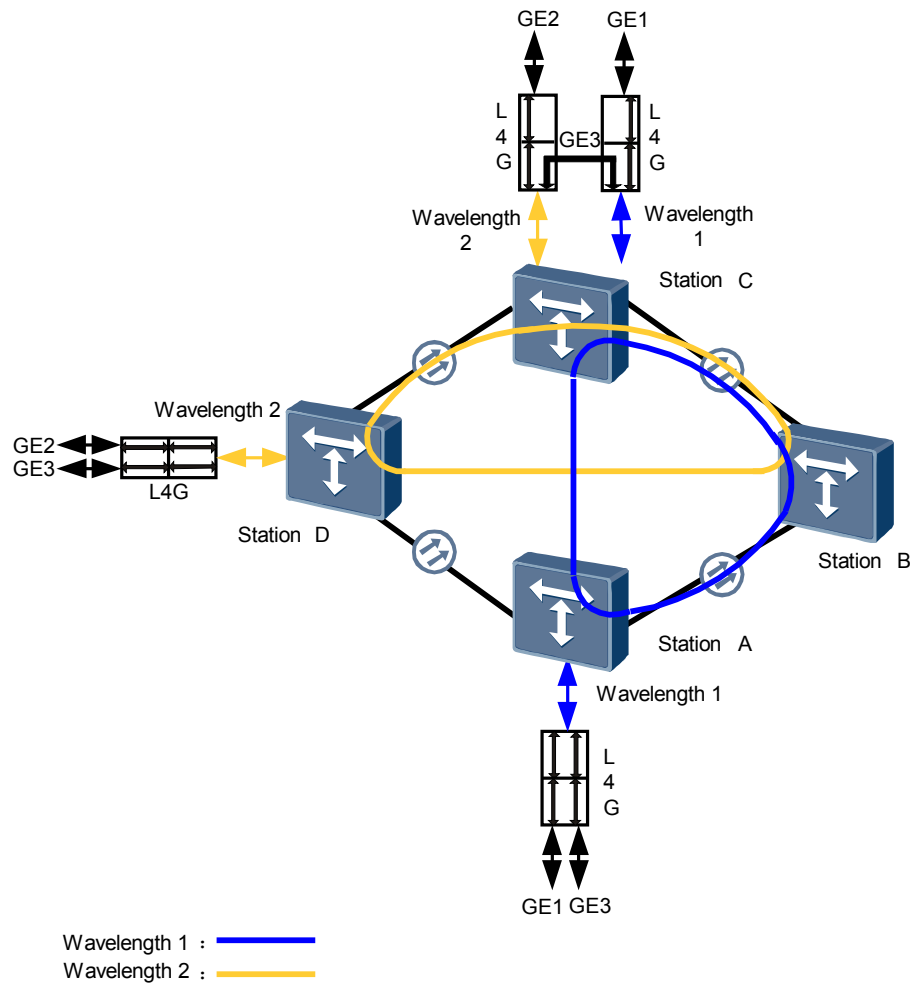
The engineering project E is the ring network carrying GE1 service signal formed by stations A, B and C and the ring network carrying GE2 service signal formed by stations B, C and D, as shown in [Figure 8-34](#). The wavelength 1 is used to connect the nodes A, B and C, and the wavelength 2 is used to connect nodes B, C and D.

Figure 8-34 Networking diagram of the engineering project E

Now the GE3 service is added between nodes A and D.

The new GE3 service can use wavelength 1 from node A to node C, wavelength 2 from node C to node D. At node C, the GE ADM service grooming for the GE3 service between the two wavelengths can be realized without affecting the running of the original GE1 and GE2 services, as shown in [Figure 8-35](#).

Figure 8-35 Networking diagram of the engineering project E after GE ADM service grooming

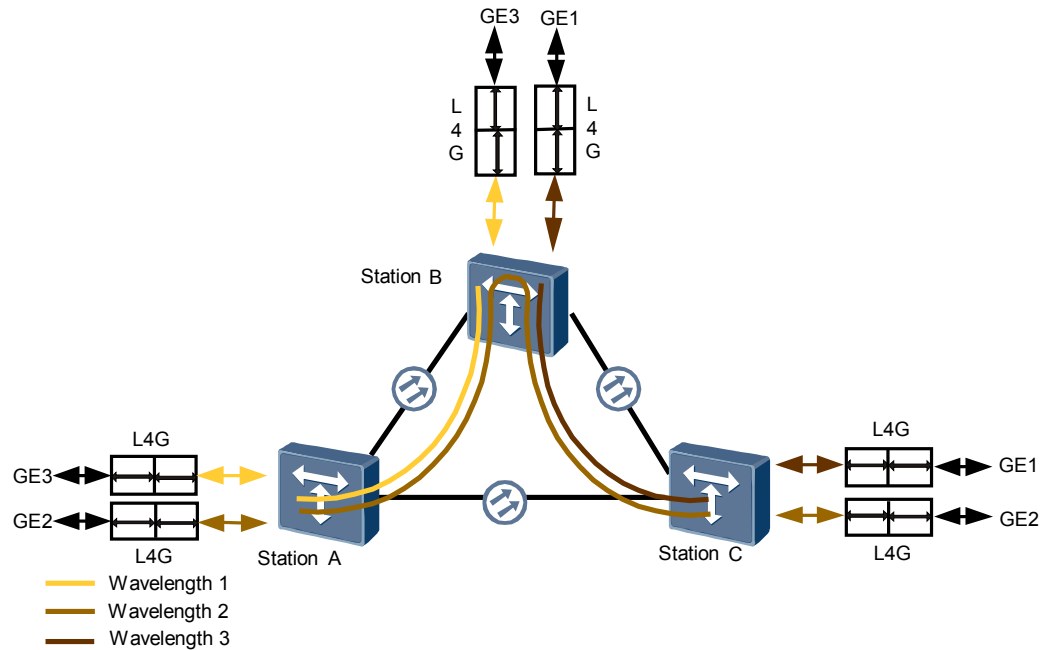


Networking Example 2

The internal service grooming of the same OTU in the same equipment can be realized through the GE ADM function. As a result, the wavelength resource can be maximally used.

The engineering project F is the ring network carrying GE1, GE2 and GE3 service signals formed by stations A, B and C, as shown in [Figure 8-36](#). The wavelength 1 is used to connect nodes A and B. The wavelength 2 is used to connect nodes A and C and it passes through the node B. The wavelength 3 is used to connect nodes B and C.

Figure 8-36 Networking diagram of the engineering project F



Now the wavelengths among the stations A, B and C are stapled to save the wavelength resource.

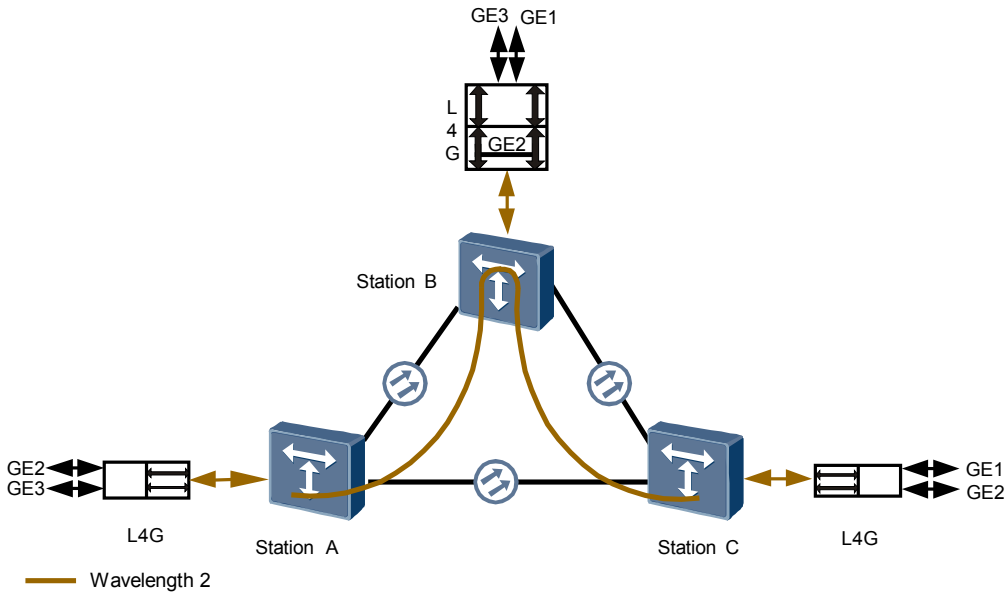
The wavelength 2 is used between the nodes A and B, and between the nodes B and C.

At node A, the GE2 and GE3 are converged in the L4G and then sent to node B through wavelength 2.

At node B, the GE2 services are sent to the L4G that adds/drops the GE1 and GE3 services. With the GE ADM, the intra-board service grooming of the GE2 is realized. Groom the GE2 service among the converged wavelengths of the GE2 and GE3 services. Then perform the convergence of the GE2 and GE1 services. After the convergence, the services are sent to node C through the wavelength 2.

In the entire process, the three channels of GE services do not influence each other, as shown in [Figure 8-37](#).

Figure 8-37 Networking diagram of the engineering project F after GE ADM service grooming



9 Protection

About This Chapter

System provides equipment level protection and network level protection.

[9.1 Equipment Level Protection](#)

The product provides various equipment level protection schemes to enhance the system reliability.

[9.2 Network Level Protection](#)

The product provides various network level protection schemes to enhance the system reliability.

[9.3 Network Management Channel](#)

The system provides protection of network management information channel and interconnection of network management information.

9.1 Equipment Level Protection

The product provides various equipment level protection schemes to enhance the system reliability.

9.1.1 DC Input Protection

The power supply system supports two $-48\text{ V}/-60\text{ V}$ DC power inputs for mutual backup. Therefore, the equipment keeps running normally in case either of the two DC inputs is faulty.

9.2 Network Level Protection

The product provides various network level protection schemes to enhance the system reliability.

9.2.1 Overview

The product provides various network level protection schemes to enhance the system reliability.

Classification

The protection schemes of the OptiX Metro 6100 system can be carried out in the following ways, as listed in [Table 9-1](#).

Table 9-1 Service protection mode supported the OptiX Metro 6100 system

Protection Classification	Protection Type
Optical line protection	Optical line protection
Optical channel protection	Intra-Board Wavelength Protection Extended Intra-Board Wavelength Protection Inter-Broad Wavelength Protection 1+1 Wavelength Protection at Client Inter-Subrack 1+1 Optical Channel Protection
Wavelength cross-connection protection	Wavelength Cross-Connection Protection
Tribute protection switching and double path protection switching	Tribute Protection Switching and Double Path Protection Switching
VLAN SNCP protection	VLAN SNCP protection
Optical wavelength shared protection	Optical Wavelength Shared Protection (OWSP) Optical Wavelength Shared Protection (DCP)

Application Scenario

The user can choose the service protection type according to the actual application, as shown in [Table 9-2](#).

Table 9-2 Application scenarios of the protection types supported by the OptiX Metro 6100

Protection Type	Application Scenario
Optical line protection	It protects the entire fiber line. It uses the dual fed selective receiving function of the OLP board and the diverse routing to provide protection for line fibers between the adjacent stations.
Intra-board wavelength protection	It protects a single OTU board with the dual fed selective receiving function. It uses the dual fed selective receiving function of a single OTU board and the diverse routing to provide protection for a service by adopting two different routings.
Extended intra-board wavelength protection	It protects a single OTU board with only one group of WDM-side transmit and receive optical interfaces. It uses the dual fed selective receiving function of the OLP/DCP/CP40 board and the diverse routing of the same wavelength to provide protection for a single OTU board with only one group of WDM-side transmit and receive optical interfaces.
Inter-board wavelength protection	It protects a single OTU board with only one group of WDM-side transmit and receive optical interfaces. The SCS board splits the signals and dually sends the service to the OTU board with only one group of WDM-side transmit and receive optical interfaces. It uses two routings for a single service to realize the protection of active path and standby path. The working OTU and the protection OTU must be in the same subrack.
Inter-subrack 1+1 optical channel protection	For the OTU board with service convergence function, it provides 1+1 protection for services on a single client side. The working OTU and the protection OTU can be in different subracks. For the OTU board without service convergence function, it provides 1+1 channel protection for a single channel. The working OTU and the protection OTU can be in different subracks.
1+1 wavelength protection at client	It protects the OTU board with service convergence function. It can protect a single client-side service. The OTU board does not support cross-connection.

Protection Type	Application Scenario
WXCP protection	It protects the OTU board with convergence and cross-connection function that can configure cross-connection and protection for a single client-side service. Dual source single sink mode does not require any SCS board. The OTU board needs to support cross-connection and to meet the requirement of slot configuration. The switching is fast. Dual source dual sink mode requires SCS board. In some networking modes, cross connection protection can be realized when the OTU board does not meet the requirement of slot configuration.
TPS protection and DPPS protection	It protects the TBE board and other OTU boards with the cross-connect grooming capacity of the GE services. The TPS protection provides the inter-board 1+1 protection for the TBE board. The DPPS protection provides the bidirectional WXCP protection for two TBE boards carrying GE services.
VLAN SNCP protection	It protects the VLAN paths of the protected client-side services on the L4G/ESG8 board. With the VLAN broadcasting function, the dual-fed and selective receiving function of VLAN paths realizes protection.
Optical wavelength shared protection (OWSP)	It protects the ring network configured with the distributed services. It provides protection for one channel of service among all stations by realizing the sharing between two wavelengths.
Optical wavelength shared protection (DCP)	It protects a ring network configured with distributed services. It uses two wavelengths to achieve shared protection for one service among all stations.

Switching Command

Arranged by priority from high to low, there are the following five protection switching commands:

- clear switching
- locked switching
- forced switching
- automatic switching
- manual switching

Automatic switching is automatically triggered by the system upon internal switching conditions.

Locked switching, forced switching, manual switching and clear switching are externally issued on the NM as means to test and maintain the system. A clear switching command can be issued on the NM to clear the preceding three external switching commands.

A clear switching command cannot be issued to automatic switching command. When the system is in the wait-to-restore (WTR) state, however, externally issuing a clear switching command can end the WTR state at once and services are switched back to the working channel.

 NOTE

The priorities of the protection switching functions are as follows:

- A protection switching fails to actuate if another protection switching of higher priority is in process.
- A protection switching succeeds to actuate if only the protection switching of lower priority is in process and the protection switching of lower priority is cleared.
- Protection switching commands of the same priority cannot be carried out at the same time. The current command must be cleared before the desired command is issued.

The protection switching commands are described as follows:

- **Automatic switching**

If the working channel is faulty while the protection channel is normal, the services are switched from working channel to the protection channel. If both channels are faulty, the services are not switched.

 NOTE

The automatic switching can be triggered by SD or SF conditions. The SF is the switching condition of the system by default. You can enable/disable the SD switching condition when creating the protection group through the T2000.

- **Lockout of protection**

This protection switching command locks the services on the working channel, no matter the working or protection channel is good or not.

- **Forced switching**

Services are switched from the current channel to the other regardless whether the working and protection channels functions normally. Services are either forcibly switched from the working channel to the protection channel or from the latter to the former.

- **Manual switching**

A manual switching command is issued to manually switch services from the working channel to the protection channel or from the latter to the former. Because the priority of manual switching is lower than that of automatic switching, the manual switching is valid only when both the working and protection channels are normal.

- **Clear switching**

A clear switching command is issued to clear any of the preceding three external commands or clear an automatic switching state at the end of the WTR time. After a clear switching command is issued, the system re-determines the service operating channel and protection switching state upon the states of the working and protection channels and the restoration mode set for the protection group.

 NOTE

When the SCC is reset, the external switching command is cleared and the switch state is the automatic switching control state. At that time, if the working and protection channels are normal or faulty, the services works in the working channel. If the working channel is faulty and the protection channel is normal, the services works in the protection channel.

Restoration Mode

The restoration mode can be set to either revertive mode or non-revertive mode.

- Revertive mode

After services are switched from the working channel to the protection channel, when the working channel is restored the services are switched back to the working channel after a certain WTR time taken to confirm that the working channel is normal.

 NOTE

- In the revertive mode, the forced switching and manual switching can switch services only from the working channel to the protection channel.
- In the revertive mode, when both the working and protection channels are normal services are switched back to the working channel after a clear switching command is issued.

- Non-revertive mode

After services are switched from the working channel to the protection channel, when the working channel is restored the services remain operating in the protection channel and are switched back to the working channel only when a fault occurs in the protection channel.

 NOTE

- Forced or manual switching commands of the same priority can be issued only once. Before such a command is re-issued, a clear switching command should be issued. This is of special concern in the case of non-revertive mode.
- In the non-revertive mode, when both the working and protection channels are normal services remain operating in the protection channel after a clear switching command is issued.

Relations of Switching States and Channel States

Table 9-3 lists the corresponding relations of switching states and channel states of the OptiX Metro 6100 system in T2000.

Table 9-3 Corresponding relations of switching states and channel states

Switching State	Channel State	Service Operating Channel
Idle	The working and protection channels are normal.	Working channel
Idle (the protection channel is working)	The working and protection channels are normal.	Protection channel
SF (SD) Switching	The working channel is normal and the protection channel is faulty.	Working channel
Auto Switching	The working channel is faulty and the protection channel is normal.	Protection channel
	The working and protection channels are faulty.	Channel to which the fault occurs in a later time
Lockout	No matter the working and protection channels are normal or not.	Working channel

Switching State	Channel State	Service Operating Channel
Forced Switching ^a	No matter the working and protection channels are normal or not.	Protection
Manual Switching ^a	The protection channel must be normal.	Protection
Forced to Working ^b	No matter the working and protection channels are normal or not.	Working channel
Forced to Protection ^b	No matter the working and protection channels are normal or not.	Protection channel
Manual to Working ^c	The working channel must be normal.	Working channel
Manual to Protection ^c	The protection channel must be normal.	Protection channel
a: When the working mode is revertive, services only can be switched from working channel to the protection channel in the forced switching or the manual switching mode. b: In the non-revertive mode, the two switching commands for forced switching have the same priority. If one command is issued, the other command cannot be issued. The current command must be cleared before the new switching command is issued. c: In the non-revertive mode, the two switching commands for manual switching have the same priority. If one command is issued, the other command cannot be issued. The current command must be cleared before the new switching command is issued.		

9.2.2 Optical Line Protection

This section describes the functionality, related units, trigger conditions, related alarms, working principle, configuration rules and application of the optical line protection.

Functionality

The optical line protection scheme adopts two pairs of fibers, one pair as the working path and the other as the protection path, to protect the line signals.

NOTE

The optical line protection belongs to the WDM-side line protection. It has no relationship with the type of the OTU adopted by the OptiX Metro 6100 system.

Related Boards

Table 9-4 lists the boards involved in the optical line protection.

Table 9-4 Boards involved in the optical line protection

Board Name	Function
OLP	Splits and couples the line signals. Detects the optical power. Performs the switching when detecting abnormal optical signals.

Trigger Conditions

The trigger conditions for the optical line protection switching are as follows:

- A MUT_LOS alarm occurs.
- A POWER_DIFF_OVER alarm occurs.

For details of the above alarms, refer to *Alarms and Performance Events Reference*.

Dependent Alarms

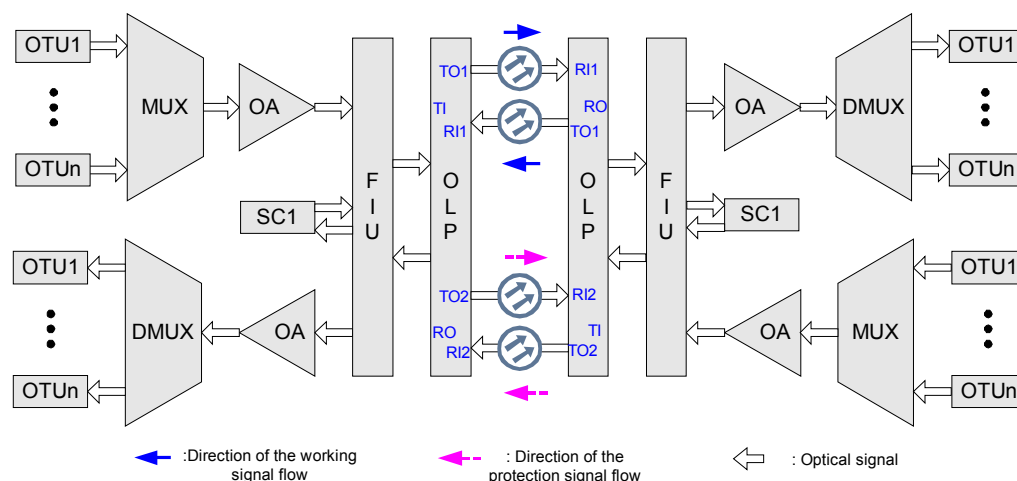
When the optical line protection succeeds, the OLP board reports the PS alarm.

Working Principle

This protection scheme adopts dual-fed selective receiving and unidirectional switching. As shown in [Figure 9-1](#), the RI1 and TO1 optical ports are connected by the working line fibers, and the RI2 and TO2 optical ports are connected by the protection line fibers.

For details about the working principle of the OLP board, refer to the *Hardware Description*.

Figure 9-1 Working principle of optical line protection



NOTE

An OTU in the transmit direction and the corresponding OTU in the receive direction at the same station are actually one physical OTU.

The OLP at the transmit end sends signals to the working line fiber and the protection line fiber at the same time. The OLP at the receive end detects the optical power of line signals, makes comparison, and sends the line signals transmitted over the working line fiber from the RI1 optical port to the FIU.

When the OLP at the receive end detects that the optical power discrepant value of the line signals transmitted is over the threshold, it sends the signals transmitted from the RI2 optical port to the FIU. The line signals are automatically switched to the protection line fiber. Although the working line fiber is abnormal, services are not interrupted.

After the recovery of the working line fiber, the OLP at the receive end detects that the optical power of the line signals transmitted over the working line fiber is normal. Based on the pre-configuration made on the NM, the line signals can be switched back to the working line fiber, or still remain in the protection line fiber.

 NOTE

The switching of the OLP is performed based on the judgment on the optical power of the paths. Thus, before using this protection function, it is required to ensure that the difference in the input optical power of the optical interfaces of the working and the protection paths is less than 3 dBm. If the power difference is more than 3 dBm and less than 5 dBm, the POWER_DIFF_DEFECT alarm is reported. If the power difference is more than 5 dBm, the POWER_DIFF_OVER alarm is reported and the switching is triggered.

Configuration Rules

The configuration rules of the optical line protection are as follows:

- The board must be configured, and the power loss of the OLP board must be considered during network design.
- After passing through different routes, the signals of a same line are received at the same receive end. Hence, during network design, ensure that the performance of the equipment providing the working route and the protection route meets the requirements of the system.
- When the power difference between the working and the protection channels exceeds the specific threshold, the system triggers the protection switching. Hence, during the network design, ensure that the input optical power values of the working and protection channels of the OLP board at the receive end are the same. If they are different, add an optical attenuator with a small attenuation value on a fiber according to the actual network situation.
- The protection can be set to be revertive or non-revertive.

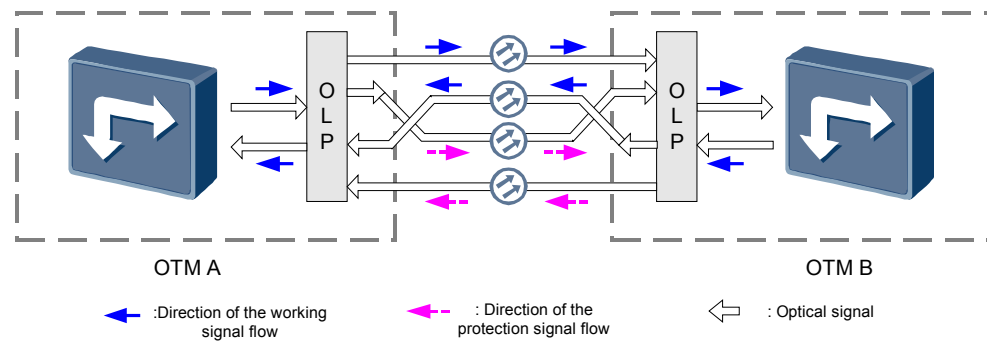
Application

With the help of the OLP, the DWDM equipment protects the transmission line on the optical layer level to improve the network performance.

The switching is finished within 100ms.

As shown in [Figure 9-2](#), station A and station B form a point-to-point network in project T. Both A and B are OTM stations. The optical line protection is adopted between the two stations. Each station is configured with an OLP board.

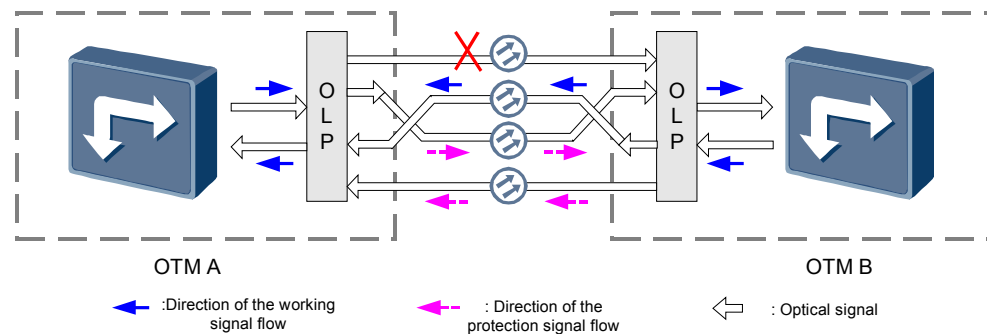
Figure 9-2 Application of optical line protection (normal)



See [Figure 9-2](#). When signals are transmitted from station A to station B, the OLP in station A sends signals over the working line fiber and the protection line fiber at the same time. The OLP in station B selects the signals transmitted over the working line fiber.

When signals are transmitted from station B to station A, the OLP in station B sends signals over the working line fiber and the protection line fiber at the same time. The OLP in station A selects the signals transmitted over the working line fiber.

Figure 9-3 Application of optical line protection (switching)



See [Figure 9-3](#). In the direction from station A to station B:

When the working line fiber breaks, the OLP in station B detects that there are no signals in the working receive direction. The board performs the switching. It selects the signals transmitted over the protection line fiber.

In the direction from station B to station A:

There is no switching because the optical power is normal, and the route of signals remains the same.

NOTE

The equipment monitors the protection line fiber in real time. When the protection line fiber breaks or degrades in performance, the equipment can detect it in time and the trouble can be shot immediately.

9.2.3 Intra-Board Wavelength Protection

This section describes the functionality, related units, trigger conditions, related alarms, working principle, configuration rules and application of the intra-board wavelength protection.

Functionality

The intra-board wavelength protection scheme adopts the OTU that has the dual-fed signal selection function to protect the client services.

Related Boards

Table 9-5 lists the boards involved in the intra-board wavelength protection.

Table 9-5 Boards involved in the intra-board wavelength protection

Board Name	Function
OTU	Splits and couples the service signals. Detects optical signals. Performs the switching when detecting abnormal optical signals.

Trigger Conditions

The trigger conditions for the intra-board wavelength protection switching are as follows:

- There is a signal failure (SF) condition and the SF is a trigger condition. SF includes the following board-side alarms:
 - R_LOS
 - R_LOF
 - OTU_LOF
 - OTU_AIS
 - ODU_AIS
 - ODU_OCI
 - ODU_LCK
- There is a signal degraded (SD) condition and the SD is a trigger condition. SD includes the following board-side alarms:
 - B1_EXC
 - SM_BIP8_OVER
 - PM_BIP8_OVER
 - B1_SD
 - SM_BIP8_SD
 - PM_BIP8_SD

Different OTUs would report different alarms. For details of the above alarms, refer to *Alarms and Performance Events Reference*.

The alarms against certain optical interface and channels of the OTU can be set on the T2000 as SD switching conditions. [Table 9-6](#) lists the alarms, against the optical interface and channels, which can be set as SD switching conditions of each OTU.

Table 9-6 Alarms relevant to SD switching conditions and the ports and channels

Board	Alarms		
	B1_SD	SM_BIP8_SD	PM_BIP8_SD
FDGD	Port 1, 2	Port 1, 2	Port 1, 2
AP8D/FCED/ C8LDG	Port 1, 2	-	-
AS8D	Port 1-10	-	-
LQS	Port 1-6	-	-
C8LQMD	Port 3-6	Port 1, 2	Port 1, 2
C9LQMD	Port 3-6	Port 1-3	Port 1-3
C9LQM2D	Port 3-6	Port 1-3, 7	Port 1-3, 7
LWX/LWM	Port 1-3	-	-
Note: If the same port supports various services, all the three alarms can be set as the SD switching conditions. When the service type is changed, the board automatically counts the corresponding bit errors and reports an SD alarm according to the actual service type.			



NOTE

If the switching is triggered by an SF condition, the switching time is 50 ms.

If the switching is triggered by an SD condition, the switching time is 50 ms. The time required for detecting SD errors are as follows:

- 90 ms when the BER is 10e-3.
- 180 ms when the BER is 10e-4.
- 1080 ms when the BER is 10e-5.

Dependent Alarms

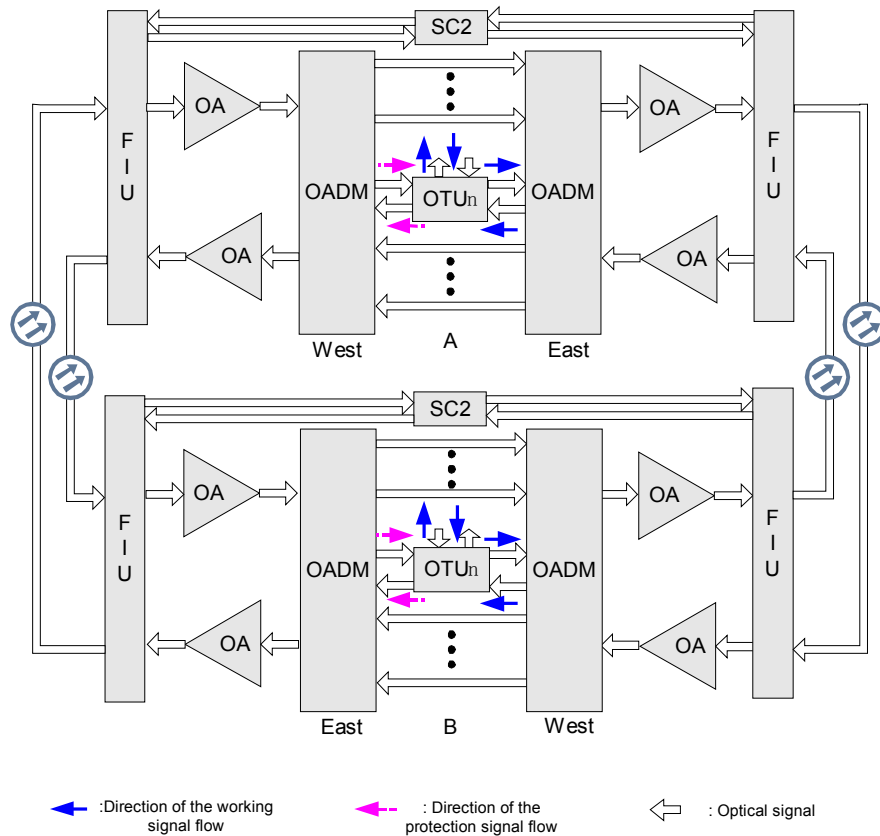
When the intra-board wavelength protection switching succeeds, the SCC board reports the OPS_PS_INDI alarm.

Working Principle

The intra-board wavelength protection is applied to protect the client service signals with different routes but in the same OTU board in a ring.

This protection scheme adopts dual-fed signal selection and unidirectional switching. It applies to ring networks. For the working principle diagram of the intra-board wavelength protection, see [Figure 9-4](#).

Figure 9-4 Working principle of intra-board wavelength protection



The OTU that has the dual-fed signal selection function divides the incoming signals at the WDM side and feeds the signals into the east OADM and the west OADM. Signals are transmitted to the receive end by different optical routes.

The WDM-side laser on the OTU that corresponds to only the working route is turned on. This OTU detects the current working channel for SF or SD and transmits signals to the client-side equipment. The WDM-side laser that corresponds to the protection route is shut down.

When the OTU at the receive end detects the failure of the signals transmitted by the working route, the corresponding client-side laser of the working route is shut down. Then the corresponding client-side laser of protection route is turned on. Only the signals transmitted by the protection route are sent to the client-side equipment.

After the recovery of the working wavelength route, service signals can be switched back to the working route or not based on the pre-configuration made on the NM.

Configuration Rules

The configuration rules of the intra-board wavelength protection are as follows:

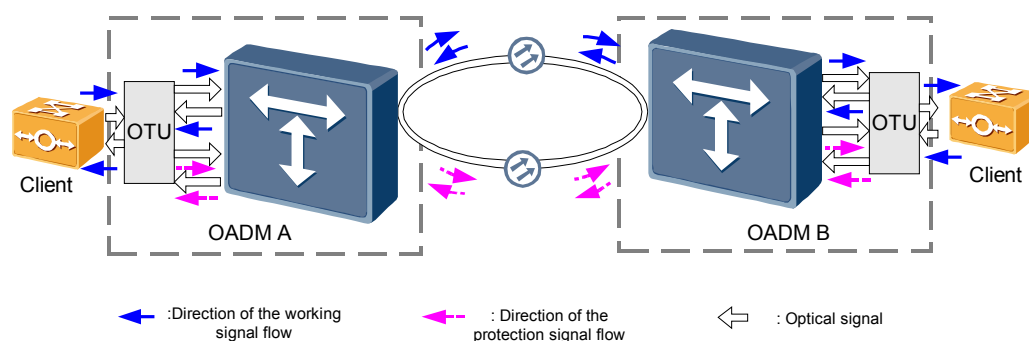
- The dual fed selective receiving OTU board must be configured.
- The protection can be set to be revertive or non-revertive.

Application

When only one wavelength is to be protected, only one OTU is required in one node. This protection scheme is of low cost. But once the OTU fails, the protection can be provided no longer.

As shown in [Figure 9-5](#), station A and station B form a ring network in project T. Both A and B are OADM stations. The intra-board wavelength protection is adopted between the two stations. Each station is configured with an OTU board (with the dual-fed signal selection function).

Figure 9-5 Application of intra-board wavelength protection (normal)

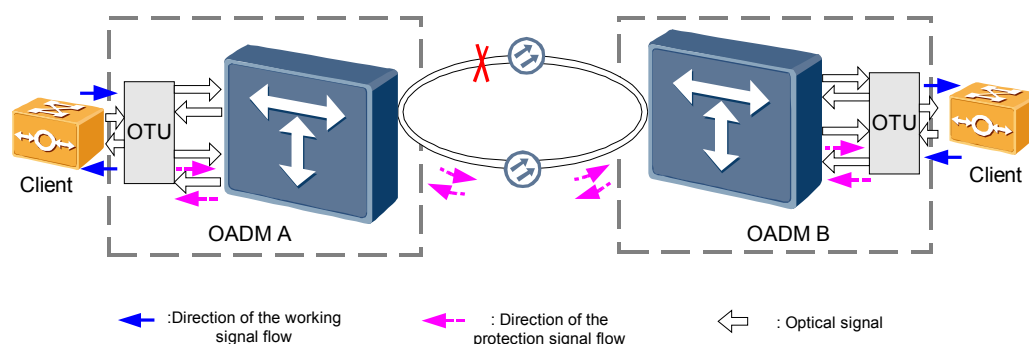


See [Figure 9-5](#).

The OTU in station A sends signals to station B by the working route and the protection route at the same time. When signals reach station B, only the signals transmitted by the working route are sent to the client-side equipment by the OTU.

Similarly, the OTU in station B sends signals to station A by two routes at the same time. In station A, only the signals transmitted by the working route are sent to the client-side equipment.

Figure 9-6 Application of intra-board wavelength protection (switching)



See [Figure 9-6](#).

When the working route between station A and station B fails, the OTU in each station shuts down the corresponding laser. Only the signals transmitted by the protection route are sent to the client-side equipment.

9.2.4 Extended Intra-Board Wavelength Protection

This section describes the functionality, related units, trigger conditions, related alarms, working principle, configuration rules and application of the extended intra-board wavelength protection.

Functionality

The extended intra-board wavelength protection adopts an OLP/DCP/CP40 board that supports dual-fed single receiving to protect the service on the WDM-side of an OTU board.

Related Boards

Table 9-7 lists the boards involved in the extended intra-board wavelength protection.

Table 9-7 Boards involved in the extended intra-board wavelength protection

Board Name	Function
OTU	Detects optical signals.
OLP/DCP ^a /CP40 ^b	Splits and couples service signals. Performs switching after the SCC issues the status of the service.
SCC	Communicates with the OTU and OLP/DCP/CP40; reports the status of the service to the OLP/DCP/CP40.
a: Only the C8DCP01 supports the extended intra-board wavelength protection. b: The CP40 board is used only for the 40 Gbit/s extended intra-board wavelength protection. The 40 Gbit/s extended intra-board wavelength protection must be configured with the CP40 board.	

Trigger Conditions

The trigger conditions for the extended intra-board wavelength protection switching are as follows:

- There is a signal failure (SF) condition and the SF is a trigger condition. SF includes the following board-side alarms:
 - R_LOS
 - R_LOF
 - OTU_LOF
 - OTU_AIS
 - ODU_AIS
 - ODU_OCI
 - ODU_LCK
 - REM_SF
- There is a signal degraded (SD) condition and the SD is a trigger condition. SD includes the following board-side alarms:

- B1_SD
- B1_EXC
- SM_BIP8_OVER
- PM_BIP8_OVER
- SM_BIP8_SD
- PM_BIP8_SD
- REM_SD

Different OTUs would report different alarms. For details of the above alarms, refer to *Alarms and Performance Events Reference*.

The alarms against certain optical interface and channels of the OTU can be set on the T2000 as SD switching conditions. [Table 9-8](#) and [Table 9-9](#) list the alarms, against the optical interface and channels, which can be set as SD switching conditions of each OTU.

Table 9-8 Alarms relevant to SD switching conditions and the Ports and channels for the less than 10 Gbit/s OTU board

Board	Alarms	
	SM_BIP8_SD	PM_BIP8_SD
L4G/LWF/LWFS/LBF/LBFS	Port 1	Port 1
C6LQG	-	-
ETMX/ETMXS	Port 1, 3-6	Port 1, 3-6
LOG/LOGS/ELOG/ELOGS/LOM/LOMS/LBE/LBES	Port 1	Port 1
TMX/TMXS	Port 1	Port 1
Note: If the same port supports various services, SM_BIP8_SD and PM_BIP8_SD can be set as the SD switching conditions. When the service type is changed, the board automatically counts the corresponding bit errors and reports an SD alarm according to the actual service type.		

Table 9-9 Alarms relevant to SD switching conditions and the ports and channels for the 40 Gbit/s OTU board

Alarms	Board	
	LU40S	TMX40S
B1_SD	Port 1	Port 3-6, channel 1 Port 1, channel 3-6
SM_BIP8_SD	Port 1	Port 3-6, channel 1 Port 1, channel 1

Alarms	Board	
	LU40S	TMX40S
PM_BIP8_SD	Port 1	Port 3–6, channel 1 Port 1, channel 1
IN_PWR_LOW	Port 1	Port 3–6, channel 1 Port 1, channel 1
IN_PWR_HIGH	Port 1	Port 3–6, channel 1 Port 1, channel 1



NOTE

If the switching is triggered by an SF condition, the switching time is 50 ms.

If the switching is triggered by an SD condition, the switching time is 50 ms. The time required for detecting SD errors are as follows:

- 90 ms when the BER is 10e-3.
- 180 ms when the BER is 10e-4.
- 1080 ms when the BER is 10e-5.

Dependent Alarms

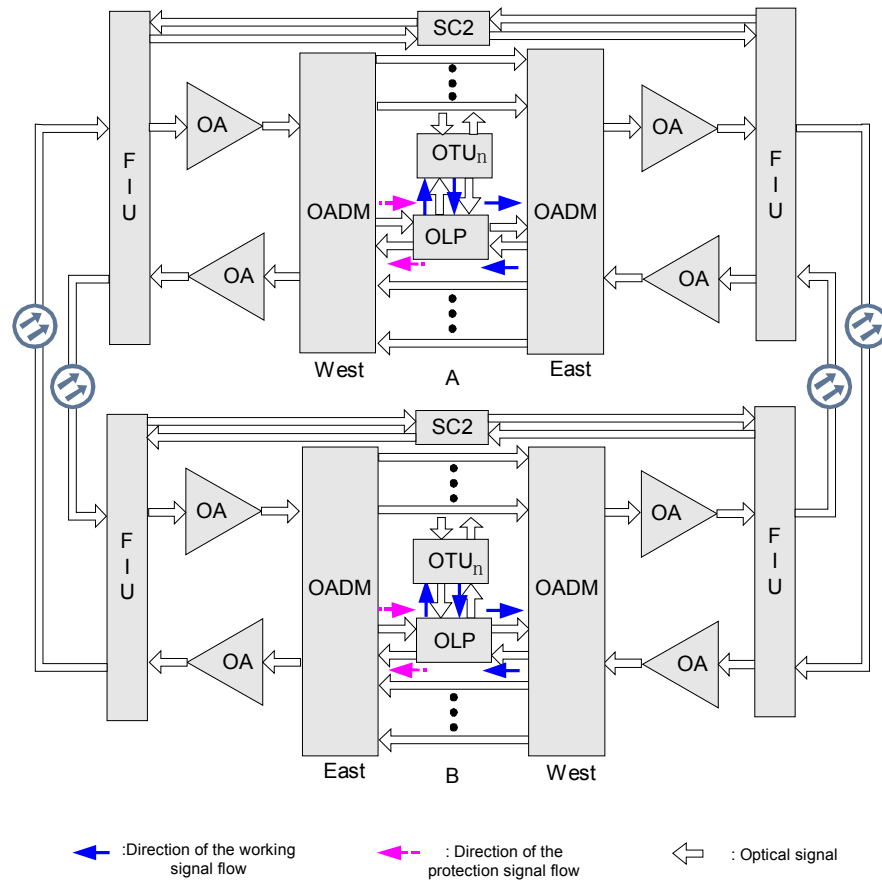
When the extended intra-board wavelength protection switching succeeds, the OLP/DCP/CP40 board reports the PS alarm.

Working Principle

Extended intra-board wavelength protection adopts dual-fed selectively receiving and unidirectional switching. As shown in [Figure 9-7](#), the RI1/TO1 optical ports of the OLP/DCP/CP40 are corresponding to the active channel and the RI2/TO2 to the standby channel.

For the working principle of the OLP/DCP/CP40, refer to the *Hardware Description*.

Figure 9-7 Principle of extended intra-board wavelength protection



NOTE

Take the OLP board as example in [Figure 9-7](#). The working principle of the DCP/CP40 board is the same as the OLP board.

In normal cases, the OLP/DCP/CP40 at the transmit end splits the services of the OTU into two and sends them respectively to the active channel and the standby channel.

The services in the active route and the standby route reach the receive end at the same time. The OLP/DCP/CP40 board at the receive end detects the status of the signals in the active channel. If the active channel is normal, the service signals travel into the OTU through the active channel and are sent to the client side.

When the OLP/DCP/CP40 board at the receive end detects that the status of the signals in the active channel is not qualified, the OLP/DCP/CP40 board automatically switches the signals to the standby channel. This ensures that the services do not fail when the active channel is abnormal.

When the power fault in the active channel is removed, the OLP/DCP/CP40 board at the receive end detects normal status in the active channel signals. According to the existing configuration in the NM system, the service signals can switch back to the active channel or remain in the standby channel.

Configuration Rules

The configuration rules of the extended intra-board wavelength protection are as follows:

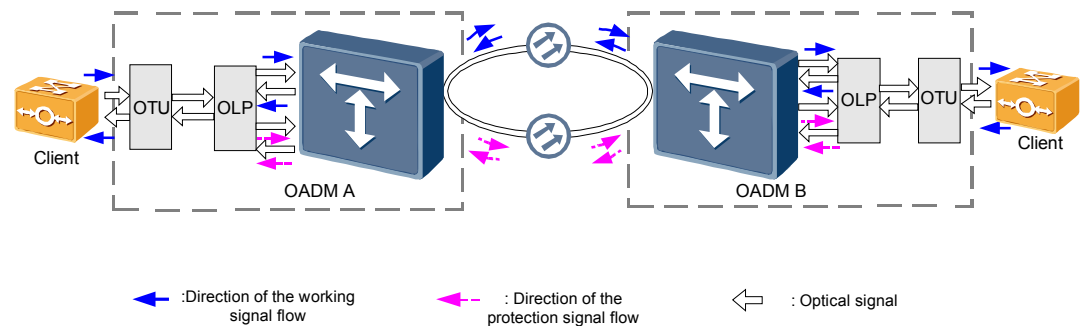
- The single fed single receiving OTU must be configured.
- The protection can be set to revertive or non-revertive.

Application

Extended intra-board wavelength protection can realize intra-board wavelength protection for the OTU that has only one group of WDM-side transmit and receive optical ports. The OLP/DCP/CP40 costs less than the OTU does, so the extended intra-board wavelength protection helps to lower the network cost.

As shown in **Figure 9-8**, Project P adopts a ring network formed by station A and B. Stations A and B are OADM stations and adopt extended intra-board wavelength protection between them. Each station is configured with an OLP and an OTU that supports dual-fed selectively receiving.

Figure 9-8 Extended intra-board wavelength protection (normal)



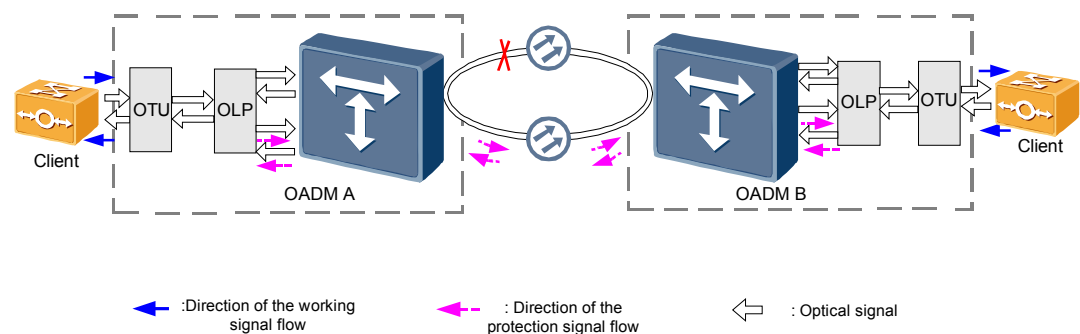
NOTE

Take the OLP board as example in **Figure 9-8**. The working principle of the DCP/CP40 board is the same as the OLP board.

As shown in **Figure 9-8**, in normal cases, the OLP of station A splits the service signals of the OTU into two, which are active signals and standby signals. They are sent to station B at the same time in an active route and a standby route. When the signals reach station B, the OLP only sends only the signals of the active route to the client side through the OTU.

In the same way, at station B, the OTU sends the signals in two routes to station A at the same time. At station A, the OLP sends only the signals of the active route to the OTU and then to the client side.

Figure 9-9 Extended intra-board wavelength protection (switched)



 NOTE

Take the OLP board as example in [Figure 9-9](#). The working principle of the DCP/CP40 board is the same as the OLP board.

As shown in [Figure 9-9](#), when the active route between station A and station B is faulty, the OLP boards of them adjust the optical switches of the OLP that are corresponding to the active routes. Only the signals from the standby routes are sent to the OTU and to the client side.

9.2.5 1+1 Wavelength Protection at Client

This section describes the functionality, related units, trigger conditions, related alarms, working principle, configuration rules and application of the 1+1 wavelength protection at client.

Functionality

To protect the client services, the 1+1 wavelength protection at client scheme adopts two OTUs that have the convergence function: one is the working OTU, and the other is the protection OTU.

Related Boards

[Table 9-10](#) lists the boards involved in the 1+1 wavelength protection at client.

Table 9-10 Boards involved in the 1+1 wavelength protection at client

Board Name	Function
SCS	Splits and couples the service signals.
OTU (OTUs support the convergence function)	Detects optical signals. Reports the information of the detected optical signals to the SCC. Turns on or shuts down the client-side laser under the control of the SCC.
SCC	Communicates with the OTU, and controls the OTU to turn on or shut down the client-side laser.

Trigger Conditions

The trigger conditions for the switching of the 1+1 wavelength protection at client are as follows:

- The board is offline, including the following situations: Removing or cold resetting the board.
- There is a signal failure (SF) condition and the SF is a trigger condition. SF includes the following board-side alarms:
 - R_LOS
 - R_LOF
 - OTU_LOF

- ODU_AIS
- ODU_OCI
- ODU_LCK
- REM_SF
- There is a signal degraded (SD) condition and the SD is a trigger condition. SD includes the following board-side alarms:
 - B1_EXC
 - SM_BIP8_OVER
 - PM_BIP8_OVER
 - REM_SD
 - B1_SD
 - SM_BIP8_SD
 - PM_BIP8_SD

Different OTUs would report different alarms. For details of the above alarms, refer to *Alarms and Performance Events Reference*.

The alarms against certain optical interface and channels of the OTU can be set on the T2000 as SD switching conditions. [Table 9-11](#) and [Table 9-12](#) list the alarms, against the optical interface and channels, which can be set as SD switching conditions of each OTU.

Table 9-11 Alarms relevant to SD switching conditions and the ports and channels

Board	Alarms		
	B1_SD	SM_BIP8_SD	PM_BIP8_SD
FDGS	Port 1	Port 1	Port 1
FDGD	Port 1, 2	Port 1, 2	Port 1, 2
AP8S/FCES	Port 1	-	-
AP8D/FCED/LDG	Port 1, 2	-	-
AS8S	Port 1, 3-10	-	-
AS8D	Port 1-10	-	-
LQS	Port 1-6	-	-
C8LQMS	Port 3-6	Port 1	Port 1
C8LQMD	Port 3-6	Port 1, 2	Port 1,2
C9LQMS	Port 3-6	Port 1, 3	Port 1, 3
C9LQMD	Port 3-6	Port 1-3	Port 1-3
C9LQM2S	Port 3-8	Port 1, 3	Port 1, 3
C9LQM2D	Port 3-8	Port 1-3, 7	Port 1-3, 7
TMX/TMXS	Port 3-6	Port 1	Port 1

Board	Alarms		
	B1_SD	SM_BIP8_SD	PM_BIP8_SD
CAETMX/ CAETMXS	Port 3-6	Port 1	Port 1
Note: If the same port supports various services, all the three alarms can be set as the SD switching conditions. When the service type is changed, the board automatically counts the corresponding bit errors and reports an SD alarm according to the actual service type.			

Table 9-12 Alarms relevant to SD switching conditions and the ports and channels for the 40 Gbit/s OTU board

Alarms	Board
	TMX40S
B1_SD	Port 3–6, channel 1 Port 1, channel 3–6
SM_BIP8_SD	Port 3–6, channel 1 Port 1, channel 1
PM_BIP8_SD	Port 3–6, channel 1 Port 1, channel 1
IN_PWR_LOW	Port 3–6, channel 1 Port 1, channel 1
IN_PWR_HIGH	Port 3–6, channel 1 Port 1, channel 1

 NOTE

If the switching is triggered by an SF condition, the switching time is 50 ms.

If the switching is triggered by an SD condition, the switching time is 50 ms. The time required for detecting SD errors are as follows:

- 90 ms when the BER is 10e-3.
- 180 ms when the BER is 10e-4.
- 1080 ms when the BER is 10e-5.

Dependent Alarms

- When the 1+1 wavelength protection switching succeeds, the SCC board reports the OPS_PS_INDI alarm.
- When the 1+1 wavelength protection switching fails, the SCC board reports the OPS_PS_FAIL alarm.

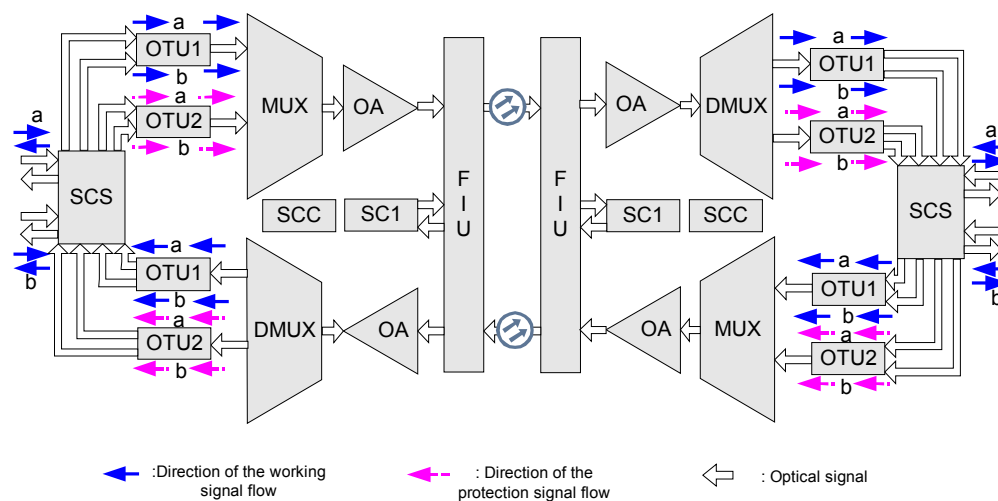
- If the attributes of the active and standby ports of the 1+1 wavelength protection group are not consistent, the SCC board reports the OPS_MAIN_BAK_ATTR_DIFF alarm.

Working Principle

The dual-fed signal selection mode is adopted. Only the receive end performs the switching. The switching of any client service will not affect other services of the same OTU. The working OTU and the protection OTU should be installed in the same subrack. For the working principle diagram of the 1+1 wavelength protection at client, see [Figure 9-10](#).

For details about the working principle of the SCS board, refer to the *Hardware Description*.

Figure 9-10 Working principle of 1+1 wavelength protection at client



NOTE

- An OTU in the transmit direction and the corresponding OTU in the receive direction at the same station, are actually one physical OTU.
- The SCS in the transmit direction and the SCS in the receive direction at the same station, are actually one physical SCS.
- The SCC board communicates with the OTU through the backplane.

The SCS board at the transmit end splits each signal into two channels. Then it sends them to the working OTU (OTU1) and the protection OTU (OTU2).

The services transmitted by the working and the protection wavelength routes reach the receive end at the same time. The SCC at the receive end controls the OTU1 and the OTU2 based on the detection information reported by the OTU. All the client-side lasers of the OTU1 function normally. All the client-side lasers of the OTU2 are shut down. Only the signals transmitted by the working wavelength channel are sent to the client-side equipment.

When the WDM-side laser of the OTU1 at the receive end detects alarms (such as LOS or LOF) that trigger the switching, the SCC shuts down all the client-side lasers that control the 1+1 wavelength protection configured on the client side of the OTU1 and turns on all the client-side lasers that control the 1+1 wavelength protection configured on the client side of the OTU2. All client services are transmitted over the protection wavelength route.

When a client-side laser of the OTU1 at the receive end detects the alarms (such as LOS or LOF) that trigger the switching, the SCC board directs the OTU1 to shut down the corresponding

client-side laser. The client-side laser of the OTU2 for this channel is enabled. Thus, for this channel of services, only the signals transmitted by the protection wavelength route are sent to the SCS board. All the other channels of signals are not switched to the protection wavelength route. They are still transmitted by the working wavelength route.

After the recovery of the working wavelength route, service signals transmitted by the protection wavelength route can be switched back to the OTU1 or not based on the pre-configuration made on the NM.

The client-side protection can be seen as a subset of inter-board wavelength protection. When a protection switching occurs, only a part of the client-side services are switched to the protection OTU. Other client-side services are not affected. As a result, there is no need to switch the services.

When a channel of client signal received by the working OTU fails, only this channel is switched. There is no switching at the WDM side.

The working OTU in the opposite end shut down the client-side transmit laser for this failed channel. The protection OTU at the opposite end turns on the corresponding client-side transmit laser. The other normal signals are still transmitted through the working OTU.

 NOTE

The channel protection pair should be set on the NM to achieve the 1+1 wavelength protection at client. The boards with intra-board or inter-board cross-connect grooming capacity must be configured with cross-connect services on the NM before performing the service grooming. So these boards must be fully configured with pass-through cross-connections when configured with the protection.

Configuration Rules

The configuration rules of the 1+1 wavelength protection at client are as follows:

- The SCS must be configured.
- The working OTU and protection OTU must be configured in the same subrack while the SCS board can be configured in another subrack.
- The convergence type of the protected services of the OTU board must be configured.
- The protection can be set to revertive or non-revertive.

Application

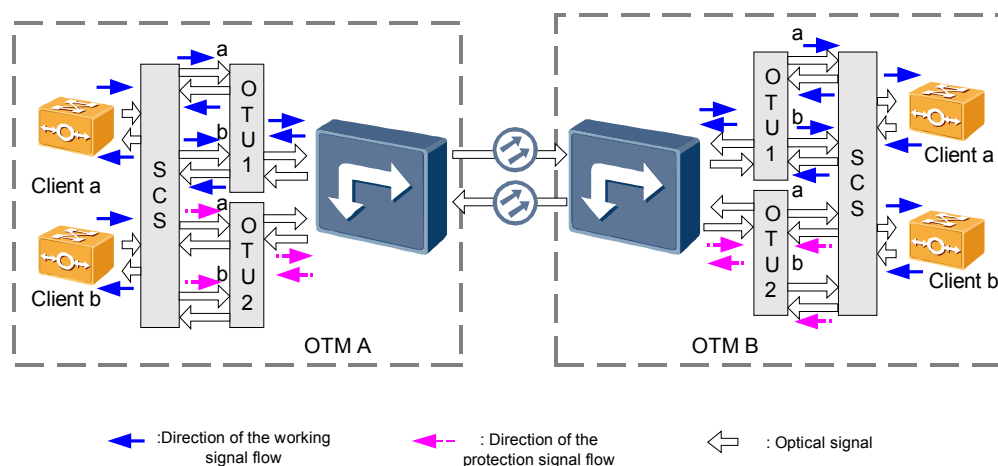
The 1+1 wavelength protection at client only applies to the OTUs that have the convergence function. When a protection switching occurs, only a part of the client-side services are switched to the protection OTU. Other client-side services are not affected. As a result, there is no need to switch the services.

Applications of the 1+1 wavelength protection at client are as follows:

- Protection for the multiple client service signals with the same route and in different OTUs in a chain.
- Protection for the multiple client service signals with different routes and in different OTUs in a ring.

As shown in [Figure 9-11](#), station A and station B form a point-to-point network in Project T. Both A and B are OTM stations. The 1+1 wavelength protection at client is adopted between the two stations. Each station is configured with one SCS board and two OTUs that have the convergence function.

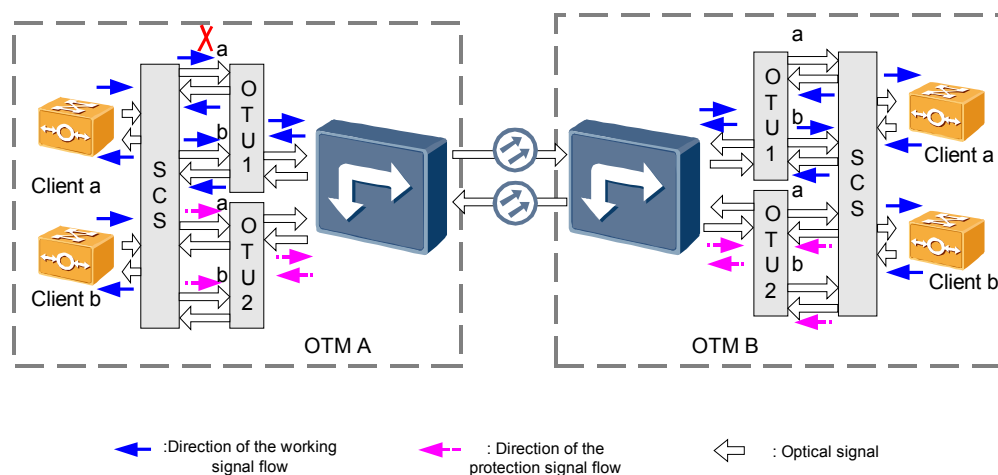
Figure 9-11 Application of 1+1 wavelength protection at client (normal)



See [Figure 9-11](#). When signals are transmitted from station A to station B, the SCS board in station A sends signals to the working OTU (OTU1) and the protection OTU (OTU2) at the same time. When signals reach the OTU1 and OTU2 in station B, only the signals sent from the OTU1 are transmitted to the client-side equipment through the SCS board.

When signals are transmitted from station B to station A, the SCS board in station B sends signals to the OTU1 and the OTU2 at the same time. In station A, only the signals sent from the OTU1 are transmitted to the client-side equipment through the SCS board.

Figure 9-12 Application of 1+1 wavelength protection at client (normal)



See [Figure 9-12](#). In the direction from station A to station B:

In station A, the fiber (used to transmit client service a) between the SCS board and the input port of the OTU1 breaks. After detection and control, only the OTU2 in station B sends the client-side signals to the client-side equipment through the SCS board. Client service b is still transmitted by the original route.

In the direction from station B to station A:

There is no switching because the working wavelength route is normal, and the route of signals remains the same.

If the carried services are GE or FC services, the protected services between station A and station B are switched at both ends. Bi-directional service routes are all switched from the working wavelength route to the protection wavelength route.

9.2.6 Inter-Board Wavelength Protection

This section describes the functionality, related units, trigger conditions, related alarms, working principle, configuration rules and application of the inter-board wavelength protection.

Functionality

The inter-board wavelength protection scheme adopts two wavelengths: one is the working wavelength, and the other is the protection wavelength. The two wavelengths adopt different routes to transmit signals to protect the client services.



NOTE

In the case of convergence boards, it is recommended that you configure the client-side 1+1 wavelength protection.

Related Boards

Table 9-13 lists the boards involved in the inter-board wavelength protection.

Table 9-13 Boards involved in the inter-board wavelength protection

Board Name	Function
SCS	Splits and couples the service signals.
OTU	Detects the optical signals. Reports the information of the detected optical signals to the SCC. Turns on or shuts down the client-side laser under the control of the SCC.
SCC	Communicates with the OTU, and controls the OTU to turn on or shut down the client-side laser.



CAUTION

When the OTUs are configured with the inter-board wavelength protection, the OTUs supporting SuperWDM and the OTUs not supporting SuperWDM cannot be set into one protection group.

Trigger Conditions

The trigger conditions for the inter-board wavelength protection switching are as follows:

- The board is offline, including the following situations: Removing or cold resetting the board.
- There is a signal failure (SF) condition and the SF is a trigger condition. SF includes the following board-side alarms:
 - R_LOS
 - R_LOF
 - OTU_LOF
 - ODU_AIS
 - ODU_OCI
 - ODU_LCK
 - REM_SF
- There is a signal degraded (SD) condition and the SD is a trigger condition. SD includes the following board-side alarms:
 - B1_EXC
 - SM_BIP8_OVER
 - PM_BIP8_OVER
 - REM_SD
 - B1_SD
 - SM_BIP8_SD
 - PM_BIP8_SD

Different OTUs would report different alarms. For details of the above alarms, refer to *Alarms and Performance Events Reference*.

The alarms against certain optical interface and channels of the OTU can be set on the T2000 as SD switching conditions. [Table 9-14](#) and [Table 9-15](#) list the alarms, against the optical interface and channels, which can be set as SD switching conditions of each OTU.

Table 9-14 Alarms relevant to SD switching conditions and the Ports and channels

Board	Alarms		
	B1_SD	SM_BIP8_SD	PM_BIP8_SD
FDGS/L4G/LWF/LWFS/LBF/LBFS/LW40	Port 1	Port 1	Port 1
LWC1	Port 1, 2	Port 1, 2	Port 1, 2
AP8S/FCES/LQG	Port 1	-	-
LDG	Port 1, 2	-	-
AS8S	Port 1, 3-10	-	-
LQS	Port 1-6	-	-
ETMX/ETMXS	Port 3-6	Port 1, 3-6	Port 1, 3-6
C8LQMS	Port 3-6	Port 1	Port 1

Board	Alarms		
	B1_SD	SM_BIP8_SD	PM_BIP8_SD
C9LQMS	Port 3-6	Port 1, 3	Port 1, 3
C9LQM2S	Port 3-8	Port 1, 3	Port 1, 3
LWX/LWM	Port 1-3	-	-
LOG/LOGS/ELOG/ ELOGS/LOM/ LOMS/LBE/LBES	-	Port 1	Port 1
TMX/TMXS	Port 3-6	Port 1	Port 1
Note: If the same port supports various services, all the three alarms can be set as the SD switching conditions. When the service type is changed, the board automatically counts the corresponding bit errors and reports an SD alarm according to the actual service type.			

Table 9-15 Alarms relevant to SD switching conditions and the ports and channels for the 40 Gbit/s OTU board

Alarms	Board	
	LU40S	TMX40S
B1_SD	Port 1	Port 3-6, channel 1 Port 1, channel 3-6
SM_BIP8_SD	Port 1	Port 3-6, channel 1 Port 1, channel 1
PM_BIP8_SD	Port 1	Port 3-6, channel 1 Port 1, channel 1
IN_PWR_LOW	Port 1	Port 3-6, channel 1 Port 1, channel 1
IN_PWR_HIGH	Port 1	Port 3-6, channel 1 Port 1, channel 1

 NOTE

If the switching is triggered by an SF condition, the switching time is 50 ms.

If the switching is triggered by an SD condition, the switching time is 50 ms. The time required for detecting SD errors are as follows:

- 90 ms when the BER is 10e-3.
- 180 ms when the BER is 10e-4.
- 1080 ms when the BER is 10e-5.

Dependent Alarms

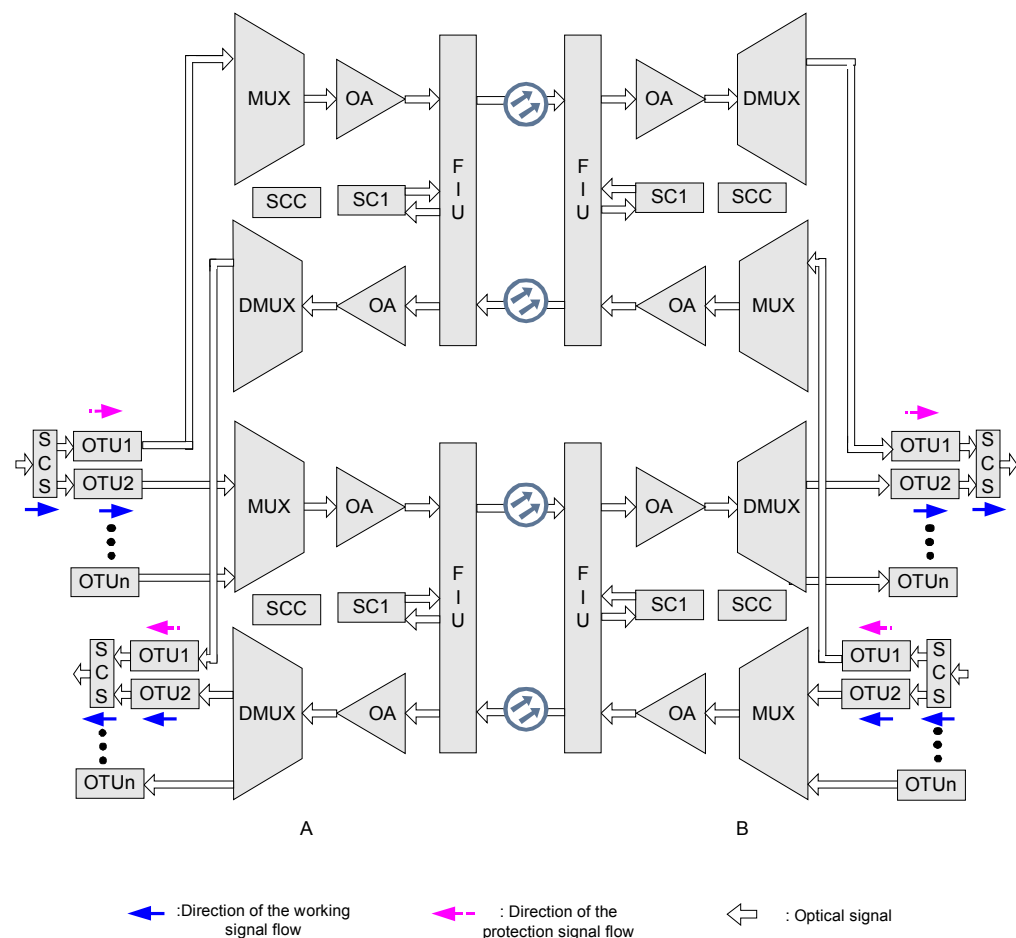
- When the inter-board wavelength protection switching succeeds, the SCC board reports the OPS_PS_INDI alarm.
- When the inter-board wavelength protection switching fails, the SCC board reports the OPS_PS_FAIL alarm.
- If the attributes of the active and standby ports of the inter-board wavelength protection group are not consistent, the SCC board reports the OPS_MAIN_BAK_ATTR_DIFF alarm.

Working Principle

This protection scheme adopts dual-fed single selection. The working OTU and the protection OTU are required to be installed in the same subrack. For the working principle diagram of the inter-board wavelength protection, see [Figure 9-13](#). The signals carried by the working wavelength and those carried by the protection wavelength reach the receive end by different routes.

For details about the working principle of the SCS, refer to the *Hardware Description*.

Figure 9-13 Working principle of the inter-board wavelength protection



 NOTE

- An OTU in the transmit direction and the corresponding OTU in the receive direction at the same station, are actually one physical OTU. An SCS in the transmit direction and the SCS in the receive direction at the same station, as shown also in this figure, are actually one physical SCS.
- The SCC communicates with the OTU through the backplane.

In normal conditions, the SCS at the transmit end divides the incoming client signals and feeds the signals into the working OTU (OTU2) and the protection OTU (OTU1).

The signals carried by the working wavelength and those carried by the protection wavelength reach the receive end at the same time. The SCC at the receive end controls the OTU1 and OTU2 based on the detection information reported by the OTU. The client-side laser of the OTU2 works normally. The client-side laser of the OTU1 is shut down. Only the signals carried by the working wavelength are transmitted to the SCS. The SCS sends the working signals to the client-side equipment.

When the OTU2 at the receive end detects the failure of the signals carried by the working wavelength, the SCC directs the OTU1 to turn on its client-side laser. The client-side laser of the OTU2 is shut down. Only the signals carried by the protection wavelength are transmitted to the SCS. The SCS sends the protection signals to the client-side equipment.

After the recovery of the working wavelength route, service signals can be switched back to the OTU2 or not based on the pre-configuration made on the NM.

**CAUTION**

The channel protection pair should be set by using the NM to achieve the inter-board wavelength protection.

The boards with intra-board or inter-board cross-connect grooming capacity must be configured with cross-connect services on the NM before performing the service grooming. So these boards must be fully configured with pass-through cross-connections when configured with the protection.

Configuration Rules

The configuration rules of the inter-board wavelength protection are as follows:

- In the case of convergence boards, it is recommended that you configure the 1+1 wavelength protection at client.
- The working OTU and protection OTU must be configured in the same subrack while the SCS board can be configured in another subrack.
- The OTU of different types cannot be configured in the same protection group.
- The OTU supporting SuperWDM is not recommended to configure with the OTU that does not support SuperWDM. For example, the LWF board and the LWFS board.
- The protection can be set to revertive or non-revertive.

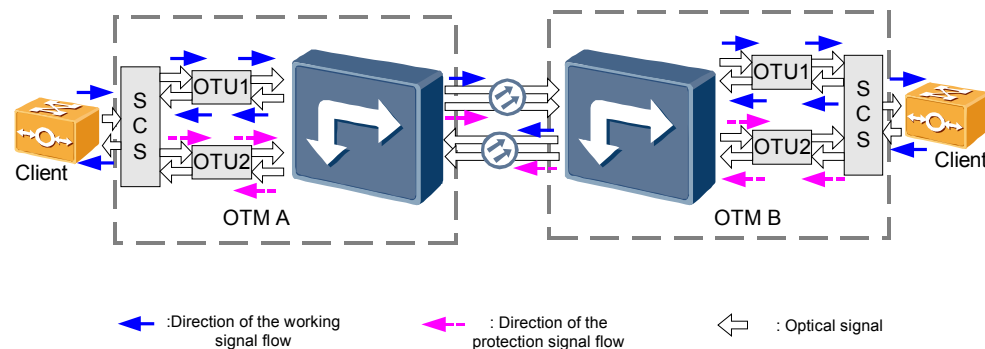
Application

The merit of this protection scheme is that even the OTU goes faulty, the protection can still function well, though double OTUs are required for protection.

The inter-board wavelength protection applies to ring and point-to-point networks.

Take that the signals carried by the working wavelength and those carried by the protection wavelength reach the receive end by different routes as an example to introduce the inter-board wavelength protection. As shown in **Figure 9-14**, station A and station B form a point-to-point network in Project T. Both A and B are OTM stations. The inter-board wavelength protection is adopted between the two stations. Each station is configured with one SCS board and two OTUs. The inter-board wavelength protection is adopted between the two stations. Each station is configured with one SCS board and two OTUs.

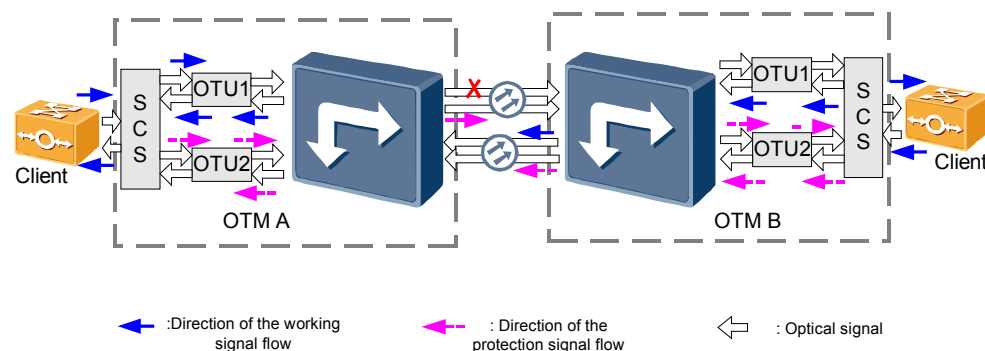
Figure 9-14 Application of inter-board wavelength protection (normal)



See **Figure 9-14**. When signals are transmitted from station A to station B, the SCS in station A sends signals to the working OTU (OTU1) and the protection OTU (OTU2) at the same time. When signals are received by the OTU1 and the OTU2 in station B, only the signals transmitted by the OTU1 are sent to the client-side equipment by the SCS.

When signals are transmitted from station B to station A, the SCS in station B sends the signals to the OTU1 and the OTU2 at the same time. In station A, only the signals transmitted by the OTU1 are sent to the client-side equipment by the SCS.

Figure 9-15 Application of inter-board wavelength protection (normal)



See **Figure 9-15**. In the direction from station A to station B:

When a fiber cut occurs to the working route from station A to station B, only the OTU2 on the protection route in station B sends the client-side signals to the client-side equipment through the SCS board.

In the direction from station B to station A:

There is no switching because the working wavelength route is normal, and the route of signals remains the same.

9.2.7 Inter-Subrack 1+1 Optical Channel Protection

This section describes the functionality, related units, trigger conditions, related alarms, working principle, configuration rules and application of the inter-subrack 1+1 optical channel protection.

Functionality

The inter-subrack 1+1 optical channel protection scheme adopts two wavelengths: one is the working wavelength, and the other is the protection wavelength. The two wavelengths adopt different routes located in the different subracks to transmit signals to protect the client services.



NOTE

The TBE does not support the inter-subrack 1+1 optical channel protection.

Related Boards

Table 9-16 lists the boards involved in the inter-subrack 1+1 optical channel protection.

Table 9-16 Boards involved in the inter-subrack 1+1 optical channel protection

Board Name	Function
OLP, DCP	Splits and couples the service signals. Detects optical power. Performs the switching when detecting abnormal optical power.
OTU	Detects optical signals. Reports the information of the detected optical signals to the SCC. Turns on or shuts down the client-side laser under the control of the SCC.
SCC	Communicates with the OTU, and controls the OTU to turn on or shut down the client-side laser.

Trigger Conditions

The trigger conditions for the inter-subrack 1+1 optical channel protection switching are as follows:

- The board is offline, including the following situations: Removing or cold resetting the board.
- A POWER_DIFF_OVER alarm occurs.
- There is a signal failure (SF) condition and the SF is a trigger condition. SF includes the following board-side alarms:
 - R_LOS
 - R_LOF

- OTU_LOF
- ODU_AIS
- ODU_OCI
- ODU_LCK
- REM_SF
- There is a signal degraded (SD) condition and the SD is a trigger condition. SD includes the following board-side alarms:
 - B1_EXC
 - SM_BIP8_OVER
 - PM_BIP8_OVER
 - REM_SD
 - B1_SD
 - SM_BIP8_SD
 - PM_BIP8_SD

Different OTUs would report different alarms. For details of the above alarms, refer to *Alarms and Performance Events Reference*.

The alarms against certain optical interface and channels of the OTU can be set on the T2000 as SD switching conditions. [Table 9-17](#) and [Table 9-18](#) list the alarms, against the optical interface and channels, which can be set as SD switching conditions of each OTU.

Table 9-17 Alarms relevant to SD switching conditions and the Ports and channels

Board	Alarms		
	B1_SD	SM_BIP8_SD	PM_BIP8_SD
FDGS/L4G/LWF/ LWFS/LBF/LBFS/ LW40	Port 1	Port 1	Port 1
FDGD/LWC1	Port 1, 2	Port 1, 2	Port 1, 2
AP8S/FCES/LQG	Port 1	-	-
AP8D/FCED/LDG	Port 1, 2	-	-
AS8S	Port 1, 3-10	-	-
AS8D	Port 1-10	-	-
LQS	Port 1-6	-	-
ETMX/ETMXS	Port 3-6	Port 1, 3-6	Port 1, 3-6
C8LQMS	Port 3-6	Port 1	Port 1
C8LQMD	Port 3-6	Port 1, 2	Port 1, 2
C9LQMS	Port 3-6	Port 1, 3	Port 1, 3
C9LQMD	Port 3-6	Port 1-3	Port 1-3

Board	Alarms		
	B1_SD	SM_BIP8_SD	PM_BIP8_SD
C9LQM2S	Port 3-8	Port 1-3	Port 1-3
C9LQM2D	Port 3-8	Port 1-3, 7	Port 1-3, 7
LWC1D	Port 1-3	Port 1-3	Port 1-3
LWX/LWM	Port 1-3	-	-
LOG/LOGS/ELOG/ ELOGS/LOM/ LOMS/LBE/LBES	-	Port 1	Port 1
TMX/TMXS	Port 3-6	Port 1	Port 1
Note: If the same port supports various services, all the three alarms can be set as the SD switching conditions. When the service type is changed, the board automatically counts the corresponding bit errors and reports an SD alarm according to the actual service type.			

Table 9-18 Alarms relevant to SD switching conditions and the ports and channels for the 40 Gbit/s OTU board

Alarms	Board	
	LU40S	TMX40S
B1_SD	Port 1	Port 3-6, channel 1 Port 1, channel 3-6
SM_BIP8_SD	Port 1	Port 3-6, channel 1 Port 1, channel 1
PM_BIP8_SD	Port 1	Port 3-6, channel 1 Port 1, channel 1
IN_PWR_LOW	Port 1	Port 3-6, channel 1 Port 1, channel 1
IN_PWR_HIGH	Port 1	Port 3-6, channel 1 Port 1, channel 1



NOTE

If the switching is triggered by an SF condition, the switching time is 50 ms.

If the switching is triggered by an SD condition, the switching time is 50 ms. The time required for detecting SD errors are as follows:

- 90 ms when the BER is $10e-3$.
- 180 ms when the BER is $10e-4$.
- 1080 ms when the BER is $10e-5$.

Dependent Alarms

- When the inter-subrack 1+1 optical channel protection switching succeeds, the OLP or DCP board reports the PS alarm.
- If the attributes of the active and standby ports of the inter-subrack 1+1 optical channel protection group are not consistent, the SCC board reports the OPS_MAIN_BAK_ATTR_DIFF alarm.

Working Principle

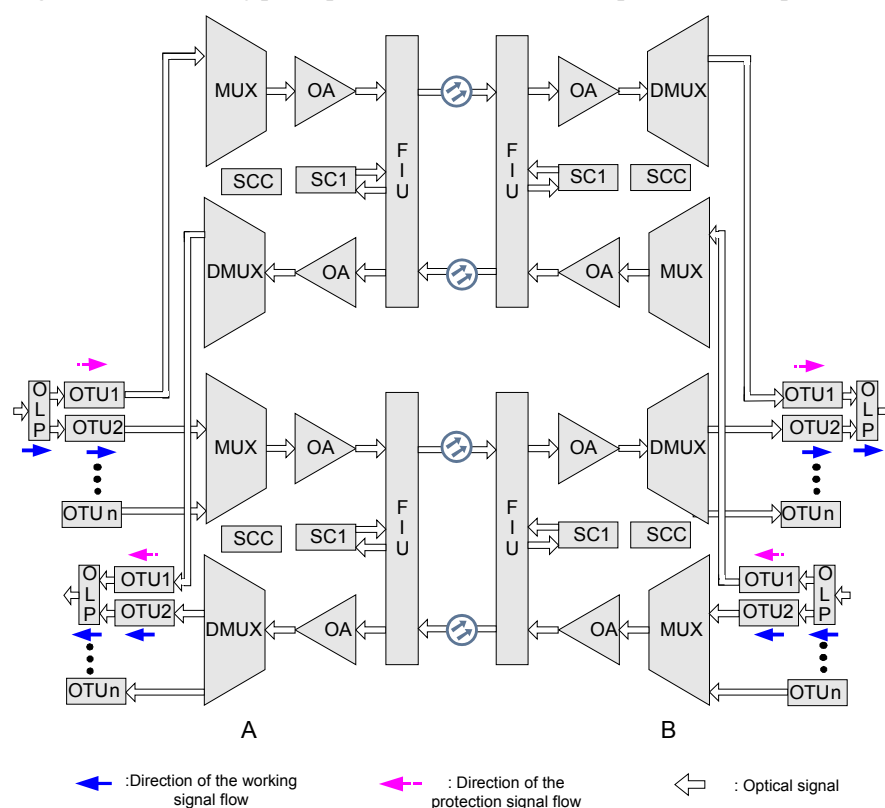
This protection scheme adopts dual-fed signal selection. The working OTU and the protection OTU can be installed in different subracks.

With either the OLP or the DCP, the working principle of the protection is the same. The difference is that, the OLP protects one channel of signals; the DCP protects two channels of signals at the same time.

For details about the working principle of the OLP and that of the DCP, refer to the *Hardware Description*.

In the following example of the inter-subrack 1+1 optical channel protection, the OLP is adopted. See [Figure 9-16](#). At that time, the working OTU and the protection OTU can be in different subracks. The signals carried by the working wavelength and those carried by the protection wavelength can reach the receive end by different routes.

Figure 9-16 Working principle of inter-subrack 1+1 optical channel protection



NOTE

- An OTU in the transmit direction and the corresponding OTU in the receive direction at the same station, are actually one physical OTU. An OLP in the transmit direction and its corresponding OLP in the receive direction at the same station, are actually one physical OLP.
- The SCC unit communicates with the OTU through the backplane.

Normally, the OLP at the transmit end divides the incoming client signals and feeds the signals into the working OTU (OTU2) and the protection OTU (OTU1).

Signals transmitted by the working and the protection wavelength routes reach the receive end. The OTU detects the signals. If the signals are normal, both the working and the protection OTUs send signals to the OLP. The OLP compares the optical power of the signals, and then transmit the signals sent from the working OTU to the client-side equipment with the optical switches of the OLP.

When the OTU2 at the receive end detects abnormal signals, the information is reported to the SCC. The SCC controls the OTU1 and the OTU2. The client-side laser of the OTU2 is shut down. The client-side laser of the OTU1 functions normally. Only the signals transmitted by the protection wavelength route are sent to the OLP board. The OLP board compares the optical power of signals and detects that no signals are transmitted by the working wavelength route. Thus, only the signals transmitted by the protection wavelength route are sent to the client-side equipment with the optical switches of the OLP.

After the recovery of the working wavelength route, service signals can be switched back to the OTU2 or not based on the pre-configuration made on the NM.

The inter-subrack channel protection supports all the OTUs that have the service cross-connection and grooming functions to realize the inter-board cross-connection. In the inter-

subrack channel protection mode, this function can realize the automatic change of the cross-connect route of the working and the protection channels or the attached route of the service.



CAUTION

The channel protection pair should be set by using the NM to achieve the inter-subrack 1+1 optical channel protection.

The boards with intra-board or inter-board cross-connect grooming capacity must be configured with cross-connect services on the NM before performing the service grooming. So these boards must be fully configured with pass-through cross-connections when configured with the protection.



NOTE

- For the configuration of the inter-subrack 1+1 optical channel protection for the OTU boards (such as the LDG, FDG, LOG, and LOGS) that carry the GE services, the optical ports of these OTU boards at the local and the opposite ends do not support the auto-negotiation function. The Auto-negotiation attribute must be disabled.
- The switching of the DCP or the OLP is performed based on the judgment on the optical power of the paths. Thus, before using this protection function, it is required to ensure that the difference in the input optical power of the optical interfaces of the working and the protection paths is less than 3 dBm. If the power difference is more than 3 dBm and less than 5 dBm, the POWER_DIFF_DEFECT alarm is reported. If the power difference is more than 5 dBm, the POWER_DIFF_OVER alarm is reported and the switching is triggered.
- The optical interface 1 of a DCP or OLP is corresponding to the working channel while the optical interface 2 to the protection channel. This principle should be strictly followed in fiber connection between the working OTU and the protection OTU.
- As for the configuration of the inter-subrack 1+1 optical channel protection, the working OTU, protection OTU and OLP/DCP can be configured in different subracks. At the same time, the access of third-party equipment or services is also supported.
- The OLP and the DCP support the subrack power protection. It is recommended to configure the working OTU and the OLP/DCP in the same subrack and configure the protection OTU in another subrack.
- One client-side service is the smallest unit of the protection service granules. For the OTU boards that have the convergence function, the protection switching is only performed to some client-side services.

Configuration Rules

The configuration rules of the inter-subrack 1+1 optical channel protection are as follows:

- The OLP board is configured to protect one channel of services.
- The DCP board is configured to protect two channels of services.
- The working OTU, protection OTU and OLP/DCP can be configured in one subrack or different subracks. It is recommended to configure the working OTU and the OLP/DCP in the same subrack and configure the protection OTU in another subrack.
- The protection can be set to revertive or non-revertive.

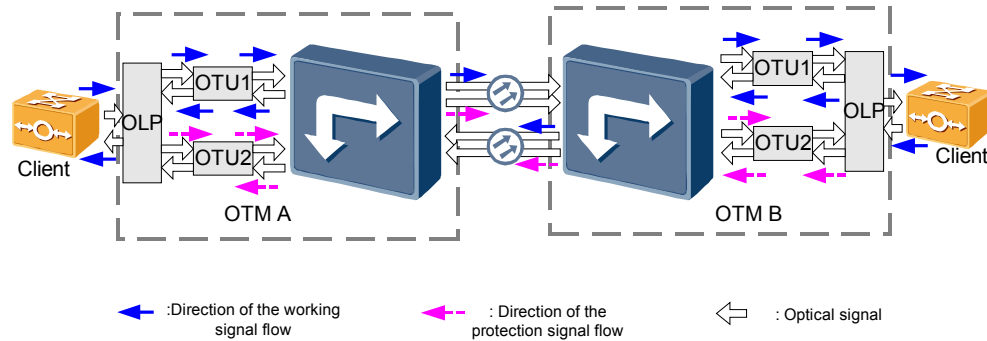
Application

Inter-subrack 1+1 optical channel protection allows more flexible configuration of boards. The working OTU and the protection OTU may reside in different subracks. Inter-subrack 1+1

optical channel protection also protects the service carried by the OTUs with the convergence function.

Take that the signals carried by the working wavelength and those carried by the protection wavelength reach the receive end by different routes as an example to introduce the inter-subrack wavelength protection. As shown in **Figure 9-17**, station A and station B form a point-to-point network in Project T. Both A and B are OTM stations. The inter-subrack 1+1 optical channel protection is adopted between the two stations. Each station is configured with one OLP and two OTUs. The two OTUs are installed in different subracks.

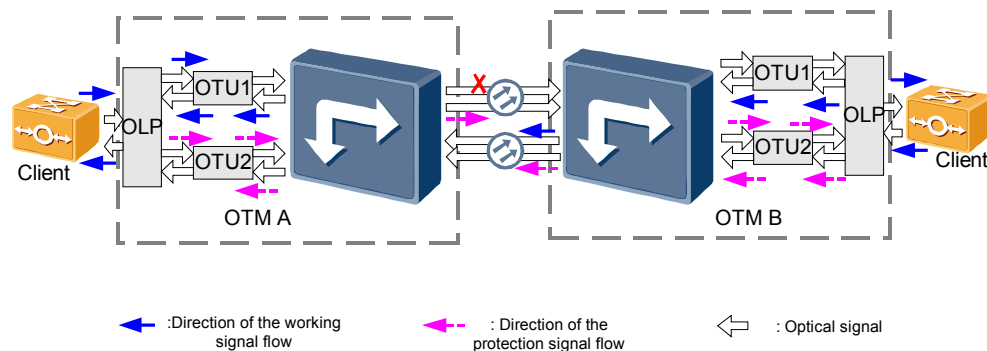
Figure 9-17 Application of inter-subrack 1+1 optical channel protection (normal)



See **Figure 9-17**. When signals are transmitted from station A to station B, the OLP in station A sends signals to the working OTU (OTU1) and the protection OTU (OTU2) at the same time. When signals reach station B, only the signals sent from the OTU1 are transmitted to the client-side equipment through the OLP.

When signals are transmitted from station B to station A, the OLP in station B sends signals to the OTU1 and the OTU2 at the same time. In station A, only the signals sent from the OTU1 are transmitted to the client-side equipment through the OLP.

Figure 9-18 Application of inter-subrack 1+1 optical channel protection (switching)



See **Figure 9-18**. In the direction from station A to station B:

When a fiber cut occurs to the working route from station A to station B, only the OTU2 on the protection route in station B sends the client-side signals to the client-side equipment through the optical switches of the OLP.

In the direction from station B to station A:

There is no switching because the working wavelength route is normal, and the route of signals remains the same.

9.2.8 Wavelength Cross-Connection Protection

This section describes the functionality, related units, trigger conditions, related alarms, working principle, configuration rules and application of the wavelength cross-connection protection.

Functionality

The wavelength cross-connection protection (WXCP) protects client services by using two (active and standby) OTUs that can cross connect and grooming services and the protected service is selected from the active or standby OTU at the receiving end.

For the cross-connection slots of the OptiX Metro 6100 subrack, refer to [8.2.1 Description](#).

Related Boards

[Table 9-19](#) lists the boards involved in the wavelength cross-connection protection.

Table 9-19 Boards involved in the wavelength cross-connection protection

Board Name	Function
SCS (optional)	Splits and couples service signals.
OTU (supporting GE signal cross-connection)	Cross connects GE signals. Detects optical signals. Reports the detected optical signal information to the SCC. Turns on/off the client-side laser according to the SCC control.
SCC	Communicates with the OTU and controls the OTU to turn on/off the client-side laser.

NOTE

WXCP can be realized in two modes. They are dual source single sink and dual source dual sink. For details, refers to details about protection principles. Only the dual source dual sink mode requires the SCS.

Trigger Conditions

The trigger conditions for the wavelength cross-connection protection switching are as follows:

- The board is offline, including the following situations: Removing or cold resetting the board.
- There is a signal failure (SF) condition and the SF is a trigger condition. SF includes the following board-side alarms:
 - R_LOS
 - R_LOF

- OTU_LOF
- ODU_AIS
- ODU_OCI
- ODU_LCK
- REM_SF
- There is a signal degraded (SD) condition and the SD is a trigger condition. SD includes the following board-side alarms:
 - B1_EXC
 - SM_BIP8_OVER
 - PM_BIP8_OVER
 - REM_SD
 - B1_SD
 - SM_BIP8_SD
 - PM_BIP8_SD

Different OTUs would report different alarms. For details of the above alarms, refer to *Alarms and Performance Events Reference*.

The alarms against certain optical interface and channels of the OTU can be set on the T2000 as SD switching conditions. [Table 9-20](#) lists the alarms, against the optical interface and channels, which can be set as SD switching conditions of each OTU.

Table 9-20 Alarms relevant to SD switching conditions and the Ports and channels

Board	Alarms		
	B1_SD	SM_BIP8_SD	PM_BIP8_SD
L4G	Port 1	Port 1	Port 1
LQG	Port 1	-	-
LOG/LOGS/ELOG/ ELOGS/LOM/ LOMS	-	Port 1	Port 1
Note: If the same port supports various services, all the three alarms can be set as the SD switching conditions. When the service type is changed, the board automatically counts the corresponding bit errors and reports an SD alarm according to the actual service type.			

Dependent Alarms

- When the wavelength cross-connection protection switching succeeds, the SCC board reports the OPS_PS_INDI alarm.
- When the wavelength cross-connection protection switching fails, the SCC board reports the OPS_PS_FAIL alarm.
- If the attributes of the active and standby ports of the wavelength cross-connection protection group are not consistent, the SCC board reports the OPS_MAIN_BAK_ATTR_DIFF alarm.

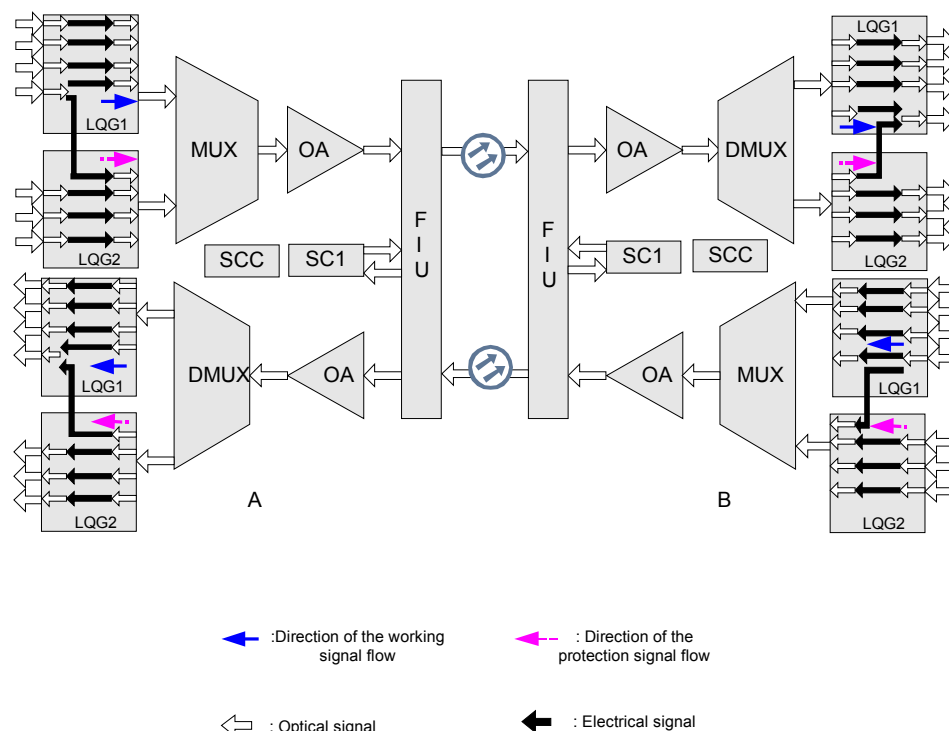
Working Principle

The wavelength cross-connection protection can be realized in two modes. They are dual source single sink and dual source dual sink. No switching of client service does not affect the other client services of the shared OTU.

- Dual source single sink

In this mode, the protection is realized by switching the service cross-connection between the active and standby OTUs. **Figure 9-19** takes the LQG as an example.

Figure 9-19 Working principle of WXCP (dual source single sink)



NOTE

- Each pair of transmit OTU and receive OTU in the same station is physically one board.
- The SCC and OTU communicate through the backplane.

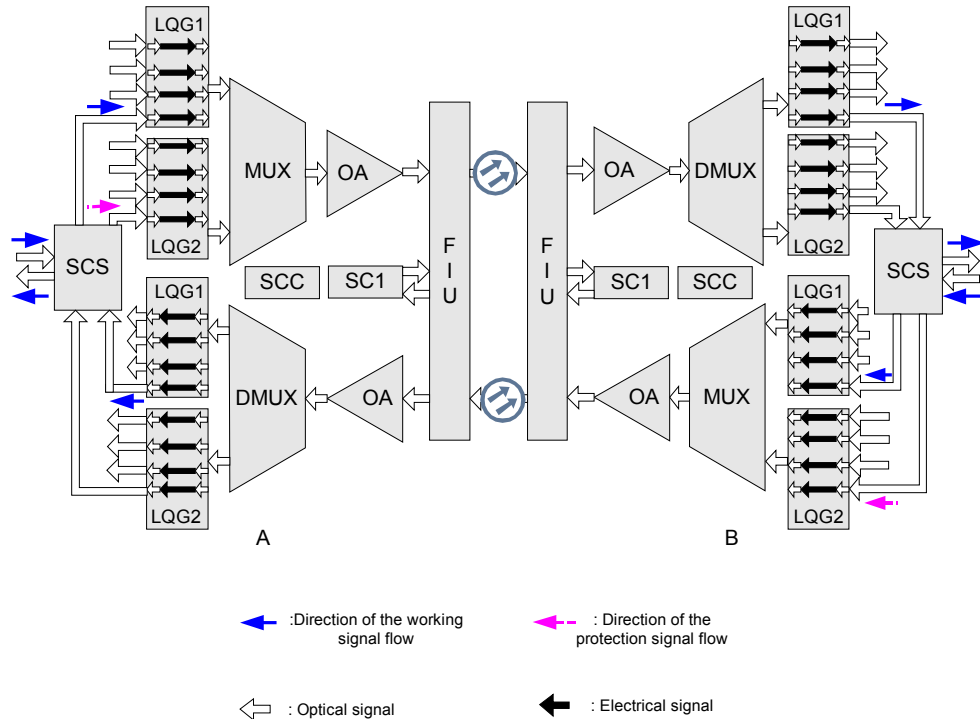
The protected client service is directly connected to the active OTU board only. The standby OTU and the client-side laser that is corresponding to the protected service do not work. Two service cross-connections are reserved during the configuration.

In normal cases, the active channel works. At the receive end, only the cross-connection that is corresponding to the active OTU is enabled. The cross-connection of the standby channel is disconnected. When the active channel is faulty, the cross-connection of the active channel at the receive end is disconnected. That of the standby OTU is enabled and the standby channel works. When the standby route returns to normal, the service signals can switch back to the OTU1 or remain in the OTU2 according to the existing configuration.

- Dual source dual sink

In this mode, the protection is realized by adopting the SCS and the OTU. **Figure 9-20** takes the LQG as an example.

Figure 9-20 Working principle of WXCP (dual source dual sink) networking mode (1)

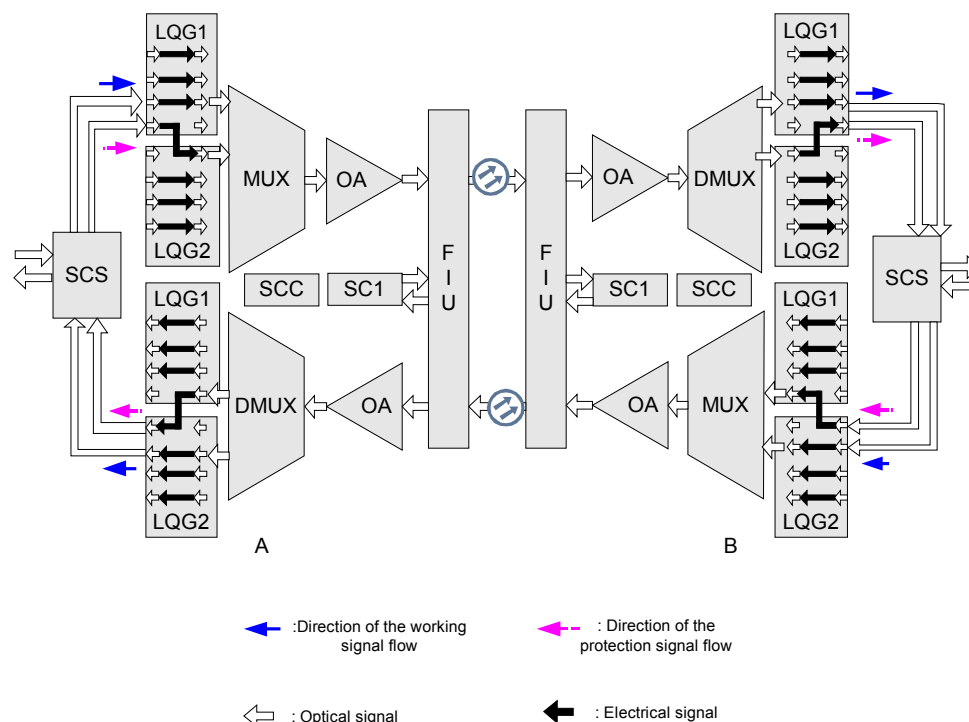


NOTE

- Each pair of transmit OTU and receive OTU in the same station is physically one board.
- The SCC and OTU communicate through the backplane.

Figure 9-21 shows the other common networking mode in dual source dual sink mode. The SCS board sends the services on the client side to the two receive interfaces of the working OTU. One channel of signals is cross-connected to the protection OTU through backplane of the subrack to realize the WXCP protection of the client-side services. The working OTU and the protection OTU should meet the requirements of slot configuration for inter-board cross-connection in this networking mode.

Figure 9-21 Working principle of WXCP (dual source dual sink) networking mode (2)



NOTE

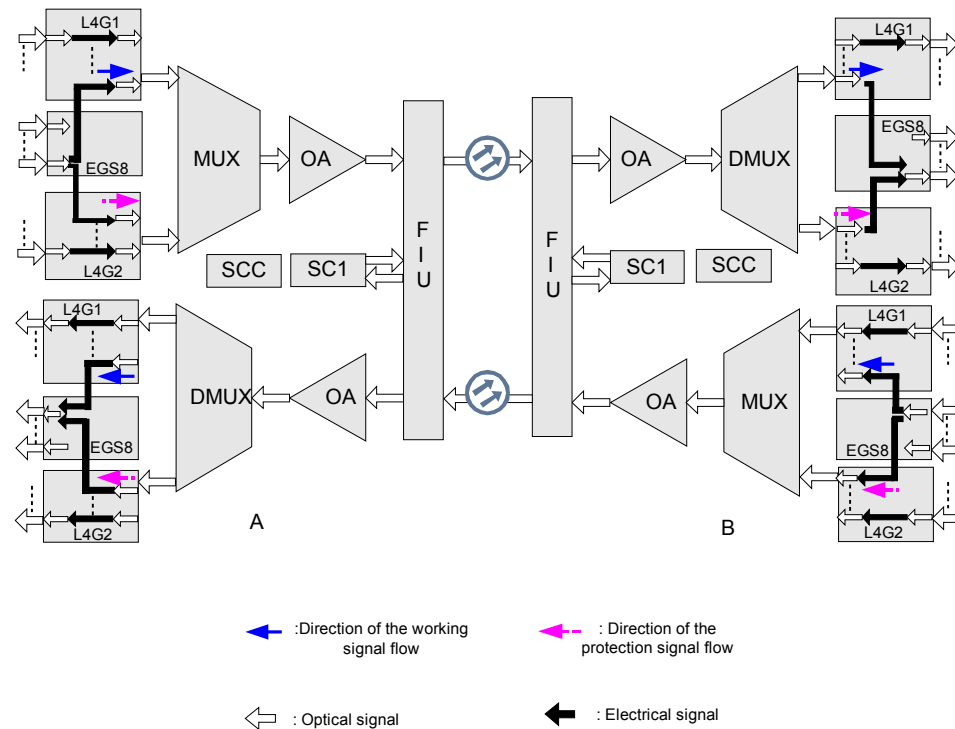
- Each pair of transmit OTU and receive OTU in the same station is physically one board.
- The SCC and OTU communicate through the backplane.

This mode is similar to the client-side 1+1 wavelength protection. In both of the two modes, client-side lasers are shut down according to the channel status. The difference between the two is that, in dual source dual sink WXCP, the client-side laser is corresponding to the cross-connection configuration while the client-side 1+1 wavelength protection fixes the corresponding relationship during fiber connection.

For the principle of 1+1 wavelength protection principle, refer to "[9.2.5 1+1 Wavelength Protection at Client](#)".

The EGS8 with two L4G boards can provide WXCP for any service that is connected to the EGS8, as shown in [Figure 9-22](#).

Figure 9-22 WXCP realized by EGS8 with L4G



Configuration Rules

The configuration rules of the WXCP are as follows:

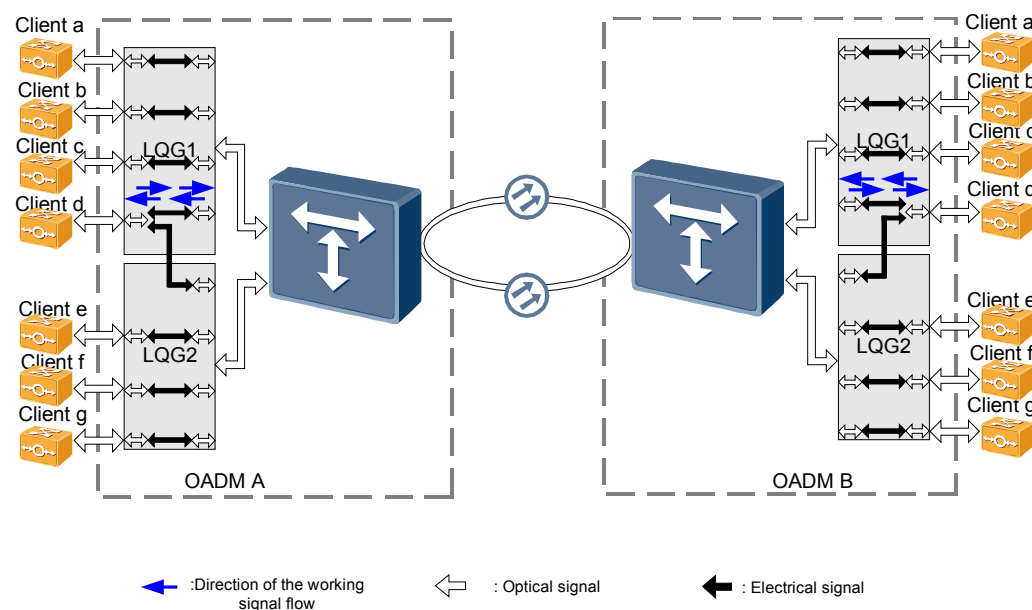
- Only the OTU board with the cross-connection function can configure the WXCP.
- The SCS board must be configured when the WXCP (dual source dual sink) is configured.
- The OTU boards in the same protection group must be configured in the same subrack.
- During configuration, the slots house the board must be considered. For details, refer to [8 Grooming of Wavelengths and Services](#).
- The WXCP can be set to be revertive or non-revertive.

Application

WXCP has advantages such as high wavelength utilization, flexible configuration, quick switching, high stability and high reliability.

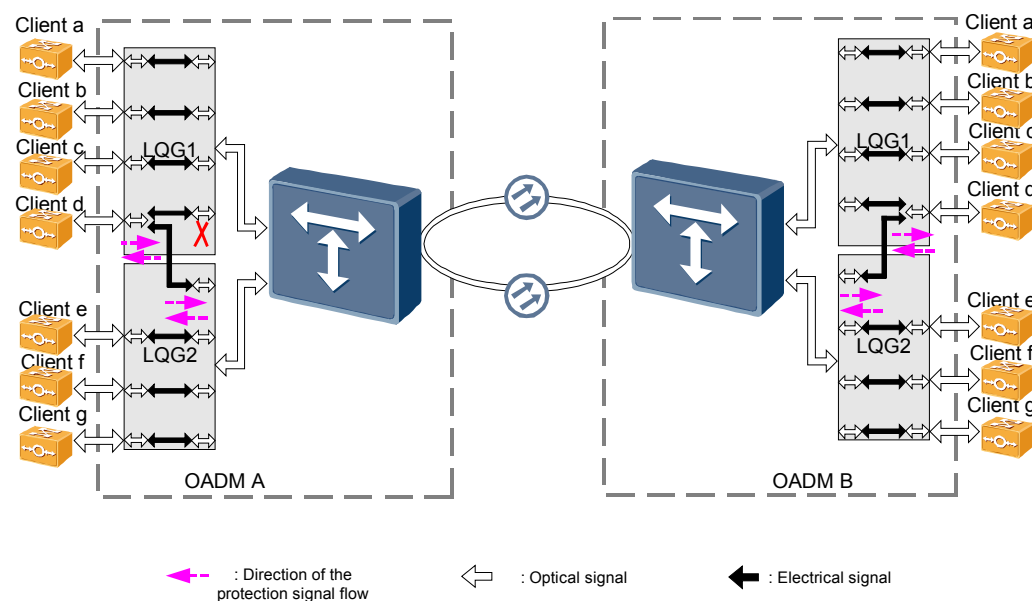
As shown in [Figure 9-23](#), Project T adopts a ring network formed by station A and B. Stations A and B are OTM stations and adopt dual source single sink WXCP between them to protect the client services d. Each station is configured with two LQG boards.

Figure 9-23 Application of WXCP (normal)



As shown in **Figure 9-23**, two cross-connections are reserved for client service d. In normal cases, the LQG1 cross-connection corresponding to client service d is enabled and the active channel works.

Figure 9-24 Application of WXCP (switched)



As shown in **Figure 9-24**, when the active channel is faulty, the LQG1 cross-connection corresponding to client service d is disconnected and the LQG2 cross-connection is enabled. The standby channel works.

9.2.9 VLAN SNCP Protection

This section describes the functionality, related units, trigger conditions, related alarms, working principle, configuration rules and application of the VLAN SNCP protection.

Functionality

In VLAN SNCP protection, Ethernet VLAN broadcasting function is used. A working and a protection channel are configured on Layer 2 of the L4G/EGS8 board. Signals are selectively received on the receive end. In this manner, the VLAN path carrying services on the L4G/EGS8 board is protected.

Related Boards

Table 9-21 lists the boards involved in the VLAN SNCP protection.

Table 9-21 Boards involved in the VLAN SNCP protection

Board Name	Function
L4G/EGS8	Performs service cross-connection on Layer 2. Broadcasts OAM test frames. Detects OAM test frames. Reports the information about the detected OAM test frame to the SCC. Follow instructions from the SCC to perform a protection switching.
SCC	Communicates with the L4G/EGS8 board, and delivers the protection switching command according to OAM test frame information.

Trigger Conditions

The trigger conditions for the VLAN SNCP protection are as follows:

- There is a signal failure (SF) condition and the SF is a trigger condition. SF includes the following board-side alarms:
 - R_LOS
 - R_LOF
 - OTU_LOF
 - ODU_AIS
 - ODU_OCI
 - ODU_LCK
 - REM_SF
- OAM frame detects abnormal.

Different OTUs would report different alarms. For details of the above alarms, refer to *Alarms and Performance Events Reference*.

Dependent Alarms

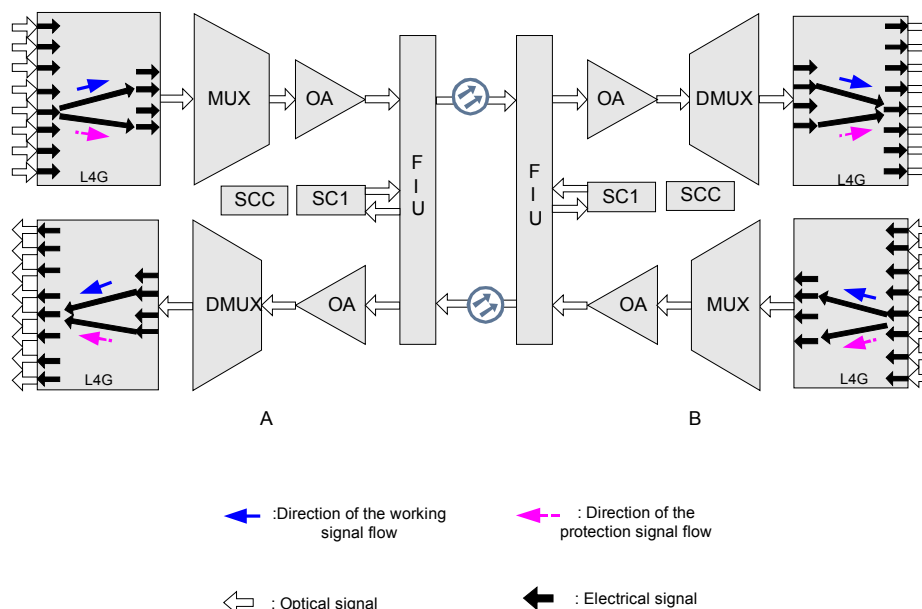
- When the VLAN SNCP protection switching succeeds, the SCC board reports the OPS_PS_INDI alarm.
- When the VLAN SNCP protection switching fails, the SCC board reports the OPS_PS_FAIL alarm.

Working Principle

VLAN SNCP protection is realized in two-source and one-sink mode. The switching of a VLAN service brings no impact on the other VLAN services on the same OTU board.

When VLAN SNCP protection is configured to a service, the protection is achieved by switching the service between the working and protection VLAN paths. **Figure 9-25** describes the VLAN SNCP protection for services on the L4G board.

Figure 9-25 Working principle of the VLAN SNCP protection principle



NOTE

Each group of transmit and receive OTU boards at a station is physically one board.

On the transmit end, the services under protection connect to the L4G/EGS8 board. The L2 processing module of the L4G/EGS8 board duplicates the VLAN path carrying the services, and sends OAM test frames periodically. If the port corresponding to the L2 processing module on the receive end detects OAM frames normally, the services in the working path are selected. If there is an abnormality in the detection of OAM frames, a switching to the protection path is triggered and the services in the protection path are selected. That is, a switching occurs in the VLAN path protection.

Before the configuration of a group of VLAN SNCP protection, two services with the same VLAN ID and sink port need be created. During the VLAN SNCP protection configuration, bind two VLAN paths into one protection group. Enable the OAM state, and thus OAM frames become the detection condition of SF alarms. The L4G/EGS8 board starts a switching according to the effectiveness of OAM frames on the receive end.

Configuration Rules

The following are the rules for configuring the VLAN SNCP protection:

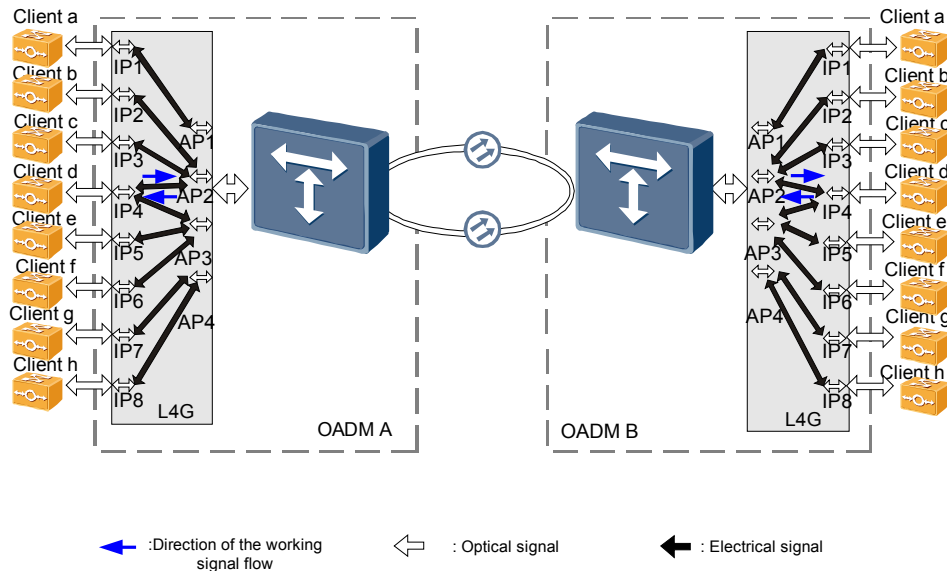
- The function is available on the L4G and EGS8 boards only.
- In one group of VLAN SNCP protection, the VLAN IDs and sink ports of the services must be the same.

Application

VLAN SNCP protection realizes the dual-fed and selective receiving operation from the VLAN path carrying client-side services to the ports with intra-board GE services. In this manner, the services in the VLAN path are protected.

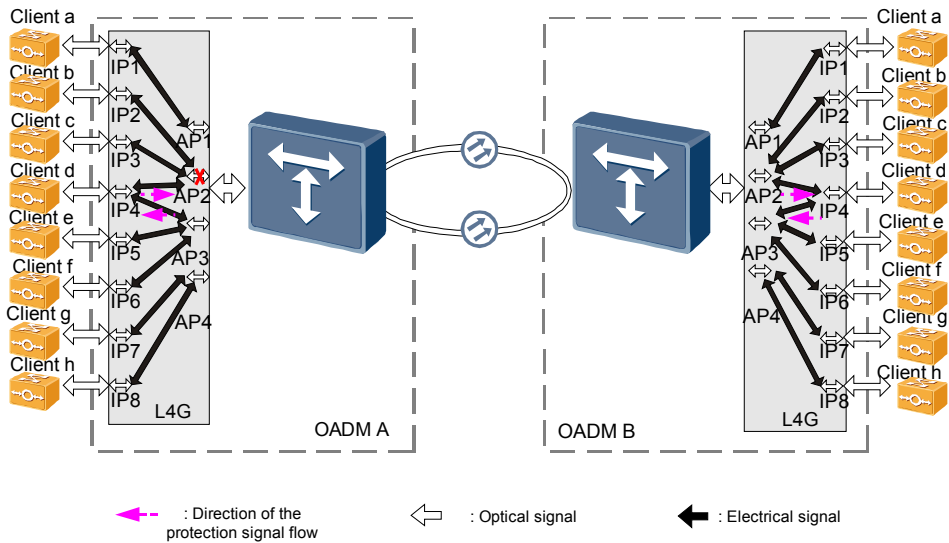
See [Figure 9-26](#). Project T is a ring network comprising stations A and B. A and B are OADM stations. Service d between A and B is configured with VLAN SNCP protection in the two-source and one-sink mode. An L4G board is installed in each station.

Figure 9-26 VLAN SNCP protection application (normal)



As shown in [Figure 9-26](#), a VLAN path is allocated for service d on the client-side port. The path is duplicated and connected to AP2 and AP3 ports. Under normal conditions, the connection between IP4 and AP2 works, and service d is in the working path.

Figure 9-27 VLAN SNCP protection application (switching)



See Figure 9-27. When the working path is faulty, the connection between IP4 and AP2 is broken. The receive end cannot detect OAM frames normally. In this case, the connection between IP4 and AP3 takes effect, and service d switches to the protection path.

9.2.10 Tribute Protection Switching and Double Path Protection Switching

This section describes the functionality, related units, trigger conditions, related alarms, working principle, configuration rules and application of the tribute protection switching and the double path protection switching.

Functionality

The tribute protection switching (TPS) and the double path protection switching (DPPS) are only for networks formed by the TBE and other OTUs that have GE service cross-connection and grooming ability. TPS and DPPS must be configured at the same time.

With TPS and DPPS, the OptiX Metro 6100 system provides double GE service protection that is corresponding to TBE boards and OTU boards. That is, the TPS provides intra-board 1+1 protection corresponding to TBE boards while the DPPS provides bidirectional WXCP corresponding to two TBE boards that bear GE services.

Related Boards

Table 9-22 lists the boards involved in the TPS and DPPS.

Table 9-22 Boards involved in the TPS and DPPS

Board Name	Function
SCS	Participates in TPS. Splits and couples service signals.
TBE	Participates in TPS. Participates in DPPS. Detects optical signals. Reports the detected optical signal information to the SCC. Turns on/off the client-side laser according to the control of the SCC.
OTU (supporting GE signal cross-connection)	Participates in DPPS. Cross connects GE signals. Detects optical signals. Reports the detected optical signal information to the SCC.
SCC	Communicates with the OTU and controls the OTU to turn on/off the client-side laser.

Trigger Conditions

The trigger conditions for the TPS are as follows:

- The board is offline, including the following situations: Removing or cold resetting the board.
- There is a signal failure (SF) condition. SF includes the following board-side alarms:
 - R_LOS
 - VCXO_LOC

The trigger conditions for the DPPS are as follows:

- The board is offline, including the following situations: Removing or cold resetting the board.
- The TPS is performed
- There is a signal failure (SF) condition. SF includes the following board-side alarms:
 - R_LOS

Different OTUs would report different alarms. For details of the above alarms, refer to *Alarms and Performance Events Reference*.

Dependent Alarms

- When the TPS and DPPS protection switching succeeds, the SCC board reports the OPS_PS_INDI alarm.
- When the TPS and DPPS protection switching fails, the SCC board reports the OPS_PS_FAIL alarm.

- If the attributes of the active and standby ports of the TPS and DPPS group are not consistent, the SCC board reports the OPS_MAIN_BAK_ATTR_DIFF alarm.

Working Principle

In TPS, a SCS splits the optical signal accessed from the client side and the signals are sent to two TBE boards at the same time. Both the two TBE boards process the signals. At the receive end, the optical signals in the active channel and the standby channel are accessed into one TBE. The working principle is similar to that of the intra-board 1+1 protection.

In DPPS, the intra-board cross-connection of the TBE is used and the corresponding OTU that has GE service cross-connection and grooming ability is also provided.

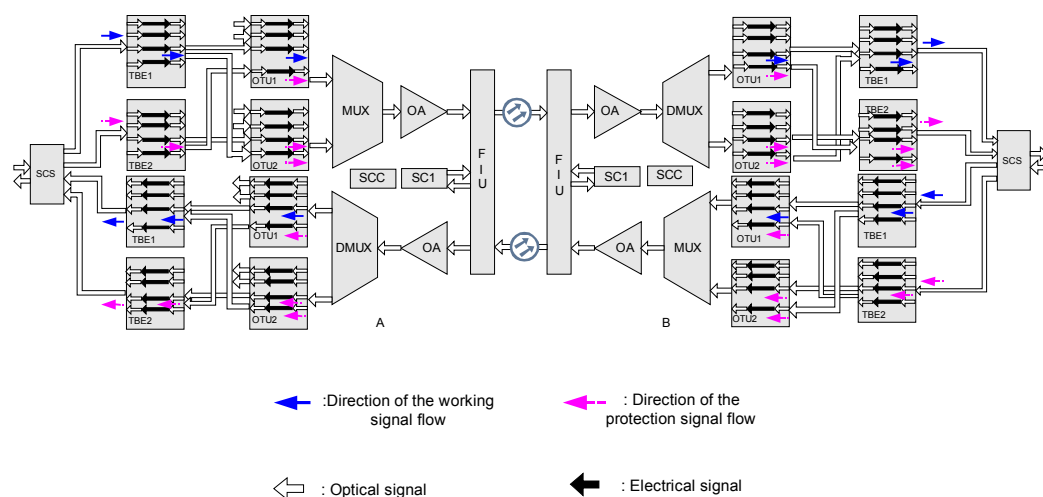
NOTE

The TPS occurs only in the local station. If the board in the opposite station is normal, the TPS does not occur in the opposite station. The DPPS is that the protection switching is performed to the board in the local station and the corresponding board in the opposite station at the same time.

Several dual source single sink WXCP protection groups are configured between the active/standby TBES and the corresponding active/standby OTUs. The protection switching is realized by adjusting the cross-connection of the active/standby TBES and the corresponding active/standby OTUs.

Figure 9-28 shows the protection. The TPS and DPPS realized by two TBE boards and two OTU boards that have GE service cross-connection and grooming ability are taken as an example.

Figure 9-28 Working principle of TPS and DPPS



The configuration of the protection is done in two steps.

- Configure the TPS of the active and standby TBE boards, that is, the protection between the TBE and the SCS.
- Configure the DPPS between the mapping ports of the active/standby OTUs and the corresponding active/standby TBES. After the configuration, the system automatically creates four dual source single sink WXCP protection groups. Two of them protect uplink services and another two protect downlink services.

NOTE

Configure the TPS of the TBE before you configure the DPPS of the corresponding OTU. Ensure that the active and standby boards of the TPS and DPPS have consistent attributes and parameters.

When an SF alarm or board absence occurs in the channels of the active OTU, DPPS is triggered and the services of the active OTU switch to the standby OTU.

When an SF alarm or board absence occurs in the active TBE or when the 10GE port of the TBE has an R_LOS alarm, the downlink TBE performs TPS and DPPS in the uplink direction. A protection switching is performed in the four WXCP groups and the services are switched to the standby TBE.

Configuration Rules

The configuration rules of the TPS and DPPS are as follows:

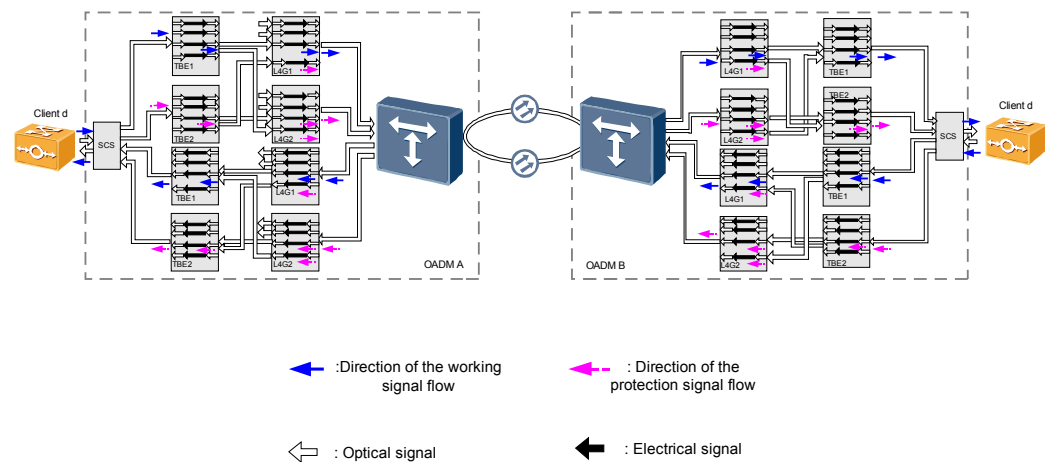
- Only the OTU board with the cross-connection function can configure the TPS and DPPS.
- The OTU boards in the same protection group must be configured in the same subrack.
- During configuration, the slots house the board must be considered. For details, refer to [8 Grooming of Wavelengths and Services](#).
- The protection can be set to revertive or non-revertive.

Application

TPS and DPPS provide double protection for GE services between the TBE and the OTU that is accessed to the WDM side with GE service cross-connection and grooming ability. The TPS and DPPS ensure that the services remain normal when the TBE or the corresponding OTU is faulty.

As shown in [Figure 9-29](#), Project T adopts a ring network formed by stations A and B. Stations A and B are OTM stations and adopt TPS and DPPS between them to protect the client service d. Each station is configured with two TBE boards, four L4G boards, and one SCS board.

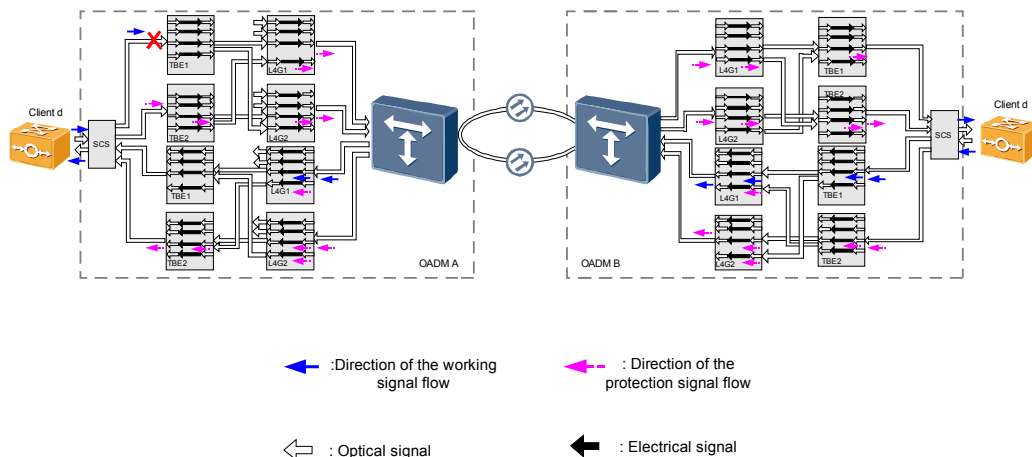
Figure 9-29 Application of TPS and DPPS (normal)



As shown in **Figure 9-29**, two cross-connections are reserved in each TBE. In normal cases, the SCS of station A sends client service d to the active TBE1 and standby TBE2 at the same time. The WXCP in the L4G1 uplink direction is enabled and the active channel works.

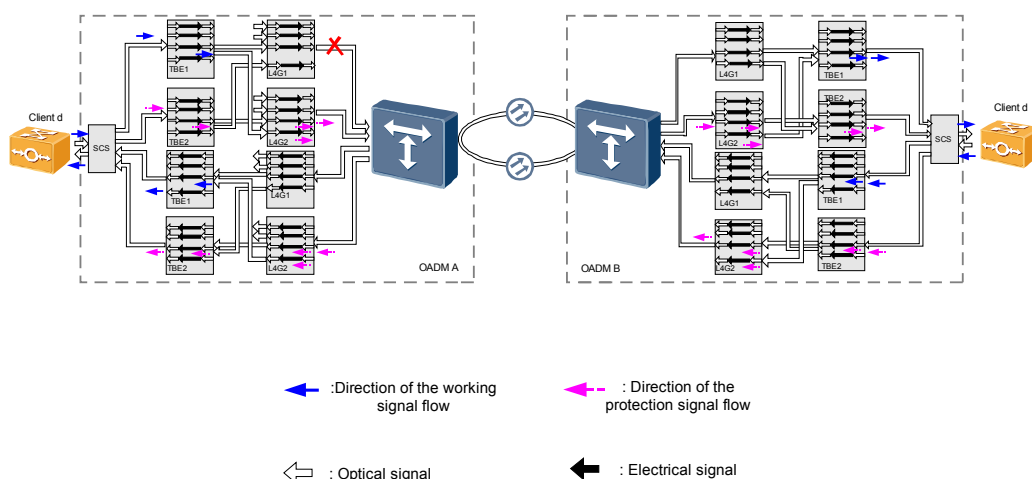
The signals reach the L4G1 of station B. Only those from the L4G1 travel through the downlink WXCP to the active TBE1 and standby TBE2 and are sent to the client side by the SCS.

Figure 9-30 Application of TPS and DPPS (TPS occurring)



As shown in **Figure 9-30**, when the TBE1 of station A is faulty, the TBE2 receives the client service d that is dual fed from the SCS. The cross-connection between the TBE2 and L4G1 +L4G2 is enabled. The standby channel works. The L4G1 and L4G2 of station B receive signals from the standby channel and cross connect them to the TBE1 and TBE2 of station B. Because of the TPS, the SCS chooses signals from the standby channel, that is, TBE2 signals, and sends them to the client side.

Figure 9-31 Application of TPS and DPPS (DPPS occurring)



As shown in **Figure 9-31**, when the L4G1 of station A is faulty, the TBE1 and TBE2 receive the client service d that is dual fed from the SCS. The cross-connection between the TBE1+TBE2

and the L4G2 is enabled. The standby channel works. Because of the TPS, the L4G2 of station B receives signals from the standby channel and cross connect them to the TBE1 and TBE2 of station B. The SCS chooses TBE1 signals, and sends them to the client side.

9.2.11 Optical Wavelength Shared Protection (OWSP)

This section describes the functionality, related units, trigger conditions, related alarms, working principle, configuration rules and application of the optical wavelength shared protection with OWSP board.

Functionality

The optical wavelength shared protection applies to ring networks. Two pairs of wavelength routes (one east, the other west) are provided for the OTU, forming a protection pair, to achieve the optical wavelength shared protection.



NOTE

Optical wavelength shared protection belongs to the WDM-side line protection. It has no relationship with the type of the OTU adopted.

Related Boards

Table 9-23 lists the units used to achieve the optical wavelength shared protection.

Table 9-23 Boards involved in the optical wavelength shared protection

Board Name	Function
OWSP	Splits and couples the service signals. Detects optical signals. Controls and performs the switching.

Trigger Conditions

The trigger condition for the optical wavelength shared protection is the R_LOS alarm.

Dependent Alarms

When the optical wavelength shared protection switching succeeds, the OWSP board reports the PS alarm.

Working Principle

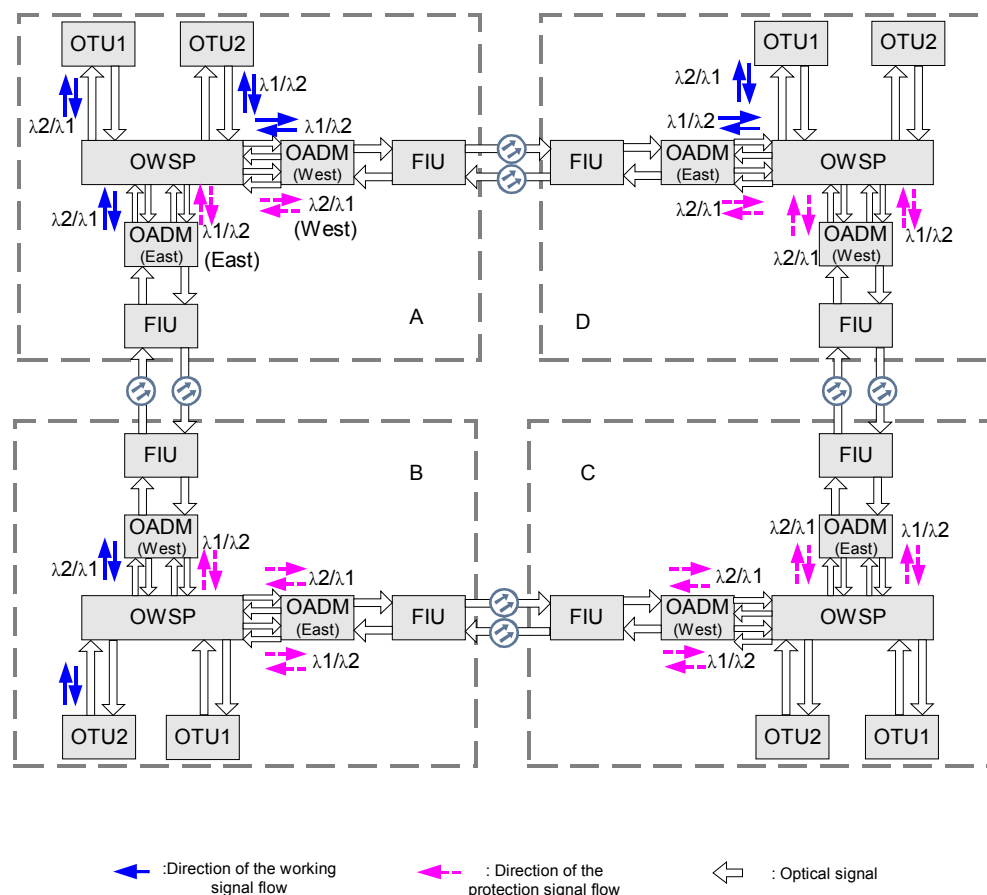
Optical wavelength shared protection applies to ring networks. The switching is performed at both the receive end and the transmit end. A failure of a wavelength in the working path can cause the signals transmitted in opposite directions to be switched to the protection path at the same time.

For details about the working principle of the OWSP board, refer to the *Hardware Description*.

As shown in **Figure 9-32**, stations A, B, C, and D form a ring network. Each station has one channel of services with its adjacent nodes.

Take one channel of services between A and B as well as A and D as an example to explain the protection scheme.

Figure 9-32 Working principle of the optical wavelength shared protection



Normally, the direction of the working route between A and B is east (A-B). The signals transmitted from A to B are carried by λ_1 . The signals transmitted from B to A are carried by λ_2 . The direction of the protection route between A and B is west (A-D-C-B). The λ_2 from A to B and the λ_1 from B to A are protection wavelengths.

Normally, the direction of the working route between A and D is west (A-D). The signals transmitted from A to D are carried by λ_2 . The signals transmitted from D to A are carried by λ_1 . The direction of the protection route between A and D is east (A-B-C-D). The λ_1 from A to D and the λ_2 from D to A are protection wavelengths.

When the OWSP unit in station A detects the failure of its east route, the services from A to B are switched to the west protection route at this station. The traffic from A to D is not affected. The signals transmitted from A to B (through D) are carried by λ_1 . The signals transmitted from B to A (through D) are carried by λ_2 . The signals transmitted from A to D are carried by λ_2 . The signals transmitted from D to A are carried by λ_1 .

After the east route of the station A recovers, the OWSP board performs the same operation and services are switched once again. The signals are carried by the original wavelengths.

Configuration Rules

The configuration rules of the optical wavelength shared are as follows:

- The OWSP board needs to be configured at the single station adding or dropping services.
- For the single OADM station that does not add or drop wavelengths, the OWSP board is not needed. Although the fiber connection exists, the wavelength can be set to pass through the single station.
- Confirm the working mode and evaluate the performance of the protection path, including OSNR, dispersion, PMD and optical power. When the protection path cannot transmit signals due to the abnormality of the previous performance items, adding the corresponding regeneration station can solve the problem.

Application

For a ring network in which service is distributed on adjacent nodes, one OWSP board in each node can realize the OWSP of one channel among all nodes by using two wavelengths only.

Since the OWSP uses the same wavelength to protect the services between nodes in the ring network, wavelength resources are saved and the cost is dropped.

NOTE

Switching time for the ring with 0–4 nodes is less than 50 ms.

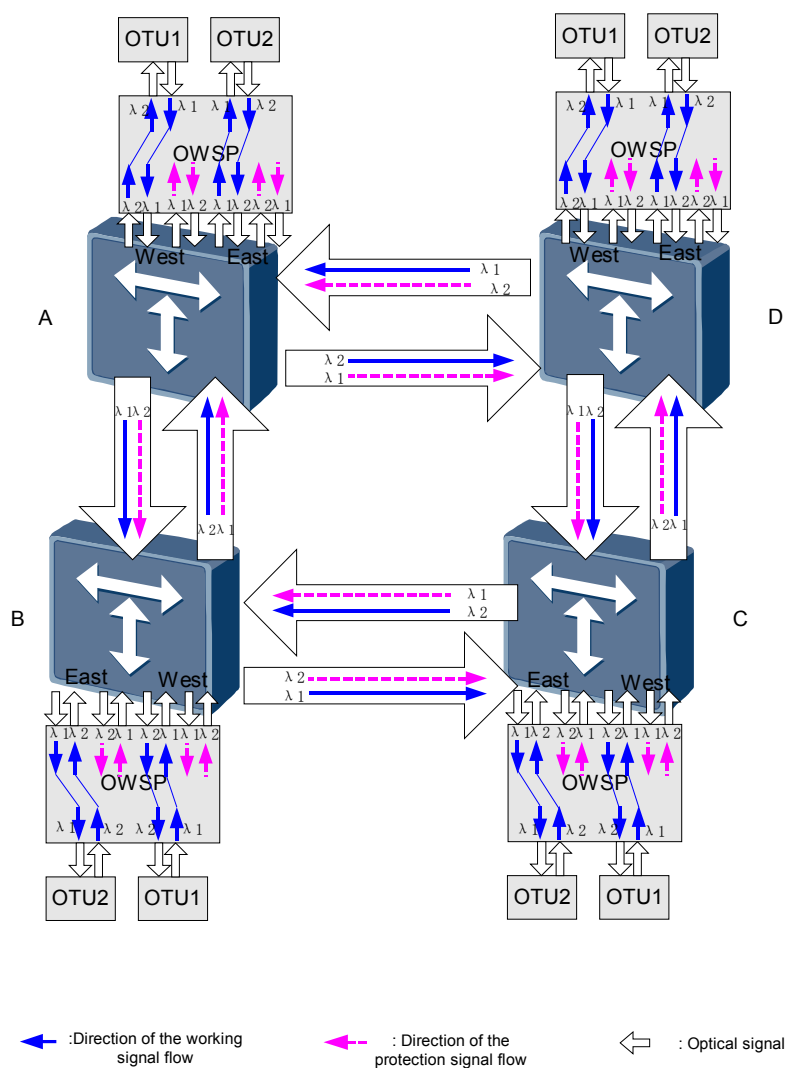
Switching time for the ring with 5–8 nodes is less than 100 ms.

The optical wavelength shared protection is applied in a ring network where the node with a large number of services to be added/dropped is located. There is high requirement on the wavelength utilization and adjacent multiplex sections share wavelengths.

As shown in [Figure 9-33](#), stations A, B, C and D form a ring network in Project T. All the four stations are OADM stations. Each station has a channel of services with its adjacent stations. The optical wavelength shared protection is adopted. Each station is configured with one OWSP board and two OTUs.

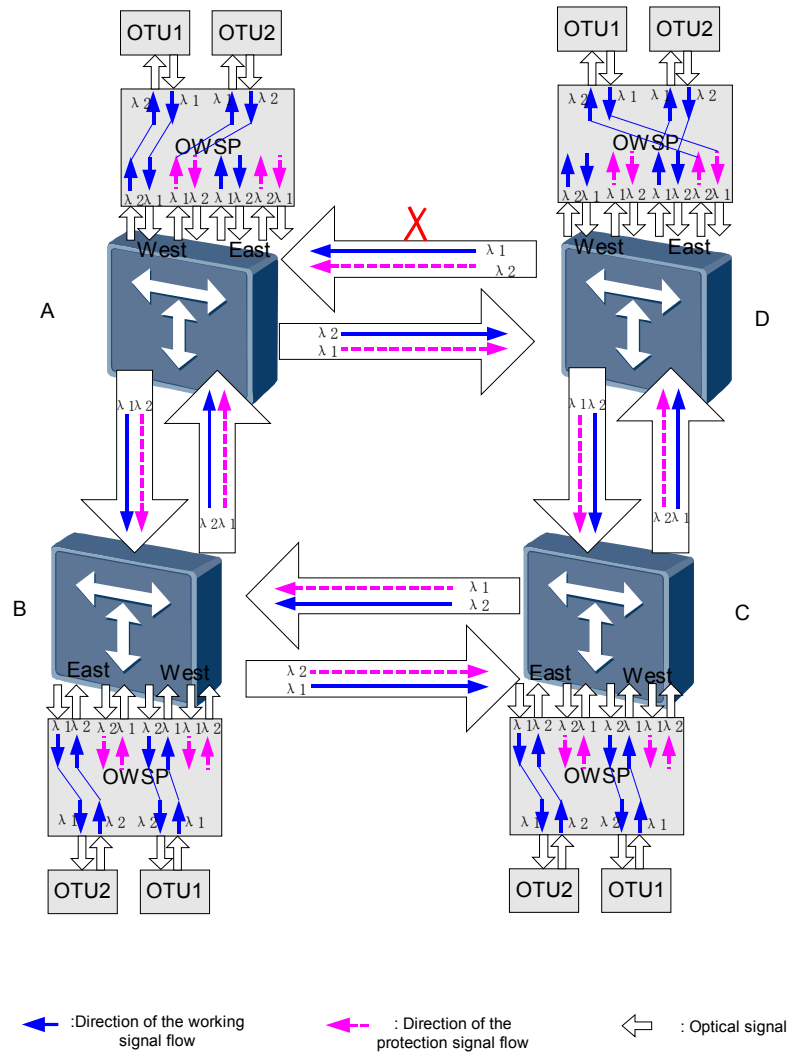
Take the protection of the services between A and D as an example.

Figure 9-33 Application of optical wavelength shared protection (normal)



See [Figure 9-33](#), two wavelengths are shared by four stations. Normally, the OTU2 in station A uses $\lambda 2$ to send the signals to station D through the OWSP board. The OTU1 in station D uses $\lambda 1$ to send the signals to station A through the OWSP board.

Figure 9-34 Application of optical wavelength shared protection (switching)



See [Figure 9-34](#). When the optical route between station A and station B (from station D to station A) fails, the OWSP board in station A detects the failure. The OTU2 in station A uses $\lambda 2$ to send signals to the OWSP board. The OWSP board performs the switching and sends these signals to station B. Signals are transmitted to station D through stations B and C. The OWSP board in station D performs the switching accordingly and sends the signals carried by $\lambda 2$ to the OTU1.

9.2.12 Optical Wavelength Shared Protection (DCP)

This section describes the functionality, related units, trigger conditions, related alarms, working principle, configuration rules and application of the optical wavelength shared protection with DCP board.

Functionality

The optical wavelength shared protection (DCP) is used in ring networks configured with distributed services. It occupies two wavelengths, realizing the shared protection for one channel of services among all stations.

The protection uses a piece of WDM equipment to achieve the ring protection realized by SDH equipment, which lowers the networking cost.

 NOTE

The optical wavelength shared protection (DCP) is applicable to subracks of the OptiX Metro 6100 with versions of V100R006 or above.

Related Boards

Table 9-24 lists the boards involved in the optical wavelength shared protection (DCP).

Table 9-24 Boards involved in the optical wavelength shared protection (DCP)

Board Name	Function
DCP	Splits and selects service signals. Detects optical power. Starts a switching when the difference of optical power between the working and protection channels exceeds the threshold or when the MUT_LOS alarm occurs in the working channel.
OTU	Detects signals. Detects SF and SD switching events and reports the events to the SCC.
ST1/ST2	Realizes the communication between NEs. Forms an OSC protocol ring, receives and transmits the automatic protection switching (APS) protocol, and separates control signaling from service signals.
SCC	Receives and processes APS bytes. Receives SF and SD switching events reported by boards, and delivers the optical switch switching command to the DCP board.

Trigger Conditions

The trigger conditions for the optical wavelength shared protection (DCP) are as follows:

- There is a signal failure (SF) condition. SF includes the following board-side alarms:
 - R_LOS
 - R_LOF
 - OTU_LOF
 - OTU_AIS
 - ODU_AIS
 - ODU_OCI
 - ODU_LCK
 - FEC_LOF
- There is a signal degraded (SD) condition and the SD is a trigger condition. SD includes the following board-side alarms:

- B1_EXC
- SM_BIP8_OVER
- PM_BIP8_OVER
- B1_SD
- SM_BIP8_SD
- PM_BIP8_SD

Different OTUs would report different alarms. For details of the above alarms, refer to *Alarms and Performance Events Reference*.

The alarms against certain optical interface and channels of the OTU can be set on the T2000 as SD switching conditions. **Table 9-25** lists the alarms, against the optical interface and channels, which can be set as SD switching conditions of each OTU.

Table 9-25 Alarms relevant to SD switching conditions and the ports and channels

Board	Alarms		
	B1_SD	SM_BIP8_SD	PM_BIP8_SD
FDGS/L4G/LWF/ LWFS/LBF/LBFS/ LW40	Port 1	Port 1	Port 1
FDGD/LWC1	Port 1, 2	Port 1, 2	Port 1, 2
AP8S/FCES/LQG	Port 1	-	-
AP8D/FCED/LDG	Port 1, 2	-	-
AS8S	Port 1, 3-10	-	-
AS8D	Port 1-10	-	-
LQS	Port 1-6	-	-
LQMS/ETMX/ ETMXS	Port 3-6	Port 1, 3-6	Port 1, 3-6
LQMD	Port 3-6	Port 1-6	Port 1-6
LWC1D	Port 1-3	Port 1-3	Port 1-3
LWX/LWM	Port 1-3	-	-
LOG/LOGS/ELOG/ ELOGS/LOM/ LOMS/LBE/LBES	-	Port 1	Port 1
TMX/TMXS	Port 3-6	Port 1	Port 1

 NOTE

IN_PWR_HIGH and IN_PWR_LOW are alarms generated on the LWX board.

Dependent Alarms

When the optical wavelength shared protection (DCP) succeeds, the SCC board reports the OWSP_PS alarm.

NOTE

The pass-through stations in the network configured with the optical wavelength shared protection (DCP) do not report switching alarms.

Working Principle

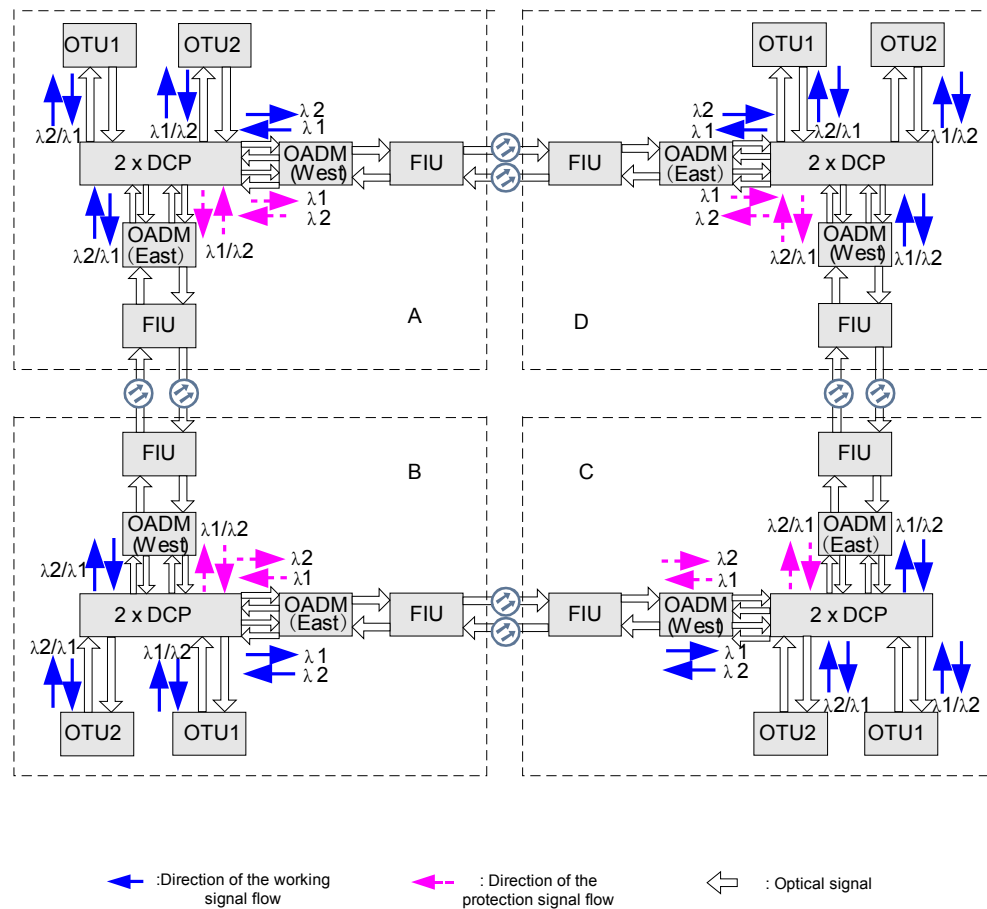
The optical wavelength shared protection (DCP) is applicable to ring networks. It requires a network protection switching protocol. It features dual-ended switching and dual-fed and selective receiving. In other words, when the wavelength for receiving signals in the working channel fails, the receiving and transmitting are switched to the protection channel.

Each station in optical wavelength shared protection (DCP) can work in two modes. In the first mode, the station serves as the source or sink end of a service. In the other mode, the station serves as a pass-through station, which helps to form a protection route upon a switching of services in other stations.

- In the service transmit direction, the signals are duplicated on the DCP board to the working route and the protection route.
- In the service receive direction, the same two channels of signals are received from the working and protection channels under normal conditions.
- Upon a failure, the receiving and transmitting are switched to the protection channel at the same time.

See [Figure 9-35](#). Stations A, B, C, and D form a ring network. There is a service between each station and the adjacent station. The optical wavelength shared protection (DCP) is configured for each service. This section describes the working principle based on a service between A and B and a service between A and D.

Figure 9-35 Working principle of the optical wavelength shared protection (DCP)



NOTE

Services are added through the OTU1 board and are dropped through the OTU2 board in the adjacent downstream station. Or, services are added through the OTU2 board and are dropped through the OTU1 board in the adjacent downstream station.

Under normal conditions, the working route from A to B is east (A-B), and $\lambda 1$ is the working wavelength. The protection route from A to B is west (A-D-C-B), and $\lambda 1$ in the protection channel is the protection wavelength. The working route from B to A is west (B-A), and $\lambda 2$ is the working wavelength. The protection route from B to A is east (B-C-D-A), and $\lambda 2$ in the protection channel is the protection wavelength.

Under normal conditions, the working route from A to D is west (A-D), and $\lambda 2$ is the working wavelength. The protection route from A to D is west (A-B-C-D), and $\lambda 2$ in the protection channel is the protection wavelength. The working route from D to A is east (D-A), and $\lambda 1$ is the working wavelength. The protection route from D to A is east (D-C-B-A), and $\lambda 1$ in the protection channel is the protection wavelength.

When station A detects a failure in the east route, the services from A to B and from B to A switch to the protection routes. There is no impact on the services between A and D. In this case, the following wavelengths are used between A and B: The services from A to B switch to the west protection route (A-D-C-B), and the protection wavelength $\lambda 1$ is used; The services from B to A switch to the east protection route (B-C-D-A), and the protection wavelength $\lambda 2$ is used. There is no impact on the bidirectional services between A and D.

When the east route at station A is recovered, a similar process is performed under OWSP protection. A switching occurs again. The original working wavelength is used.

Configuration Rules

The following are rules for configuring the optical wavelength shared protection (DCP):

- The optical wavelength shared protection (DCP) supports a maximum of eight stations. Each station supports a maximum of 40 protection groups. Each protection group uses the dedicated OSC protocol channel. Each OSC protocol loop supports 40 channels.
- The OSC board must be installed in the main subrack.
- Two DCP boards are required in the station where wavelengths are to be added or dropped.
- The DCP board is not required in the OADM station where no wavelengths are to be added or dropped.



NOTE

The revertive mode of the switching in the optical wavelength shared protection (DCP) can be set to revertive only.

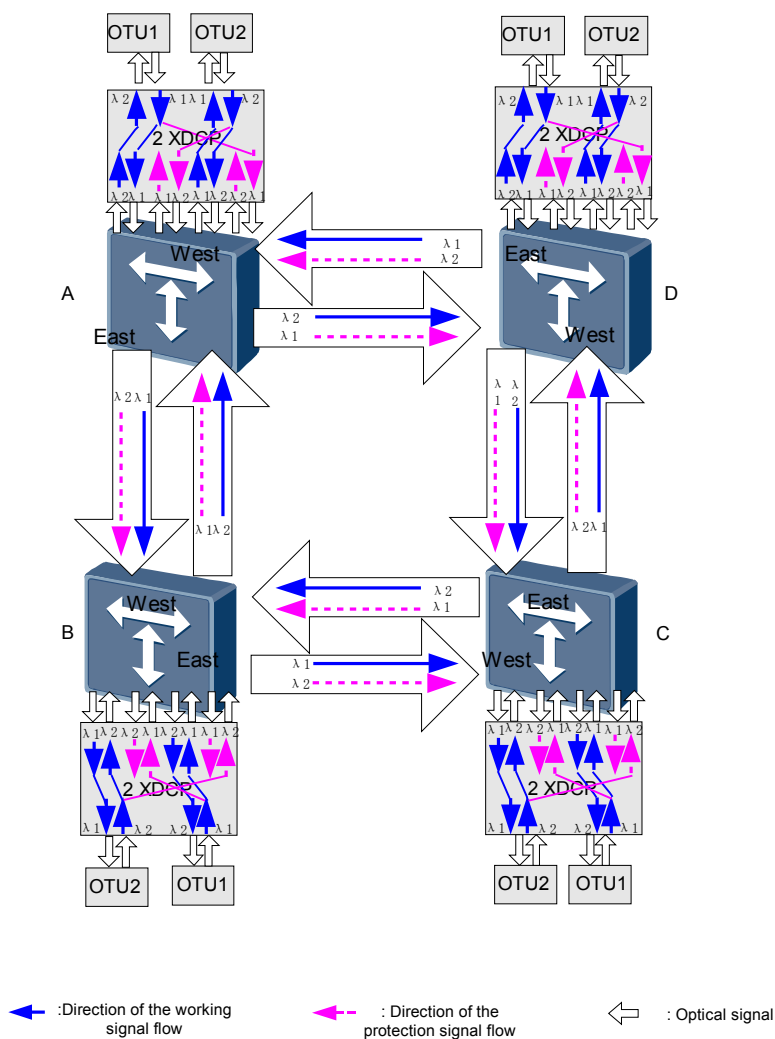
Application

In the optical wavelength shared protection (DCP), different services between stations in a ring can be protected by using the same wavelength. In this manner, the wavelength is shared, wavelength resource is saved, and the spare part cost is reduced.

See [Figure 9-36](#). Project T is a ring network comprising stations A, B, C and D. A, B, C and D are OADM stations. There is a service between each station and the adjacent station. The optical wavelength shared protection (DCP) is configured for each service. Two DCP boards and two OTU boards are configured at each station.

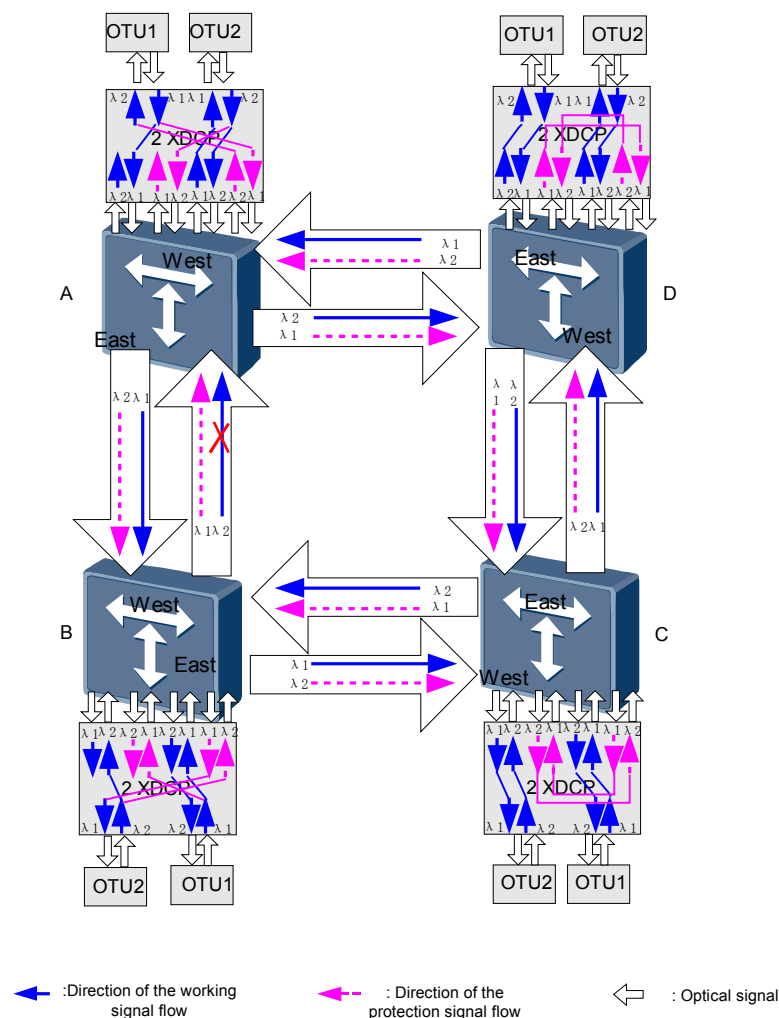
This section describes the protection of the services between A and B.

Figure 9-36 Application of optical wavelength shared protection (DCP) (normal)



See [Figure 9-36](#). Two wavelengths are shared by four stations. Under normal conditions, the working route from A to B is east (A–B), and the OTU1 board at A uses the working wavelength λ_1 to transmit service signals to B. The working route from B to A is west (B–A), and the OTU2 board at B uses the working wavelength λ_2 to transmit service signals to A.

Figure 9-37 Application of optical wavelength shared protection (DCP) (switching)



See [Figure 9-36](#). When a failure in the optical route from B is detected at A, the services from A to B and from B to A switch to protection routes.

The OTU1 board at A uses $\lambda 1$ to send the services from A to B to the DCP board. The DCP board performs a switching, and sends the services to B through the protection route (A-D-C-B). The DCP board at B performs a switching accordingly, and sends the services in the protection wavelength $\lambda 1$ from C to the OTU2. The protection wavelength passes through D and C, and uses the proper protection channels.

The OTU2 board at B uses $\lambda 2$ to send the services from B to A to the DCP board. The DCP board performs a switching, and sends the services to A through the protection route (B-C-D-A). The DCP board at A performs a switching accordingly, and sends the services in the protection wavelength $\lambda 2$ from D to the OTU1. The protection wavelength passes through C and D, and uses the proper protection channels.

9.3 Network Management Channel

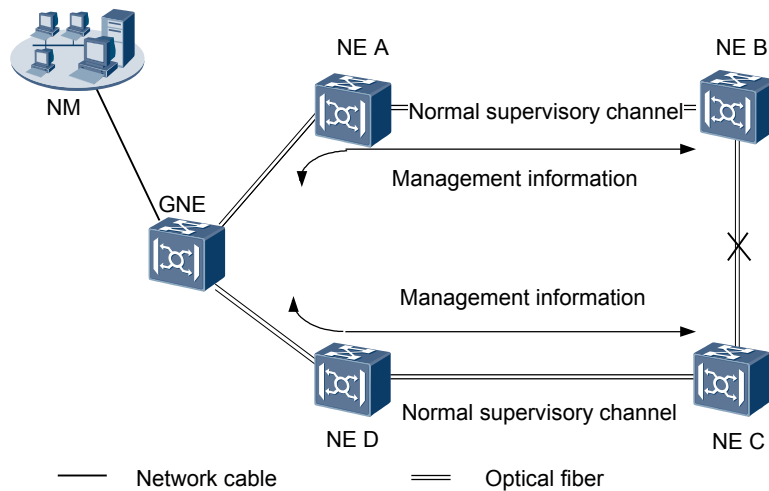
The system provides protection of network management information channel and interconnection of network management information.

9.3.1 Protection of Network Management Information Channel

In a DWDM system, network management information is transmitted over an optical supervisory channel, which shares the same physical channel with the working path. Any anomaly or failure in the working path can affect the supervisory channel. Therefore, a backup supervisory channel must be provided.

In a ring network, when fiber cut occurs in a certain direction, network management information is automatically switched to the optical supervisory channel in the other direction of the ring, as shown in [Figure 9-38](#). This does not affect the management of the entire network.

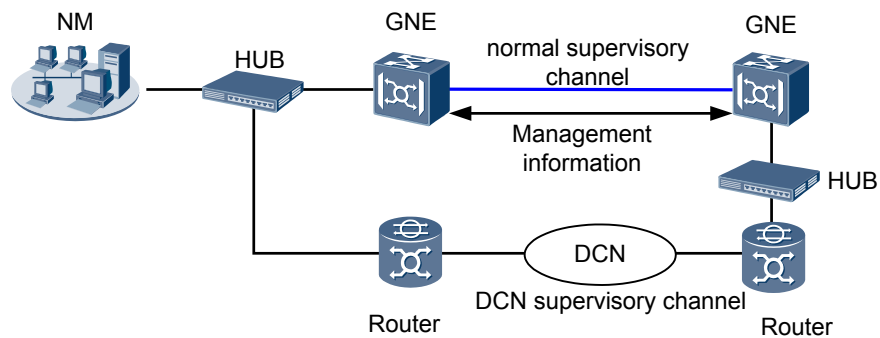
Figure 9-38 Network management protection in ring network (a certain section fails)



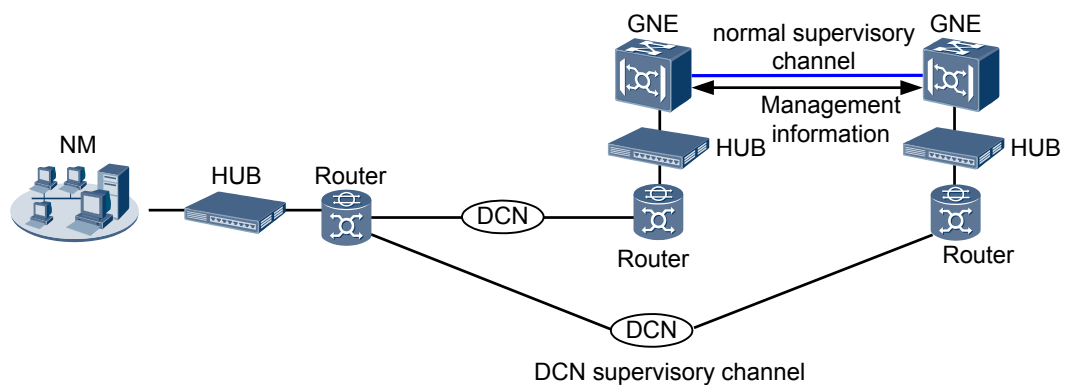
With data communication network (DCN), the system also provides network management information channel. The user can choose a method to use the channel based on the networking and spanning. In the point-to-point networking and chain networking, when both the fiber transmission and the supervisory channel fail, the network becomes unmanageable. This can be prevented by the network management information channel in DCN mode. The system NE can provide network management information channel by the DCN.

To set up a DCN network management channel, access the DCN between the two NEs through a router. With initial configuration, network management information is transmitted over the normal supervisory channel when the network is normal. See [Figure 9-39](#).

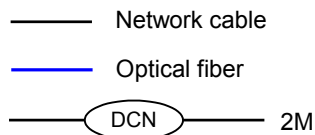
Figure 9-39 Network management through the normal supervisory channel



(1) The NM and the GNE at the same station

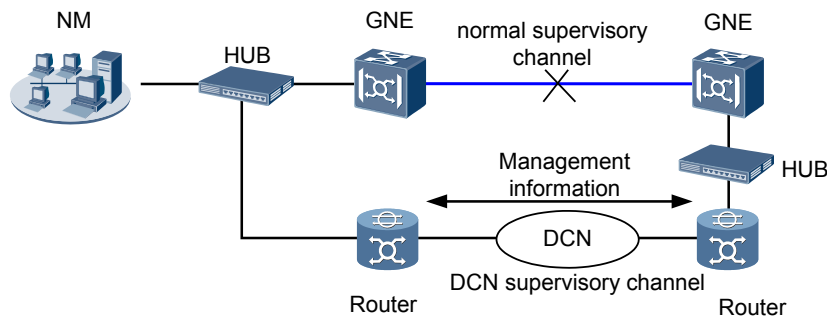


(2) The NM and the GNE at the different station

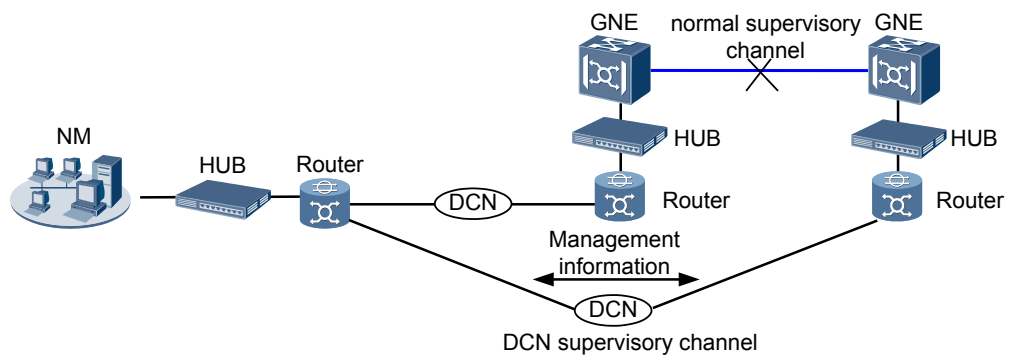


When the normal supervisory channel fails, network elements automatically switch the management information to the DCN supervisory channel to ensure the supervision and operation on the entire network, as shown in [Figure 9-40](#).

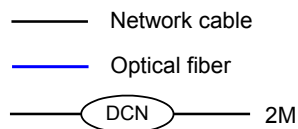
Figure 9-40 Network management through the DCN supervisory channel



(1) The NM and the GNE at the same station



(2) The NM and the GNE at the different station

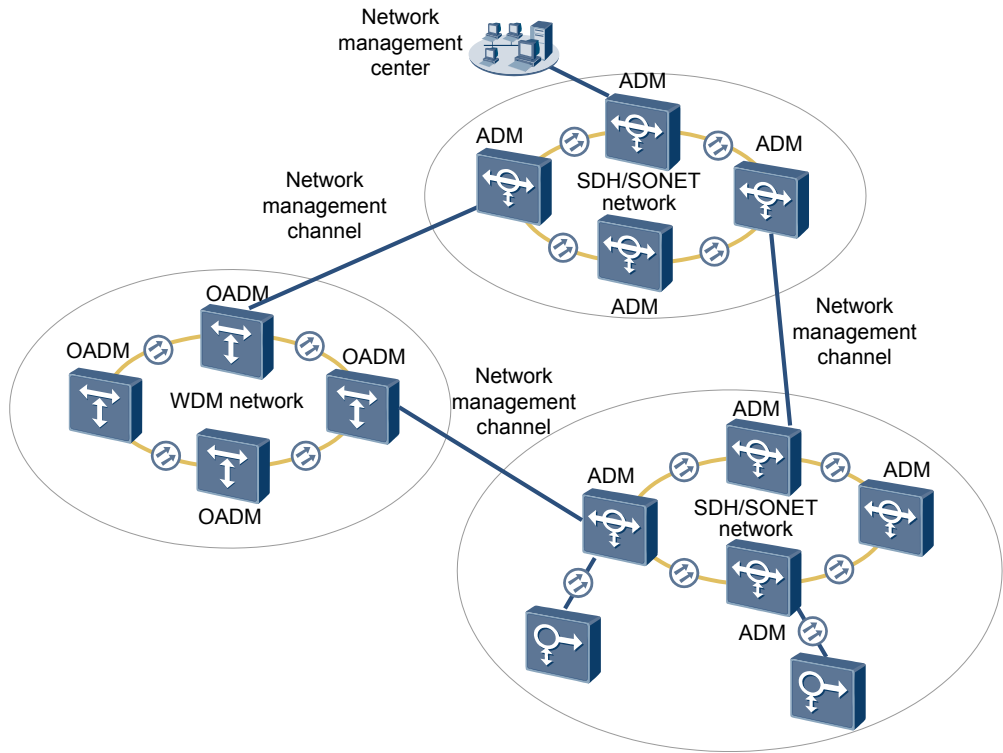


It is important to select different routes for the DCN supervisory channel and normal channel during network planning. Otherwise, the backup function does not take effect.

9.3.2 Interconnection of Network Management Channel

The system provides various data interfaces (for example Ethernet interface) for the interconnection of network management channels among different DWDM networks, or between a DWDM network and a SONET network, as shown in [Figure 9-41](#). It enables unified management of different transmission equipment.

Figure 9-41 Supervision over OptiX transmission network



10 Management of Optical Power

About This Chapter

Optical power management consists of intelligent optical power adjustment, automatic optical power control, and automatic optical power equalization.

10.1 Intelligent Power Adjustment

The system provides the intelligent power adjustment (IPA) function. When there is a fiber break on the line, the downstream optical amplifier is shutdown to prevent exposed optical fibers hurting human body.

10.2 Intelligent Power Adjustment of Raman System

The power of the pump light from Raman amplifiers is very high. Before you turn on a Raman amplifier, you must configure and enable IPA function. After a fiber cut is detected, shut down the Raman amplifier, so that the optical power of the entire line is on a safety level.

10.3 Automatic Level Control

The system provides the automatic level control (ALC) function. As the attenuation on a line segment is increased, the output power as well as the input and output powers of other downstream amplifiers will not be changed. Hence there will be much less influence on OSNR. The optical power received by the receiver will not be changed.

10.4 Automatic Power Equilibrium

The system provides the automatic power equilibrium (APE). With the APE function, you can enable the system to automatically adjust the optical power of the transmit end of each channel to keep the flatness of the optical power of the receive end to maintain the OSNR. OptiX Metro 6040 DWDM system can realize this function with the OptiX Metro 6100 DWDM system.

10.1 Intelligent Power Adjustment

The system provides the intelligent power adjustment (IPA) function. When there is a fiber break on the line, the downstream optical amplifier is shutdown to prevent exposed optical fibers hurting human body.

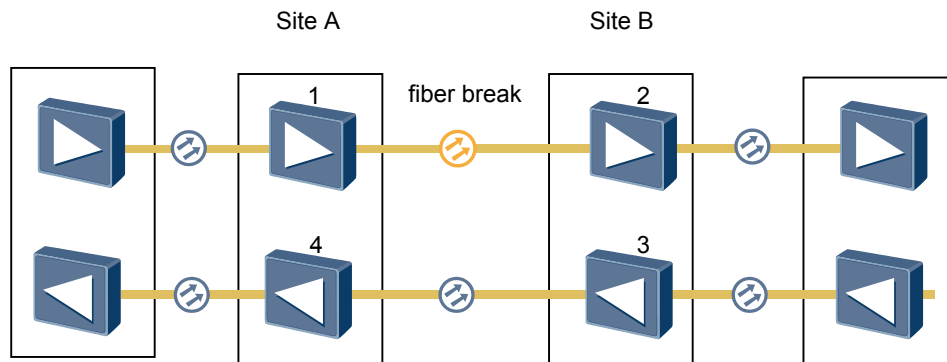
10.1.1 Function Description

To prevent exposed optical fibers hurting human body, especially eyes, the system provides the IPA functions.

In the DWDM system, optical fiber break, equipment failure or optical connector removal may lead to the loss of optical signals on the main optical channel and on the optical auxiliary channels. To prevent exposed optical fibers hurting human body, especially eyes, and to avoid surge of the optical amplifier, the system provides the IPA functions. Where the loss of optical power signals happens on one or more optical trunk sections on the main optical channel and the optical supervisory channels, the system can detect the loss of optical signals on the link and instantly shut down the upstream optical amplifier. When optical signals of the system are recovered, normal operation of the optical amplifier is restored.

Figure 10-1 shows how to achieve the IPA function. When there is a fiber break on the line, the amplifier 3 and 1 are shut down. Then all the downstream optical amplifiers have no optical power output due to the amplifier feature.

Figure 10-1 Function description of the IPA



When the optical signals are restored to normal, the optical amplifier works again. Amplifiers 3 and 1 are restarted.

NOTE

In the DWDM system, the IPA function is started only when optical signals of the active optical path are lost. When this function is executed, only the lasers on the main path are shut down. No operation will be implemented on the optical supervisory channel. Hence the functions of all optical supervisory channels will not be affected.

10.1.2 Function Implementation

IPA function is implemented by various boards with different functions.

Fiber Break Detection

There are two methods used to detect the fiber break to achieve the IPA function.

- Detecting the optical power of the optical amplifier
- Detecting the optical power of the D40
- Detecting the optical power of the C6VA4 board

Through a combination of these methods, the fiber break can be judged more correctly. Following is the logic of problem handling.

- If all the configured detection items meet the fiber break condition at the same time, initiate the shut down process of the IPA.
- If one of the detection conditions recovers to normality, initiate the recovery process of the IPA.

Table 10-1 lists the related alarms of the detection board that trigger the IPA:

Table 10-1 Related alarms of the detection board that trigger the IPA

Detection Board/Auxiliary Detection Board	Alarm Trigger Condition of the IPA
OAU, OBU, OPU	MUT_LOS
D40	MUT_LOS
C6VA4	R_LOS, MUT_LOS

Involved Boards

The boards of the following types are involved in realizing the IPA:

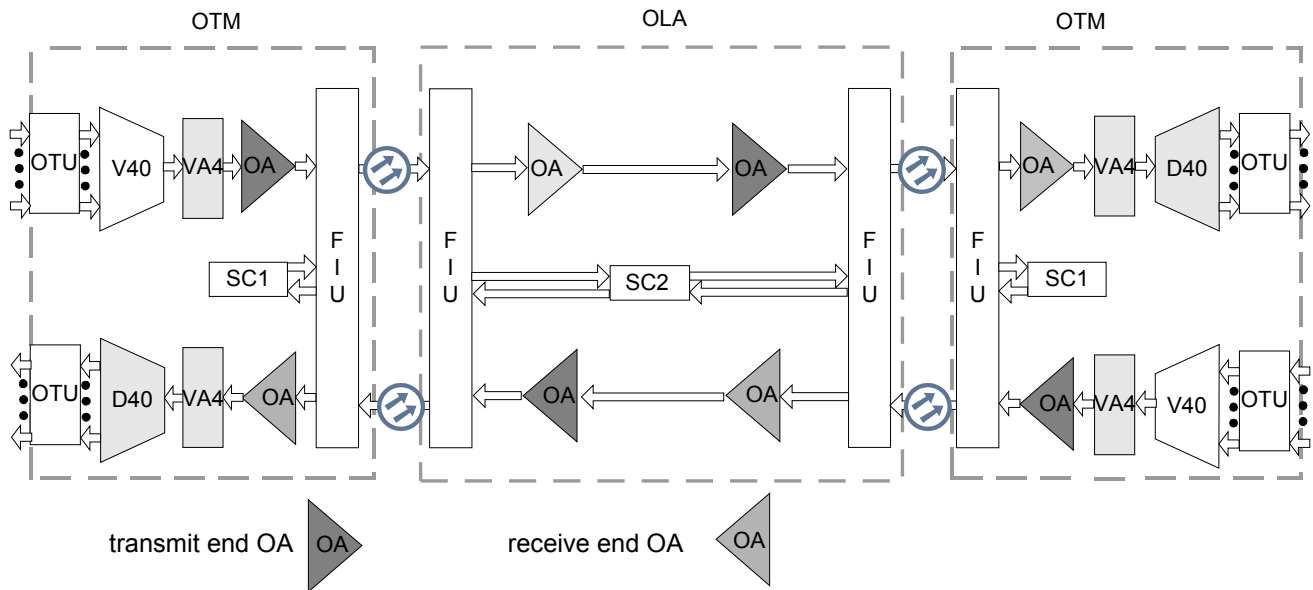
- Detection board
 - Functional unit: OAU, OBU, OPU, C6VA4, D40
 - It must be configured. Normally, an amplifier board serves as the detection board. In this case, it is used mainly to detect the optical power. The threshold of the detection board is adjustable. Normally, the default optical power threshold of the optical amplifier board does not need manual adjustment.
 - If the system is not configured with an optical amplifier board, the D40 board on the receive end or the C6VA4 board on the transmit end can be used to detect the optical power.
 - For how to set the threshold of the detection board, refer to *Configuration Guide*.
- Control implementation board
 - Functional unit: OAU, OBU, VOA
 - It must be configured. Normally, an amplifier board serves as the control implementation board. In this case, it provides only the shutdown function.
 - If the system is not configured with an optical amplifier board, you can also configure the L2VOA board to realize the optical source shutdown function at the transmit end and the detection of the transmitted light by adjusting the attenuation.

10.1.3 Networking Application

This section describes an example of the networking application of the IPA function.

Figure 10-2 is the typical networking application diagram of the IPA function. The transmit end OA in each station serves as a control implementation board, and the receive end OA in each station serves as a detection board. The control implementation board and detection in each station form an IPA pair.

Figure 10-2 IPA networking



10.1.4 Configuration Principle

The IPA function can be configured according to customers' requirements.

Network-Level Configuration Rules

- In a simple application scenario, you just need to configure one optical amplifier board as the detection board and another optical amplifier board as the shutdown board (namely, the control implementation board), and set the relevant control parameters.
- In a complex application scenario, when required, you can set the VA4 at the transmit end or the D40 at the receive end as the detection board.
- When the optical amplifier board serves as a fiber-cut detection board, the IPA function can be configured at only the ingress and egress nodes. In the case of a fiber cut, the optical amplifier boards at the ingress and egress nodes are shut down.
- To accurately locate any fiber cut to realize the IPA function, it is recommended to configure various detection units. In that case, the IPA function need be configured at each node. In the case of a fiber cut, the optical amplifier boards before and after the location of the fiber cut are shut down.
- After one IPA pair is configured at one end of the line, another IPA pair can be configured at the other end or any node of the line. The two IPA pairs jointly carry out the IPA process.

NE-Level Configuration Rules

- One IPA pair is configured at each of the transmit and receive ends. The two IPA pairs form an IPA group.

Board-Level Configuration Rules

- In a simple application scenario, configure the OAU, OBU or OPU as the detection board; configure the OAU or OBU as the control implementation board.
- In an application scenario without any optical amplifier board, configure the D40 or VA4 board as the detection board and configure the VOA board as the control implementation board.
- The optical signal flow direction of the D40 or VA4 board must be the same as that of the main path. This is a prerequisite to detection.

10.2 Intelligent Power Adjustment of Raman System

The power of the pump light from Raman amplifiers is very high. Before you turn on a Raman amplifier, you must configure and enable IPA function. After a fiber cut is detected, shut down the Raman amplifier, so that the optical power of the entire line is on a safety level.

10.2.1 Function Description

To prevent exposed optical fibers hurting human body, especially eyes, the system provides the IPA functions.

In the DWDM system, optical fiber break, equipment failure or optical connector removal may lead to the loss of optical signals of the optical channel. To prevent exposed optical fibers hurting human body, especially eyes, and to avoid surge of the optical amplifier, the system provides the IPA functions. Where the loss of optical power signals happens on one or more optical trunk sections on the main optical channel and the optical supervisory channels, the system can detect the loss of optical signals on the link and instantly shut down the upstream optical amplifier.

The power of the pump light from Raman amplifiers is very high. Hence, in a system configured with Raman amplifiers, you must configure and enable IPA function before you turn on a Raman amplifier. After a fiber cut is detected, shut down the Raman amplifier. Then, there is no strong pump light sent from the LINE interface on the amplifier and thus the optical power of the entire line is on a safety level. After the optical signals in the system become normal, make the amplifier work normally.

NOTE

In the first deployment, the lasers of Raman amplifiers are disabled. Users need to configure IPA function on the T2000. Otherwise, the lasers of Raman amplifiers cannot be enabled.

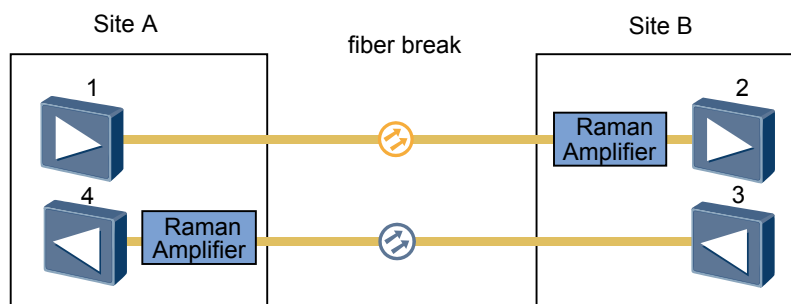
For a link with Raman amplifiers, it is not allowed that you disable or delete IPA function. During the commissioning, maintenance and replacement of a Raman amplifier, when you need to remove the fiber from the LINE interface on the amplifier, you can shut down the Raman amplifier on the T2000 and then disable IPA function. Before removing the fiber, make sure that the Raman amplifier is shut down.

Only NM users that are assigned to the **Device Operation Set** have the authority to manage Raman amplifiers and IPA function.

If the Raman board is rebooted while the IPA function is working (enabled), and this configuration was previously saved on the SCC board, then IPA function is recovered to its original status (enabled) once the reboot process finishes.

Figure 10-3 shows how to achieve the IPA function. When there is a fiber break on the line, the amplifier 3 and 1 are shut down. And the system adopts Raman amplifier(s), shut it down upon any fiber break to lower the optical power of the entire system to a safe level because the optical power of the pump is high.

Figure 10-3 Function description of the IPA



When the optical signals are restored to normal, the optical amplifier will work again. And the amplifier 3 and 1 will be restarted. The Raman amplifier will be restarted also.

NOTE

During the restarting of IPA, Raman amplifiers are shut down. Raman amplifiers are enabled automatically after the link is recovered and IPA returns into the normal working state.

NOTE

In the DWDM system, the IPA function is started only when optical signals of the active optical path are lost. When this function is executed, only the lasers on the main path are shut down. No operation will be implemented on the optical supervisory channel. Hence the functions of all optical supervisory channels will not be affected.

10.2.2 Function Implementation

IPA function is implemented by various boards with different functions.

Fiber Break Detection

There are three methods used to detect the fiber break to achieve the IPA function:

- Detect the optical power of the optical amplifier unit
- Detect the signals of the auxiliary detection unit
- Detect the optical power of the Raman amplifier

Through a combination of the three methods, a fiber break can be determined precisely.

Following is the logic of problem handling:

- If the configured detection item meets the fiber break condition, initiate the shut down process of the IPA.
- If the detection condition recovers to normality, initiate the recovery process of the IPA.

 NOTE

In a system with Raman amplifier(s), do not use the optical amplifier as the only detection tool. The backward pump of the Raman amplifier has so much optical scattering power that the downstream receive site still detects some noise input power even if there is complete fiber break. This brings difficult to the judgment on optical fiber break. Especially when there are limited signal channels, it is not possible to determine fiber break only by the detection of the power of the optical amplifier.

Table 10-2 lists the related alarms of the Raman amplifier that trigger the IPA:

Table 10-2 Related alarms that trigger the IPA restart with Raman amplifier

Detection Board/Auxiliary Detection Board	Alarm Trigger Condition of the IPA
OAU, OBU, OPU	MUT_LOS
LWF, LWFS, LRF, LRFS, TMX, TMXS, ETMX, ETMXS, LBF, LBE, LBES, TMR, TMRS, LOM, LOG, LOGS, LWC1, TRC1, TRC2, FDG	R_LOS, R_LOC, R_LOF, OTU_LOF
TMR, TMRS, LWM, LWMR, LWX, LWXR, LDG, FDG, LQG, ELOG, LQS, AP8, LQM, AS8, FCE, L4G	R_LOS, R_LOC, R_LOF
SC1, SC2, TC1, TC2, ST1, ST2	OSC_LOS, R_LOF, OSC_RDI
FIU	MUT_LOS
D40	MUT_LOS
C6VA4	R_LOS, MUT_LOS
RPC	The board detects the input optical power of the Raman board. If the optical power is less than the lower threshold, a LOS event is reported to the SCC board. The SCC board determines whether to enable the IPA function.

Involved Units

The IPA involves boards of the following types:

- Detection board
 - Functional unit: OAU, OBU, OPU, C6VA4, D40
 - It is optional. The detection board detects the optical power. The threshold of the detection board is configurable. For details on how to set the threshold of the detection board in a Raman system, refer to the *Configuration Guide*.
- Control implementation board
 - Functional unit: OAU, OBU, L2VOA
 - It must be configured. Generally the optical amplifier board performs only the shutdown function. You can also configure the L2VOA board at the transmit end to realize the

optical source shutdown at the transmit end and the detection to the transmitted signals by adjusting the attenuation value. In an IPA pair, the detection board and the shutdown board cannot be the same board. In different IPA pairs, the detection board and the shutdown board cannot be used repeatedly.

- Raman amplifier
 - Functional unit: RPC
 - It must be configured. The Raman amplifier by hardware supports optical power loss detection. The alarm can be used to judge whether there is an IPA fiber break after the threshold of the Raman amplifier is set and the alarm engagement of the Raman amplifier is configured. For details on how to set the threshold of the Raman board, refer to the *Configuration Guide*.
- Auxiliary detection board
 - Functional unit: all OTUs, SC1, SC2, TC1, TC2, ST1, ST2, D40 and FIU
 - It must be configured. The auxiliary detection board detects service signals. The LOS alarm of the auxiliary detection board can be a condition to determine an IPA fiber break.
 - In an OptiX Metro 6100 DWDM system, a maximum of four auxiliary detection boards can be configured. The alarms on each board are not reported directly. After being combined by the logical relation, the alarms serve as a condition for IPA to determine a fiber break.

 NOTE

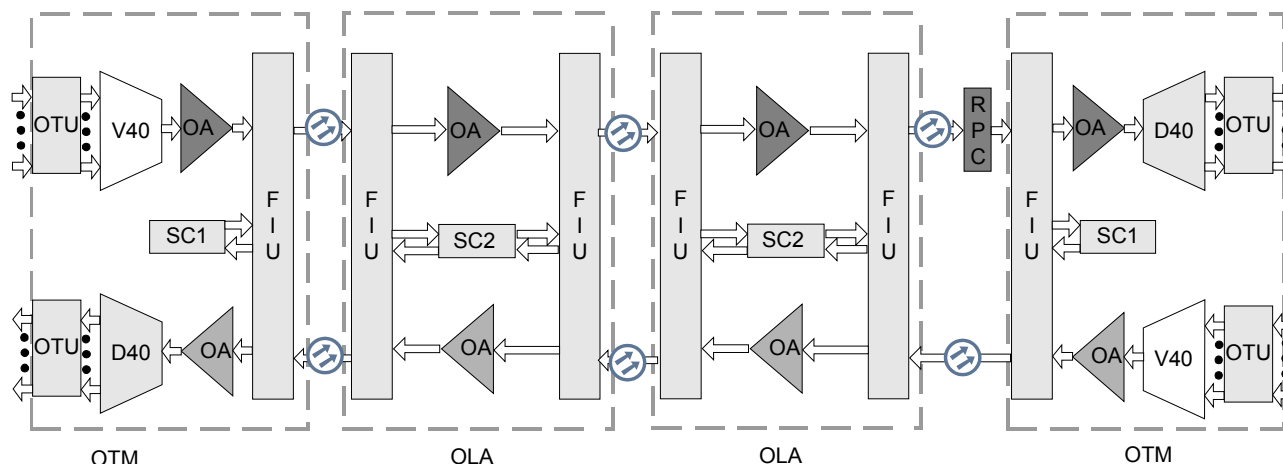
If there is only one Raman board in one IPA pair, the IPA pair can be configured as **Raman amplifier** and it has the shutdown and fiber-break detection functions at the same time.

10.2.3 Networking Application

This section describes an example of the networking application of the IPA with Raman function.

Figure 10-4 is the typical networking application diagram of the IPA with Raman function. The west-to-east OA in each station serves as a control implementation board, and the east-to-west OA in each station serves as a detection board. The control implementation board and detection board in each station form an IPA pair. The RPC board at an endpoint of the line can also be set to perform the shutdown and detection function. All the OTUs, FIU, OSC and D40 boards in each station can serve as the auxiliary detection boards which facilitate the accurate determination of fiber cuts.

Figure 10-4 IPA networking in a Raman system



10.2.4 Configuration Principle

IPA function can be configured according to customers' requirements.

Network-Level Configuration Rules

- To ensure the normal communication between NEs, the software versions of all NEs involved in inter-NE communication must be the same.
- When the optical amplifier board serves as a fiber cut detection board, the IPA function can be configured at only the ingress and egress nodes. In the case of a fiber cut, the optical amplifier boards at the ingress and egress nodes are shut down.
- To accurately locate any fiber cut to realize the IPA function, configure both the detection unit and the auxiliary unit. In that case, the IPA function need be configured at each node. In the case of a fiber cut, the optical amplifier boards before and after the location of the fiber cut are automatically shut down or the VOA shuts down the optical signals on the line.
- After one IPA pair is configured at one end of the line, another IPA pair can be configured at the other end or any node of the line. The two IPA pairs jointly carry out the IPA process.



NOTE

IPA pairs must be configured in a pair.

NE-Level Configuration Rules

- When the optical amplifier board serves as a fiber cut detection board, the IPA function can be configured at only the ingress and egress nodes. In the case of a fiber cut, the optical amplifier boards at the ingress and egress nodes are shut down.
- To accurately locate any fiber cut to realize the IPA function, configure both the detection unit and the auxiliary unit. In that case, the IPA function need be configured at each node. In the case of a fiber cut, the optical amplifier boards before and after the location of the fiber cut are shut down.
- Only the Raman amplifier supports cross-subrack configuration. Other boards involved in an IPA protection group must be configured in the same subrack.

- When you configure IPA, make sure that inter-subrack communication is normal. If the communication is abnormal, an error is returned when you configure cross-subrack IPA pairs. During IPA operation, IPA stops if the inter-subrack communication is abnormal. When the inter-subrack communication is restored, IPA is restarted.
- Generally, the OTU board is not planned for any optical amplifier node. If you need to enable the auxiliary detection board, make sure that the local subrack is configured with the OTU.
- The optical signal flow direction of auxiliary detection boards must be the same as that of the main path.

Board-Level Configuration Rules

- Configure the OAU or OBU as the detection board; configure the OAU or OBU as the control implementation board.
- Configure various OTU boards, SC1, SC2, TC1, TC2, ST1, ST2, FIU and D40 as auxiliary detection boards.
- To ensure reliable detection, avoid the case that there is only one type of auxiliary detection board (for example, there is only the OSC board).

10.3 Automatic Level Control

The system provides the automatic level control (ALC) function. As the attenuation on a line segment is increased, the output power as well as the input and output powers of other downstream amplifiers will not be changed. Hence there will be much less influence on OSNR. The optical power received by the receiver will not be changed.

10.3.1 Function Description

When ALC function is enabled, the increase in the line loss in a section causes the decrease in the input power of the amplifier in that section. Its output power and the input and output power of the downstream amplifiers remain the same.

In a DWDM system, optical fiber aging, optical connector aging or manual factors may lead to abnormal loss of transmission lines. In case the loss on a line segment increases, all input and output power is reduced on all downstream amplifiers. The system OSNR deteriorates. At the same time, the received optical power will also be reduced. Receiving performance will be greatly affected. The closer the attenuated segment is to the transmission end, the greater is the influence on OSNR, as shown in [Figure 10-5](#).

If ALC function is activated, this effect can be minimized. As the loss on a line segment is increased, the input power on the amplifier is reduced. But due to ALC, the output power as well as the input and output powers of other downstream amplifiers will not be changed. Hence there will be much less influence on OSNR. The optical power received by the receiver will not be changed. [Figure 10-6](#) shows the power changes on optical line amplification regenerators in the ALC mode in case of abnormal loss on optical fiber lines.

Figure 10-5 System power with ALC inactivated

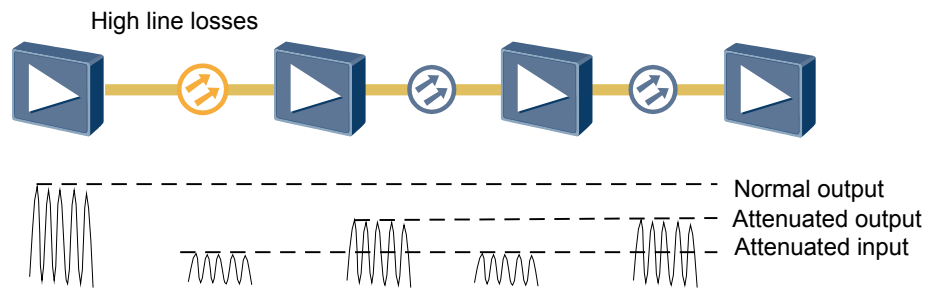
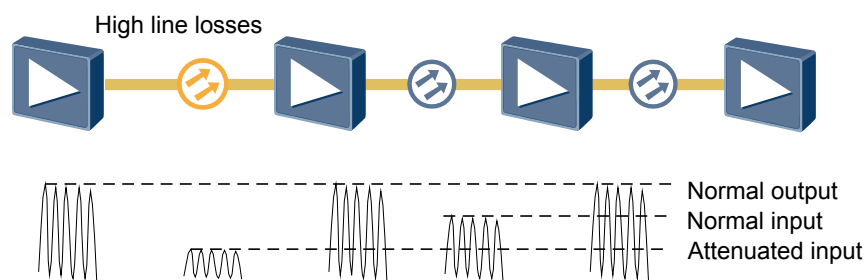


Figure 10-6 System power with ALC activated



NOTE

Normally, two elements might cause the input power change in the optical amplifier:

- The addition/reduction of access channels (multiple channels might be added or dropped at the same time).
- The abnormal loss in the physical media.

10.3.2 Function Implementation

ALC function is implemented by various boards with different functions.

Realization Principle

There are two ways to achieve the ALC function: reference power detection and channel amount detection. The realization principle of the channel amount detection mode is relatively simple, and the multichannel spectrum analyzer unit (MCA) needs to be configured. Although the reference power detection mode does not need to be configured with MCA, the realization principle of it is complicated because many parameters need to be set. The link attenuation adjustment mode is optimized on the basis of the reference power detection mode.

- Channel amount detection

Prerequisite: One MCA needs to be configured on the ALC link.

Realization: The optical amplifier works in automatic gain control (AGC) mode and realizes ALC function with the MCA. The MCA analyzes the amount of working channels. Based on the amount of channels and the output power, the optical amplifier determines the working status and adjusts the attenuation to keep the output power stable (the absolute value of total power remains unchanged).

- Reference power detection

Prerequisite: The output optical power of the first node on the ALC link is taken as a reference value.

Realization: The optical amplifier works in AGC mode, by adjusting the attenuation to keep the output optical power stable. (The absolute value of total power remains unchanged.)

ALC detects gain exceptions for a reference node to check whether the output power is abnormal. The input and output power of the reference node are checked in a scheduled manner, to obtain the actual gain value. The gain value is compared with the configured standard gain. If the gain change exceeds the exception threshold, a gain exception is reported to the T2000 to prompt the user to start ALC adjustment.

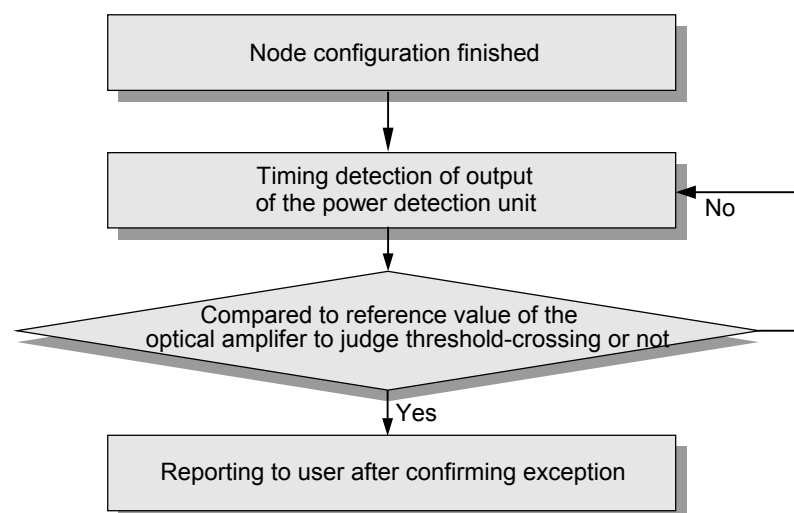
Single-Site ALC Detection Flow

- Channel amount detection and reference power detection

Figure 10-7 shows the ALC exception detection flow of channel amount detection and reference power detection. To implement the ALC at a single site, follow the steps below:

- The output optical power of the first site on the ALC link is taken as a reference value.
- The SCC periodically checks the output optical power of the optical amplifier unit, and compares it with the standard output power.
- If the difference between them is beyond a certain range, and it is not power fluctuation caused by adding or dropping wavelengths, output power exceptions are reported to the T2000.
- User can manually restart or the system can automatically triggers the ALC power adjustment process according to the configuration.

Figure 10-7 ALC exception detection flow



Calculation of the standard output power is as follows. For the detailed definitions of the parameters, refer to **Table 10-3**.

- Channel Amount Detection

$$P = \text{StdPower} + \text{Offset} + 10\lg N$$

N = number of channels

- Reference Power Detection + Noise compensation

When $P_{ref} \geq 14\text{dBm}$:

$$P = P_{ref} + \text{offset} + P_{offset} \quad (1)$$

When $P_{ref} < 14\text{dBm}$:

$$P = P_{ref} + \text{offset} + P_{offset} + (14 - P_{ref}) \times P_{ase} / (14 - \text{StdPower}) \quad (2)$$

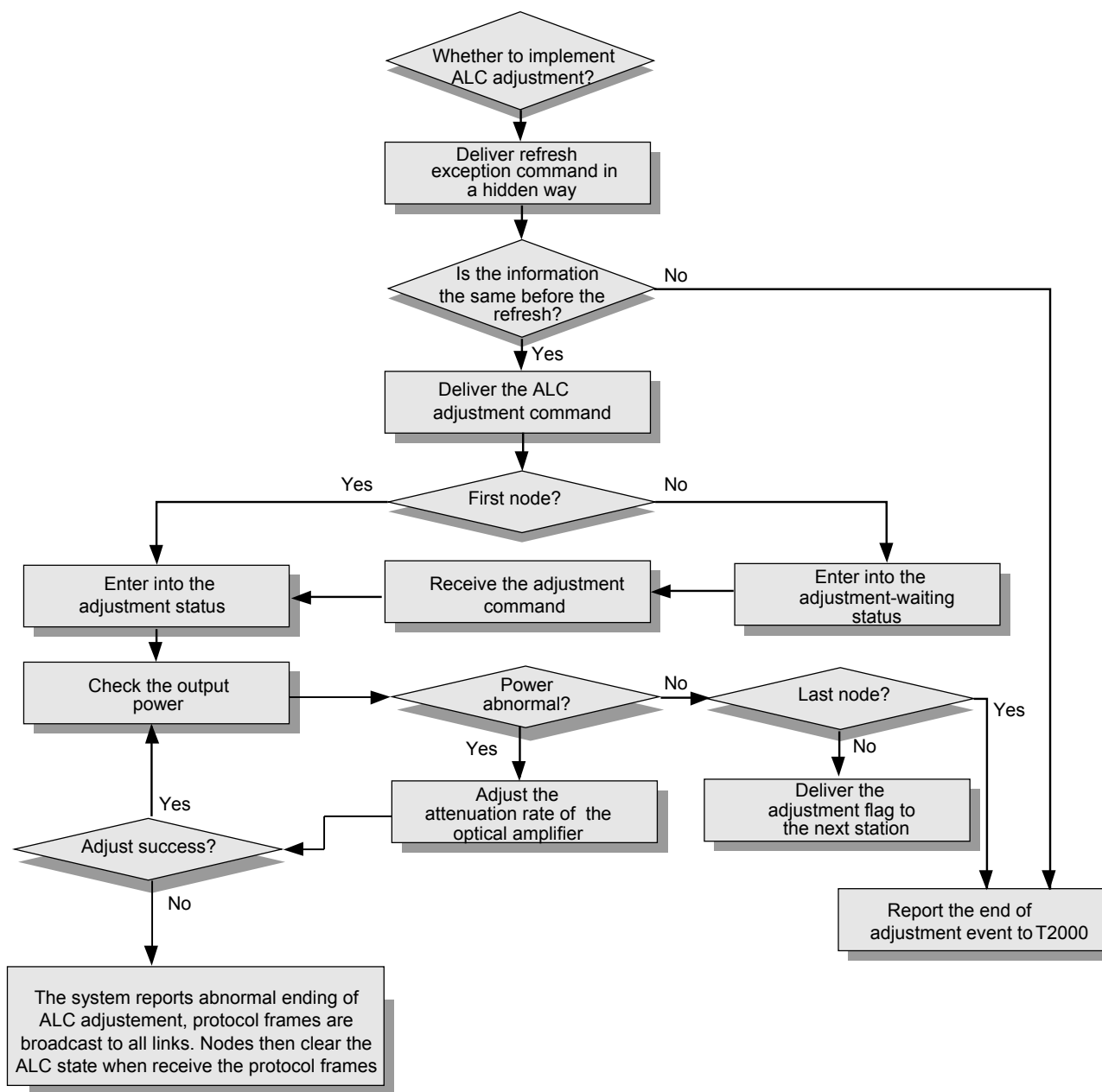
Table 10-3 Definition of parameters for the calculation of the standard output power

Symbol	Definition	Description	Value
P	Standard output power	The ALC function judges whether the output optical power at a node is normal by comparing it with the standard output optical power. If the detected output optical power value is not within the standard output optical power range, the output power value is considered abnormal.	Measured value (the output optical power of the optical amplifier unit when the system operates normally)
P _{ref}	Output power of the reference node	Detected output optical power of the reference node (usually the first node of the ALC link).	Measured value (the output optical power at the reference point)
Offset	Standard power offset	The difference between the standard optical power of a single wavelength of the amplifier at the detection point and that of the amplifier at the reference point.	Counted value ^a
P _{offset} ^c	Overall optical power offset compensation rate	When the total power of the system reaches a certain value, the accumulated noise can be regarded fixed. Adopt a fixed P _{offset} to compensate for it.	Counted value (formula 1, 2)
P _{ase} ^c	Single-wavelength ASE noise compensation rate	P _{ase} mainly applies to the situation where there are only a few system wavelengths. The noise caused by the amplifier is rather big and requires extra compensation.	Counted value (formula 1, 2)
StdPower _r	Standard output power for single wavelength	The standard output optical power of a single wavelength of the optical amplifier (noise influence is not considered).	1 dBm - 7 dBm (parameter of the optical amplifier unit) ^b

Symbol	Definition	Description	Value
		a: For channel amount detection, when you decide the value of the Offset, take into consideration the power difference caused by noise compensation besides the single-channel standard optical power difference. The deviation caused by noise also needs to be set because the output optical power of the optical amplifier board can be different from the standard output. For instance, the output optical power can be raised a bit considering the ASE noise of the optical amplifier. The user can set it based on the actual situation. The deviation range is from -3 dB to 5 dB. b: The value of Stdpower can be selected from 1 dBm to 7 dBm for different systems. c: The value of the Poffset and the Pase can be calculated according to formula.	

ALC Link Adjustment Flow

Figure 10-8 ALC adjustment flow



The T2000, after receiving the power exception events reported by the NE software, performs the adjustment as shown in [Figure 10-8](#).

- After the T2000 reports the ALC power exceptions, the service channel can determine whether to execute the ALC adjustment command by referring to other system information.
- When the ALC adjustment command is executed on the T2000, the T2000 delivers a command to update the ALC exception information in a hidden way. Then, the NE software,

after receiving the command to update the exception information from the T2000, checks the link reference. After that, each node directly judges if the output optical power is abnormal based on the latest reference obtained (namely, with no need to confirm). If yes, the exceptions are reported to the T2000. Otherwise, no processing is made.

- If the exception information returned from the host is consistent with that before the user delivers the adjustment command, the T2000 formally delivers the ALC adjustment command. If the exception information returned changes, the event indicating the end of adjustment is reported. Thus, it can be ensured what the system adjusts after the user starts the ALC adjustment is what the user can see and wants to adjust.
- During the ALC adjustment on the ALC link, the first site reports the ALC adjustment start event when the ALC adjustment begins and the last site reports the ALC adjustment end event when the ALC adjustment ends. Each site reports the VOA adjustment event when the VOA adjustment begins on it or ends on it.
- During the adjustment, in case of an abnormal event (for example, the adjustment scope of the system exceeds the adjustable scope of VOA) that stops the normal ALC adjustment an ALC abnormal pause event is reported. Besides, a protocol frame is broadcasted to the entire link, requiring the link nodes that receive the protocol frame to clear their ALC states.

Involved Boards and Ports

- Power detection unit
 - Function unit: OAU, OBU, OPU, L2VOA, C6VA4
 - It monitors the output power of the power detection unit to determine whether the link is normal.
- Reference unit
 - Function unit when using the wavelength count detection mode: The MCA board detects signals in each channel on the receive end and provides the ALC with the wavelength count information.
 - Function unit when using the power reference mode: The ALC detects the output power of the amplifier unit, such as the OAU, OBU, or OPU. The output power is used as the reference for the downstream node.
- Power adjustment unit
 - Function unit: OAU, VOA, and VA4 (Although the OAU is an amplifier unit, it can serve as an adjustment unit because it is equipped with a VOA.)
 - It adjusts the line attenuation.
- Optical supervisory channel unit and clock transmitting unit
 - Function unit: SC1, SC2, ST1, ST2
 - It provides a supervisory channel connection that serves as a physical channel for transporting protocol frames.
- System control & communication unit
 - Function unit: SCC
 - It is the system that performs the ALC function.
- Ethernet interface
 - Function unit: ETHERNET1 and ETHERNET2 on the subrack interface area are used for the connection between subracks.

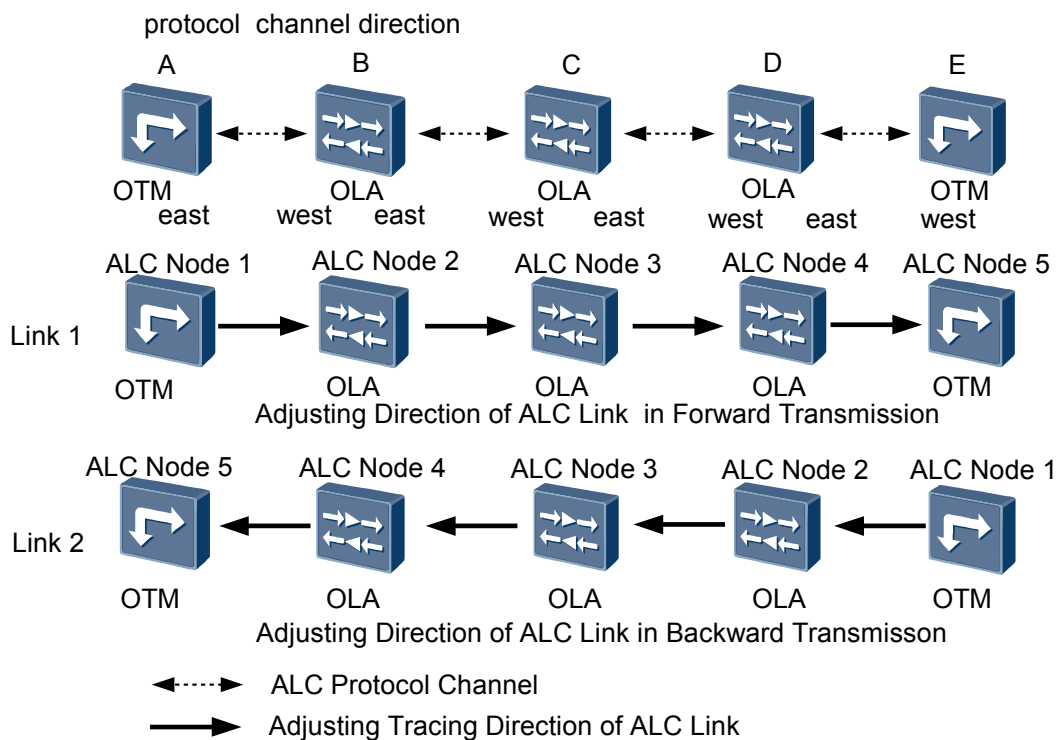
- It enables the inter-subrack communication, achieves the communication with the subrack where the supervisory channel board is located, and transmits the ALC protocol frames.

10.3.3 Networking of Application

It requires creating ALC link on the line to perform the adjustment of the ALC.

To fulfill the ALC application, the ALC link must be created first. Take the ALC networking as an example. See [Figure 10-9](#).

Figure 10-9 Networking of ALC application



Creating the ALC Protocol Channel

The information exchange between ALC link nodes needs to be conducted by using the ALC protocol frames transmitted on the supervisory channel. Configuring a protocol channel is the precondition for providing the normal ALC adjustment function. The units that can provide the supervisory channel are SC1/SC2. The track direction of the protocol channel can be set to eastward or westward.

 NOTE

The direction configured for the ALC protocol channels must be correct, and consistent with the physical cabling direction of the supervisory channel unit. If you do it the opposite way, the protocol channels may work normally, but the protocol frames are transceived in the wrong direction. Thus, normal information exchange cannot be implemented, as well as the ALC function.

Creating ALC Link

It requires creating ALC link on the line to perform the adjustment of the ALC. The ALC link includes the nodes that take part in the adjustment of the ALC. To determine the ALC link direction, refer to [Figure 10-9](#). For link 1, as its direction is consistent with the protocol channel direction, it is in the forward direction. For link 2, as its direction is contrary to the protocol channel direction, it is in the backward direction.

NOTE

As for the line that has the OADM stations, regard the transmit end and the receive end of the OADM as two ALC links. The optical amplifier at the receive end of the OADM site is regarded as the last node in the former ALC link. The optical amplifier at the transmit end of the OADM site is regarded as the first node in the next ALC link.

10.3.4 Configuration Principle

ALC is optional and configured according to users' requirement.

Network-Level Configuration Rules

- The egress node must not be set as the reference node; the ingress node must be configured with the reference unit.
- When you configure the ingress node, do not configure the detection unit and the reference unit at the same optical interface.
- A node without an optical amplifier can serve as a passthrough node. Such a node provides only a protocol channel.
- It is recommended to configure a node that adds/drops wavelengths as the reference node.

NE-Level Configuration Rules

- To realize the ALC function, the optical power adjustment board need be configured at the adjustment node; the optical amplifier unit need be configured at the reference node

NOTE

The optical power is usually detected at the ingress node of an ALC link. In the case of the MCA board, the wavelength count can be detected at any node.

- To realize the ALC function, it is recommended to configure the optical amplifier unit, the OSC unit and attenuation adjustment unit in the same subrack.
- If the optical amplifier unit, the OSC processing unit and attenuation adjustment unit are configured in different subracks, the ETHERNET1 and ETHERNET2 network interfaces in the subrack area of the OptiX Metro 6100 must be connected to set up inter-subrack communication.

Board-Level Configuration Rules

- Configure the OAU, OBU, OPU as the reference unit.
- Configure the OAU, OBU, OPU, VOA, VA4 as the detection unit.
- Configure the OAU, VOA, VA4 as the attenuation adjustment unit.
- When the detection board is the OBU, the attenuation adjustment unit must not be null.

10.4 Automatic Power Equilibrium

The system provides the automatic power equilibrium (APE). With the APE function, you can enable the system to automatically adjust the optical power of the transmit end of each channel to keep the flatness of the optical power of the receive end to maintain the OSNR. OptiX Metro 6040 DWDM system can realize this function with the OptiX Metro 6100 DWDM system.

10.4.1 Function Description

With the APE function, you can retain the flatness of the optical power of the receive end and to maintain the OSNR.

In a DWDM system, the variety of the optical fiber condition in the running of the system may change the flatness of a channel's power from that in the commissioning, and degrade the optical signal noise ratio (OSNR) of signals at the receive end, as shown in . With the APE function provided by the system, you can enable the system to automatically adjust the optical power of the transmit end of each channel to retain the flatness of the optical power of the receive end close to that in the commissioning and to maintain the OSNR, as shown in .

The application of the APE streamlines the operation of the network maintenance for the operator. The design of starting regulation manually facilitates you to determine whether to adjust the optical power according to the actual status of the network.

10.4.2 Function Implementation

The APE function is implementation by the service unit and the SCC unit.

Implementation Principles

To implement the APE, follow the steps below:

- During the commissioning, apply manual adjustment on the power regulating unit to ensure that each channel is working normally and that bit error rate and OSNR meet the requirement.
- After the commissioning, save the power curve of the receive end as the standard power curve.
- Detect optical power of every channel received by the power monitoring unit through the optical port at the receive end.
- According to the detected optical power of every channel, adjust the attenuation rate of the according channel of the power regulating unit, so as to maintain the optical signal-to-noise ratio (OSNR) of every channel at the receive end by keeping the flatness of the optical power of every channel.

NOTE

During the running of the equipment, the power monitoring unit analyzes the data scanned in a spectral scanning period, which is set in the power monitoring unit configuration and is not provided in the APE. If the power offset exceeds the threshold configured, the system reports the event of optical power unbalance. The user can enable the automatic adjustment or determine whether to adjust it based on the network condition.

Involved Boards

The APE functions through the service board and the SCC board. The APE involves boards of the following types:

- Power monitoring unit
 - It detects the signal power of the channels at the receive end and reports an APE uneven event.
 - Board supporting this function include: MCA.
- Power regulating unit
 - It is the adjusting entity of the APE and adjusts the attenuation of channels.
 - Boards supporting this function include: V40.
- System control and communication board
 - It is the executive entity of the APE.
 - Boards supporting this function include: SCC.

NOTE

As for the APE function, the APE protocol frames can be transmitted to each node once the physical communications route is available. Besides the OSC, the APE protocol frames can be transmitted also through ESC, interconnection of Ethernet interfaces or DCN.

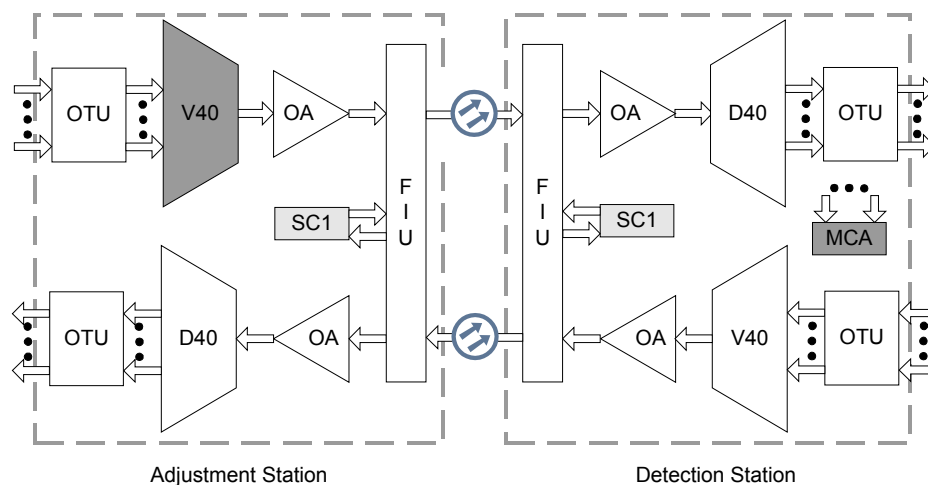
10.4.3 Networking Application

Introduces the networking for this application, illustrating only the unidirectional APE pair.

Figure 10-10 shows the networking for this application, illustrating only the unidirectional APE pair.

As for the APE function, the optical interface of the MCA at the receive site detects the optical power equilibrium of each channel. Then the VMUX (V40) at the transmit end adjusts the optical power attenuation value to balance the OSNR of each channel according to the detection. The communication between sites is achieved through the supervisory channel.

Figure 10-10 APE networking



10.4.4 Configuration Principle

APE is optional and configured according to users' requirement.

Network-Level Configuration Rules

- The APE function can be used during deployment commissioning and is optional at the customer's requirement.
- To avoid errors during an APE adjustment process, do not configure an APE pair in an inter-span overlap manner.
- APE requires that the adjustment board is configured in the transmit-end node and the detection board is configured in the receive-end node.
- To enable the APE function, one optical interface of the adjustment board at the adjustment node must be paired with one optical interface of the detection board at the detection node, and the APE function must be set to **Enabled**.
- The OADM adds/drops some wavelengths whose optical power cannot be equalized when the two end nodes are configured with APE. As a result, the wavelengths to be added/dropped to/from the OADM node need be set with monitoring flags, and the APE function at the two nodes need be set to **Disabled**.
- The DWDM system is a two-fiber bidirectional system. Hence, one detection board and one adjustment board are configured at each of the two ends of one APE pair.

NE-Level Configuration Rules

- To realize the APE function in a system that adopts OSC communication, it is recommended to configure the MCA board and the OSC processing board in the same subrack. If the two boards are configured in different subracks, inter-subrack communication must be set up.
- To realize the APE function in a system that adopts OSC communication, it is recommended to configure the optical power adjustment board and the OSC processing board in the same subrack. If the two boards are configured in different subracks, inter-subrack communication must be set up.
- To realize the APE function, the transmit-end OTM node need be configured with the V40 board and the receive-end OTM node need be configured with the MCA board. Each of the two ends of one APE pair need be configured with one MCA and one V40.
- To realize the APE function, it is recommended to configure the MCA board and the OSC processing board (including the SC1/SC2 or ST1/ST2) in the same subrack. If the MCA board and the OSC processing board are configured in different subracks, inter-subrack communication must be set up.
- To realize the APE function, it is recommended to configure the V40 board and the OSC processing board (including the SC1/SC2 or ST1/ST2) in the same subrack. If the V40 board and the OSC processing board are configured in different subracks, inter-subrack communication must be set up.
- The system adopts the master-slave control mode. The node configured with the MCA is the master node that provides core functions such as detection, judgement, computation and communication. The node configured with the V40 just serves to adjust each channel according to protocol frames. You just need to configure at the MCA node like this: Specify the V40 node as an even node, and set the transmit-end supervisory subrack and the receive-end supervisory subrack to **Null**.
- An OADM node might exist between the adjustment node and the detection node. This OADM node locally adds/drops wavelengths. In this case, APE cannot equalize the optical

power of the wavelengths added/dropped to/from this OADM node. Hence, the monitoring flags of the wavelengths to be added/dropped to/from this OADM node should be set to **Disabled**.

Board-Level Configuration Rules

- Configure the MCA as the detection unit.
- Configure the V40 as the adjustment unit.
- Configure the SC1, SC2, ST1 or ST2 as the OSC unit.

11 Operation, Administration and Maintenance

About This Chapter

This chapter describes the operation, administration and maintenance of the product.

[11.1 System Operation](#)

In order to improve the WDM system, the system adopts various technologies to control, adjust and manage the system. This is to ensure the normal and effective running of the system.

[11.2 Administration and Maintenance](#)

The design of the cabinet and boards and the configuration of the system embody the requirements on easy and effective operation, administration, and maintenance of the equipment.

[11.3 NE Security Management Features](#)

Security management is to prevent illegal users from logging in to the network. It is an important feature to ensure the network security.

11.1 System Operation

In order to improve the WDM system, the system adopts various technologies to control, adjust and manage the system. This is to ensure the normal and effective running of the system.

- The system provides the automatic level control (ALC) function. When the attenuation of a certain fiber span of the line increases, this function keeps the output power of the downstream optical amplifiers from changing to minimize the impact on the OSNR. In addition, the optical power received by the receiver does not change.
- The system provides the intelligent power adjustment (IPA) function. When there is a fiber cut on the line, the downstream optical amplifier is shut down to prevent exposed optical fibers hurting human body. In addition, to avoid surge of the optical amplifier, the system provides the IPA functions. Where the loss of optical power signals happens on one or more optical trunk sections on the main optical channel, the system can detect the loss of optical signals on the link and instantly shut down the upstream optical amplifier. Then, all the downstream optical amplifiers have no optical power output due to the amplifier feature. This keeps the optical power of the line at a safe level.
- The system fully considers the demands for optical fiber management on the design of the cabinet and subrack. Various cabling channels are available to facilitate the fiber management in a cabinet and between cabinets. This simplifies the installation and maintenance of the system.
- The system is designed with intelligent system for ambient temperature monitoring, reporting, and alarming. This ensures that the normal running of the system is under a stable temperature.

11.2 Administration and Maintenance

The design of the cabinet and boards and the configuration of the system embody the requirements on easy and effective operation, administration, and maintenance of the equipment.

11.2.1 Supervision and Administration Module

The system control and communication (SCC) board monitors and manages the system NE. The power monitoring unit (PMU) in the power box of the system fulfills the input of external environment alarms and the output and concatenation of various alarms.

SCC Board

SCC board is fixed into the IU7 slot. (This slot is only for the SCC board.)

The SCC board collects the state information, alarm and performance parameters from the functional modules of each board.

Then the SCC converts, processes and stores the information and parameters. At the same time, it sends the control and administration information to the other functional modules.

OptiX Metro 6100 provides two versions that involve four types of SCC boards. [Table 11-1](#) provides the differences between these four types of SCCs.

Table 11-1 Function differences between four types of SCCs

Board Type	Function Difference
C8SCC01	The board supports the LAN Switch module to cascade the network interfaces of subracks, and supports the HWECC, IP over DCC and OSI over DCC protocols. The board does not support the CF card.
C8SCC02	The board supports the LAN Switch module to cascade the network interfaces of subracks, and supports the HWECC, IP over DCC and OSI over DCC protocols. The board supports the CF card.
C6SCC01	The board does not have the LAN Switch module and supports only the HWECC protocol. The board does not support the CF card.
C6SCC02	The board does not have the LAN Switch module and supports only the HWECC protocol. The board supports the CF card.

The SCC provides functional interfaces to facilitate the communication between the functional modules of each board and the NM, as shown in [Table 11-2](#).

Table 11-2 Description of the functional interfaces of the SCC in the OptiX Metro 6100 system

Functional Interface	Description
F&f (Note)	Connect the RS-232 interface to a PC or a workstation for commissioning.
OAM (Note)	The operation, administration and maintenance interface The X.25 interface is provided to communicate with the terminal through the public packet switched network. OAM interface locates in the interface area of the OptiX Metro 6100 subrackOptiX Metro 6040 chassis.
F1 (Note)	Provides three orderwire phones and a 64 Kbit/s co-directional data channel.
F2 (Note)	Uses the F2 byte of the supervisory channel and possesses the features of both RS-232 and RS-422 interfaces. This interface can be used for express orderwire. The maximum rate is 19.2 Kbit/s.
F3 (Note)	Uses the F3 byte of the supervisory channel and possesses the features of both RS-232 and RS-422 interfaces. This interface can be used for express orderwire. The maximum rate is 19.2 Kbit/s.

Functional Interface	Description
DCC communication	Provides the data communication channel (DCC) of the supervisory link.
Communication module	Communicates with other boards in the subrack, collects performance data, and delivers the configuration.
Qx	Network management communication interface. Qx interface locates in the interface area of the OptiX Metro 6100 subrackOptiX Metro 6040 chassis.

The SCC monitors the running status of the boards in the OptiX Metro 6100 NE.

The main monitoring parameters include:

- Input optical power
- Output optical power
- Laser temperature
- B1 performance parameter
- FEC performance parameter
- Ethernet performance parameter
- OTN performance parameter

PMU Board

PMU board is fixed into the IU14 slot. (The slot is only for the PMU board.)

OptiX Metro 6100 provides two versions that involve five types of PMU boards. That is C8PMU and C6PMU.

The front panel of the PMU board provides four indicators for the status of the subrack and seven connectors of RJ-45.

Table 11-3 Description of the functional interfaces of the PMU in the OptiX Metro 6100 system

Interface	Connector Type	Description
F&f	RJ-45	Bears features of RS-232 interfaces.
Serial 1	RJ-45	Uses F2 bytes of the supervisory channel. Provides features of RS-232 and RS-422 interfaces. The maximum rate is 19.2 Kbit/s.
Serial 2	RJ-45	Uses F3 bytes of the supervisory channel. Provides features of RS-232 and RS-422 interfaces. The maximum rate is 19.2 Kbit/s.
F1	RJ-45	Co-directional 64 Kbit/s data interface.

Interface	Connector Type	Description
LAMP1/LAMP2	RJ-45	Output alarm driving signals to cabinet indicators. It is used to concatenate equipment alarms among subracks.
ETH ^a	RJ-45	Ethernet communication interface, realising communication between boards in the subrack and external equipment.
a: Only the C6PMU provides the ETH interface.		

Functions of PMU Board

The functions of the C8PMU board are listed as follows:

- Monitors the voltage of the two powers inputs of the subrack. Reports over-voltage alarms, under-voltage alarms and the detected voltage to the SCC.
- Inputs 16 external alarm values to realize remote monitoring of external alarms.
- Outputs four alarm values. Outputs alarms to the DC power distribution cabinet or to the integrated alarm control device.
- Monitors the board temperature. Reports temperature performance and alarms to the SCC.
- Instructed by the SCC, drives the cabinet indicators and the four subrack status indicators on the front panels. Output the status of four status indicators to the alarm cascading interfaces.
- Receives the status signals of the lightening protection circuit of the power filter unit (DPFU). Reports lightening protection alarms to the SCC.
- Provides the OADM with 5 V power supply.
- Provides audible alarms and the alarm test switch.
- Provides the in-service loading of the board software (needs to cooperate with the C8SCC) and the querying of the production information.

Besides the preceding six interfaces, the C6PMU also provides an RJ-45 ETH electrical interface (an Ethernet interface through which external devices communicate with each board in the subrack). The C6PMU, however, supports neither the in-service loading of board software nor the query of board manufacturing information.

11.2.2 Optical Supervisory Channel Administration

The management information of the stations in the system is transmitted through the optical supervisory channel (OSC).

- Functions of the OSC

The optical supervisory channel (OSC) serves to transmit the monitoring and management information among the stations in the OptiX Metro 6100 system. The channel of the OSC is at 1510 nm.

The OSC boards include the SC1/SC2, TC1/TC2 and ST1/ST2.

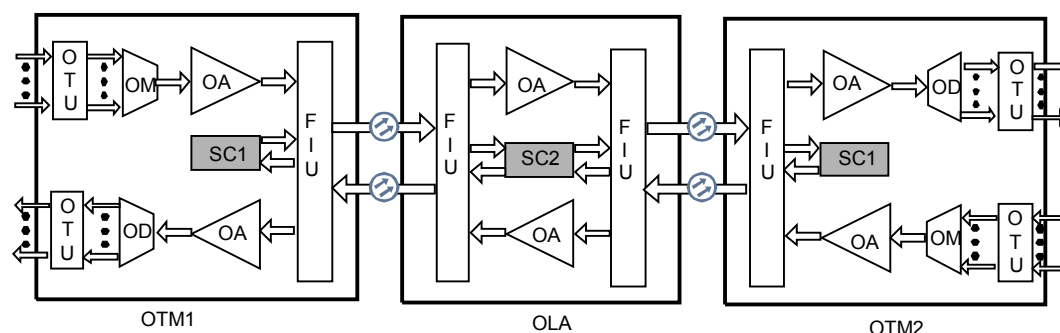
The 2M CMI encoding is adopted.

- Working of the OSC

Figure 11-1 shows the signal flow of the OSC among three stations. The signals of the OSC and the service signals are independent from each other. The supervisory signals are not amplified. They are terminated and regenerated in a station.

Here is an example to explain the communication process of the OSC. The communication between the optical terminal multiplexer (OTM) and the optical line amplifier (OLA) is taken as the example.

Figure 11-1 Signal flow of the OSC among three stations in chain networking



In the transmit direction of the OTM1, the SC1 in OTM1 receives the overhead data frames from the SCC. After processing of the signals, E/O conversion is performed by the optical transmit module. This is to modulate the supervisory data frames to the wavelength of the OSC (1510 nm). The wavelength of the OSC is multiplexed with the service signals by the multiplexer of the FIU.

Then signals are transmitted to the OLA. The demultiplexer of the FIU demultiplexes the signals into service signals and OSC signals. The service signals are transmitted to the east after they are regenerated and amplified by the optical amplifier unit (OAU).

The optical receive module of the SC2 in the OLA performs O/E conversion on the OSC signals and the supervisory data frames are recovered. After being processed, the supervisory data frames are sent to the SCC of the OLA to exchange data.

In the receive direction of OTM1, the SCC in the OLA transmits the data to the SCC in OTM1 in a similar process. The OSC is divided in sections. The communication between the OLA and OTM2 is the same as that between the OLA and OTM1.

The OSC working modes of the ST1/ST2/TC1/TC2 is the same with the SC1/SC2.

- **Frame Structure of the OSC Signals**

Figure 11-2 shows the timeslots of the E1 frame adopted by the OSC signals. There are 32 timeslots in a frame, numbered 0 to 31.

Figure 11-2 Timeslot assignment diagram of the OSC overhead

0	1	2	3	...	14	15	16	...	31
---	---	---	---	-----	----	----	----	-----	----

For the definition and functions of the timeslots in the E1 frame of the OSC, refer to [Table 11-4](#).

Table 11-4 Functions of the timeslots in the E1 frame of the OSC

Timeslot Number	Name	Function
0	Frame alignment signal	Locates the starting point of each E1 frame.
1	E1 byte	Provides the path for orderwire phone. Orderwire phone interface locates in the interface area, named as PHONE1 and PHONE2.
2	F1 byte	Co-directional 64 Kbit/s data interface
3–13, 15	D1–D12 bytes	DCC channel Used to transmit the OAM data information, such as the issued commands and the data of the queried alarms and performances. The OSC board extracts relevant bytes and sends them to the SCC for processing. The D4 to D12 bytes can be configured to transmitted the ASON management information transparently.
14	ALC byte	Provides the channel for the transmission of ALC protocol byte.
17	F2 byte	Reserved for the user (usually, the network provider) for the temporary orderwire communication with the purpose of specific maintenance.
18	F3 byte	Reserved for the user (usually, the network provider) for the temporary orderwire communication with the purpose of specific maintenance.
19	E2 byte	Provides the path for orderwire phone.
20	APE byte	Provides the channel for the transmission of APE protocol byte.
Other	Reserved	-

11.2.3 Networking Management

The system supports the management of the network management tool over equipment.

Any NM tool in compliance with the ITU-T Recommendations can be used to manage the system. Through the connection between the NM tool and the SCC of the NE, the user can:

- Perform a centralized management of multiple systems at a station.
- Monitor the state of the equipment and the network by fault reports and alarm monitoring.
- Configure and plan multiple NEs.

11.2.4 Alarm and Performance Event Management

The system uses the alarm and performance event monitoring function for administration and maintenance.

System Alarm Function

The system supports the alarm management function. This enables the set and query of alarm level, automatic report of alarms, and deletion of history alarms. These help the user to monitor and maintain the system in real time.

It records all the alarms that have happened, including the alarms that have been cleared or not and all the performance events.

System Performance Monitoring Function

Performance event is a key parameter that reflects the working performance

The knowledge of the causes that lead to the performance events, the relevant boards, and alarms helps locate the faults during routine maintenance and analyze the faults when they occur.

The performance events are related with the alarms. If the performance event value exceeds the preset threshold value, relevant alarm will be generated. Thus, when a performance event occurs, check whether the relevant alarm is generated.

System Monitoring Items

The system provides the following system monitoring information.

- Temperature of the running board
- In-position status of the physical board
- Management function of the boards in different functional units
- Management function of the fan module
- Management function of the power module
- Optical power of the OTU
- Current of the laser
- Temperature of the laser

11.3 NE Security Management Features

Security management is to prevent illegal users from logging in to the network. It is an important feature to ensure the network security.

11.3.1 Basic and Advanced ACL Access Control

The system supports basic and advanced access control list (ACL) to realize access control.

ACL provides basic data stream filtering function. The NE configured with ACL can determine whether to filter the IP packet when the packet passes the NE. ACL determines that a specified data stream can be transmitted into or out from a network.

ACL is configured to ensure network security. If ACL is configured properly, a network has high security even when it is under attack. ACL provides the basic traffic control function.

ACL determines whether an NE receives specified IP packets. The NE configured with ACL checks each IP packet arrived and determines whether to receive the IP packet based on the ACL.

The OptiX Metro 6100 supports basic and advanced ACL to ensure the security of each NE.

- Basic ACL

It realizes the access control through source IP. The NE requiring normal security level can be configured with basic ACL, which enables the NE to check the source address of IP packets. Basic ACL occupies less resources.

- Advanced ACL

It realizes the access control through source IP, destination IP, source port, destination port and ICMP protocol type. The NE requiring high security level can be configured with advanced ACL, which enables the NE to check the source address, destination address, source port, destination port and protocol type of IP packets. Advanced ACL occupies a lot of resources. When advanced ACL and basic ACL coexist, the system performs verification based on advanced ACL rules.

For detailed ACL configuration procedures, refer to the *Configuration Guide*.

11.3.2 Query of Security Log

The system supports the query of all security events and logs. The security events and logs of a specified NE can be uploaded to the T2000.

The following are the security logs that can be queried.

- Start and stop events of the system and applications
- Successful login and logout records of an NE user
- Failed login records
- New user and user authority change records
- Security strategy and configuration change records of an NE
- NE and board software upgrade (local or remote upgrade) records

11.3.3 NE User Management

The system supports the security management to the NE user that has logged in to the NE.

The management covers:

- Creation and authority assignment of an NE user
- Change of an NE user password
- Query of NE security parameters, including the user expiration date and password change time
- Management to all NE users by using the T2000, including the display of users, the change of passwords and the setting of security log query authority
- Cancellation and deletion of long term unused user accounts

**CAUTION**

- When there is only one user with the authority of administrator on the NE, the user cannot be deleted, and the authority level and valid period of the user cannot be changed.
- When there is only one user with the authority of administrator on the NE, the password of the user can be changed. Keep the user account and password safe. Without the password or user account, the operations corresponding to the authority level cannot be performed. The initialization of the password can only be realized by replacing the SCC board.

11.3.4 Syslog Protocol

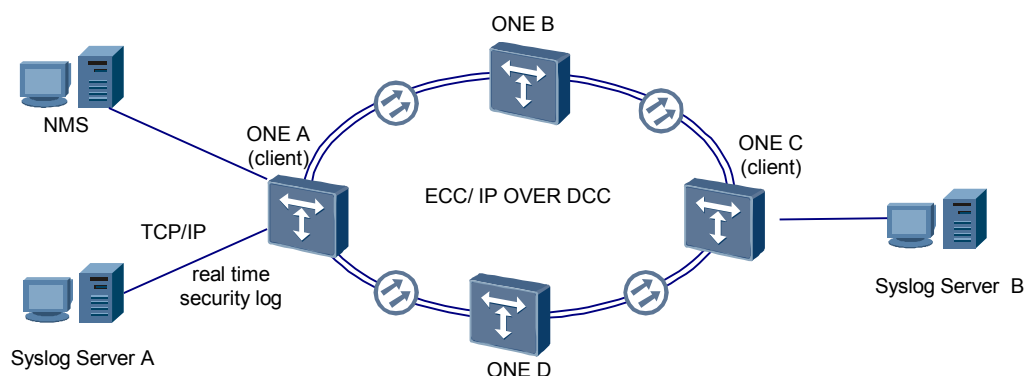
The system log service (Syslog service) is used for the security management on an NE. For a unified control by maintenance engineers, all types of information are transmitted to the log server in the format complying with the system log (Syslog) protocol.

The system supports:

- Enabling and disabling of the Syslog protocol
- Setting of the Syslog protocol transmit modes : UDP (by default) and TCP
- Adding and deletion of the Syslog servers
- Coexisting of multiple Syslog servers and the sending of logs to multiple servers at the same time
- Reporting of alarms upon the communication disconnection between the Syslog server and the NE

Figure 11-3 shows how the Syslog protocol is transmitted in a network. To ensure the security of system logs, make sure that at least two system log servers are available in a network. Normally, IP protocol is used for the communication between the NE and the system log servers. The communication between NEs can be realized through several methods, for example, ECC mode or IP OVER DCC mode.

Figure 11-3 Schematic diagram of the Syslog protocol transmitting



 NOTE

Normally, a system log server is a workstation or server that is dedicated to storing the system logs of all NEs in a network.

A forwarding gateway NE receives the system logs of other NEs and forwards the logs to the system log server. In [Figure 11-3](#), ONE A and ONE C are forwarding gateway NEs.

When the IP protocol is adopted on each NE for communication, every NE can directly communicate with the two system log servers through the IP protocol. Hence, configure the IP addresses and port numbers on the NE, and the system is able to transmit the NE logs to the two Syslog servers through the auto addressing function of IP protocol. No forwarding gateway NE is required.

When ECC mode is adopted on each NE for communication, the NE that does not directly connect to the Syslog servers cannot communicate with the servers. The logs of the NE must be transmitted to a gateway NE that directly communicates with the Syslog servers through ECC. Then, the logs are forwarded to the Syslog servers by the gateway NE. Hence, the forwarding gateway NE must be configured, for example, configure NE A as the forwarding gateway NE for NE D.

For detailed Syslog configuration procedures, refer to the *Configuration Guide*.

11.3.5 Control of Logical Ports

The system supports the disabling of idle logical ports (including management ports, signaling ports and client-side ports).

11.3.6 Control of Physical Ports

The system supports the disabling of idle physical ports (including commissioning network ports and commissioning serial ports).

The commissioning network ports and serial ports in the system are disabled by default.

11.3.7 Setting Warning Screen Information

The system supports to set and query the Warning Screen information.

You can issue tip information to the NE by properly setting the **Warning Screen** information of the NE on the T2000 LCT. If the Warning Screen is enabled, the system collects and reports the **Warning Screen** information at the time you access the NE on the T2000 LCT. The **Warning Screen** information is displayed in the T2000 LCT login interface. **Warning Screen** information setting involves the following operations:

- User sets and queries the **Warning Screen** information.
- User sets and queries the switch status of the **Warning Screen**.

For the configuration steps of the Warning Screen information in details, refer to *OptiX iManager T2000 LCT User Guide*.

11.3.8 SSL Protocol

The Secure Socket Layer (SSL) protocol is a kind of security communication protocol. The data packet sent by the NE to the network management system can be verified in terms of integrity and encrypted by using the SSL protocol. Integrity verification ensures that the user data is not altered maliciously. Data encryption prevents the transmitted information from un-authorized capture. In this manner, security communication is achieved.

The OptiX Metro 6100 supports the following protocol or functions:

- The OptiX Metro 6100 communicates with other NEs through the security channel of the SSL protocol.
- Users enable or disable the security channel of the SSL protocol on the T2000.
- Users query how the NE is connected to the T2000 on the T2000. Two connection ways are supported: Security SSL/Common.

For the SSL protocol configuration steps, refer to the *Configuration Guide*.

11.3.9 Username and Password Encryption

The system supports encryption of the username and password for an NE. The username and password is in the form of encrypted messages during transmission. On the T2000, the system encrypts the username and password, and then issues them to the corresponding NE.

11.3.10 NTP Authentication

The system supports the NTP authentication. When using the NTP, the equipment needs to perform NTP authentication.

When adding an NTP server, you must set the IP address and key of the NTP server on the T2000. In addition, you need to configure and start the NTP authentication function. Set the ID authentication key and specify it as a reliable key. Configure on the T2000 that the key created by you previously is reliable by default. The T2000 client is synchronous to the NTP server that provides reliable key. Otherwise, the T2000 client is not synchronous to the NTP server.

For the NTP authentication configuration steps, refer to the *Configuration Guide*.

12 Networking and Design Considerations

About This Chapter

This chapter describes the factors that should be considered during the networking, planning and set-up of the product.

12.1 Optical Power Budget

In a DWDM system, the power loss introduced by line fibers, optical modules, and optical components need be compensated by using an optical amplifier (Erbium-doped fiber amplifier or Raman amplifier). The process of making the power budget is a process of configuring amplifiers. It is required that the optical power at the transmit end meets the requirement of incident optical power, and the optical power at the receive end is within the working range of the receiver.

12.2 Dispersion

Fiber dispersion is another factor that affects the high-speed and long-distance transmission of a WDM system.

12.3 Span Specification

During network planning, you need to plan the downstream amplifiers based on the line loss of each span.

12.4 OSNR Budget

The planning of the OSNR budget includes the planning of the OSNR of OTUs and that of the cascaded amplifiers.

12.5 Non-Linear Requirement

Optical power, OSNR and dispersion are of main concern in the networking planning. Besides, many non-linear factors also have impact on system networking.

12.6 Impact of PMD

Due to circular asymmetry of a fiber, differential group delay (DGD) is generated when two different orthogonal polarizations travel over the fiber. The reciprocal of the ratio of the DGD to time is called polarization mode dispersion (PMD). The PMD results in a system penalty, and eventually causes signal degrade at the receive end.

12.7 Wavelength Allocation

The wavelength allocation is one of the factors being considered seriously during the network planning.

12.8 Networking Mode

The service models on the network are the main factors for the selection of networking modes. The physical topology of a network is also helpful in deciding a networking mode.

12.9 Station Configuration

The station configuration should be planned according to the network requirements.

12.10 NE Type

The NE types that the product supports are the OTM, FOADM, ROADM, OLA and REG.

12.11 NE Communication

The T2000 communicates, manages and maintains NEs by using a data communication network (DCN). NEs in a DCN communicate with each other by using the data communication channel (DCC).

12.12 Protection Mode

Planning the protection mode includes the planning of equipment-level protection and that of network-level protection.

12.13 Optical Power Management

Planning the optical power management includes the planning of three functions: automatic level control (ALC), intelligent power adjustment (IPA), and automatic power equilibrium (APE).

12.14 Hardware Planning

Hardware planning includes the planning of cabinets, planning of subracks and frames, and planning of boards.

12.15 Optical Attenuators

Optical attenuator planning includes the planning of fixed optical attenuators and planning of variable optical attenuators.

12.16 Ambient Conditions

The safe operation of the product requires good ambient conditions.

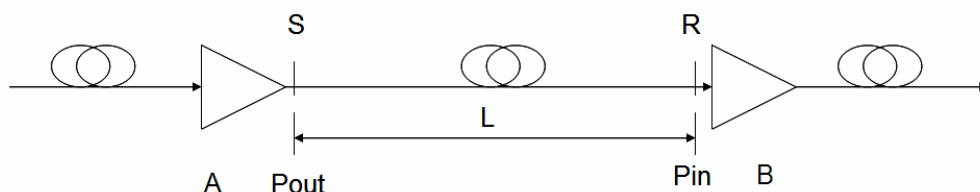
12.17 Power Supply and Power Consumption

The product requires the equipment room to provide two $-48\text{ V}/-60\text{ V}$ DC power supplies of mutual backup.

12.1 Optical Power Budget

In a DWDM system, the power loss introduced by line fibers, optical modules, and optical components need be compensated by using an optical amplifier (Erbium-doped fiber amplifier or Raman amplifier). The process of making the power budget is a process of configuring amplifiers. It is required that the optical power at the transmit end meets the requirement of incident optical power, and the optical power at the receive end is within the working range of the receiver.

Figure 12-1 Loss of the regenerating section



In the network design phase, amplifiers should be configured after the fiber loss of the entire link is calculated and the system margin is considered. Set the system margin to 3 dB when there is no special requirement. Then, make proper adjustment based on the configuration of dispersion compensation modules.

As shown in [Figure 12-1](#), the transmit reference point at station A is S, and the receive reference point at station B is R. L indicates the transmission distance between point S and point R. The loss of the regenerating section is calculated as follows:

$$\text{Loss of the regenerating section (dB)} = L \text{ (km)} \times a \text{ (dB/km)} + b \text{ (dB)}$$

a is the attenuation coefficient (dB/km). According to the ITU-T recommendations, its value is usually 0.275 dB/km in the engineering design (end-of-life value). Long-haul transmission of optical signals requires that the power of signals is great enough to offset the attenuation of fibers. The attenuation coefficient of the ordinary G.652 and G.655 fibers at the 1550 nm window is generally about 0.22 dB/km. To take the factors such as optical connectors and fiber redundancy into consideration, the comprehensive fiber attenuation coefficient is default to 0.275 dB/km. b is the insertion loss of the fiber connectors at the jump stations. It is 1 dB by default unless otherwise specified.

If the accurate actual loss of each line is known, you can directly add the system engineering margin to the measured value during the networking planning. Usually, the system engineering margin is 3 dB. If the optical supervisory channel (OSC) is adopted, you need to consider the extra power of fiber line units. Usually, the extra power is considered as 3 dB (the insertion loss of the FIUs at the two ends). If the electric supervisory channel (ESC) is used, the extra power can be ignored.

12.2 Dispersion

Fiber dispersion is another factor that affects the high-speed and long-distance transmission of a WDM system.

The fiber dispersion is of the following two types:

- Chromatic dispersion
- Polarization mode dispersion (PMD)

For a system at 10 Gbit/s, the dispersion usually refers to chromatic dispersion. For a system at the rate lower than 10 Gbit/s, the influence of the PMD is not as obvious as the chromatic dispersion. Hence, chromatic dispersion is of main concern in real engineering.

Currently, systems are configured with optical sources with high dispersion tolerance (such as the EML laser and M-Z externally modulated laser). In long-distance transmission, you can use dispersion compensation modules (DCMs).

In the engineering design, the dispersion limit of the system is the main factor. The formula is as follows:

Dispersion limit = (Dispersion tolerance/Dispersion coefficient) + DCM compensation – (10 km to 30 km)

NOTE

Ensure that the system has a 10 km to 30 km margin when you consider the dispersion limit of the system.

The OptiX Metro 6100 provides dispersion tolerance for various OTUs. Take the LWF board that is configured with 800 ps/nm WDM-side modules as an example. Its dispersion tolerance on the WDM side is 800 ps/nm. The dispersion limit in G.652 fibers is about 40 km. If services are transmitted over G.652 fibers and the system has a margin of 10 km to 30 km, the transmission distance after the LWF is added with DCMs is $L = 40 + \text{DCM} - (10 \text{ km to } 30 \text{ km})$.

In the case of G.652 fibers, if the line transmission distance is about 80 km, then

$$\text{DCM} = L - 40 + (10 \text{ to } 30) = 80 - 40 + (10 \text{ to } 30) = 50 \text{ to } 70 \text{ (km)}$$

According to the result, the 60 km DCM can meet the requirement.

When you set the dispersion compensation point, consider the concentrated dispersion compensation so that the number of DCMs can be reduced. Besides, the residual dispersion at the add/drop nodes should be taken into account. If the residual dispersion exceeds the adjustment range, dispersion compensation cannot be performed.

It is recommended to configure DCMs from the OTM station. Configure DCMs when the residual dispersion exceeds 80 km, to achieve the effects similar to the dispersion pre-compensation.

The 20 km or 40 km DCMs are usually configured for dispersion pre-compensation. The power budget decides which type of optical modules to be selected.

Among the commonly used transmission fibers, the G.652 fiber has a dispersion coefficient of 17 ps/nm*km. The dispersion coefficient of the G.655 fiber is 4.5 ps/nm*km. If the G.655 fiber is used, certain conversion from G.655 to G.652 can be made for the calculation. For example, the transmission distance of a G.655 fiber is 300 km. Then, the transmission distance of a G.652 fiber is $300 \times 4.5 / 17 = 79 \text{ (km)}$. Thus, a 60 km DCM module is needed for dispersion

compensation. When the G.655 fiber is used for long-distance transmission, however, the DCM that matches the G.655 fiber should be selected for dispersion compensation. In this case, do not use the DCM that matches the G.652 fiber.

12.3 Span Specification

During network planning, you need to plan the downstream amplifiers based on the line loss of each span.



NOTE

Amplifiers are not used in the OptiX Metro 6100 CWDM system. Hence, the OTUs of different specifications are required. Select the OTUs based on the link power budget and the number of wavelengths (affecting the power budget of multiplexers and demultiplexers).

During the networking planning, nodes are classified into four types: optical line amplifier (OLA) station with DCM compensation, OLA station without DCM compensation, OADM station with DCM compensation, and OADM station without DCM compensation. Make the inter-station power budget first and then the intra-station power budget. Then, configure amplifiers based on the inter-station and intra-station power budget.

The OptiX Metro 6100 provides four types of amplifiers: OAU, OBU, OPU, and RPC. There are many application scenarios of the OAU, OBU, and OPU. [Table 12-1](#) lists their specifications.

Table 12-1 Specifications of the OAU, OBU, and OPU

Board Name	Operating Wavelength Range	Input Power Range (dBm)	Typical Per-Channel Input Power (dBm)	Maximum Nominal Per-Channel Output Power (dBm)	Gain (dB)
C6OAU01A C6OAU01B C9OAU01	1529 to 1561	−32 to 0	−22 to −16 ^a	4	20 to 31
C6OAU02A C6OAU02B C9OAU02	1529 to 1561	−32 to −3	−25 to −19 ^a	1	20 to 31
C6OAU03A C6OAU03B C9OAU03	1529 to 1561	−32 to −6	−28 to −22 ^a	4	26 to 32

Board Name	Operating Wavelength Range	Input Power Range (dBm)	Typical Per-Channel Input Power (dBm)	Maximum Nominal Per-Channel Output Power (dBm)	Gain (dB)
C6OAU05A C9OAU05A	1529 to 1561	-32 to 0	-32 to -16 ^a	7	23 to 34
C6OBU01	1529 to 1561	-32 to -6	-22	1	23 ±1
C6OBU03 C8OBU03 C9OBU03	1529 to 1561	-24 to -3	-19	4	23 ±1
C6OBU05 C9OBU05	1529 to 1561	-24 to 0	-16	7	23
C6OPU01	1529 to 1561	-32 to -6	-22	-2	20 ±2
C6OPU02	1529 to 1561	-32 to -6	-22	-2	20 ±1
C8OPU02	1529 to 1561	-32 to -4	-20	0	20 ±1
C6OPU03 C9OPU03	1529 to 1561	-32 to -8	-24	-1	23 ±2
C8OPU04	1529 to 1561	-32 to -1	-17	0	17±1
a: In the case of the OAU01, OAU02, OAU03, and OAU05, make the per-channel input power close to the maximum value, that is, -16 dBm, -19 dBm, -22 dBm, and -16 dBm respectively.					

In the case of an OLA station without DCM compensation, [Table 12-2](#) lists the span design details.

Table 12-2 Span design for an OLA station without DCM compensation

Span Specification (dB)	Configuration of Amplifiers	Equivalent Span (dB)
<20	OPU	20

Span Specification (dB)	Configuration of Amplifiers	Equivalent Span (dB)
20–23	OBU	23
23–31	OAU	Actual loss of the span

In the case of an OLA station with DCM compensation, [Table 12-3](#) lists the span design details.

Table 12-3 Span design for an OLA station with DCM compensation

Span Specification (dB)	Configuration of Amplifiers	Equivalent Span (dB)
<20	OAU	20
20–23	OAU	23
24–29	OBU+OPU	Actual loss of the span
29–36	OBU+OAU	Actual loss of the span
Note: "+" indicates that amplifiers of several types are used in cooperation.		

As for an OADM station, the loss of the span should be considered first in configuring amplifiers. Different combinations of amplifiers are decided based on the compensation requirement of the intra-station DCM and OADM.

In the case of an OADM station without DCM compensation, [Table 12-4](#) lists the span design details. The intra-station budget is the power budget of the OADM.

Table 12-4 Span design for an OADM station without DCM compensation

Span Specification (dB)	Intra-Station Budget (dB)	Configuration of Amplifiers	Equivalent Span (dB)
<20	20 or 24	OBU+OPU	20
21–24	19 or 23	OBU+OPU	24
24–31	23	OAU+OPU	Actual loss of the span
32–36	18–22	OAU+OPU	Actual loss of the span
Note: "+" indicates that amplifiers of several types are used in cooperation.			

In the case of an OADM station with DCM compensation, [Table 12-5](#) lists the span design details. The intra-station budget consists of the power budget of DCMs and that of the OADM.

Table 12-5 Span design for an OADM station with DCM compensation

Span Specification (dB)	Intra-Station Budget (dB)		Configuration of Amplifiers	Equivalent Span (dB)
	DCM	OADM Station		
<23	8	23	OAU+OPU	23
24–33	10	23	OBU+OB +OPU	Actual loss of the span
33–36	10	19	OBU+OAU +OPU	Actual loss of the span
Note: "+" indicates that amplifiers of several types are used in cooperation.				

 NOTE

When the span specification is more than 36 dB, the RPC board is required.

Select amplifiers based on the above rules. Then, calculate the OSNR of signals based on the route of each wavelength. Evaluate the factors such as dispersion, power budget, and line penalty, and select proper OTUs to meet the transmission requirements.

12.4 OSNR Budget

The planning of the OSNR budget includes the planning of the OSNR of OTUs and that of the cascaded amplifiers.

12.4.1 OSNR Requirement of OTUs

Radiation noise of amplifiers is introduced when optical amplifiers are used for power compensation for line loss. Accordingly, OSNR decreases and the transmission performance degrades.

In a long-haul or multi-node transmission system, more than one optical amplifier is adopted and cascaded to extend the transmission distance. Although the optical power is compensated, the accumulative radiation noise of amplifiers leads to the decrease in OSNR and the quality of the signals at the receive end. When the OSNR degrades to a certain degree, the performance of the system is seriously affected.

Table 12-6 lists the reference values of the OSNR designed for signals at different rates.

Table 12-6 Requirements of OSNR

Rate	FEC Mode	Code Pattern	Requirement of OSNR _{ab}
2.5 Gbit/s	FEC	NRZ	15
	Without FEC	NRZ	21

Rate	FEC Mode	Code Pattern	Requirement of OSNR ^{ab}
5 Gbit/s	FEC	NRZ	22
10 Gbit/s	FEC	NRZ	20
	AFEC	NRZ	18
	AFEC	DRZ	16
40Gbit/s	AFEC	DPQSK	18
a: Refer to this table to learn the OSNR requirement of signals at different rates, to ensure that the system BER after error correction is 1.0×10^{-15}. b: The value is for reference only. In the actual network designing, different OSNR requirements may be adopted.			

12.4.2 OSNR Requirement of the Cascaded Amplifiers

For various network applications, the OSNR requirement of the cascaded amplifiers at different rates is similar to that of the OTUs.

The OSNR requirement of the OTUs is listed in [Table 12-6](#). In a DWDM system that adopts multiple cascaded line optical amplifiers, the optical power of noise is dominated by the inherent radiation noise of amplifiers. In an actual DWDM system, if the gain of an EDFA is unbalanced, the output power and noise coefficient of each channel are different. Hence, during networking planning, the OSNR of the worst channel must be considered. That is, the OSNR of the worst channel should meet requirements and have enough margins.

12.5 Non-Linear Requirement

Optical power, OSNR and dispersion are of main concern in the networking planning. Besides, many non-linear factors also have impact on system networking.

The following are non-linear factors: stimulated brillouin scattering (SBS), stimulated Raman scattering (SRS), self phase modulation (SPM), cross phase modulation (XPM), four-wave mixing (FWM), polarization mode dispersion (PMD), and polarization dependent loss (PDL).

The impact brought by non-linear effects is usually considered as accessorial penalty. Hence, reserve 2 dB margin for the OSNR that the system allows when designing the system.

12.6 Impact of PMD

Due to circular asymmetry of a fiber, differential group delay (DGD) is generated when two different orthogonal polarizations travel over the fiber. The reciprocal of the ratio of the DGD to time is called polarization mode dispersion (PMD). The PMD results in a system penalty, and eventually causes signal degrade at the receive end.

To measure the PMD of different transmission media and components and thus to assess the impact of PMD, a PMD coefficient is adopted to quantify the PMD of a transmission medium or component.

PMD coefficient unit: $\text{ps}/\text{km}^{1/2}$

DGD is the delay differential between two polarizations and is used to measure the phase difference between two orthogonal polarizations. The DGD varies randomly with time and frequency.

The DGD unit is ps.

Cause of PMD

Before a signal generated by a single-mode laser is transmitted to the WDM side, the signal is modulated in a specified manner and then converted into an optical signal, which consists of two different orthogonal polarizations. In an ideal fiber, the fiber has a perfect circular cross-section and the two orthogonal polarizations travel at the same speed and with the same phase. In a real fiber, however, the fiber fails to retain a circular cross-section either temporarily or in a long term due a production process problem or because the fiber is pressed seriously, causing the two orthogonal polarizations to travel at different speeds and with different phases.

PMD Impact on Systems

Factors such as fiber production process problems and other engineering factors can break the circular symmetry of a fiber. When a signal travels along a fiber, the two orthogonal polarizations of the signal travel with different phases after they travel over certain distance. As a result, the orthogonal polarization (the red portion) in the X direction travels at a slower speed than the orthogonal polarization (the blue portion) in the Y direction. That is, the two orthogonal polarizations gradually have different phases due to the PMD impact.

After the two polarizations present different phases as shown in [Figure 12-2](#) and [Figure 12-3](#), the polarization in the X direction is located in the position where signal "0" should be located. That is, this position originally corresponds to signal "0" either in the X or Y direction. However, the polarization adjacent to signal "1" in the X direction is moved to the position intended for signal "0" because of phase delay. In this case, signal "0" exhibits certain strength that can be detected by the receiver. In the worst case, the receiver may consider signal "0" as signal "1" by mistake. In addition, the strength of signal "1" decreases at the receive end because the polarization in the X direction is moved to another position. In the worst case, the receiver may consider signal "1" as signal "0" by mistake.

Currently, new code patterns and new fibers with low PMD coefficients are widely used to reduce the impact of PMD.

Figure 12-2 Orthogonal polarizations in the case of signal "1"

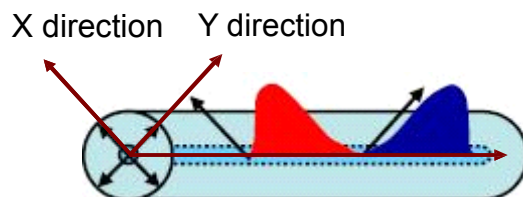
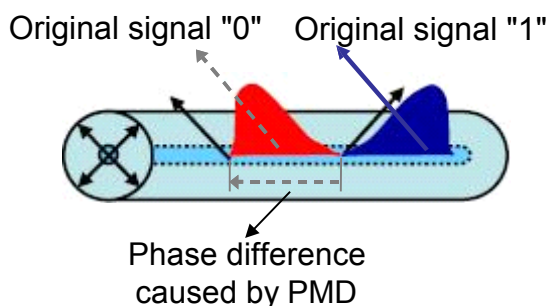


Figure 12-3 Orthogonal polarizations in case of PMD



Relationship Between Transmission Rate and PMD Tolerance

According to ITU-T Recommendations, the PMD, transmission rate, and transmission distance must satisfy the $B_{[\text{Gbit/s}]} \cdot \text{PMD}_{[\text{ps/km}^{1/2}]} \cdot (L_{[\text{km}]})^{1/2} = 100$ equation. In the case of a fiber with a specific PMD coefficient, the transmission distance decreases sharply when the transmission rate increases. For details, see [Table 12-7](#). When the PMD coefficient is greater than the specified value, the transmission performance of a system will be affected. This is especially the case with the fibers routed before mid-nineties in the 20th century. To upgrade the transmission rate of a network using those fibers to 10 Gbit/s or higher, you need to take into account the PMD of a fiber.

Table 12-7 Relationship between transmission rate and PMD tolerance

PMD Coefficient Unit: $\text{ps/km}^{1/2}$	Maximum Transmission Distance		
	2.5 Gbit/s	10 Gbit/s	40 Gbit/s
3.0	180	11	< 1
1.0	1600	100	6
0.5	6400	400	25
0.1	160000	10000	625

12.7 Wavelength Allocation

The wavelength allocation is one of the factors being considered seriously during the network planning.

The basic principles of wavelength allocation are described as follows:

- Primary principle: Priority is given to the allocation of long wavelength resources.
- Usually, there are several phases in network construction. In the initial phase, there are a few wavelength resources. Long wavelengths are better choice. On the one hand, the performance of long-wavelength channels is better than that of short-wavelength channels when there are a few wavelengths. On the other hand, the power budget of long-wavelength channels is stable after expansion and is not affected easily.

- Secondary principle: The same wavelength resources are allocated to the services that may not cause span conflicts, such as protection services and distributed services.
 - As for the intra-board 1+1 protection and the client-side 1+1 protection, the same wavelength resources are allocated to the transmit and the receive end. This is to ensure that the same wavelength on the entire ring is used by the same service. If different wavelength resources are allocated to the protected services, some wavelengths in the long trail and the short trail will be in the idle state. When other services are allocated with wavelength resources, conflict of wavelengths may happen.
 - As for the optical channel shared protection, the allocation of paired wavelengths is considered first. In the case of ring services, such as SDH or GE ADM services, the same wavelength resources should be allocated because no segmental spatial reuse is possible. In this manner, the possibility of wavelength conflict is reduced and the wavelength utilization is improved.
- Third principle: Wavelengths are allocated first for the service groups that have restrictions in wavelength allocation.
- A service group consists of services whose routes are exclusive from each other. The protection or ring services can be regarded as a service group. Wavelengths are allocated for the service group where there are great difficulties in allocation, to lighten the wavelength congestion in the future.
- Fourth principle: As for the wavelength allocation for the services with a regenerator, the same wavelength resource should be allocated to the services. The routes of the services before and after the regenerator are exclusive from each other. Hence, if different wavelength resources are used, wavelength conflict with other services may occur.

12.8 Networking Mode

The service models on the network are the main factors for the selection of networking modes. The physical topology of a network is also helpful in deciding a networking mode.

Following are the main service models in MANs:

- Point-to-point services: Voice services, data private services and storage services are point-to-point services.
- Convergent services (also called star services): Most of the broadband Internet surfing and video on demand services in Triple Play are convergent services.
- Broadcast services: The IPTV in Triple Play is a kind of broadcast service. It is unidirectional and features drop and continue.

Following are the rules for selecting a networking mode:

- Point-to-point networking mode can be adopted for simple point-to-point services. When there are several groups of different point-to-point services, ring networking is recommended. Make use of the ADM function of sub-wavelengths to have wavelength resources shared by several services.
- In convergent services, it is a point-to-point service from the central node to each edge node. The service type is usually Ethernet service (mainly GE service). In this case, point-to-point networking wastes fiber resources. It is recommended to select ring networking and make use of the wavelength grooming function to have wavelength resources shared.
- In broadcast service model, the central node sends services to several edge nodes at the same time. The services can be wavelength services or sub-wavelength services. Ring

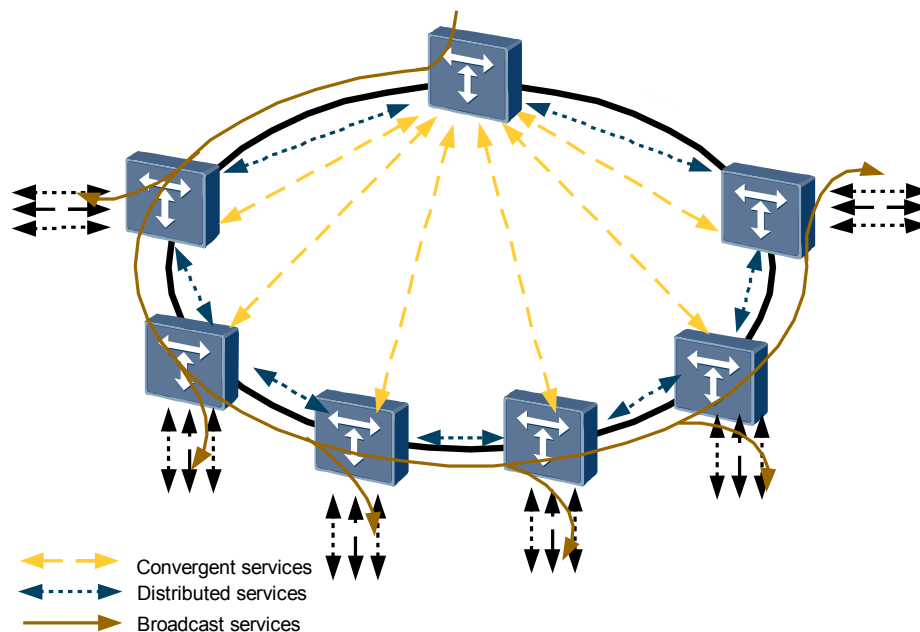
networking is recommended. Make use of the ADM function of wavelengths and sub-wavelengths to have services dropped or pass through at each edge node.

The OptiX Metro 6100 supports the following functions:

- Supports the wavelength add/drop function provided in traditional WDM equipment. The product can realize ring, ring with chain and mesh networking at a wavelength level.
- Supports the add/drop multiplexing of sub-wavelength services such as Ethernet services. The product supports the arbitrary grooming of sub-wavelength services, such as GE services, and realizes ring, ring with chain and mesh networking at a wavelength level. In this manner, the requirements of point-to-point services, convergent services and broadcast services are satisfied.

Figure 12-4 shows the distributed services, convergent services and broadcast services in ring networking.

Figure 12-4 Example of distributed services, convergent services and broadcast services



12.9 Station Configuration

The station configuration should be planned according to the network requirements.

The basic rules for configuring stations are described as follows:

- Check whether the transmission line needs a starting point or an end point. At the starting or end point, an optical terminal station must be configured.
- Check whether there are services to be added or dropped. At the service adding or dropping point, an optical add/drop multiplexing station must be configured.
- If no services are to be added or dropped, it is needed to make the power budget, dispersion budget and noise budget. Then, decide whether to set stations and the type of stations based on the budgets.

- If the optical power budget cannot satisfy the system requirement, an OLA station is required to amplify the weak signals from the line and make dispersion compensation. Thus, the transmission distance is extended without using electrical regenerators.
- After passing multiple OLA stations and having a long distance transmission, services have accumulated optical noise, non-linear effects and polarization mode dispersion that exceed the range acceptable to the receiver. In this case, an electrical REG station (that is, to configure regenerating OTUs at the OADM station) is required to reshape, retime, and regenerate the electrical signals.
- If optical power, dispersion, and optical noise meet the system requirements, no station is required.

12.10 NE Type

The NE types that the product supports are the OTM, FOADM, ROADM, OLA and REG.

For details, refer to [6 DWDM System Configuration](#) and [7 CWDM System Configuration](#).

The following are the rules for configuring the OTM stations:

- If a large number of services are added and dropped on a node, set the node as an OTM station.
- Set the end nodes in point-to-point, chain and ring with chain networks as OTM stations.
- If the node is the starting or end node of a large number of services in a ring, set the node as a back-to-back OTM station.

The following are the rules for configuring the FOADM and ROADM stations:

- If a small number of services are added and dropped on a node, set the node as an FOADM station. The station is used to add or drop the services terminated or generated at the local station. Other services pass through the local station after being processed.
- Set the intermediate nodes in a chain and the nodes where a small number of services are added and dropped in a ring as FOADM stations.
- The factors that affect the transmission performance are dispersion, power, optical noise, non-linear effects, or polarization mode dispersion in the optical regenerating section. When the optical transmission line is rather long and one or several factors affecting the transmission performance make the line extension impossible, use the M40 and D40 boards (not OADM boards) and set the node as the FOADM station. Besides, configure regenerating OTUs to reshape, retime and regenerate electrical signals.
- If a node requires flexible and dynamic allocation of service wavelengths, set the node as an ROADM station.

The following are the rules for configuring an OLA station:

- If a node requires transmission distance extension, set the node as an OLA station to amplify the weak signals from the line and make dispersion compensation. Thus, the transmission distance is extended without using electrical regenerators

For the setting of OLA stations, refer to [12.1 Optical Power Budget](#).

The following are the rules for configuring an REG station:

- If the distance is longer or such factors as dispersion, optical noise, non-linear effect, or PMD affect the transmission performance, the REG equipment will be used to further extend the optical transmission distance. An REG implements the 3R function: reshaping,

re-timing and regenerating. This is to improve the signal quality and to extend the transmission distance.

12.11 NE Communication

The T2000 communicates, manages and maintains NEs by using a data communication network (DCN). NEs in a DCN communicate with each other by using the data communication channel (DCC).

The OptiX Metro 6100 supports the following DCN construction modes:

- HWECC: The DCC is used to transmit the HWECC protocol. The HWECC protocol is a private protocol developed by Huawei to support the DCN networking of OptiX equipment.
- IP over DCC: The DCC is used to transmit the data supporting the transmission control protocol/internet protocol (TCP/IP).
- OSI over DCC: The DCC is used to transmit the data supporting the open systems interconnection (OSI) protocol.

12.11.1 Basic Rules

When establishing a DCN, select a proper communication protocol based on the networking mode.

The following are the basic rules for planning a DCN:

- When the OptiX Metro 6100 interworks with other Huawei optical network equipment, HWECC protocol or IP over DCC is recommended. The entire DCN network should use the same communication protocol.
- When the OptiX Metro 6100 interworks with third-party equipment, select IP over DCC or OSI over DCC if it is supported by the third-party equipment.
- When the OptiX Metro 6100 interworks with third-party equipment, use DCC bytes to realize transparent transmission if the third-party equipment does not support IP over DCC or OSI over DCC.
- No matter which protocol is used to establish a DCN network, configure the scale of the DCN network properly. Divide the network into areas based on the network situation, to prevent the impact of large scale on the DCN network.
- To ensure the reliability of a communication network, configure the DCN network as a ring, to ensure that route protection is available when a fiber cut or NE abnormality is detected. If the equipment fibers cannot form a ring, build extra DCN paths to make a ring, to ensure the route protection.

12.11.2 General Rules for Gateway NE Planning

The GNE is a kind of special NE to achieve communications between the T2000 and NEs.

The management information from the T2000 to NEs must be forwarded by the GNE. When creating an NE on the T2000, you need to specify the GNE first. In the network-wide planning, select the GNE based on the actual networking and the communication protocol.

**CAUTION**

Improper GNE brings impact on the communication efficiency of the network.

The following are the general rules for planning the GNE:

- There should not be too many GNEs in a network. Otherwise, the network performance might be affected. It is suggested that the number of GNEs on a NM does not exceed 100. If the number exceeds 100, use extended ECC to combine the GNEs.
- Each GNE should not be connected with too many non-gateway NEs (including extended ECC). The number should not exceed 60. If the number exceeds 60, create more GNEs.
- Normally, multiple independent NEs in a network topology (for example, the NEs in a ring network) should belong to the same GNE.
- When choosing a GNE from several NEs, choose the NE that is close to the T2000 server in the DCN.

12.11.3 ID Planning Rules

The T2000 uses NE ID as the equipment identifier. Hence, no matter which protocol (HWECC, IP over DCC or OSI over DCC) is used in a DCN, NE ID is required.

The following are the rules for planning an NE ID:

- An NE ID must be unique in the DCN network.
- An NE ID is a 24-bit binary digit, and is divided into high order 8 bits and low order 16 bits. The high order 8 bits represent the extended ID (default to 9). It is also called subnet number, because it identifies different subnets. The number of a subnet cannot be 0 or 0xFF. The low order 16 bits represent the basic ID. The value cannot be 0 or be larger than or equal to 0xBFEF.
- In a ring network, the ID of NEs should increase one by one along the direction of the primary ring.
- In a ring network, it is suggested to configure the ID of every NE in a station, and then configure the ID of every NE in the next station.
- In the case of a complex network, divide the network into rings and chains. Set the ID of the NEs on the ring to a number from 1 to N, and the ID of the NEs on the chain to a number from N+1.

12.11.4 HWECC Planning Rules

By default, after every NE in a network is allocated with an NE ID, ECC communication can be achieved without any more configuration.

Supporting Capability

HWECC uses D1–D3 bytes as the physical transmission path. D4–D12 or D1–D12 bytes can also be used. The OptiX Metro 6100 supports the following modes:

- 22 channels of D1–D3, four channels of D4–D12, two channels of D1–D12 (default)
- 22 channels of D1–D3, six channels of D4–D12
- 22 channels of D1–D3, five channels of D1–D12

HWECC supports the communication by using fibers or Ethernet cables. When no optical path is available between nodes, set the extended ECC by using Ethernet cables.

Basic Planning Rules

HWECC is used to establish a DCN between Huawei transmission equipment. The following are the rules for planning HWECC:

- In a network where the OptiX Metro 6100 serves as the GNE, the number of the NEs in a HWECC subnet is recommended not to exceed 50.
- Configure the HWECC subnet network as a ring, to ensure that route protection is available when a fiber cut or NE abnormality is detected.
- Do not enable ECC communication between different HWECC subnets. That is,
 - Do not use network cables or fibers to make physical connection between any NEs in two HWECC subnets.
 - Use the T2000 to disable the ECC communication between NEs in different HWECC subnets.
- The OptiX Metro 6100 allocates ECC channels for the interfaces on each board automatically. Disable unnecessary ECC channels according to the number limit of ECCs in the equipment.
- Since the number of NEs managed by a GNE is limited, multiple GNEs are required when the number of NEs is large.
- In the application where extended ECC communication is required, manual extended ECC is recommended. Do not use automatic extended ECC, so that the bandwidth between NEs using extended ECC for communication is saved.

IP Planning Rules

An IP address is used in the communication between a GNE and the T2000. Hence, a GNE requires an IP address. Besides, the NE that requires the extended ECC function should be provided with an IP address.

Normally, the IP address of an NE requires no manual setting. It varies with the NE ID. The format of an IP address is 129.E.A.B. E is the extended ID of the NE, which does not vary with the NE ID. The default value is 9. A.B is the high order 8 bits and low 8 bits in an NE ID. When you set an IP address for an NE manually, the relation between the IP address and the NE ID is removed.

The default subnet is 129.9.0.0, and the subnet mask is 255.255.0.0.

General Rules for Planning a GNE

The following are the rules for planning a GNE:

- The IP address and subnet mask of the GNE must be set correctly.
- Only the equipment that connects to the T2000 by using network cables can serve as a GNE.
- In the actual networking, the GNE has the largest traffic volume. To ensure stable communication, select the equipment with strong ECC processing ability as the GNE. The GNE and other NEs should form a star network, to reduce the traffic volume in other NEs.
- To ensure reliable connection between the T2000 and the network, configure a backup GNE. The rules for selecting a backup GEN is similar to that of selecting a main GNE. In

addition, the backup GNE can be used to manage some NEs. In this manner, the two GNEs are of mutual backup, which improves the network stability.

12.11.5 IP over DCC Planning Rules

IP over DCC is used to manage NEs when the product interworks with third-party equipment, or when the product interworks with Huawei transmission equipment.

Supporting Capability

The TCP/IP protocol is used to realize IP over DCC:

- NEs can be connected to the T2000 directly or through the GNE.
- Supports the application layer protocols of the TCP/IP, such as FTP, Telnet, and SNMP.
- Supports the dynamic route protocol and static route protocol of open shortest path first (OSPF).

Through IP over DCC, the OptiX Metro 6100 can interwork with third-party equipment that supports IP over DCC.

Basic Planning Rules

The following are the rules for planning IP over DCC.

- The number of NEs in the same OSPF area cannot exceed 60.
- When the T2000 is used to manage NEs, the number of non-gateway NEs accessed through one GNE cannot exceed 60.

IP Address Planning Rules

The following are the rules for planning the IP address:

- Each NE has a unique IP address.
- NEs can use standard IP addresses (types A, B and C), that is, the IP addresses ranging from 1.0.0.1 to 223.255.255.254. The broadcast address, network address and address 127.x.x.x cannot be used. The subnet address 192.168.x.x and 192.169.x.x cannot be used.
- The IP address must be used with the subnet mask, and must support the subnet mask with variable length.
- When NEs use static route protocol to connect to the T2000 directly, the GNE and non-gateway NEs must use different IP subnets.
- In the case of two networks connected through Ethernet, they should be divided into different IP subnets. In this manner, every NE can be accessed to the T2000 during network area dividing.

12.11.6 OSI over DCC Planning Rules

OSI over DCC is used to manage NEs when the product interworks with third-party equipment that supports OSI over DCC.

Supporting Capability

The protocols on the lower four layers in OSI are used to realize OSI over DCC.

- Simple network service access point (NSAP) address is used as the protocol identifier of a node.
- Supports three types of network nodes: end system (ES), level 1 intermediate system (L1-IS), and level 2 intermediate system (L2-IS).
- The IS-IS protocol is used between intermediate systems to exchange dynamic route information.
- The ES-IS protocol is used between the end system and intermediate system to realize neighbor discovery and route information exchanging.
- Supports the IS-IS Level 2 protocol which realizes route layering.
- Supports the TP4 protocol.
- NEs can be connected to the T2000 directly or through the GNE.

Through OSI over DCC, the OptiX Metro 6100 can interwork with third-party equipment that supports OSI over DCC.

Basic Planning Principles

When a network comprises only Huawei equipment, OSI over DCC is not recommended. The following are the rules for planning OSI over DCC:

- Only the node at the end of a network can be configured as an ES. The limited route resources of an ES affect the network expansion. Hence, it is not suggested to configure the equipment as an ES. The T2000 works as an ES.
- L1-IS is the default node type of Huawei products. It supports only Level 1 routing.
- If Level 2 routing is required, set the network node type of the equipment to L2-IS. L2-IS maintains two route tables at the same time: one is used in intra-area routing and the other in inter-area routing.
- The OptiX Metro 6100 supports IS-IS Level 2 routing. When the OSI communication protocol is used, you need to divide the network into areas according to the network size. The number of areas in a DCN cannot exceed 32. The number of NEs in an area cannot exceed 50.
- Configure the DCN network as a ring, to ensure that route protection is available when a fiber cut or NE abnormality is detected.
- When the OptiX Metro 6100 interworks with third-party equipment, follow the design rules for the third-party equipment as well during network planning.

Network Area Dividing

The OSI protocol supports layered routing function. It uses the SYS ID of the system to achieve intra-area routing, and uses the AREA ID to achieve inter-area routing. In the DCN planning, divide the network into areas properly and decide the number of NEs in each area according to the network topology.

If the number of NEs in a network is smaller than 50, there is no need to divide the network into areas. In this case, set the node type of all NEs to L1-IS, and set the AREA ID in the NSAP area address of all NEs to the same value.

In the case of a large scale network, follow the rules below to divide the network:

- Divide the DCN network into several areas to manage.

- Set several NEs in each area to L2-IS. Two NEs in each are recommended because the two can be of mutual backup.
- In the DCN network, all L2 equipment must be arranged in a continuous manner.

Rules for Planning a GNE

When OSI over DCC is used to establish a DCN, the TP4 connection is required between the T2000 and the GNE. The management data that the T2000 sends to non-gateway NEs is forwarded by the GNE. When creating a GNE on the T2000, enter the NE ID and the NSAP address. When creating a non-gateway NE, enter the NE ID and allocate a GNE for the NE.

When all nodes in a DCN network run the OSI protocol stack, do not set all NEs as GNEs. Set some NEs as GNEs. Set the other NEs as non-gateway NEs and allocate GNEs for non-gateway NEs. The number of non-gateway NEs under one GNE should not be larger than 64. Otherwise, the GNE is overloaded, and the performance of the T2000 is decreased.

Set an NE that is close to the T2000 as a GNE. In this case, the communication between the T2000 and the GNE requires less overheads and provides higher efficiency.

When dividing a network to support layered routes, set one or several NEs in each area as GNEs. When creating a non-gateway NE, allocate a GNE in the local area for the NE.

To ensure the reliable communication between the T2000 and non-gateway NE, a backup GNE is normally allocated for the non-gateway NE.

Setting LAPD Roles on Optical Interfaces

When the OSI protocol is used to achieve the communication through optical interfaces, the LAPD protocol is used on the optical interface link layer. According to the requirements of the LAPD protocol, set different LAPD roles for the two ends of the interconnecting optical interface.

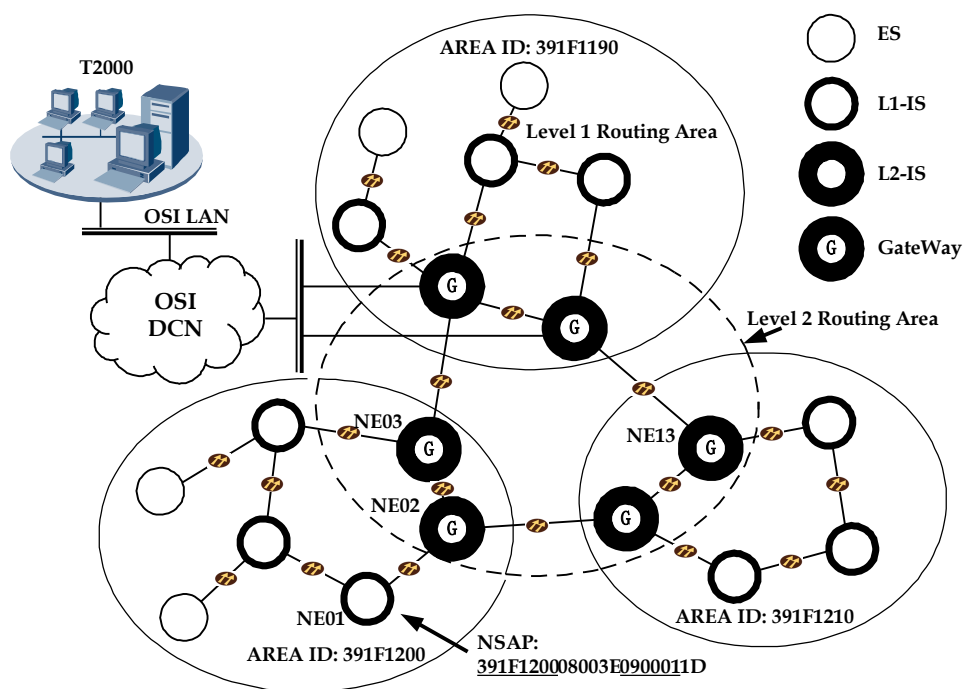
There are two LAPD roles: user and network. The optical interfaces connected by a fiber must be set to different rules. That is, the interface at one end is set to user, and that at the other end is set to network.

Example

This section describes how to select a GNE, to plan node types and to plan network area address in a DCN through an example.

See [Figure 12-5](#). The network comprises Huawei equipment and third-party equipment. It requires OSI over DCC to form a DCN.

Figure 12-5 DCN network planning in OSI over DCC mode



When planning the DCN for this network, divide the network into three areas. The AREA IDs of the three areas are hexadecimal digits: 391F1190, 391F1200, and 391F1210. Set the equipment on the core layer as L2-IS, and that on the edge layer as L1-IS. Configure the NE that is close to the T2000 as the GNE.

After the AREA ID and NE ID are defined, the NSAP address of the NE is determined. For example, the NSAP address of NE01 is 391F120008003E0900011D.

NOTE

Although the GNE in [Figure 12-5](#) is L2 equipment, it does not indicate that only L2 equipment can be a GNE. L1 equipment can also serve as a GNE.

The OSI protocol requires that L2-IS nodes in the network must be arranged in a continuous manner. Hence, select L2-IS nodes properly when dividing a network. If NE03 and NE13 in [Figure 12-5](#) are configured as L1-IS, the network communication goes abnormal before L2 is not arranged in a continuous manner. In this case, the T2000 is unable to manage the NEs in the area with AREA ID of 391F1200 and 391F1210.

12.12 Protection Mode

Planning the protection mode includes the planning of equipment-level protection and that of network-level protection.

12.12.1 Selecting Equipment-Level Protection

The equipment-level protection modes supported by the product are mainly the board power backup protection that must be configured.

12.12.2 Selecting Network-Level Protection

The various network level protection schemes must be planned according to the networking requirements.

The OptiX Metro 6100 provides the following protection modes.

Protection Type	Protection Range
Optical line protection	Line fiber
Intra-board wavelength protection.	WDM-side wavelengths of the OTU
Extended intra-board wavelength protection	WDM-side wavelengths of the OTU
Inter-board wavelength protection.	WDM-side wavelengths of the OTU
Inter-subrack wavelength protection	WDM-side wavelengths of the OTU
Client-side 1+1 wavelength protection	Client-side services of the OTU
Wavelength cross-connection protection (WXCP)	Client-side services of the OTU that provides the convergence and cross-connection functions
TPS protection and DPPS protection	Client-side services of the OTU that provides the convergence and cross-connection functions
VLAN SNCP protection	VLAN channel carried service in the L4G/EGS8 board
Optical wavelength shared protection (OWSP) Optical wavelength shared protection (DCP)	Line fiber

For details of network-level protection, refer to [9 Protection](#).

12.13 Optical Power Management

Planning the optical power management includes the planning of three functions: automatic level control (ALC), intelligent power adjustment (IPA), and automatic power equilibrium (APE).

12.13.1 Automatic Level Control (ALC)

The product provides the ALC function to compensate for the impact of factors such as fiber aging on the optical power budget for the line.

When the attenuation of a certain fiber span of the line increases, ALC limits the impact to this span and thus maintains the output and input optical power of the downstream amplifier. In a real application, if the margin within a span is not enough to compensate for the impact brought by fiber aging on optical power budget, the optical power budget margin of adjacent spans are used to compensate for the aging effects. This limits the dependent power change to a

controllable range to avoid the impact on the optical power budget and OSNR budget of the entire network. ALC is achieved through gain adjustment.

The downstream amplifier of each span obtains the output optical power value of its upstream amplifier in the same span; compares the obtained value with its own output optical power; and calculates the actual loss of the span. After that, the downstream amplifier compares the actual loss with the standard span loss and calculates the deviation in loss of this span.

At last, the downstream amplifier compares the accumulated loss deviation of the upstream spans with the adjustment threshold. If the accumulated loss deviation crosses the threshold, the downstream amplifier initiates ALC of this span. If the downstream amplifier of this span can completely compensate the deviation, the compensation is complete. Otherwise, this amplifier sends the remaining deviation downstream for further compensation until completion.

By default, the ALC function must be configured. For the rules for configuring the ALC function, refer to [10.3 Automatic Level Control](#).

12.13.2 Automatic Power Equilibrium (APE)

The product provides the APE function.

An MCA board configured on the receive side detects the OSNR of each channel on the receive side; compares the detection results with the provisioned target values; and determines the optical power of which channels need be adjusted and to what degree they are to be adjusted. After that, the MCA sends the requests for power adjustment to the transmit side through the OSC or ESC. Receiving the requests, the VMUX, VOA or OTU on the transmit side adjusts the optical power of different channels based on the requests. Through this process, the OSNR of different channels are equalized.

It shall be noted that the units that perform the APE function can be installed in different stations. Hence, the optical power can be adjusted wavelength by wavelength.

When the transmission distance exceeds 400 km or the OSNR of different channels in a span exceeds 2 dB, it is recommended to configure the APE function. For the rules for configuring the APE function, refer to [10.4 Automatic Power Equilibrium](#).

12.13.3 Intelligent Power Adjustment (IPA)

The product provides the IPA function.

In a DWDM system, a fiber cut, equipment degradation or connector disconnection might lead to the loss of optical signals in optical channels. To avoid direct exposure of human body, especially eyes, to laser leakage from a fiber cut, and to avoid the power surge of the optical amplifier, the system provides the IPA function. When the optical power signals on one or more optical regenerator sections are lost, the system can detect the loss of optical signals on the link and shuts down upstream optical amplifiers at once. When the optical signals are restored, the optical amplifiers resume normal operation.

The IPA function must be configured for optical amplifier stations. For the rules for configuring the IPA function, refer to [10.1 Intelligent Power Adjustment](#).

12.13.4 Intelligent Power Adjustment with Raman

The product provides the IPA function for the RPC board.

When the DWDM system is configured with the Raman amplifier, the optical power that overflows from the fiber cut end face is too large, normal IPA cannot properly ensure the safety

of maintainers during operations on this system. Hence, the Raman system IPA function derives from the normal IPA function. In the Raman system IPA function based on the normal IPA function, when detecting a fiber cut, the system shuts down the Raman amplifier to ensure that the optical power of the entire line is on a safe level. When the optical signals are restored, the optical amplifier configured with IPA is restarted and is back to normal functioning.

The IPA function must be configured for optical amplifier nodes. For the rules to configure the Intelligent Power Adjustment in a Raman System, refer to [10.2 Intelligent Power Adjustment of Raman System](#).

12.14 Hardware Planning

Hardware planning includes the planning of cabinets, planning of subracks and frames, and planning of boards.

12.14.1 Planning Cabinets

The product can be installed in an ETSI 600 mm cabinet, an ETSI 300 mm cabinet, a 19- or 23-inch cabinet, or a 19-inch open rack.

Typically, the OptiX Metro 6100 subrack is installed in the ETSI 300 mm cabinet.

Generally, cabinets are installed in a row inside a room. They are arranged in a face-to-face or back-to-back mode. Their positions are shown in [Figure 12-6](#) and [Figure 12-7](#).

To meet the requirements of heat dissipation and maintenance, the space around a cabinet should be reserved according to the following rules:

- The space reserved before a cabinet should not be less than 1000 mm.
- The space reserved beside both sides of a cabinet should not be less than 800 mm.
- The space reserved behind a cabinet should not be less than 50 mm. (In back-to-back mode, this rule is not required.)

Figure 12-6 Top view of cabinets in face-to-face arrangement

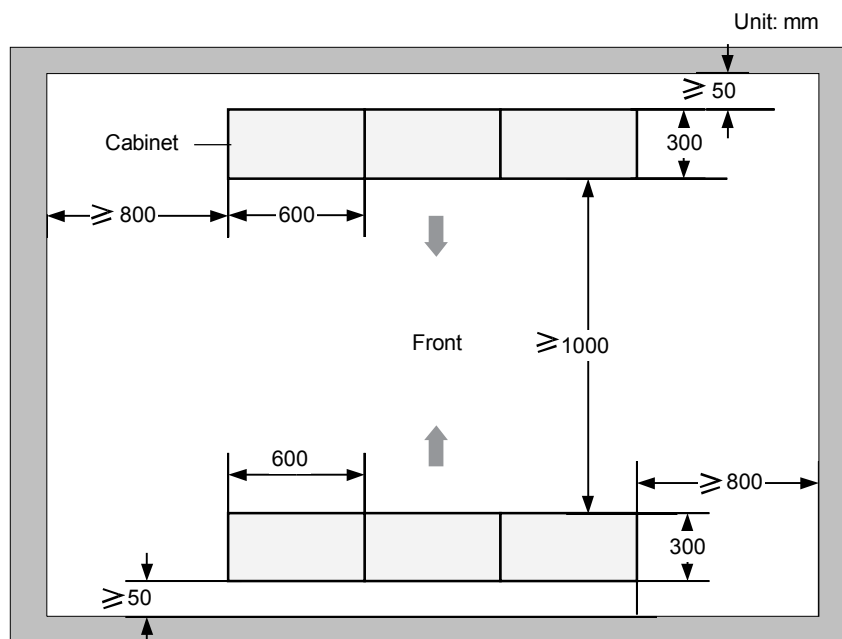
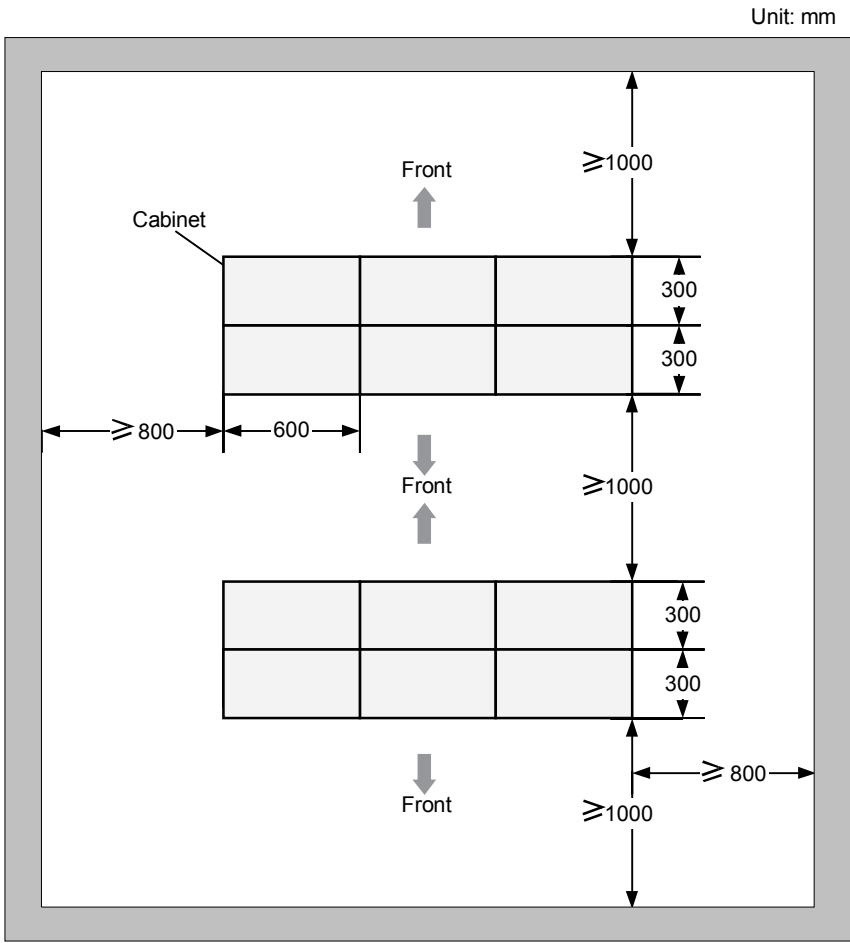


Figure 12-7 Top view of cabinets in back-to-back arrangement



12.14.2 Planning Subracks

By height, the ETSI 300 mm cabinet can be classified into two types: the 2.2 m cabinet and the 2.6 m cabinet. The product is highly compact because of the compact mechanical structure design of the cabinet.

Table 12-8 provides the configuration of the ETSI 300 mm cabinets fully loaded with subracks. If the cabinet is not fully loaded, install subracks from the bottom up.

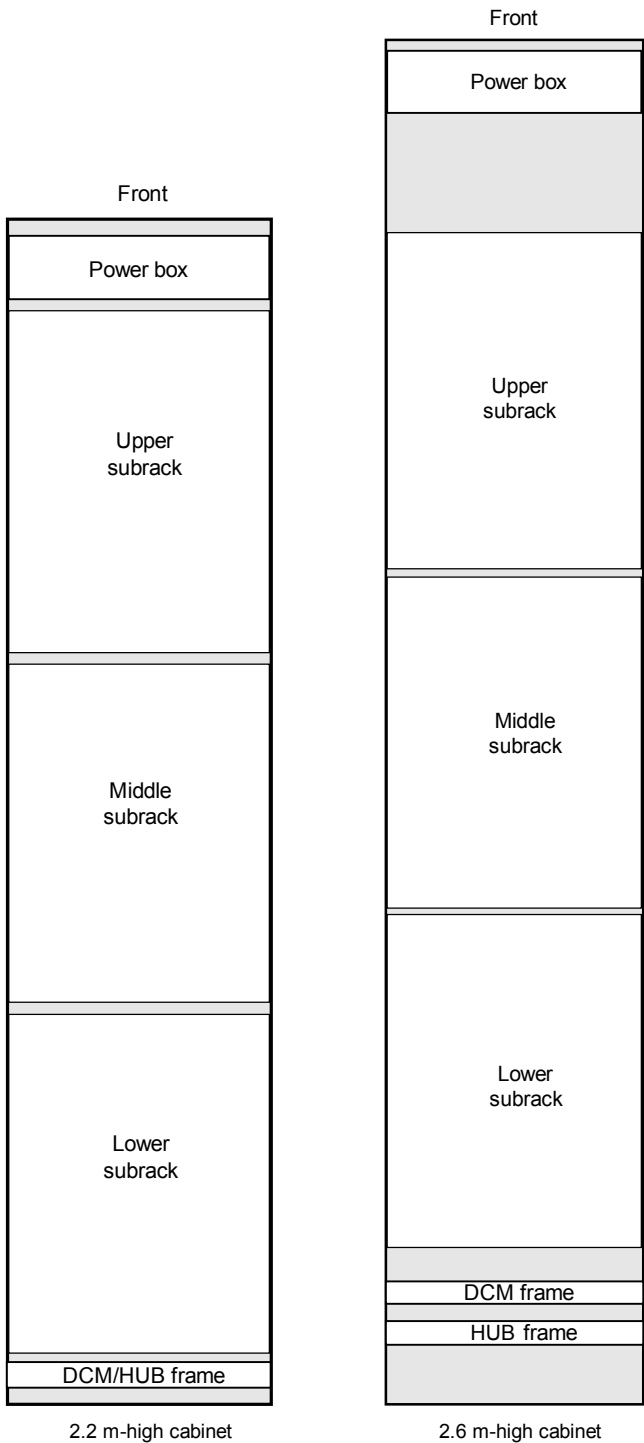
Table 12-8 ETSI Configuration of the fully loaded ETSI 300 mm cabinet of the two heights

Cabinet Height	Power Box Count	Subrack Count	DCM Frame Count	HUB Frame Count
2.2 m	1	3 (or two subracks and two OADM frames)	1 ^a (Or 4 in special cases)	1 ^a

Cabinet Height	Power Box Count	Subrack Count	DCM Frame Count	HUB Frame Count
2.6 m	1	3 (or two subracks and two OADM frames)	1 (Or 4 in special cases)	1
a: The 2.2 m cabinet can only contain either one DCM frame or one HUB frame.				

Figure 12-8 shows the positions of the OptiX Metro 6100 subracks and other mechanical parts installed in a 2.2 m cabinet and those in a 2.6 m cabinet. When multiple subracks are to be installed, install them from bottom up one by one. When only one subrack is to be installed, install it in the lower position. When two subracks are to be installed, install them in the lower and middle positions.

Figure 12-8 Positions of the mechanical parts in a 2.2 m cabinet and those in a 2.6 m cabinet (three subracks)

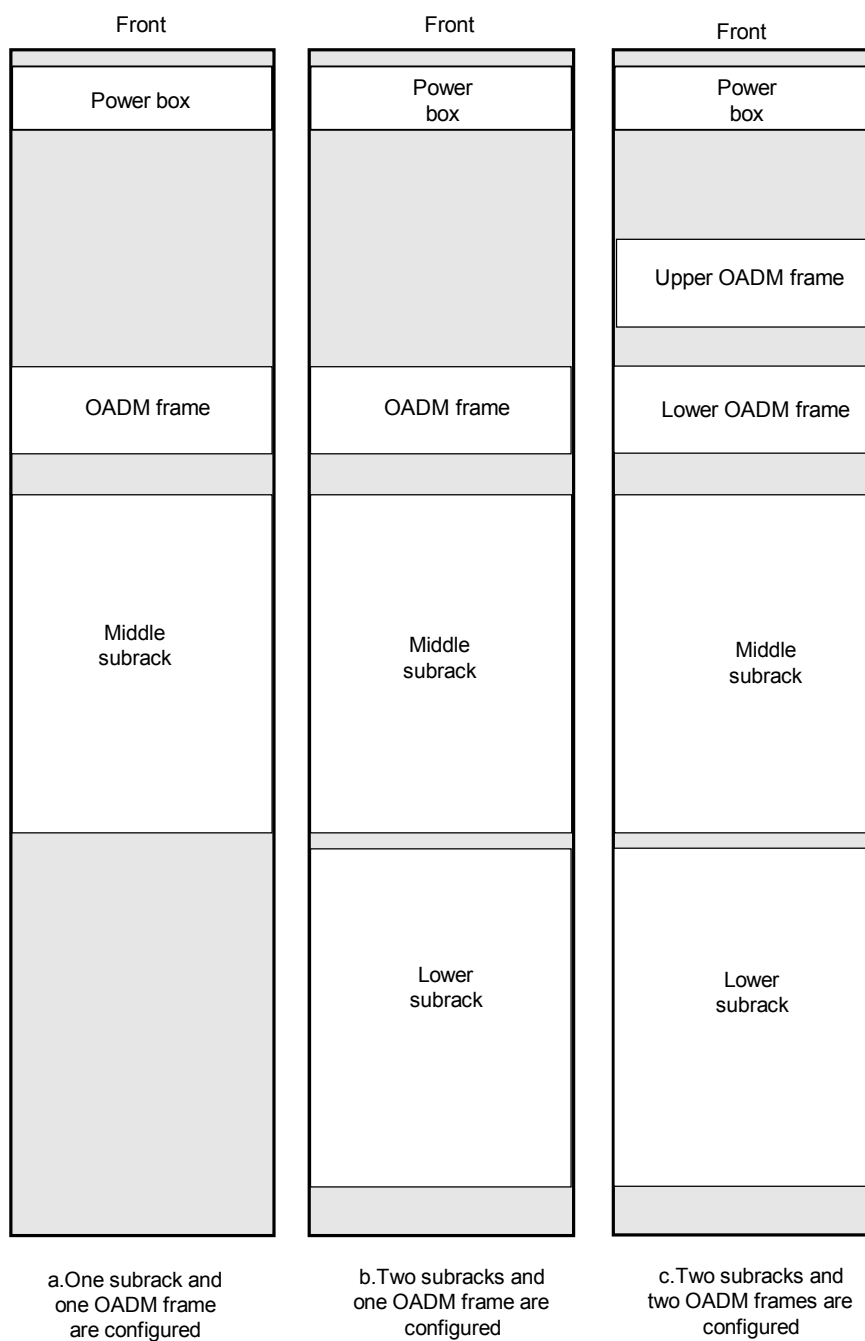


12.14.3 Planning OADM Frames

Based on the number of OADM boards to be seated and the number of subracks to be installed, the OADM frames planning has the various modes.

Three planning modes of the OADM frame is shown in [Figure 12-9](#).

Figure 12-9 Modes of installing OADM frames



NOTE

In configuration mode b, the OADM frame communicates with the middle subrack.

NOTE

In configuration mode c, the lower OADM frame communicates with the middle subrack; the upper OADM frame communicates with the lower subrack.

12.14.4 Planning Boards

The planning of boards of the product slightly varies with the supported WDM specifications.

For details, refer to the configuration rules described in [6 DWDM System Configuration](#) and [7 CWDM System Configuration](#).

12.15 Optical Attenuators

Optical attenuator planning includes the planning of fixed optical attenuators and planning of variable optical attenuators.

12.15.1 Fixed Optical Attenuators (FOAs)

The FOA of the product is an inline optical attenuator of LC/PC type. FOAs are generally installed at the receive-end optical interfaces of a board.

General Rules for Configuring FOAs

The input optical power of a board ranges from (receiver sensitivity + 3) dBm to (overload point – 5) dBm.

For a positive intrinsic negative (PIN) receiver optical module, after an FOA is added:

- The optical power of a 2.5 Gbit/s PIN receiver optical module should be adjustable from –15 dBm to –5 dBm (the normal operating optical power of a 2.5 Gbit/s PIN receiver optical module ranges from –18 dBm to 0 dBm).
- The optical power of a 10 Gbit/s PIN receiver optical module should be adjustable from –11 dBm to –5 dBm (the normal operating optical power of a 10 Gbit/s PIN receiver optical module ranges from –14 dBm to 0 dBm).

For an avalanche photo diode (APD) receiver optical module, after an FOA is added:

- The optical power of a 2.5 Gbit/s APD receiver optical module should be adjustable from –22 dBm to –14 dBm (the normal operating optical power of a 10 Gbit/s PIN receiver optical module ranges from –25 dBm to –9 dBm).
- The optical power of a 10 Gbit/s APD receiver optical module should be adjustable from –18 dBm to –14 dBm (the normal operating optical power of a 10 Gbit/s APD receiver optical module ranges from –26 dBm to –9 dBm).

Rules for Configuring FOAs on the Client Side of the OTU

The rules for configuring FOAs on the client side of the OTU are as follows:

- If the client side of the OTU is a multimode optical module, do not configure an FOA at the optical interface on the module.
- If the client side of the OTU is a single-mode PIN receiver optical module, configure a 7 dB FOA at the optical interface on the module.
- If the client side of a 2.5Gbit/s or 5Gbit/s OTU is a single-mode APD receiver optical module, configure a 15 dB FOA at the optical interface on the module.
- If the client side of a 10Gbit/s OTU is a single-mode APD receiver optical module, configure a 10 dB FOA at the optical interface on the module.

 NOTE

The preceding configuration rules are on a basis that the optical module configured on client equipment is the same as that on the OTU and the OptiX Metro 6100 is connected to the client equipment across a short distance. Otherwise, change or remove the FOA according to the requirement of optical power receiving.

Rules for Configuring FOAs on the WDM Side of the OTU

The requirements of configuring FOAs on the WDM side of the OTU depend on the following three factors:

- Type of the optical demultiplexing board or optical add/drop multiplexing board
- Type of the receive-end optical amplifier board
- Type of the receive-end optical interfaces on the WDM side of the OTU

 NOTE

- The rules for configuring the FOAs on the regenerating OTU are the same as those on the WDM side of the OTU.
- FOAs must be placed at the WDM-side receive end of each OTU and must not be placed in the main optical path.

Table 12-9 provides the requirements of configuring FOAs on the WDM side of the OTU in a station that uses the D40 board as the optical demultiplexing board.

Table 12-9 Rules for configuring FOAs in a station that uses the D40 board

Type of the Receive-End Optical Amplifier Board	Type of WDM-Side Receive-End Optical Interfaces of the OTU	Recommended FOAs
C6OAU01 C9OAU01 C6OAU03 C9OAU03 C6OBU03 C9OBU03	PIN	5 dB
	APD	15 dB
C6OAU02 C9OAU02 C6OBU01	PIN	5 dB
	APD	15 dB
OPU or no optical amplifier board	PIN	None
	APD	15 dB

Table 12-10 provides the requirements of configuring FOAs on the WDM side of the OTU in a station that uses the OADM or ROADM board.

Table 12-10 Rules for configuring FOAs in a station that uses the OADM or ROADM board

Type of the Receive-End Optical Amplifier Board	Type of WDM-Side Receive-End Optical Interfaces of the OTU	Recommended FOAs
C6OAU01	PIN	10 dB
C9OAU01	APD	15 dB + 5 dB
C6OAU03		
C9OAU03		
C6OBU03		
C9OBU03		
C6OAU02	PIN	7 dB
C9OAU02	APD	15 dB
C6OBU01		
OPU or no optical amplifier board	PIN	7 dB
	APD	15 dB



NOTE

The 15dB+5dB configuration mode is as follows: Configure a 15 dB FOA at the receive-end optical interface of the OTU; configure a 5 dB FOA, if necessary, at the corresponding transmit-end optical interface of the OADM board according to the actual optical power.

For the station whose line-side receive end has no optical amplifier, if the WDM side of the OTU is a single-mode PIN receiver optical module, FOAs need not be configured. If the WDM side of the OTU is a single-mode APD receiver optical module, a 10 dB FOA need be configured.

Special Rules for Configuring FOAs at the WDM-Side Transmit-End Optical Interfaces of the OTU

In the OTM station, if the V40 is configured, the WDM side of the OTU no longer requires any manual optical attenuator by default. If the V40 is configured and the OPU is configured at the receive end, configure one 5 dB FOA at the transmit-end optical interface on the WDM side of each tunable OTU.

Rules for Configuring FOA for Pass-Through Wavelengths in a 40-Channel OADM Station

When a 40-channel OADM station is configured with pass-through wavelengths, do not configure a 7 dB FOA between the receive-end optical amplifier and the D40. Add a 7 dB FOA to the PIN receiver optical module on the WDM side of the OTU or add a 15 dB FOA to the APD receiver optical module on the WDM side of the OTU.

12.15.2 Variable Optical Attenuators (VOAs)

A DWDM system uses two types of VOAs: the manual VOA (MVOA) and the electrical VOA (EVOA). Both the MVOA and EVOA are used to adjust the optical power of optical signals. The two differ in that the MVOA requires screwdrivers for manual adjustment while the EVOA

supports remote adjustment by using the NE software or T2000. This section describes the rules for configuring VOAs in various scenarios.

OTM Station

The rules for configuring VOAs at the OTM station are as follows:

- The input optical interface of the optical amplifier at the receive end and that at the transmit end should be configured with one VOA each.
- The rules for configuring VOAs for add channels are as follows:
 - For the station that adopts the M40 or OADM board, each WDM-side output optical interface of the OTU should be configured with one VOA.
 - For the station that adopts the V40 board, WDM-side output optical interfaces of the OTU do not require VOAs because each input optical interface of the V40 is configured with a VOA.



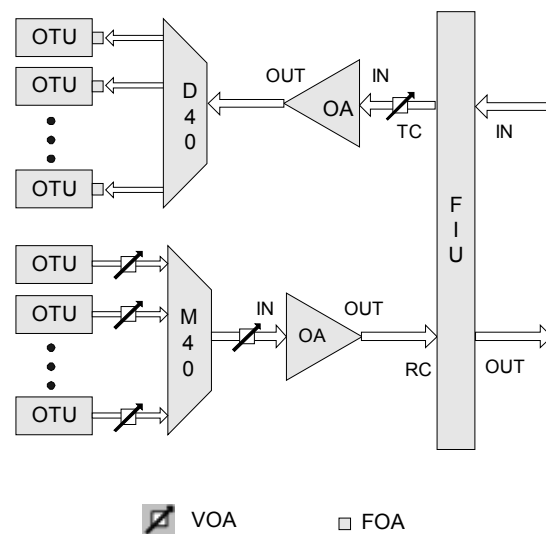
NOTE

Each transmit-end optical interface of the OTU whose WDM side dual feeds and selectively receives signals should be configured with one VOA.

- For drop channels, VOAs are not required; FOAs are used instead.

Figure 12-10 shows how to configure VOAs at the OTM station with the M40 and D40 boards.

Figure 12-10 Diagram of configuring VOAs at the OTM station



FOADM Station

The rules for configuring VOAs at the FOADM station are as follows:

- The input optical interface of the optical amplifier at the receive end should be configured with one VOA.
- The rules for configuring VOAs for add channels are as follows:

- For the station that adopts the M40 or OADM board, each WDM-side output optical interface of the OTU should be configured with one VOA.
- For the station that adopts the V40 board, WDM-side output optical interfaces of the OTU do not require VOAs because each input optical interface of the V40 is configured with a VOA.

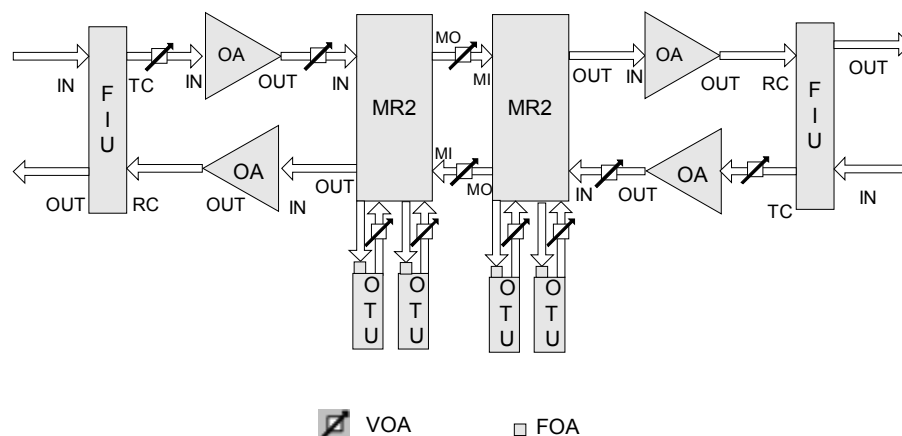
NOTE

Each transmit-end optical interface of the OTU whose WDM side dual feeds and selectively receives signals should be configured with one VOA.

- For drop channels, VOAs are not required; FOAs are used instead.
- The rules for configuring VOAs for pass-through channels are as follows:
 - For the station that adopts the M40 and D40 boards, the pass-through wavelength in the east and that in the west should be configured with one VOA each.
 - For the station that adopts the FOADM board, the pass-through multiplexed wavelength in the east and that in the west should be configured with one VOA each.
 - For the station that adopts the V40 board, pass-through wavelengths do not require VOAs.

Figure 12-11 shows how to configure VOAs at the OADM station with the MR2 boards.

Figure 12-11 Diagram of configuring VOAs at the OADM station



ROADM Station

There are four types of ROADM stations:

- ROADM station with the DWC boards
- ROADM station with the WSD9 and WSM9 boards
- ROADM station with the WSD9 and RMU9 boards
- ROADM station with the WSMD4 boards

The rules for configuring VOAs at the ROADM station with the DWC boards are as follows:

- The input optical interface of the optical amplifier at the receive end should be configured with one VOA.

- For add channels, VOAs are not required.
- For drop channels, VOAs are not required; FOAs are used instead.
- For pass-through channels, VOAs are not required.

Figure 12-12 shows how to configure VOAs at the ROADM station with the DWC, D40 and V40 boards.

Figure 12-12 Diagram of configuring VOAs at the ROADM station with the DWCs (1)

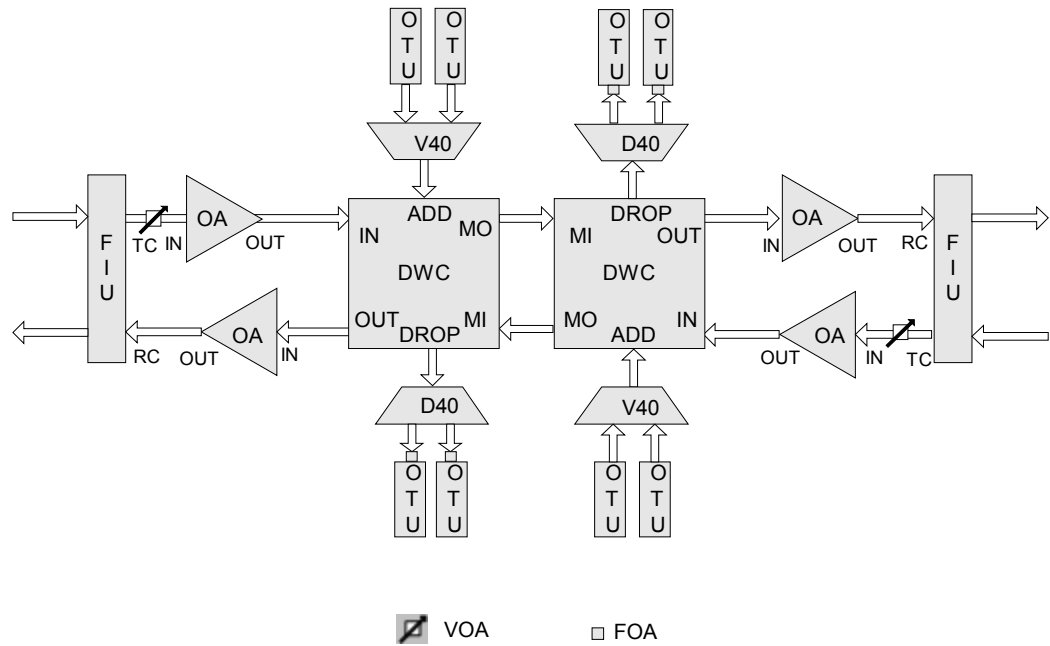


Figure 12-13 shows how to configure VOAs at the ROADM station with the DWC, D40 and M40 boards.

Figure 12-13 Diagram of configuring VOAs at the ROADM station with the DWCs (2)

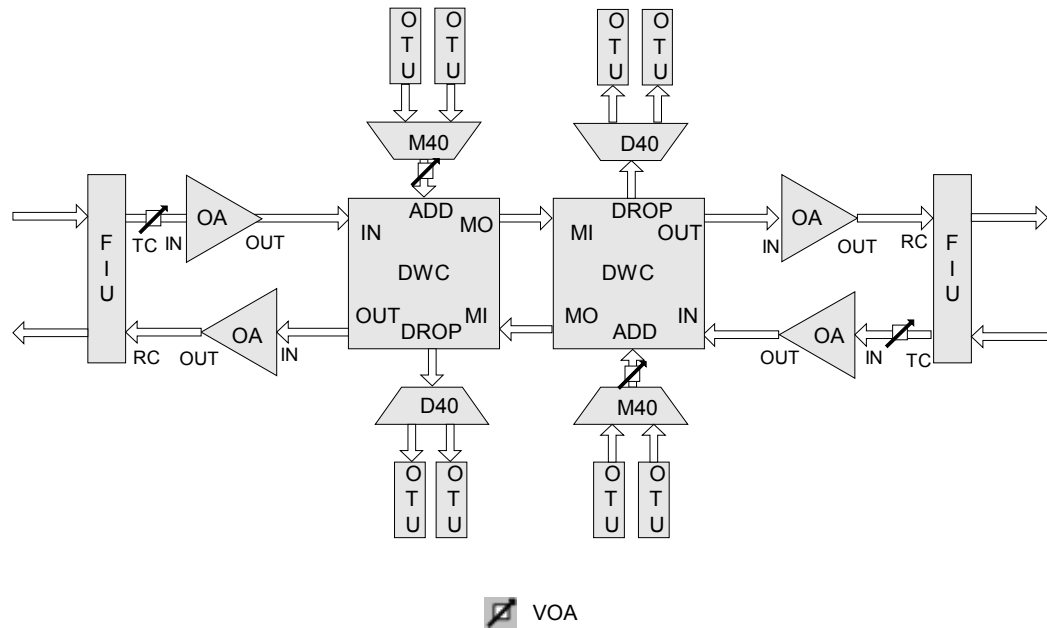
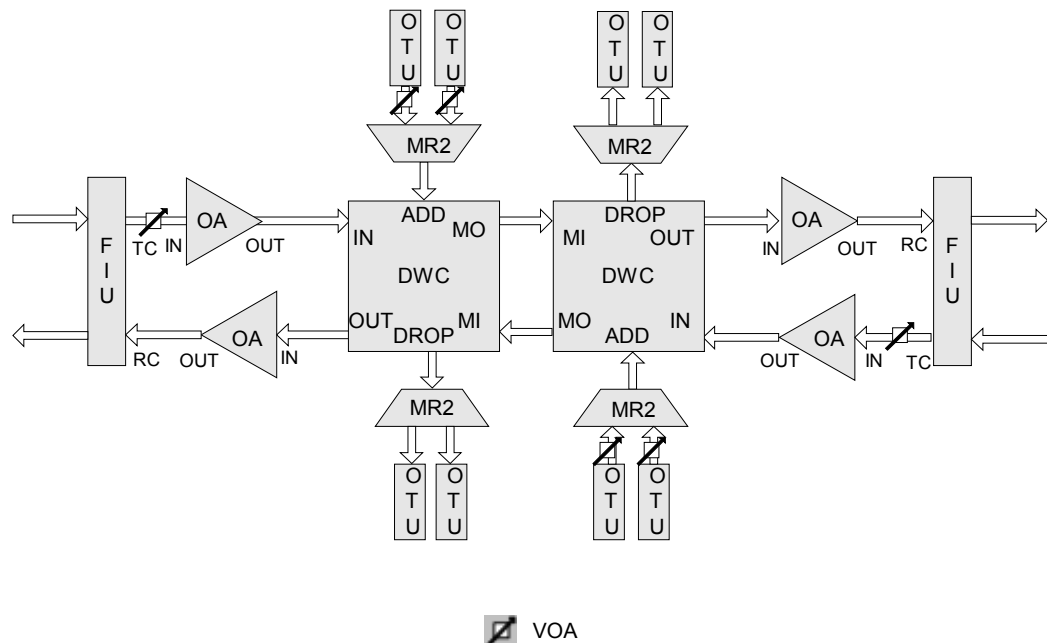


Figure 12-14 shows how to configure VOAs at the ROADM station with the DWC and MR2 boards.

Figure 12-14 Diagram of configuring VOAs at the ROADM station with the DWCs (3)

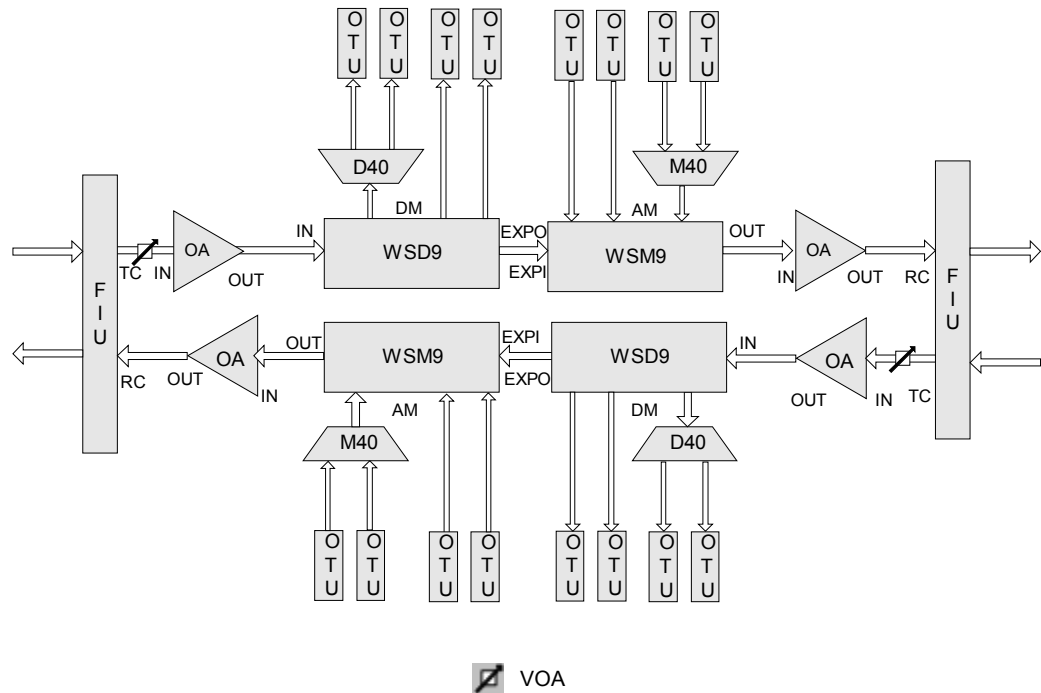


The rules for configuring VOAs at the ROADM station with the WSD9 and WSM9 are as follows:

- The input optical interface of the optical amplifier at the receive end should be configured with one VOA.
- VOAs are required in none of add, drop or pass-through channels.

Figure 12-15 shows how to configure VOAs at the ROADM station with the WSD9, WSM9, D40 and M40 boards.

Figure 12-15 Diagram of configuring VOAs at the ROADM station with the WSD9 and WSM9



The rules for configuring VOAs at the ROADM station with the WSD9 and RMU9 are as follows:

- When the WSM9 and RMU9 are combined with the OADM, optical multiplexer (OM) or optical demultiplexer (OD), the input optical interface of the optical amplifier at the receive end and that at the transmit end should be configured with one VOA each. If the WSM9 and RMU9 are directly connected to the OTU, the input optical interface of the optical amplifier at only the receive end should be configured with one VOA.
- The rules for configuring VOAs for add channels are as follows:
 - If an optical amplifier is configured between the TOA and ROA optical interfaces of the RMU9, a VOA should be configured between ROA and the optical amplifier.
 - When the WSM9 and RMU9 are directly connected to the OTU, VOAs are not required.
 - VOAs are not required between the OADM or OM and RMU9.
 - When the WSM9 and RMU9 are combined with the M40, if optical power equilibrium is not required for add channels, VOAs are not required at output optical interfaces on the WDM side of the OTU; if optical power equilibrium is required for add channels, VOAs should be configured at output optical interfaces on the WDM side of the OTU.
 - When the WSM9 and RMU9 are combined with the OADM, VOAs should be configured at output optical interfaces on the WDM side of the OTU.

- When the WSM9 and RMU9 are combined with the V40, VOAs are not required at output optical interfaces on the WDM side of the OTU.

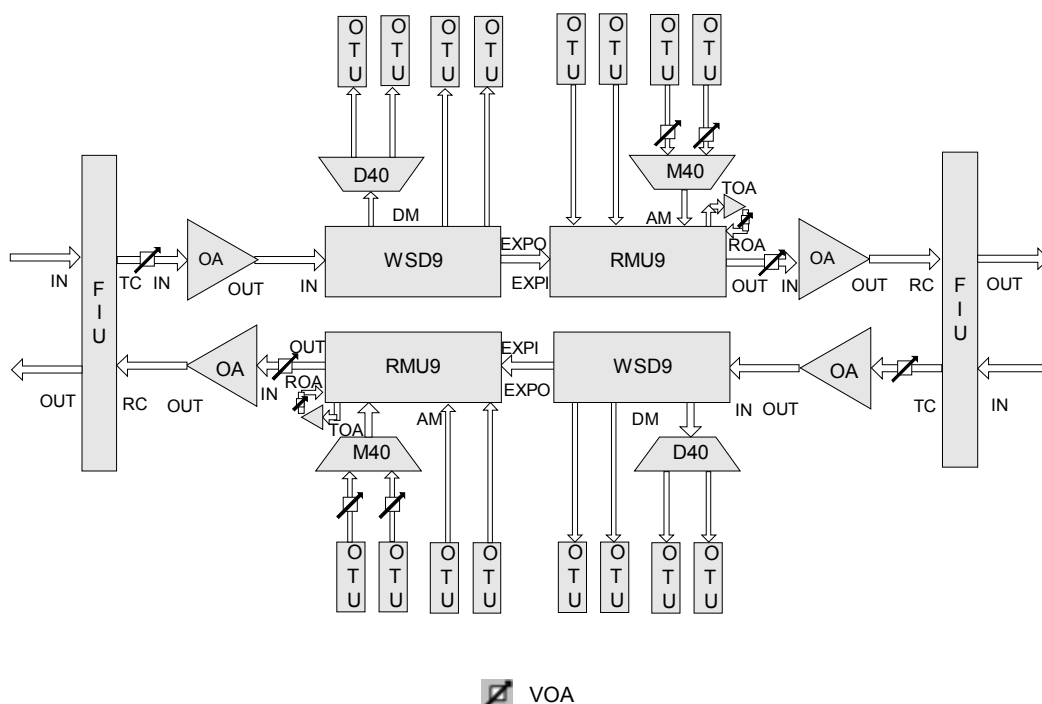
NOTE

Each transmit-end optical interface of the OTU whose WDM side dual feeds and selectively receives signals should be configured with one VOA.

- For drop channels, VOAs are not required.
- For pass-through channels, VOAs are not required.

Figure 12-16 shows how to configure VOAs at the ROADM station with the WSD9, RMU9, M40 and D40 boards.

Figure 12-16 Diagram of configuring VOAs at the ROADM station with the WSD9 and RMU9



The rules for configuring VOAs at the ROADM station with the WSMD4 boards are as follows:

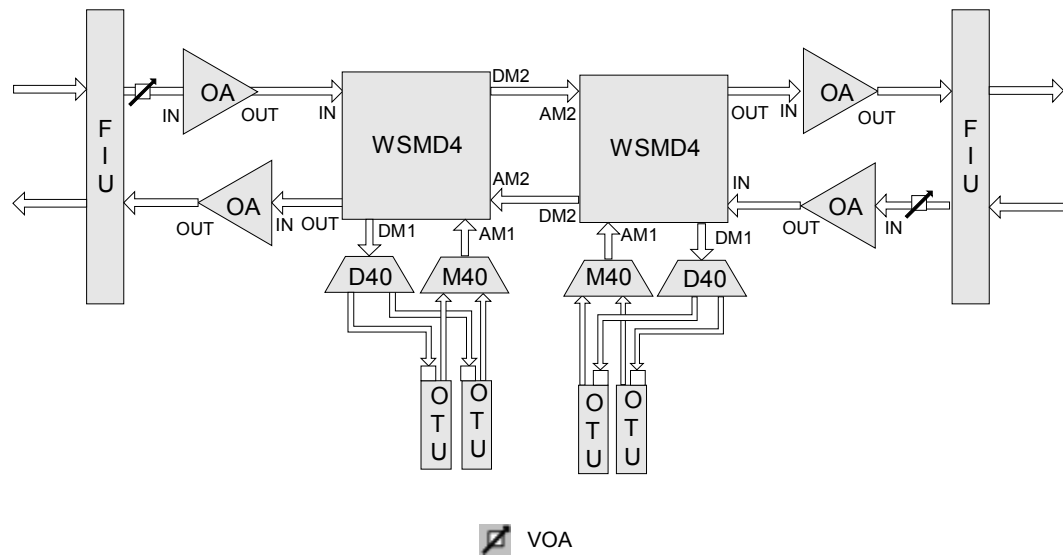
- The input optical interface of the optical amplifier at the receive end should be configured with one VOA.
- In add channels, VOAs are not required.
- In drop channels, VOAs are not required; FOAs are used instead.
- In pass-through channels, VOAs are not required.

NOTE

The WSMD4 board owns the EVOA in the direction of the add channel. You can adjust the optical power of single wavelength remotely by using the T2000.

Figure 12-17 shows how to configure VOAs at the ROADM station with the WSMD4 boards.

Figure 12-17 Diagram of configuring VOAs at the ROADM station with the WSMD4

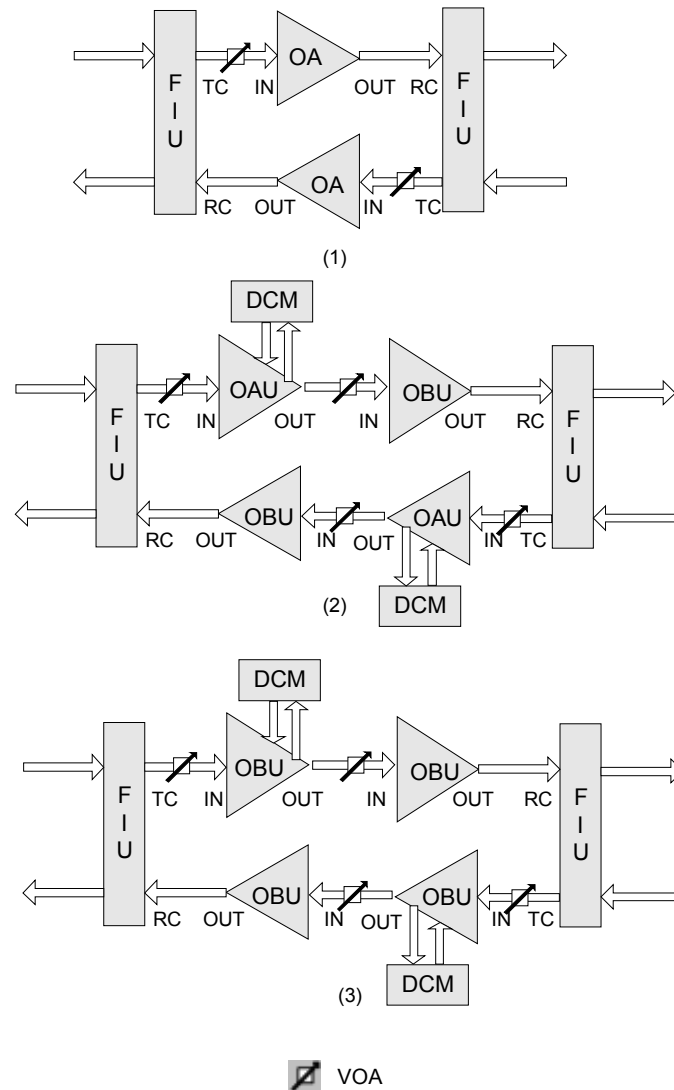


OLA Station

The rules for configuring VOAs at the OLA station are as follows:

- When one direction of the OLA station is configured with only one OA, one VOA should be configured both between the east FIU and the input optical interface of the OA and between the west FIU and the input optical interface of the OA. See [Figure 12-18 \(1\)](#).
- When one direction of the OLA station is configured with two optical amplifiers (as shown in [Figure 12-18 \(2\) and \(3\)](#)), the configuration rules are as follows:
 - One VOA should be configured both between the east FIU and the input optical interface of the optical amplifier and between the west FIU and the input optical interface of the optical amplifier.
 - A VOA should be configured between the two optical amplifiers.
 - It is recommended to configure a VOA between the RDC and TDC optical interfaces of the DCM.

Figure 12-18 Diagram of configuring VOAs at the OLA station



12.16 Ambient Conditions

The safe operation of the product requires good ambient conditions.

When you plan the ambient conditions of the OptiX Metro 6100, consider the following factors:

- Location of the equipment room
- Interior layout of the equipment room
- Architecture of the equipment room
- Cleanness of the equipment room
- Humidity and temperature requirements of the equipment
- ESD protection
- Lightning protection grounding

- Power supply of the equipment
- Fire protection

For details, refer to the *Installation Guide*.

12.17 Power Supply and Power Consumption

The product requires the equipment room to provide two –48 V/–60V DC power supplies of mutual backup.

Table 12-11 and **Table 12-12** list the specifications of power supply and power consumption of the OptiX Metro 6100.

Table 12-11 Specifications of power supply and power consumption of the OptiX Metro 6100 standard subrack

Items	Specification
Nominal working voltage	–48 V DC/–60 V DC
Working voltage range	–38.4 V DC to –72 V DC
Maximum power consumption of a fully loaded cabinet	2000 W
Maximum power consumption of a fully loaded subrack	650 W
Rated current	20 A

Table 12-12 Specifications of power supply and power consumption of the OptiX Metro 6100 enhanced subrack

Items	Specification
Nominal working voltage	–48 V DC/–60 V DC
Working voltage range	–38.4 V DC to –72 V DC
Maximum power consumption of a fully loaded cabinet	2000 W
Maximum power consumption of a fully loaded subrack	800 W
Rated current	30 A

The power consumption of an OptiX Metro 6100 subrack can be obtained by summing up the power consumption of each board in the subrack.

For the data of power consumption of each board, refer to the *Hardware Description*.

13 Technical Specifications

About This Chapter

Technical specifications include general specifications, main optical path, wavelength and frequency of optical channels, laser class and board specifications.

[13.1 General Specifications of OptiX Metro 6100](#)

General specifications include cabinet specifications, power box specifications, standard subrack specifications, independent OLA subrack specifications, auxiliary interface, DCM and DCM frame specifications and HUB and HUB frame specifications.

[13.2 Main Optical Path](#)

The characteristic of the optical interface at points MPI-S or S' and MPI-R or R' as well as the main optical path parameters are shown in the following tables. In this section, the span specifications are provided when FEC technology is adopted and the Raman technology is not used.

[13.3 Wavelength and Frequency of Optical Channels](#)

The system uses the frequencies and wavelengths in the C band.

[13.4 Optical Transponder Board Specifications](#)

The specifications of the OTU boards include the specifications of the optical modules at the client and WDM sides, mechanical specifications, and power consumption.

[13.5 Optical Multiplexer and Demultiplexer Board Specifications](#)

The specifications of optical multiplexers and demultiplexers include the optical specifications, mechanical specifications, and power consumption of the EFIU/FIU/M40/V40/D40 boards.

[13.6 Optical Add and Drop Multiplexing Board Specifications](#)

The specifications of optical add/drop multiplexing boards include the optical specifications, mechanical specifications, and power consumption of the MR2/MR4/SBM1/SBM2 boards and DWC/RMU9/WDM9/WSD9/WSMD4 boards.

[13.7 Optical Amplifier Board Specifications](#)

The specifications of optical amplifier boards include the optical specifications, mechanical specifications, and power consumption of the OAU/OBU/OPU/PC boards.

[13.8 System Control, Supervision and Communication Board Specifications](#)

The specifications of system control, supervision and communication boards include the mechanical specifications and power consumption of the SCC and PMU boards.

13.9 Optical Supervisory Channel and Timing Transmission Board Specifications

The specifications of optical supervisory channels and timing transmission boards include the optical specifications, mechanical specifications, and power consumption of the SC1/SC2/TC1/TC2/ST1/ST2 boards.

13.10 Optical Protection Board Specifications

The specifications of protection boards include the optical specifications, mechanical specifications, and power consumption of the DCP/OLP/OWSP/SCS boards.

13.11 Spectrum Analyzer Board Specifications

The specifications of spectrum analyzer boards include the optical specifications, mechanical specifications, and power consumption of the MCA board.

13.12 Variable Optical Attenuator Board Specifications

The specifications of variable optical attenuator boards include the optical specifications, mechanical specifications, and power consumption of the VA2/VA4/VOA boards.

13.1 General Specifications of OptiX Metro 6100

General specifications include cabinet specifications, power box specifications, standard subrack specifications, independent OLA subrack specifications, auxiliary interface, DCM and DCM frame specifications and HUB and HUB frame specifications.

13.1.1 Cabinet Specifications

The technical specifications of the cabinet include dimensions, weight, power consumption and power supply.

Table 13-1 Cabinet specifications

Item	2.2m Cabinet Specifications	2.6m Cabinet Specifications
Dimensions	2200 mm (H) x 600 mm (W) x 300 mm (D)	2600 mm (H) x 600 mm (W) x 300 mm (D)
Weight	69 kg	80 kg
Maximum power consumption	2000 W	2000 W
Power supply	-48 V/-60 V DC	-48 V/-60 V DC

13.1.2 Subrack Specifications

Specifications include dimensions, power consumption, and power supply.

Table 13-2 shows the technical parameters of the OptiX Metro 6100 standard subrack.

Table 13-2 Technical parameters of the standard subrack

Item	Parameter
Dimensions	625.0 mm (H) x 440.0 mm (W) x 290.0 mm (D) ^a
Weight (empty subrack)	18.0 kg
Maximum power consumption (full configuration)	650.0 W
Minimum power consumption (only configuring with the SCC, the PMU and the fan tray assembly)	65.5 W
Power consumption of a fan tray assembly	43.0 W
Rated working current	16 A
Nominal working voltage	-48 V DC or -60 V DC

Item	Parameter
Working voltage range	–38.4 V to –72 V DC
Fuse capacity	20 A
a: H = Height, W = Width, D = Depth	

Table 13-3 shows the technical parameters of the OptiX Metro 6100 enhanced subrack.

Table 13-3 Technical parameters of the enhanced subrack

Item	Parameter
Dimensions	625.0 mm (H) x 440.0 mm (W) x 290.0 mm (D) ^a
Weight (empty subrack)	18.0 kg
Maximum power consumption (full configuration)	800.0 W
Minimum power consumption (only configuring with the SCC, the PMU and the fan tray assembly)	65.5 W
Power consumption of a fan tray assembly	43.0 W
Rated working current	16 A
Nominal working voltage	–48 V DC or –60 V DC
Working voltage range	–38.4 V to –72 V DC
Fuse capacity	30 A
a: H = Height, W = Width, D = Depth	

Table 13-4 shows the technical parameters of the common units.

Table 13-4 Power consumption of the common units

Unit Name	Maximum Power Consumption	Remarks
OTU subrack	449.5W	It is the power consumption when the subrack is installed with twelve LWMs (single-fed board), one SCC, one PMU, and one fan tray assembly.
OTM subrack	361.5W	It is the power consumption when the subrack is installed with eight LWMs (single-fed board), one M40, one D40, one SCC, one PMU, and one fan tray assembly.

Unit Name	Maximum Power Consumption	Remarks
OLA subrack	179.5W	It is the power consumption when the subrack is installed with two OAUs, two OBUs, two FIUs, one SC2, one SCC, one PMU, and one fan tray assembly.

13.2 Main Optical Path

The characteristic of the optical interface at points MPI-S or S' and MPI-R or R' as well as the main optical path parameters are shown in the following tables. In this section, the span specifications are provided when FEC technology is adopted and the Raman technology is not used.

Table 13-5 Main optical path parameters of the OptiX Metro 6100 DWDM system (G.652 fiber, NRZ)

Item		Unit	Performance Parameter	
-		-	FEC	Without FEC
Span of line		-	16 x 22dB	8 x 22dB
Number of channels		-	40	40
Maximum bit rate of channel		Gbit/s	10	2.5
Optical interface at points MPI-S and S'				
Channel output power (output port of amplifiers)	Average	dBm	+4.0	+4.0
	Maximum	dBm	+7.0	+7.0
	Minimum	dBm	+1.0	+1.0
Maximum total output power		dBm	+20.0	+20.0
Maximum channel power difference at point MPI-S		dB	6	6
Optical path (MPI-S - MPI-R)				
Maximum optical path penalty		dB	≤ 2	≤ 2
Maximum dispersion		ps/nm	25600	12800
Maximum discrete reflectance		dB	-27	-27
Maximum average differential group delay (DGD)		ps	15	60
Optical interface at points MPI-R and R'				

Item		Unit	Performance Parameter	
Channel input power (input port of amplifiers)	Average	dBm	-19	-19
	Maximum	dBm	-16	-16
	Minimum	dBm	-24	-24
Minimum channel optical signal-to-noise ratio at point MPI-R		dB	17.5	21.5
Maximum channel power difference at point MPI-R		dB	8	8

Table 13-6 Main optical path parameters of the OptiX Metro 6100 DWDM system (G.652 fiber, DRZ)

Item		Unit	Performance Parameter
-		-	FEC, SuperWDM
Span of line		-	20 x 22dB
Number of channels		-	40
Maximum bit rate of channel		Gbit/s	10
Optical interface at points MPI-S and S'			
Channel output power (output port of amplifiers)	Average	dBm	4
	Maximum	dBm	7
	Minimum	dBm	1
Maximum total output power		dBm	+20.0
Maximum channel power difference at point MPI-S		dB	6
Optical path (MPI-S - MPI-R)			
Maximum optical path penalty		dB	
Maximum dispersion		ps/nm	32000
Maximum discrete reflectance		dB	-27
Maximum average differential group delay (DGD)		ps	18
Optical interface at points MPI-R and R'			
Channel input power (input port of amplifiers)	Average	dBm	-20
	Maximum	dBm	-16
	Minimum	dBm	-24

Item	Unit	Performance Parameter
Minimum channel optical signal-to-noise ratio at point MPI-R	dB	14.5
Maximum channel power difference at point MPI-R	dB	8

Table 13-7 Main optical path parameters of the OptiX Metro 6100 DWDM system (G.652 fiber, DRZ, single span)

Item		Unit	Performance Parameter
-		-	FEC, Raman, SuperWDM
Span of line		-	1 x 46dB
Number of channels		-	40
Maximum bit rate of channel		Gbit/s	10
Optical interface at points MPI-S and S'			
Channel output power (output port of amplifiers)	Average	dBm	+7dBm
	Maximum	dBm	+9dBm
	Minimum	DBm	+4dBm
Maximum total output power		dBm	23dBm
Maximum channel power difference at point MPI-S		dB	3
Optical path (MPI-S - MPI-R)			
Maximum optical path penalty		Db	2
Maximum dispersion		ps/nm	3500
Maximum discrete reflectance		dB	-27
Maximum average differential group delay (DGD)		Ps	18
Optical interface at points MPI-R and R'			
Channel input power (input port of amplifiers)	Average	dBm	-30dBm
	Maximum	dBm	-27dBm
	Minimum	dBm	-32dBm
Minimum channel optical signal-to-noise ratio at point MPI-R		dB	17.5
Maximum channel power difference at point MPI-R		dB	5

Table 13-8 Main optical path parameters of the OptiX Metro 6100 CWDM system

Item		Unit	Performance Parameter
Span of line		-	1 x 22dB
Number of channels		-	16
Maximum bit rate of channel		Gbit/s	2.5
Optical interface at points MPI-S and S'			
Channel output power	Average	dBm	+3
	Maximum	dBm	+5
	Minimum	dBm	0
Maximum total output power		dBm	15
Maximum channel power difference at point MPI-S		dB	5
Optical path (MPI-S - MPI-R)			
Maximum discrete reflectance		dB	-24
Maximum average differential group delay (DGD)		ps	60
Optical interface at points MPI-R and R'			
Channel input power	Average	dBm	-19
	Maximum	dBm	-17
	Minimum	dBm	-22
Maximum channel power difference at point MPI-R		dB	5

13.3 Wavelength and Frequency of Optical Channels

The system uses the frequencies and wavelengths in the C band.

13.3.1 Nominal Central Wavelength and Frequency of DWDM System

The minimum channel spacing in C band is 100 GHz. The available number of available wavelengths is 40. The operating wavelength range is from 192.10 THz to 196.00 THz (1529.55 nm to 1560.61 nm).

Table 13-9 Nominal central wavelength and frequency of DWDM system

Frequency (THz)	Wavelength (nm)	Frequency (THz)	Wavelength (nm)
192.1	1560.61	194.1	1544.53
192.2	1559.79	194.2	1543.73
192.3	1558.98	194.3	1542.94
192.4	1558.17	194.4	1542.14
192.5	1557.36	194.5	1541.35
192.6	1556.56	194.6	1540.56
192.7	1555.75	194.7	1539.77
192.8	1554.94	194.8	1538.98
192.9	1554.13	194.9	1538.19
193.0	1553.33	195.0	1537.40
193.1	1552.52	195.1	1536.61
193.2	1551.72	195.2	1535.82
193.3	1550.92	195.3	1535.04
193.4	1550.12	195.4	1534.25
193.5	1549.32	195.5	1533.47
193.6	1548.51	195.6	1532.68
193.7	1547.72	195.7	1531.90
193.8	1546.92	195.8	1531.12
193.9	1546.12	195.9	1530.33
194.0	1545.32	196.0	1529.55

13.3.2 Nominal Central Wavelengths of CWDM System

The minimum channel spacing is 20 nm. The operating wavelength range is from 1311 nm to 1611 nm.

Table 13-10 Nominal central wavelengths of CWDM system

Wavelength (nm)
1611
1591

Wavelength (nm)
1571
1551
1531
1511
1491
1471
1451
1431
1411
1391
1371
1351
1331
1311

13.4 Optical Transponder Board Specifications

The specifications of the OTU boards include the specifications of the optical modules at the client and WDM sides, mechanical specifications, and power consumption.

13.4.1 AP8 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-11](#), [Table 13-12](#), [Table 13-13](#), [Table 13-14](#), [Table 13-15](#) and [Table 13-16](#) list the optical specifications of the C6AP8.

Table 13-11 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-12 Specifications of optical module for FC service at client side

Item	Unit	Value	
		FC 100/FC 200	
Line code format	-	NRZ	NRZ

Item	Unit	Value	
		FC 100/FC 200	
Target distance	km	2	0.5
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	770 to 860
Maximum mean launched power	dBm	-3	-2.5
Minimum mean launched power	dBm	-10	-9.5
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1580	770 to 860
Receiver sensitivity	dBm	-18	-17
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	NA	NA

Table 13-13 Specifications of optical module for ESCON and other services at client side

Item	Unit	Value	
Target distance	km	2	15
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	
Maximum mean launched power	dBm	-14	-8
Minimum mean launched power	dBm	-19	-15
Minimum extinction ratio	dB	8.2	8.2
Maximum -20 dB spectral width	nm	NA	NA

Item	Unit	Value	
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-27	-28
Minimum receiver overload	dBm	-14	-8
Maximum reflectance	dB	NA	NA
Note	-	The ESCON, DVB-ASI and FE services can be accessed to this optical module.	

Table 13-14 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					
Maximum mean launched power ^a	dBm	-1	-1	3	3
Minimum mean launched power ^a	dBm	-5	-5	-2	-2
Typical value of mean launched power ^a	dBm	-2	-2	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Dispersion tolerance	ps/nm	12800	12800	6500	3200
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.					

Table 13-15 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		
Maximum mean launched power ^a	dBm	3
Minimum mean launched power ^a	dBm	-1
Typical value of mean launched power ^a	dBm	1
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	12800
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		

Item	Unit	Value
		12800 ps/nm-tunable
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.		

Table 13-16 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ±6.5	≤ ±6.5
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD

Item	Unit	Value	
		1600 ps/nm-4mW	
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.1 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 52.6 W
- Maximum power consumption at 55°C : 58.0 W

13.4.2 AS8 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-17, **Table 13-18**, **Table 13-19** and **Table 13-20** list the optical specifications of the L2AS8 and C7AS8.

Table 13-17 Specifications of optical module for STM-1/OC-3/STM-4/OC-12 service at client side

Item	Unit	Value	
		S-4.1	L-4.1
Line code format	-	NRZ	NRZ
Optical source type	-	MLM	SLM
Target distance	km	15	40
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1274 to 1356	1280 to 1335
Maximum mean launched power	dBm	-8	2
Minimum mean launched power	dBm	-15	-3
Minimum extinction ratio	dB	8.2	10
Maximum -20 dB spectral width	nm	NA	1
Minimum side mode suppression ratio	dB	NA	30
Eye pattern mask	-	G.957-compliant	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-28	-28
Minimum receiver overload	dBm	-8	-8
Maximum reflectance	dB	-27	-14

Table 13-18 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					
Maximum mean launched power ^a	dBm	-1	-1	3	3

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Minimum mean launched power ^a	dBm	-5	-5	-2	-2
Typical value of mean launched power ^a	dBm	-2	-2	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.					

Table 13-19 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		

Item	Unit	Value
		12800 ps/nm-tunable
Maximum mean launched power ^a	dBm	3
Minimum mean launched power ^a	dBm	-1
Typical value of mean launched power ^a	dBm	1
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	12800
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.		

Table 13-20 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-2	-2

Item	Unit	Value	
		1600 ps/nm-4mW	
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	$\leq \pm 6.5$	$\leq \pm 6.5$
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.2 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 36.0 W
- Maximum power consumption at 55°C : 38.5 W

13.4.3 ELOG Specifications

ELOG board specifications include specifications of optical module on the client and WDM sides, laser safety level, mechanical specifications and power consumption.

Optical Interface Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

The following tables gives the details about the optical specifications for the C8ELOG and C9ELOG.

Table 13-21 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-22 Specifications of optical module for FC service at client side

Item	Unit	Value	
		FC 100/FC 200	
Line code format	-	NRZ	NRZ
Target distance	km	2	0.5
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	770 to 860
Maximum mean launched power	dBm	-3	-2.5
Minimum mean launched power	dBm	-10	-9.5
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1580	770 to 860
Receiver sensitivity	dBm	-18	-17
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	NA	NA

Table 13-23 Specifications of optical module at DWDM side

Item	Unit	Value	
Line code format	-	NRZ	NRZ-tunable
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	2	2
Minimum mean launched power	dBm	-3	-3
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	
Maximum -20 dB spectral width	nm	0.3	0.3
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	800	1100
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-16	-16
Minimum receiver overload	dBm	0	0
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.1 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 46.0 W
- Maximum power consumption at 55°C : 48.1 W

13.4.4 ELOGS Specifications

ELOGS board specifications include specifications of optical module on the client and WDM sides, laser safety level, mechanical specifications and power consumption.

Optical Interface Specifications

 NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

The following table gives the details about the optical specifications for the ELOGS.

Table 13-24 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-25 Specifications of optical module for FC service at client side

Item	Unit	Value	
		FC 100/FC 200	
Line code format	-	NRZ	NRZ
Target distance	km	2	0.5
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	770 to 860
Maximum mean launched power	dBm	-3	-2.5
Minimum mean launched power	dBm	-10	-9.5
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1580	770 to 860
Receiver sensitivity	dBm	-18	-17
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	NA	NA

Table 13-26 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.1kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C8ELOGS: 54.0 W
 - C9ELOGS: 51.3 W

- CBELOGS: 42.5 W
- Maximum power consumption at 55°C :
 - C8ELOGS: 58.0 W
 - C9ELOGS: 53.8 W
 - CBELOGS: 46.8 W

13.4.5 EGS8 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-27 lists the details about the optical specifications for the C7EGS8.

Table 13-27 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 2.0 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 36.5 W
- Maximum power consumption at 55°C : 38.0 W

13.4.6 ETMX Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-28, **Table 13-29**, and **Table 13-30** list the optical specifications on the client and WDM side of the C8ETMX.

Table 13-28 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-29 Specifications of optical module for the OTU1 service at client side

Item	Unit	Value			
		P1I1-1D1	P1S1-1D1	P1L1-1D1	P1L1-1D2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.959.1-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650

Item	Unit	Value			
		P1I1-1D1	P1S1-1D1	P1L1-1D1	P1L1-1D2
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-30 Specifications of optical module at DWDM side

Item	Unit	Value		
Line code format	-	NRZ	NRZ-tunable	NRZ
Transmitter parameter specifications at point S				
Maximum mean launched power	dBm	-1	0	4
Minimum mean launched power	dBm	-5	-5	0
Minimum extinction ratio	dB	10	10	10
Central frequency	THz	192.10 to 196.00		
Central frequency deviation	GHz	±10		
Maximum -20 dB spectral width	nm	0.3	0.3	0.3
Minimum side mode suppression ratio	dB	35	35	35
Dispersion tolerance	ps/nm	800	800	1600
Receiver parameter specifications at point R				
Receiver type	-	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-16	-16	-26
Minimum receiver overload	dBm	0	0	-9
Maximum reflectance	dB	-27	-27	-27

Table 13-31, **Table 13-32** and **Table 13-33** list the optical specifications on the client and WDM side of the C9ETMX and CAETMX.

Table 13-31 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-32 Specifications of optical module for the OTU1 service at client side

Item	Unit	Value			
		P1I1-1D1	P1S1-1D1	P1L1-1D1	P1L1-1D2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.959.1-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD

Item	Unit	Value			
		P1I1-1D1	P1S1-1D1	P1L1-1D1	P1L1-1D2
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-33 Specifications of optical module at DWDM side

Item	Unit	Value	
Line code format	-	NRZ	NRZ-tunable
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	2	2
Minimum mean launched power	dBm	-3	-3
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	
Maximum -20 dB spectral width	nm	0.3	0.3
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	800	1100
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-16	-16
Minimum receiver overload	dBm	0	0
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C8ETMX/C9ETMX/CAETMX: 1.1 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C:
 - C8ETMX: 34.6W
 - C9ETMX/CAETMX: 32.2 W
- Maximum power consumption at 55°C:
 - C8ETMX: 38.1 W
 - C9ETMX/CAETMX: 35.4 W

13.4.7 ETMXS Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-34](#), [Table 13-35](#) and [Table 13-36](#) list the optical specifications on the client and WDM side of the C8ETMXS.

Table 13-34 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-35 Specifications of optical module for the OTU1 service at client side

Item	Unit	Value			
		P1I1-1D1	P1S1-1D1	P1L1-1D1	P1L1-1D2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.959.1-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-36 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	-1
Minimum mean launched power	dBm	-4
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Table 13-37, **Table 13-38** and **Table 13-39** list the optical specifications on the client and WDM side of the C9ETMXS/CAETMXS/CBETMXS.

Table 13-37 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-38 Specifications of optical module for the OTU1 service at client side

Item	Unit	Value			
		P1I1-1D1	P1S1-1D1	P1L1-1D1	P1L1-1D2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.959.1-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-39 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.3 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C:
 - C8ETMXS: 43.6 W
 - C9ETMXS/CAETMXS: 34.5 W

- CBETMXS: 38.8 W
- Maximum power consumption at 55°C:
 - C8ETMXS: 47.9 W
 - C9ETMXS/CAETMXS: 37.9 W
 - CBETMXS: 42.7 W

13.4.8 FCE Specifications

The following table gives the details about the electrical specifications for the FCE.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-40, **Table 13-41**, **Table 13-42** and **Table 13-43** list the optical specifications on the client and WDM side of the C6FCE.

Table 13-40 Specifications of optical module for FC service at client side

Item	Unit	Value	
		FC 100/FC 200	
Line code format	-	NRZ	NRZ
Target distance	km	2	0.5
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	770 to 860
Maximum mean launched power	dBm	-3	-2.5
Minimum mean launched power	dBm	-10	-9.5
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1580	770 to 860
Receiver sensitivity	dBm	-18	-17

Item	Unit	Value	
		FC 100/FC 200	
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	NA	NA

Table 13-41 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					
Maximum mean launched power ^a	dBm	-1	-1	3	3
Minimum mean launched power ^a	dBm	-5	-5	-2	-2
Typical value of mean launched power ^a	dBm	-2	-2	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.					

Table 13-42 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		
Maximum mean launched power ^a	dBm	3
Minimum mean launched power ^a	dBm	-1
Typical value of mean launched power ^a	dBm	1
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	12800
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.		

Table 13-43 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ± 6.5	≤ ± 6.5
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.1 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C: 32.0 W
- Maximum power consumption at 55°C: 35.2 W

13.4.9 FDG Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-44](#), [Table 13-45](#), [Table 13-46](#) and [Table 13-47](#) list the details about the optical specifications for the FDG.

Table 13-44 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-45 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2mW-APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					
Maximum mean launched power ^a	dBm	-1	-1	3	3
Minimum mean launched power ^a	dBm	-5	-5	-2	-2
Typical value of mean launched power ^a	dBm	-2	-2	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.					

Table 13-46 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		
Maximum mean launched power ^a	dBm	3
Minimum mean launched power ^a	dBm	-1
Typical value of mean launched power ^a	dBm	1
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35

Item	Unit	Value
		12800 ps/nm-tunable
Dispersion tolerance	ps/nm	12800
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.		

Table 13-47 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	$\leq \pm 6.5$	$\leq \pm 6.5$
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	

Item	Unit	Value	
		1600 ps/nm-4mW	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C6FDG: 1.0kg
 - C8FDG: 1.1kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6FDG: 34.5W
 - CWFDG: 28.0W
- Maximum power consumption at 55°C :
 - C6FDG: 38.0W
 - C8FDG: 30.8W

13.4.10 L4G Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-48](#), [Table 13-49](#), and [Table 13-50](#) list the details about the optical specifications for the C7L4G.

Table 13-48 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-49 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value	
		3400 ps/nm	6400 ps/nm
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	2	-1
Minimum mean launched power	dBm	-2	-5
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	
Maximum -20 dB spectral width	nm	0.3	0.3
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	3400	6400
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-25	-25
Minimum receiver overload	dBm	-9	-9
Maximum reflectance	dB	-27	-27

Table 13-50 Specifications of the tunable wavelength optical module at DWDM side

Item	Unit	Value
		3400 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00

Item	Unit	Value
		3400 ps/nm-tunable
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.3
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	3400
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-25
Minimum receiver overload	dBm	-9
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 2.5 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 38.4 W
- Maximum power consumption at 55°C : 50.0 W

13.4.11 LAM Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-51 to **Table 13-59** list the optical specifications of the C7LAM.

Table 13-51 Specifications of optical module for FE service at client side

Item	Unit	Value
		100 BASE-FX
Target distance	km	40
Transmitter parameter specifications at point S		
Operating wavelength range	nm	1310
Maximum mean launched power	dBm	0
Minimum mean launched power	dBm	-5
Minimum extinction ratio	dB	10
Maximum -20 dB spectral width	nm	1.0
Minimum side mode suppression ratio	dB	NA
Eye pattern mask	-	IEEE802.3z-compliant
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1310
Receiver sensitivity	dBm	-30
Minimum receiver overload	dBm	-10

Table 13-52 Specifications of optical module for GE service at client side

Item	Unit	Value	
		100BASE-SX	100BASE-SX
Target distance	km	10, 40, 80	0.55
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1260 to 1360	770 to 860
Maximum mean launched power	dBm	0	-3
Minimum mean launched power	dBm	-9.5	-11
Minimum extinction ratio	dB	9	9
Maximum -20 dB spectral width	nm	1.0	1.0
Minimum side mode suppression ratio	dB	NA	NA

Item	Unit	Value	
		100BASE-SX	100BASE-SX
Eye pattern mask	-	IEEE802.3z-compliant	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1355	770 to 860
Receiver sensitivity	dBm	-20	-17
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	-12	-12

Table 13-53 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-54 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-55 Specifications of optical module for STM-1/OC-3/STM-4/OC-12 service at client side

Item	Unit	Value	
		S-4.1	L-4.1
Line code format	-	NRZ	NRZ
Optical source type	-	MLM	SLM
Target distance	km	15	40
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1274 to 1356	1280 to 1335
Maximum mean launched power	dBm	-8	2
Minimum mean launched power	dBm	-15	-3
Minimum extinction ratio	dB	8.2	10
Maximum -20 dB spectral width	nm	NA	1
Minimum side mode suppression ratio	dB	NA	30
Eye pattern mask	-	G.957-compliant	

Item	Unit	Value	
		S-4.1	L-4.1
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-28	-28
Minimum receiver overload	dBm	-8	-8
Maximum reflectance	dB	-27	-14

Table 13-56 Specifications of optical module for FC service at client side

Item	Unit	Value	
		FC 100/FC 200	
Line code format	-	NRZ	NRZ
Target distance	km	2	0.5
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	770 to 860
Maximum mean launched power	dBm	-3	-2.5
Minimum mean launched power	dBm	-10	-9.5
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1580	770 to 860
Receiver sensitivity	dBm	-18	-17
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	NA	NA

Table 13-57 Specifications of optical module for ESCON and other services at client side

Item	Unit	Value	
Target distance	km	2	15
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	
Maximum mean launched power	dBm	-14	-8
Minimum mean launched power	dBm	-19	-15
Minimum extinction ratio	dB	8.2	8.2
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-27	-28
Minimum receiver overload	dBm	-14	-8
Maximum reflectance	dB	NA	NA
Note	-	The ESCON, DVB-ASI and FE services can be accessed to this optical module.	

Table 13-58 Specifications of optical module for any service at DWDM side

Item	Unit	Value
Target distance	km	120
Transmitter parameter specifications at point S		
Operating rate range	Gbit/s	0.125 - 2.5
Maximum mean launched power	dBm	3
Minimum mean launched power	dBm	-1
Minimum extinction ratio	dB	8.2
Central frequency	THz	192.10 - 196.00

Item	Unit	Value
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	NA
Minimum side mode suppression ratio	dB	30
Dispersion tolerance	ps/nm	2400
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating rate range	Gbit/s	0.125 - 2.5
Operating wavelength range	nm	1260 - 1570
Receiver sensitivity	dBm	-28
Minimum receiver overload	dBm	-8
Maximum reflectance	dB	-27

Table 13-59 Specifications of optical module for any service at CWDM side

Item	Unit	Value
Target distance	km	80
Transmitter parameter specifications at point S		
Operating rate range	Gbit/s	0.125 - 2.5
Maximum mean launched power	dBm	5
Minimum mean launched power	dBm	0
Minimum extinction ratio	dB	9
Central Wavelength	nm	1471, 1491, 1511, 1531, 1551, 1571, 1591, 1611
Central frequency deviation	nm	±6.5
Maximum -20 dB spectral width	nm	1
Minimum side mode suppression ratio	dB	30
Dispersion tolerance	ps/nm	1600
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		

Item	Unit	Value
Receiver type	-	APD
Operating rate range	Gbit/s	0.125 - 2.5
Operating wavelength range	nm	1270 - 1620
Receiver sensitivity	dBm	-28
Minimum receiver overload	dBm	-9
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.0 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 26.7W
- Maximum power consumption at 55°C : 29.4 W

13.4.12 LBE Specifications

LBE board specifications include specifications of optical module on the client and WDM sides, laser safety level, mechanical specifications and power consumption.

Optical Specifications



NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-60 and **Table 13-61** list the details about the optical specifications for the C6LBE.

Table 13-60 Specifications of optical module for 10G-LAN service at client side

Item	Unit	Value	
		10G BASE-LR	10G BASE-ER
Line code format	-	NRZ	NRZ
Optical source type	-	SLM	SLM
Target distance	km	10	40
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1290 to 1330	1530 to 1565
Maximum mean launched power	dBm	-1	2
Minimum mean launched power	dBm	-6	-4.7
Minimum extinction ratio	dB	6	8.2
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	NA	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-14.4	-15.8
Minimum receiver overload	dBm	+0.5	-1
Maximum reflectance	dB	-27	-27

Table 13-61 Specifications of optical module at DWDM side

Item	Unit	Value		
Line code format	-	NRZ	NRZ-tunable	NRZ
Transmitter parameter specifications at point S				
Maximum mean launched power	dBm	-1	0	4
Minimum mean launched power	dBm	-5	-5	0
Minimum extinction ratio	dB	10	10	10

Item	Unit	Value		
Central frequency	THz	192.10 to 196.00		
Central frequency deviation	GHz	±10		
Maximum -20 dB spectral width	nm	0.3	0.3	0.3
Minimum side mode suppression ratio	dB	35	35	35
Dispersion tolerance	ps/nm	800	800	1600
Receiver parameter specifications at point R				
Receiver type	-	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-16	-16	-26
Minimum receiver overload	dBm	0	0	-9
Maximum reflectance	dB	-27	-27	-27

Table 13-62 and **Table 13-63** list the details about the optical specifications for the C8LBE and CALBE.

Table 13-62 Specifications of optical module for 10G-LAN service at client side

Item	Unit	Value			
Supported optical interface type	-	10G Base-SR	10G Base-LR	10G Base-ER	10G Base-ZR
Optical line code	-	NRZ	NRZ	NRZ	NRZ
Light source type	-	MLM	SLM	SLM	SLM
Intended transmission distance	km	0.3	10	40	80
Transmitter characteristics at point S					
Operating wavelength range	nm	840 to 860	1290 to 1330	1530 to 1565	1530 to 1565
Maximum mean launched optical power	dBm	-1.3	-1	2	4
Minimum mean launched optical power	dBm	-7.3	-6	-4.7	0

Item	Unit	Value			
Minimum extinction ratio	dB	3	6	8.2	9
Maximum -20 dB spectrum width	nm	NA	NA	NA	NA
Minimum side-mode suppression ratio	dB	30	30	30	30
Eye pattern	-	Compliant with G.691 template			
Receiver characteristics at point R					
Receiver type	-	PIN	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650			
Receiver sensitivity	dBm	-7.5	-14.4	-15.8	-24
Minimum overload point	dBm	-1	+0.5	-1	-7
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-63 Specifications of optical module at DWDM side

Item	Unit	Value	
Line code format	-	NRZ	NRZ-tunable
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	2	2
Minimum mean launched power	dBm	-3	-3
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	
Maximum -20 dB spectral width	nm	0.3	0.3
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	800	1100
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	

Item	Unit	Value	
Receiver sensitivity	dBm	-16	-16
Minimum receiver overload	dBm	0	0
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C6LBE/C8LBE: 1.1 kg
 - CALBE: 1.2 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C:
 - C6LBE/C8LBE: 44.3 W
 - CALBE: 43.7 W
- Maximum power consumption at 55°C:
 - C6LBE/C8LBE: 48.7 W
 - CALBE: 48.1 W

13.4.13 LBES Specifications

LBES board specifications include specifications of optical module on the client and WDM sides, laser safety level, mechanical specifications and power consumption.

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Optical Specifications

[Table 13-64](#) and [Table 13-65](#) list the details about the optical specifications for the C6LBES.

Table 13-64 Specifications of optical module for 10G-LAN service at client side

Item	Unit	Value	
		10G BASE-LR	10G BASE-ER
Line code format	-	NRZ	NRZ
Optical source type	-	SLM	SLM
Target distance	km	10	40
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1290 to 1330	1530 to 1565
Maximum mean launched power	dBm	-1	2
Minimum mean launched power	dBm	-6	-4.7
Minimum extinction ratio	dB	6	8.2
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	NA	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-14.4	-15.8
Minimum receiver overload	dBm	+0.5	-1
Maximum reflectance	dB	-27	-27

Table 13-65 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	-1
Minimum mean launched power	dBm	-4
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10

Item	Unit	Value
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Table 13-66 and **Table 13-67** list the details about the optical specifications for the C8LBES and CALBES.

Table 13-66 Specifications of optical module for 10G-LAN service at client side

Item	Unit	Value			
Supported optical interface type	-	10G Base-SR	10G Base-LR	10G Base-ER	10G Base-ZR
Optical line code	-	NRZ	NRZ	NRZ	NRZ
Light source type	-	MLM	SLM	SLM	SLM
Intended transmission distance	km	0.3	10	40	80
Transmitter characteristics at point S					
Operating wavelength range	nm	840 to 860	1290 to 1330	1530 to 1565	1530 to 1565
Maximum mean launched optical power	dBm	-1.3	-1	2	4
Minimum mean launched optical power	dBm	-7.3	-6	-4.7	0
Minimum extinction ratio	dB	3	6	8.2	9
Maximum -20 dB spectrum width	nm	NA	NA	NA	NA

Item	Unit	Value			
Minimum side-mode suppression ratio	dB	30	30	30	30
Eye pattern	-	Compliant with G.691 template			
Receiver characteristics at point R					
Receiver type	-	PIN	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650			
Receiver sensitivity	dBm	-7.5	-14.4	-15.8	-24
Minimum overload point	dBm	-1	+0.5	-1	-7
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-67 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C6LBES/C8LBES: 1.1 kg
 - CALBES: 1.5 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25 °C:
 - C6LBES/C8LBES: 48.0 W
 - CALBES: 52.1 W
- Maximum power consumption at 55 °C:
 - C6LBES/C8LBES: 53.0 W
 - CALBES: 57.3 W

13.4.14 LBF Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-68, **Table 13-69** and **Table 13-70** list the optical specifications of the LBF.

Table 13-68 Specifications of optical module for 10G-LAN service at client side

Item	Unit	Value			
Supported optical interface type	-	10G Base-SR	10G Base-LR	10G Base-ER	10G Base-ZR
Optical line code	-	NRZ	NRZ	NRZ	NRZ
Light source type	-	MLM	SLM	SLM	SLM
Intended transmission distance	km	0.3	10	40	80

Item	Unit	Value			
Transmitter characteristics at point S					
Operating wavelength range	nm	840 to 860	1290 to 1330	1530 to 1565	1530 to 1565
Maximum mean launched optical power	dBm	-1.3	-1	2	4
Minimum mean launched optical power	dBm	-7.3	-6	-4.7	0
Minimum extinction ratio	dB	3	6	8.2	9
Maximum -20 dB spectrum width	nm	NA	NA	NA	NA
Minimum side-mode suppression ratio	dB	30	30	30	30
Eye pattern	-	Compliant with G.691 template			
Receiver characteristics at point R					
Receiver type	-	PIN	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650			
Receiver sensitivity	dBm	-7.5	-14.4	-15.8	-24
Minimum overload point	dBm	-1	+0.5	-1	-7
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-69 Specifications of optical module for 10G-WAN/STM-64/OTU2 service at client side

Item	Unit	Specifications		
Supported optical interface type	-	I-64.1/ P1I1-2D1	S-64.2b/ P1S1-2D2b	L64.2/ P1L1-2D2
Optical line code	-	NRZ	NRZ	NRZ
Light source type	-	SLM	SLM	SLM
Intended transmission distance	km	2	40	80
Transmitter characteristics at point S				

Item	Unit	Specifications		
Operating wavelength range	nm	1290 to 1330	1530 to 1565	1530 to 1565
Maximum mean launched optical power	dBm	-1	+2	+4
Minimum mean launched optical power	dBm	-6	-1	0
Minimum extinction ratio	dB	6	8.2	9
Maximum -20 dB spectrum width	nm	1	0.3	0.3
Minimum side-mode suppression ratio	dB	30	30	30
Eye pattern	-	Compliant with G.691 template		
Receiver characteristics at point R				
Receiver type	-	PIN	PIN	PIN
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-11	-14	-24
Minimum overload point	dBm	-1	-1	-7
Maximum reflectance	dB	-27	-27	-27

Table 13-70 Specifications of optical module at DWDM side

Item	Unit	Value	
Line code format	-	NRZ	NRZ-tunable
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	2	2
Minimum mean launched power	dBm	-3	-3
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	
Maximum -20 dB spectral width	nm	0.3	0.3

Item	Unit	Value	
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	800	1100
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-16	-16
Minimum receiver overload	dBm	0	0
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C8LBF: 1.1 kg
 - C9LBF: 0.95 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C8LBF: 47.7 W
 - C9LBF: 22.0 W
- Maximum power consumption at 55°C :
 - C8LBF: 53.0 W
 - C9LBF: 26.0 W

13.4.15 LBFS Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-71, Table 13-72, Table 13-73 and Table 13-74 list the optical specifications of the LBFS.

Table 13-71 Specifications of optical module for 10G-LAN service at client side

Item	Unit	Value			
Supported optical interface type	-	10G Base-SR	10G Base-LR	10G Base-ER	10G Base-ZR
Optical line code	-	NRZ	NRZ	NRZ	NRZ
Light source type	-	MLM	SLM	SLM	SLM
Intended transmission distance	km	0.3	10	40	80
Transmitter characteristics at point S					
Operating wavelength range	nm	840 to 860	1290 to 1330	1530 to 1565	1530 to 1565
Maximum mean launched optical power	dBm	-1.3	-1	2	4
Minimum mean launched optical power	dBm	-7.3	-6	-4.7	0
Minimum extinction ratio	dB	3	6	8.2	9
Maximum -20 dB spectrum width	nm	NA	NA	NA	NA
Minimum side-mode suppression ratio	dB	30	30	30	30
Eye pattern	-	Compliant with G.691 template			
Receiver characteristics at point R					
Receiver type	-	PIN	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650			
Receiver sensitivity	dBm	-7.5	-14.4	-15.8	-24

Item	Unit	Value			
Minimum overload point	dBm	-1	+0.5	-1	-7
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-72 Specifications of optical module for 10G-WAN/STM-64/OTU2 service at client side

Item	Unit	Specifications		
Supported optical interface type	-	I-64.1/ P1I1-2D1	S-64.2b/ P1S1-2D2b	L64.2/ P1L1-2D2
Optical line code	-	NRZ	NRZ	NRZ
Light source type	-	SLM	SLM	SLM
Intended transmission distance	km	2	40	80
Transmitter characteristics at point S				
Operating wavelength range	nm	1290 to 1330	1530 to 1565	1530 to 1565
Maximum mean launched optical power	dBm	-1	+2	+4
Minimum mean launched optical power	dBm	-6	-1	0
Minimum extinction ratio	dB	6	8.2	9
Maximum -20 dB spectrum width	nm	1	0.3	0.3
Minimum side-mode suppression ratio	dB	30	30	30
Eye pattern	-	Compliant with G.691 template		
Receiver characteristics at point R				
Receiver type	-	PIN	PIN	PIN
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-11	-14	-24
Minimum overload point	dBm	-1	-1	-7

Item	Unit	Specifications		
Maximum reflectance	dB	-27	-27	-27

Table 13-73 Specifications of optical module for FC 10G service at client side

Parameters	Unit	Specifications	
Optical Interface type	-	-	-
Line code format	-	NRZ	NRZ
Optical source type	-	SLM	SLM
Target distance	-	2 km (1.2 mi.)	0.3 km (0.18 mi.)
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1290-1330	840-860
Maximum mean launched power	dBm	-1	-1.3
Minimum mean launched power	dBm	-6	-7.3
Minimum extinction ratio	dB	6	3
Maximum -20 dB spectral width	nm	1	-
Minimum SMSR	dB	30	30
Eye pattern mask	-	Compliant with the parameter template of Fiber Channel physical interface (FC-PI-2)	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Receiver sensitivity	dBm	-11	-7.5
Receiver overload	dBm	-1	-1
Maximum reflectance	dB	-27	-12

Table 13-74 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		

Item	Unit	Value
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C8LBFS: 1.1 kg
 - C9LBFS: 0.95 kg
 - CBLBFS: 1.0 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C8LBFS: 53.6 W
 - C9LBFS: 26.0 W
 - CBLBFS: 29.3 W

- Maximum power consumption at 55°C :
 - C8LBFS: 58.9 W
 - C9LBFS: 28.6 W
 - CBLBFS: 32.2 W

13.4.16 LDG Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications



NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-75, **Table 13-76**, **Table 13-77** and **Table 13-78** list the optical specifications of the LDG.

Table 13-75 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-76 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2mW-APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					
Maximum mean launched power ^a	dBm	-1	-1	3	3
Minimum mean launched power ^a	dBm	-5	-5	-2	-2
Typical value of mean launched power ^a	dBm	-2	-2	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.					

Table 13-77 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		
Maximum mean launched power ^a	dBm	3
Minimum mean launched power ^a	dBm	-1
Typical value of mean launched power ^a	dBm	1
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	12800
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD

Item	Unit	Value
		12800 ps/nm-tunable
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.		

Table 13-78 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ± 6.5	≤ ± 6.5
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650

Item	Unit	Value	
		1600 ps/nm-4mW	
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C6LDG: 1.0kg
 - C8LDG: 1.1kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6LDG: 29.5W
 - C8LDG: 28.0W
- Maximum power consumption at 55°C :
 - C6LDG: 33.0W
 - C6LDG: 30.8W

13.4.17 LOG Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-79, **Table 13-80** and **Table 13-81** list the optical specifications on the client and WDM side of the C6LOG.

Table 13-79 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-80 Specifications of optical module for FC service at client side

Item	Unit	Value	
		FC 100/FC 200	
Line code format	-	NRZ	NRZ
Target distance	km	2	0.5
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	770 to 860
Maximum mean launched power	dBm	-3	-2.5
Minimum mean launched power	dBm	-10	-9.5
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1580	770 to 860
Receiver sensitivity	dBm	-18	-17
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	NA	NA

Table 13-81 Specifications of optical module at DWDM side

Item	Unit	Value		
Line code format	-	NRZ	NRZ-tunable	NRZ
Transmitter parameter specifications at point S				
Maximum mean launched power	dBm	-1	0	4
Minimum mean launched power	dBm	-5	-5	0
Minimum extinction ratio	dB	10	10	10
Central frequency	THz	192.10 to 196.00		

Item	Unit	Value		
Central frequency deviation	GHz	±10		
Maximum -20 dB spectral width	nm	0.3	0.3	0.3
Minimum side mode suppression ratio	dB	35	35	35
Dispersion tolerance	ps/nm	800	800	1600
Receiver parameter specifications at point R				
Receiver type	-	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-16	-16	-26
Minimum receiver overload	dBm	0	0	-9
Maximum reflectance	dB	-27	-27	-27

[Table 13-81](#), [Table 13-82](#) and [Table 13-83](#) list the optical specifications on the client and WDM side of the C9LOG.

Table 13-82 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-83 Specifications of optical module for FC service at client side

Item	Unit	Value	
		FC 100/FC 200	
Line code format	-	NRZ	NRZ
Target distance	km	2	0.5
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	770 to 860
Maximum mean launched power	dBm	-3	-2.5
Minimum mean launched power	dBm	-10	-9.5
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			

Item	Unit	Value	
		FC 100/FC 200	
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1580	770 to 860
Receiver sensitivity	dBm	-18	-17
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	NA	NA

Table 13-84 Specifications of optical module at DWDM side

Item	Unit	Value	
Line code format	-	NRZ	NRZ-tunable
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	2	2
Minimum mean launched power	dBm	-3	-3
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	
Maximum -20 dB spectral width	nm	0.3	0.3
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	800	1100
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-16	-16
Minimum receiver overload	dBm	0	0
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: 1.5 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6LOG: 55.0 W
 - C9LOG: 44.0 W
- Maximum power consumption at 55°C :
 - C6LOG: 60.0 W
 - C9LOG: 53.6 W

13.4.18 LOGS Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-85](#), [Table 13-86](#) and [Table 13-87](#) list the optical specifications on the client and WDM side of the C6LOGS.

Table 13-85 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-86 Specifications of optical module for FC service at client side

Item	Unit	Value	
		FC 100/FC 200	
Line code format	-	NRZ	NRZ
Target distance	km	2	0.5
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	770 to 860
Maximum mean launched power	dBm	-3	-2.5
Minimum mean launched power	dBm	-10	-9.5

Item	Unit	Value	
		FC 100/FC 200	
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1580	770 to 860
Receiver sensitivity	dBm	-18	-17
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	NA	NA

Table 13-87 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	-1
Minimum mean launched power	dBm	-4
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0

Item	Unit	Value
Maximum reflectance	dB	-27

Table 13-88, Table 13-89 and **Table 13-90** list the optical specifications on the client and WDM side of the C9LOGS.

Table 13-88 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-89 Specifications of optical module for FC service at client side

Item	Unit	Value	
		FC 100/FC 200	
Line code format	-	NRZ	NRZ
Target distance	km	2	0.5
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	770 to 860
Maximum mean launched power	dBm	-3	-2.5
Minimum mean launched power	dBm	-10	-9.5
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	-
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1580	770 to 860
Receiver sensitivity	dBm	-18	-17
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	NA	NA

Table 13-90 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00

Item	Unit	Value
Central frequency deviation	GHz	± 10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: 1.5 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6LOGS: 58.0 W
 - C9LOGS: 47.3 W
- Maximum power consumption at 55°C : 64.0 W
 - C6LOGS: 64.0 W
 - C9LOGS: 56.5 W

13.4.19 LOM Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-91](#), [Table 13-92](#), [Table 13-93](#) and [Table 13-94](#) list the optical specifications on the client and WDM side of the C8LOM.

Table 13-91 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-92 Specifications of optical module for FC and FICON service at client side

Item	Unit	Value			
		FC 400 module		FC 100/ FICON module /FC 200/ FICON Express module	
		Multi-mode	Single-mode	Multi-mode	Single-mode
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.3	10	0.5	2
Transmitter parameter specifications at point S					
Operating wavelength range	nm	830 to 860	1270 to 1355	830 to 860	1266 to 1360
Maximum mean launched power	dBm	-1	-2	-2.5	-3
Minimum mean launched power	dBm	-9	-8	-9.5	-10
Eye pattern mask	-	Compliant with Fiber Channel-Physical interface (FC-PI-2)			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1260 to 1600	770 to 860	1270 to 1580
Receiver sensitivity	dBm	-14	-16	-17	-18
Minimum receiver overload	dBm	0	0	0	-3
Maximum reflectance	dB	-12	-12	NA	NA

Table 13-93 Specifications of optical module at DWDM side

Item	Unit	Value	
Line code format	-	NRZ	NRZ-tunable
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	2	2

Item	Unit	Value	
Minimum mean launched power	dBm	-3	-3
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	
Maximum -20 dB spectral width	nm	0.3	0.3
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	800	1100
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-16	-16
Minimum receiver overload	dBm	0	0
Maximum reflectance	dB	-27	-27

Table 13-94 Specifications of optical module at DWDM side

Parameters	Unit	Specifications
Optical line code	-	Tunable ODB
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Nominal central frequency	THz	192.10 to 196.05
Central frequency deviation	GHz	±2.5
Maximum -20 dB spectral width	nm	0.6
Minimum SMSR	dB	30
Dispersion tolerance	ps/nm	400
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1529 to 1561

Parameters	Unit	Specifications
Receiver sensitivity (With FEC open) EOL	dBm	-28
Receiver overload (With FEC open)	dBm	-9
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: 1.8 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 71.6 W
- Maximum power consumption at 55°C : 78.8 W

13.4.20 LOMS Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-95](#), [Table 13-96](#) and [Table 13-97](#) list the optical specifications on the client and WDM side of the C8LOMS.

Table 13-95 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-96 Specifications of optical module for FC and FICON service at client side

Item	Unit	Value			
		FC 400 module		FC 100/ FICON module /FC 200/ FICON Express module	
		Multi-mode	Single-mode	Multi-mode	Single-mode
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.3	10	0.5	2
Transmitter parameter specifications at point S					
Operating wavelength range	nm	830 to 860	1270 to 1355	830 to 860	1266 to 1360
Maximum mean launched power	dBm	-1	-2	-2.5	-3
Minimum mean launched power	dBm	-9	-8	-9.5	-10
Eye pattern mask	-	Compliant with Fiber Channel-Physical interface (FC-PI-2)			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1260 to 1600	770 to 860	1270 to 1580
Receiver sensitivity	dBm	-14	-16	-17	-18
Minimum receiver overload	dBm	0	0	0	-3
Maximum reflectance	dB	-12	-12	NA	NA

Table 13-97 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Minimum extinction ratio	dB	13

Item	Unit	Value
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height×Width×Depth): 345.0 mm×64.0 mm×218.5 mm
- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: 1.8 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 71.6 W
- Maximum power consumption at 55°C : 78.8 W

13.4.21 LQG Specifications

LQG board specifications include specifications of optical module on the client and WDM sides, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-98, **Table 13-99** and **Table 13-100** list the optical specifications on the client and WDM side of the C6LQG and C9LQG.

Table 13-98 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-99 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value	
		3400 ps/nm	6400 ps/nm
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	2	-1
Minimum mean launched power	dBm	-2	-5
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	
Maximum -20 dB spectral width	nm	0.3	0.3
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	3400	6400
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-25	-25
Minimum receiver overload	dBm	-9	-9
Maximum reflectance	dB	-27	-27

Table 13-100 Specifications of the tunable wavelength optical module at DWDM side

Item	Unit	Value
		3400 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00

Item	Unit	Value
		3400 ps/nm-tunable
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.3
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	3400
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-25
Minimum receiver overload	dBm	-9
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.0 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6LQG: 48.0 W
 - C9LQG: 30.0 W
- Maximum power consumption at 55°C :
 - C6LQG: 54.0 W
 - C9LQG: 35.0 W

13.4.22 LQM Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-101 to **Table 13-108** list the optical specifications on the client and WDM side of the LQM.

Table 13-101 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-102 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-103 Specifications of optical module for STM-1/OC-3/STM-4/OC-12 service at client side

Item	Unit	Value	
		S-4.1	L-4.1
Line code format	-	NRZ	NRZ
Optical source type	-	MLM	SLM
Target distance	km	15	40
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1274 to 1356	1280 to 1335
Maximum mean launched power	dBm	-8	2
Minimum mean launched power	dBm	-15	-3
Minimum extinction ratio	dB	8.2	10
Maximum -20 dB spectral width	nm	NA	1
Minimum side mode suppression ratio	dB	NA	30
Eye pattern mask	-	G.957-compliant	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-28	-28
Minimum receiver overload	dBm	-8	-8
Maximum reflectance	dB	-27	-14

Table 13-104 Specifications of optical module for FC service at client side

Item	Unit	Value	
		FC 100/FC 200	
Line code format	-	NRZ	NRZ
Target distance	km	2	0.5
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	770 to 860
Maximum mean launched power	dBm	-3	-2.5
Minimum mean launched power	dBm	-10	-9.5
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1580	770 to 860
Receiver sensitivity	dBm	-18	-17
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	NA	NA

Table 13-105 Specifications of optical module for ESCON and other services at client side

Item	Unit	Value	
Target distance	km	2	15
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	
Maximum mean launched power	dBm	-14	-8
Minimum mean launched power	dBm	-19	-15
Minimum extinction ratio	dB	8.2	8.2

Item	Unit	Value	
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-27	-28
Minimum receiver overload	dBm	-14	-8
Maximum reflectance	dB	NA	NA
Note	-	The ESCON, DVB-ASI and FE services can be accessed to this optical module.	

Table 13-106 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2mW-APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					
Maximum mean launched power ^a	dBm	-1	-1	3	3
Minimum mean launched power ^a	dBm	-5	-5	-2	-2
Typical value of mean launched power ^a	dBm	-2	-2	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Minimum side mode suppression ratio	dB	35	35	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.					

Table 13-107 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		
Maximum mean launched power ^a	dBm	3
Minimum mean launched power ^a	dBm	-1
Typical value of mean launched power ^a	dBm	1
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	12800

Item	Unit	Value
		12800 ps/nm-tunable
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.		

Table 13-108 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ±6.5	≤ ±6.5
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			

Item	Unit	Value	
		1600 ps/nm-4mW	
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.1kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 64.0 W
- Maximum power consumption at 55°C : 69.0 W

13.4.23 LQM2 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-109 to **Table 13-116** list the optical specifications on the client and WDM side of the C9LQM2.

Table 13-109 Specifications of optical module for GE service at client side

Item	Unit	Value			
		1000BASE-SX	1000BASE-LX-10km	1000BASE-LX-40km	1000BASE-ZX-80km
Line code format	-	NRZ	NRZ	NRZ	NRZ
Target distance	km	0.5	10	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Maximum mean launched power	dBm	-2.5	-3	3	5
Minimum mean launched power	dBm	-9.5	-11.5	-4.5	-2
Minimum extinction ratio	dB	9	9	9	9
Eye pattern mask	-	IEEE802.3z-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	PIN	PIN
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355	1500 to 1580
Receiver sensitivity	dBm	-17	-19	-21	-23
Minimum receiver overload	dBm	0	-3	-3	-3

Table 13-110 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-111 Specifications of optical module for STM-1/OC-3/STM-4/OC-12 service at client side

Item	Unit	Value	
		S-4.1	L-4.1
Line code format	-	NRZ	NRZ
Optical source type	-	MLM	SLM
Target distance	km	15	40
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1274 to 1356	1280 to 1335
Maximum mean launched power	dBm	-8	2
Minimum mean launched power	dBm	-15	-3
Minimum extinction ratio	dB	8.2	10
Maximum -20 dB spectral width	nm	NA	1
Minimum side mode suppression ratio	dB	NA	30
Eye pattern mask	-	G.957-compliant	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-28	-28
Minimum receiver overload	dBm	-8	-8
Maximum reflectance	dB	-27	-14

Table 13-112 Specifications of optical module for FC service at client side

Item	Unit	Value	
		FC 100/FC 200	
Line code format	-	NRZ	NRZ
Target distance	km	2	0.5
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	770 to 860
Maximum mean launched power	dBm	-3	-2.5
Minimum mean launched power	dBm	-10	-9.5
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1270 to 1580	770 to 860
Receiver sensitivity	dBm	-18	-17
Minimum receiver overload	dBm	-3	0
Maximum reflectance	dB	NA	NA

Table 13-113 Specifications of optical module for ESCON and other services at client side

Item	Unit	Value	
Target distance	km	2	15
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1266 to 1360	
Maximum mean launched power	dBm	-14	-8
Minimum mean launched power	dBm	-19	-15
Minimum extinction ratio	dB	8.2	8.2

Item	Unit	Value	
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	-	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-27	-28
Minimum receiver overload	dBm	-14	-8
Maximum reflectance	dB	NA	NA
Note	-	The ESCON, DVB-ASI and FE services can be accessed to this optical module.	

Table 13-114 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2mW-APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					
Maximum mean launched power ^a	dBm	-1	-1	3	3
Minimum mean launched power ^a	dBm	-5	-5	-2	-2
Typical value of mean launched power ^a	dBm	-2	-2	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Minimum side mode suppression ratio	dB	35	35	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.					

Table 13-115 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		
Maximum mean launched power ^a	dBm	3
Minimum mean launched power ^a	dBm	-1
Typical value of mean launched power ^a	dBm	1
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	12800

Item	Unit	Value
		12800 ps/nm-tunable
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.		

Table 13-116 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ±6.5	≤ ±6.5
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			

Item	Unit	Value	
		1600 ps/nm-4mW	
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.5 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 64.0 W
- Maximum power consumption at 55°C : 69.0 W

13.4.24 LQS Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-117](#), [Table 13-118](#), [Table 13-119](#) and [Table 13-120](#) list the optical specifications on the client and WDM side of the C6LQS and C7LQS.

Table 13-117 Specifications of optical module for STM-1/OC-3/STM-4/OC-12 service at client side

Item	Unit	Value	
		S-4.1	L-4.1
Line code format	-	NRZ	NRZ
Optical source type	-	MLM	SLM
Target distance	km	15	40
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1274 to 1356	1280 to 1335
Maximum mean launched power	dBm	-8	2
Minimum mean launched power	dBm	-15	-3
Minimum extinction ratio	dB	8.2	10
Maximum -20 dB spectral width	nm	NA	1
Minimum side mode suppression ratio	dB	NA	30
Eye pattern mask	-	G.957-compliant	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-28	-28
Minimum receiver overload	dBm	-8	-8
Maximum reflectance	dB	-27	-14

Table 13-118 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					
Maximum mean launched power ^a	dBm	-1	-1	3	3

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Minimum mean launched power ^a	dBm	-5	-5	-2	-2
Typical value of mean launched power ^a	dBm	-2	-2	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.					

Table 13-119 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		

Item	Unit	Value
		12800 ps/nm-tunable
Maximum mean launched power ^a	dBm	3
Minimum mean launched power ^a	dBm	-1
Typical value of mean launched power ^a	dBm	1
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	12800
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.		

Table 13-120 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-2	-2

Item	Unit	Value	
		1600 ps/nm-4mW	
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ± 6.5	≤ ± 6.5
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.2 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 30.0 W
- Maximum power consumption at 55°C : 33.0 W

13.4.25 LRF Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-121 list the optical specifications on WDM side of the L2LRF.

Table 13-121 Specifications of optical module at DWDM side

Item	Unit	Value		
Line code format	-	NRZ	NRZ-tunable	NRZ
Transmitter parameter specifications at point S				
Maximum mean launched power	dBm	-1	0	4
Minimum mean launched power	dBm	-5	-5	0
Minimum extinction ratio	dB	10	10	10
Central frequency	THz	192.10 to 196.00		
Central frequency deviation	GHz	±10		
Maximum -20 dB spectral width	nm	0.3	0.3	0.3
Minimum side mode suppression ratio	dB	35	35	35
Dispersion tolerance	ps/nm	800	800	1600
Receiver parameter specifications at point R				
Receiver type	-	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-16	-16	-26
Minimum receiver overload	dBm	0	0	-9
Maximum reflectance	dB	-27	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.25 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 24.0 W
- Maximum power consumption at 55°C : 26.4 W

13.4.26 LRFS Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-122 list the optical specifications on WDM side of the C6LRFS.

Table 13-122 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	-1
Minimum mean launched power	dBm	-4
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000

Item	Unit	Value
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.25 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 37.0 W
- Maximum power consumption at 55°C : 40.7 W

13.4.27 LU40S Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-123 and **Table 13-124** list the optical specifications on the client and WDM side of the C9LU40S.

Table 13-123 Specifications of optical module for STM-256/OC-768 service at client side

Parameters	Unit	Specifications
Supported optical interface type	-	VSR2000-3R2

Parameters	Unit	Specifications
Optical line code	-	NRZ
Target distance	km	2
Transmitter parameter specifications at point S		
Operating wavelength range	nm	1530 to 1565
Maximum mean launched power	dBm	3
Minimum mean launched power	dBm	0
Minimum extinction ratio	dB	8.2
Maximum -20 dB spectrum width	nm	1
Minimum side-mode suppression ratio (SMSR)	dB	35
Eye pattern mask	NA	Compliant with G.693
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1290 to 1570
Receiver sensitivity	dBm	-6
Receiver overload	dBm	3
Maximum reflectance	dB	-27

Table 13-124 Specifications of optical module at DWDM side

Parameter	Unit	Specification
Line code format	-	DQPSK
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	0
Minimum power	dBm	-5
Minimum extinction ratio	dB	NA
Nominal central frequency	THz	192.10 - 196.00
Central frequency deviation	GHz	±2.5
Maximum -20 dB spectral width	nm	NA
Minimum SMSR	dB	35
Maximum dispersion	ps/nm	-500 to +500

Parameter	Unit	Specification
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1529 - 1561
Receiver sensitivity	dBm	-16
Receiver overload	dBm	0
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: 4.55 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25 °C: 84.0 W
- Maximum power consumption at 55 °C: 92.4 W

13.4.28 LUR40S Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-125 list the optical specifications on the WDM side of the C9LUR40S.

Table 13-125 Specifications of optical module at DWDM side

Parameter	Unit	Specification	
Line code format	-	DQPSK	nDQPSK

Parameter	Unit	Specification	
Channel spacing	GHz	50	50
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	0	0
Minimum power	dBm	-5	-5
Minimum extinction ratio	dB	NA	NA
Nominal central frequency	THz	192.10 - 196.05	192.10 - 196.05
Central frequency deviation	GHz	±2.5	±2.5
Maximum -20 dB spectral width	nm	NA	NA
Maximum -3 dB spectral width	nm	NA	NA
Minimum SMSR	dB	35	35
Maximum dispersion (back to back)	ps/nm	-500 to +500	-800 to +800
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1529 - 1561	1529 - 1561
Receiver sensitivity	dBm	-16	-16
Receiver overload	dBm	0	0
Maximum reflectance	dB	-27	-27
a: For a system with a 100 GHz channel spacing, do not use optical modules of the ODB code pattern. Instead, use optical modules of the DQPSK or nDQPSK code pattern.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: 3.75 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25 °C: 74.0 W
- Maximum power consumption at 55 °C: 81.4 W

13.4.29 LWC1 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

 NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-126](#), [Table 13-127](#), [Table 13-128](#) and [Table 13-129](#) list the optical specifications on the client and WDM side of the C6LWC1.

Table 13-126 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-127 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2mW-APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					
Maximum mean launched power ^a	dBm	-1	-1	3	3
Minimum mean launched power ^a	dBm	-5	-5	-2	-2
Typical value of mean launched power ^a	dBm	-2	-2	0	0

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.					

Table 13-128 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		
Maximum mean launched power ^a	dBm	3
Minimum mean launched power ^a	dBm	-1
Typical value of mean launched power ^a	dBm	1
Minimum extinction ratio	dB	10

Item	Unit	Value
		12800 ps/nm-tunable
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	12800
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.		

Table 13-129 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ±6.5	≤ ±6.5

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

[Table 13-130](#), [Table 13-131](#), [Table 13-132](#), [Table 13-133](#) and [Table 13-134](#) list the optical specifications on the client and WDM side of the C8LWC1.

Table 13-130 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-131 Specifications of optical module for the OTU1 service at client side

Item	Unit	Value			
		P1I1-1D1	P1S1-1D1	P1L1-1D1	P1L1-1D2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM

Item	Unit	Value			
		P1I1-1D1	P1S1-1D1 1	P1L1-1D1	P1L1-1D2
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.959.1-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-132 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					
Maximum mean launched power ^a	dBm	-1	-1	3	3
Minimum mean launched power ^a	dBm	-5	-5	-2	-2
Typical value of mean launched power ^a	dBm	-2	-2	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.					

Table 13-133 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		
Maximum mean launched power ^a	dBm	3
Minimum mean launched power ^a	dBm	-1
Typical value of mean launched power ^a	dBm	1
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	12800
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.		

Table 13-134 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ± 6.5	≤ ± 6.5
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.1 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C:
 - C6LWC1: 21.5 W
 - C8LWC1: 21.5 W
 - C9LWC1: 13.5 W
- Maximum power consumption at 55°C:
 - C6LWC1: 23.6 W
 - C8LWC1: 23.6 W
 - C9LWC1: 14.6 W

13.4.30 LWF Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-135 and **Table 13-136** list the optical specifications of the L2LWF and C7LWF.

Table 13-135 Specifications of optical module for STM-64 service at client side

Item	Unit	Value				
		I-64.1	I-64.2	S-64.2b	Se-64.2a	Le-64.2
Line code format	-	NRZ	NRZ	NRZ	NRZ	NRZ
Optical source type	-	SLM	SLM	SLM	SLM	SLM
Target distance	km	2	25	40	40	60
Transmitter parameter specifications at point S						
Operating wavelength range	nm	1290 to 1330	1530 to 1565	1530 to 1565	1530 to 1565	1530 to 1565
Maximum mean launched power	dBm	-1	-1	2	2	4
Minimum mean launched power	dBm	-6	-5	-1	-1	2
Minimum extinction ratio	dB	6	8.2	8.2	8.2	8.2

Item	Unit	Value				
		I-64.1	I-64.2	S-64.2b	Se-64.2a	Le-64.2
Maximum -20 dB spectral width	nm	1	0.3	0.3	0.3	0.3
Minimum side mode suppression ratio	dB	30	30	30	30	30
Eye pattern mask	-	G.691-compliant				
Receiver parameter specifications at point R						
Receiver type	-	PIN	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-11	-14	-14	-18	-18
Minimum receiver overload	dBm	-1	-1	-1	-8	-8
Maximum reflectance	dB	-27	-27	-27	-27	-27
Jitter characteristics	-	Compliant with G.783				

Table 13-136 Specifications of optical module at DWDM side

Item	Unit	Value		
Line code format	-	NRZ	NRZ-tunable	NRZ
Transmitter parameter specifications at point S				
Maximum mean launched power	dBm	-1	0	4
Minimum mean launched power	dBm	-5	-5	0
Minimum extinction ratio	dB	10	10	10
Central frequency	THz	192.10 to 196.00		
Central frequency deviation	GHz	±10		
Maximum -20 dB spectral width	nm	0.3	0.3	0.3
Minimum side mode suppression ratio	dB	35	35	35
Dispersion tolerance	ps/nm	800	800	1600
Receiver parameter specifications at point R				

Item	Unit	Value		
Receiver type	-	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-16	-16	-26
Minimum receiver overload	dBm	0	0	-9
Maximum reflectance	dB	-27	-27	-27

Table 13-137 and **Table 13-138** list the optical specifications of the C8LWF, C9LWF and CALWF.

Table 13-137 Specifications of optical module for STM-64 service at client side

Item	Unit	Specifications		
		I-64.1	S-64.2b	L64.2
Optical line code	-	NRZ	NRZ	NRZ
Light source type	-	SLM	SLM	SLM
Intended transmission distance	km	2	40	80
Transmitter characteristics at point S				
Operating wavelength range	nm	1290 to 1330	1530 to 1565	1530 to 1565
Maximum mean launched optical power	dBm	-1	+2	+4
Minimum mean launched optical power	dBm	-6	-1	0
Minimum extinction ratio	dB	6	8.2	9
Maximum -20 dB spectrum width	nm	1	0.3	0.3
Minimum side-mode suppression ratio	dB	30	30	30
Eye pattern	-	Compliant with G.691 template		
Receiver characteristics at point R				
Receiver type	-	PIN	PIN	PIN
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-11	-14	-24
Minimum overload point	dBm	-1	-1	-7

Item	Unit	Specifications		
		I-64.1	S-64.2b	L64.2
Maximum reflectance	dB	-27	-27	-27

Table 13-138 Specifications of optical module at DWDM side

Item	Unit	Value	
Line code format	-	NRZ	NRZ-tunable
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	2	2
Minimum mean launched power	dBm	-3	-3
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	
Maximum -20 dB spectral width	nm	0.3	0.3
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	800	1100
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-16	-16
Minimum receiver overload	dBm	0	0
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:

- L2LWF/C7LWF/C8LWF: 1.55 kg
- C9LWF/CALWF: 0.95 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C:
 - L2LWF/C7LWF/C8LWF: 38.0 W
 - C9LWF: 28.4 W
 - CALWF: 30.0 W
- Maximum power consumption at 55°C :
 - L2LWF/C7LWF/C8LWF: 42.0 W
 - C9LWF: 30.2 W
 - CALWF: 34.0 W

13.4.31 LWFS Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-139 and **Table 13-140** list the optical specifications of the C6LWFS and C7LWFS.

Table 13-139 Specifications of optical module for STM-64 service at client side

Item	Unit	Value				
		I-64.1	I-64.2	S-64.2b	Se-64.2a	Le-64.2
Line code format	-	NRZ	NRZ	NRZ	NRZ	NRZ
Optical source type	-	SLM	SLM	SLM	SLM	SLM
Target distance	km	2	25	40	40	60
Transmitter parameter specifications at point S						
Operating wavelength range	nm	1290 to 1330	1530 to 1565	1530 to 1565	1530 to 1565	1530 to 1565
Maximum mean launched power	dBm	-1	-1	2	2	4
Minimum mean launched power	dBm	-6	-5	-1	-1	2

Item	Unit	Value				
		I-64.1	I-64.2	S-64.2b	Se-64.2a	Le-64.2
Minimum extinction ratio	dB	6	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	1	0.3	0.3	0.3	0.3
Minimum side mode suppression ratio	dB	30	30	30	30	30
Eye pattern mask	-	G.691-compliant				
Receiver parameter specifications at point R						
Receiver type	-	PIN	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-11	-14	-14	-18	-18
Minimum receiver overload	dBm	-1	-1	-1	-8	-8
Maximum reflectance	dB	-27	-27	-27	-27	-27
Jitter characteristics	-	Compliant with G.783				

Table 13-140 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	-1
Minimum mean launched power	dBm	-4
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN

Item	Unit	Value
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Table 13-141 and **Table 13-142** list the optical specifications of the C8LWFS.

Table 13-141 Specifications of optical module for STM-64 service at client side

Item	Unit	Specifications		
		I-64.1	S-64.2b	L64.2
Optical line code	-	NRZ	NRZ	NRZ
Light source type	-	SLM	SLM	SLM
Intended transmission distance	km	2	40	80
Transmitter characteristics at point S				
Operating wavelength range	nm	1290 to 1330	1530 to 1565	1530 to 1565
Maximum mean launched optical power	dBm	-1	+2	+4
Minimum mean launched optical power	dBm	-6	-1	0
Minimum extinction ratio	dB	6	8.2	9
Maximum -20 dB spectrum width	nm	1	0.3	0.3
Minimum side-mode suppression ratio	dB	30	30	30
Eye pattern	-	Compliant with G.691 template		
Receiver characteristics at point R				
Receiver type	-	PIN	PIN	PIN
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-11	-14	-24
Minimum overload point	dBm	-1	-1	-7
Maximum reflectance	dB	-27	-27	-27

Table 13-142 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Table 13-143 and **Table 13-144** list the optical specifications of the C9LWFS and CALWFS.

Table 13-143 Specifications of optical module for STM-64 service at client side

Item	Unit	Specifications		
		I-64.1	S-64.2b	L64.2
Optical line code	-	NRZ	NRZ	NRZ
Light source type	-	SLM	SLM	SLM
Intended transmission distance	km	2	40	80
Transmitter characteristics at point S				
Operating wavelength range	nm	1290 to 1330	1530 to 1565	1530 to 1565
Maximum mean launched optical power	dBm	-1	+2	+4

Item	Unit	Specifications		
		I-64.1	S-64.2b	L64.2
Minimum mean launched optical power	dBm	-6	-1	0
Minimum extinction ratio	dB	6	8.2	9
Maximum -20 dB spectrum width	nm	1	0.3	0.3
Minimum side-mode suppression ratio	dB	30	30	30
Eye pattern	-	Compliant with G.691 template		
Receiver characteristics at point R				
Receiver type	-	PIN	PIN	PIN
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-11	-14	-24
Minimum overload point	dBm	-1	-1	-7
Maximum reflectance	dB	-27	-27	-27

Table 13-144 Specifications of optical module at DWDM side

Item	Unit	Value	
Line code format	-	RZ	DRZ-tunable
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	0	0
Minimum mean launched power	dBm	-5	-5
Minimum extinction ratio	dB	12	+13
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	±3
Maximum -20 dB spectral width	nm	0.3	0.3
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	500	1000
Eye pattern mask	-	NA	NA
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN

Item	Unit	Value	
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-16	-16
Minimum receiver overload	dBm	0	0
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C6LWFS/C7LWFS/C8LWFS: 1.55 kg
 - C9LWFS: 1.4 kg
 - CALWFS: 0.95 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6LWFS/C7LWFS/C8LWFS: 51.0 W
 - C9LWFS: 40.0 W
 - CALWFS: 30.0 W
- Maximum power consumption at 55°C :
 - C6LWFS/C7LWFS/C8LWFS: 56.0 W
 - C9LWFS: 44.0 W
 - CALWFS: 34.0 W

13.4.32 LWM Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-145, **Table 13-146**, **Table 13-147**, **Table 13-148** and **Table 13-149** list the details about the optical specifications for the C6LWM and C8LWM.

Table 13-145 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value		
		I-16	S-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM
Target distance	km	2	15	80
Transmitter parameter specifications at point S				
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1500 to 1580
Maximum mean launched power	dBm	-3	0	3
Minimum mean launched power	dBm	-10	-5	-2
Minimum extinction ratio	dB	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1
Minimum side mode suppression ratio	dB	NA	30	30
Eye pattern mask	-	G.957-compliant		
Receiver parameter specifications at point R				
Receiver type	-	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-28
Minimum receiver overload	dBm	-3	0	-9
Maximum reflectance	dB	-27	-27	-27
Jitter characteristics	-	Compliant with G.783		

Table 13-146 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value				
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2mW-APD	3200 ps/nm-2mW-APD
Line code format	-	NRZ	NRZ	NRZ	NRZ	NRZ

Item	Unit	Value				
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2mW-APD	3200 ps/nm-2mW-APD
Transmitter parameter specifications at point S						
Maximum mean launched power ^a	dBm	-1	-1	3	3	3
Minimum mean launched power ^a	dBm	-5	-5	-2	-2	-2
Typical value of mean launched power ^a	dBm	-2	-2	0	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2	8.2
Central frequency	THz	192.10 to 196.00				
Central frequency deviation	GHz	±10				
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200	3200
Eye pattern mask	-	G.957-compliant				
Receiver parameter specifications at point R						
Receiver type	-	PIN	APD	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26	-26
Minimum receiver overload	dBm	0	-9	0	-9	-10
Maximum reflectance	dB	-27	-27	-27	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.						

Table 13-147 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value	
		12800 ps/nm-tunable	
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-1	-1
Typical value of mean launched power ^a	dBm	1	1
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	±10
Maximum -20 dB spectral width	nm	0.2	0.2
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	12800	12800
Eye pattern mask	-	G.957-compliant	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-26
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

Table 13-148 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8

Item	Unit	Value	
		1600 ps/nm-4mW	
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ± 6.5	≤ ± 6.5
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.			

Table 13-149 Specifications of optical module at CWDM side

Item	Unit	Value
		1600 ps/nm-4mW
Line code format	-	NRZ
Maximum wavelength count	-	16

Item	Unit	Value
		1600 ps/nm-4mW
Transmitter parameter specifications at point S		
Maximum mean launched power ^a	dBm	5
Minimum mean launched power ^a	dBm	2.5
Minimum extinction ratio	dB	8.2
Central wavelength	nm	1311 to 1611
Central wavelength deviation	nm	≤ ±6.5
Maximum -20 dB spectral width	nm	1
Minimum side mode suppression ratio	dB	30
Dispersion tolerance	ps/nm	1600
Eye pattern mask	-	Compliant with G.957
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27
a: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.		

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 0.9 kg

Power Consumption

The power consumption of the board is as follows:

- single-fed board
 - Maximum power consumption at 25°C :

- C6LWM: 32.0W
- C8LWM: 33.5W
- Maximum power consumption at 55°C :
 - C6LWM: 35.5W
 - C8LWM: 37.0W
- dual-fed board
 - Maximum power consumption at 25°C :
 - C6LWM: 32.0W
 - C8LWM: 33.5W
 - Maximum power consumption at 55°C :
 - C6LWM: 35.5W
 - C8LWM: 37.0W

13.4.33 LWMR Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-150, **Table 13-151**, **Table 13-152** and **Table 13-153** list the details about the optical specifications for the C6LWMR and C8LWMR.

Table 13-150 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value				
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2m W-APD	3200 ps/nm-2m W-APD
Line code format	-	NRZ	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S						
Maximum mean launched power	dBm	-1	-1	3	3	3
Minimum mean launched power	dBm	-5	-5	-2	-2	-2
Typical value of mean launched power	dBm	-2	-2	0	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2	8.2

Item	Unit	Value				
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2m W-APD	3200 ps/nm-2m W-APD
Central frequency	THz	192.10 to 196.00				
Central frequency deviation	GHz	±10				
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200	3200
Eye pattern mask	-	G.957-compliant				
Receiver parameter specifications at point R						
Receiver type	-	PIN	APD	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26	-26
Minimum receiver overload	dBm	0	-9	0	-9	-10
Maximum reflectance	dB	-27	-27	-27	-27	-27

Table 13-151 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value	
		12800 ps/nm-tunable	
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-1	-1
Typical value of mean launched power ^a	dBm	1	1
Minimum extinction ratio	dB	10	10

Item	Unit	Value	
		12800 ps/nm-tunable	
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	±10
Maximum -20 dB spectral width	nm	0.2	0.2
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	12800	12800
Eye pattern mask	-	G.957-compliant	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-26
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27

Table 13-152 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	3	3
Minimum mean launched power	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ±6.5	≤ ±6.5

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27

Table 13-153 Specifications of optical module at CWDM side

Item	Unit	Value
		1600 ps/nm-4mW
Line code format	-	NRZ
Maximum wavelength count	-	16
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	5
Minimum mean launched power	dBm	2.5
Minimum extinction ratio	dB	8.2
Central wavelength	nm	1311 to 1611
Central wavelength deviation	nm	≤ ±6.5
Maximum -20 dB spectral width	nm	1
Minimum side mode suppression ratio	dB	30
Dispersion tolerance	ps/nm	1600
Eye pattern mask	-	Compliant with G.957

Item	Unit	Value
		1600 ps/nm-4mW
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.0 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 43.0 W
- Maximum power consumption at 55°C : 47.5 W

13.4.34 LWX Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-154](#), [Table 13-155](#), [Table 13-156](#), [Table 13-157](#) and [Table 13-158](#) list the optical specifications on the client and WDM side of the C6LWX and C8LWX.

Table 13-154 Specifications of optical module at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.2	1000 BASE-SX
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	-
Target distance	km	2	15	80	0.5
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1500 to 1580	770 to 860
Maximum mean launched power	dBm	-3	0	3	-2.5
Minimum mean launched power	dBm	-10	-5	-2	-9.5
Minimum extinction ratio	dB	8.2	8.2	8.2	9
Maximum -20 dB spectral width	nm	NA	1	1	NA
Minimum side mode suppression ratio	dB	NA	30	30	NA
Eye pattern mask	-	G.957-compliant			IEEE802.3z-compliant
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	PIN
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	770 to 860
Receiver sensitivity	dBm	-18	-18	-28	-17
Minimum receiver overload	dBm	-3	0	-9	0
Maximum reflectance	dB	-27	-27	-27	NA
Note	-	The STM-16/OC-48, FC200, FC100, GE, STM-4/OC-12, ESCON, STM-1/OC-3, DVB-ASI, and FE services can be accessed to this optical module.			The FC200 and FC100 services can be accessed to this optical module.

Table 13-155 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value				
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2m W-APD	3200 ps/nm-2m W-APD
Line code format ^a	-	NRZ	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S						
Maximum mean launched power ^b	dBm	-1	-1	3	3	3
Minimum mean launched power ^b	dBm	-5	-5	-2	-2	-2
Typical value of mean launched power ^b	dBm	-2	-2	0	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2	8.2
Central frequency	THz	192.10 to 196.00				
Central frequency deviation	GHz	±10				
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200	3200
Eye pattern mask ^c	-	G.957-compliant				
Receiver parameter specifications at point R						
Receiver type	-	PIN	APD	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26	-26
Minimum receiver overload	dBm	0	-9	0	-9	-10
Maximum reflectance	dB	-27	-27	-27	-27	-27

Item	Unit	Value				
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2m W-APD	3200 ps/nm-2m W-APD
a: For this board, the line code format at the WDM side is adopted according to that at the client side. The line code format is NRZ if the signal is the SDH signal or the OTN signal.						
b: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower.						
c: For this board, the eye pattern at the WDM side is adopted based on the type of the services at the client side. The eye pattern mask is compliant with G.957 if the signal is the SDH signal or the OTN signal.						

Table 13-156 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value	
		12800 ps/nm-tunable	
Line code format ^a	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^b	dBm	3	3
Minimum mean launched power ^b	dBm	-1	-1
Typical value of mean launched power ^b	dBm	1	1
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	±10
Maximum -20 dB spectral width	nm	0.2	0.2
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	12800	12800
Eye pattern mask ^c	-	G.957-compliant	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-26
Minimum receiver overload	dBm	-10	-9

Item	Unit	Value	
		12800 ps/nm-tunable	
Maximum reflectance	dB	-27	-27
a: For this board, the line code format at the WDM side is adopted according to that at the client side. The line code format is NRZ if the signal is the SDH signal or the OTN signal. b: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower. c: For this board, the eye pattern at the WDM side is adopted based on the type of the services at the client side. The eye pattern mask is compliant with G.957 if the signal is the SDH signal or the OTN signal.			

Table 13-157 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format ^a	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^b	dBm	3	3
Minimum mean launched power _b	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ± 6.5	≤ ± 6.5
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask ^c	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum reflectance	dB	-27	-27
a: For this board, the line code format at the WDM side is adopted according to that at the client side. The line code format is NRZ if the signal is the SDH signal or the OTN signal. b: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower. c: For this board, the eye pattern at the WDM side is adopted based on the type of the services at the client side. The eye pattern mask is compliant with G.957 if the signal is the SDH signal or the OTN signal.			

Table 13-158 Specifications of optical module at CWDM side

Item	Unit	Value
		1600 ps/nm-4mW
Line code format ^a	-	NRZ
Maximum wavelength count	-	16
Transmitter parameter specifications at point S		
Maximum mean launched power ^b	dBm	5
Minimum mean launched power ^b	dBm	2.5
Minimum extinction ratio	dB	8.2
Central wavelength	nm	1311 to 1611
Central wavelength deviation	nm	≤ ±6.5
Maximum -20 dB spectral width	nm	1
Minimum side mode suppression ratio	dB	30
Dispersion tolerance	ps/nm	1600
Eye pattern mask ^c	-	Compliant with G.957
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27

Item	Unit	Value
		1600 ps/nm-4mW
a: For this board, the line code format at the WDM side is adopted according to that at the client side. The line code format is NRZ if the signal is the SDH signal or the OTN signal. b: This value is for the single-fed board, while that for the dual-fed board is 3 dB lower. c: For this board, the eye pattern at the WDM side is adopted based on the type of the services at the client side. The eye pattern mask is compliant with G.957 if the signal is the SDH signal or the OTN signal.		

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 0.9 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C:
 - The LWX with a pair of input and output optical interfaces:
 - C6LWX: 32.0W
 - C8LWX: 33.5W
 - The LWX with two pairs of input and output optical interfaces:
 - C6LWX: 35.5W
 - C8LWX: 37.0W
- Maximum power consumption at 55°C:
 - The LWX with a pair of input and output optical interfaces:
 - C6LWX: 32.0W
 - C8LWX: 33.5W
 - The LWX with two pairs of input and output optical interfaces:
 - C6LWX: 35.5W
 - C8LWX: 37.0W

13.4.35 LWXR Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-159](#), [Table 13-160](#), [Table 13-161](#) and [Table 13-162](#) list the optical specifications on the client and WDM side of the C6LWXR and C8LWXR.

Table 13-159 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value				
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2m W-APD	3200 ps/nm-2m W-APD
Line code format	-	NRZ	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S						
Maximum mean launched power	dBm	-1	-1	3	3	3
Minimum mean launched power	dBm	-5	-5	-2	-2	-2
Typical value of mean launched power	dBm	-2	-2	0	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2	8.2
Central frequency	THz	192.10 to 196.00				
Central frequency deviation	GHz	±10				
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200	3200
Eye pattern mask	-	G.957-compliant				
Receiver parameter specifications at point R						
Receiver type	-	PIN	APD	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26	-26

Item	Unit	Value				
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2m W-APD	3200 ps/nm-2m W-APD
Minimum receiver overload	dBm	0	-9	0	-9	-10
Maximum reflectance	dB	-27	-27	-27	-27	-27

Table 13-160 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value	
		12800 ps/nm-tunable	
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power ^a	dBm	3	3
Minimum mean launched power ^a	dBm	-1	-1
Typical value of mean launched power ^a	dBm	1	1
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	±10
Maximum -20 dB spectral width	nm	0.2	0.2
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	12800	12800
Eye pattern mask	-	G.957-compliant	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-26
Minimum receiver overload	dBm	-10	-9

Item	Unit	Value	
		12800 ps/nm-tunable	
Maximum reflectance	dB	-27	-27

Table 13-161 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	3	3
Minimum mean launched power	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	$\leq \pm 6.5$	$\leq \pm 6.5$
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27

Table 13-162 Specifications of optical module at CWDM side

Item	Unit	Value
		1600 ps/nm-4mW
Line code format	-	NRZ
Maximum wavelength count	-	16
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	5
Minimum mean launched power	dBm	2.5
Minimum extinction ratio	dB	8.2
Central wavelength	nm	1311 to 1611
Central wavelength deviation	nm	$\leq \pm 6.5$
Maximum -20 dB spectral width	nm	1
Minimum side mode suppression ratio	dB	30
Dispersion tolerance	ps/nm	1600
Eye pattern mask	-	Compliant with G.957
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.0 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 43.0 W
- Maximum power consumption at 55°C : 47.5 W

13.4.36 TBE Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-163 and **Table 13-164** list the optical specifications of the C8TBE.

Table 13-163 Specifications of optical module for GE service

Item	Unit	Specifications		
		1000 BASE-SX	1000 BASE-LX-10 km	1000 BASE-ZX-40 km
Optical line code	-	NRZ	NRZ	NRZ
Intended transmission distance	km	0.5	10	40
Transmitter characteristics at point S				
Operating wavelength range	nm	770 to 860	1270 to 1355	1270 to 1355
Maximum mean launched optical power	dBm	-2.5	-3	3
Minimum mean launched optical power	dBm	-9.5	-11.5	-4.5
Minimum extinction ratio	dB	9	9	9
Eye pattern	-	Compliant with IEEE802.3z template		
Receiver characteristics at point R				
Receiver type	-	PIN	PIN	PIN
Received signal wavelength range	nm	830 to 860	1270 to 1355	1270 to 1355
Receiver sensitivity	dBm	-17	-19	-21
Minimum overload point	dBm	0	-3	-3

Table 13-164 Specifications of optical module for 10G-LAN service at client side

Item	Unit	Value	
		10G BASE-LR	10G BASE-ER
Line code format	-	NRZ	NRZ
Optical source type	-	SLM	SLM
Target distance	km	10	40
Transmitter parameter specifications at point S			
Operating wavelength range	nm	1290 to 1330	1530 to 1565
Maximum mean launched power	dBm	-1	2
Minimum mean launched power	dBm	-6	-4.7
Minimum extinction ratio	dB	6	8.2
Maximum -20 dB spectral width	nm	NA	NA
Minimum side mode suppression ratio	dB	NA	NA
Eye pattern mask	-	NA	
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-14.4	-15.8
Minimum receiver overload	dBm	+0.5	-1
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 0.9 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 29W
- Maximum power consumption at 55°C : 32W

13.4.37 TMR Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-165 lists the optical specifications of the C6TMR.

Table 13-165 Specifications of optical module at DWDM side

Item	Unit	Value		
Line code format	-	NRZ	NRZ-tunable	NRZ
Transmitter parameter specifications at point S				
Maximum mean launched power	dBm	-1	0	4
Minimum mean launched power	dBm	-5	-5	0
Minimum extinction ratio	dB	10	10	10
Central frequency	THz	192.10 to 196.00		
Central frequency deviation	GHz	±10		
Maximum -20 dB spectral width	nm	0.3	0.3	0.3
Minimum side mode suppression ratio	dB	35	35	35
Dispersion tolerance	ps/nm	800	800	1600
Receiver parameter specifications at point R				
Receiver type	-	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-16	-16	-26
Minimum receiver overload	dBm	0	0	-9
Maximum reflectance	dB	-27	-27	-27

Table 13-166 lists the optical specifications of the C8TMR and C9TMR.

Table 13-166 Specifications of optical module at DWDM side

Item	Unit	Value	
Line code format	-	NRZ	NRZ-tunable
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	2	2
Minimum mean launched power	dBm	-3	-3
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	
Maximum -20 dB spectral width	nm	0.3	0.3
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	800	1100
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-16	-16
Minimum receiver overload	dBm	0	0
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C6TMR/C8TMR: 0.9 kg
 - C9TMR: 0.85 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 35.0 W
 - C6TMR/C8TMR: 35.0 W
 - C9TMR: 19.5 W
- Maximum power consumption at 55°C :
 - C6TMR/C8TMR: 38.5 W
 - C9TMR: 21.5 W

13.4.38 TMRS Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

 NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-167 lists the optical specifications of the C6TMRS.

Table 13-167 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	-1
Minimum mean launched power	dBm	-4
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Table 13-168 lists the optical specifications of the C8TMRS, C9TMRS and CBTMRS.

Table 13-168 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C6TMRS/C8TMRS: 0.9 kg
 - C9TMRS/CBTMRS: 0.85 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6TMRS/C8TMRS: 42.0 W
 - C9TMRS: 23.5 W
 - CBTMRS: 24.1 W
- Maximum power consumption at 55°C :
 - C6TMRS/C8TMRS: 46.0 W
 - C9TMRS: 25.5 W
 - CBTMRS: 26.5 W

13.4.39 TMX Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

 NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-169 and **Table 13-170** list the optical specifications of the C6TMX.

Table 13-169 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-170 Specifications of optical module at DWDM side

Item	Unit	Value		
Line code format	-	NRZ	NRZ-tunable	NRZ
Transmitter parameter specifications at point S				
Maximum mean launched power	dBm	-1	0	4
Minimum mean launched power	dBm	-5	-5	0
Minimum extinction ratio	dB	10	10	10
Central frequency	THz	192.10 to 196.00		
Central frequency deviation	GHz	±10		

Item	Unit	Value		
Maximum -20 dB spectral width	nm	0.3	0.3	0.3
Minimum side mode suppression ratio	dB	35	35	35
Dispersion tolerance	ps/nm	800	800	1600
Receiver parameter specifications at point R				
Receiver type	-	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-16	-16	-26
Minimum receiver overload	dBm	0	0	-9
Maximum reflectance	dB	-27	-27	-27

Table 13-171 and **Table 13-172** list the optical specifications of the C7TMX.

Table 13-171 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-172 Specifications of optical module at DWDM side

Item	Unit	Value		
Line code format	-	NRZ	NRZ-tunable	NRZ
Transmitter parameter specifications at point S				
Maximum mean launched power	dBm	-1	0	4
Minimum mean launched power	dBm	-5	-5	0
Minimum extinction ratio	dB	10	10	10
Central frequency	THz	192.10 to 196.00		
Central frequency deviation	GHz	±10		

Item	Unit	Value		
Maximum -20 dB spectral width	nm	0.3	0.3	0.3
Minimum side mode suppression ratio	dB	35	35	35
Dispersion tolerance	ps/nm	800	800	1600
Receiver parameter specifications at point R				
Receiver type	-	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-16	-16	-26
Minimum receiver overload	dBm	0	0	-9
Maximum reflectance	dB	-27	-27	-27

Table 13-173 and **Table 13-174** list the optical specifications of the C8TMX and C9TMX.

Table 13-173 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-174 Specifications of optical module at DWDM side

Item	Unit	Value	
Line code format	-	NRZ	NRZ-tunable
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	2	2
Minimum mean launched power	dBm	-3	-3
Minimum extinction ratio	dB	10	10
Central frequency	THz	192.10 to 196.00	
Central frequency deviation	GHz	±10	
Maximum -20 dB spectral width	nm	0.3	0.3

Item	Unit	Value	
Minimum side mode suppression ratio	dB	35	35
Dispersion tolerance	ps/nm	800	1100
Receiver parameter specifications at point R			
Receiver type	-	PIN	PIN
Operating wavelength range	nm	1200 to 1650	
Receiver sensitivity	dBm	-16	-16
Minimum receiver overload	dBm	0	0
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight:
 - C6TMX/C7TMX/C9TMX: 1.2 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6TMX: 34.6 W
 - C7TMX/C8TMX/C9TMX: 32.2 W
- Maximum power consumption at 55°C :
 - C6TMX: 38.1 W
 - C7TMX/C8TMX/C9TMX: 35.4 W

13.4.40 TMXS Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

 NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-175 and **Table 13-176** list the optical specifications of the C6TMXS and C7TMXS.

Table 13-175 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-176 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	-1
Minimum mean launched power	dBm	-4
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Table 13-177 and **Table 13-178** list the optical specifications of the C8TMXS and C9TMXS.

Table 13-177 Specifications of optical module for STM-16/OC-48 service at client side

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Line code format	-	NRZ	NRZ	NRZ	NRZ
Optical source type	-	MLM	SLM	SLM	SLM
Target distance	km	2	15	40	80
Transmitter parameter specifications at point S					
Operating wavelength range	nm	1266 to 1360	1260 to 1360	1280 to 1335	1500 to 1580
Maximum mean launched power	dBm	-3	0	3	3
Minimum mean launched power	dBm	-10	-5	-2	-2
Minimum extinction ratio	dB	8.2	8.2	8.2	8.2
Maximum -20 dB spectral width	nm	NA	1	1	1
Minimum side mode suppression ratio	dB	NA	30	30	30
Dispersion tolerance	ps/nm	NA	NA	NA	NA
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	PIN	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-18	-27	-28

Item	Unit	Value			
		I-16	S-16.1	L-16.1	L-16.2
Minimum receiver overload	dBm	-3	0	-9	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-178 Specifications of optical module at DWDM side

Item	Unit	Value
Line code format	-	DRZ-tunable
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	2
Minimum mean launched power	dBm	-3
Minimum extinction ratio	dB	13
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.5
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	1000
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-16
Minimum receiver overload	dBm	0
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight:
 - C6TMXS/C7TMXS/C9TMXS: 1.5 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6TMXS: 43.6 W
 - C7TMXS/C8TMXS/C9TMXS: 34.5 W
- Maximum power consumption at 55°C :
 - C6TMXS: 47.9 W
 - C7TMXS/C8TMXS/C9TMXS: 37.9 W

13.4.41 TMX40S Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

Table 13-179, **Table 13-180** and **Table 13-181** list the optical specifications of the TMX40S.

Table 13-179 Specifications of optical module for 10G-LAN service at client side

Item	Unit	Value			
Supported optical interface type	-	10G Base-SR	10G Base-LR	10G Base-ER	10G Base-ZR
Optical line code	-	NRZ	NRZ	NRZ	NRZ
Light source type	-	MLM	SLM	SLM	SLM
Intended transmission distance	km	0.3	10	40	80
Transmitter characteristics at point S					
Operating wavelength range	nm	840 to 860	1290 to 1330	1530 to 1565	1530 to 1565
Maximum mean launched optical power	dBm	-1.3	-1	2	4

Item	Unit	Value			
Minimum mean launched optical power	dBm	-7.3	-6	-4.7	0
Minimum extinction ratio	dB	3	6	8.2	9
Maximum -20 dB spectrum width	nm	NA	NA	NA	NA
Minimum side-mode suppression ratio	dB	30	30	30	30
Eye pattern	-	Compliant with G.691 template			
Receiver characteristics at point R					
Receiver type	-	PIN	PIN	PIN	APD
Operating wavelength range	nm	1200 to 1650			
Receiver sensitivity	dBm	-7.5	-14.4	-15.8	-24
Minimum overload point	dBm	-1	+0.5	-1	-7
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-180 Specifications of optical module for 10G-WAN/STM-64/OTU2 service at client side

Item	Unit	Specifications		
Supported optical interface type	-	I-64.1/ P1I1-2D1	S-64.2b/ P1S1-2D2b	L64.2/ P1L1-2D2
Optical line code	-	NRZ	NRZ	NRZ
Light source type	-	SLM	SLM	SLM
Intended transmission distance	km	2	40	80
Transmitter characteristics at point S				
Operating wavelength range	nm	1290 to 1330	1530 to 1565	1530 to 1565
Maximum mean launched optical power	dBm	-1	+2	+4
Minimum mean launched optical power	dBm	-6	-1	0

Item	Unit	Specifications		
Minimum extinction ratio	dB	6	8.2	9
Maximum -20 dB spectrum width	nm	1	0.3	0.3
Minimum side-mode suppression ratio	dB	30	30	30
Eye pattern	-	Compliant with G.691 template		
Receiver characteristics at point R				
Receiver type	-	PIN	PIN	PIN
Operating wavelength range	nm	1200 to 1650		
Receiver sensitivity	dBm	-11	-14	-24
Minimum overload point	dBm	-1	-1	-7
Maximum reflectance	dB	-27	-27	-27

Table 13-181 Specifications of optical module at DWDM side

Parameter	Unit	Specification
Line code format	-	DQPSK
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	0
Minimum power	dBm	-5
Minimum extinction ratio	dB	NA
Nominal central frequency	THz	192.10 - 196.00
Central frequency deviation	GHz	±2.5
Maximum -20 dB spectral width	nm	NA
Minimum SMSR	dB	35
Maximum dispersion	ps/nm	-500 to +500
Receiver parameter specifications at point R		
Receiver type	-	PIN
Operating wavelength range	nm	1529 - 1561
Receiver sensitivity	dBm	-16

Parameter	Unit	Specification
Receiver overload	dBm	0
Maximum reflectance	dB	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: 4.75 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 85.0 W
- Maximum power consumption at 55°C : 93.5 W

13.4.42 TRC1 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-182](#), [Table 13-183](#) and [Table 13-184](#) list the optical specifications of the C6TRC1 and C8TRC1.

Table 13-182 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/nm-PIN	12800 ps/nm-APD	6500 ps/nm-PIN	3200 ps/nm-2mW-APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Maximum mean launched power	dBm	-1	-1	3	3
Minimum mean launched power	dBm	-5	-5	-2	-2
Typical value of mean launched power	dBm	-2	-2	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-183 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		

Item	Unit	Value
		12800 ps/nm-tunable
Maximum mean launched power	dBm	3
Minimum mean launched power	dBm	-1
Typical value of mean launched power	dBm	1
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	12800
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27

Table 13-184 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	3	3
Minimum mean launched power	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611

Item	Unit	Value	
		1600 ps/nm-4mW	
Central wavelength deviation	nm	≤ ± 6.5	≤ ± 6.5
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C6TRC1/C8TRC1: 1.0kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6TRC1: 21.5W
 - C8TRC1: 23.0W
- Maximum power consumption at 55°C :
 - C6TRC1: 21.5W

- C8TRC1: 23.0W

13.4.43 TRC2 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NOTE

A margin of the lower threshold of input optical power compared with the receiver sensitivity of the board and a margin of the upper threshold of output optical power compared with the overload point of the board are reserved on the T2000 for precaution.

[Table 13-185](#), [Table 13-186](#) and [Table 13-187](#) list the optical specifications of the C8TRC2.

Table 13-185 Specifications of fixed wavelength optical module at DWDM side

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Line code format	-	NRZ	NRZ	NRZ	NRZ
Transmitter parameter specifications at point S					
Maximum mean launched power	dBm	-1	-1	3	3
Minimum mean launched power	dBm	-5	-5	-2	-2
Typical value of mean launched power	dBm	-2	-2	0	0
Minimum extinction ratio	dB	10	10	8.2	8.2
Central frequency	THz	192.10 to 196.00			
Central frequency deviation	GHz	±10			
Maximum -20 dB spectral width	nm	0.2	0.2	0.5	0.5
Minimum side mode suppression ratio	dB	35	35	30	30
Dispersion tolerance	ps/nm	12800	12800	6500	3200
Eye pattern mask	-	G.957-compliant			
Receiver parameter specifications at point R					
Receiver type	-	PIN	APD	PIN	APD

Item	Unit	Value			
		12800 ps/ nm-PIN	12800 ps/ nm-APD	6500 ps/ nm-PIN	3200 ps/ nm-2mW- APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-18	-28	-18	-26
Minimum receiver overload	dBm	0	-9	0	-9
Maximum reflectance	dB	-27	-27	-27	-27

Table 13-186 Specifications of tunable wavelength optical module at DWDM side

Item	Unit	Value
		12800 ps/nm-tunable
Line code format	-	NRZ
Transmitter parameter specifications at point S		
Maximum mean launched power	dBm	3
Minimum mean launched power	dBm	-1
Typical value of mean launched power	dBm	1
Minimum extinction ratio	dB	10
Central frequency	THz	192.10 to 196.00
Central frequency deviation	GHz	±10
Maximum -20 dB spectral width	nm	0.2
Minimum side mode suppression ratio	dB	35
Dispersion tolerance	ps/nm	12800
Eye pattern mask	-	G.957-compliant
Receiver parameter specifications at point R		
Receiver type	-	APD
Operating wavelength range	nm	1200 to 1650
Receiver sensitivity	dBm	-26
Minimum receiver overload	dBm	-10
Maximum reflectance	dB	-27

Table 13-187 Specifications of optical module at CWDM side

Item	Unit	Value	
		1600 ps/nm-4mW	
Maximum wavelength count	-	8	8
Line code format	-	NRZ	NRZ
Transmitter parameter specifications at point S			
Maximum mean launched power	dBm	3	3
Minimum mean launched power	dBm	-2	-2
Minimum extinction ratio	dB	8.2	8.2
Central wavelength	nm	1471 to 1611	1471 to 1611
Central wavelength deviation	nm	≤ ± 6.5	≤ ± 6.5
Maximum -20 dB spectral width	nm	1	1
Minimum side mode suppression ratio	dB	30	30
Dispersion tolerance	ps/nm	1600	
Eye pattern mask	-	Compliant with G.957	
Receiver parameter specifications at point R			
Receiver type	-	APD	APD
Operating wavelength range	nm	1200 to 1650	1200 to 1650
Receiver sensitivity	dBm	-26	-28
Minimum receiver overload	dBm	-10	-9
Maximum reflectance	dB	-27	-27

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.0 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 21.5 W
- Maximum power consumption at 55°C : 23.0 W

13.4.44 Jitter Transfer Characteristics

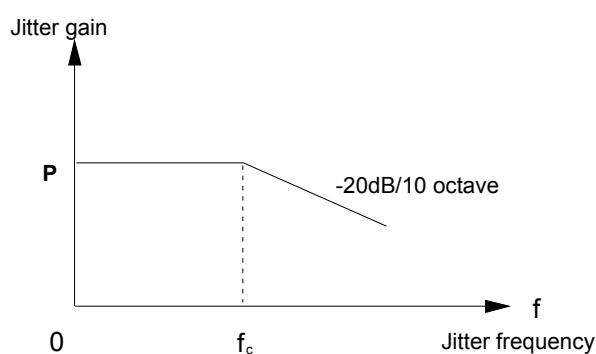
Specifications include jitter transfer characteristics specifications of OTU.

The OTU has the jitter transfer characteristics. Its jitter transfer function should be under the curve. See [Figure 13-1](#). For its specifications, refer to [Table 13-188](#).

Table 13-188 Jitter transfer characteristics specifications

STM Level	f_c (k Hz)	P(dB)
STM-1(A)	130	0.1
STM-4(A)	500	0.1
STM-16(A)/OTU1	2000	0.1
STM-64(A)/OTU2	1000	0.1
STM-256(A)/OTU3	4000	0.1

Figure 13-1 Jitter transfer characteristics



13.4.45 Input Jitter Tolerance

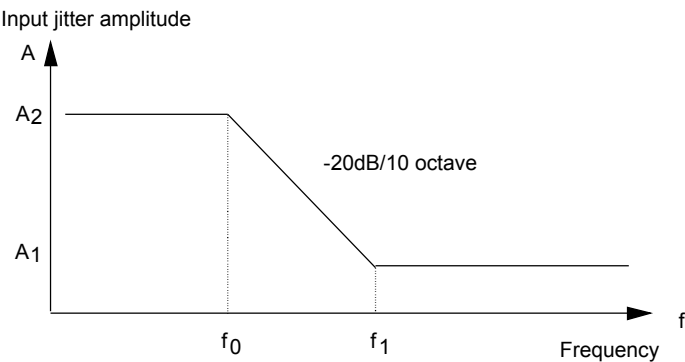
Specifications include input jitter tolerance specifications of OTU.

The OTU is able to tolerate the input jitter pattern that is shown in [Figure 13-2](#). The specifications are given in [Table 13-189](#).

Table 13-189 Input jitter tolerance specifications

STM Level	$f_0(\text{kHz})$	$f_1(\text{kHz})$	$A_1(\text{UI}_{\text{p-p}})$	$A_2(\text{UI}_{\text{p-p}})$
STM-1(A)	6.5	65	0.15	1.5
STM-4(A)	25	250	0.15	1.5
STM-16(A)/ OTU1	100	1000	0.15	1.5
STM-64(A)/ OTU2	400	4000	0.15	1.5
STM-256(A)	1920	16000	0.18	1.5
OTU3	480	16000	0.18	6.0

Figure 13-2 Input jitter tolerance



13.4.46 Output Jitter

Specifications include output jitter specifications of OTU.

The specifications of OTU output jitter are given in [Table 13-190](#).

Table 13-190 Output jitter specifications

STM Level	Interface Measurement Band		Peak-Peak Amplitude (UI)
	High-Pass (KHz)	Low-Pass (MHz)	
STM-1	0.5	1.3	0.3
	65	1.3	0.1
STM-4	1	5	0.3
	250	5	0.1

STM Level	Interface Measurement Band		Peak-Peak Amplitude (UI)
	High-Pass (KHz)	Low-Pass (MHz)	
STM-16/OTU1	5	20	0.3
	1000	20	0.1
STM-64/OTU2	20	80	0.3
	4000	80	0.1
STM-256	80	320	0.3
	16000	320	0.14
OTU3	20	320	1.2
	16000	320	0.14

13.5 Optical Multiplexer and Demultiplexer Board Specifications

The specifications of optical multiplexers and demultiplexers include the optical specifications, mechanical specifications, and power consumption of the EFIU/FIU/M40/V40/D40 boards.

13.5.1 D40 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-191 lists the optical specifications of the D40.

Table 13-191 D40 board specifications

Item	Unit	Value
Adjacent channel spacing	GHz	100
Insertion loss	dB	< 6.5
Optical return loss	dB	> 40
Adjacent channel isolation	dB	> 25
Non-adjacent channel isolation	dB	> 25
Polarization dependent loss	dB	< 0.5
Temperature characteristics	nm/°C	< 0.002

Item	Unit	Value
Maximum channel insertion loss difference	dB	< 3
-1 dB bandwidth	nm	> 0.2
-20 dB bandwidth	nm	< 1.4

Wavelength and Frequency of Optical Interfaces

Wavelength and frequency of each optical interface on the D40 is shown in [Table 13-192](#).

Table 13-192 Wavelength and frequency of each optical interface on the D40

No.	Frequency (THz)	Wavelength (nm)	No.	Frequency (THz)	Wavelength (nm)
01	192.1	1560.61	21	194.1	1544.53
02	192.2	1559.79	22	194.2	1543.73
03	192.3	1558.98	23	194.3	1542.94
04	192.4	1558.17	24	194.4	1542.14
05	192.5	1557.36	25	194.5	1541.35
06	192.6	1556.55	26	194.6	1540.56
07	192.7	1555.75	27	194.7	1539.77
08	192.8	1554.94	28	194.8	1538.98
09	192.9	1554.13	29	194.9	1538.19
10	193.0	1553.33	30	195.0	1537.40
11	193.1	1552.52	31	195.1	1536.61
12	193.2	1551.72	32	195.2	1535.82
13	193.3	1550.92	33	195.3	1535.04
14	193.4	1550.12	34	195.4	1534.25
15	193.5	1549.32	35	195.5	1533.47
16	193.6	1548.51	36	195.6	1532.68
17	193.7	1547.72	37	195.7	1531.90
18	193.8	1546.92	38	195.8	1531.12
19	193.9	1546.12	39	195.9	1530.33
20	194.0	1545.32	40	196.0	1529.55

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight:
 - C6D40: 1.2 kg
 - C9D40: 1.6 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6D40: 20.0 W
 - C9D40: 12.0 W
- Maximum power consumption at 55°C :
 - C6D40: 22.0 W
 - C9D40: 14.0 W

13.5.2 EFIU Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-193 lists the optical specifications of the C6EFIU.

Table 13-193 board specifications

Corresponding Interfaces	Item	Unit	Value
-	Operating wavelength range of main path	nm	1529 to 1561
-	Operating wavelength range of optical supervisory channel	nm	1500 to 1520
-	Optical return loss	dB	> 40
IN-TM RM-OUT	Insertion loss	dB	≤ 1.5

Corresponding Interfaces	Item	Unit	Value
IN-TC RC-OUT	Insertion loss	dB	≤ 1.0
IN-TM	Isolation	dB	> 40
IN-TC	Isolation	dB	> 12

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 142.6 mm x 24.0 mm x 27.5 mm
- Weight: 0.7 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 0.5 W
- Maximum power consumption at 55°C : 0.6 W

13.5.3 FIU Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-194 lists the optical specifications of the FIU.

Table 13-194 board specifications

Corresponding Interfaces	Item	Unit	Value
-	Operating wavelength range of main path	nm	1529 to 1561
-	Operating wavelength range of optical supervisory channel	nm	1500 to 1520
-	Optical return loss	dB	> 40

Corresponding Interfaces	Item	Unit	Value
IN-TM RM-OUT	Insertion loss	dB	≤ 1.5
IN-TC RC-OUT	Insertion loss	dB	≤ 1.0
IN-TM	Isolation	dB	> 40
IN-TC	Isolation	dB	> 12

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - C6FIU/C7FIU/C9FIU: 0.9 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6FIU/C7FIU: 4.3 W
 - C9FIU: 2.1 W
- Maximum power consumption at 55°C :
 - C6FIU: 4.8 W
 - C7FIU: 4.7 W
 - C9FIU: 2.5 W

13.5.4 M40 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-195 lists the optical specifications of the M40.

Table 13-195 M40 board specifications

Item	Unit	Value
Adjacent channel spacing	GHz	100
Insertion loss	dB	< 6.5
Optical return loss	dB	> 40
Operating wavelength range	nm	1529 to 1561
Adjacent channel isolation	dB	> 22
Non-adjacent channel isolation	dB	> 25
Polarization dependence loss	dB	< 0.5
Temperature characteristics	nm/°C	< 0.002
Maximum channel insertion loss difference	dB	< 3

Wavelength and Frequency of Optical Interfaces

Wavelength and frequency of each optical interface on the M40 is shown in [Table 13-196](#).

Table 13-196 Wavelength and frequency of each optical interface on the M40

No.	Frequency (THz)	Wavelength (nm)	No.	Frequency (THz)	Wavelength (nm)
01	192.1	1560.61	21	194.1	1544.53
02	192.2	1559.79	22	194.2	1543.73
03	192.3	1558.98	23	194.3	1542.94
04	192.4	1558.17	24	194.4	1542.14
05	192.5	1557.36	25	194.5	1541.35
06	192.6	1556.55	26	194.6	1540.56
07	192.7	1555.75	27	194.7	1539.77
08	192.8	1554.94	28	194.8	1538.98
09	192.9	1554.13	29	194.9	1538.19
10	193.0	1553.33	30	195.0	1537.40
11	193.1	1552.52	31	195.1	1536.61
12	193.2	1551.72	32	195.2	1535.82
13	193.3	1550.92	33	195.3	1535.04
14	193.4	1550.12	34	195.4	1534.25

No.	Frequency (THz)	Wavelength (nm)	No.	Frequency (THz)	Wavelength (nm)
15	193.5	1549.32	35	195.5	1533.47
16	193.6	1548.51	36	195.6	1532.68
17	193.7	1547.72	37	195.7	1531.90
18	193.8	1546.92	38	195.8	1531.12
19	193.9	1546.12	39	195.9	1530.33
20	194.0	1545.32	40	196.0	1529.55

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: C6M40/C9M40: 1.6 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6M40: 20.0 W
 - C9M40: 12.0 W
- Maximum power consumption at 55°C :
 - C6M40: 22.0 W
 - C9M40: 14.0 W

13.5.5 V40 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

[Table 13-197](#) lists the optical specifications of the V40.

Table 13-197 V40 board specifications

Item	Unit	Value
Adjacent channel spacing	GHz	100
Insertion loss	dB	< 8 ^a
Optical return loss	dB	> 40
Operating wavelength range	nm	1529 to 1561
Adjacent channel isolation	dB	> 22
Non-adjacent channel isolation	dB	> 25
Attenuation range	dB	0 to 15 ^b
Polarization dependence loss	dB	< 0.5
Temperature characteristics	nm/°C	< 0.002
Maximum channel insertion loss difference	dB	< 3
a. Before delivery, the VOA value of each channel is set to 3 dB. Therefore, the value of insertion loss may be 11 dB in testing. The VOA value can be adjusted as required. b. Boards of different versions use different optical modules. For details on the maximum attenuation value that can be set, query it on the NMS.		

Wavelength and Frequency of Optical Interfaces

Wavelength and frequency of each optical interface on the V40 is shown in [Table 13-198](#).

Table 13-198 Wavelength and frequency of each optical interface on the V40

No.	Frequency (THz)	Wavelength (nm)	No.	Frequency (THz)	Wavelength (nm)
01	192.1	1560.61	21	194.1	1544.53
02	192.2	1559.79	22	194.2	1543.73
03	192.3	1558.98	23	194.3	1542.94
04	192.4	1558.17	24	194.4	1542.14
05	192.5	1557.36	25	194.5	1541.35
06	192.6	1556.55	26	194.6	1540.56
07	192.7	1555.75	27	194.7	1539.77
08	192.8	1554.94	28	194.8	1538.98
09	192.9	1554.13	29	194.9	1538.19
10	193.0	1553.33	30	195.0	1537.40

No.	Frequency (THz)	Wavelength (nm)	No.	Frequency (THz)	Wavelength (nm)
11	193.1	1552.52	31	195.1	1536.61
12	193.2	1551.72	32	195.2	1535.82
13	193.3	1550.92	33	195.3	1535.04
14	193.4	1550.12	34	195.4	1534.25
15	193.5	1549.32	35	195.5	1533.47
16	193.6	1548.51	36	195.6	1532.68
17	193.7	1547.72	37	195.7	1531.90
18	193.8	1546.92	38	195.8	1531.12
19	193.9	1546.12	39	195.9	1530.33
20	194.0	1545.32	40	196.0	1529.55

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: C6V40/C9V40: 2.2 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6V40: 46.0 W
 - C9V40: 24.0 W
- Maximum power consumption at 55°C :
 - C6V40: 50.0 W
 - C9V40: 26.0 W

13.6 Optical Add and Drop Multiplexing Board Specifications

The specifications of optical add/drop multiplexing boards include the optical specifications, mechanical specifications, and power consumption of the MR2/MR4/SBM1/SBM2 boards and DWC/RMU9/WDM9/WSD9/WSMD4 boards.

13.6.1 DWC Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-199 lists the optical specifications of the C6DWC.

Table 13-199 C6DWC board specifications

Item		Unit	Value
Operating wavelength range of main path		nm	1529 to 1561
Adjacent channel spacing		GHz	100
Channel attenuation range		dB	0 to 15
Insertion loss	Corresponding interface: IN-DROP	dB	< 8.0
	Corresponding interface: IN-MO	dB	< 12.0
	Corresponding interface: MI-OUT	dB	< 4.0
	Corresponding interface: ADD-OUT	dB	< 4.0
Insertion loss flatness		dB	1.0
0.5 dB spectral width		GHz	> 50
Block extinction ratio		dB	> 35
PMD		ps	< 0.5
PDL		dB	< 0.7
Optical return loss		dB	> 40
Maximum input optical power		dBm	25
Module response time		ms	< 50

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: 2.7 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 30.0 W
- Maximum power consumption at 55°C : 33.0 W

13.6.2 MR2 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-200 lists the optical specifications of the C6MR2.

Table 13-200 C6MR2 board specifications

Corresponding interfaces	Item		Unit	Value
-	Operating wavelength range	DWDM	nm	1529 - 1561
		CWDM	nm	1311 - 1611
-	Adjacent channel spacing	DWDM	GHz	100
		CWDM	nm	20
IN-D1 IN-D2	0.5 dB spectral width	DWDM	nm	$\geq \pm 0.11$
		CWDM	nm	$\geq \pm 6.5$
	Drop channel insertion loss		dB	≤ 1.5
	Adjacent isolation		dB	> 25
	Non-adjacent channel isolation		dB	> 35
A1-OUT A2-OUT	0.5 dB spectral width	DWDM	nm	$\geq \pm 0.11$
		CWDM	nm	$\geq \pm 6.5$
	Add channel insertion loss		dB	≤ 1.5

Corresponding interfaces	Item	Unit	Value
In-MO MI-Out	Insertion loss	dB	≤ 1.0
	Isolation	dB	> 13
-	Optical return loss	dB	> 40

Rules of Adding/Dropping Wavelength

In DWDM and CWDM systems, the MR2 supports adding/dropping of any wavelength with no wavelength distribution rules.

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 142.6 mm x 24.0 mm x 27.5 mm
- Weight: 0.7 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 0.5 W
- Maximum power consumption at 55°C : 0.6 W

13.6.3 MR4 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-201 lists the optical specifications of the CM6MR4.

Table 13-201 CM6MR4 board specifications

Corresponding interfaces	Item		Unit	Value
-	Operating wavelength range	DWDM	nm	1529 - 1561
		CWDM	nm	1311 - 1611

Corresponding interfaces	Item		Unit	Value
-	Adjacent channel spacing	DWDM	GHz	100
		CWDM	nm	20
In-D1 In-D2 In-D3 In-D4	0.5 dB spectral width	DWDM	nm	$\geq \pm 0.11$
		CWDM	nm	$\geq \pm 6.5$
	Drop channel insertion loss	DWDM	dB	≤ 2.5
		CWDM	dB	≤ 1.5
	Adjacent channel isolation		dB	> 25
	Non-adjacent channel isolation		dB	> 35
A1-Out A2-Out A3-Out A4-Out	0.5 dB spectral width	DWDM	nm	$\geq \pm 0.11$
		CWDM	nm	$\geq \pm 6.5$
	Add channel insertion loss	DWDM	dB	≤ 2.5
		CWDM	dB	≤ 1.5
In-MRO MRI-Out	Insertion loss		dB	≤ 2.0
	Isolation		dB	> 13
-	Optical return loss		dB	> 40

Rules of Adding/Dropping Wavelength

In a DWDM system, the MR4 supports adding/dropping of four channels in the same band of the same board. [Table 13-202](#) shows the rules of wavelength distribution.

Table 13-202 Wavelength distribution rules of the CM6MR4 in a DWDM system

Band Group	Wavelength (nm)			
	A1/D1	A2/D2	A3/D3	A4/D4
The first group	1560.61	1559.79	1558.98	1558.17
The second group	1556.56	1555.75	1554.94	1554.13
The third group	1552.52	1551.72	1550.92	1550.12
The fourth group	1548.51	1547.72	1546.92	1546.12
The fifth group	1544.53	1543.73	1542.94	1542.14
The sixth group	1540.56	1539.77	1538.98	1538.19

Band Group	Wavelength (nm)			
	A1/D1	A2/D2	A3/D3	A4/D4
The seventh group	1536.61	1535.82	1535.04	1534.25
The eighth group	1532.68	1531.90	1531.12	1530.33

In a CWDM system, the MR4 supports adding/dropping of four channels in the same board. [Table 13-203](#) shows the rules of wavelength distribution.

Table 13-203 Wavelength distribution rules of the CM6MR4 in a CWDM system

Wavelength (nm)			
A1/D1	A2/D2	A3/D3	A4/D4
1551	1571	1591	1611
1471	1491	1511	1531
1391	1411	1431	1451
1311	1331	1351	1371

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 142.6 mm x 24.0 mm x 27.5 mm
- Weight: 0.7 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 0.5 W
- Maximum power consumption at 55°C : 0.6 W

13.6.4 SBM1 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-204 lists the optical specifications of the C6SBM1.

Table 13-204 C6SBM1 board specifications

Corresponding interfaces	Item	Unit	Value
-	Operating wavelength range	nm	1311 to 1611
-	Adjacent channel spacing	nm	20
LINE-D1 A1-LINE	0.5 dB spectral width	nm	$\geq \pm 6.5$
	Insertion loss	dB	≤ 2.0
	Adjacent isolation	dB	> 30
LINE-EXT	Insertion loss	dB	≤ 1.5
-	Optical return loss	dB	> 40

Rules of Adding/Dropping Wavelength

In CWDM system, the SBM1 supports adding/dropping of any wavelength with no wavelength distribution rules.

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 142.6 mm x 24.0 mm x 27.5 mm
- Weight: 0.7 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 0.5 W
- Maximum power consumption at 55°C : 0.6 W

13.6.5 SBM2 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-205 lists the optical specifications of the C6SBM2.

Table 13-205 C6SBM2 board specifications

Corresponding interfaces	Item	Unit	Value
-	Operating wavelength range	nm	1311 to 1611
-	Adjacent channel spacing	nm	20
LINE-D1	0.5 dB spectral width	nm	$\geq \pm 6.5$
LINE-D2	Insertion loss	dB	≤ 3.0
A1-LINE	Adjacent isolation	dB	> 30
A2-LINE			
LINE-EXT	Insertion loss	dB	≤ 2.0
-	Optical return loss	dB	> 40

Rules of Adding/Dropping Wavelength

In CWDM system, the SBM2 supports adding/dropping of any wavelength with no wavelength distribution rules.

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 142.6 mm x 24.0 mm x 27.5 mm
- Weight: 0.7 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 0.5 W
- Maximum power consumption at 55°C : 0.6 W

13.6.6 RMU9 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-206 lists the optical specifications of the C8RMU9.

Table 13-206 C8RMU9 board specifications

Item		Unit	Value
Insertion loss	EXPI - OUT	dB	≤ 8.5
	AMx ^a - TOA	dB	≤ 12.5 ^b
	ROA-OUT	dB	≤ 1.5
Operating wavelength range		nm	1529 to 1561
Optical return loss		dB	> 40
Attenuation range		dB	0 to 15
Polarization dependence loss		dB	≤ 0.5
Attenuation accuracy		dB	≤ 1
a: AMx represents the AM1 - AM8 interface.			
b: Tested value when the attenuation of the VOA is set to 0 dB.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 0.9 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 7.2 W
- Maximum power consumption at 55°C : 7.9 W

13.6.7 WSD9 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-207 lists the optical specifications of the C8WSD9.

Table 13-207 Display of the C8WSD9 optical interfaces

Item		Unit	Value
Adjacent channel spacing		GHz	100
Insertion loss	IN-DMx ^a IN-EXPO	dB	≤ 8 ^b
Operating wavelength range		nm	1529 to 1561
Extinction ratio		dB	≥ 35
Attenuation range		dB	0 to 15
Dimension		-	1 x 9
a: DMx represents the DM1 - DM8 interface.			
b: Tested value when the attenuation of the VOA is set to 0 dB.			

Table 13-208 lists the optical specifications of the C9WSD9.

Table 13-208 Optical interface parameter specifications of the C9WSD9

Parameters	Unit	Specifications
Channel spacing	GHz	100/50
Operating wavelength range (Comply with ITU-T Grid)	nm	1529 to 1561
Operating wavelength number	-	40
0.5 dB spectral width	nm	≥ 0.2
Insertion loss ^a	dB	< 8.0
Insertion loss uniformity	dB	1.5
Optical reflectance	dB	< -40
Extinction ratio	dB	≥ 35
Directivity	dB	35
Reconfiguration time	second	≤ 3
Attenuation range of each of dropping wavelengths	dB	0 to 15
Attenuation precision of each of dropping wavelengths	dB	≤1 (0 to 10 dB)
		≤1.5 (>10 dB)
a: This is the insertion loss when the build-in VOA is set to 0.		

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: 2.8 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C8WSD9: 13.0 W
 - C9WSD9: 23.2 W
- Maximum power consumption at 55°C :
 - C8WSD9: 15.0 W
 - C9WSD9: 25.5 W

13.6.8 WSM9 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-209 lists the optical specifications of the C8WSM9.

Table 13-209 Display of the C8WSM9 optical interfaces

Item		Unit	Value
Adjacent channel spacing		GHz	100
Insertion loss	Mx ^a -OUT EXPI-OUT	dB	≤ 8 ^b
Operating wavelength range		nm	1529 to 1561
Extinction ratio		dB	≥ 35
Attenuation range		dB	0 to 15
Dimension		-	9 x 1
a: AMx represents the AM1-AM8 interface.			
b: Tested value when the attenuation of the VOA is set to 0 dB.			

Table 13-210 lists the optical specifications of the C9WSM9.

Table 13-210 Optical interface parameter specifications of the C9WSM9

Parameters	Unit	Specifications
Channel spacing	GHz	100/50
Operating wavelength range (Comply with ITU-T Grid)	nm	1529 to 1561
Operating wavelength number	-	40
0.5 dB spectral width	nm	≥ 0.2
Insertion loss ^a	dB	< 8.0
Insertion loss uniformity	dB	1.5
Optical reflectance	dB	< -40
Extinction ratio	dB	≥ 35
Directivity	dB	35
Reconfiguration time	second	≤ 3
Attenuation range of each of dropping wavelengths	dB	0 to 15
Attenuation precision of each of dropping wavelengths	dB	<1 (0 to 10 dB)
		<1.5 (>10 dB)
Polarization dependence loss	dB	
a: This is the insertion loss when the build-in VOA is set to 0.		

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: 2.8 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :

- C8WSM9: 13.0 W
- C9WSM9: 23.2 W
- Maximum power consumption at 55°C :
 - C8WSM9: 15.0 W
 - C9WSM9: 25.5 W

13.6.9 WSMD4 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-211 lists the optical specifications of the C9WSMD4.

Table 13-211 Display of the C9WSMD4 optical interfaces

Parameters		Unit	Indices
Channel spacing		GHz	100
Operating wavelength range		nm	1529 to 1561
Operating wavelength number		-	40
Channel attenuation range		dB	0 to 15
Insertion loss	AMx ^a -OUT	dB	< 8.0 ^b
	IN-DMx ^a	dB	< 7.5
Insertion loss uniformity		dB	1.5
Return loss		dB	> 35
Maximum input optical power		dBm	23
a: AMx represents the AM1 - AM4 interface. DMx represents the DM1 - DM4 interface. b: This is the insertion loss when the build-in VOA is set to 0.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight: 2.7 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 11.7 W
- Maximum power consumption at 55°C : 12.9 W

13.7 Optical Amplifier Board Specifications

The specifications of optical amplifier boards include the optical specifications, mechanical specifications, and power consumption of the OAU/OBU/OPU/PC boards.

13.7.1 OAU Specifications

OAU board specifications include specifications of optical module on the client and WDM sides, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-212 shows the details about the optical specifications of the OAU.

Table 13-212 OAU board specifications

Item	Unit	Value			
		C6OAU01A C6OAU01B C9OAU01	C6OAU02A C6OAU02B C9OAU02	C6OAU03A C6OAU03B C9OAU03	C6OAU05A C9OAU05A
Operating wavelength range	nm	1529 - 1561	1529 - 1561	1529 - 1561	1529 - 1561
Total input power range	dBm	-32 to 0	-32 to -3	-32 to -6	-32 to 0
Total output power range	dBm	-12 to 20	-12 to 17	-6 to 20	-9 to 23
Input power of single wavelength	dBm	-22 to -16	-25 to -19	-28 to -22	-32 to -16
Maximum output power of single wavelength	dBm	4	1	4	7
Channel gain	dB	20 - 31	20 - 31	26 - 32	23 - 34

Item	Unit	Value			
		C6OAU01A C6OAU01B C9OAU01	C6OAU02A C6OAU02B C9OAU02	C6OAU03A C6OAU03B C9OAU03	C6OAU05A C9OAU05A
Noise figure	dB	≤ 6 (when the gain is 31 dB)	≤ 6 (when the gain is 26 dB)	≤ 6 (when the gain is 32 dB)	≤ 6 (when the gain is 34 dB)
		≤ 7 (when the gain is 26 dB)	≤ 7 (when the gain is 23 dB)	≤ 7 (when the gain is 29 dB)	≤ 7 (when the gain is 30 dB)
		≤ 9 (when the gain is 20 dB)	≤ 8 (when the gain is 20 dB)	≤ 8 (when the gain is 26 dB)	≤ 9 (when the gain is 23 dB)
Gain flatness	dB	< 2.0	< 2.0	< 2.0	< 2.0
Gain spectrum-shape pre-tilt	dB	1.0±0.2	0.5±0.2	1.0±0.2	1.0±0.2

Laser Safety Level

- OAU01,OAU02 and OAU03

The laser safety level of the optical interface is CLASS 1M.

Maximum optical power output from an optical interface of the board: between 10 dBm and 22.15 dBm.

- OAU05

The laser safety level of the optical interface is CLASS 3B.

Maximum optical power output from an optical interface of the board: between 22.15 dBm and 27 dBm.

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
- Weight:
 - C6OAU/C9OAU: 2.4 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - C6OAU: 30.0 W

- C9OAU: 24.0 W
- Maximum power consumption at 55°C :
 - C6OAU: 50.0 W
 - C9OAU: 28.0 W

13.7.2 OBU Specifications

OBU board specifications include specifications of optical module on the client and WDM sides, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-213 shows the details about the optical specifications of the OBU.

Table 13-213 OBU board specifications

Item	Unit	Value		
		OBU01	OBU03	OBU05
Operating wavelength range	nm	1529 - 1561	1529 - 1561	1529 - 1561
Total input power range	dBm	-32 to -6	-24 to -3	-24 to 0
Total output power range	dBm	-9 to 17	-1 to 20	-1 to 23
Typical input power of single wavelength	dBm	-22	-19	-16
Maximum output power of single wavelength	dBm	1	4	7
Noise figure	dB	≤ 6.0	≤ 6.0	≤ 7.0
Channel gain	dB	23±1	23±1	23
Gain flatness	dB	< 2.0	< 2.0	< 2.0
Gain spectrum-shape pre-tilt	dB	0.5±0.2	1.0±0.2	1.0±0.2

Laser Safety Level

- OBU01 and OBU03
The laser safety level of the optical interface is CLASS 1M.
Maximum optical power output from an optical interface of the board: between 10 dBm and 22.15 dBm.
- OBU05
The laser safety level of the optical interface is CLASS 3B.

Maximum optical power output from an optical interface of the board: between 22.15 dBm and 27 dBm.

Mechanical Specifications

The mechanical specifications of the board are as follows:

- OBU01
 - Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
 - Weight: 2.2 kg
- OBU03 and OBU05
 - Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
 - Weight: 2.4 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 23 W
- Maximum power consumption at 55°C : 30 W

13.7.3 OPU Specifications

OPU board specifications include specifications of optical module on the client and WDM sides, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-214 shows the details about the optical specifications of the OPU.

Table 13-214 OPU board specifications

Item	Unit	Value				
		C6OPU 01 (Not support GFF)	C6OPU 02	C8OPU 02	C6OPU 03 C9OPU 03	C8OPU 04
Operating wavelength range	nm	1529 - 1561	1529 - 1561	1529 - 1561	1529 - 1561	1529 - 1561
Total input power range	dBm	-32 to -6	-32 to -6	-32 to -4	-32 to -8	-32 to -1
Total output power range	dBm	-12 to 14	-12 to 14	-12 to 16	-9 to 15	-15 to 16
Typical input power of single wavelength	dBm	-22	-22	-20	-24	-17

Item	Unit	Value				
		C6OPU 01 (Not support GFF)	C6OPU 02	C8OPU 02	C6OPU 03 C9OPU 03	C8OPU 04
Maximum output power of single wavelength	dBm	-2	-2	0	-1	0
Noise figure	dB	≤ 5.5	≤ 5.5	≤ 5.5	≤ 5.5	≤ 5.5
Channel gain	dB	20±2	20±1	20±1	23±2	17±1
Gain flatness	dB	< 3.0	< 2.0	< 1.5	< 2.0	< 1.5

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

Specifications the mechanical specifications of the board are as follows:

- OPU01, OPU02 and OPU04
 - Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
 - Weight: 2.0 kg
- OPU03
 - Dimensions (Height x Width x Depth): 345.0 mm x 64.0 mm x 218.5 mm
 - Weight: 2.0 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 20 W
- Maximum power consumption at 55°C : 22 W

13.7.4 RPC Specifications

RPC board specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Interface Specifications

Table 13-215 provides the version mapping of the C8RPC board.

Table 13-215 C8RPC board specifications

Item	Unit	Performance Specifications
Pump wavelength	nm	1427, 1457
Pump count	-	≤ 4
Operating wavelength type	-	Band C
Maximum optical power of pump	dBm	29
G.652 fiber gain ^a	dB	> 10
Equivalent noise figure of G.652 fiber	dB	≤ 1
Polarization dependent loss	dB	≤ 0.5
Temperature characteristic	nm/oC	≤ 1
Output connector type	-	LSH/APC
a: The gain in this table is the switch gain, namely, the difference between the optical power when powering on the RPC and that when powering off the RPC.		

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 96.0 mm x 218.5 mm
- Weight: 2.4 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 70 W
- Maximum power consumption at 55°C : 77 W

13.8 System Control, Supervision and Communication Board Specifications

The specifications of system control, supervision and communication boards include the mechanical specifications and power consumption of the SCC and PMU boards.

13.8.1 SCC Specifications

Specifications include the mechanical specifications and power consumption.

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 24.0 mm x 218.5 mm
- Weight: 0.8 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 10.5 W
- Maximum power consumption at 55°C : 11.5 W

13.8.2 PMU Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

NA

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 24.0 mm x 218.5 mm
- Weight: 1.1kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25 °C:
 - C6PMU/C8PMU: 12.0W
- Maximum power consumption at 25 °C:
 - C6PMU/C8PMU: 13.2W

13.9 Optical Supervisory Channel and Timing Transmission Board Specifications

The specifications of optical supervisory channels and timing transmission boards include the optical specifications, mechanical specifications, and power consumption of the SC1/SC2/TC1/TC2/ST1/ST2 boards.

13.9.1 SC1 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-216 list the optical specifications on the client and WDM side of the SC1.

Table 13-216 Optical interface parameter specifications of the board

Item	Unit	Specifications	
		Normal power	High power
Operating wavelength range	nm	1500-1520	1500-1520
Signal rate	Mbit/s	2.048 ^a	2.048 ^a
Line code format	-	CMI	CMI
Launched power	dBm	-7 to 0	5 to 10
Optical source type	-	MLM LD	MLM LD
Minimum receiver sensitivity (BER=1x10 ⁻¹²)	dBm	-48	-48
Receiver overload	dBm	-3	
a: It is the signal rate before CMI encoding.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 0.9 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - L2SC1: 4.0 W
 - C9SC1: 6.5 W
- Maximum power consumption at 55°C : 4.4 W

- L2SC1: 4.4 W
- C9SC1: 7.2 W

13.9.2 SC2 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-217 list the optical specifications on the client and WDM side of the SC2.

Table 13-217 Optical interface parameter specifications of the board

Item	Unit	Specifications	
		Normal power	High power
Operating wavelength range	nm	1500-1520	1500-1520
Signal rate	Mbit/s	2.048 ^a	2.048 ^a
Line code format	-	CMI	CMI
Launched power	dBm	-7 to 0	5 to 10
Optical source type	-	MLM LD	MLM LD
Minimum receiver sensitivity (BER=1x10 ⁻¹²)	dBm	-48	-48
Receiver overload	dBm	-3	
a: It is the signal rate before CMI encoding.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.0 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :

- L2SC1: 7.0 W
- C9SC1: 8.0 W
- Maximum power consumption at 55°C :
 - L2SC1: 7.7 W
 - C9SC1: 9.6 W

13.9.3 TC1 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-218 lists the optical specifications on the client and WDM side of the L2TC1.

Table 13-218 Optical interface parameter specifications of the board

Item	Unit	Specifications	
		Normal Power	High Power
Operating wavelength range	nm	C band: 1500 to 1520 or L band: 1615–1635	C band: 1500 to 1520
Signal rate	Mbit/s	8.192	8.192
Line code format	-	CMI	CMI
Launched power	dBm	-7 to 0	5 to 10
Optical source type	-	MLM LD	MLM LD
Minimum receiver sensitivity (BER=1×10 ⁻¹²)	dBm	-48	-48
Minimum overload	dBm	-3	-3

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 0.9 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 8.5 W
- Maximum power consumption at 55°C : 9.4 W

13.9.4 TC2 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-219 lists the optical specifications on the client and WDM side of the L2TC2.

Table 13-219 Optical interface parameter specifications of the board

Item	Unit	Specifications	
		Normal Power	High Power
Operating wavelength range	nm	C band: 1500 to 1520 or L band: 1615–1635	C band: 1500 to 1520
Signal rate	Mbit/s	8.192	8.192
Line code format	-	CMI	CMI
Launched power	dBm	-7 to 0	5 to 10
Optical source type	-	MLM LD	MLM LD
Minimum receiver sensitivity (BER=1×10 ⁻¹²)	dBm	-48	-48
Minimum overload	dBm	-3	-3

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.1 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 11.5 W
- Maximum power consumption at 55°C : 12.7 W

13.9.5 ST1 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-220 list the optical specifications on the client and WDM side of the C9ST1.

Table 13-220 Optical interface parameter specifications of the board

Item	Unit	Specification	
		Normal Power	High Power
Operating wavelength range	nm	1500 to 1520	1500 to 1520
Bit rate	Mbit/s	8.448 ^a	8.448 ^a
Code pattern	-	CMI	CMI
Launched optical power	dBm	-7 to 0	5 to 10
Light source type	-	MLM LD	MLM LD
Receiver sensitivity (BER = 1 x 10 ⁻¹²)	dBm	≤ -48	≤ -48
Minimum overload	dBm	-3	-3
a: It is the signal rate before CMI encoding. After CMI encoding the signal rate on the line would be 16 Mbit/s.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.0 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 28.5 W
- Maximum power consumption at 55°C : 31.4 W

13.9.6 ST2 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-221 list the optical specifications on the client and WDM side of the C9ST2.

Table 13-221 Optical interface parameter specifications of the board

Item	Unit	Specification	
		Normal Power	High Power
Operating wavelength range	nm	1500 to 1520	1500 to 1520
Bit rate	Mbit/s	8.448 ^a	8.448 ^a
Code pattern	-	CMI	CMI
Launched optical power	dBm	-7 to 0	5 to 10
Light source type	-	MLM LD	MLM LD
Receiver sensitivity (BER = 1 x 10 ⁻¹²)	dBm	≤ -48	≤ -48
Minimum overload	dBm	-3	-3
a: It is the signal rate before CMI encoding. After CMI encoding the signal rate on the line would be 16 Mbit/s.			

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.2 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 31.5 W
- Maximum power consumption at 55°C : 34.7 W

13.10 Optical Protection Board Specifications

The specifications of protection boards include the optical specifications, mechanical specifications, and power consumption of the DCP/OLP/OWSP/SCS boards.

13.10.1 CP40 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-222 lists the optical specifications on the client and WDM side of the CP40.

Table 13-222 Optical interface parameter specifications of the CP40

Parameters		Unit	Value
Insertion loss at the transmit end	TI - TO1, TI - TO2	dB	< 4
Insertion loss at the receive end	RI1 - RO, RI2 - RO	dB	< 1.5
Range of the input optical power		dBm	-13 to 0
Dispersion tuning range		ps/nm	-400 to +400
Dispersion tuning resolution		ps/nm	10
Dispersion accuracy		ps/nm	10
Tuning Stability		ps/nm	5
Return loss		dB	45

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.8 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 28.0 W
- Maximum power consumption at 55°C : 30.8 W

13.10.2 DCP Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-223 and **Table 13-224** list the optical specifications on the client and WDM side of the C8DCP.

Table 13-223 Optical interface parameter specifications of the C8DCP (single mode)

Corresponding interfaces	Item	Unit	Value
TI1-TO11 TI1-TO12 TI2-TO21 TI2-TO22	Insertion loss on the transmit end	dB	< 4
RI11-RO1 RI12-RO1 RI21-RO2 RI22-RO2	Insertion loss on the receive end	dB	< 1.5
Range of the input optical power		dBm	-35 to 7
Switching time		ms	50
Alarm threshold of optical power difference		dB	3
Switching threshold of optical power difference		dB	5

Table 13-224 Optical interface parameter specifications of the C8DCP (multimode)

Corresponding interfaces	Item	Unit	Value
TI1-TO11 TI1-TO12 TI2-TO21 TI2-TO22	Insertion loss on the transmit end	dB	< 4.5

Corresponding interfaces	Item	Unit	Value
RI11-RO1 RI12-RO1 RI21-RO2 RI22-RO2	Insertion loss on the receive end	dB	< 3
Range of the input optical power		dBm	7 to -28
Switching time		ms	50
Alarm threshold of optical power difference		dB	3
Switching threshold of optical power difference		dB	5

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.0 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 6.0 W
- Maximum power consumption at 55°C : 6.6 W

13.10.3 OLP Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

[Table 13-225](#), [Table 13-226](#), [Table 13-227](#), [Table 13-228](#), [Table 13-229](#), and [Table 13-230](#) list the optical specifications on the client and WDM side of the OLP.

Table 13-225 Optical interface parameter specifications of the L2OLP01 (single mode)

Corresponding interfaces	Item	Unit	Value
TI-TO1 TI-TO2	Signal splitter insertion loss	dB	< 4

Corresponding interfaces	Item	Unit	Value
RI1-RO RI2-RO	Signal selection insertion loss	dB	< 1.5
Range of the input optical power		dBm	7 to -40
Alarm threshold of optical power difference		dB	3
Switching threshold of optical power difference		dB	5

Table 13-226 Optical interface parameter specifications of the L2OLP02 board specifications (single mode)

Corresponding interfaces	Item	Unit	Value
TI-TO1 TI-TO2	Signal splitter insertion loss	dB	< 4
RI1-RO RI2-RO	Signal selection insertion loss	dB	< 1.5
Range of the input optical power		dBm	7 to -40
Alarm threshold of optical power difference		dB	3
Switching threshold of optical power difference		dB	5

Table 13-227 Optical interface parameter specifications of the C6OLP01 board specifications (single mode)

Corresponding interfaces	Item	Unit	Value
TI-TO1 TI-TO2	Signal splitter insertion loss	dB	< 4
RI1-RO RI2-RO	Signal selection insertion loss	dB	< 1.5
Range of the input optical power		dBm	7 to -40
Alarm threshold of optical power difference		dB	3
Switching threshold of optical power difference		dB	5

Table 13-228 Optical interface parameter specifications of the C8OLP01 board specifications (multimode)

Corresponding interfaces	Item	Unit	Value
TI-TO1 TI-TO2	Signal splitter insertion loss	dB	< 4.5
RI1-RO RI2-RO	Signal selection insertion loss	dB	< 3
Range of the input optical power		dBm	7 to -35
Alarm threshold of optical power difference		dB	3
Switching threshold of optical power difference		dB	5

Table 13-229 Optical interface parameter specifications of the C8OLP02 board specifications (single mode)

Corresponding interfaces	Item	Unit	Value
TI-TO1 TI-TO2	Signal splitter insertion loss	dB	< 4
RI1-RO RI2-RO	Signal selection insertion loss	dB	< 1.5
Range of the input optical power		dBm	7 to -35
Alarm threshold of optical power difference		dB	3
Switching threshold of optical power difference		dB	5

Table 13-230 Optical interface parameter specifications of the C8OLP03 board specifications (single mode)

Corresponding interfaces	Item	Unit	Value
TI-TO1 TI-TO2	Signal splitter insertion loss	dB	< 4
RI1-RO RI2-RO	Signal selection insertion loss	dB	< 1.5
Range of the input optical power		dBm	-30 to 23
Alarm threshold of optical power difference		dB	3

Corresponding interfaces	Item	Unit	Value
	Switching threshold of optical power difference	dB	5

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight:
 - L2OLP/C6OLP/C8OLP: 0.8kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C :
 - L2OLP: 7.0W
 - C6OLP/C8OLP: 6.0W
- Maximum power consumption at 55°C :
 - L2OLP: 7.7W
 - C6OLP/C8OLP: 6.6W

13.10.4 OWSP Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-231 lists the optical specifications of the OWSP.

Table 13-231 OWSP board specifications

Corresponding interfaces	Item	Unit	Value
WWI-WDRP EWI-EDRP	Insertion loss	dB	< 1.5

Corresponding interfaces	Item	Unit	Value
EPI-WDRP EPI-WPO WPI-EPO WPI-EDRP	Insertion loss	dB	< 2.5
EADD-WPO EADD-EWO WADD-WWO WADD-EPO	Insertion loss	dB	< 4.5

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 3.3 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25 °C: 6.5 W
- Maximum power consumption at 55 °C: 7.0 W

13.10.5 SCS Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-232 list the optical specifications on the client and WDM side of the SCS.

Table 13-232 Optical interface parameter specifications of the SCS board specifications

Item	Unit	Value
Single-mode insertion loss	dB	< 4
Multimode insertion loss	dB	< 4.5

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 0.7 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 4.3 W
- Maximum power consumption at 55°C : 4.7 W

13.11 Spectrum Analyzer Board Specifications

The specifications of spectrum analyzer boards include the optical specifications, mechanical specifications, and power consumption of the MCA board.

13.11.1 MCA Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-233 list the optical specifications of the MCA.

Table 13-233 Optical interface parameter specifications of the L2MCA and C7MCA board specifications

Item	Unit	Value
Operating wavelength range	nm	1529 - 1561
Detect range for single channel optical power	dBm	-30 to -10
Detect accuracy for optical power	dBm	±1.5
Detect accuracy for OSNR ^a	dB	±1.5
Detect accuracy for central wavelength	nm	±0.1
a: The detect range for OSNR is 13 dB to 23 dB, and the wavelength spacing is 100 GHz.		

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.70 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 7.0 W
- Maximum power consumption at 55°C : 7.7 W

13.12 Variable Optical Attenuator Board Specifications

The specifications of variable optical attenuator boards include the optical specifications, mechanical specifications, and power consumption of the VA2/VA4/VOA boards.

13.12.1 VA2 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-234 list the optical specifications on the client and WDM side of the C9VA2.

Table 13-234 Optical interface parameter specifications of the VA2

Parameters	Unit	Specifications
Attenuation range	dB	1.5 - 20
Adjustment accuracy	dB	0.7 (attenuation $\leq$ 10 dB) 1.0 (attenuation $\leq$ 15 dB) 1.5 (attenuation $\leq$ 20 dB)

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.5 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 10.0 W
- Maximum power consumption at 55°C : 11.0 W

13.12.2 VA4 Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-235 lists the optical specifications on the client and WDM side of the C6VA4.

Table 13-235 Optical interface parameter specifications of the board

Item	Unit	Value
Attenuation range	dB	2 - 20
Adjustment accuracy	dB	0.5

Table 13-236 lists the optical specifications on the client and WDM side of the C9VA4.

Table 13-236 Optical interface parameter specifications of the VA4

Parameters	Unit	Specifications
Attenuation range	dB	1.5 - 20
Adjustment accuracy	dB	0.7 (attenuation $\leq$ 10 dB) 1.0 (attenuation $\leq$ 15 dB) 1.5 (attenuation $\leq$ 20 dB)

Laser Safety Level

The laser safety level of the optical interface is CLASS 1M (The maximum output optical power of each optical interface ranges from 10 dBm (10 mW) to 22.15 dBm (164 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 1.5 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 10.0 W
- Maximum power consumption at 55°C : 11.0 W

13.12.3 VOA Specifications

Specifications include optical specifications, laser safety level, mechanical specifications and power consumption.

Optical Specifications

Table 13-237 lists the optical specifications on the client and WDM side of the L2VOA.

Table 13-237 Optical interface parameter specifications of the board

Item	Unit	Value
Attenuation range	dB	2 - 20
Adjustment accuracy	dB	0.5

Table 13-238 lists the optical specifications on the client and WDM side of the C9VOA.

Table 13-238 Optical interface parameter specifications of the VOA

Parameters	Unit	Specifications
Attenuation range	dB	1.5 - 20
Adjustment accuracy	dB	0.7 (attenuation $\leq$ 10 dB) 1.0 (attenuation $\leq$ 15 dB) 1.5 (attenuation $\leq$ 20 dB)

Laser Safety Level

The laser safety level of the optical interface is CLASS 1 (The maximum output optical power of each optical interface is lower than 10 dBm (10 mW)).

Mechanical Specifications

The mechanical specifications of the board are as follows:

- Dimensions (Height x Width x Depth): 345.0 mm x 32.0 mm x 218.5 mm
- Weight: 0.8 kg

Power Consumption

The power consumption of the board is as follows:

- Maximum power consumption at 25°C : 6.5 W
- Maximum power consumption at 55°C :
 - L2VOA: 7.1 W
 - C9VOA: 7.2 W

A Equipment Specifications and Environment Requirements

Equipment specifications and environment requirements include: performance specifications for optical interfaces, power supply requirements, reliability specifications, electromagnetic compatibility, safety certifications and environment requirement.

[A.1 Performance Specifications for Optical Interfaces](#)

[A.2 Power Supply Requirements](#)

[A.3 Electromagnetic Compatibility \(EMC\)](#)

[A.4 Environment Requirement](#)

A.1 Performance Specifications for Optical Interfaces

SDH optical interfaces: ITU-T G.691-compliant and ITU-T G.957-compliant

10GE optical signal interface: 10GBASE-LR and 10GBASE-ER

GE optical interfaces: IEEE 802.3z-compliant

ESCON optical interfaces: ANSI X3.296, ANSI X3.230-compliant

FC optical interfaces: ANSI X3.303, ANSI X3.230-compliant

Optical fiber connector: LC/PC/LSH/APC

Laser security: In compliance with ITU-T G.664 (with ALS function)

A.2 Power Supply Requirements

DC input voltage: -48V DC/-60V DC

Voltage range: -38.4 V to -72.0 V DC

A.3 Electromagnetic Compatibility (EMC)

The system is in compliance with ETS 300 386 and ETS 300 127, including:

- Radiated Emission: EN55022
- Conducted Emission: EN55022
- Electrostatic Discharge: IEC61000-4-2
- Conducted Sensitivity: IEC61000-4-6
- Electrical Fast Transient/Burst: IEC61000-4-4
- Radiated Sensitivity: IEC61000-4-3
- Surge: IEC61000-4-5
- Voltage Dips and Short Interruption: IEC61000-4-29
- Immunity to radiated electromagnetic fields: ENV50140

A.4 Environment Requirement

Environment requirement includes the requirement on storage, transport, and operation.

A.4.1 Storage Environment

The storage environment complies with ETSI EN 300 019-1-1 and NEBS GR-63-CORE.

Climate Environment

Table A-1 Requirements on climate environment

Item	Range
Altitude	≤ 5000 m
Air pressure	70 kPa to 106 kPa
Temperature	-40°C to $+70^{\circ}\text{C}$
Temperature change rate	$\leq 1^{\circ}\text{C} / \text{min}$
Relative humidity	10% to 100%
Solar radiation	$\leq 1120 \text{ W/s}^2$
Heat radiation	$\leq 600 \text{ W/s}^2$
Wind speed	≤ 30 m/s

Waterproof Requirement

- Equipment storage requirements at the customer site: Generally the equipment is stored indoors.
- There should be no water on the floor and no water leakage on the packing boxes of the equipment. The equipment should not be stored in places where there is possible leakage, such as near the auto firefighting and heating facilities.
- If the equipment is required to be stored outdoors, the following four conditions should be met at the same time:
 - The packing boxes are intact.
 - Necessary rainproof measures should have been taken to prevent rainwater from entering the packing boxes.
 - There is no water on the ground where the packing boxes are stored, let alone water entering into the packing boxes.
 - The packing boxes are not directly exposed to the sun.

Biologic Environment

- Avoiding the reproduction of animalcule, such as epiphyte and mildew.
- Getting rid of rodent (such as mice).

Clarity of Air

- No explosive, conductive, magnetic conductive or corrosive dust.
- The density of mechanically active substance meets the requirements listed in [Table A-2](#).
- The density of chemical active substance meets the requirements listed in [Table A-3](#).

Table A-2 Requirements on the density of mechanically active substance

Mechanically Active Substance	Content
Suspending dust	$\leq 5.00 \text{ mg/m}^3$
Precipitable dust	$\leq 20.0 \text{ mg/m}^2 \cdot \text{h}$
Sand	$\leq 300 \text{ mg/m}^3$

Table A-3 Requirements on the density of chemical active substance

Chemical Active Substance	Content
SO ₂	$\leq 0.30 \text{ mg/m}^3$
H ₂ S	$\leq 0.10 \text{ mg/m}^3$
NO ₂	$\leq 0.50 \text{ mg/m}^3$
NH ₃	$\leq 1.00 \text{ mg/m}^3$
Cl ₂	$\leq 0.10 \text{ mg/m}^3$
HCl	$\leq 0.10 \text{ mg/m}^3$
HF	$\leq 0.01 \text{ mg/m}^3$
O ₃	$\leq 0.05 \text{ mg/m}^3$

Mechanical Stress

Table A-4 Requirements on mechanical stress

Item	Sub-item	Range	
Sinusoidal vibration	Displacement	≤ 7.0 mm	–
	Acceleration	–	≤ 20.0 m/s ²
	Frequency range	2 Hz to 9 Hz	9 Hz to 200 Hz
Non-steady impact	Impact response spectrum II	≤ 250 m/s ²	
	Static load	≤ 5 kPa	
NOTE Impact response spectrum: the curve of the maximum acceleration response generated by the equipment under the stipulated impact motivation. Impact response spectrum II indicates the duration of semi sinusoidal impact spectrum is 6ms. Static load: The pressure from upside, that the equipment with package can endure when the equipment is piled as per stipulation.			

A.4.2 Transport Environment

The transport environment complies with ETSI EN 300 019-1-2 and NEBS GR-63-CORE.

Climate Environment

Table A-5 Requirements on climate environment

Item	Range
Altitude	≤ 5000 m
Air pressure	70 kPa to 106 kPa
Temperature	-40°C to +70°C
Temperature change rate	$\leq 3^{\circ}\text{C} / \text{min}$
Relative humidity	10% to 100%
Solar radiation	$\leq 1120 \text{ W/s}^2$
Heat radiation	$\leq 600 \text{ W/s}^2$
Wind speed	≤ 30 m/s

Waterproof Requirement

The following conditions should be met during the transportation:

- The packing boxes are intact.
- Necessary rainproof measures should be taken for the means of transport to prevent rainwater from entering the packing boxes.
- There is no water in the means of transportation.

Biologic Environment

- Avoiding the reproduction of animalcule, such as epiphyte, mildew.
- Getting rid of rodent (such as mice).

Clarity of Air

- No explosive, conductive, magnetic conductive or corrosive dust.
- The density of mechanically active substance complies with the requirements of [Table A-6](#).
- The density of chemical active substance complies with the requirements of [Table A-7](#).

Table A-6 Requirements on the density of mechanically active substance

Mechanically Active Substance	Content
Suspending dust	No requirement
Precipitable dust	$\leq 3.0 \text{ mg/m}^2\cdot\text{h}$
Sand	$\leq 100 \text{ mg/m}^3$

Table A-7 Requirements on the density of chemical active substance

Chemical Active Substance	Content
SO ₂	$\leq 0.30 \text{ mg/m}^3$
H ₂ S	$\leq 0.10 \text{ mg/m}^3$
NO ₂	$\leq 0.50 \text{ mg/m}^3$
NH ₃	$\leq 1.00 \text{ mg/m}^3$
Cl ₂	$\leq 0.10 \text{ mg/m}^3$
HCl	$\leq 0.10 \text{ mg/m}^3$
HF	$\leq 0.01 \text{ mg/m}^3$
O ₃	$\leq 0.05 \text{ mg/m}^3$

Mechanical Stress

Table A-8 Requirements on mechanical stress

Item	Sub-item	Range		
Sinusoidal vibration	Displacement	$\leq 7.5 \text{ mm}$	–	–
	Acceleration	–	$\leq 20.0 \text{ m/s}^2$	$\leq 40.0 \text{ m/s}^2$
	Frequency range	2 Hz to 9 Hz	9 Hz to 200 Hz	200 Hz to 500 Hz
Random vibration	Acceleration spectrum density	$10 \text{ m}^2/\text{s}^3$	$3 \text{ m}^2/\text{s}^3$	$1 \text{ m}^2/\text{s}^3$
	Frequency range	2 Hz to 9 Hz	9 Hz to 200 Hz	200 Hz to 500 Hz
Non-steady impact	Impact response spectrum II	$\leq 300 \text{ m/s}^2$		

Item	Sub-item	Range
	Static load	≤ 10 kPa
NOTE Impact response spectrum: the curve of the maximum acceleration response generated by the equipment under the stipulated impact motivation. Impact response spectrum II indicates the duration of semi sinusoidal impact spectrum is 6ms. Static load: The pressure from upside, that the equipment with package can endure when the equipment is piled as per stipulation.		

A.4.3 Operation Environment

The operation environment complies with ETSI EN 300 019-1-3 and NEBS GR-63-CORE.

The product needs to be installed indoors and the surrounding temperature should be maintained around 25°C.

Climate Environment

Table A-9 Requirements on temperature, humidity

Equipment Name	Temperature		Relative Humidity	
	Long-term Operation	Short-term Operation	Long-term Operation	Short-term Operation
	0°C to 45°C	-5°C to 50°C	10% to 90%	5% to 95%
NOTE Testing point of product temperature and humidity: when the cabinet of the product has no protection board in the front and at the back, the value is tested 1.5 meters above the floor and 0.4 meter in front of the cabinet. Short-term working condition means that the successive working time does not exceed 96 hours and the accumulated time every year does not exceed 15 days.				

Table A-10 Other requirements on climate environment

Item	Range
Altitude	≤ 4000 m
Air pressure	70 kPa to 106 kPa
Temperature change rate	$\leq 5^{\circ}\text{C/h}$
Solar radiation	≤ 700 W/s ²
Heat radiation	≤ 600 W/s ²
Wind speed	≤ 1 m/s

Biologic Environment

- Avoiding the reproduction of animalcule, such as epiphyte, mildew.
- Getting rid of rodent (such as mice).

Clarity of Air

- No explosive, conductive, magnetic conductive or corrosive dust.
- The density of mechanically active substance meets the requirements listed in [Table A-11](#).
- The density of chemical active substance meets the requirements listed in [Table A-12](#).

Table A-11 Requirements on the density of mechanically active substance

Mechanically Active Substance	Content
Dust particle	$\leq 3 \times 10^5$ particle/m ³
Suspending dust	≤ 0.4 mg/m ³
Precipitable dust	≤ 15 mg/m ² •h
Sand	≤ 100 mg/m ³

Table A-12 Requirements on the density of chemical active substance

Chemical Active Substance	Content
SO ₂	≤ 0.20 mg/m ³
H ₂ S	≤ 0.006 mg/m ³
NH ₃	≤ 0.05 mg/m ³
Cl ₂	≤ 0.01 mg/m ³
HCl	≤ 0.10 mg/m ³
HF	≤ 0.01 mg/m ³
O ₃	≤ 0.005 mg/m ³
CO	≤ 5.0 mg/m ³

Mechanical Stress

Table A-13 Requirements on mechanical stress

Item	Sub-item	Range	
Sinusoidal vibration	Displacement	≤ 3.5mm	-
	Acceleration	-	≤ 10.0 m/s ²
	Frequency range	2 Hz to 9 Hz	9 Hz to 200 Hz
Non-steady impact	Impact response spectrum II	≤ 100 m/s ²	
	Static load	0	
NOTE Impact response spectrum: the curve of the maximum acceleration response generated by the equipment under the stipulated impact motivation. Impact response spectrum II indicates the duration of semi sinusoidal impact spectrum is 6ms. Static load: The pressure from upside, that the equipment with package can endure when the equipment is piled as per stipulation.			

B Power Consumption, Weight and Slots of Boards

This chapter describes the power consumption, weight and slots of board.

Board		Maximum power consumption at 25°C (W)	Maximum power consumption at 55°C (W)	Weight (kg)	Number of slots occupied	Available slots
LWF	L2LWF C7LWF C8LWF	38.0	42.0	1.55	1	IU1-IU6, IU8-IU13
	C9LWF	28.4	30.2	0.95		
	CALWF	30.0	34.0	0.95		
LWFS	C6LWFS C7LWFS C8LWFS	51.0	56.0	1.55	1	IU1-IU6, IU8-IU13
	C9LWFS	40.0	44.0	1.4		
	CALWFS	30.0	34.0	0.95		
L2LRF		24.0	26.4	1.25	1	IU1-IU6, IU8-IU13

Board		Maximum power consumption at 25°C (W)	Maximum power consumption at 55°C (W)	Weight (kg)	Number of slots occupied	Available slots
C6LRFS		37.0	40.7	1.25	1	IU1-IU6, IU8-IU13
C9TMX40S		85.0	93.5	4.75	2	IU1-IU5, IU8-IU12
C9LU40S		84.0	92.4	4.55	2	IU1-IU5, IU8-IU12
C9LUR40S		74.0	81.4	3.75	2	IU1-IU5, IU8-IU12
LBE	C6LBE C8LBE	44.3	48.7	1.1	1	IU1-IU6, IU8-IU13
	CALBE	43.7	48.1	1.2		
LBES	C6LBES C8LBES	48.0	53.0	1.1	1	IU1-IU6, IU8-IU13
	CALBES	52.1	57.3	1.5		
LBF	C8LBF	47.7	53.0	1.1	1	IU1-IU6, IU8-IU13
	C9LBF	30.0	34.0	0.95		
LBFS	C8LBFS	53.6	58.9	1.1	1	IU1-IU6, IU8-IU13
	C9LBFS	30.0	34.0	0.95		
	CBLBFS	23.9	26.3	0.95		
ETMX	C8ETMX	34.6	38.1	1.1	1	IU1-IU6, IU8-IU13
	C9ETMX CAETMX	32.2	35.4	1.1		
ETMXS	C8ETMXS	43.6	47.9	1.3	1	IU1-IU6, IU8-IU13

Board		Maximum power consumption at 25°C (W)	Maximum power consumption at 55°C (W)	Weight (kg)	Number of slots occupied	Available slots
	C9ET MXS CAET MXS	34.5	37.9	1.3		
	CBET MXS	38.8	42.7	1.3		
TMX	C6TM X	34.6	38.1	1.2	2	IU1-IU5, IU8-IU12
	C7TM X	32.2	35.4	1.2		
	C8TM X					
	C9TM X					
TMXS	C6TM XS	43.6	47.9	1.5	2	IU1-IU5, IU8-IU12
	C7TM XS	34.5	37.9	1.5		
	C8TM XS					
	C9TM XS					
TMR	C6TM R	35.0	38.5	0.9	1	IU1-IU6, IU8-IU13
	C8TM R					
	C9TM R	19.5	21.5	0.85		
TMRS	C6TM RS	42.0	46.0	0.9	1	IU1-IU6, IU8-IU13
	C8TM RS					
	C9TM RS	23.5	25.5	0.85		
	CBTM RS	24.1	26.5	0.85		

Board		Maximum power consumption at 25°C (W)	Maximum power consumption at 55°C (W)	Weight (kg)	Number of slots occupied	Available slots
LWC1	C6LWC1	21.5	23.6	1.1	1	IU1-IU6, IU8-IU13
	C8LWC1	21.5	23.6	1.1		
	C9LWC1	13.5	14.6	1.1		
TRC1	C6TRC1	21.5	23.0	1.0	1	IU1-IU6, IU8-IU13
	C8TRC1	21.5	23.0	1.0		
C8TRC2		21.5	23.0	1.0	1	IU1-IU6, IU8-IU13
C6LWM	Single-fed board	32.0	35.5	0.9	1	IU1-IU6, IU8-IU13
	Dual-fed board	33.5	37.0			
C8LWM	Single-fed board	32.0	35.5	0.9	1	IU1-IU6, IU8-IU13
	Dual-fed board	33.5	37.0			
LWMR	C6LWMR	43.0	47.5	1.0	1	IU1-IU6, IU8-IU13
	C8LWMR	43.0	47.5	1.0		
C6LWX	Single-fed board	32.0	35.5	0.9	1	IU1-IU6, IU8-IU13
	Dual-fed board	33.5	37.0			
C8LWX	Single-fed board	32.0	35.5	0.9	1	IU1-IU6, IU8-IU13
	Dual-fed board	33.5	37.0			
LWXR	C6LWXR	43.0	47.5	1.0	1	IU1-IU6, IU8-IU13
	C8LWXR	43.0	47.5	1.0		
LQG	C6LQG	48.0	54.0	1.0	1	IU1-IU6, IU8-IU13

Board		Maximum power consumption at 25°C (W)	Maximum power consumption at 55°C (W)	Weight (kg)	Number of slots occupied	Available slots
	C9LQG	48.0	54.0	1.0		
LDG	C6LDG	29.5	33.0	1.0	1	IU1-IU6, IU8-IU13
	C8LDG	28.0	30.8	1.1		
FDG	C6FDG	34.5	38.0	1.0	1	IU1-IU6, IU8-IU13
	C8FDG	28.0	30.8	1.1		
C8ELOG C9ELOG		46.0	48.1	1.1	1	IU1-IU6, IU8-IU13
C8ELOGS		54.0	58.0	1.1	1	IU1-IU6, IU8-IU13
C9ELOGS		51.3	53.8	1.1	1	IU1-IU6, IU8-IU13
CBELOGS		42.5	46.8	1.1	1	IU1-IU6, IU8-IU13
C6LOG		55.0	60.0	1.5	2	IU1-IU5, IU8-IU12
C9LOG		44.0	53.6	1.5	2	IU1-IU5, IU8-IU12
C6LOGS		58.0	64.0	1.5	2	IU1-IU5, IU8-IU12
C9LOGS		47.3	56.5	1.5	2	IU1-IU5, IU8-IU12
C8LOM		71.6	78.8	1.8	2	IU2-IU6, IU9-IU13
C8LOMS		71.6	78.8	1.8	2	IU2-IU6, IU9-IU13
LQS	C6LQS	30	33	1.2	1	IU1-IU6, IU8-IU13
	C7LQS	30	33	1.2		
AS8	L2AS8	36	38.5	1.2	1	IU1-IU6, IU8-IU13
	C7AS8	36	38.5	1.2		
C6AP8		52.6	58.0	1.3	1	IU1-IU6, IU8-IU13

Board		Maximum power consumption at 25°C (W)	Maximum power consumption at 55°C (W)	Weight (kg)	Number of slots occupied	Available slots
LQM	C8LQM	64.0	69.0	1.1	1	IU1-IU6, IU8-IU13
	C9LQM	64.0	69.0	1.1		
C9LQM2		62.7	69.0	1.1	1	IU1-IU6, IU8-IU13
C6FCE		52.6	58.0	1.1	1	IU1-IU6, IU8-IU13
C7L4G		38.4	50.0	2.5	1	IU1-IU6, IU8-IU13
C7EGS8		36.5	38.0	2.0	1	IU1-IU6, IU8-IU13
C8TBE		29.0	32.0	0.9	1	IU1-IU6, IU8-IU13
C7LAM		26.7	29.4	1.0	1	IU1-IU6, IU8-IU13
M40	C6M40	20.0	22.0	1.6	2	IU2-IU6, IU9-IU13
	C9M40	12.0	14.0	1.6		
V40	C6V40	46.0	50.0	2.2	2	IU2-IU6, IU9-IU13
	C9V40	24.0	26.0	2.2		
D40	C6D40	20.0	22.0	1.2	2	IU2-IU6, IU9-IU13
	C9D40	12.0	14.0	1.6		
FIU	C6FIU	4.3	4.8	0.9	1	IU1-IU6, IU8-IU13
	C7FIU	4.3	4.7	0.9		
	C9FIU	2.1	2.5	0.9		
C6EFIU		0.5	0.6	0.7	1	IU15-IU22 (ODM frame)
C6ACS		0.5	0.6	0.7	1	IU15-IU22 (ODM frame)
C6DWC		30.0	33.0	2.7	2	IU1-IU5, IU8-IU12
WSD9	C8WSD9	13.0	15.0	2.8	2	IU2-IU6, IU9-IU13

Board		Maximum power consumption at 25°C (W)	Maximum power consumption at 55°C (W)	Weight (kg)	Number of slots occupied	Available slots
	C9WSD9	23.2	25.5	2.8		
WSM9	C8WSM9	13.0	15.0	2.8	2	IU2-IU6, IU9-IU13
	C9WSM9	23.2	25.5			
C8RMU9		7.2	7.9	0.9	1	IU1-IU6, IU8-IU13
C9WSMD4		11.7	12.9	2.8	2	IU1-IU5, IU8-IU12
CM6MB4		0.5	0.6	0.7	1	IU15-IU22 (OADM frame)
C6MB2		0.5	0.6	0.7	1	IU15-IU22 (OADM frame)
C6MR2		0.5	0.6	0.7	1	IU15-IU22 (OADM frame)
CM6MR4		0.5	0.6	0.7	1	IU15-IU22 (OADM frame)
C6SBM2		0.5	0.6	0.7	1	IU15-IU22 (OADM frame)
C6SBM1		0.5	0.6	0.7	1	IU15-IU22 (OADM frame)
OAU	C6OAU	30.0	50.0	2.4	2	IU1-IU5, IU8-IU12
	C9OAU	24.0	28.0	2.4		
OBU	C6OBU01	23.0	30.0	2.2	1	IU1-IU6, IU8-IU13
	C6OBU03	23.0	30.0	2.4	2	IU1-IU5, IU8-IU12
	C6OBU05					
	C8OPU03					
	C9OPU03					
	C9OPU05					

Board		Maximum power consumption at 25°C (W)	Maximum power consumption at 55°C (W)	Weight (kg)	Number of slots occupied	Available slots
OPU	C6OPU01 C6OPU02	20.0	22.0	2.0	1	IU1-IU6, IU8-IU13
	C8OPU02 C8OPU04	20.0	22.0	2.0		
	C6OPU03 C9OPU03	20.0	22.0	2.0	2	IU1-IU5, IU8-IU12
C8RPC		70.0	77.0	2.4	3	IU1-IU4, IU8-IU11
L2SC1 C9SC1		4.0	4.4	0.9	1	IU6, IU8
L2SC2 C9SC2		7.0	7.7	1.0	1	IU6, IU8
SCC	C6SCC	10.5	11.5	0.8	1	IU7
	C8SCC	10.5	11.5	0.8		
C9ST1		28.5	31.4	1.0	1	IU6, IU8
C9ST2		31.5	34.7	1.2	1	IU6, IU8
C9CP40		28.0	30.8	1.8	1	IU1-IU6, IU8-IU13
C8DCP		6.0	6.6	1.0	1	IU1-IU6, IU8-IU13
OLP	L2OLP	7.0	7.7	0.8	1	IU1-IU6, IU8-IU13
	C6OLP	6.0	6.6	0.8		
	C8OLP	6.0	6.6	0.8		
L2OWSP		6.5	7.0	1.0	1	IU1-IU6, IU8-IU13
L2SCS		4.3	4.7	0.7	1	IU1-IU6, IU8-IU13
VOA	L2VOA	6.5	7.1	0.8	1	IU1-IU6, IU8-IU13
	C9VOA		7.2			

Board		Maximum power consumption at 25°C (W)	Maximum power consumption at 55°C (W)	Weight (kg)	Number of slots occupied	Available slots
VA4	C6VA4	10.0	11.0	1.5	1	IU1-IU6, IU8-IU13
	C9VA4	10.0	11.0	1.5		
C9VA2		9.0	9.9	0.8	1	IU1-IU6, IU8-IU13
MCA	L2MCA	7.0	7.7	1.7	2	IU1-IU5, IU8-IU12
	C7MCA	7.0	7.7	1.7		
PMU	C6PMU	12.0	13.2	1.1	1	IU14
	C8PMU	12.0	13.2	1.1		
CTL		0.5	-	0.2	1	-
FAN		40	-	6.5	-	-

C Technical Fundamental

The following technologies are widely used: OTN technology, erbium doped fiber amplification technology, Raman amplification technology, and jitter suppression technology.

C.1 OTN Technology

Optical transport network (OTN) is a brand-new optical transport technical system defined by Recommendations such as ITU-T G.872, G.798, and G.709.

C.2 FEC and AFEC

The optical wavelength conversion units have forward error correction (FEC) function and advanced forward error correction (AFEC).

C.3 Erbium-Doped Fiber Amplifier

The system uses an advanced erbium-doped fiber amplifier (EDFA) technology to amplify C-band optical signals and thus achieves long haul transmission without electrical regenerators.

C.4 Raman Amplification

The Raman amplifier is an important application of stimulated Raman scattering (SRS).

C.5 Jitter Suppression

The optical transponder unit (OTU) of the system employs the jitter suppression and clock extraction technology.

C.6 CWDM Technology

The DWDM technology is preferred among the existing fiber application technologies; however, it is highly expensive. Hence, the industry requires a low-cost WDM technology. The CWDM technology meets this requirement.

C.1 OTN Technology

Optical transport network (OTN) is a brand-new optical transport technical system defined by Recommendations such as ITU-T G.872, G.798, and G.709.

C.1.1 Technical Background

In the OTN, the operability and manageability of the SDH/SONET are applied to the WDM system. As a result, the OTN integrates the advantages of the SDH/SONET and WDM.

The SDH/SONET and WDM are the major and advanced technologies used in current transport network. The SDH/SONET mainly helps to process the electrical layer of services, which features VC cross-connect grooming, synchronization, and single-channel line. The SDH/SONET provides access, multiplexing, transport, flexible grooming, management and protection for sub-rate services such as E1, T1, E3, T3 and STM-N. The WDM mainly serves to process the optical layer of services, which features multichannel multiplexing and demultiplexing and long haul transmission, and thus provides low-cost transport for services of wavelength level.

The SDH/SONET network based on VC grooming lacks expandability, whereas the requirements for network bandwidth increase continuously. The traditional WDM technology uses a method where client signals are directly mapped to an optical path and thus is limited to the point-to-point application.

Then, the OTN emerges as the times require. The OTN is based on the SDH/SONET (mapping, multiplexing, flexible cross-connection, embedded overhead, concatenation, protection, and FEC). The operability and manageability of the SDH/SONET are applied to the WDM system. As a result, the OTN combines the advantages of SDH/SONET and WDM. In addition, the OTN defines complete system architecture. In an OTN, each network is specified with a management and monitoring mechanism; both the optical and electrical layers have network survivability mechanism. This completely meets the operators' requirements for operation and maintenance.

C.1.2 OTN Criteria

The OTN system must comply with certain standards.

The OTN standard system is mainly based on the following ITU-T Recommendations:

- G.805: General functional structure of a transport network, applicable to the SDH and ATM.
- G.871: The framework that apply to the OTN.
- G.806: Equipment feature description methods and general functions, applicable to the SDH and OTN.
- G.872: Defines networks at three layers including the OCh, OMS and OTS and describes the functions of network at each layer. G.872 also divides the OCh layer into three sublayers including OTU, ODU and OPU.
- G.798: Specifies each atomic functional module of OTN; specifies the processing of each layer of OTN, including the adaption function of customer/service layer and termination and connection functions of each layer of OTN. G.798 plays a similar role to that of G.783.
- G.709: Defines the OTN frame structure and the overhead function of each layer; defines the mapping processing of client services into OTN, including VC mapping and OTN multiplexing processing. G.709 plays a similar role to that of G.707.

- G.7710: General equipment management function requirements that apply to the SDH and OTN.
- G.874: OTN management information model and function requirements that describes based on G.7710 the five special management functions of OTN (FCAPS).
- G.808.1: General protection switching that applies to the SDH and OTN.
- G.873.1: Defines the linear ODUk protection in OTN domain.
- ITU-T Recommendations such as G.959.1 and G.664 specify the physical-layer characteristics of OTN. Some other recommendations are under establishment, such as G.808.2 (general protection switching) and G.873.2 (OTN domain ring network ODUk protection).

C.1.3 Features of OTN Technology

The OTN technology involves several types of technology.

The OTN technology has the following features:

- The OTN uses the OPUk container to transparently adapt and transport any client service without changing any of the payload or overhead information; provides effective management and QoS monitoring; and is compatible with any new services in the future.
- The OTN adopts asynchronous mapping and multiplexing so that networkwide synchronization is no longer required. This eliminates the limitation from synchronization and simplifies the system design.
- By cross-connecting and multiplexing the ODU1 channel, the OTN enables sub-rate services to be flexibly groomed between the OCh and client-side port, which balances the high utilization of wavelength bandwidth and flexible end-to-end grooming.
- The OTN provides the standard FEC function to achieve a maximum coding gain of 6.2 dB ($\text{BER} = 10\text{E}^{-15}$). This decreases the OSNR tolerance of optical channels; stretches the electrical regeneration distance; reduces the number of system stations; and lowers the total cost for networking.
- With different TCM monitoring initiation points, different carriers and customers can monitor the transmission quality of the same service. This enables easy maintenance and fault locating.

C.1.4 Frame Structure of OTN

The OTN consists of optical layers (OTS_n, OMS_n and OCh) and electrical layers (OTU_k, ODU_k and OPU_k) according to network layer definition.

Figure C-1 shows the OTN layer structure.

Figure C-1 OTN layer structure

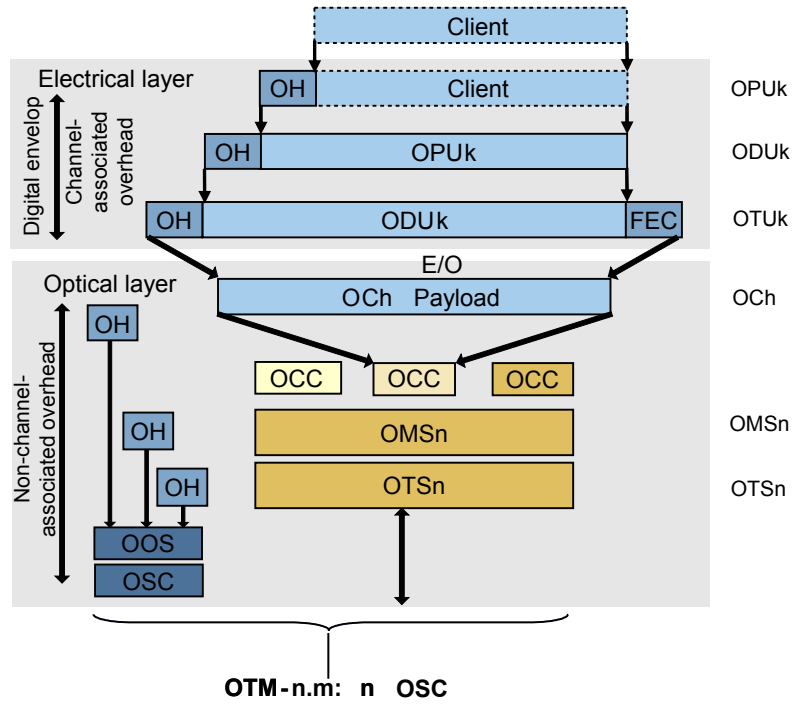


Figure C-2 shows the structure of the overheads at the OTN optical layer. **Table C-1** provides the abbreviations of the overheads.

Figure C-2 OTN optical-layer overhead structure

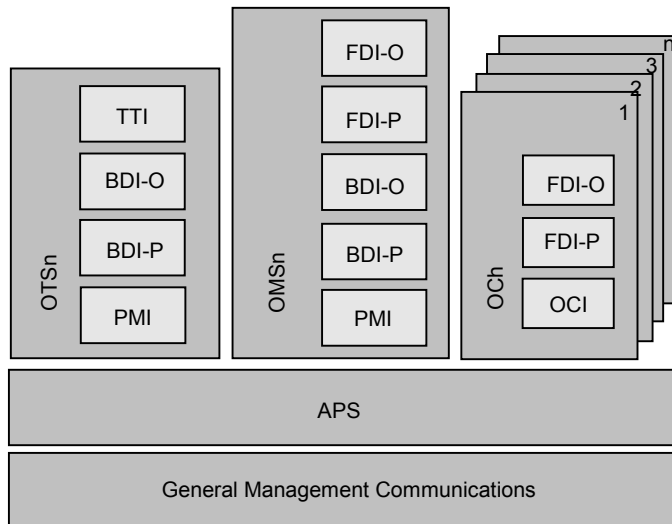


Table C-1 Abbreviations of the overheads at the OTN optical layer

Abbreviation	Expansion Name
OTSn	Optical Transmission Section with n wavelengths
OMSn	Optical Multiplex Section with n wavelengths
OCh	Optical Channel
TTI	Trail Trace Identifier
BDI-P	Backward Defect Identifier-Payload
BDI-O	Backward Defect Identifier-Overhead
FDI-P	Forward Defect Identifier-Payload
FDI-O	Forward Defect Identifier-Overhead
PMI	Payload Mismatch Identifier
OCI	Open Connection Identifier
APS	Automatic Protection Switching
Note: The OTSn layer is the server layer of the OMSn layer, whereas the OMSn layer is the server layer of the OCh layer.	

C.2 FEC and AFEC

The optical wavelength conversion units have forward error correction (FEC) function and advanced forward error correction (AFEC).

The FEC technology is the error correction technology. The OTU adopts Reed-Solomon Coding. It can correct eight byte errors at most in any location for 255 bytes, and has a fairly powerful capability of error correction. Because the redundancy codes are added, the digital rate is increased. The FEC complies with the ITU-T G.975.1 or G.975 and supports the processing of overhead as stated in the ITU-T G.709.

The FEC function can improve the OSNR budget of the DWDM transmission system and increase the transmission distance. In addition, the FEC function can reduce bit error rate in line transmission, and alleviate the effects on the signal transmission quality caused by the aging components or deterioration of fiber performance, thus improving the communication quality of the DWDM transmission network.

The AFEC is a new error correction technique. It adopts two-level encoding, increases encoding gain, and equally distributes the burst errors. AFEC is more powerful than FEC.

C.2.1 Types of the FEC Coding

The main types of the FEC coding include BCH code and RS code.

In all codes, the Bose, Ray-Chaudhuri, Hocquenghem (BCH) code and Reed-solomom (RS) code are regarded as the code that is most likely to be used in the optical transmission system. They both belong to the cyclic error correction code.

The BCH code can correct the random bit error. The BCH code is the cyclic code that can correct multiple random errors respectively put forwarded by three inventors. As a result, it is named by combining the initial letters of their names.

The RS code can correct the random symbol error. The RS code regards one octet or adjacent two bytes as one symbol and can correct a symbol error or more in a code set.

C.2.2 FEC Classification

The FEC is classified into in-band and out-band FEC coding according to the different positions for storing the overhead in the FEC code.

The in-band FEC coding is to carry the supervisory element of the FEC code by using a part of overhead bytes in the SDH/SONET signal frame. The coding mode can reduce the bit error ratio on the premise that the code rate is not increased. The in-band FEC coding must use the byte in the overhead. It is restricted by the available bytes in the frame overhead and the length of the frame. As a result, the bit error ratio cannot be greatly reduced.

Currently the in-band FEC coding is mainly compliant with the ITU-T G.707. It adopts the BCH3 code with the proper complexity and good coding performance. Generally the in-band FEC coding is referred as the defined solution adopting the BCH3 coding.

The out-band FEC coding is compliant with the ITU-T G.975. It uses the RS alternative encoding and decoding and puts the check character to the end of frame. Because some overhead is inserted, the line rate increases. The coding redundancy is 7% and the corresponding rate is increased by 7%. For example, for the STM-16/OC-48, the signal rate is increased from 2.48832 Gbit/s to 2.666 Gbit/s. The coding redundancy is large. As a result, the error correction capacity is powerful and not restricted by the format of the SDH/SONET frame. The out-band FEC coding has a great flexibility.

The features of the RS code are described as follows:

- Powerful error correction capacity. The RS code can correct eight symbol (byte) errors per character (255 bytes).
- Low complexity of FEC encoding and decoding. There is no need for processing the complex frame overhead.
- Small code overhead. The code rate is rather high and the extra overhead is 7%.

The OTU of the OptiX WDM products adopts the out-band FEC coding that is compliant with the ITU-T G.975.1 or G.975 and supports the processing of overhead stated in the ITU-T G.709.

C.2.3 FEC and AFEC Scheme

The FEC and AFEC schemes include the RS (255, 239) coding stated in the ITU-T G.709 and the frame structure of the OTUkV stated in the ITU-T G.709.

The first FEC scheme is the RS (255, 239) coding stated in the ITU-T G.709, which is called as the completely standard OTUk frame. The frame structure is shown in [Figure C-3](#).

Figure C-3 Structure of the standard OTUk frame using the FEC scheme stated in the ITU-T G.709

		Column#					
		1	14	15 16 17	3824	3825	4080
R O W #	1	FA OH	OTUk OH	OTUk payload=ODUk		ROW 1 RS(255,239)FEC redundancy	
	2					ROW 2 RS(255,239)FEC redundancy	
	3					ROW 3 RS(255,239)FEC redundancy	
	4					ROW 4 RS(255,239)FEC redundancy	

The FEC scheme can improve the BER performance. If the signals with the BER 1.0E-4 are input, the BER performance can be improved to 5.0E-15 after error correction by the FEC technology.

The second FEC scheme is the frame structure of the OTUkV stated in the ITU-T G.709, as shown in [Figure C-4](#). The overhead structure of the OTUkV frame is the same as that of the OTUk frame. The only difference is that the FEC in the OTUkV frame is alternative FEC. The FEC coding can be defined by the vendor itself. It is encouraged that the vendor researches the FEC coding whose coding gain capacity is more powerful than that of the RS (255, 239). Some units of the OptiX WDM products of Huawei use the self-defined FEC coding, called AFEC (Advanced FEC).

Figure C-4 Structure of the standard OTUk frame using the AFEC scheme stated in the ITU-T G.709

		Column#							
		1	——	14	15 16 17	3824	3825	——	4080
R o w #	1	FA OH		OTUKV OH		OTUKV payload=ODUk	OTUKV FEC		
	2								
	3								
	4								

The [Table C-2](#) lists the differences and the features of the FEC and the AFEC schemes.

Table C-2 Comparison between the FEC and AFEC schemes

Item	FEC	AFEC
Overhead frame format	OTUk	OTUkV
Coding redundancy ^a	7%	7%
Coding delay	<20 us	<150 us
Coding gain	6.2 dB	≥7 dB
a: When the LBF or LBFS transparently transmits the 10GE-LAN service signals and outputs 10.71 Gbit/s OTU2 signals on its WDM side, the FEC coding redundancy is 4%		

 **NOTE**

The coding algorithm of the FEC and AFEC schemes are different. If two OTUs that have the WDM-side signals of the same rate use the different coding scheme, the two OTUs cannot be interconnected. In the actual engineering, ensure that the FEC coding modes of the upstream OTU and the downstream OTU are the same.

C.3 Erbium-Doped Fiber Amplifier

The system uses an advanced erbium-doped fiber amplifier (EDFA) technology to amplify C-band optical signals and thus achieves long haul transmission without electrical regenerators.

The EDFA integrates a gain lock technology and a transient control technology to disassociate the signal gain of each channel with the total number of channels in a fiber. In addition, the

EDFA prevents the burst bit errors from occurring in the existing channels when channels are increased or decreased.

The optical amplifiers adopted by the system are capable of amplifying the signals that are spaced at 100 GHz and 50 GHz in the C band and comply with ITU-T G.694.1. The system configured with optical amplifiers, runs stably about 20 seconds after being powered on.

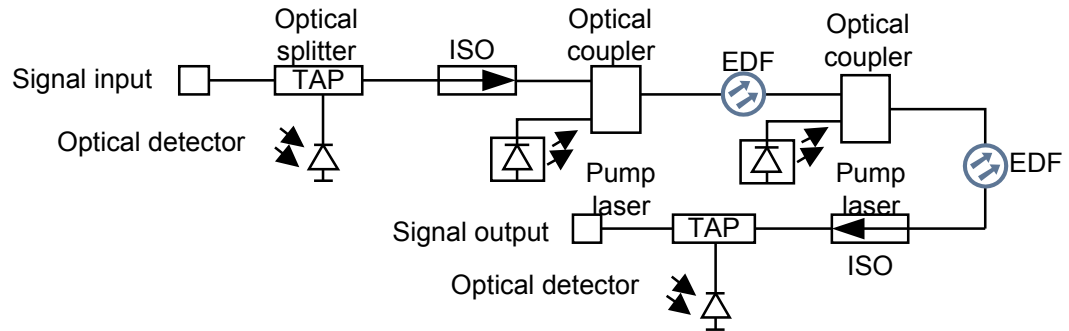
C.3.1 Working Principle of the EDFA

The EDFA is mainly constituted by the optical passive devices, pump source, and erbium doped fiber in the specific optical structure.

The **Figure C-5** shows the optical structure of a typical EDFA with double pump sources.

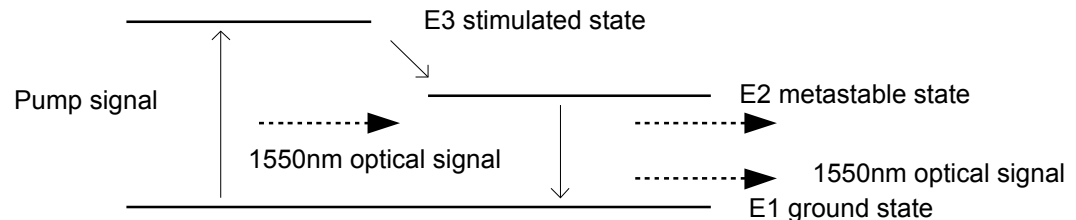
As shown in **Figure C-5**, the signal light and pump light emitted by the pump laser are multiplexed by the WDM component and then transmitted on the erbium doped fiber. The two pump lasers constitute a two-level pump. The erbium doped fiber can perform the amplification after stimulated by the pump light. Hence, the optical signals can be amplified.

Figure C-5 Typical structure inside the EDFA



The erbium doped fiber is the core of the fiber amplifier. It is the fiber doped a certain concentration of Er^{3+} ions. The states of the erbium ion are illustrated to describe the working principle of amplification. The external electrons of the erbium ion are in three states (E1, E2 and E3). The E1 is the ground state, E2 is the metastable state and the E3 is the stimulated state, as shown in **Figure C-6**.

Figure C-6 State figure of the erbium ion



When the erbium doped fiber is stimulated by the high-energy pump laser, the erbium ion can be stimulated from the ground state to the high-energy state E3. The high-energy state is not

stable. Hence, the erbium ion experiences the transition without radiation (no photons are released) and falls into the metastable state E2. The E2 state is metastable. In this state, the particles have a long life expectancy. After stimulated by the pump light, particles continuously influx into the E2 state in the way of transition without radiation to realize the reversal distribution. When the optical signals of 1550 nm wavelength are transmitted on the erbium doped fiber, the particles in metastable state transit to the ground state in the way of stimulated radiation. At the same time, the photons that are the same as those in the incident signal light. Hence, the number of photons in signal light is greatly increased. In this way, the signal light is continuously amplified during transmission on the erbium doped fiber.

C.3.2 Application of the EDFA

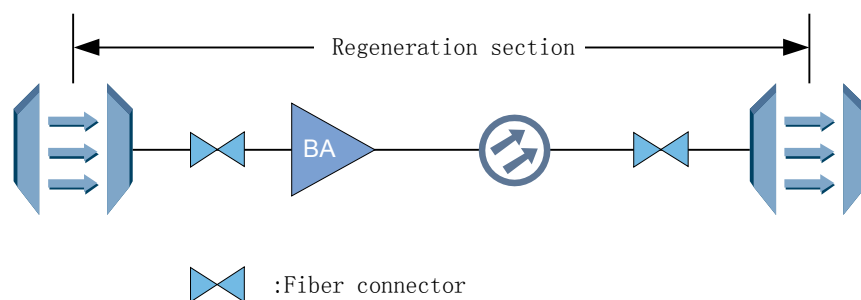
According to the location in the DWDM optical transmission network, the EDFA can be classified into the following three types: Booster Amplifier (BA), Line Amplifier (LA), Preamplifier (PA).

The basic function and the application scenarios are described as follows.

Booster Amplifier

The booster amplifier (BA) is used after optical transmitter of the terminal multiplexer or electrical regenerator, as shown in [Figure C-7](#). The BA is mainly used to increase the transmitted optical power. It increases the incident optical power (generally the power is higher than 10 dBm) to extend the transmission distance. The BA needs stringent requirements for linear amplification of the power but not noise. The BA generally works in the gain or input power saturation interval to increase the efficiency of the transition from pump source power to optical signal power.

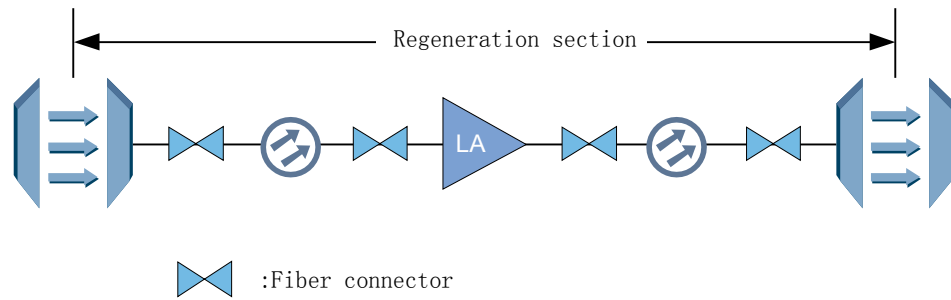
Figure C-7 Location of the BA in the DWDM system



Line Amplifier

The line amplifier (LA) is used in the intermediate repeater section, as shown in [Figure C-8](#). The EDFA is directly put into the fiber transmission line to realize the direction amplification of the signals. One repeater section can be configured with multiple LAs as required. The LA is mainly used for long-haul communication. At that time, the high gain for small signals and small noise figure are required for the EDFA.

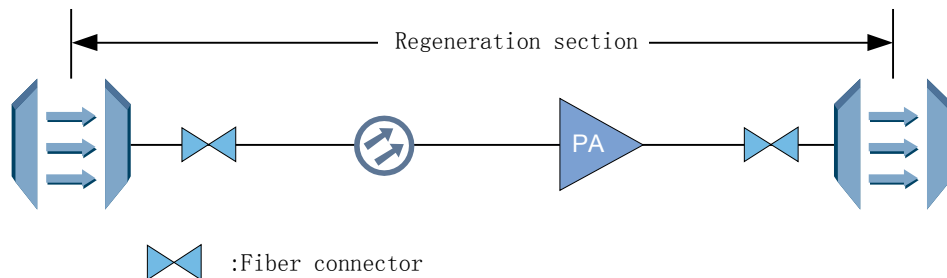
Figure C-8 Location of the LA in the DWDM system



Preamplifier

The preamplifier (PA) is used before the optical receiver at the end of the repeater section, as shown in [Figure C-9](#). The PA is mainly used to amplify the small signals experiencing the line attenuation so as to increase the receiver sensitivity of the optical receiver. The major problem at that time is the noise problem. The PA greatly improves the sensitivity of the direct detection receiver. The sensitivity of the receiver with the EDFA of 2.5 Gbit/s can up to -43.3dBm that is lower than that of the direct detection receiver without the EDFA by about 10 dB.

Figure C-9 Location of the PA in the DWDM system



NOTE

The BA, PA and LA differ in location in the DWDM network, input optical power and gain.

- BA: High input optical power, low gain.
- PA: Low input optical power. The gain is near to that of the BA.
- LA: The input optical power is near to that of the PA. The gain is higher than that of the BA.

C.3.3 Limitation of the EDFA

The EDFA solves the line attenuation problem of the DWDM system, and brings some new problems at the same time.

Non-Linear Problem

The use of the EDFA can increase the optical power; however, the optical power should be moderate. If the optical power is up to a certain value, the non-linear effect (including stimulated

Raman scattering and stimulated Brillouin scattering) occurs to the fiber. Especially, the stimulated Brillouin scattering (SBS) has a strong impact on the EDFA. The non-linear effect greatly restricts the amplification performance of the EDFA and the realization of the long-haul transmission without electrical regenerators.

Optical Surge Problem

Due to the slow gain dynamics of the EDFA, the optical surge occurs when the power of input signals rapidly increases or decreases. That is, peaks are generated in the output optical power. Especially when the EDFA is cascaded, the optical surge problem is more obvious. The peak optical power can up to several watts, which may damage the O/E converter and the surface of the optical connector.

Dispersion Problem

When the EDFA is used, the attenuation limitation problem for the long-haul transmission without electrical regenerators is solved. However, the total dispersion of the fiber is increased accompanying with the distance increase. The original attenuation-limited system becomes the dispersion-limited system.

Optical Signal-to-Ratio Problem

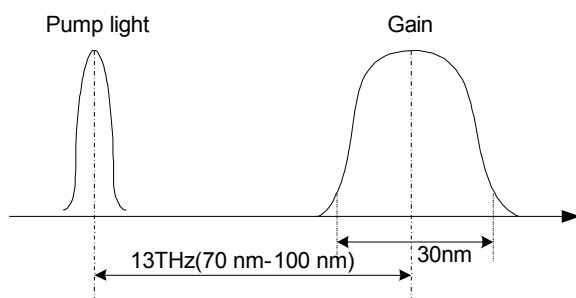
The EDFA generates the amplified spontaneous emission (ASE) in the optical spectrum of several decades of nanometers. The beat noise related to the ASE leads to the degrade of the OSNR at the receive end. Accompanying with the increased number of cascaded amplifiers, this beat noise is linearly increased. Hence, the bit error ratio is degraded. In addition, the noise is cumulative in an index manner accompanying with the gain amplitude of the noise.

C.4 Raman Amplification

The Raman amplifier is an important application of stimulated Raman scattering (SRS).

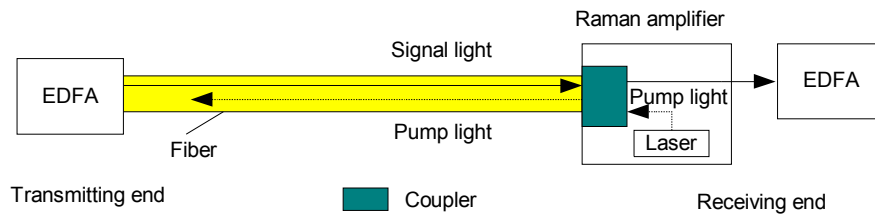
Quartz fiber has a very broad SRS gain spectrum. It has a broad peak near the frequency of 13 THz. If a weak signal and a strong pump light are transmitted in the fiber simultaneously, and their frequency difference is within the range of Raman gain spectrum, the weak signal beam can be amplified. The gain spectrum of the fiber Raman amplifier is shown in the [Figure C-10](#).

Figure C-10 Raman amplifier gain spectrum



The fiber Raman amplifier is always used with the EDFA amplifier at the receive end. It adopts distributed amplification mechanism for extra long haul and extra long span applications, as shown in [Figure C-11](#).

Figure C-11 Raman amplification application



Usually the optical fiber Raman amplifier is used at the receive end of DWDM system to amplify optical signals. The Raman amplifier, which is mainly composed of pumping lasers, works by counter pumping.

NOTE

Counter pumping means the pump light is injected at the fiber end and the direction is opposite to the main signals. This kind of pumping achieves a big phase difference between the main signals and the pump light. The Raman pump power vibration is leveled in the direction opposite to signal transmission, thus effectively suppressing the noise created by the pump.

C.4.1 Principle of Raman Amplification

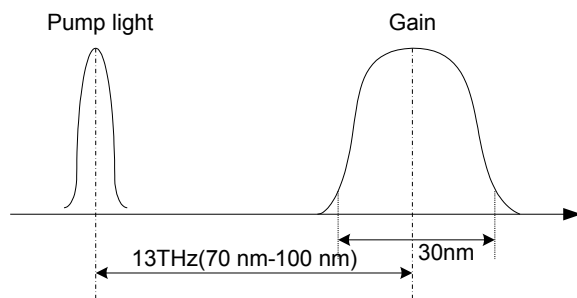
The Raman amplifier uses the Raman effect in the fiber to realize the amplifying process.

In the general fiber system, the optical power is low and the fiber has the linear transmission feature. When the incident optical power of the light into the fiber (non-linear optical medium) is rather high, the high-capacity pump light (of short wavelength) scatters to transfer a portion of their power to the beam whose frequency is down shifted. The amount of frequency shift depends on the vibration mode. This process is called Raman Effect. In the quantum mechanics, the process is described as that a photon of the incident wavelength is scattered to a low-frequency photon and the molecule performs the transition between the vibration states. The incident photons are called pump light and the low-frequency frequency-shift photons are called the Stokes wavelength. The common Raman scattering needs a very strong laser power. In fiber communication, as the non-linear medium, the single-mode fiber has the core of a very small diameter (generally less than 10 μ m). Hence, the single-mode fiber can restrict the interaction of the strong laser field and medium to a very small cross section and greatly increase the optical power intensity of the incident light field. In low-loss fiber, the interaction of the light field and medium can influence the fiber over a very long distance. The power of this fiber span is well coupled. Hence, the stimulated Raman scattering (SRS) can be used in the fiber.

As shown in experience, quartz fiber has a very broad SRS gain spectrum. It has a broad peak at the point where the frequency of the pump light is down shifted about 13 THz. If a weak signal and a strong pump light are transmitted in the fiber simultaneously, and the wavelength of the weak signal is within the range of Raman gain bandwidth of the pump light, the weak signal beam can be amplified. The optical amplifier based on the SRS scheme is called the Raman amplifier. The Stokes wavelength can be described in the following physical figure. A loss of the incident photon leads to the generation of a photon whose frequency is down shifted about 13 THz (that is the Stokes wavelength). The other power is absorbed by the medium in a molecule vibration manner to perform the transition between the vibration states.

Figure C-12 shows the gain spectrum of the Raman amplifier. For the pump light of a certain wavelength (1440 nm for example), a gain spectrum generates at the point where the frequency is down shifted about 13 THz (In the 1550 nm band. The wavelength is up shifted about 100 nm). For the 500 mW pump light in the common single-mode fiber, the gain bandwidth of about 30 nm can be generated.

Figure C-12 Gain spectrum of the Raman amplifier



The gain of the Raman amplifier is on-off gain. That is the difference of the output power values when the amplifier is open and close.

C.4.2 Classification of Raman Amplifiers

The Raman amplifier can be classified into two types: discrete Raman amplifier and distributed Raman amplifier.

- Discrete Raman amplifier

The fiber gain medium used is rather short and generally within ten kilometers. The discrete Raman amplifier has stringent requirements for the pump power. Generally the pump power is from several watts to a value less than 20 watts. The pump can generate the high gain more than 40 dB. It also can perform the centralized amplification to the signals as the EDFA does. Hence, the discrete Raman amplifier is mainly used to amplify the band that cannot be amplified by the EDFA.

- Distributed Raman amplifier

The fiber used is rather long and generally several decades of kilometers. The pump power can be decreased to several hundred of milliwatts. The distributed Raman amplifier is mainly used to assist the EDFA to improve the performance of the DWDM communication system, suppress the non-linear effect and enhance the OSNR. In DWDM system, the transmission capacities, especially the number increase of the multiplexed wavelengths, leads to the power increase and a more severe non-linear effect in fiber communication. As a result, the channel crosstalk and signal distortion are incident. The distributed Raman amplifier can greatly decrease the incident power of the signals and keep a proper OSNR, which is widely used in the long-haul system.

C.4.3 Feature of Raman Amplifiers

The Raman amplifier has three prominent features: the gain wavelength is decided by the wavelength of pump light, the gain medium is the transmission fiber and the noise figure is small.

- The gain wavelength is decided by the wavelength of pump light.

If the wavelength of the pump source is proper, the signal of any wavelength can be amplified in theory, as shown in Figure C-9. The dotted line is the gain spectrum generated by the three pump sources. The Raman amplifier can amplify the band that cannot be amplified by the EDFA due to the feature. If multiple pump sources are used, a greatly wider gain bandwidth can be obtained than that by using the EDFA (The gain bandwidth of the latter one is only 80 nm due to the limitation of the state transition scheme). Hence, the Raman amplifier is irreplaceable in developing the entire low-loss area of 1270-1670 nm of the fiber.

- The gain medium is the transmission fiber.

The Raman amplifier can amplify the optical signals on line and form the distributed amplification. With the Raman amplifier, the long-haul transmission without an REG and remotely operated pump (ROP) can be realized. The Raman amplifier is especially applicable to the place where the REG is not likely to be set such as the submarine communication. In addition, the amplification is distributed along the fiber but not centralized. The optical power of signals on the entire fiber is rather low. Hence, the interference of the non-linear effect especially the four-wavelength mixing (FWM) effect is decreased.

- The noise figure is small.

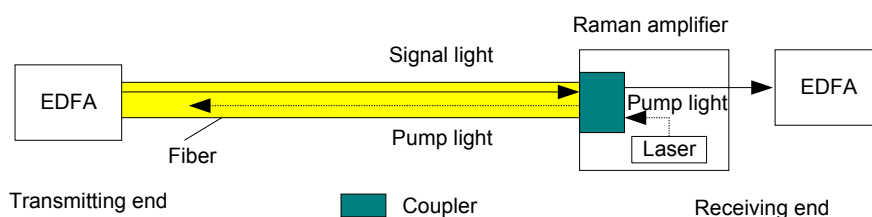
The cooperation of the Raman amplifier and the EDFA can greatly decrease the noise figure of the system and increase the transmission span distance.

C.4.4 Application of Raman Amplifiers

Generally the Raman amplifier works with the EDFA and locates at the frontmost end of the receive station in the DWDM system.

The Raman amplifier sends the pump light to the transmission fiber and synchronously separates the multiplexed signals and then outputs them to the sequent EDFA. The Raman amplifier uses the distributed amplification of the reverse pump to amplify the optical signals during transmission. The realization scheme is shown in [Figure C-13](#).

Figure C-13 Application of Raman amplifier



The transmission distance of the pump light and signal light are different. This reverse pump mode leads to the big phase difference of the signal light and pump light. The power fluctuation of the Raman pump is averaged in the reverse transmission. As a result, the noise generated by the pump can be effectively suppressed.

C.4.5 Strength and Weakness of Raman Amplifiers

Raman Amplifier owns several strength and weakness.

Main Strengths

- The gain is generated on each type of fibers. The gain wavelength is decided on the pump wavelength.
- The structure of the amplifier is simple.
- The non-linear effect can be suppressed.
- The gain is flat in the broadband range (30 nm).
- More pump wavelengths can be chosen, and the bandwidth and gain flatness can be increased.

Main Weaknesses

- The photon efficiency of the pump is rather low, and high-power pump is needed.
- The polarization dependent gain is strong. Hence, the quadrature pump mode is recommended.
- The transient gain is generated. Hence, the backward pump mode is recommended.
- The optical components and the fiber of the system need to carry the high optical power. Hence, the component encapsulation must be improved. The reliable laser should be used and the fiber connector should be clean.
- The gain feature of the fiber on site is not consistent. The pump control technology must be used.

C.4.6 Precautions of Raman Amplifiers

Raman amplifier must be used carefully.

Clean Fiber Surface

The output optical power of the Raman amplifier is rather high. If the surface of the fiber jumper is dirty, the filth of the fiber surface absorbs the energy and heats. As a result, the jumper is easy to be damaged, burned, and the system performance is affected.

Performance Guarantee of Cables

The gain medium of the Raman amplifier is the transmission cable. Hence, the type and quality of the transmission cable have great influence on the performance of the Raman amplifier. If the fiber, especially the end near the Raman amplifier has the poor quality (big loss point or large reflection factor), the system performance is greatly influenced, and even the line would be burned. Hence, testing the cable before enabling the Raman amplifier is necessary.

Dedicated APC Fiber Connector

The reverse output optical power of the Raman amplifier reaches 30 dBm. Hence, the fiber connector must be the dedicated APC fiber connector. If the PC fiber connector is used, large reflection is formed, which damages the fiber connector.

Prohibition of Inserting and Removing a Fiber When the Raman Laser Is Enabled

When the laser of the Raman amplifier is enabled, do not insert or remove the fiber connector to avoid the eye damage caused by the strong laser.

**NOTE**

The OptiX WDM system uses the Raman amplifier of reverse pump structure. The strong pump light is accessed into the fiber through the input interface but not the output interface.

Prohibition of Greatly Bending the Fiber

The bend radius of the fiber jumper of the Raman amplifier should meet the requirements and cannot be greatly bent. Otherwise the fiber jumper would be burned.

Enabling the Laser of the Raman Amplifier on the NM

For security consideration, if the laser is disabled after the Raman amplifier is normally working, the Raman amplifier would stop working. You can issue the corresponding command on the NM to enable the laser of the Raman amplifier.

Confirm the Jumper Connection Before Enabling the Laser

Before enabling the laser of the Raman amplifier, you must well connector the jumper at the input interface and the corresponding ODF subrack jumper.

C.5 Jitter Suppression

The optical transponder unit (OTU) of the system employs the jitter suppression and clock extraction technology.

Hence, the jitter performance of the system is better than the related WDM technology standard. The OTU extracts the B1, B2 and J0 byte to locate the bit errors. It determines that the bit errors are on the client side or on the DWDM side, and then analyses the cause of bit errors. The jitter suppression function is very important when the system is connected to SDH equipment of other vendors.

C.6 CWDM Technology

The DWDM technology is preferred among the existing fiber application technologies; however, it is highly expensive. Hence, the industry requires a low-cost WDM technology. The CWDM technology meets this requirement.

CWDM is different from DWDM as follows:

- The carrier channel spacing of a CWDM system is large; hence, one fiber supports the multiplexing of only 2 to 16 wavelengths. "Course" and "dense" are derived from different channel spaces.
- The modulated laser of a CWDM system is uncooled while the modulated laser of a DWDM system is cooled. The cooled laser requires the cooling technique to stabilize wavelengths, which is difficult to achieve; hence, the DWDM technology is of high cost.

The DFB laser used in a CWDM system requires no cooling; hence, the cost of a CWDM system is approximately 30% less than the cost of a DWDM system. Many MAN operators demand a rational transmission solution, hence, the CWDM system is widely accepted in the industry.

A DWDM and a CWDM system mainly differ in channel spacing of different wavelengths transmitted in the same fiber. Currently, a CWDM system operates at 1260 nm through 1620

nm with a channel spacing of 20 nm and can multiplex 16 wavelengths. The 1400 nm band is rarely used because this band causes large optical power loss.

Compared with a DWDM system, a CWDM system provides a certain number of wavelengths; supports the transmission distance within 100 km; greatly lowers the system cost; and is highly flexible. Hence, the CWDM system mainly applies to MANs. The CWDM system provides high bandwidth with low cost and thus applies to various prevalent networks such as the point-to-point, Ethernet, and SONET ring. The CWDM system is especially applicable for communication scenarios with short distance, high bandwidth and dense access points, such as intra- or inter-building network communication.

The CWDM technology is of low cost and thus has certain limitations in performance. Industry experts consider that currently, the CWDM technology mainly has the following disadvantages:

- The CWDM system can multiplex less number of wavelengths in a single fiber. Thus, the cost of expanding the system in future is high.
- The cost of equipment such as multiplexer and multiplexing modulator needs to be reduced; such equipment cannot be remodeled from the corresponding equipment of the DWDM system.
- Currently, the CWDM technology is not standardized.

D Complied Standards

The product complies with the related standards and regulations.

[D.1 ITU-T Recommendations](#)

[D.2 IEEE Standards](#)

[D.3 Laser Security Standards](#)

[D.4 Security Standards](#)

[D.5 EMC Standards](#)

[D.6 Environment Related Standards](#)

[D.7 International Standards](#)

D.1 ITU-T Recommendations

Recommendation	Describes
G.661	Definition and test methods for the relevant generic parameters of optical fiber amplifiers
G.662	Generic characteristics of optical fiber amplifier devices and sub-systems
G.663	Application related aspects of optical fiber amplifier devices and sub-systems
G.664	Optical safety procedures and requirements for optical transport systems
G.665	Definitions and Test Methods for Generic Characteristics of Raman Amplifiers and Raman Amplified Subsystems
G.691	Optical interfaces for single channel STM-64 and other SDH systems with optical amplifiers
G.692	Optical interfaces for multichannel systems with optical amplifiers
G.693	Optical interfaces for intra-office systems
G.694.2	Spectral grids for WDM applications: CWDM frequency grid
G.696.1	Intra-Domain DWDM applications
G.697	Optical monitoring for DWDM systems
G.702	Digital hierarchy bit rates
G.703	Physical/electrical characteristic of hierarchical digital interfaces
G.704	Synchronous frame structures used at 1544, 6312, 2048, 8448 and 44736kbit/s hierarchical levels
G.707	Network node interface for the synchronous digital hierarchy (SDH)
G.709	Interfaces for the Optical Transport Network
G.7710	Equipment Management Function (EMF) requirements that are common to multiple transport technologies
G.775	Loss of signal (LOS) and alarm indication signal (AIS) defect detection and clearance criteria
G.773	Protocol suites for Q-interfaces for management of transmission systems

Recommendation	Describes
G.774 1 G.774 2 G.774 3 G.774 4 G.774 5	Synchronous Digital Hierarchy (SDH) management information model for the network element view
G.783	Characteristics of Synchronous Digital Hierarchy (SDH) equipment functional blocks
G.784	Synchronous Digital Hierarchy (SDH) management
G.798	Characteristics of optical transport network hierarchy equipment functional blocks
G.803	Architectures of transport networks based on the Synchronous Digital Hierarchy (SDH)
G.808.1	The generic functional models, characteristics and processes associated with various linear protection schemes for connection-oriented layer networks
G.813	Timing characteristics of SDH equipment slave clocks (SEC)
G.823	The control of jitter and wander within digital networks which are based on the 2048kbit/s hierarchy
G.824	The control of jitter and wander within digital networks which are based on the 1544kbit/s hierarchy
G.825	The control of jitter and wander within digital networks which are based on the Synchronous Digital Hierarchy (SDH)
G.826	Error performance parameters and objectives for international, constant bit rate digital paths at or above the primary rate
G.831	Management capabilities of transport networks based on the Synchronous Digital Hierarchy (SDH)
G.841	Types and characteristics of SDH network protection architectures
G.842	Cooperation of the SDH network protection structures
G.870	Terms and definitions for Optical Transport
G.871	Framework for optical transport network (OTN)
G.872	The functional architecture of optical transport networks using the modelling methodology described in ITU-T Rec. G.805
G.873.1	The APS protocol and protection switching operation for the linear protection schemes for the Optical Transport Network at the Optical Channel Data Unit (ODUk) level

Recommendation	Describes
G.874	Management aspects of the Optical Transport Network Element containing transport functions of one or more of the layer networks of the optical transport network.
G.875	Optical transport network (OTN) management information model for the network element view
G.957	Optical interfaces of equipments and systems relating to the synchronous digital hierarchy
G.959.1	Optical transport network physical layer interfaces
G.975	Forward error correction for submarine systems
G.975.1	Forward error correction for high bit rate DWDM submarine systems
M.2401	Error Performance Limits and Procedures for Bringing-Into-Service and Maintenance of multi-operator international paths and sections within Optical Transport Networks
M.3010	Principles for a telecommunication management network
G.8201	Error performance parameters and objectives for multi-operator international paths within the Optical Transport Network (OTN)
G.8251	The control of jitter and wander within the optical transport network (OTN) Series

D.2 IEEE Standards

Standard	Description
IEEE Std 802.3	Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specification
IEEE 802.3z	Media Access Control (MAC) parameters, physical Layer, repeater and management parameters for 1000 Mb/s operation
IEEE 802.3ae	Media Access Control (MAC) parameters, physical Layer, and management parameters for 10Gb/s operation

D.3 Laser Security Standards

Standard	Description
IEC 60825-1	Safety of laser products-Part 1: Equipment classification, requirements and user's guide

Standard	Description
IEC 60825-2	Safety of laser products-Part 2: Safety of optical fiber communication systems

D.4 Security Standards

Standard	Description
IEC 60215	Safety requirements for radio transmitting equipment
EN 60950-1	Safety of Information Technology Equipment. Including Electrical Business Equipment
IEC 60950-1	Safety of Information Technology Equipment. Including Electrical Business Equipment
CAN/CSA-C22.2 No 60950-1	Safety of Information Technology Equipment Including Electrical Business Equipment
UL 60950-1	3:rd edition Safety of Information Technology Equipment Including Electrical Business Equipment

D.5 EMC Standards

Standard	Description
IEC Publication 1000-4-2	Testing and measurement techniques of electrostatic discharge immunity test
IEC Publication 1000-4-3	Radiated RF electromagnetic field immunity test
IEC Publication 1000-4-4	Testing and measurement techniques of electrical fast transients/burst immunity test
IEC Publication 1000-4-6	Immunity to conducted disturbances
EN 55022	Information technology equipment-Radio disturbance characteristics-Limits and methods of measurement
EN 55024	Information technology equipment-Immunity characteristics-Limits and methods of measurement
IEC 61000-4-2	Testing and measurement techniques -Electrostatic discharge immunity test
IEC 61000-4-3	Testing and measurement techniques –Radiated, radio-frequency, electromagnetic field immunity test

Standard	Description
IEC 61000-4-4	Testing and measurement techniques –Electrical fast transient/burst immunity test
IEC 61000-4-5	Testing and measurement techniques –Surge immunity test
IEC 61000-4-6	Testing and measurement techniques –Immunity to conducted disturbances, induced by radio-frequency fields
IEC 61000-4-11	Testing and measurement techniques –Voltage dips, short interruptions and voltage variations immunity tests
IEC 61000-4-29	Testing and measurement techniques –Voltage dips, short interruptions
ETSI EN 300 386	Electromagnetic compatibility and Radio spectrum Matters (ERM); Telecommunication network equipment; Electro Magnetic Compatibility (EMC) requirements
GR-1089-CORE	Electromagnetic compatibility and electrical safety - generic criteria for network telecommunications equipment

D.6 Environment Related Standards

Standard	Description
ETSI EN 300 019-1-1	Class 1.1: weather-protected, partly temperature-controlled storage locations Class 1.2: weather-protected, not temperature-controlled storage locations
ETSI EN 300 019-1-2	Class 2.2: Careful transportation
ETSI EN 300 019-1-3	Class 3.2 Partly temperature-controlled location
NEBS GR-63-CORE	Network Equipment-Building System (NEBS) Requirements: Physical Protection
RoHS	Restriction of the use of certain hazardous substance in electrical and electronic equipment

D.7 International Standards

Standard	Description
IEC 61291-1	Optical amplifiers - Part 4: Multichannel Applications Performance specification Template
CAN/CSA-C22.2 No 1-M94	Audio, Video and Similar Electronic Equipment
73/23/EEC	Low Voltage Directive
IEC 529	Classification of degrees of protection provided by enclosures (IP Code)
SMPTE 259M	Television — SDTV1 Digital Signal/Data — Serial Digital Interface
SMPTE 424M	Television — 3 Gb/s Signal/Data Serial Interface
SMPTE 292M	Television ---- Bit-Serial Digital Interface forHigh-Definition Television Systems
CENELEC EN 50083-9	Cable networks for television signals, sound signals and interactive services - Part 9: Interfaces for CATV/SMATV headends and similar professional equipment for DVB/MPEG-2 transport streams
ISO 9314	Fiber Distributed Data Interface (FDDI)

E Glossary

A	
Add/Drop multiplexer	A multiplexer capable of extracting and inserting lower-rate signals from a higher-rate multiplexed signal without completely demultiplexing the signal.
Add/drop wavelength	In the OADM equipment, the MR2 board carries the wavelength that directly adds or drops services.
ADM	See add/drop multiplexer.
Administrator	A user who has authority to access all the Management Domains of the EML Core product. He has access to the whole network and to all the management functionalities.
AIS	Alarm Indication Signal. A signal sent downstream in a digital network if an upstream failure has been detected and persist for a certain time.
Alarm cascading	The shunt-wound output of the alarm signals of several subracks or cabinets.
Alarm correlation analysis	A process where in alarm is raised within five seconds after alarm is raised, and alarm complies with the conditions defined in the alarm correlation analysis rule, you can either suppress the alarm or raise its severity level according to the behavior defined in the alarm correlation rule.
Alarm indication signal	A code sent downstream in a digital network as an indication that an upstream failure has been detected. It is associated with multiple transport layers.
Alarm indication	On the cabinet of an NE, there are three indicators with different colors indicating the current status of the NE. You can stop the NE alarm indication through the T2000.
Alarm	A visible or an audible indication to notify the person concerned that a failure or an emergency has occurred. See also Event.

ALC link	A piece of end-to-end configuration information, which exists in the equipment (single station) as an ALC link node. Through the ALC function of each node, it fulfils optical power control on the line that contains the link.
ALC	Automatic Level Control. The technique supports the adjustment of optical power aimed to restrain the output power to be inferior on the downstream and keep the optical power to be within a certain working range.
APD	Avalanche Photodiode. A semiconductor photodetector with integral detection and amplification stages. Electrons generated at a p/n junction are accelerated in a region where they free an avalanche of other electrons. APDs can detect faint signals but require higher voltages than other semiconductor electronics.
Asynchronous	A network where transmission system payloads are not synchronized and each network terminal runs on its own clock.
Attenuation	Reduction of signal magnitude or signal loss, usually expressed in decibels.
Attenuator	A passive component that attenuates an electrical or optical signal.
Automatic gain control	A technique which is used to adjust the gain of each wavelength signal within allowed range.
Auto-negotiation	The rate/work mode of the communication party set as self-negotiation is specified through negotiation according to the transmission rate of the opposite party.
B	
Back up	A method to copy the important data into a backing storage in case that the original is damaged or corrupted.
Backplane	A PCB circuit board in the subrack, which is connected with all the boards in position.
Bandwidth	Information-carrying capacity of a communication channel. Analog bandwidth is the range of signal frequencies that can be transmitted by a communication channel or network.
Bit error rate	The number of coding violations detected in a unit of time, usually one second. Bit error rate (BER) is calculated with this formula: BER = errored bits received/total bits sent
Bit error	An error occurs to some bits in the digital code stream after being received, judged, and regenerated, thus damaging the quality of the transmitted information.
bit/s	The number of bits passing a point every second. The transmission rate for digital information.

C	
Cabling	The methods to route the cables or fibers.
Centralized alarm system	The system that gathers all the information about alarms into a certain terminal console.
Chain network	One type of network that all network nodes are connected one after one to be in series.
Client	A kind of terminal (PC or workstation) connected to a network that can send instructions to a server and get results through a user interface. See also server.
Configuration management	Configuration management enables inventory query of network configuration resources, including relevant configuration of NMS or SNMS, NE, subnet, links, SNC, route, TP, edge point, equipment, and so on. Real-time inventory change report can also be provided through this resource, it will be timely reported to the upper NMS to notify the carrier of the current network operation status and ensure data consistency of the upper NMSs.
Configure	To set the basic parameters of an operation object.
Connection point	A reference point where the output of a trail termination source or a connection is bound to the input of another connection, or where the output of a connection is bound to the input of a trail termination sink or another connection. The connection point is characterized by the information which passes across it. A bidirectional connection point is formed by the association of a contradirectional pair.
Connection	A "transport entity" which consists of an associated pair of "unidirectional connections" capable of simultaneously transferring information in opposite directions between their respective inputs and outputs.
D	
DCF	Dispersion Compensation Fiber. A kind of fiber which uses negative dispersion to compensate for the positive dispersion of transmitting fiber to maintain the original shape of the signal pulse.
DCM	Dispersion Compensation Module. A module, which contains dispersion compensation fibers to compensate for the positive dispersion of transmitting fiber.
DCN	Data Communication Network. A communication network within a TMN or between TMNs which supports the data communication function (DCF).
Defect	A limited interruption in the ability of an item to perform a required function.
Demultiplexing	A process applied to a multiplex signal for recovering signals combined within it and for restoring the distinct individual channels of the signals.

Dense wavelength Division multiplexing	The higher capacity version of WDM, which is a means of increasing the capacity of fiber-optic data transmission systems through the multiplexing of multiple wavelengths of light. Commercially available DWDM systems support the multiplexing of from 8 to 40 wavelengths of light.
Distributed service	The transmitting services are distributed between each neighboring nodes connected over a ring network.
Domain	The domain of the T2000 specifies the scope of address or functions which are available to a certain user.
Dual-Fed	A description of a ring that has entry nodes that add traffic to the ring through the bridging function.
DWDM	Dense Wavelength Division Multiplexing. The technology utilizes the characteristics of broad bandwidth and low attenuation of single mode optical fiber, employs multiple wavelengths with specific frequency spacing as carriers, and allows multiple channels to transmit simultaneously in the same fiber.
E	
ECC	Embedded Control Channel. An ECC provides a logical operations channel between SDH NEs, utilizing a data communications channel (DCC) as its physical layer.
EDFA	Erbium-Doped Fiber Amplifier. The optical amplifier that its fiber doped with the rare earth element erbium, which can amplify at 1530 to 1610 nm when the optical amplifier is pumped by an external light source.
ESC	Electric Supervisory Channel. A technology realizes the communication among all the nodes and transmits the monitoring data in the optical transmission network. The monitoring data of ESC is introduced into DCC service overhead and is transmitted with service signals.
ESCON	Enterprise System Connection. A path protocol which connects the host with various control units in a storage system. It is a serial bit stream transmission protocol. The transmission rate is 200 Mbit/s.
ESD	Electrostatic Discharge. The phenomena the energy being produced by electrostatic resource discharge instantly.
Ethernet	A data link level protocol comprising the OSI model's bottom two layers. It is a broadcast networking technology that can use several different physical media, including twisted pair cable and coaxial cable. Ethernet usually uses CSMA/CD. TCP/IP is commonly used with Ethernet networks.
ETSI	European Telecommunications Standards Institute
Eye pattern	A graphic presentation formed by the superimposition of the waveforms of all possible pulse sequences.

F	
F1 byte	The user path byte, which is reserved for the user, but is typically special for network providers. The F1 byte is mainly used to provide the temporary data or voice path for special maintenance objectives. It belongs to the regenerator section overhead byte.
Fan tray assembly	A module which contains fans used for heat dissipation.
Fault	A fault is the inability of a function to perform a required action. This does not include an inability due to preventive maintenance, lack of external resources, or planned actions.
FC	Fiber Channel. A standard of data storage network for transmitting signals at 100 Mbit/s to 4.25Gbit/s over fiber or (at slow speeds) copper.
FDDI	Fiber Distributed Data Interface. A standard for a 100 Mbit/s fiber-optic local-area network.
Fiber channel	The channel which is used for fiber routing.
Fiber connector	A device mounted on the end of a fiber-optic cable, light source, receiver, or housing that mates to a similar device to couple light into and out of optical fibers. A connector joins two fiber ends, or one fiber end and a light source or detector.
Fiber jumper	The fiber which is used to connect the subrack with the ODF.
Fiber spool box	A box which is used to spool the fiber.
Fiber spool	The spool on the side of a subrack which is used for fiber routing.
FICON	Fiber Connect. A new generation connection protocol which connects the host with various control units. It carries single byte command protocol through the physical path of fiber channel, and provides higher rate and better performance than ESCON.
Frame	A cyclic set of consecutive time slots in which the relative position of each time slot can be identified.
G	
Gain spectrum-shape pre-tilt	The technology to keep the gain into being a basically fixed value.
Gain	The ratio between the optical power from the input optical interface of the optical amplifier and the optical power from the output optical interface of the jumper fiber, which expressed in dB.
H	
History alarms	Alarms that have been cleared and acknowledged.

History performance data	The performance data that is stored in the history register and the auto-report performance data that is stored on the T2000.
I	
Input jitter tolerance	For STS-N electrical interfaces input jitter tolerance is the maximum amplitude of sinusoidal jitter at a given jitter frequency, which when modulating the signal at an equipment input port, results in no more than two errored seconds cumulative, where these errored seconds are integrated over successive 30 second measurement intervals.
IP over DCC	The IP Over DCC follows TCP/IP telecommunications standards and controls the remote NEs through the Internet. The IP Over DCC means that the IP over DCC uses overhead DCC byte (the default is D1-D3) for communication.
IPA	Intelligent Power Adjustment. The technology that the system reduces the optical power of all the amplifiers in an adjacent regeneration section in the upstream to a safety level if the system detects the loss of optical signals on the link. The loss of optical signals may due to the fiber is broken, the performance of equipments trend to be inferior or the connector is not plugged well. Thus, the maintenance engineers are not hurt by the laser being sent out from the slice of broken fiber.
J	
Jitter tolerance	For STS-N electrical interfaces, input jitter tolerance is the maximum amplitude of sinusoidal jitter at a given jitter frequency, which results in no more than two errored seconds cumulative, when the signal is modulated at an equipment input port. These errored seconds are integrated over successive 30 second measurement intervals. Requirements on input jitter tolerance as just stated, are specified in terms of compliance with a jitter mask, which represents a combination of points. Each point corresponds to a minimum amplitude of sinusoidal jitter at a given jitter frequency which results in two or fewer errored seconds in a 30 second measurement interval when the signal is modulated at the equipment input port. For the OC-N optical interface, it is defined as the amplitude of the peak-to-peak sinusoidal jitter applied at the input of an OC-N interface that causes a 1 dB power penalty.
Jitter transfer	The physical relationship between jitter applied at the input port and the jitter appearing at the output port.
Jitter	Short waveform variations caused by vibration, voltage fluctuations, control system instability, and so on.
L	

Laser	The device that generates the directional light covering a narrow range of wavelengths. Laser light is more coherent than ordinary light. Semiconductor diode lasers are the used light source in fiber-optic system.
Layer	A concept used to allow the transport network functionality to be described hierarchically as successive levels; each layer being solely concerned with the generation and transfer of its characteristic information.
Link	A "topological component" that provides transport capacity between two endpoints in different subnetworks through a fixed (that is inflexible routing) relationship. The endpoints are "subnetwork termination point pools" for SONET, and link termination points for ATM. Multiple links may exist between a pair of subnetworks. A link also represents a set of "link connections".
Loopback	The fault of each path on the optical fiber can be located by setting loopback for each path of the line. There are three kinds of loopback modes: No loopback, Outloop, Inloop.
Lower subrack	The subrack close to the bottom of the cabinet when a cabinet contains several subracks.
M	
MAC	Media Access Control. The data link sublayer that is responsible for transferring data to and from the Physical Layer.
MAN	Metropolitan Area Network. An IEEE-approved network that supports high speeds over a metropolitan area.
Mean launched power	The average power of a pseudo-random data sequence coupled into the fiber by the transmitter.
MF	Mediation Function. A function that routes or acts on information passing between network elements and network operations in telecommunications network management.
Mounting ear	A component on the side of a subrack, which is used to install the subrack in a cabinet.
Multiplex	To transmit two or more signals over a single channel.
Multiplexer	An equipment which combines a number of tributary channels onto a fewer number of aggregate bearer channels, the relationship between the tributary and aggregate channels being fixed.
Multiplexing	A procedure by which multiple lower order path layer signals are adapted into a higher order path or the multiple higher order path layer signals are adapted into a multiplex section.
N	

NE ID	A unique identifier to which each NE corresponds in a network. In the OptiX transmission equipment, it is specified that the NE ID is a 24-bit binary digit, that is, three bytes. The DIP switch on the SCC board of the NE constitutes the lower 16 bits of the NE ID. The higher eight bits of the NE ID is the extended ID (default value: 9), which is also called the subnet number. The extended ID is typically used to identify different subnets.
Noise figure	The specification to scale the random signal in the system presenting in addition to any wanted signal.
NRZ	Non Return to Zero. A digital code in which the signal level is low for 0 bit and high for 1 bit and dose not return to 0 between successive 1 bits.
O	
OADM	Optical Add/Drop Multiplexer. A device that can be used to add the optical signals of various wavelengths to one channel and drop the optical signals of various wavelengths from one channel.
OCP	Optical Channel Protection. A protection mechanism supports working channels with multiple wavelengths and protection one in order to be against the situation that there is any fault in the working channel.
OLA	Optical Line Amplifier. A device that amplifies an optical signal in the transmitting link without converting it into electrical form.
OLP	Optical Line Protection. A protection mechanism supports a working path and a protection path with dual-fed signal selection function. Normally, the working path carries the traffic. The protection path will work to be against the situation that there is any fault in the working link.
ONE	Optical Network Element. A stand-alone physical entity in an optical transmission network that supports at least network element functions.
Online help	An indexed collection of information on all aspects of the T2000. They can be accessed at any time from the Help menu or by pressing the F1 key.
Optical amplifier	A device or subsystem in which optical signals can be amplified by means of stimulated emission taking place in a suitable active medium. It is used to amplify the optical signal of the optical transmission system.
Optical connector	A component normally attached to an optical cable or piece of apparatus for the purpose of providing frequent optical interconnection/disconnection of optical fibers or cables.
Optical demultiplexer	A device which performs the inverse operation of a wavelength multiplexer, where the input is an optical signal comprising two or more wavelength ranges and the output of each port is endowed with the different and preselected wavelength.

Optical interface	A device to allow two or more corresponding optical transmitting units to be connected.
Optical multiplexer	A branching device with two or more input ports and one output port where the light in each input port is restricted to a preselected wavelength range and the output is the combination of light from the input ports.
Optical spectrum analyzer	An instrument that scans the spectrum to record power, measures the value of loss insertion and tests the performance of the wavelength and optical signal noise ratio (OSNR) of each channel.
Optical switch	A passive component possessing two or more ports which selectively transmits, redirects, or blocks optical power in an optical fiber transmission line.
OSC	Optical Supervisory Channel. A technology realizes communication among nodes in optical transmission network and transmits the monitoring data in a certain channel (the wavelength of the working channel for it is 1510 nm and that of the corresponding protection one is 1625 nm).
OSNR	Optical Signal-to-Noise Ratio. Ratio of the optical power of the transmitted optical signal to the noise on the received signal.
OTDR	Optical Time Domain Reflectometer. An instrument that measures transmission characteristics by sending a short pulse of light down a fiber and observing backscattered light.
OTM	Optical Terminal Multiplexer. A device that multiplex or demultiplex optical signals into a transmission link or into the client side.
OTU	Optical Transponder board. A device that access service signals compliant with standards at the client side and convert them into standard DWDM or CWDM wavelengths.
Output optical power	The ranger of optical energy level of output signals.
Overhead	Extra bits in a digital stream used to carry information besides traffic signals. Orderwire, for example, would be considered overhead information.
P	
Path	A logical connection between the point at which a standard frame format for the signal at the given rate is assembled, and the point at which the standard frame format for the signal is disassembled.
PDH	Plesiochronous Digital Hierarchy. PDH is the digital networking hierarchy that was used before the advent of SONET/SDH.

Performance threshold	Performance events usually have upper and lower thresholds. When the performance event count value exceeds the upper threshold, a performance threshold-crossing event is generated; when the performance event count value is below the upper threshold for a period of time, the performance threshold-crossing event is ended. In this way, performance jitter caused by some sudden events can be shielded.
PIN	Photodiode. A semiconductor detector with an intrinsic region separating the p- and n-doped regions. It has fast linear response and is used in fiber-optic receivers.
Plesiochronous	A network with nodes timed by separate clock sources with almost the same timing.
PMU	Power Monitor Unit. One type of power and environment monitoring unit.
Polarization dependence loss	The maximum variation of loss result from a variation of the state of polarization of the input signal at nominal operating conditions.
Power and environment monitoring unit	The power and environment monitoring unit is installed at the top of the cabinet of the SDH equipment and is used to monitor the environment variables, such as the power supply and temperature. With external signal input through the relay, fire alarm, smoke alarm, burglary alarm, and so on. can be monitored as well.
Power box	A direct current power distribution box at the upper part of a cabinet, which supplies power for the subracks in the cabinet.
Procedure	A generic term for an action.
Process	A generic term for a collection of actions.
R	
Receiver overload	Receiver overload is the maximum acceptable value of the received average power at point R to achieve a 1×10^{-10} BER.
Receiver sensitivity	Receiver sensitivity is defined as the minimum acceptable value of average received power at point R to achieve a 1×10^{-10} BER.
Reflection coefficient	The difference between the amount of light incident and the amount that is reflected back from a surface.
REG	A device that performs regeneration.
Regeneration	The process of receiving and reconstructing a digital signal so that the amplitudes, waveforms and timing of its signal elements are constrained within specified limits.
Regenerator section overhead	The regenerator section overhead comprises rows 1 to 3 of the SOH of the STM-N signal.
Ring network	One type of network that all network nodes are connected one after one to be a cycle.

S	
SDH	Synchronous Digital Hierarchy. A hierarchical set of digital transport structures, standardized for the transport of suitably adapted payloads over physical transmission networks.
Service protection	The measures to make sure the service transmitting not to be damaged or corrupted.
Severity	See Alarm severity.
Side mode suppression ratio	The ratio of the largest peak of the total source spectrum to the second largest peak.
Span	The set of SONET lines between two adjacent nodes on a ring.
Splitter	A device that divides incident light into two separate beams.
Star network	Network of several nodes where each terminal is linked individually to a central node.
STM-N	Synchronous Transport Module. An STM is the information structure used to support section layer connections in the SDH. It consists of information payload and Section Overhead (SOH) information fields organised in a block frame structure which repeats every 125 ms. The information is suitably conditioned for serial transmission on the selected media at a rate which is synchronised to the network. A basic STM is defined at 155.520kbit/s.
SuperWDM	A technical solution can extend effectively the transmitting distance of DWDM system with the application of Super CRZ encoding and the advanced phase modulation capability.
Support	The frame on the bottom of a cabinet, when installing the cabinet on the antistatic floor.
Synchronous digital hierarchy	A hierarchical set of digital transport structures, standardized for the transportation of suitably adapted payloads over physical (primarily optical) transmission networks.
Synchronous	A network where transmission system payloads are synchronized to a master (network) clock and traced to a reference clock.
T	
T2000	The T2000 is a subnet management system (SNMS). In the telecommunication management network architecture, the T2000 is located between the NE level and network level, which can supports all NE level functions and part of the network level management functions. See also NM.

TCP/IP	Transmission Control Protocol/Internet Protocol. Common name for the suite of protocols developed to support the construction of worldwide internetworks.
Telecom management network	The entity which provides the means used to transport and process information related to management functions for the telecommunications network.
Timeslot	Single timeslot on an E1 digital interface—that is, a 64-kbps, synchronous, full-duplex data channel, typically used for a single voice connection.
TL1	Transaction Language 1, A Telcordia Technologies machine-to-machine communications language that is a subset of ITU-TSS, formerly CCITT's, human-machine language.
TMN	Telecommunications Management Network. The entity which provides the means used to transport and process information related to management functions for the telecommunications network.
Tray	A discal component in the cabinet, which is used to place the chassis or other equipment.
U	
Unit management layer	Designates the management functions performed on units assembled in a network.
Unprotected	Services transmitted through an ordinary way, once a failure or interruption occurs, the data cannot be restored for lack of protection mechanism.
User	The user of the T2000 client, and the user and password define the corresponding authority of operation and management of the T2000.
V	
VOA	Variable Optical Attenuator. An attenuator in which the attenuation can be varied.
W	
Wander	The long-term variations of the significant instants of a digital signal from their ideal position in time (where long-term implies that these variations are of frequency less than 10Hz).
Wavelength division multiplexing	A means of increasing the capacity of fiber-optic data transmission systems through the multiplexing of multiple wavelengths of light. WDM systems support the multiplexing of as many as four wavelengths.

WDM	Wavelength-Division Multiplexing. WDM technology utilizes the characteristics of broad bandwidth and low attenuation of single mode optical fiber, employs multiple wavelengths as carriers, and allows multiple channels to transmit simultaneously in a single fiber.
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F Acronyms and Abbreviations

A	
ADM	Add and drop multiplexer
AGC	Automatic gain control
ALC	Automatic level control
ALS	Automatic laser shutdown
APE	Automatic power equilibrium
APS	Automatic protection switching
ASE	Amplified spontaneous emission
AWG	Arrayed waveguide grating
B	
BA	Booster amplifier
BER	Bit error ratio
C	
CLNS	Connectionless network service
CMI	Coded mark inversion
CPU	Central processing unit
CRC	Cyclical redundancy check
CRZ	Chirped return to zero
CSES	Continuous severely errored second
CWDM	Coarse Wavelength Division Multiplex

D	
DCC	Data communication channel
DCF	Dispersion compensation fiber
DCM	Dispersion compensation module
DCN	Data communication network
DDN	Digital data network
DFB	Distributed feedback
DSP	Digital signal processing
DSCR	Dispersion slope compensation rate
DWDM	Dense wavelength division multiplexing
DRZ	Differential phase return to zero
E	
ECC	Embedded control channel
EDFA	Erbium-doped fiber amplifier
EFEC	Enhanced forward error correction
ELH	Extra long haul
EMC	Electromagnetic compatibility
ETSI	European Telecommunication Standards Institute
F	
FEC	Forward error correction
FWM	Four-wave mixing
G	
GE	Gigabit Ethernet
GFF	Gain flattening filter
GUI	Graphic user interface
I	

IEEE	Institute of Electrical and Electronic Engineers
IPA	Intelligent power adjustment
ITU-T	International Telecommunication Union-Telecommunication Standardisation Sector
L	
LAN	Local area network
LCN	Local communication network
LCT	Local craft terminal
LD	Laser diode
LHP	Long Hop
M	
MCF	Message communication function
MD	Mediation device
MPI-R	Main path interface at the receiver
MPI-S	Main path interface at the transmitter
MQW	Multi-quantum well
N	
NE	Network element
NF	Noise figure
NRZ	Non return to zero
O	
OA	Optical amplifier
OADM	Optical add and drop multiplexer
OAM	Operation, administration and maintenance
OAMS	Optical fiber line automatic monitoring system
OD	Optical demultiplexing
ODF	Optical distribution frame

OEQ	Optical equaliser
OHP	Overhead processing
OLA	Optical line amplifier
OM	Optical multiplexing
OMS	Optical multiplex section
ORL	Optical return loss
OS	Operations system
OSI	Open systems interconnection
OSNR	Optical signal to noise ratio
OTDR	Optical time domain reflectometer
OTM	Optical terminal multiplexer
OTS	Optical transmission section
OTT	Optical tunable transponder
OTU	Optical transponder board
P	
PDH	Plesiochronous digital hierarchy
PDL	Polarization dependent loss
PIN	Positive intrinsic negative
PMD	Polarization mode dispersion
POS	Packet Over SDH/SONET
R	
RS	Reed-Solomon
RTU	Remote test unit
Q	
QA	Q adaptation
S	
SBS	Stimulated Brillouin Scattering

SCC	System control & communication
SDH	Synchronous digital hierarchy
SLIP	Serial line internet protocol
SLM	Single longitudinal mode
SONET	Synchronous optical network
SPM	Self phase modulation
SRS	Stimulated Raman Scattering
STM	Synchronous transport module
Super CRZ	Super chirped return to zero
T	
TCP/IP	Transport control protocol / Internet protocol
TDM	Time division multiplexing
TEC	Thermoelectric cool
TMN	Telecommunication management network
TTL	Transistor-transistor logic
X	
XPM	Cross phase modulation
W	
WDM	Wavelength division multiplexing
WS	Work station